
Drug Discovery & Development Basics

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26-10-02

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Ten-page summary: the essential mental model

The central idea

Drug discovery is the process of finding and developing interventions that change disease biology enough to help patients, at an acceptable level of risk.

The central chain is:

```
Disease mechanism
→ therapeutic hypothesis
→ target or biological intervention
→ therapeutic modality and product
→ exposure at the relevant tissue
→ target engagement and pathway change
→ patient benefit and toxicity
```

Every arrow is a possible point of failure. A molecule can bind a protein strongly and still fail as a medicine because the target is not sufficiently important in the relevant patients, the intervention cannot reach the relevant tissue, cells compensate through alternative pathways, tumour cells evolve resistance, the required exposure is toxic, or the clinical benefit is too small relative to existing treatment.

Drug discovery is therefore not simply the search for a molecule that binds a protein. It is an attempt to find a practical intervention that produces a sufficiently large and durable clinical benefit, in an appropriate patient population, without causing unacceptable harm.

The most useful high-level question is:

```
What biological state causes or maintains disease,
and can it be changed safely enough to help patients?
```

A useful medicine must satisfy the whole causal chain. Strong evidence at one step does not remove uncertainty at the next.

Biology: cells, genes, proteins, and pathways

A cell is the basic living unit of the body. Cells take up nutrients, use energy, make molecules, sense their environment, communicate with other cells, move, divide, repair damage, and die when appropriate.

Different cell types usually contain nearly the same DNA but behave differently because they use different subsets of genes. A neuron, immune cell, liver cell, and cancer cell can therefore have very different functions despite sharing most of the same genetic instructions.

The simplified flow of biological information is:

```
DNA
→ RNA
→ protein
→ cell behaviour
```

DNA stores genetic information. A gene is a region of DNA that can produce a functional product. Many genes are transcribed into messenger RNA, or mRNA. The mRNA is translated by ribosomes into an amino-acid chain, which folds into a protein.

Proteins carry out much of the molecular work of cells. They can act as enzymes, receptors, transporters, transcription factors, structural components, signalling molecules, antibodies, and components of the immune system.

A protein is not merely an amino-acid sequence. Its function also depends on its three-dimensional structure, dynamic conformations, location, interactions with other molecules, chemical modifications, and cellular context.

A mutation is a change in DNA sequence. Some mutations have little effect. Others can change an amino-acid sequence, alter how much protein is made, change how a protein is regulated, or affect whether a gene is used at all. In disease, mutations can create an important biological dependency, remove a normal biological control, or alter sensitivity to treatment.

Signals, receptors, enzymes, and pathways

Cells receive information from their environment through signalling systems. A *receptor* is a protein that detects a particular signal. A *ligand* is a molecule that binds a receptor. Ligands include hormones, growth factors, cytokines, neurotransmitters, metabolites, and therapeutic drugs.

A simplified signalling sequence is:

```
Ligand binds receptor
→ receptor changes state
→ intracellular signalling proteins are activated
→ gene expression or protein activity changes
→ cell behaviour changes
```

An *enzyme* is usually a protein that accelerates a chemical reaction. The molecule on which an enzyme acts is called its *substrate*. The part of an enzyme that binds the substrate and carries out the reaction is often called the *active site*.

Kinases are especially important enzymes in cell signalling. A kinase commonly transfers a phosphate group from ATP to another protein. This process is called phosphorylation.

```
Kinase + ATP + substrate protein
→ kinase + ADP + phosphorylated substrate protein
```

Phosphorylation can change a protein's activity, shape, location, stability, or interactions with other proteins. It is therefore a common way for cells to transmit and regulate information.

A signalling pathway is a connected set of molecular events through which one signal changes cell behaviour. Real pathways are not simple straight lines. They include feedback loops, redundancy, cross-talk, and context dependence. Blocking one pathway component can work well when disease cells depend strongly on it, but alternative routes can sometimes compensate.

Chemistry: molecules, water, and molecular properties

Many medicines are *small molecules*: discrete chemical compounds that can often enter cells, bind proteins, and alter biological processes. A small molecule is composed of atoms joined by covalent bonds.

A molecule can be represented as a graph:

```
Atoms
→ graph nodes
Covalent bonds
→ graph edges
Atom and bond properties
→ node and edge features
```

A molecular graph is useful, but it is not the whole molecule. Drug-relevant behaviour can also depend on stereochemistry, protonation state, tautomeric form, three-dimensional conformation, interactions with water, and the biological environment.

A *functional group* is a chemically meaningful arrangement of atoms that influences molecular behaviour. Examples include amines, alcohols, carboxylic acids, amides, esters, aromatic rings, heterocycles, and halogens.

Functional groups can affect:

```
Hydrogen bonding
→ protein binding and water solubility
Ionisation and charge
→ permeability, distribution, and binding
Chemical reactivity
→ stability and possible toxicity
Metabolic susceptibility
→ clearance and half-life
```

Most conventional protein–ligand binding depends mainly on *non-covalent interactions*. Important examples include electrostatic attraction or repulsion between charged groups, hydrogen bonds, van der Waals interactions, hydrophobic interactions, and interactions involving aromatic electron systems.

Binding does not depend simply on the number of favourable contacts visible in a protein–ligand structure. It depends on the total balance of favourable and unfavourable effects:

```
Favourable molecular interactions
- desolvation costs
- conformational strain
- loss of flexibility
- unfavourable interactions
→ net binding affinity
```

Solubility is the extent to which a compound can dissolve under defined conditions. Poor solubility can prevent adequate absorption, complicate formulation, produce misleading assay results, and limit the concentration that can be tested or administered.

Lipophilicity is the tendency of a compound to partition into non-polar environments rather than water. Greater lipophilicity can sometimes improve membrane crossing and hydrophobic interactions in protein pockets. Excessive lipophilicity can reduce solubility, increase non-specific binding, increase metabolic or toxicity risk, and make optimisation more difficult.

The aim is not to maximise one property. It is to achieve an appropriate balance for the target, disease, route of administration, and desired tissue exposure.

Charge, stereochemistry, and molecular conformation

Many drug-like molecules can gain or lose protons depending on pH. This changes their charge state. The *pKa* of an ionisable group is related to the pH at which that group is approximately half protonated and half deprotonated.

A basic group, such as many amines, is more likely to be protonated and positively charged at lower pH. An acidic group, such as a carboxylic acid, is more likely to be deprotonated and negatively charged at higher pH.

```
Charge state
→ solubility
→ membrane permeability
→ tissue distribution
→ protein binding
→ target interaction
```

A compound may be potent against a purified protein but fail in cells if its dominant charge state prevents sufficient entry across the cell membrane.

Stereochemistry describes the three-dimensional arrangement of atoms in a molecule. Molecules can have the same atoms and connectivity but differ in their three-dimensional arrangement. These forms can have very different potency, selectivity, metabolism, toxicity, and clinical effects.

Molecules are also flexible. Rotation around many single bonds produces alternative three-dimensional arrangements called *conformations*, or *conformers*. The collection of conformations accessible to a molecule under particular conditions is its *conformational ensemble*.

Binding may depend on:

```
Accessible ligand conformation
+ compatible protein conformation
+ favourable interaction pattern
→ possible productive binding
```

Flexibility can help a molecule adapt to a binding site, but excessive flexibility can create an energetic penalty when the molecule becomes constrained on binding.

Protein structure, binding sites, and therapeutic opportunities

Proteins are chains of amino acids that fold into three-dimensional structures. Protein structure is often described at four levels:

```
Primary structure
→ amino-acid sequence
Secondary structure
→ local helices and beta sheets
Tertiary structure
→ overall fold of one protein chain
Quaternary structure
→ arrangement of multiple protein chains
```

Proteins are dynamic rather than rigid. They can move, fluctuate, rearrange side chains, bind cofactors, interact with other proteins, and sample active and inactive conformations.

A structure obtained by X-ray crystallography, cryo-electron microscopy, NMR, or computational prediction is highly informative, but it usually represents one or a limited number of states under particular conditions. It does not necessarily show every biologically relevant conformation.

A *binding site* is a region of a protein where another molecule can bind. An *active site* is the region of an enzyme where its normal chemical reaction occurs. A ligand can bind an active site, an allosteric site, a protein–protein interaction surface, or another regulatory region.

An *orthosteric site* is the main functional site normally used by a substrate, cofactor, or endogenous ligand. An *allosteric site* is a distinct site elsewhere on the protein where binding changes protein shape, dynamics, interactions, or activity.

A *cryptic pocket* is a binding pocket that is not obvious in one static protein structure but becomes accessible in another protein conformation.

Protein changes conformation
→ new pocket becomes accessible
→ ligand may bind and stabilise that state
→ protein function may change

Structural information can help identify possible binding sites, selectivity opportunities, antibody epitopes, resistance mechanisms, and plausible therapeutic mechanisms. It generates hypotheses; it does not prove that a target is clinically valid or that a therapeutic molecule will work in patients.

Several concepts should remain separate:

Biological importance
→ does this protein matter in disease?
Ligandability
→ does it contain a useful site for ligand binding?
Druggability
→ can a modality modulate it safely and effectively?
Clinical validation
→ does changing it help patients?

Disease biology and cancer as an example

Disease occurs when cells, tissues, organs, pathogens, or immune systems enter an inappropriate biological state. A medicine aims to change some part of that state.

Cancer is a useful example because it combines genetics, cell signalling, molecular targets, drug resistance, biomarkers, immune biology, and clinical development. Cancer is not one disease. It is a collection of diseases in which cells acquire abnormal capabilities that allow them to grow, survive, spread, and evade normal biological control.

Cancer can be understood as an evolutionary process:

Cells acquire biological alterations
→ some alterations create growth or survival advantage
→ advantaged cells produce descendants
→ further alterations accumulate
→ tumour-cell populations compete and evolve

An *oncogene* promotes cancer when abnormally activated, overexpressed, amplified, or rearranged. A *tumour suppressor gene* normally restrains inappropriate cell growth, supports DNA repair, or promotes removal of damaged cells.

A *driver mutation* contributes causally to tumour development, maintenance, or progression. A *passenger mutation* is present but does not meaningfully drive the cancer phenotype.

Some tumours are strongly dependent on one pathway or oncogenic signal. This is often called *oncogene addiction*. Targeting that dependency can produce substantial clinical responses.

However, tumours are heterogeneous. Different cells within one tumour, different lesions in one patient, and the same tumour at different times can differ in mutations, gene expression, cell state, immune interactions, and drug sensitivity.

Treatment creates selection pressure:

```
Treatment kills sensitive cells
→ resistant cells survive
→ resistant clones expand
→ disease can progress
```

Resistance can arise through target mutations, target loss, bypass signalling, altered drug exposure, changes in cell state, immune escape, or other mechanisms.

Biomarkers, targets, and therapeutic modalities

A *biomarker* is a measurable biological characteristic relevant to disease or treatment. Biomarkers can help diagnose disease, classify disease subtypes, estimate prognosis, select treatments, monitor response, or detect toxicity.

```
Diagnostic biomarker
→ helps identify or classify disease
Prognostic biomarker
→ provides information about likely outcome
regardless of treatment
Predictive biomarker
→ identifies patients more or less likely to benefit
from a particular treatment
```

A *target* is a biological entity whose modulation is intended to create therapeutic benefit. Targets are often proteins, but can also be RNAs, protein–protein interactions, cell-surface antigens, cells, or broader biological pathways.

Therapeutic modalities include small molecules, antibodies, antibody–drug conjugates, peptides, nucleic-acid medicines, protein degraders, vaccines, and cell therapies.

The location of a target matters. Intracellular proteins are often more accessible to small molecules. Extracellular and cell-surface proteins are often more accessible to antibodies, antibody–drug conjugates, and engineered immune-cell therapies. These are useful heuristics rather than absolute rules.

Modality should be chosen according to the biology and therapeutic goal:

```
Intracellular enzyme or signalling protein
→ small molecule or degrader may be feasible
Extracellular ligand or cell-surface receptor
→ antibody may block, activate, or recruit immunity
Internalising cell-surface antigen
→ ADC may enable preferential payload delivery
Tumour mutation presented through MHC
→ vaccine or conventional T-cell response may be relevant
Accessible cell-surface antigen
→ CAR-T or other cell therapy may be possible
```

Pharmacology: from binding to effect

Pharmacology studies how drugs interact with biological systems and how those interactions create benefit and harm.

A drug can act through different mechanisms:

```
Agonist
→ activates a receptor
Antagonist
→ prevents receptor activation
Inhibitor
→ reduces target function
Neutralising antibody
→ blocks a ligand or receptor interaction
Degradar
→ reduces the amount of target protein
Allosteric modulator
→ changes target behaviour from a distinct binding site
```

Affinity describes how strongly a molecule binds a target under defined conditions. Potency describes the concentration or dose needed to produce a specified effect in a particular assay. Efficacy describes the maximum effect achievable in that system. Selectivity describes the degree to which a drug acts on its intended target rather than other targets.

These concepts are related but not interchangeable:

```
High affinity
≠ high efficacy
High potency in one assay
≠ high potency in every biological setting
Target engagement
≠ clinical benefit
```

The central pharmacological chain is:

```
Drug binds or alters target
→ target function changes
→ pathway changes
→ cell behaviour changes
→ tissue or organism changes
→ clinical benefit and/or toxicity
```

A compound can bind a purified protein strongly but fail clinically because it does not reach the relevant tissue, does not alter the target enough, affects unintended targets, is too toxic, or acts on a target that is not sufficiently important in the relevant patients.

PK, PD, exposure, and dose

A *dose* is the amount of drug administered. *Exposure* is the concentration of drug in the body over time. Two patients can receive the same dose but experience different exposure because of differences in absorption, metabolism, clearance, organ function, body size, diet, genetics, disease state, or interactions with other medicines.

Pharmacokinetics, or PK, describes what the body does to the drug:

Absorption

→ entry into systemic circulation

Distribution

→ movement from blood into tissues

Metabolism

→ chemical modification of the drug

Excretion

→ removal of drug or metabolites

AUC, or area under the concentration–time curve, describes total systemic exposure over a specified period. $C_{\max}$ describes the highest observed concentration after dosing. Half-life describes the approximate time required for concentration to decrease by half during an appropriate elimination phase.

Pharmacodynamics, or PD, describes what the drug does to the body. PD can include target occupancy, pathway-biomarker change, altered cell behaviour, tumour shrinkage, symptom improvement, and toxicity.

The practical chain is:

Dose

→ systemic exposure

→ tissue exposure

→ target engagement

→ pathway modulation

→ cellular and tissue effect

→ clinical outcome

Plasma exposure is informative but incomplete. A drug may reach high concentration in blood while having insufficient free concentration in a tumour, brain, inflamed tissue, or intracellular compartment.

The *therapeutic window* is the range of exposure expected to provide useful benefit without unacceptable toxicity:

Too little exposure

→ inadequate target modulation or efficacy

Appropriate exposure

→ useful biological effect and acceptable safety

Too much exposure

→ unacceptable toxicity

Small molecules: from a hit to a credible candidate

Small molecules are particularly useful for many intracellular targets because some can cross cell membranes. They can inhibit enzymes, bind allosteric pockets, alter receptors, modulate ion channels or transporters, form covalent bonds with selected targets, or recruit protein-degradation machinery.

Their small size does not guarantee useful PK, tissue penetration, selectivity, or safety.

A *hit* is an initial molecule with reproducible evidence of desired activity. A *lead* is a more mature compound or chemical series with enough evidence of target-relevant activity and developability to justify systematic optimisation.

Small-molecule discovery is an iterative process:

Define a biological and chemical hypothesis

- design or select compounds
- synthesise and purify compounds
- test in relevant assays
- analyse activity, selectivity, PK, safety, and structure
- update the hypothesis
- repeat

Medicinal chemistry is a multi-objective optimisation problem:

Potency

- + selectivity
- + cellular activity
- + solubility
- + permeability
- + metabolic stability
- + tissue exposure
- + safety
- + synthetic feasibility
- + formulation and manufacturability

Improving one property can worsen another. The objective is not the best value in one assay; it is a balanced overall profile suitable for a defined therapeutic use.

KRAS illustrates this principle. KRAS was historically considered difficult to target because it appeared to have few suitable pockets, binds GDP and GTP tightly, and functions through a dynamic conformational cycle. The KRAS G12C mutation created an opportunity for covalent inhibitors to bind a pocket accessible when KRAS visits its inactive GDP-bound state.

KRAS G12C enters GDP-bound state

- inhibitor binds switch II pocket and mutant cysteine
- inhibitor traps KRAS in an inactive state
- downstream signalling is reduced

This shows that a target once considered difficult can become tractable through structural insight, chemical design, and mutation-specific biology. It also shows that resistance, incomplete pathway suppression, tumour heterogeneity, and safety remain important after a successful targeting strategy is found.

Antibodies and antibody–drug conjugates

Antibodies are proteins that recognise exposed molecular targets. A monoclonal antibody is a population of identical antibodies designed to bind the same target region, called an *epitope*.

Fab regions

- bind the target epitope

Fc region

- interacts with immune-system components
- and influences antibody behaviour in the body

Conventional antibodies usually do not enter living cells efficiently. They are therefore especially useful for soluble extracellular proteins and cell-surface targets.

Antibodies can neutralise a ligand, block or activate a receptor, alter receptor internalisation, recruit immune effector cells, or deliver a payload. Two antibodies against the same target can have different biological effects because they may bind different epitopes or have different Fc properties.

HER2 is a cell-surface receptor tyrosine kinase. In some cancers, *ERBB2* gene amplification or HER2 overexpression produces excessive HER2-associated signalling. HER2 can therefore be both a therapeutic target and a predictive biomarker.

Trastuzumab is a HER2-directed antibody. Its activity is not simply “switching off HER2.” It can perturb HER2-associated biology and recruit immune-mediated killing through *antibody-dependent cellular cytotoxicity*, or ADCC.

```
Trastuzumab binds HER2 on tumour cell
→ Fc region is recognised by immune effector cell
→ immune cell becomes activated
→ tumour cell may be attacked and killed
```

An *antibody-drug conjugate*, or ADC, combines a targeting antibody, linker, and potent cytotoxic payload.

```
Targeting antibody
+
Linker
+
Cytotoxic payload
```

An ADC is not merely “an antibody plus chemotherapy.” Its activity depends on target expression and accessibility, tumour penetration, internalisation, intracellular trafficking, linker processing, payload release, payload sensitivity, and safety.

Trastuzumab deruxtecan, or T-DXd, is a HER2-directed ADC. Its trastuzumab-based antibody recognises HER2, while its DXd payload inhibits topoisomerase I after intracellular release.

```
T-DXd binds HER2
→ HER2-ADC complex is internalised
→ linker is processed and DXd is released
→ topoisomerase-I inhibition causes replication stress and DNA damage
→ susceptible cell may die
```

DXd can diffuse locally from targeted cells into nearby cells. This *bystander effect* may help treat heterogeneous tumours, but it also reduces the precision of payload exposure and can contribute to toxicity.

The common HER2 target does not mean that trastuzumab and T-DXd have the same pharmacology, efficacy, safety profile, or resistance mechanisms.

T cells, immune checkpoints, and pembrolizumab

The adaptive immune system can recognise particular molecular features and retain immunological memory. T cells are central to this system.

T cells generally do not recognise whole proteins directly. Their T-cell receptors recognise short peptides presented at the cell surface by *major histocompatibility complex*, or MHC, molecules.

```
Protein is processed into peptide
→ peptide is loaded onto MHC
→ peptide-MHC complex is displayed at cell surface
→ T cell can inspect it
```

MHC class I is present on most nucleated cells and commonly presents peptides to CD8 T cells. MHC class II is mainly present on specialised antigen-presenting cells and commonly presents peptides to CD4 T cells.

```
CD8 T cells
→ can directly kill recognised abnormal cells
CD4 T cells
→ coordinate and sustain immune responses
```

Cancer mutations can create *neoantigens*: mutation-derived peptides that may be presented by MHC and recognised by T cells. Not every mutation produces a useful neoantigen. The altered peptide must be produced, processed, bind the patient's MHC molecules, be displayed, and be recognised by an appropriate T-cell receptor.

T-cell activation requires more than peptide–MHC recognition:

```
Signal 1
→ T-cell receptor recognises peptide-MHC
Signal 2
→ co-stimulation indicates an appropriate immune context
Signal 3
→ cytokines and local environment shape T-cell behaviour
Inhibitory signals
→ restrain activation and limit tissue damage
```

PD-1 is an inhibitory receptor expressed on activated T cells. PD-L1 and PD-L2 can bind PD-1 and reduce T-cell activity. This pathway normally helps limit harmful immune responses, but tumours can exploit it to suppress anti-tumour immunity.

Pembrolizumab is a PD-1-blocking antibody.

```
Without pembrolizumab:
PD-L1 or PD-L2 binds PD-1
→ inhibitory T-cell signalling
With pembrolizumab:
Pembrolizumab binds PD-1
→ reduced PD-1-mediated inhibition
→ tumour-reactive T cells may sustain or regain activity
```

Pembrolizumab does not directly kill tumour cells. It is most likely to help when tumour-reactive T cells are present or can develop, tumour antigens can be presented, T cells can enter the tumour, and PD-1-mediated inhibition is an important barrier to effective immunity.

PD-L1 can be informative but is an imperfect predictive biomarker. Response also depends on antigenicity, MHC presentation, T-cell priming, immune-cell infiltration, other suppressive pathways, tumour heterogeneity, and sensitivity to immune killing.

Because PD-1 also helps restrain immune responses against normal tissues, blocking it can cause *immune-related adverse events*. These can affect multiple organs and illustrate the central trade-off: stronger anti-tumour immunity can also increase harmful immune activity.

Cancer vaccines, personalised neoantigen therapy, and CAR-T cells

A therapeutic cancer vaccine aims to stimulate or strengthen an immune response against cancer-associated antigens. Unlike preventive vaccines against infection, it is generally given to people who already have cancer or who are at risk of recurrence after curative-intent treatment.

A personalised neoantigen vaccine uses information from an individual patient's tumour:

Tumour and matched normal samples

- identify tumour-specific mutations
- infer patient-specific MHC type
- predict and prioritise candidate neoantigens
- manufacture personalised vaccine
- stimulate tumour-reactive T-cell responses

mRNA neoantigen vaccines contain messenger RNA encoding selected neoantigen sequences, commonly delivered in lipid nanoparticles. The mRNA provides temporary instructions for cells to produce antigenic material; it does not alter the patient's DNA.

mRNA enters suitable cells

- encoded antigenic material is produced
- peptides are processed and displayed by MHC
- neoantigen-reactive T cells can be activated or expanded

Combining a neoantigen vaccine with pembrolizumab has a clear biological rationale:

Neoantigen vaccine

- aims to generate or expand tumour-specific T cells

Pembrolizumab

- reduces PD-1-mediated inhibition of those T cells

Combination

- aims to improve recognition and sustained anti-tumour activity

As of **26 August 2026**, intismeran autogene, also called mRNA-4157 or V940, remains an *investigational* individualised mRNA neoantigen therapy. On 19 August 2026, Merck and Moderna announced positive topline results from the Phase 3 *INTerpath-001* trial of intismeran autogene plus pembrolizumab in completely resected stage IIB–IV melanoma. The companies reported that the trial met its primary endpoint of recurrence-free survival and a key secondary endpoint of distant-metastasis-free survival versus pembrolizumab alone.

These findings are important but should be interpreted appropriately: they are topline company-reported results, not yet a complete peer-reviewed evidence package. Detailed effect sizes, follow-up, subgroup findings, safety data, quality-of-life outcomes, and overall-survival results remain necessary for full clinical interpretation.

Cell therapy uses living cells as the therapeutic product. CAR-T therapy modifies T cells to express a *chimeric antigen receptor*, or CAR.

Extracellular recognition domain

- often antibody-derived and binds cell-surface antigen

Transmembrane region

- anchors receptor in T-cell membrane

Intracellular signalling domains

- activate T cell after antigen binding

Unlike a conventional T-cell receptor, a CAR can recognise a selected cell-surface antigen directly without requiring peptide processing and MHC presentation.

CAR-T cell binds target antigen

- CAR signalling activates T cell
- cytotoxic molecules and cytokines are released
- target cell may be killed

CAR-T target selection is crucial. An ideal antigen is strongly and consistently expressed on tumour cells, accessible at the cell surface, absent from essential normal tissues, and difficult for tumour cells to lose.

This is especially difficult in solid tumours. Many antigens are tumour-associated rather than truly tumour-specific, tumour expression is heterogeneous, and the tumour microenvironment can limit T-cell entry and function.

On-target, off-tumour toxicity occurs when the therapy correctly recognises its intended antigen, but the same antigen is also expressed on healthy tissue. CAR-T therapy can also cause systemic immune toxicities, including cytokine release syndrome and neurological toxicity.

Cancer vaccines and CAR-T therapy therefore use immune recognition differently:

Neoantigen vaccine

→ aims to teach or strengthen peptide-MHC-specific immunity

CAR-T therapy

→ engineers T cells to recognise a chosen cell-surface antigen directly

Assays, measurements, and experimental evidence

An *assay* is an experimental system used to measure a chemical, biological, pharmacological, or clinical property. Assays can measure binding, enzyme activity, cell viability, gene expression, protein phosphorylation, immune-cell activation, drug concentration, toxicity, or clinical outcomes.

Different assays answer different questions:

Biochemical assay

→ does an intervention bind or alter purified target activity?

Cell-based assay

→ can it produce the expected effect in living cells?

Preclinical model

→ does it affect disease-relevant biology in a more complex system?

Clinical study

→ does it provide meaningful patient benefit?

Results from one system do not automatically transfer to another. A compound can inhibit a purified enzyme but fail in cells because it cannot enter cells, is unstable, is exported by transporters, is metabolised, or encounters a different biological context.

Common measurements include K_d , K_i , IC50, and EC50. These measurements depend on assay conditions, including protein construct, substrate concentration, cell type, incubation time, readout technology, temperature, buffer composition, and data processing.

Experimental data can contain measurement noise, batch effects, assay drift, missing values, censored values, systematic bias, and irreproducible findings. A measurement is evidence about a particular system under particular conditions; it is not automatically a timeless fact about a molecule or target.

From therapeutic hypothesis to medicine

A candidate must survive many stages beyond early evidence of binding or cell activity.

```
Unmet medical need
→ disease hypothesis
→ target identification and validation
→ assay development
→ hit discovery
→ optimisation
→ candidate selection
→ preclinical development
→ clinical trials
→ regulatory review
→ lifecycle management
```

The process is iterative rather than strictly linear. New evidence can require changes to the target hypothesis, modality, molecule, dose, biomarker strategy, patient population, manufacturing approach, or trial design. It can also justify stopping the programme.

A development candidate needs a coherent overall package:

```
Plausible mechanism
+
Adequate pharmacology and selectivity
+
PK and tissue exposure
+
Safety margin
+
Manufacturing quality and formulation
+
Biomarker strategy
+
Credible clinical-development plan
```

Before clinical testing, preclinical development assesses pharmacology, PK/PD, tissue distribution, toxicology, safety pharmacology, formulation, and manufacturing quality. The objective is not to prove clinical benefit, but to establish a sufficiently credible package for carefully controlled testing in humans.

Clinical development is often described in phases:

```
Phase I
→ safety, tolerability, PK, PD, target engagement,
dose, schedule, and early activity in humans
Phase II
→ evidence of activity, dose refinement, safety,
biomarker strategy, and patient-population selection
Phase III
→ confirmatory evidence of benefit-risk
relative to an appropriate comparator
```

The highest tolerable dose is not automatically the best dose. Dose selection should integrate safety, PK, exposure, target engagement, biological activity, preliminary efficacy, long-term tolerability, quality of life, and treatment burden.

Clinical evidence: what ultimately matters

Clinical development asks:

For which patients,
compared with what standard of care,
at which dose and schedule,
does the treatment provide clinically meaningful benefit
that justifies its risks?

An *indication* defines the specific disease setting for which a therapy is studied or approved. In oncology, it may include cancer type, disease stage, biomarker status, resectability, treatment line, prior therapy, and combination context.

A well-designed clinical trial requires appropriate eligibility criteria, comparator, endpoints, dose and schedule, safety monitoring, and statistical plan.

Randomisation helps make treatment groups comparable on average. Control groups show what outcomes might have occurred without the investigational treatment. Blinding, when feasible, can reduce bias in outcome assessment and clinical decision-making.

Endpoints may include tumour response, duration of response, progression-free survival, recurrence-free survival, distant-metastasis-free survival, overall survival, symptoms, function, quality of life, or other patient-relevant outcomes.

Objective response rate
→ proportion of patients with predefined tumour shrinkage
Duration of response
→ how long those responses last
Progression-free survival
→ time until progression or death
Overall survival
→ time until death from any cause
Recurrence-free survival
→ time until recurrence or death after curative-intent treatment
Distant-metastasis-free survival
→ time until distant recurrence or death

A *surrogate endpoint* is expected to predict clinical benefit but is not itself necessarily a direct patient-relevant outcome. Tumour shrinkage, for example, can be important evidence of activity but does not automatically prove longer survival, improved symptoms, or improved quality of life.

Patient-reported outcomes are important because they measure symptoms, function, wellbeing, and treatment burden directly from patients. A statistically significant result is not automatically clinically meaningful if the benefit is small, short-lived, offset by toxicity, or does not improve outcomes that matter to patients.

Safety must be interpreted in context. Higher toxicity may be acceptable in an aggressive, otherwise lethal disease with few alternatives, but the same toxicity may be unacceptable in earlier-stage disease with curative options or effective standard treatment.

Benefit-risk is therefore context-specific:

```
Magnitude and durability of benefit
+
Survival, recurrence, symptoms, and quality of life
+
Frequency, severity, and reversibility of harms
+
Treatment burden and feasibility
+
Available alternatives
→ overall benefit-risk
```

Approval is not the end of evidence generation. Ongoing safety monitoring and real-world evidence can clarify effectiveness, rare adverse events, use in broader populations, treatment sequencing, adherence, and opportunities for new indications or improved dosing.

The essential integrated picture

A useful medicine must satisfy the entire causal chain:

```
Right disease hypothesis
+ right patient population
+ right target or biological intervention
+ right therapeutic modality
+ suitable chemical, biological, and pharmaceutical properties
+ sufficient exposure at the relevant tissue
+ adequate target engagement and pathway modulation
+ meaningful efficacy
+ acceptable safety and treatment burden
+ feasible manufacture and delivery
```

The most important distinctions to retain are:

```
DNA stores instructions;
proteins perform much of the molecular work.
A target can be biologically important
without being ligandable, druggable, or clinically validated.
Protein structures are dynamic models;
a plausible binding site or docking pose is a hypothesis,
not proof of therapeutic activity.
Binding affinity is not the same as potency;
potency is not the same as efficacy;
target engagement is not the same as patient benefit.
Dose is not exposure;
plasma exposure is not tissue exposure;
tissue exposure is not necessarily target engagement.
A compound or biologic active in one assay
is not automatically active in cells, animal models, or patients.
A successful molecule or engineered cell product
is not automatically a medicine;
it must also be safe, deliverable, manufacturable,
and beneficial in the intended patient population.
A statistically significant clinical result
is not automatically clinically meaningful;
benefit must justify toxicity, burden, and available alternatives.
```

The broad objective of drug discovery is therefore to connect molecular understanding with patient benefit. Biology identifies possible disease mechanisms. Chemistry and biotechnology provide ways to create interventions. Pharmacology explains how interventions alter biological systems. PK and PD connect dose to exposure and biological effect. Clinical development determines whether the intervention genuinely improves patient outcomes.

Part I The minimum biological and chemical language

1 Cells, proteins, genes, and pathways

1.1 What a cell is doing

A *cell* is the basic living unit of the body. Humans are made of trillions of cells, and different groups of cells work together to form tissues and organs. For example, the brain contains neurons, the immune system contains many types of immune cells, and the liver contains specialised liver cells.

A cell is surrounded by a thin boundary called the *cell membrane*. This membrane separates the inside of the cell from its surroundings, while also controlling which molecules can enter or leave. Cells need a controlled internal environment to function: they take up nutrients and signals from their surroundings, use energy, make new molecules, remove waste, and communicate with other cells.

A simplified picture of a typical human cell is:

Cell membrane = boundary between cell and surroundings
Cytoplasm = fluid-filled interior containing many molecules and structures
Nucleus = compartment containing most of the cell's DNA

Cells are not passive. They constantly sense information from their environment, such as nutrients, oxygen, stress, infection, or signals sent by nearby cells. They then change their behaviour in response. A cell might divide, stop dividing, move, produce a particular protein, release a signalling molecule, repair damage, or die.

Different cell types generally contain the same DNA, but they use different parts of it. A neuron, an immune cell, and a liver cell behave differently because they produce different sets and amounts of proteins. This is one of the main ways that cells acquire specialised functions.

For drug discovery, the important idea is that disease usually arises because cells, tissues, or immune systems are in an inappropriate state. A medicine aims to change some part of the underlying biological system: for example, reduce an overactive growth signal, block an inflammatory pathway, kill infected or cancerous cells, restore a missing protein function, or help immune cells recognise cancer.

1.2 DNA, genes, and mutations

DNA, short for deoxyribonucleic acid, is the molecule that stores genetic information. In most human cells, most DNA is stored in the *nucleus*, a compartment inside the cell. DNA is tightly packaged into structures called *chromosomes*. Humans usually have 23 pairs of chromosomes, with one copy of each pair inherited from each parent.

DNA is a long chemical sequence built from four types of nucleotide bases, usually represented by the letters A, C, G, and T. The particular order of these letters encodes biological information. A useful analogy is that DNA is written in a four-letter alphabet, and its sequence provides instructions that cells can use to build and regulate molecular machinery.

A *gene* is a region of DNA that contains instructions for making a functional product. In many cases, this product is a protein. Some genes instead produce functional RNA molecules that are not translated into proteins.

The central simplified flow of molecular biology is:

DNA → RNA → protein

This is often called the *central dogma*. DNA is the relatively stable long-term store of information. RNA is usually a shorter-lived working copy or functional molecule. Proteins are the molecular machines that perform much of the cell's work.

The central dogma is useful, but it is not a complete description of biology. Cells regulate many steps in this process. They can control whether a gene is used at all, how much RNA is produced, how RNA is processed, how much protein is made, where the protein is sent, how long it survives, and whether it is active.

A *mutation* is a change in the DNA sequence. For example, one DNA base might be replaced by another, a short section might be deleted, or a larger region of DNA might be duplicated or rearranged. Mutations can be inherited from parents, acquired during life, or arise when individual cells divide.

Many mutations have little or no functional consequence. Others can change a protein's amino-acid sequence, alter how much of a gene is expressed, or disrupt DNA regions that regulate gene activity. In cancer, some acquired mutations give a cell a growth or survival advantage. These may become important *driver mutations*, discussed in Chapter 3.

1.3 Transcription and RNA

Transcription is the process through which a cell copies the information in a gene from DNA into *RNA*, short for ribonucleic acid. RNA is chemically related to DNA, but it is usually single-stranded and is often used as a more temporary or mobile form of genetic information.

The main type of RNA relevant to protein production is *messenger RNA*, or *mRNA*. An mRNA molecule contains a copy of the instructions from a particular gene that can later be used to produce a protein.

A useful analogy is:

```
DNA = long-term master archive
Gene = a particular instruction or recipe within the archive
mRNA = a temporary working copy of that recipe
Protein = the functional machine built from the recipe
```

A cell does not use every gene all the time. For example, an immune cell needs to express genes involved in immune recognition and signalling, whereas a liver cell needs to express genes involved in metabolism and detoxification. The process of controlling which genes are used, and to what extent, is called *gene regulation*.

Cells regulate transcription in response to developmental stage, tissue identity, nutrients, stress, infection, and signals from other cells. This allows the same DNA sequence to support many different cell types and behaviours.

A *transcription factor* is a protein that binds specific DNA sequences and influences transcription. Some transcription factors increase the production of mRNA from particular genes, while others reduce it. Transcription factors are important because they connect upstream signals to longer-lasting changes in cell state.

For example, an activated receptor at the cell surface can trigger an intracellular signalling pathway. This pathway may eventually activate a transcription factor in the nucleus, which then changes expression of genes involved in cell growth, inflammation, metabolism, survival, or immune function.

1.4 Translation, amino acids, and proteins

Translation is the process through which a cell reads an mRNA sequence and uses it to construct a protein. Translation takes place using molecular machinery called *ribosomes*, which read the mRNA sequence and link together the appropriate amino acids.

Amino acids are small chemical building blocks. Proteins are long chains of amino acids linked together in a particular order. There are roughly 20 standard amino acids used in human proteins, and each has different

chemical properties. Some are charged, some are hydrophobic, some are polar, and some can participate in particular chemical reactions or molecular interactions.

The sequence of an mRNA determines the sequence of amino acids in the resulting protein:

```
DNA sequence
→ mRNA sequence
→ amino-acid sequence
→ folded protein
```

After being produced, an amino-acid chain usually folds into a particular three-dimensional shape. This shape is crucial: it determines which molecules the protein can bind, which chemical reactions it can perform, where it is located in the cell, and how it responds to signals.

A *protein* is therefore not merely an amino-acid sequence. It is a dynamic three-dimensional molecular object. Its function depends on its sequence, three-dimensional structure, movement between different conformational states, location in or outside the cell, binding partners, and chemical modifications such as phosphorylation.

Proteins carry out much of the molecular work of cells. They can act as enzymes that catalyse chemical reactions, receptors that detect external signals, transporters that move molecules across membranes, structural elements, molecular motors, signalling molecules, transcription factors, antibodies, and components of immune systems.

A mutation can alter a protein by changing one or more amino acids, changing the amount of protein produced, or preventing the protein from folding or functioning correctly. In some diseases, including cancer, such changes can make a protein abnormally active, inactive, unstable, or incorrectly regulated.

1.5 Enzymes, substrates, and active sites

An *enzyme* is a protein that accelerates a particular chemical reaction. Many reactions would occur far too slowly under normal physiological conditions without enzymes. Enzymes make these reactions faster by lowering the *activation energy*: the energy barrier that must be overcome for a reaction to occur.

The molecule on which an enzyme acts is called its *substrate*. A simplified picture is:

```
Enzyme + substrate → enzyme-substrate complex → product
```

The enzyme is usually not consumed by the reaction. It can repeatedly bind substrate molecules and convert them into products.

An enzyme generally contains an *active site*: a small region whose shape and chemical environment allow it to bind a substrate and catalyse the relevant reaction. Active sites are not necessarily rigid locks; both the enzyme and substrate can change shape slightly during binding.

A *growth factor* is a signalling protein that can influence cell growth, survival, division, movement, or repair. Growth factors usually act by binding specific receptors on the surface of target cells, thereby activating intracellular signalling pathways.

For example, *epidermal growth factor*, or *EGF*, is a growth factor that can bind the *epidermal growth factor receptor*, or *EGFR*. EGFR is a receptor tyrosine kinase. In normal tissues, EGF–EGFR signalling helps regulate processes such as cell growth and tissue repair. In some cancers, EGFR signalling becomes abnormally active because of mutations, excessive EGFR production, or other alterations. Tumour cells can then receive excessive growth or survival signals.

An especially important class of enzymes in cell signalling is *kinases*. A kinase transfers a *phosphate group* from one molecule to another, often to a protein. This process is called *phosphorylation*.

The phosphate group is commonly donated by *ATP*, short for adenosine triphosphate. *ATP* is a small molecule used throughout cells as a source of chemical energy. It contains three phosphate groups; when a kinase transfers its outermost phosphate group to a substrate protein, *ATP* becomes *ADP*, or adenosine diphosphate:

Kinase + ATP + substrate protein → kinase + ADP + phosphorylated substrate protein

Adding a phosphate group can change a protein's shape, activity, location, stability, or interactions with other proteins. Phosphorylation therefore acts as a molecular switch that cells use to transmit and regulate information.

A *receptor tyrosine kinase* is a protein that combines two functions. First, it is a *receptor*: part of the protein is located on the cell surface and can bind an external signalling molecule, such as a growth factor. Second, it is a *tyrosine kinase*: another part of the protein extends inside the cell and can add phosphate groups to particular amino acids called *tyrosines* on proteins.

When a growth factor binds the outer part of a receptor tyrosine kinase, it can cause the receptor to change shape or bind another receptor molecule. This activates the kinase region inside the cell. The receptor can then add phosphate groups to itself, a process called *autophosphorylation*, and to other proteins. These phosphorylation events trigger a signalling cascade that can ultimately change cell behaviour:

Growth factor binds the outer part of receptor
→ kinase region inside the cell becomes active
→ receptor and downstream proteins are phosphorylated
→ signalling pathway is activated
→ cell behaviour changes

Kinases are important in cancer because mutations, gene amplification, or other alterations can make growth-promoting kinase signalling abnormally active. If a tumour depends on this signalling, inhibiting the relevant kinase may reduce tumour-cell growth or survival.

Many kinase inhibitors are *ATP-competitive inhibitors*. They bind in or near a kinase's ATP-binding pocket, preventing ATP from binding normally. The kinase can then no longer efficiently phosphorylate its substrates, reducing downstream signalling. Designing selective inhibitors is challenging because ATP is abundant in cells and many kinases have related ATP-binding pockets.

Not all inhibitors bind active sites or compete with ATP. Some bind *allosteric sites*: distinct regions of a protein where binding changes its shape, dynamics, or probability of becoming active. This is discussed further in Chapter 4.

1.6 Cell-surface and intracellular proteins

A protein's location strongly affects both its biological function and which types of drugs can act on it. Recall that a cell is surrounded by a thin boundary called the *cell membrane*, also called the *plasma membrane*. This membrane separates the inside of the cell from the outside environment.

Some proteins are found on the outside surface of this membrane. Others are found inside the cell. This distinction is important because a drug must physically reach a protein in order to bind or alter it.

A *cell-surface protein* is embedded in, or attached to, the cell membrane. Part of the protein faces the outside of the cell, where it can interact with molecules in the blood, surrounding tissue, or neighbouring cells. Cell-surface proteins include receptors, transporters, adhesion molecules, and immune markers.

For example, HER2 is a cell-surface receptor found at high levels on some cancer cells. Because part of HER2 is exposed outside the cell, it can be recognised by an antibody such as trastuzumab.

Cell-surface proteins can often be targeted by *biologics*: medicines made from, or closely based on, biological molecules. Important examples include:

- *Monoclonal antibodies*, which can bind an exposed part of a cell-surface protein.
- *Antibody–drug conjugates*, or ADCs, which use an antibody to deliver a cytotoxic payload preferentially to cells expressing a particular surface target.
- *CAR-T cells*, which are engineered immune cells designed to recognise a chosen cell-surface antigen.

An *intracellular protein* is located inside the cell. It may be found in the *cytoplasm*, the fluid-filled interior of the cell; the *nucleus*, where most DNA is stored; the *mitochondria*, which help produce cellular energy; or other specialised cellular compartments.

Intracellular proteins include many enzymes, kinases, transcription factors, structural proteins, and signalling proteins. For example, KRAS is an intracellular signalling protein, while many kinases act inside the cell to transmit signals from cell-surface receptors.

Small molecules are often especially useful for intracellular targets because some are small enough and chemically suitable enough to cross the cell membrane. Once inside, they can bind an intracellular protein and alter its function.

Conventional antibodies are much larger proteins and generally cannot enter living cells efficiently. They are therefore usually not suitable for directly targeting intracellular proteins, although there are specialised strategies that attempt to overcome this limitation.

This distinction helps explain why different therapeutic modalities exist:

```
Extracellular or cell-surface target
→ antibodies, ADCs, or cell therapies may be feasible
Intracellular target
→ small molecules may be especially useful
Protein with no accessible binding pocket
→ conventional small-molecule inhibition may be difficult
```

Importantly, these are not absolute rules. Some intracellular proteins can be targeted indirectly by antibodies, and some cell-surface proteins can also be targeted by small molecules. Nevertheless, protein location is one of the first things to ask when considering whether a target is therapeutically tractable.

1.7 Receptors, ligands, and signalling

Cells need to receive information from their surroundings. They need to know, for example, whether nutrients are available, whether neighbouring cells are present, whether the body is injured or infected, or whether they should grow, divide, move, or stop functioning.

A *receptor* is a protein that detects a particular signal and helps convert that signal into a biological response. Receptors are often found on the cell surface, where they can detect molecules outside the cell, but some receptors are located inside cells.

A *ligand* is a molecule that binds to a receptor. Many ligands are naturally produced by the body. Examples include:

- *Hormones*, which coordinate processes across different organs.
- *Growth factors*, which can promote cell growth, survival, or division.
- *Cytokines*, which are signalling molecules used especially by immune cells.
- *Neurotransmitters*, which allow nerve cells to communicate.

Therapeutic drugs can also act as ligands. Depending on the drug and receptor, a drug may activate the receptor, block it, reduce its activity, or otherwise change its behaviour.

For a cell-surface receptor, the simplified sequence is often:

```
Ligand binds receptor
→ receptor changes shape or forms a complex
→ intracellular signalling proteins are activated
→ signalling reaches the nucleus or other cellular machinery
→ cell behaviour changes
```

The word *signalling* refers to the process by which a cell receives, transmits, processes, and responds to information. A receptor on the cell surface can detect an external signal, but the consequences of that signal often occur inside the cell. The receptor must therefore communicate with intracellular proteins.

For example, a *growth factor* can bind a *receptor tyrosine kinase*. This activates the receptor's intracellular kinase region, which can then phosphorylate itself and other proteins. These proteins may activate additional signalling molecules, eventually changing gene expression, cell growth, metabolism, survival, movement, or division.

A *signalling pathway* is a connected series of molecular events that transmits and transforms information within a cell. A pathway is often drawn as a simple sequence:

```
Receptor → kinase A → kinase B → transcription factor → gene expression
```

This type of diagram is useful for learning because it identifies major components and the general direction of information flow. However, real signalling pathways are not simple straight lines. They are interconnected networks. A protein may participate in several pathways, one pathway may activate or inhibit another, and the same signal may have different effects in different cell types.

Real signalling networks often include:

- *Feedback loops*, where downstream signalling increases or decreases earlier steps in the pathway.
- *Redundancy*, where multiple proteins or pathways can produce similar effects.
- *Cross-talk*, where one pathway influences another pathway.
- *Context dependence*, where the outcome depends on cell type, tissue, mutations, and the surrounding environment.

A pathway may therefore be strongly active in one tissue but not another. The same receptor can also produce different outcomes depending on which other proteins are present, whether the cell contains mutations, and which signals it receives at the same time.

This complexity is important in drug discovery. Blocking a single receptor or kinase may have a strong effect if the disease depends on that pathway, but cells can sometimes compensate by using alternative signalling routes.

1.8 Phosphorylation and kinases

As introduced in the previous subsection, *phosphorylation* is the addition of a *phosphate group* to a molecule, often a protein. A phosphate group is a small chemical group containing one phosphorus atom and several oxygen atoms. When attached to a protein, it usually adds substantial negative charge to a particular location on that protein.

This added charge can change the protein's shape, alter how strongly it interacts with other molecules, create a new binding site for another protein, change where the protein is located in the cell, or increase or reduce its activity. Phosphorylation therefore acts as a flexible and reversible molecular switch that cells use to regulate protein function and transmit information.

A *kinase* is an enzyme that adds a phosphate group to a target molecule. In many signalling pathways, kinases transfer a phosphate group from ATP to a protein. A *phosphatase* is an enzyme that removes phosphate groups from proteins.

The balance between kinase and phosphatase activity helps control whether a signalling protein is active, inactive, or ready to interact with another molecule:

```
Kinase activity
→ adds phosphate group
Phosphatase activity
→ removes phosphate group
```

For example, the following sequence can occur after a growth factor binds a receptor:

```
Growth factor binds receptor
→ receptor kinase becomes active
→ downstream proteins are phosphorylated
→ signalling pathway is activated
→ cell receives a pro-growth or pro-survival signal
```

This is simplified, but it captures the central idea: phosphorylation allows a signal at the cell surface to be passed from one protein to another inside the cell.

Kinases are particularly important in cancer because mutations, gene amplification, or other alterations can make growth-promoting kinase signalling abnormally active. A kinase that should only be active briefly in response to a normal signal may instead become continuously active, helping cancer cells grow or survive when they should not.

Many cancer drugs are therefore *kinase inhibitors*: drugs designed to reduce the activity of a particular kinase or group of kinases. Examples include inhibitors targeting EGFR, BRAF, ALK, HER2, and CDK4/6.

A kinase inhibitor can reduce signalling by preventing the kinase from phosphorylating its normal substrate proteins. If a tumour depends strongly on that kinase-driven pathway, this may reduce tumour-cell growth, promote cell death, or make the cancer more sensitive to another treatment.

However, kinase inhibition does not always produce a durable response. Tumours may contain mutations that prevent drug binding, activate alternative pathways, or allow resistant cells to survive and expand. These mechanisms of resistance are discussed further in Chapter 3.

1.9 Protein–protein interactions

A *protein–protein interaction*, often abbreviated *PPI*, occurs when two proteins bind to each other. Proteins rarely act entirely alone. Much of cell biology depends on proteins forming temporary or stable interactions with other proteins.

A PPI can have many purposes. One protein may activate another, inhibit another, help transport it to a particular location, stabilise its structure, or bring it into a larger molecular complex.

A *peptide* is a short chain of amino acids. Peptides can be produced when cells break down proteins into smaller fragments. Some peptide fragments can be displayed on the surface of a cell by specialised proteins called *major histocompatibility complex*, or *MHC*, molecules. This allows T cells to inspect fragments of proteins produced within, or taken up by, another cell.

Examples include:

- A receptor may need to bind another receptor to form an active signalling complex.
- A transcription factor may need to bind a *cofactor*, another protein that helps regulate gene expression.

- An enzyme may need to bind a regulatory protein that activates or inhibits it.
- A T-cell receptor can recognise a peptide only when that peptide is displayed by an MHC molecule on another cell.
- PD-1 on a T cell can bind PD-L1 on a tumour cell or immune cell, sending an inhibitory signal to the T cell.

PPIs are often important in disease, making them attractive potential drug targets. For example, a cancer cell may rely on an interaction between two signalling proteins, or a tumour may evade immune attack through an inhibitory interaction such as PD-1–PD-L1.

However, PPIs can be difficult to target using conventional small molecules. Many small-molecule drugs bind in relatively deep pockets on proteins, such as enzyme active sites. In contrast, protein–protein interfaces are often broad, relatively flat, and spread across many contact points.

A useful, though simplified, comparison is:

```

Enzyme active site
→ often a relatively defined pocket
→ can be suitable for small-molecule binding
Protein-protein interface
→ often broad and relatively flat
→ may be more difficult for a conventional small molecule to disrupt
  
```

This does not mean PPIs are impossible to target. Some protein interfaces contain smaller pockets or “hot spots” where binding has a particularly strong functional effect. In addition, other modalities may be useful. Antibodies can block extracellular PPIs, peptides and engineered proteins can mimic natural binding partners, and some small molecules can act as *molecular glues* that stabilise a useful interaction rather than disrupting one.

The key idea is that a drug can work not only by binding one protein in isolation, but also by changing the way proteins interact with each other.

1.10 Key takeaways

The central biological flow introduced in this chapter is:

```

DNA stores instructions
→ genes are selectively transcribed into RNA
→ RNA is translated into proteins
→ proteins interact in signalling and functional networks
→ networks determine cell behaviour
→ drugs alter selected proteins, interactions, cells, or pathways
  
```

Several distinctions are especially useful to retain:

- DNA stores genetic information, while proteins perform much of the cell’s molecular work.
- A gene can be thought of as one set of instructions within DNA; whether that gene is used depends on cell type and context.
- Proteins are dynamic three-dimensional molecules whose location, shape, interactions, and chemical modifications affect their function.
- Enzymes catalyse chemical reactions; kinases are enzymes that add phosphate groups and are particularly important in signalling.

- Cell-surface proteins are often accessible to antibodies and cell therapies, while intracellular proteins are often more accessible to small molecules.
- Receptors detect signals, ligands bind receptors, and signalling pathways convert these events into changes in cell behaviour.
- Proteins often work by interacting with other proteins, so therapeutically useful interventions may target a protein, a binding pocket, or an interaction between proteins.

When reading a target-biology paper, a useful set of questions is:

What is this protein or pathway doing?
 Where is the protein located: outside the cell, on the cell surface, or inside the cell?
 What normally activates, inhibits, or regulates it?
 Which other proteins, pathways, or cell types does it interact with?
 Why might changing this protein or pathway help in a specific disease?
 Which therapeutic modality could physically reach and modulate it?

1.11 Super-short summary

A cell uses proteins to sense, process, and respond to information. DNA stores genetic instructions; selected genes are transcribed into RNA, and mRNA is translated into proteins. Proteins act as enzymes, receptors, signalling molecules, and structural components.

Many proteins communicate through signalling pathways. Kinases add phosphate groups to proteins, often switching signalling events on or off. Drugs can alter disease biology by binding proteins, inhibiting enzymes, blocking receptors, changing protein–protein interactions, or otherwise modifying signalling pathways.

A target’s location matters: cell-surface proteins are often accessible to antibodies and cell therapies, while intracellular proteins are often more accessible to small molecules. The central question is always: what does the target do, where is it located, how is it regulated, and why would changing it help in disease?

1.12 Practice questions

How to use these questions. Try to answer each question without looking at the answer. The goal is to explain the concepts in your own words and understand the causal chain, rather than memorise exact wording.

1. **What is the basic role of a cell, and why can different cell types behave differently despite containing nearly the same DNA?**

Answer: A cell is the basic living unit of the body. It senses and responds to its environment, uses energy, makes molecules, communicates with other cells, and can grow, divide, move, repair damage, or die. Different cell types use different subsets of genes and therefore produce different sets and amounts of proteins, giving them specialised functions.

2. **Explain the central dogma: how does information flow from DNA to a functional protein?**

Answer:

DNA stores genetic instructions
 → a gene is transcribed into RNA
 → mRNA is translated by ribosomes into an amino-acid chain
 → the chain folds into a functional protein

3. What is a mutation, and how can it change a cell's behaviour?

Answer: A mutation is a change in DNA sequence. It can have little effect, but it can also alter a protein's amino-acid sequence, change how much of a protein is produced, or disrupt gene regulation. In cancer, some mutations create growth or survival advantages and become driver mutations.

4. Why is a protein more than an amino-acid sequence?

Answer: A protein folds into a dynamic three-dimensional structure. Its function depends on its shape, movement between conformations, location in or outside the cell, binding partners, and chemical modifications, as well as its amino-acid sequence.

5. What is an enzyme, what is a substrate, and what is an active site?

Answer: An enzyme is usually a protein that accelerates a chemical reaction. The substrate is the molecule on which the enzyme acts. The active site is the region of the enzyme that binds the substrate and enables the reaction.

6. What are ATP, kinases, and phosphorylation, and how are they related?

Answer: ATP is a small molecule that can donate a phosphate group. A kinase is an enzyme that commonly transfers a phosphate group from ATP to another protein. This process, phosphorylation, can change the target protein's activity, shape, location, stability, or interactions, and is widely used for cell signalling.

7. Why are kinases important drug targets in cancer, and how do ATP-competitive kinase inhibitors work?

Answer: Cancer-associated mutations or gene amplification can make growth-promoting kinase signalling abnormally active. ATP-competitive kinase inhibitors bind in or near the kinase's ATP-binding pocket, preventing normal ATP binding and reducing phosphorylation of downstream proteins. This can reduce tumour-promoting signalling when the tumour depends on that pathway.

8. Why does the location of a target protein matter for drug discovery?

Answer: A therapeutic must physically reach its target. Cell-surface or extracellular proteins are often accessible to antibodies, ADCs, and cell therapies. Intracellular proteins are often more accessible to small molecules that can cross the cell membrane. A protein with no useful binding pocket may be difficult to target with a conventional small molecule.

9. What are receptors, ligands, and signalling pathways?

Answer: A receptor is a protein that detects a signal. A ligand is a molecule that binds the receptor. A signalling pathway is a connected set of molecular events through which receptor activation changes cell behaviour, for example through kinase activity, phosphorylation, transcription factors, and altered gene expression.

10. Why are signalling pathways and protein-protein interactions important, but also challenging, in drug discovery?

Answer: Proteins work in interconnected networks rather than isolated linear pathways. A disease may depend on an abnormal pathway or protein-protein interaction, making it a possible therapeutic target. However, pathways can contain feedback, redundancy, and alternative routes that create resistance, while protein-protein interfaces are often broad and relatively flat, making them difficult for conventional small molecules to disrupt.

2 Chemical language for molecular drug discovery

2.1 Why chemistry matters for drug discovery

Many medicines are *small molecules*: discrete chemical compounds that can often enter cells, reach a target protein, and change its behaviour. Their usefulness depends not only on whether they bind a target, but also

on whether they can dissolve, reach the relevant tissue, enter the relevant cells, remain present long enough, avoid problematic off-target effects, and be manufactured practically.

A useful causal chain is:

```
Chemical structure
→ shape, charge, flexibility, and reactivity
→ interactions with proteins, membranes, and water
→ target modulation, exposure, safety, and developability
```

A molecule can bind a target strongly and still be a poor drug candidate. It may be insoluble, unable to enter cells, rapidly metabolised, unable to reach the relevant tissue, chemically unstable, toxic, or too difficult to manufacture.

Drug discovery is therefore not a search for the strongest possible binder. It is an attempt to find a molecule with a sufficiently favourable overall profile for a particular therapeutic use.

The central practical question is:

```
Can this molecule reach the relevant target,
modulate it in the desired way,
and do so at a tolerable exposure?
```

2.2 Molecules as connected atoms and three-dimensional objects

A small molecule consists of atoms held together by *covalent bonds*. The atoms most commonly found in drug-like molecules include carbon, hydrogen, nitrogen, oxygen, sulfur, phosphorus, fluorine, chlorine, bromine, and iodine.

The particular atoms present, the way they are connected, and their arrangement in space determine many important properties. These include molecular shape, charge, flexibility, reactivity, solubility, membrane permeability, metabolism, and protein binding.

In computational chemistry, a molecule is often represented as a graph:

```
Atoms
→ graph nodes
Covalent bonds
→ graph edges
Atom and bond properties
→ node and edge features
```

This representation is useful, but it is incomplete. A molecular graph does not automatically specify the molecule's three-dimensional conformation, stereochemistry, protonation state, interactions with water, or the biological environment in which it is measured.

A useful distinction is:

```
Two-dimensional structure
→ which atoms are connected to which atoms
Three-dimensional structure
→ how those atoms are arranged in space
```

Both matter because a molecule must adopt a physically compatible shape and chemical interaction pattern to bind a protein productively.

2.3 Functional groups and molecular recognition

A *functional group* is a chemically meaningful arrangement of atoms that influences a molecule's properties and reactivity. Functional groups provide a practical language for medicinal chemistry because changing one group can alter several important properties at once.

Common examples include:

- *Amines*, which contain nitrogen and are often basic, meaning that they can gain a proton and become positively charged.
- *Alcohols*, which contain an oxygen–hydrogen group and can often participate in hydrogen bonding.
- *Carboxylic acids*, which can lose a proton and become negatively charged.
- *Amides*, which are common in proteins and drug molecules and can participate in hydrogen bonding.
- *Esters*, which can sometimes be chemically or enzymatically cleaved.
- *Aromatic rings* and *heterocycles*, which can influence molecular shape, lipophilicity, binding interactions, and metabolic stability.
- *Halogens*, such as fluorine or chlorine, which can alter lipophilicity, metabolic stability, and protein-binding interactions.

Most conventional protein–ligand binding depends mainly on *non-covalent interactions*: individually reversible interactions between the ligand and the protein.

Important non-covalent interactions include:

- *Electrostatic interactions*: attraction or repulsion between charged groups.
- *Hydrogen bonds*: directional interactions between suitable donor and acceptor groups.
- *van der Waals interactions*: short-range interactions that depend strongly on atomic distance and shape complementarity.
- *Hydrophobic interactions*: the tendency of non-polar groups to avoid water and associate with non-polar environments.
- π *interactions*: interactions involving aromatic electron systems.

A ligand does not bind strongly simply because it makes many favourable contacts with a protein. Binding depends on the overall balance of favourable and unfavourable effects:

Favourable interactions with target
– loss of favourable interactions with water
– conformational strain
– loss of molecular flexibility
– unfavourable contacts
→ net binding affinity

Medicinal chemistry therefore involves more than adding interactions to a molecule. A modification that improves one interaction may worsen solubility, permeability, selectivity, stability, or another property needed for a viable medicine.

2.4 Water, lipophilicity, ionisation, and cell entry

Drug molecules must behave appropriately in both water-rich and non-polar biological environments.

Blood, intracellular fluid, and most assay buffers are largely water. In contrast, the interior of cell membranes and some regions of protein binding sites are relatively *non-polar*. A non-polar environment has little separation of electrical charge and interacts relatively poorly with water. Fats, oils, and lipid cell membranes are familiar examples.

Solubility is the extent to which a compound can dissolve in a particular solvent under defined conditions. Poor solubility can limit absorption, complicate formulation, produce unreliable assay results, and prevent testing or administration at a sufficiently high concentration.

Lipophilicity is the tendency of a compound to partition into non-polar environments, such as fats, oils, lipid cell membranes, or hydrophobic protein pockets, rather than into water.

The practical trade-off is:

```
Greater lipophilicity
→ may improve membrane crossing
→ may strengthen hydrophobic protein-binding interactions
Excessive lipophilicity
→ may reduce aqueous solubility
→ may increase non-specific binding
→ may increase metabolic or toxicity risk
```

Many drug-like molecules also contain groups that can gain or lose protons. This changes their *charge state*. For example, many amines can become positively charged, whereas carboxylic acids can become negatively charged.

The *pKa* of an ionisable group is related to the pH at which that group is approximately half protonated and half deprotonated. The practical importance is that the charge state of a compound can change across environments such as assay buffer, blood, intracellular fluid, and acidic cellular compartments.

Charge can affect:

```
Charge state
→ solubility
→ membrane permeability
→ tissue distribution
→ protein binding
→ target interaction
```

A molecule that is mostly charged may dissolve well in water but cross lipid cell membranes poorly. A molecule that is mostly neutral may cross membranes more readily but be less soluble. The goal is not to maximise either property independently, but to achieve an appropriate balance for the target, tissue, route of administration, and intended use.

For intracellular targets, this balance is especially important. A compound may bind purified protein strongly in a biochemical assay but show weak cellular activity if it cannot enter cells or cannot achieve sufficient free concentration inside them.

2.5 Stereochemistry, conformation, and flexibility

Two molecules can have the same atoms and the same two-dimensional connectivity but differ in their three-dimensional arrangement. This is called *stereochemistry*.

A molecule is *chiral* if it cannot be superimposed on its mirror image. The two mirror-image forms are called *enantiomers*. Other stereochemical differences can produce *diastereomers*.

Proteins are themselves chiral three-dimensional objects. As a result, stereoisomers can differ substantially in:

- Binding affinity and functional potency.
- Selectivity for one protein over another.
- Metabolism and pharmacokinetics.
- Toxicity and clinical effect.

Molecules are also not rigid objects. Rotation around many single bonds allows a molecule to adopt multiple three-dimensional shapes, called *conformations* or *conformers*. The set of conformations accessible under particular conditions is the molecule's *conformational ensemble*.

Binding depends partly on:

Whether the ligand can adopt a binding-compatible conformation
Whether that conformation is energetically accessible
Whether the protein can adapt during binding
How water molecules and charge states change on binding

Flexibility can help a ligand adapt to a binding site. However, a highly flexible molecule may pay an energetic cost when it becomes constrained in one bound shape. Conversely, a more rigid molecule may bind efficiently when its preferred shape already resembles the bound conformation.

This is why molecular structure is not fully described by a single two-dimensional chemical drawing.

2.6 Chemical stability, metabolism, and reactivity

A useful drug candidate must remain chemically intact long enough to be manufactured, formulated, administered, absorbed, distributed, and act on its target.

Chemical stability concerns whether a molecule reacts or degrades under relevant conditions, such as exposure to water, oxygen, light, acid, base, or heat.

Metabolic stability concerns whether enzymes in the body rapidly modify or remove the molecule. Rapid metabolism can reduce drug exposure below the level needed for useful target modulation. Conversely, metabolic products can sometimes contribute to activity or toxicity.

Some molecules are intentionally reactive. For example, a *covalent inhibitor* is designed to form a covalent bond with a target protein. This can produce durable target engagement, but it also requires careful control of selectivity. Unwanted reactivity can create safety risks if a compound reacts with many unintended proteins or other biological molecules.

The key question is:

Is the molecule sufficiently stable and sufficiently selective
for its intended mechanism and clinical use?

2.7 Key takeaways

Small-molecule drug discovery requires understanding how chemical structure influences biological behaviour. Several distinctions are especially useful to retain:

-
- A molecular graph describes atom connectivity, but it may not fully capture three-dimensional structure, stereochemistry, charge state, conformation, or biological context.
 - Functional groups influence protein binding, water solubility, ionisation, permeability, metabolism, and chemical reactivity.
 - Protein binding depends on the net balance of favourable and unfavourable interactions, not simply on the number of contacts visible in a structural model.
 - Solubility, lipophilicity, and charge must be balanced. A molecule that crosses membranes efficiently may still be too insoluble, non-specific, or toxic to become useful.
 - Stereoisomers and conformations can have substantially different biological properties.
 - Chemical and metabolic stability determine whether a molecule can achieve useful exposure in the body.
 - Small-molecule design is a multi-objective optimisation problem rather than a search for the highest possible binding score.

When reading a small-molecule discovery paper, ask:

Which structural features are thought to drive target binding?
Can the molecule dissolve and reach the relevant tissue?
Can it enter the relevant cells, if the target is intracellular?
Are stereochemistry and relevant conformations specified?
Could metabolism, instability, or unwanted reactivity limit its use?

2.8 Super-short summary

Small molecules are three-dimensional chemical objects whose structure determines how they interact with proteins, water, membranes, and metabolic enzymes.

A useful drug must do more than bind a target. It must also have suitable solubility, cell entry, tissue exposure, stability, selectivity, safety, and manufacturability. Drug discovery therefore seeks an appropriate overall balance rather than the best value for one isolated property.

2.9 Practice questions

How to use these questions. Try to answer each question without looking at the answer. Focus on the causal chain from chemical structure to molecular properties, biological behaviour, and drug-discovery decisions.

1. **Why is strong binding to a target protein insufficient for a molecule to become a useful medicine?**

Answer: A strongly binding molecule can still fail if it is insoluble, cannot enter relevant cells, does not reach the target tissue, is rapidly metabolised, is chemically unstable, causes toxicity, binds problematic off-targets, or cannot be manufactured practically. A medicine requires a sufficiently favourable overall profile.

2. **What is the difference between a molecular graph and a three-dimensional molecular structure?**

Answer: A molecular graph describes which atoms are connected by covalent bonds. A three-dimensional structure describes how those atoms are arranged in space. The graph alone may not fully specify stereochemistry, charge state, conformation, or interactions with the biological environment.

3. **What is a functional group, and why is it useful in medicinal chemistry?**

Answer: A functional group is a chemically meaningful arrangement of atoms, such as an amine, alcohol, carboxylic acid, amide, aromatic ring, or halogen-containing group. Functional groups influence protein binding, charge, solubility, metabolism, reactivity, and permeability, making them useful units for molecular design.

4. **Why can a ligand with many favourable interactions still bind a target weakly?**

Answer: Binding depends on the net balance of favourable and unfavourable effects. A ligand may lose favourable interactions with water, adopt a strained shape, lose flexibility, or make unfavourable contacts that offset apparently favourable interactions with the target.

5. **What is the relationship between solubility, lipophilicity, and membrane permeability?**

Answer: Greater lipophilicity can help a molecule enter lipid cell membranes and bind hydrophobic regions of proteins, but excessive lipophilicity can reduce aqueous solubility and increase non-specific binding, metabolic risk, or toxicity. The goal is an appropriate balance rather than maximal lipophilicity.

6. **Why can charge state affect whether a molecule works in cells?**

Answer: Charge affects solubility, membrane permeability, tissue distribution, protein binding, and target interaction. A compound may bind purified protein strongly but show weak cellular activity if its dominant charge state prevents sufficient entry into cells.

7. **Why can stereoisomers have different pharmacological effects?**

Answer: Stereoisomers have the same atoms and connectivity but different three-dimensional arrangements. Because proteins are chiral, one stereoisomer may bind a target strongly while another binds weakly, engages another target, is metabolised differently, or causes different toxicity.

8. **What is a conformational ensemble?**

Answer: A conformational ensemble is the set of three-dimensional shapes accessible to a molecule under particular conditions. Molecules can rotate around many bonds and therefore usually do not exist in only one fixed shape.

9. **Why can molecular flexibility be both useful and costly for protein binding?**

Answer: Flexibility can allow a ligand to adapt to a binding site. However, a flexible molecule may lose substantial conformational freedom when it binds, creating an energetic cost that can weaken binding or complicate optimisation.

10. **Why are chemical stability, metabolic stability, and reactivity important in drug discovery?**

Answer: A molecule must remain intact long enough to be manufactured, administered, reach the target, and act on it. Rapid metabolism can prevent sufficient exposure, while uncontrolled chemical reactivity can cause degradation or harmful reactions with unintended biological molecules.

3 What cancer is, mechanistically

3.1 Cancer is an evolutionary disease of cells

Cancer is not a single disease. It is a large collection of diseases in which cells acquire abnormal capabilities that allow them to grow, survive, spread, and evade normal biological control.

Normal tissues are carefully regulated. Cells divide when new cells are needed, stop dividing when appropriate, respond to signals from neighbouring cells, repair damage when possible, and die when they are severely damaged or no longer needed. This coordination allows tissues and organs to maintain their normal structure and function.

Cancer develops when a cell gradually escapes some of these controls. It may acquire mutations that make a growth-promoting signalling pathway too active, remove a normal brake on cell division, prevent cell death, or allow the cell to ignore signals from its surroundings.

Cancer can therefore be understood as an evolutionary process occurring inside the body:

```
Cells acquire genetic or other biological alterations
→ some alterations give a growth or survival advantage
→ these cells produce descendants
→ further alterations accumulate
→ different tumour-cell populations compete and evolve
```

A cell population that has acquired cancer-associated properties is called a *tumour*. A tumour may begin from one abnormal cell, but it can eventually contain millions or billions of related cells with different mutations and behaviours.

The evolutionary process is shaped by many pressures. These include limited nutrients and oxygen, attack by the immune system, competition with neighbouring tumour cells, interactions with surrounding tissue, and treatment. A drug can kill sensitive cancer cells, but this may create an opportunity for pre-existing resistant cells to survive and expand.

This evolutionary perspective is important because cancer is not a static target. The biology of a tumour can change over time, particularly after treatment.

3.2 Hallmarks and capabilities of cancer

Cancer cells can acquire several broad capabilities that help them survive and progress. These are often described as the *hallmarks of cancer*. The hallmarks are not a checklist that every tumour satisfies in exactly the same way. Instead, they provide a useful framework for understanding the different biological problems that cancer cells solve.

Important examples include:

- *Sustained proliferative signalling*: Cancer cells continue to receive or generate signals that promote cell growth and division.
- *Evading growth suppressors*: Cancer cells ignore normal signals that would stop inappropriate division.
- *Resisting cell death*: Cancer cells avoid *apoptosis*, a regulated form of programmed cell death that can remove severely damaged or unnecessary cells.
- *Enabling replicative immortality*: Cancer cells maintain the ability to divide repeatedly, whereas many normal cells eventually stop dividing after a limited number of divisions.
- *Inducing angiogenesis*: Cancer cells encourage the formation of new blood vessels, allowing the tumour to receive oxygen and nutrients.
- *Activating invasion and metastasis*: Cancer cells gain the ability to invade nearby tissue and spread to distant parts of the body.
- *Evading immune destruction*: Cancer cells avoid, suppress, or resist attack by immune cells.
- *Reprogramming metabolism*: Cancer cells alter how they obtain and use energy and molecular building blocks in order to support growth and division.

These capabilities are connected. For example, a tumour that grows rapidly needs nutrients and oxygen, which can increase its dependence on blood-vessel formation. A tumour that contains many abnormal proteins may be more visible to the immune system, creating pressure to evolve immune-evasion mechanisms.

Cancer treatment aims to identify a particular vulnerability within this complex system. A tumour may depend unusually strongly on one growth pathway, one DNA-repair mechanism, one cell-surface target, or one immune-evasion strategy. If that dependency can be targeted safely, it may create a therapeutic opportunity.

3.3 Oncogenes and tumour suppressor genes

Some genes are especially important in cancer because they normally regulate cell growth, survival, DNA repair, or cell death.

An *oncogene* is a gene whose abnormal activation promotes cancer. The normal, non-cancer-associated version of such a gene is often called a *proto-oncogene*. Proto-oncogenes can have important normal roles in cell growth, division, or signalling. However, mutation, gene amplification, rearrangement, or excessive expression can make their activity abnormally strong or persistent.

For example, a mutation may cause a signalling protein to remain active even when the normal external signal is absent. Alternatively, a cell may acquire many copies of a gene, causing it to produce too much of a growth-promoting protein. This increase in gene copy number is called *gene amplification*.

Examples of important cancer-associated oncogenes include KRAS, EGFR, BRAF, ALK, MYC, and ERBB2/HER2.

A *tumour suppressor gene* has the opposite broad role: its normal function helps prevent inappropriate cell growth or preserves normal cellular control. Tumour suppressor genes can restrain cell division, detect and repair DNA damage, promote cell death after severe damage, or help maintain *genomic stability*, meaning the ability of a cell to preserve its DNA accurately over time.

Cancer can arise when tumour suppressor genes lose function. For example, a mutation may prevent a tumour-suppressor protein from being produced, make the protein non-functional, or reduce its expression.

Examples of important tumour suppressor genes include TP53, RB1, BRCA1, BRCA2, and PTEN. The biological effect depends on the specific gene and the disease context.

For example, BRCA1 and BRCA2 help cells repair serious forms of DNA damage, particularly *double-strand breaks*, in which both strands of the DNA molecule are broken. They contribute to a normally accurate repair process called *homologous recombination*. This process uses an intact matching DNA sequence as a template to repair the damaged DNA accurately.

When BRCA1 or BRCA2 function is lost, a tumour cell may be unable to repair double-strand DNA damage adequately through homologous recombination. This broader state is called *homologous recombination deficiency*, or *HRD*. BRCA1 and BRCA2 loss are important causes of HRD, but alterations affecting other DNA-repair genes or regulatory processes can also produce it.

HRD can create a therapeutic vulnerability. Tumour cells with impaired homologous recombination become more dependent on alternative processes that help manage DNA damage. One important example involves *PARP*, short for *poly(ADP-ribose) polymerase*. PARP proteins help detect and coordinate repair of certain forms of DNA damage. *PARP inhibitors* reduce this PARP-associated repair activity, and some also trap PARP proteins on damaged DNA, further interfering with DNA replication and repair.

The simplified logic is:

BRCA-associated homologous-recombination repair is impaired

- tumour cell depends more strongly on alternative DNA-damage responses
- PARP inhibition further limits the cell's ability to manage DNA damage
- damage accumulates
- tumour cell may die

This is an example of *synthetic lethality*: disruption of either one of two biological processes may be tolerated, but disrupting both together is lethal to the cell. This strategy can create a therapeutic window because

many normal cells retain functional BRCA-associated repair, although treatment can still cause important toxicity and not every HRD-positive tumour responds equally.

The distinction is simplified but useful:

```
Oncogene activation
→ cancer-promoting activity is turned up
Tumour-suppressor loss
→ a normal restraint on cancer is removed
```

In practice, cancer usually involves multiple alterations. A tumour may contain an activated oncogene, loss of one or more tumour suppressors, changes in gene expression, and changes in the surrounding tissue environment.

3.4 Driver and passenger mutations

A tumour can contain many DNA mutations. However, not every mutation contributes meaningfully to the cancer.

A *driver mutation* contributes causally to tumour development, progression, or maintenance. It gives the affected cell a selective advantage, for example by increasing growth signalling, preventing apoptosis, improving survival under stress, or helping the cell evade immune attack.

A *passenger mutation* is present in a tumour but is not thought to make a meaningful contribution to the cancer phenotype. Passenger mutations accumulate because cancer cells often divide repeatedly, can have impaired DNA repair, and may become genomically unstable.

This distinction matters because a tumour may contain hundreds or thousands of mutations, but only a small subset may represent useful therapeutic targets.

A tumour can become strongly dependent on one driver pathway. This is often called *oncogene addiction*. Despite having many abnormalities, the cancer relies disproportionately on one dominant cancer-promoting signal for continued growth or survival.

For example, if a tumour is highly dependent on signalling from a mutant kinase, inhibiting that kinase can cause a substantial response. However, the presence of a mutation alone does not prove that it is a driver or that targeting it will help.

The key question is:

```
Is this mutation actually driving the disease in this patient,
and does it create an actionable vulnerability?
```

Here, *actionable* means that a practical intervention exists, or could plausibly be developed, to exploit the vulnerability. A mutation may be biologically important but not currently actionable if no suitable therapeutic modality can safely target it.

3.5 Clonal evolution and tumour heterogeneity

A *clone* is a group of cells descended from a common ancestor and sharing particular features, often including one or more mutations. A tumour may begin from one transformed cell, but as it grows, different descendants can acquire additional alterations.

A *subclone* is a smaller population within a tumour that shares the earlier alterations of its ancestral clone but has acquired additional distinctive changes.

Clonal evolution describes the emergence, competition, and expansion of these genetically and functionally different tumour-cell populations. A subclone that grows more quickly, survives better under limited oxygen, or resists treatment may gradually become more common.

Tumour heterogeneity refers to the differences among cancer cells within the same patient. Heterogeneity can occur in several ways:

- *Within one tumour lesion:* Different regions of the same tumour can contain different subclones.
- *Between tumour sites:* A metastasis in the liver may differ from one in the lung or brain.
- *Over time:* A tumour can change as it grows, spreads, or is exposed to treatment.
- *At multiple biological levels:* Cancer cells can differ in DNA mutations, gene expression, protein abundance, cell state, immune interactions, and drug sensitivity.

This heterogeneity is a major reason cancer treatment is difficult. A biopsy samples only a small region of one lesion at one point in time. It may not capture every subclone present elsewhere in the patient.

A therapy may kill the dominant population of sensitive cancer cells but spare a small population of resistant cells. Those resistant cells can then expand, eventually causing disease progression.

3.6 Metastasis

Metastasis is the spread of cancer cells from the original site of the tumour to distant parts of the body. For example, breast cancer can metastasise to bone, liver, lung, or brain.

Metastasis is not simply passive movement. For cancer cells to establish a distant tumour, they generally need to complete several difficult steps:

Cancer cells invade nearby tissue

- enter blood or lymphatic vessels
- survive transport through the body
- leave the circulation at a distant site
- adapt to the new tissue environment
- grow into a new tumour lesion

The *lymphatic system* is a network of vessels and organs involved in fluid balance and immune function. Cancer cells can spread through either blood vessels or lymphatic vessels.

Metastatic disease is often more difficult to cure because it can be widespread, heterogeneous, and able to evolve independently across different sites. A metastasis in one organ may have different biological features from the original tumour or from metastases elsewhere.

Tissue-specific factors also matter for treatment. A drug may reach a liver metastasis differently from a brain metastasis. The brain is protected by the *blood–brain barrier*, a specialised barrier that limits entry of many molecules from the bloodstream into brain tissue. This can make drug delivery to brain tumours or brain metastases especially challenging.

3.7 The tumour microenvironment

The *tumour microenvironment*, often abbreviated *TME*, is the local ecosystem surrounding cancer cells. A tumour is not composed only of malignant cells. It also contains many non-cancer cells, molecules, and physical structures that influence tumour behaviour.

The TME can include:

- *Immune cells*, including cells that may attack tumour cells and cells that may suppress immune responses.

- *Fibroblasts*, cells that help produce and organise structural material in tissues.
- *Blood vessels*, which deliver oxygen, nutrients, immune cells, and drugs.
- *Extracellular matrix*, the network of proteins and other molecules outside cells that provides structural support.
- *Signalling molecules*, including growth factors and cytokines that affect cell behaviour.

The TME can influence tumour growth, drug delivery, immune response, and treatment resistance.

For example, tumour cells and surrounding cells can release cytokines or growth factors that support cancer-cell survival. Abnormal tumour blood vessels and dense extracellular matrix can make it difficult for drugs to penetrate throughout a tumour. Some immune cells in the TME can suppress anti-tumour immune responses.

The TME can also help tumours evade immune attack. A tumour may contain T cells capable of recognising cancer cells, but those T cells may be suppressed through inhibitory signals such as PD-1/PD-L1 signalling, excluded from the tumour tissue, deprived of nutrients, or inhibited by other immune-regulatory mechanisms.

The key point is that cancer cells do not act alone. Their behaviour depends partly on interactions with the surrounding tissue and immune system.

3.8 Biomarkers

A *biomarker* is a measurable biological characteristic that provides information relevant to disease or treatment. Biomarkers can help classify disease, estimate likely outcome, identify suitable treatments, monitor response, or detect toxicity.

Biomarkers can include:

- DNA mutations or gene amplifications.
- Gene-expression patterns.
- Protein abundance or protein modification.
- Cell-surface markers.
- Pathology features observed in tissue samples.
- Molecules measured in blood or other body fluids.
- Imaging measurements.
- Clinical characteristics, such as disease stage or prior treatment.

Three distinctions are especially useful:

Biomarker type	Meaning
Diagnostic biomarker	Helps identify, classify, or confirm a disease or disease subtype.
Prognostic biomarker	Provides information about likely outcome regardless of which treatment is received.
Predictive biomarker	Indicates whether a patient is more or less likely to benefit from a particular treatment.

For example, HER2 overexpression or ERBB2 gene amplification can act as a predictive biomarker for HER2-directed therapy. It identifies a group of patients whose tumours are more likely to benefit from

treatments such as trastuzumab or trastuzumab deruxtecan, although biomarker-positive status does not guarantee response in every individual patient.

A biomarker is rarely perfect. A patient with a positive biomarker may not respond, and a patient without it may occasionally benefit. Biomarkers must therefore be evaluated carefully in the relevant disease and treatment setting.

3.9 Resistance to therapy

Resistance means that a tumour does not respond to treatment, responds only partially, or initially responds but later progresses.

Primary resistance means that the tumour is resistant from the outset. The treatment may be given, but there is little or no meaningful response.

Acquired resistance means that a tumour initially responds but later becomes resistant. This often occurs because treatment kills sensitive cells while resistant cells survive and expand.

Resistance can arise through several mechanisms:

```
Target alteration
→ the drug no longer binds or inhibits the target effectively
Target loss
→ the target is no longer present or sufficiently expressed
Bypass signalling
→ the tumour uses an alternative pathway to maintain growth or survival
Downstream mutation
→ signalling remains active despite inhibition of the original target
Drug-efflux or PK changes
→ insufficient effective drug exposure inside tumour cells
Phenotypic change
→ tumour cells adopt a different cell state with reduced drug sensitivity
Immune escape
→ tumour cells reduce antigen presentation or suppress immune attack
```

Drug efflux refers to the active export of a drug from a cell, often through transporter proteins. If cancer cells export a drug efficiently, intracellular drug concentration may become too low for effective target inhibition.

PK stands for pharmacokinetics: the study of how a drug is absorbed, distributed, metabolised, and excreted by the body. It is introduced in detail in Chapter 5. Here, the key point is that inadequate drug exposure at the tumour can contribute to resistance.

A targeted therapy can therefore produce a dramatic early response and later fail as resistant clones expand. This does not necessarily mean that the original target hypothesis was wrong. It may show that the tumour has evolved around the intervention.

3.10 Why targeted therapy can work spectacularly, then fail

Molecularly targeted therapy is most effective when a tumour has a strong and actionable dependency. For example, if a cancer is highly dependent on signalling from a mutant form of EGFR, inhibiting EGFR can rapidly reduce tumour growth and cause substantial tumour shrinkage.

This can produce a striking clinical response because the treatment is exploiting a real biological weakness. The drug is not simply toxic to rapidly dividing cells in general; it is preferentially disrupting a pathway on which the cancer depends.

However, cancers are adaptive systems. A tumour may contain a small number of cells that are less dependent on the target, have a mutation preventing drug binding, or can use an alternative pathway. These cells may survive treatment.

The typical pattern is:

```
Tumour initially dominated by target-dependent cells
→ targeted therapy suppresses or kills sensitive cells
→ rare resistant cells survive
→ resistant cells expand over time
→ disease progresses despite continued treatment
```

This pattern explains why treatment combinations, sequential therapies, biomarker monitoring, and resistance research are central to modern oncology. The goal is not only to identify an initial dependency, but also to understand which patients are likely to benefit, how long benefit may last, and how the tumour may eventually escape.

3.11 Key takeaways

Cancer is not merely “cells dividing too quickly.” It is an evolving and heterogeneous biological system in which cells acquire capabilities that support abnormal growth, survival, spread, and immune evasion.

The central causal picture is:

```
Cells acquire cancer-promoting alterations
→ advantageous clones expand
→ tumours become heterogeneous
→ tumour cells interact with their microenvironment
→ treatment kills sensitive cells
→ resistant cells may survive and expand
```

Several distinctions are especially useful to retain:

- Oncogenes promote cancer when abnormally activated, while tumour suppressor genes normally restrain cancer-associated processes.
- Driver mutations contribute to cancer development or maintenance, whereas passenger mutations are present but do not meaningfully drive the disease.
- Tumours are heterogeneous: different cancer cells, lesions, and time points can have different mutations, cell states, and treatment sensitivities.
- Metastasis is the spread of cancer cells to distant organs and is a complex multi-step biological process.
- The tumour microenvironment includes immune cells, blood vessels, fibroblasts, extracellular matrix, and signalling molecules that can influence growth, immunity, drug delivery, and resistance.
- Biomarkers can help identify disease subtypes, estimate prognosis, or predict benefit from a particular therapy.
- Resistance can be present before treatment or emerge during treatment through target changes, bypass pathways, inadequate drug exposure, phenotypic change, or immune escape.

The core questions in oncology drug discovery are:

```
Which cancer cells depend on which pathway or biological process?
Which patients have that dependency?
Can the dependency be targeted safely and effectively?
How will tumour heterogeneity and resistance affect the durability of response?
```

3.12 Super-short summary

Cancer develops when cells acquire alterations that let them grow, survive, spread, and evade normal control. These altered cells evolve: advantageous clones expand, additional changes accumulate, and tumours become heterogeneous.

A tumour may depend strongly on a particular driver mutation or pathway, making targeted therapy possible. However, treatment often removes sensitive cells while resistant cells survive and expand. Cancer therapy therefore requires understanding not only the initial target, but also the patient population, tumour microenvironment, heterogeneity, and likely resistance mechanisms.

3.13 Practice questions

How to use these questions. Try to answer each question without looking at the answer. Focus on explaining the causal logic in your own words: how cancer begins, evolves, responds to treatment, and develops resistance.

1. Why is cancer described as an evolutionary disease of cells?

Answer: Cancer cells acquire genetic and other biological alterations. Some alterations give cells a growth or survival advantage, allowing those cells to produce more descendants. Further changes can accumulate, creating different tumour-cell populations that compete and evolve under pressures such as limited nutrients, immune attack, and treatment.

2. What are the broad capabilities, or hallmarks, that cancer cells can acquire?

Answer: Important capabilities include sustained growth signalling, evasion of growth suppressors and cell death, repeated division, blood-vessel formation, invasion and metastasis, immune evasion, and metabolic reprogramming. Different tumours rely on these capabilities to different degrees.

3. What is the difference between an oncogene and a tumour suppressor gene?

Answer: An oncogene promotes cancer when abnormally activated or overexpressed. A tumour suppressor gene normally restrains inappropriate growth, supports DNA repair, or promotes removal of damaged cells; cancer can arise when it loses function.

4. What are driver and passenger mutations, and why does the distinction matter?

Answer: A driver mutation contributes causally to cancer growth, survival, or progression, whereas a passenger mutation is present without meaningfully driving the cancer. The distinction matters because a tumour can contain many mutations, but only a subset are useful therapeutic targets.

5. What are homologous recombination deficiency, PARP inhibition, and synthetic lethality?

Answer: Homologous recombination deficiency, or HRD, means that a cell has impaired ability to repair certain serious forms of DNA damage through the normally accurate homologous recombination pathway. Loss of BRCA1 or BRCA2 is an important cause of HRD. PARP inhibitors further interfere with DNA-damage responses, which can cause DNA damage to accumulate in HRD tumour cells. This is an example of synthetic lethality: loss of either homologous-recombination repair or PARP-associated repair may be tolerated, but disruption of both can kill the cell.

6. What is oncogene addiction?

Answer: Oncogene addiction is a state in which a tumour depends disproportionately on one dominant cancer-promoting gene or pathway. Inhibiting that dependency can produce a substantial response, even though the tumour contains many other alterations.

7. What are clonal evolution and tumour heterogeneity?

Answer: Clonal evolution is the emergence and expansion of different tumour-cell populations descended from a common ancestor. Tumour heterogeneity is the resulting variation among cancer

cells within one lesion, between lesions, and over time, including differences in mutations, cell states, and drug sensitivity.

8. What is metastasis, and why is metastatic disease difficult to treat?

Answer: Metastasis is the spread of cancer cells from the original tumour to distant organs. It requires cancer cells to invade tissue, enter and survive in circulation, establish themselves in a new organ, and grow there. Metastatic disease is difficult because it is widespread, heterogeneous, and can evolve differently across sites.

9. What is the tumour microenvironment, and why does it matter?

Answer: The tumour microenvironment is the local ecosystem around cancer cells, including immune cells, fibroblasts, blood vessels, extracellular matrix, and signalling molecules. It can influence tumour growth, drug delivery, immune suppression, and resistance to treatment.

10. What is the difference between diagnostic, prognostic, and predictive biomarkers?

Answer: A diagnostic biomarker helps identify or classify disease. A prognostic biomarker provides information about likely outcome regardless of treatment. A predictive biomarker indicates whether a patient is more or less likely to benefit from a particular therapy.

11. Why can a molecularly targeted therapy produce a major initial response and then later fail?

Answer: The treatment can effectively kill tumour cells that depend on the target pathway, producing an initial response. However, resistant cells may already exist or arise during treatment. They can survive through target changes, bypass signalling, insufficient drug exposure, cell-state changes, or immune escape, and then expand until the disease progresses.

4 Pharmacology in one chapter

4.1 What pharmacology is trying to describe

Pharmacology is the study of how drugs interact with biological systems. It connects the molecular action of a drug, such as binding to a protein, to its effects in cells, tissues, organs, and ultimately patients.

At its core, pharmacology asks:

What does the drug bind or alter?
What happens because of that interaction?
How does this create clinical benefit, toxicity, or both?

A *drug target* is a biological entity whose modulation is intended to produce a useful therapeutic outcome. As introduced in Chapter 1, targets are often proteins, but they need not always be individual proteins.

A target can be:

- An *enzyme*, such as a kinase that transfers phosphate groups to other proteins.
- A *receptor*, which detects a signal and changes cell behaviour.
- An *ion channel*, a membrane protein that controls the movement of ions across a cell membrane.
- A *transporter*, a protein that moves particular molecules across a cell membrane.
- A *protein–protein interaction*, such as an interaction between an immune receptor and its ligand.
- A *nucleic acid*, such as DNA or RNA.

-
- A *cell-surface antigen*, meaning a molecular feature on a cell that can be recognised by an antibody or immune cell.
 - An *immune pathway*, such as a pathway that suppresses or activates immune cells.

The term *modulation* is deliberately broad. A drug might reduce a target's activity, activate it, block its interaction with another molecule, remove the target protein from the cell, or change its response to a natural signal.

A medicine does not necessarily “fix” a disease completely. Depending on the disease and treatment setting, it may reduce symptoms, slow disease progression, suppress abnormal activity, kill a subset of pathological cells, prevent cancer recurrence, or reduce the risk of a future event.

For example, a cancer drug may not eliminate every tumour cell. It may instead reduce tumour burden, delay disease progression, extend survival, or make surgery more likely to succeed. Whether this represents a useful treatment depends on the size and durability of the benefit, as well as the treatment's risks.

4.2 Binding is not the same as efficacy

Drug binding is often the first event in a causal chain, but binding alone is not the same as a useful treatment effect.

A simplified causal chain is:

```
Drug binds target
→ target function changes
→ pathway changes
→ cellular behaviour changes
→ tissue or organism changes
→ clinical benefit and/or toxicity
```

For example, a kinase inhibitor may bind a kinase and reduce its catalytic activity. This can reduce phosphorylation of downstream proteins, alter a signalling pathway, slow tumour-cell growth, and potentially shrink a tumour. However, each step in this chain must actually occur in the relevant patient for the treatment to help.

A compound can bind strongly to a target but still fail to provide clinical benefit. There are several possible reasons:

- The target may not be important enough for the disease in the relevant patients.
- The drug may not reach the target tissue at a sufficient concentration.
- The drug may bind the target but not change its function enough.
- The required dose may cause unacceptable toxicity.
- Other pathways may compensate for the effect of target modulation.
- The disease may be too heterogeneous for one target to control all relevant cells.

This distinction is central to drug discovery. A successful biochemical experiment showing that a molecule binds a purified protein is useful evidence, but it is only an early part of the overall therapeutic argument.

4.3 Agonists

An *agonist* is a molecule that binds to a receptor and activates it, thereby increasing receptor-mediated signalling.

Recall from Chapter 1 that a receptor is a protein that detects a signal and changes cell behaviour. The body's natural signalling molecule for a receptor is called an *endogenous ligand*. An agonist may mimic the effect of this endogenous ligand.

For example, insulin is the endogenous agonist of the insulin receptor. When insulin binds its receptor, it activates signalling pathways involved in the regulation of blood glucose and metabolism.

A therapeutic agonist is a drug designed to activate a receptor in order to produce a desired effect. Depending on the receptor and disease, agonists may be used to alter metabolism, relax airway muscles, reduce pain, regulate hormones, or modify other biological processes.

Two related terms are useful:

- A *full agonist* can produce the maximum response that is achievable in a particular biological system.
- A *partial agonist* activates the receptor but produces a smaller maximum response, even when it occupies a large proportion of the available receptors.

The difference is illustrated conceptually below:

```
Full agonist
→ can produce maximal receptor response
Partial agonist
→ activates the same receptor, but with a lower maximal response
```

Whether a response is “maximal” depends on the particular cell type, tissue, receptor expression level, and assay. Thus, agonism is not only a property of the molecule itself; it is also measured in a particular biological context.

4.4 Antagonists

An *antagonist* is a molecule that prevents a receptor from being activated by its endogenous ligand or another agonist.

In the normal state, an endogenous ligand binds its receptor and triggers a biological response. An antagonist reduces this response by preventing activation of the receptor.

A simplified example is:

```
Normal state:
Ligand binds receptor
→ receptor activates signalling
With antagonist:
Antagonist blocks receptor
→ ligand cannot activate it normally
```

A *competitive antagonist* binds at, or close to, the receptor's *orthosteric site*: the main binding site used by the endogenous ligand. The antagonist and ligand compete for access to this site.

A *non-competitive antagonist* reduces receptor signalling by another mechanism. For example, it may bind elsewhere on the receptor, reduce the receptor's ability to activate, or bind irreversibly so that the receptor can no longer function normally.

An antagonist does not necessarily reduce the amount of receptor protein in the cell. Instead, it reduces the receptor's ability to produce its usual response when a ligand is present.

This distinction is important:

Antagonist

- receptor may still be present
- receptor activation is reduced

Degrader

- amount of receptor protein is reduced

4.5 Inhibitors

An *inhibitor* is a molecule that decreases the functional activity of a target. The term is especially common for enzymes, but it can be used more broadly whenever a drug reduces a target's biological function.

For example, a *kinase inhibitor* reduces the activity of a kinase. Since kinases add phosphate groups to proteins, kinase inhibition can reduce phosphorylation of downstream proteins and thereby alter cell signalling.

When someone says “EGFR inhibition”, they usually mean that signalling mediated by the epidermal growth factor receptor, or EGFR, is being reduced. This does not necessarily mean that the amount of EGFR protein has decreased.

EGFR signalling could be inhibited through several different mechanisms:

- A small molecule might bind EGFR's intracellular kinase domain and reduce its catalytic activity.
- An antibody might bind the extracellular part of EGFR and reduce ligand-driven receptor activation.
- A protein degrader might reduce the total amount of EGFR protein.
- Another intervention might prevent an interaction that EGFR requires for normal signalling.

The word *inhibition* therefore describes the functional consequence—reduced target activity or signalling—rather than one universal molecular mechanism.

4.6 Enzyme inhibitors

An *enzyme inhibitor* reduces an enzyme's ability to carry out its normal chemical reaction.

As introduced in Chapter 1, an enzyme acts on a substrate at its active site. One common mechanism of enzyme inhibition is *competitive inhibition*. Here, the inhibitor competes with the enzyme's normal substrate, or with a required cofactor, for access to the active site.

A *cofactor* is a non-protein molecule or ion required for an enzyme to function properly. ATP is a cofactor for many kinases because the kinase uses ATP as the source of the phosphate group it transfers to its substrate.

Many kinase inhibitors are ATP-competitive inhibitors:

Normal kinase reaction:

Kinase + ATP + substrate

- phosphorylated substrate

With ATP-competitive inhibitor:

Inhibitor occupies ATP-binding pocket

- ATP binding is reduced
- substrate phosphorylation is reduced

However, enzyme inhibitors can act through many other mechanisms. They may:

- Bind an allosteric site rather than the active site.
- Form a covalent bond with the enzyme.

- Interfere with binding of a required cofactor.
- Stabilise an inactive protein conformation.
- Prevent formation of an active protein complex.

The key concept is:

An enzyme inhibitor reduces the enzyme's ability to carry out its relevant chemical reaction.

4.7 Receptor antagonists and neutralising antibodies

A *receptor antagonist* prevents receptor activation. It may do this by blocking the binding of the receptor's natural ligand, preventing the receptor from adopting its active shape, or stabilising an inactive receptor state.

A *neutralising antibody* is an antibody that prevents a biologically important interaction. It can bind either a soluble signalling molecule or a receptor.

For example, a *cytokine* is a signalling molecule often used by immune cells. If a cytokine promotes harmful inflammation, an antibody can reduce the inflammatory pathway in two broad ways:

- It can bind the cytokine itself, preventing the cytokine from reaching its receptor.
- It can bind the cytokine's receptor, preventing the cytokine from activating that receptor.

The distinction is:

Ligand-neutralising antibody
→ binds the signal itself
Receptor-blocking antibody
→ binds the receiving protein
Both can reduce pathway activation

These mechanisms can have different PK, tissue-distribution, selectivity, and safety properties. For example, a ligand-neutralising antibody may block signalling through every receptor that uses that ligand, whereas a receptor-targeting antibody affects signalling specifically through the receptor it binds.

4.8 Protein degraders

A *protein degrader* reduces the amount of a target protein rather than merely blocking one of its functions.

Cells continuously make, repair, recycle, and remove proteins. One important protein-removal system uses a small molecular label called *ubiquitin*. Proteins labelled with appropriate ubiquitin chains can be recognised and broken down by a large cellular complex called the *proteasome*.

Some therapeutic strategies exploit this natural system. A degrader can bring a target protein close to an *E3 ubiquitin ligase*, a protein complex that helps attach ubiquitin to selected proteins. The target can then be labelled and destroyed by the proteasome.

The simplified logic is:

Degrader brings target protein near E3 ubiquitin ligase
→ target is labelled with ubiquitin
→ proteasome recognises the labelled target
→ target protein is degraded

This can be useful when simple inhibition is insufficient. A protein may have several functions. For example, it may have an enzyme active site but also act as a *scaffold*, meaning that it physically brings other proteins together into a functional complex. Blocking the active site may leave the scaffolding function intact, whereas degrading the whole protein can remove both functions.

Protein degradation is not automatically better than inhibition. It creates a different biological intervention and introduces different questions:

- Is the target sufficiently degraded in the relevant tissue?
- Does degradation affect unintended proteins?
- Does the degrader have suitable pharmacokinetics and tissue distribution?
- Does removing the whole protein create more toxicity than partial inhibition?
- Can the target or cell state change to prevent effective degradation?

4.9 Orthosteric and allosteric modulation

The *orthosteric site* is the main functional binding site of a protein. For an enzyme, this is often the active site where the substrate binds. For a receptor, it is often the site bound by the endogenous ligand.

An *allosteric site* is a distinct binding site elsewhere on the protein. Binding at an allosteric site can change the protein's shape, movement, interactions, or responsiveness.

An *allosteric modulator* is therefore a molecule that alters target behaviour without necessarily competing directly with the substrate or endogenous ligand.

An allosteric inhibitor may:

- Stabilise an inactive protein conformation.
- Reduce catalytic efficiency.
- Change protein dynamics.
- Reduce the probability that a receptor becomes activated.
- Weaken a protein–protein interaction.
- Prevent formation of a required active protein complex.

A *negative allosteric modulator*, or *NAM*, reduces the response of a receptor to an agonist. It often binds outside the orthosteric site and may preserve some baseline receptor function while limiting excessive signalling.

A *positive allosteric modulator*, or *PAM*, increases the response of a receptor to its agonist. A PAM may be useful when it is desirable to strengthen the body's normal signal without activating the receptor continuously on its own.

A common oversimplification is that an allosteric inhibitor must physically close an active site so that the substrate can no longer fit. This is not generally correct. An allosteric inhibitor may instead change the probability of productive catalysis, alter protein dynamics, disrupt an interaction, or stabilise a less active protein state.

4.10 Affinity, potency, efficacy, and selectivity

Several related pharmacological terms are often confused. They refer to different parts of the drug-action process.

Term	Meaning
Affinity	How strongly a molecule binds a target under defined conditions.
Potency	How much drug is needed to produce a specified effect in a particular assay or biological system.
Efficacy	The maximum effect a drug can produce in a particular system.
Selectivity	The degree to which a drug acts on its intended target rather than other targets.

Affinity concerns binding. It is often described using the *dissociation constant*, written K_d . Under defined conditions, a lower K_d generally indicates stronger binding.

Potency concerns the concentration or dose needed to produce a defined effect. For example, an *IC50* is the concentration that produces 50% inhibition in a particular assay, while an *EC50* is the concentration that produces 50% of the maximum measured effect. Lower IC50 or EC50 values generally indicate greater potency in that specific assay.

An *inhibition constant*, written K_i , is another measure used to describe inhibitor binding under a specified mechanistic model. Unlike IC50, K_i is intended to account more directly for the interaction between an inhibitor, enzyme, substrate, and assay conditions. In practice, the interpretation of both IC50 and K_i depends on the experimental system.

Efficacy concerns the size of the largest effect that the drug can produce. A partial agonist, for example, may have high affinity for a receptor but lower efficacy than a full agonist.

Selectivity concerns unintended targets. A drug may bind its intended target but also bind other proteins, particularly at higher concentrations. Such *off-target* activity can cause toxicity or make the observed biology difficult to interpret.

These concepts should not be treated as interchangeable:

High affinity does not guarantee high efficacy.
 High potency in one assay does not guarantee high potency in another.
 High selectivity is valuable, but perfect selectivity is not always required.

Measured values depend on context. For example, IC50 can change with assay design, substrate concentration, protein construct, cell type, incubation time, and readout. A compound may be potent against purified protein but less potent in cells because it enters cells poorly, is removed by transporters, is metabolised, encounters a different target conformation, or must compete with abundant cellular ATP.

Perfect selectivity is not always necessary or even desirable. In some diseases, carefully understood activity at multiple targets may be helpful. The important question is whether the complete pattern of target activity produces an acceptable balance of benefit and risk.

4.11 Target engagement

Target engagement means that a drug physically interacts with its intended target in the relevant biological context.

This is a stronger claim than showing that a compound binds purified protein in a test tube. It asks whether the drug reaches the target in living cells, relevant tissues, or ideally patients, at a meaningful concentration and for a sufficient duration.

The key sequence is:

```
Drug is administered
→ drug reaches relevant tissue
→ drug engages intended target
→ target biology changes
```

Target engagement can be measured directly or indirectly.

Direct measures may assess *target occupancy*, meaning the proportion of available target molecules bound by the drug, or may directly detect drug–target complexes.

Indirect measures assess a biological consequence expected to occur after target engagement. For example, reduced phosphorylation of a known kinase substrate can provide indirect evidence that a kinase inhibitor is acting on its intended pathway.

Target engagement is a crucial bridge between molecular pharmacology and patient outcomes. If there is no evidence that a drug reaches and modulates its target in patients, it is difficult to interpret either a negative or a positive clinical result.

However, target engagement alone does not prove clinical benefit. A drug can engage its target successfully while failing because the target is not sufficiently important, compensatory pathways overcome the effect, or toxicity prevents adequate treatment.

4.12 Key takeaways

Pharmacology connects a drug’s molecular interaction with its target to its biological and clinical consequences.

The central causal chain is:

```
Drug binds or alters target
→ target function changes
→ pathway and cell behaviour change
→ tissue or organism changes
→ clinical benefit and/or toxicity
```

Several distinctions are especially useful to retain:

- An agonist activates a receptor, whereas an antagonist prevents receptor activation.
- An inhibitor reduces target function; for enzymes, this usually means reducing catalytic activity.
- A neutralising antibody can bind a signalling ligand or its receptor to prevent their interaction.
- A degrader reduces the amount of target protein, potentially removing both catalytic and non-catalytic functions.
- Orthosteric drugs bind the main functional site of a protein, while allosteric modulators bind elsewhere and alter target behaviour indirectly.
- Affinity describes binding strength, potency describes the amount of drug needed for an effect, efficacy describes the maximum effect, and selectivity describes preference for intended versus unintended targets.
- Target engagement shows that a drug reaches and interacts with its intended target in the relevant biological context, but it does not by itself prove patient benefit.

When encountering a statement such as “drug X inhibits target Y”, ask:

Does drug X bind target Y directly?
Which site does it bind?
Does it inhibit, activate, antagonise, neutralise, degrade,
or otherwise modulate Y?
What downstream pathway or biomarker changes?
Why should that change help patients, and what toxicity might it cause?

4.13 Super-short summary

Pharmacology explains how a drug's interaction with a target leads—or fails to lead—to patient benefit. Binding is only the beginning: the drug must change target function, alter relevant biology, reach the correct tissue at a tolerable exposure, and overcome compensatory mechanisms.

Drugs can activate receptors, block receptors, inhibit enzymes, neutralise ligands, degrade proteins, or modulate targets allosterically. To understand a drug, distinguish its molecular mechanism from its downstream biological and clinical effects.

4.14 Practice questions

How to use these questions. Try to answer each question without looking at the answer. Focus on the chain from drug binding to clinical effect, and on the difference between related pharmacological terms.

1. What is pharmacology, and what is a drug target?

Answer: Pharmacology studies how drugs interact with biological systems and how those interactions produce benefit or harm. A drug target is a biological entity, often a protein, whose modulation is intended to create a useful therapeutic outcome.

2. Why is target binding not the same as clinical efficacy?

Answer: Binding is only the first step. The drug must change target function, alter a disease-relevant pathway, reach the relevant tissue at sufficient exposure, avoid unacceptable toxicity, and overcome compensatory biology. A drug can bind strongly but still fail to help patients.

3. What is the difference between an agonist and an antagonist?

Answer: An agonist binds and activates a receptor, increasing receptor-mediated signalling. An antagonist prevents the receptor from being activated by its endogenous ligand or another agonist.

4. What is the difference between a full agonist and a partial agonist?

Answer: A full agonist can produce the maximum response achievable in a particular biological system. A partial agonist activates the same receptor but produces a lower maximum response, even when it occupies many receptors.

5. What does it mean to say that a drug inhibits EGFR?

Answer: It means that the drug reduces EGFR-mediated signalling. This could occur through direct kinase inhibition, extracellular antibody binding, protein degradation, or disruption of an interaction needed for EGFR function; it does not necessarily mean that less EGFR protein is present.

6. How can an enzyme inhibitor reduce enzyme activity?

Answer: It can compete with a substrate or cofactor at the active site, bind an allosteric site, form a covalent bond with the enzyme, stabilise an inactive conformation, or prevent formation of an active protein complex.

7. What is the difference between a ligand-neutralising antibody and a receptor-blocking antibody?

Answer: A ligand-neutralising antibody binds the signalling molecule itself and prevents it from reaching its receptor. A receptor-blocking antibody binds the receptor and prevents the signalling molecule from activating it. Both can reduce pathway activation.

8. How does a protein degrader differ from a conventional inhibitor?

Answer: A conventional inhibitor usually blocks one function of a target protein, such as catalytic activity. A degrader reduces the amount of the target protein itself, potentially removing catalytic activity and non-catalytic functions such as scaffolding.

9. What is the difference between orthosteric and allosteric modulation?

Answer: Orthosteric modulation involves binding the target's main functional site, such as an enzyme active site or a receptor's endogenous ligand-binding site. Allosteric modulation involves binding a different site that changes the target's shape, dynamics, interactions, or responsiveness.

10. Distinguish affinity, potency, efficacy, selectivity, and target engagement.

Answer: Affinity is binding strength to a target, often measured using quantities such as K_d . Potency is the concentration or dose required to produce a specified effect in a particular system; IC50 and EC50 are common potency measurements. Efficacy is the maximum effect the drug can produce. Selectivity is preference for intended rather than unintended targets. Target engagement is evidence that the drug physically interacts with its intended target in the relevant biological context.

5 PK, PD, exposure, and dose

5.1 From an administered dose to a patient outcome

A *dose* is the amount of drug given to a patient. However, receiving a dose is not the same as having a biological effect. After administration, the drug must be absorbed or delivered into the bloodstream, remain in the body long enough, reach the relevant tissue, engage enough of its target, and alter disease biology without causing unacceptable toxicity.

A useful causal chain is:

```
Dose
→ systemic exposure
→ tissue exposure
→ target engagement
→ pathway modulation
→ cellular effect
→ tumour or organism effect
→ clinical outcome
```

Each arrow represents a separate scientific question. A drug might bind its target very well in a laboratory experiment but fail in patients because it is poorly absorbed, rapidly removed from the body, unable to reach the tumour, or too toxic at the exposure required to affect the target.

For example, an oral kinase inhibitor may be potent against purified kinase protein and cancer cells in a dish. In a patient, however, it must survive the gastrointestinal tract and liver, reach the bloodstream, enter tumour tissue, enter tumour cells if the target is intracellular, and maintain a sufficiently high concentration to inhibit the kinase.

This is why drug discovery cannot focus only on whether a molecule binds a target. It must also consider whether the molecule can produce useful and tolerable drug exposure in the relevant patients.

5.2 Pharmacokinetics: what the body does to the drug

Pharmacokinetics, abbreviated *PK*, describes how the body handles a drug over time. It asks questions such as: How much drug enters the bloodstream? How long does it remain there? Which tissues does it reach? How is it chemically modified? How is it removed?

PK is commonly organised around *ADME*:

Term	Meaning
Absorption	Entry of a drug into systemic circulation after administration.
Distribution	Movement of a drug from blood into tissues and body compartments.
Metabolism	Chemical modification of a drug, often but not only in the liver.
Excretion	Elimination of a drug or its metabolites, often through urine or bile.

Systemic circulation refers broadly to the bloodstream that distributes substances throughout the body. A drug that reaches systemic circulation can potentially travel to many tissues, although it will not necessarily reach every tissue equally well.

PK asks whether sufficient active drug reaches the desired location for long enough. This is particularly important because the target may be located in a tumour, the brain, the lung, an inflamed joint, an immune cell, or another specific tissue or cell type.

The phrase “what the body does to the drug” is a useful simplification. In reality, the drug and body influence each other: drug properties influence where it goes and how long it remains, while patient-specific factors such as organ function, age, other medicines, and disease state can alter PK.

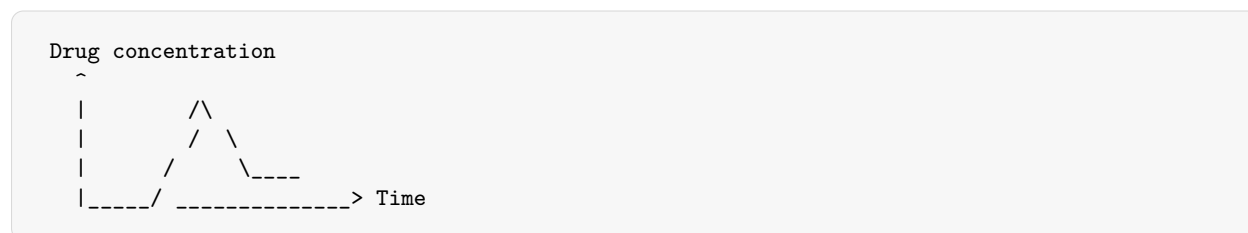
5.3 Dose and exposure

The *dose* is the amount of drug administered at one time or according to a schedule. Drugs can be administered by different *routes of administration*, including:

- *Oral*: swallowed as a tablet, capsule, or liquid.
- *Intravenous*, or *IV*: delivered directly into a vein.
- *Subcutaneous*: injected into the tissue beneath the skin.
- *Inhaled*: delivered into the lungs.
- *Topical*: applied to the skin or another local surface.

Exposure is the concentration of drug in the body over time. Exposure is commonly measured in *plasma*, the liquid component of blood. Plasma sampling is relatively practical, but plasma concentration is not always identical to concentration at the target tissue.

A *concentration–time curve* shows how drug concentration changes after administration:



After oral administration, concentration often rises as the drug is absorbed from the gastrointestinal tract into the bloodstream. After IV administration, concentration may rise quickly or be maintained during an

infusion. Concentration then usually declines as the drug distributes into tissues, is metabolised, and is excreted.

Two patients can receive the same dose but experience different exposure. One patient may absorb the drug less efficiently, metabolise it more rapidly, have impaired kidney function, or take another medication that changes drug metabolism. This is one reason why exposure can often be more informative than dose alone.

5.4 AUC, $C_{\max}$, and half-life

Several PK quantities are commonly used to summarise a concentration–time curve.

AUC, short for *area under the concentration–time curve*, summarises total systemic drug exposure over a specified time interval. Conceptually, it is the area below the concentration–time curve:

Higher AUC
→ more total drug exposure over time

AUC is useful because a drug’s biological effect may depend not only on its highest concentration, but also on how much drug exposure accumulates over time.

$C_{\max}$ is the maximum observed drug concentration after dosing. It is the highest point on the concentration–time curve. $C_{\max}$ can matter because some therapeutic effects or toxicities are particularly related to high peak exposure.

For example, two dosing schedules might produce similar AUC but different $C_{\max}$. Giving a total daily dose all at once may create a higher peak concentration than dividing it into smaller doses. If toxicity is driven mainly by high peaks, the divided schedule may be better tolerated even if total exposure is similar.

The *half-life*, often written as $t_{1/2}$, is the approximate time required for drug concentration to decrease by half during an appropriate phase of elimination.

For example, if a drug concentration is 100 units and has a half-life of 12 hours, it would decrease approximately as follows:

Start: 100 units
After 12 h: 50 units
After 24 h: 25 units
After 36 h: 12.5 units

A longer half-life can allow less frequent dosing because the drug remains in the body for longer. However, a long half-life is not always beneficial. If a serious adverse effect occurs, a long-persisting drug may be difficult to remove quickly. Half-life can also contribute to *accumulation*: repeated dosing may gradually increase drug concentration until a relatively stable average level is reached.

5.5 Clearance and bioavailability

Clearance describes how efficiently the body removes a drug from systemic circulation. It is not simply the amount of drug removed at one moment; rather, it describes the body’s overall capacity to eliminate drug relative to its concentration.

All else being equal:

High clearance

- drug concentration falls more quickly
- lower systemic exposure

Low clearance

- drug remains in circulation longer
- higher systemic exposure

Drugs can be cleared through metabolism, excretion into urine, excretion into bile, or other processes. The liver and kidneys are especially important organs for clearance of many medicines.

Bioavailability is the fraction of an administered dose that reaches systemic circulation in active form. For IV administration, bioavailability is generally treated as complete because the drug is delivered directly into the bloodstream.

Oral drugs often have lower bioavailability. Some of the swallowed drug may not be absorbed through the gastrointestinal tract. In addition, drug absorbed from the gut may be metabolised in the gut wall or liver before reaching general systemic circulation. This is called *first-pass metabolism*.

The simplified sequence for an oral drug is:

Oral dose

- absorption from gastrointestinal tract
- possible metabolism in gut wall and liver
- drug reaches systemic circulation

An oral drug may therefore fail not because it lacks potency against its target, but because it cannot achieve sufficient exposure in the body. Conversely, low clearance or unexpectedly high bioavailability can lead to excessive exposure and toxicity.

5.6 Distribution and tissue penetration

A drug must reach the tissue where its target is located. This is called *distribution* or *tissue penetration*.

For a cancer drug, plasma exposure may appear adequate while tumour exposure is too low. A drug may remain mostly in blood, distribute preferentially to some tissues rather than others, bind strongly to plasma proteins, or be unable to cross a tissue barrier.

For a brain disease or brain metastasis, crossing the *blood-brain barrier* can be a major challenge. This specialised barrier protects the brain by limiting entry of many molecules from the bloodstream. It is biologically important, but it can also prevent potentially useful drugs from reaching brain tissue.

Antibodies can have especially complex tissue-penetration behaviour. They are much larger than typical small molecules and may penetrate dense solid tumours slowly or unevenly. Their distribution can depend on tumour blood vessels, pressure inside the tumour, target abundance, target internalisation, and the density of the surrounding extracellular matrix.

Distribution depends on many drug and tissue properties, including:

- *Molecular size*: Smaller molecules can often move through tissues and cross membranes more readily than large molecules.
- *Lipophilicity*: A drug's tendency to dissolve in fats or lipid membranes, which can influence membrane crossing and tissue distribution.
- *Charge*: Electrically charged molecules may cross lipid membranes less easily.
- *Protein binding*: Drugs can bind proteins in blood and tissues. Only the unbound, or *free*, fraction is generally available to cross membranes, bind targets, and be cleared, although protein binding does not by itself determine whether a drug will achieve useful tissue exposure.

- *Transporters*: Membrane proteins that can move drugs into or out of cells.
- *Blood flow*: Tissues with greater blood flow may receive drug more rapidly.
- *Tissue barriers*: Physical and biological barriers, such as the blood–brain barrier.
- *Target binding*: Strong binding to a target in one tissue can affect how much drug remains there.

The main practical point is that a plasma concentration is informative but incomplete. For many programmes, the critical question is whether sufficient free, active drug reaches the relevant target in the relevant tissue.

5.7 Pharmacodynamics: what the drug does to the body

Pharmacodynamics, abbreviated *PD*, describes the biological effects caused by drug exposure. Whereas PK asks what the body does to the drug, PD asks what the drug does to the body.

PD effects can occur at several levels:

```
Drug binds target
→ target occupancy changes
→ pathway biomarker changes
→ cell state changes
→ tissue or tumour changes
→ clinical outcome changes
```

A *biomarker* is a measurable characteristic that provides information about disease or treatment. A *PD biomarker* is a biomarker that changes in response to the drug and provides evidence that the drug is affecting the intended biology.

For example, a kinase inhibitor may reduce phosphorylation of a downstream protein. If that phosphorylation event is known to depend on the targeted kinase pathway, reduced phosphorylation can be a PD biomarker showing that the pathway has been modulated.

PD does not necessarily mean clinical benefit. A drug may clearly change a pathway biomarker but fail to improve patient outcomes if the pathway is not sufficiently important for the disease, if pathway inhibition is incomplete, or if tumour cells compensate through another mechanism.

PK and PD are closely connected:

```
PK
→ how much drug reaches the body or tissue over time
PD
→ what biological effects occur at that exposure
```

5.8 Target occupancy

Target occupancy is the fraction of available target molecules bound by a drug at a given time and location.

In simplified terms:

```
0% occupancy
→ no target is bound
50% occupancy
→ roughly half of available target molecules are bound
100% occupancy
→ nearly all accessible target molecules are bound
```


Occupancy is related to drug concentration and binding affinity. In general, higher local drug concentration and stronger binding tend to increase occupancy, although the actual relationship can be more complicated in living systems.

The required occupancy depends on the target biology. Some targets may require near-complete and sustained inhibition before a meaningful biological effect occurs. Other targets may respond to partial modulation. For example, blocking a pathway that has substantial spare signalling capacity may require high occupancy, whereas a highly sensitive pathway may respond to lower occupancy.

Occupancy can vary across tissues and over time. A drug might produce high target occupancy in blood but lower occupancy in a solid tumour. It might also bind most available target shortly after dosing but much less just before the next dose.

This is why the relevant question is not simply “does the drug bind the target?” but rather:

Does the drug achieve sufficient target occupancy,
for long enough, in the tissue where the disease is located?

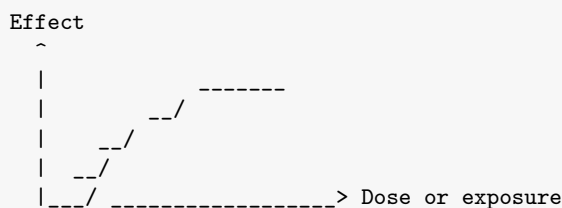
5.9 Dose–response and exposure–response relationships

A *dose–response relationship* describes how a measured effect changes as the administered dose changes.

An *exposure–response relationship* describes how an effect changes as actual drug exposure changes. Exposure–response analysis is often more informative because patients receiving the same dose can have different drug concentrations due to differences in absorption, metabolism, clearance, body size, organ function, or interactions with other medications.

Possible effects include target engagement, biomarker change, tumour shrinkage, symptom improvement, survival outcomes, and adverse events.

A simplified dose– or exposure–response curve may look like:



At low exposure, there may be little effect because insufficient drug reaches or occupies the target. As exposure increases, the effect may increase. Eventually, the desired effect may approach a plateau: further exposure produces little additional benefit because much of the achievable biological response has already been reached.

Toxicity can behave differently. It may continue to rise even after efficacy has plateaued:

Increasing exposure
→ efficacy may plateau
→ toxicity may continue to increase

This is one reason that the highest possible dose is not necessarily the best dose. Dose selection aims to identify an exposure range that provides sufficient benefit while limiting unnecessary toxicity.

5.10 Therapeutic window and therapeutic index

The *therapeutic window* is the range of drug exposure in which a medicine is expected to provide useful benefit without unacceptable toxicity.

A simplified picture is:

```
Too little exposure
→ inadequate target modulation or efficacy
Appropriate exposure
→ useful target modulation and acceptable safety
Too much exposure
→ unacceptable toxicity
```

The therapeutic window is not necessarily identical for every patient. One patient may tolerate a particular exposure well, while another may experience toxicity because of differences in age, organ function, genetics, other treatments, disease severity, or baseline health.

The *therapeutic index* is a related concept describing the separation between exposures or doses associated with desired effects and those associated with harmful effects. In simple experimental settings, it may be represented as a ratio between a toxic dose and an effective dose. In real clinical development, however, benefit and risk are more complex than a single number.

A drug with a wide therapeutic window is often easier to dose because moderate differences in patient exposure are less likely to cause either loss of efficacy or severe toxicity. A drug with a narrow therapeutic window requires more careful dose selection, monitoring, and patient management.

The therapeutic window is therefore one of the central practical constraints in drug development. A compound may be potent and effective in principle but still unsuitable if the exposure required for efficacy overlaps too strongly with exposure causing unacceptable harm.

5.11 Proof of mechanism

Proof of mechanism, often abbreviated *PoM*, is evidence in humans that a candidate drug has engaged its intended target at a predefined and measurable level and produced a functional consequence consistent with the proposed mechanism.

PoM is stronger than evidence from cells or animal models because it asks whether the expected mechanism occurs in patients at clinically feasible doses.

For an oncology kinase inhibitor, PoM might include:

```
Drug exposure at a clinically feasible dose
+
Evidence of target engagement in tumour or relevant tissue
+
Reduced downstream pathway biomarker
```

For example, a study might show that patients achieve plasma concentrations predicted to inhibit a kinase, that drug is present in tumour tissue, and that phosphorylation of a known downstream protein is reduced after treatment.

PoM does not prove that the drug will improve survival, reduce symptoms, or provide an acceptable benefit–risk balance. However, it reduces an important source of uncertainty. If a drug fails clinically despite strong PoM, the problem may be that the target is not sufficiently important, the pathway is compensated for, or the disease is too heterogeneous. If a drug fails without evidence of PoM, it may simply not have reached or modulated its intended target adequately.

5.12 Key takeaways

A dose is only the start of the causal chain. A drug must reach the correct tissue, engage its target at sufficient level and duration, change disease-relevant biology, and remain tolerable.

The central PK/PD chain is:

```
Dose
→ systemic exposure
→ tissue exposure
→ target engagement
→ PD biomarker and pathway modulation
→ cellular and tumour effect
→ clinical outcome
```

Several distinctions are especially useful to retain:

- PK describes what the body does to a drug: absorption, distribution, metabolism, and excretion.
- PD describes what a drug does to the body: target engagement, pathway effects, cellular changes, and clinical effects.
- Dose is the amount administered, while exposure is drug concentration in the body over time.
- AUC measures total exposure, $C_{\max}$ measures peak concentration, and half-life describes how quickly concentration declines.
- Clearance tends to reduce exposure, while bioavailability determines how much administered drug reaches systemic circulation.
- Plasma exposure does not guarantee sufficient exposure in the relevant tissue or tumour.
- The therapeutic window is the exposure range that balances useful efficacy with acceptable safety.
- PoM provides evidence in humans that a drug has achieved target engagement and a downstream effect consistent with its proposed mechanism.

When reading a drug paper, ask:

```
At the dose given, did the drug achieve sufficient systemic exposure?
Did it reach the relevant tissue?
Was there evidence of target engagement and pathway modulation?
Was the exposure sufficient for efficacy but tolerable for patients?
Could differences in PK explain variable response or toxicity?
```

5.13 Super-short summary

PK describes how the body handles a drug, including absorption, distribution, metabolism, and excretion. PD describes the biological effects caused by drug exposure. A dose is not enough on its own: the drug must reach the relevant tissue, engage its target for long enough, change the pathway, and remain within a tolerable therapeutic window.

AUC, $C_{\max}$, half-life, clearance, and bioavailability describe different aspects of exposure. The key practical question is whether the achieved exposure produces sufficient target engagement and benefit without unacceptable toxicity.

5.14 Practice questions

How to use these questions. Try to answer each question without looking at the answer. Focus on the causal chain from administered dose to patient outcome, and distinguish carefully between dose, exposure, PK, PD, and target engagement.

1. Why is an administered dose not the same thing as drug activity or clinical benefit?

Answer: After administration, a drug must achieve sufficient systemic exposure, reach the relevant tissue, engage enough target for long enough, alter disease-relevant biology, and remain tolerable. Failure at any step can prevent clinical benefit.

2. What is pharmacokinetics, and what does ADME stand for?

Answer: Pharmacokinetics, or PK, describes how the body handles a drug over time. ADME stands for absorption, distribution, metabolism, and excretion.

3. What is the difference between dose and exposure?

Answer: Dose is the amount of drug administered. Exposure is the concentration of drug in the body over time. Two patients can receive the same dose but have different exposures because of differences in absorption, metabolism, clearance, or other factors.

4. What do AUC, $C_{\max}$, and half-life describe?

Answer: AUC describes total drug exposure over time. $C_{\max}$ is the highest observed drug concentration after dosing. Half-life is the approximate time required for drug concentration to decrease by half during an appropriate elimination phase.

5. What are clearance and bioavailability?

Answer: Clearance describes how efficiently the body removes drug from systemic circulation. Bioavailability is the fraction of an administered dose that reaches systemic circulation in active form. High clearance tends to reduce exposure, while low oral bioavailability can prevent adequate exposure even for a potent drug.

6. Why does adequate plasma exposure not necessarily mean adequate tumour or tissue exposure?

Answer: Drugs distribute unevenly across tissues. Tissue penetration can be limited by molecular size, charge, protein binding, transporters, blood flow, tissue barriers such as the blood–brain barrier, and tumour properties such as abnormal vessels or dense extracellular matrix.

7. What is pharmacodynamics, and give an example of a PD biomarker for a kinase inhibitor.

Answer: Pharmacodynamics, or PD, describes the biological effects caused by drug exposure. For a kinase inhibitor, reduced phosphorylation of a known downstream substrate can be a PD biomarker indicating that the kinase pathway has been modulated.

8. What is target occupancy, and why is it not necessarily the same in every tissue or at every time?

Answer: Target occupancy is the fraction of available target molecules bound by a drug. It varies with local drug concentration, binding affinity, target abundance, tissue distribution, and time since dosing, so occupancy may differ between blood and tumour or between peak and trough exposure.

9. What is the difference between a dose–response and an exposure–response relationship?

Answer: A dose–response relationship links administered dose to effect. An exposure–response relationship links actual drug concentration over time to effect. Exposure–response is often more informative because patients receiving the same dose can have different exposures.

10. **What are the therapeutic window and proof of mechanism?**

Answer: The therapeutic window is the exposure range that provides useful benefit without unacceptable toxicity. Proof of mechanism, or PoM, is evidence in humans that a drug has engaged its intended target at a predefined measurable level and produced a downstream effect consistent with the proposed mechanism.

Part II Targets, mechanisms, and therapeutic modalities

6 Target identification, validation, tractability, and druggability

6.1 Four concepts that should be kept separate

Several target-related terms are often used loosely, but they answer different questions. Keeping them separate prevents an important mistake: assuming that a protein strongly associated with disease is automatically a good drug target.

Concept	Central question
Target identification	What biological entity might be worth modulating in this disease?
Target validation	What evidence supports the idea that changing this target will help patients?
Tractability	Can the target be pursued practically with a feasible development strategy?
Druggability	Can the target be modulated by a therapeutic modality at a tolerable exposure?
Ligandability	Does the target contain a physical site that can bind a ligand, especially a drug-like small molecule?

A target can score well on one dimension and poorly on another. For example, a protein may have strong human-genetic evidence for contributing to disease, but have no obvious pocket for a conventional small molecule. It may nevertheless be addressable by an antibody, RNA-based therapy, protein degrader, cell therapy, or another modality.

The central idea is:

Biological importance
≠ validated clinical target
≠ tractable drug-discovery programme
≠ successful medicine

6.2 Target identification

Target identification is the process of proposing biological targets that may be linked to a disease mechanism.

Possible evidence sources include:

- *Human genetics*, such as inherited variants associated with disease risk.
- *Tumour sequencing*, which can identify recurrent mutations, gene amplifications, and rearrangements.
- *Gene-expression studies*, which measure which genes are active in disease tissue.
- *CRISPR screens*, which systematically disable genes and observe which are required for cell survival or another phenotype.
- *RNA interference experiments*, which reduce expression of selected genes.
- *Proteomics*, which measures proteins and protein modifications.
- *Pathology*, including tissue morphology and protein staining.

-
- *Pathway knowledge*, scientific literature, clinical observations, and animal models.

A target hypothesis might be:

A subset of tumours depends on a particular signalling pathway for growth or survival
→ inhibiting a key protein in that pathway may selectively impair tumour cells

The word *may* matters. Early target identification is hypothesis generation, not proof. A gene may be highly expressed in a tumour because the tumour is growing rapidly, rather than because the gene is required for tumour growth. Similarly, a mutation may be present in many tumours but be a passenger rather than a driver mutation.

6.3 Target validation

Target validation is the accumulation of evidence that changing a target is likely to produce a useful effect in patients.

The evidence can be organised as a causal chain:

Target is altered, expressed, or active in disease
→ changing target function affects disease-relevant biology
→ effect is reproducible across relevant models
→ a feasible modality can produce sufficient modulation
→ mechanism is likely to translate to patients

Human genetics can provide particularly strong target-supporting evidence. For example, if naturally occurring human variants that reduce a protein's function protect against disease without causing unacceptable harm, this can support the hypothesis that pharmacologically reducing that protein may be beneficial.

However, no evidence source is perfect:

- A CRISPR screen may identify a dependency in one cell line that does not generalise to patient tumours.
- An animal model may not reproduce important features of human disease, especially human immune biology.
- A protein may be overexpressed in disease but not causally required.
- Genetic loss of a target throughout life may not produce the same effect as pharmacological inhibition later in life.
- A target may be important in disease but also essential in normal tissues, producing an unacceptable safety risk.

Strong target validation therefore usually comes from multiple independent lines of evidence rather than one experiment alone.

6.4 Tractability and modality

Tractability refers broadly to whether a target can be pursued practically. It includes more than whether a molecule can bind the target.

A tractable programme requires consideration of:

- Whether a suitable therapeutic modality exists.
- Whether useful assays can be developed.
- Whether disease-relevant experimental models are available.
- Whether selectivity is plausible.
- Whether a biomarker strategy can identify likely responders.
- Whether sufficient target modulation can be achieved safely.

The target should never be considered separately from the modality. For example:

```

Selective cell-surface tumour antigen
→ antibody, ADC, bispecific antibody, or CAR-T therapy may be feasible
Intracellular enzyme with a defined catalytic pocket
→ small-molecule inhibition may be feasible
Broad transcription-factor protein-protein interface
→ direct conventional small-molecule targeting may be difficult

```

A target can become more tractable as technology changes. A protein once considered inaccessible to small molecules may become addressable through discovery of an allosteric or cryptic pocket. Alternatively, a difficult intracellular protein may become accessible through a protein-degradation strategy.

6.5 Druggability

Druggability is the potential to discover a therapeutic intervention that modulates a target in the desired manner at a tolerable dose or exposure.

For small molecules, druggability depends partly on whether:

- The protein has an accessible and exploitable binding site.
- Binding at that site changes target function in a useful way.
- Selectivity over related proteins is possible.
- The resulting compound can achieve useful PK, tissue exposure, and safety.

For antibodies, the relevant considerations differ. Antibody druggability depends more on whether the target is extracellular or cell-surface accessible, whether target expression in normal tissues allows an acceptable therapeutic window, and whether antibody binding can produce the desired effect.

Druggability is not binary. It is better understood as a spectrum:

```

Biologically compelling but currently difficult
→ potentially tractable with specialised modality
→ tractable with conventional drug-discovery methods

```

A target can be biologically compelling but practically unsuitable if the required level of modulation cannot be achieved safely in the relevant tissue.

6.6 Ligandability

Ligandability refers to whether a target has a physical site that can bind a ligand. A ligand is simply a molecule that binds a biological target.

In small-molecule discovery, ligandability usually means that a protein contains a pocket with suitable size, shape, chemical properties, and accessibility to bind a drug-like molecule. Such a pocket may be an enzyme active site, an allosteric site, a pocket at a protein-protein interface, or a cryptic pocket that becomes accessible only in certain protein conformations.

A ligandable pocket is not automatically druggable. A molecule might bind strongly to a pocket but produce no useful change in the protein's function. The pocket must therefore be *functionally coupled* to the desired biological effect.

The distinction is:

```
Ligandable
→ molecule can bind a physical site
Druggable
→ binding can produce useful target modulation
at a feasible and tolerable exposure
```

Conversely, a protein may be druggable using a modality other than a conventional small molecule even if it lacks a conventionally ligandable pocket.

Small molecules are relatively small and often require a sufficiently defined pocket in which to bind. In contrast, an antibody is a much larger protein whose antigen-binding region can recognise an exposed three-dimensional surface on a target protein. This target region is called an *epitope*.

Antibody binding still depends on molecular complementarity. The antibody-binding surface and target epitope interact through many non-covalent interactions, including hydrogen bonds, electrostatic interactions, van der Waals interactions, and hydrophobic interactions. However, the antibody can recognise a relatively broad exposed surface rather than fitting into a deep small-molecule-sized pocket.

This means that an antibody does not require a conventional small-molecule binding pocket. Instead, it requires an accessible extracellular or cell-surface epitope with suitable shape and chemical properties.

For example, trastuzumab binds an exposed extracellular region of the cell-surface protein HER2. Conventional antibodies generally cannot enter living cells efficiently, so this strategy is usually restricted to soluble extracellular proteins and cell-surface targets:

```
Small-molecule target
→ often requires an accessible binding pocket
→ can often reach intracellular proteins
Antibody target
→ requires an accessible extracellular or cell-surface epitope
→ does not require a deep small-molecule-sized pocket
```

This is why a target that appears difficult for conventional small-molecule discovery may still be highly tractable for an antibody, ADC, bispecific antibody, or cell-therapy approach.

6.7 Key takeaways

A strong target is not merely a protein associated with disease. It is a target for which one can make a credible case that:

```
Modulating it, with a feasible modality, in a defined patient population,
is likely to create a clinically useful effect with an acceptable safety margin.
```

Target identification proposes a disease-relevant target. Target validation tests whether changing it is likely to help. Tractability considers whether a practical development strategy exists. Druggability asks whether

useful modulation can be achieved by a therapeutic modality, while ligandability focuses more narrowly on whether a target contains a molecule-binding site.

For conventional small molecules, ligandability usually concerns the presence of a suitable binding pocket. For antibodies, the equivalent structural requirement is an accessible extracellular or cell-surface epitope rather than a deep pocket.

6.8 Super-short summary

A disease-associated protein is not automatically a useful drug target. A target must be biologically important, supported by evidence that changing it will help, accessible to a feasible modality, and modulated safely enough to produce clinical benefit.

Ligandability means that a target can bind a molecule. In conventional small-molecule discovery, this usually requires a suitable pocket. Druggability means that such binding can be turned into a useful medicine. Antibodies can instead bind exposed extracellular or cell-surface epitopes, allowing some targets without conventional small-molecule pockets to be therapeutically addressed.

6.9 Practice questions

1. What is the difference between target identification and target validation?

Answer: Target identification proposes a biological entity that may be relevant to disease. Target validation accumulates evidence that changing that target is likely to produce a useful effect in patients.

2. Why can a disease-associated protein still be a poor drug target?

Answer: It may not be causally important, may not be required by the disease, may be difficult to modulate, may lack a suitable therapeutic modality, or may cause unacceptable toxicity when altered.

3. What is tractability?

Answer: Tractability is the broader practical feasibility of pursuing a target. It includes modality choice, assays, models, selectivity, biomarker strategy, PK, safety, and the ability to achieve useful modulation.

4. What is the difference between ligandability and druggability?

Answer: Ligandability means that a target contains a site able to bind a ligand. In small-molecule discovery, this usually means a suitable binding pocket. Druggability means that binding can produce useful target modulation at a feasible and tolerable exposure.

5. How can an antibody bind a target that lacks a conventional small-molecule pocket?

Answer: An antibody is much larger than a small molecule and can recognise an exposed three-dimensional protein surface called an epitope. Its binding region interacts with the epitope through non-covalent interactions and does not require a deep pocket. Conventional antibodies generally require an extracellular or cell-surface accessible target.

6. Why should target and modality be considered together?

Answer: A target's location, structure, biology, and tissue expression affect which modalities can reach and modulate it. A cell-surface target may suit an antibody, whereas an intracellular enzyme may suit a small molecule.

7. How can human genetics support target validation?

Answer: If naturally occurring variants that alter a target's function change disease risk in humans, especially without severe harmful effects, they can support the hypothesis that pharmacologically modulating the target may be beneficial.

8. Why is no single validation experiment definitive?

Answer: Cell lines, genetic screens, animal models, and expression studies all have limitations and may not generalise to patients. Strong validation usually requires convergent evidence from several sources.

7 Protein structure, binding sites, and molecular recognition

7.1 Why protein structure matters

Chapters 1, 2, 4, and 6 introduced proteins, molecular interactions, pharmacological mechanisms, and the distinction between ligandability and druggability. This chapter focuses on a more specific question:

How can structural information help determine whether and how a target can be modulated?

A protein's three-dimensional structure influences its function, which molecules it can bind, and which therapeutic modalities may be feasible. Structural information can help identify possible small-molecule pockets, accessible antibody epitopes, protein-protein interaction surfaces, and conformations associated with active or inactive states.

For drug discovery, structure can help answer questions such as:

Where could a therapeutic molecule bind?
Would binding at that site be likely to change protein function?
Could the site support selectivity over related proteins?
How could a disease-associated mutation alter function or drug binding?
Which experiments could test the proposed mechanism?

Structural information is valuable, but it does not establish that a target is clinically valid. A well-defined pocket does not prove that modulating the protein will help patients, and a plausible binding pose does not prove that a compound binds, enters cells, reaches the relevant tissue, or has acceptable safety. Structure is therefore one source of evidence within a wider target-to-medicine argument.

7.2 Proteins have structure, domains, and functional states

Proteins are chains of amino acids that fold into three-dimensional molecular structures. Their structure is often described at several levels:

Primary structure
→ amino-acid sequence
Secondary structure
→ local structures such as alpha helices and beta sheets
Tertiary structure
→ overall three-dimensional fold of one protein chain
Quaternary structure
→ arrangement of multiple protein chains in a complex

These levels are useful descriptions, but a protein's function usually depends on more than one static fold. Many proteins contain distinct *domains*: structural regions with particular functions. For example, one domain may bind another protein, another may bind DNA, and another may catalyse a chemical reaction.

Proteins can also contain smaller recurring sequence or structural features called *motifs*. For example, kinases contain conserved motifs that help bind ATP and transfer phosphate groups to substrate proteins.

A mutation can affect protein function in several ways. It may alter a catalytic site, weaken the stability of the folded protein, disrupt a protein–protein interaction, change the abundance of the protein, or favour an abnormally active or inactive state.

The most useful structural question is therefore not simply “what does this protein look like?” It is:

Which structural features determine this protein’s function, regulation, interactions, and potential therapeutic vulnerabilities?

7.3 Proteins are dynamic rather than rigid

A protein is not a rigid object with one permanently fixed shape. It moves, fluctuates, rearranges side chains, binds other molecules, and samples multiple *conformations*. Some conformations may be more active, inactive, accessible, or suitable for binding than others.

A useful simplified picture is:

Inactive protein state
 $\rightleftharpoons$
active protein state
Ligand binding
→ may stabilise one state
Disease-associated mutation
→ may favour one state or alter the balance between states

For example, a kinase may alternate between conformations that are more or less able to bind ATP, phosphorylate substrate proteins, or interact with regulatory proteins. A kinase inhibitor may preferentially bind one of these conformations. A mutation can alter this balance and may either activate the kinase or reduce inhibitor binding.

Experimental and predicted structures are therefore best understood as highly informative representations of one or a limited number of states under particular conditions. They may not capture every biologically relevant state in a living cell.

This matters especially when:

- A target is activated or inhibited through a conformational change.
- A potential pocket is present only in some protein states.
- A resistance mutation changes the population of accessible conformations.
- A ligand is designed to stabilise an inactive or otherwise non-functional state.

A structural model can be useful even when incomplete, but its limitations should be considered when drawing conclusions about binding and mechanism.

7.4 Binding sites and therapeutic opportunities

A *binding site* is a region of a protein where another molecule can bind. Binding sites can interact with small molecules, substrates, cofactors such as ATP, peptides, proteins, nucleic acids, metal ions, or other biological molecules.

An *active site* is the specific region of an enzyme where its normal chemical reaction occurs. A binding site need not be an active site. Therapeutically useful binding can occur at several types of site:

Orthosteric site

→ main functional site used by a substrate, cofactor, or endogenous ligand

Allosteric site

→ distinct site where binding changes protein behaviour indirectly

Protein-protein interaction site

→ surface through which proteins recognise or regulate one another

Cryptic pocket

→ pocket that becomes accessible only in some protein conformations

Orthosteric and allosteric modulation were introduced in Chapter 4. Their structural importance is that they provide different ways of changing the same target. Binding an enzyme active site may directly block catalysis, whereas binding an allosteric site may stabilise an inactive conformation or disrupt a required interaction.

A *binding pocket* is a three-dimensional cavity, groove, or cleft where a small molecule may bind. Such pockets can contain hydrophobic regions, polar groups, charged amino acids, cofactors, metal ions, and water molecules. Their size, shape, flexibility, solvent exposure, and chemical composition influence which ligands can bind.

As discussed in Chapter 6, a useful pocket can make a protein *ligandable* for conventional small-molecule discovery, but ligandability is not the same as druggability or clinical validation:

Useful binding pocket

✧ useful functional modulation

✧ adequate selectivity and exposure

✧ patient benefit

Conversely, the lack of a conventional small-molecule pocket does not mean that a target is impossible to address. An exposed extracellular protein surface may be suitable for an antibody, and some difficult intracellular targets may be addressable through protein degradation or other specialised modalities.

7.5 Cryptic pockets, selectivity, and resistance

Some proteins appear to lack an appropriate small-molecule pocket when viewed in one structure. However, protein motion can sometimes reveal a previously hidden or poorly formed site. Such a site is called a *cryptic pocket*.

The simplified logic is:

Protein changes conformation

→ new pocket becomes accessible

→ ligand binds and stabilises that state

→ protein function may change

Cryptic pockets can create therapeutic opportunities for targets that were previously considered difficult to drug. However, their existence must be supported by appropriate experimental or computational evidence. A predicted cavity alone is not enough to establish that a pocket can bind drug-like molecules with useful affinity and functional effect.

Structure can also help explain and guide *selectivity*. Many proteins belong to families with related sequences and similar folds. For example, many kinases have related ATP-binding pockets. A compound designed to bind one family member may therefore also bind others.

Selectivity can sometimes arise by:

- Interacting with amino acids that differ between related proteins.

-
- Exploiting a less conserved allosteric or cryptic pocket.
 - Binding a conformation preferentially sampled by the intended target.
 - Extending into a region outside the most conserved part of a binding site.
 - Exploiting differences in protein dynamics, accessibility, or cellular context.

Structural information can also help interpret *drug resistance*. For example, a mutation may directly alter an amino acid that contacts a drug, change the shape or accessibility of a pocket, alter the balance of active and inactive states, or activate downstream signalling independently of the original target.

However, resistance is not always structural. It can also arise through bypass signalling, target loss, inadequate tumour exposure, altered cell state, or immune escape. Structural analysis should therefore be integrated with functional and clinical evidence rather than treated as a complete explanation.

7.6 What determines productive protein–ligand binding?

Protein–ligand binding is not determined simply by whether a molecule appears to fit into a pocket. It reflects a balance between favourable and unfavourable energetic contributions.

As introduced in Chapter 2, relevant factors include hydrogen bonds, electrostatic interactions, hydrophobic interactions, van der Waals contacts, molecular charge, flexibility, and interactions with water.

A useful conceptual summary is:

```
Favourable protein–ligand interactions
– ligand conformational strain
– protein rearrangement costs
– loss of favourable interactions with water
– loss of conformational freedom
– unfavourable interactions
→ net binding affinity
```

A ligand may appear to make favourable contacts in a structural model but still bind weakly. For example, it may need to adopt an energetically unfavourable shape, displace strongly bound water molecules, or cause an unfavourable protein rearrangement. Conversely, a small structural change to a ligand can substantially improve or worsen binding by changing several of these factors at once.

Binding can also be time-dependent. The *affinity* of a ligand describes its equilibrium tendency to bind a target under defined conditions. Binding kinetics describe how quickly it associates with and dissociates from the target. *Residence time* is related to how long a ligand remains bound once the complex has formed.

Two compounds can have similar affinity but different binding and unbinding rates. These differences can matter when drug concentration changes over time in patients, although long residence time is not automatically beneficial and must be interpreted in the relevant biological setting.

7.7 Experimental and predicted structures

Structural information can come from experimental methods or computational prediction.

Experimental structural methods include:

- *X-ray crystallography*, which has historically provided many high-resolution protein and protein–ligand structures.
- *Cryo-electron microscopy*, or *cryo-EM*, which is especially valuable for large protein complexes and can capture multiple structural states.

- *Nuclear magnetic resonance*, or *NMR*, which can provide information about structures and dynamics in solution.

Other experiments can provide complementary evidence about binding and conformation. These include biophysical binding assays, mutagenesis, mass spectrometry, hydrogen–deuterium exchange experiments, and functional assays.

Computational structure prediction can provide useful hypotheses when an experimental structure is unavailable. However, a predicted structure should not automatically be treated as a complete representation of the biologically relevant binding state.

Before using a predicted or experimental structure for ligand design, useful questions include:

```
Does this structure represent the biologically relevant protein state?
Is the local pocket geometry sufficiently reliable?
Are relevant mutations, cofactors, metal ions, lipids, and water molecules represented?
Are flexible loops, side chains, and disordered regions important?
Does the structure correspond to the correct species, isoform,
protein construct, and binding partner?
```

An experimentally determined structure can also have limitations. Crystal packing, engineered protein constructs, missing flexible regions, non-physiological buffer conditions, or absent binding partners can affect interpretation. Predicted and experimental structures should both be viewed as evidence with particular strengths and limitations.

7.8 Structure-based drug design as hypothesis generation

Structure-based drug design uses structural information to guide the discovery or optimisation of therapeutic molecules. It is most useful when it supports an iterative experimental process rather than being treated as a one-step solution.

A simplified workflow is:

```
Structure or structural model
→ binding-site and mechanism hypothesis
→ design, dock, screen, or prioritise candidate molecules
→ synthesise and test compounds
→ measure binding, functional activity, selectivity, and exposure
→ update the structural and pharmacological hypothesis
```

Common approaches include *virtual screening*, in which compounds are computationally prioritised from a library; *molecular docking*, which predicts plausible ligand poses in a pocket; *fragment-based drug discovery*, which starts from small weakly binding fragments; and *structure–activity relationship analysis*, which links chemical modifications to measured changes in activity.

Structure can also guide experiments beyond small-molecule design. It can help select residues for mutagenesis, identify antibody epitopes, interpret a resistance mutation, choose a target construct for an assay, or suggest a biomarker of target modulation.

The key limitation is that structural models generate hypotheses. A docking score, predicted pose, or apparent pocket does not establish binding, selectivity, cell activity, tissue exposure, safety, or clinical benefit. Those claims require experimental testing in progressively more relevant systems.

7.9 Key takeaways

Protein structure provides a bridge between target biology, therapeutic modality, molecular design, and experimental testing.

Several distinctions are especially useful to retain:

- Proteins have distinct structural levels, domains, and functional states, but they are dynamic molecular ensembles rather than rigid objects.
- Structural information can help identify possible small-molecule pockets, antibody epitopes, protein–protein interaction sites, and conformations relevant to target function.
- Orthosteric, allosteric, protein–protein interaction, and cryptic sites can provide different opportunities to modulate the same target.
- A pocket can support ligand binding without proving that binding will create useful functional modulation, acceptable exposure, or clinical benefit.
- Structure can help explain selectivity and resistance, but resistance can also arise through non-structural mechanisms such as bypass signalling or inadequate drug exposure.
- Productive ligand binding depends on the net balance of favourable interactions and energetic costs, including conformational strain, water displacement, and loss of flexibility.
- Experimental and predicted structures are valuable but incomplete models that should be interpreted in their biological context.
- Structure-based design supports iterative hypothesis generation and experimental testing; it does not replace pharmacological, PK, safety, or clinical evidence.

When evaluating a structural drug-discovery claim, ask:

Which protein state is represented?
What site is proposed to bind the therapeutic molecule?
What functional change is expected after binding?
What experimental evidence supports the binding mode and mechanism?
How does the structural hypothesis connect to selectivity, cellular activity, tissue exposure, and patient benefit?

7.10 Super-short summary

Protein structures reveal possible ways to alter target function, but proteins are dynamic and may contain multiple relevant conformations. Structural information can help identify binding pockets, antibody epitopes, allosteric sites, cryptic pockets, selectivity opportunities, and mechanisms of resistance.

Structure-based drug design is an iterative process: structural models generate hypotheses about how a molecule might bind and act, while experiments determine whether those hypotheses lead to useful pharmacology and, ultimately, clinical benefit.

7.11 Practice questions

How to use these questions. Try to answer each question without looking at the answer. Focus on how structural information supports decisions about target modulation, therapeutic modality, selectivity, resistance, and experimental design.

1. **Why is protein structure useful in drug discovery, but insufficient on its own to establish a useful medicine?**

Answer: Structure can identify possible binding sites, help explain function, guide ligand or antibody design, and suggest ways to achieve selectivity. However, it does not prove that the target is important in disease, that a molecule reaches the target in cells or patients, that binding changes biology in the desired way, or that treatment is safe and clinically beneficial.

2. Why should proteins be viewed as dynamic rather than rigid structures?

Answer: Proteins move, rearrange side chains, bind other molecules, and sample multiple conformations. Ligands and mutations can stabilise particular states. A single structure usually represents one or a limited number of states and may not capture every biologically relevant conformation.

3. What are orthosteric, allosteric, protein–protein interaction, and cryptic binding sites?

Answer: An orthosteric site is the main functional site used by a substrate, cofactor, or endogenous ligand. An allosteric site is a distinct site where binding changes protein behaviour indirectly. A protein–protein interaction site is a surface through which proteins bind and regulate one another. A cryptic pocket is a pocket that becomes accessible only in certain protein conformations.

4. How can a cryptic pocket create a therapeutic opportunity?

Answer: A protein can change conformation and transiently expose a pocket that is not visible in another structure. If a ligand can bind and stabilise that state, it may alter protein function and provide a way to target a protein that initially appeared to lack a suitable small-molecule pocket.

5. Why can a visually plausible docking pose fail to predict strong binding?

Answer: Binding depends on more than visible contacts. A ligand may need to adopt an unfavourable conformation, displace stabilising water molecules, cause an unfavourable protein rearrangement, lose flexibility, or encounter incorrect protonation states. The predicted pose itself may also be wrong.

6. How can structural information help achieve selectivity between related proteins?

Answer: It can identify amino acids, pockets, conformations, or dynamic features that differ between related proteins. A ligand may exploit a less conserved allosteric site, extend into a variable region, or preferentially bind a conformation sampled more often by the intended target. Experimental profiling is still needed to establish selectivity.

7. How can structural analysis help explain drug resistance?

Answer: A resistance mutation may directly disrupt a drug contact, alter the shape or accessibility of a binding site, or favour a protein conformation that binds the drug poorly. However, resistance can also arise through bypass signalling, target loss, inadequate exposure, or cell-state changes that are not explained by protein structure alone.

8. What limitations should be considered when using an experimental or predicted protein structure?

Answer: Consider whether it represents the relevant protein state, species, isoform, mutation status, binding partners, cofactors, metal ions, water molecules, flexible regions, and local pocket geometry. Experimental structures can be influenced by sample preparation and construct design, while predicted structures may not capture the correct pocket state or dynamics.

9. What is the role of structure-based drug design in the design–make–test–analyse cycle?

Answer: Structure-based design generates hypotheses about binding sites, ligand poses, chemical modifications, selectivity, and mechanism. These hypotheses guide compound selection and design, but compounds must still be synthesised and tested for binding, functional activity, cellular activity, selectivity, exposure, safety, and clinical relevance.

8 Small molecules

8.1 What small molecules are good at, and where they struggle

Small molecules are discrete chemical compounds that are much smaller than antibodies or most other protein-based therapies. Many can be formulated as tablets or capsules, although oral delivery is not possible for every small molecule.

Their most important practical advantage is that some small molecules can cross cell membranes. This allows them to reach many *intracellular* targets, including enzymes, kinases, signalling proteins, nuclear receptors, and some protein–protein interactions. In contrast, conventional antibodies generally act on soluble extracellular proteins or exposed cell-surface targets.

Small molecules can alter biology through several mechanisms, including:

- Reversible inhibition of an enzyme or receptor.
- Covalent inhibition, in which a molecule forms a durable bond with a target protein.
- Orthosteric or allosteric modulation.
- Receptor agonism or antagonism.
- Modulation of ion channels or transporters.
- Recruitment of protein-degradation machinery.

However, small size is not automatically an advantage. A molecule may be too insoluble to formulate, too charged to enter cells, too rapidly metabolised to maintain exposure, insufficiently selective, unable to reach the relevant tissue, or toxic at the concentration needed to modulate its target.

The central question is therefore:

Can a molecule reach the relevant cells,
modulate the intended target sufficiently,
and do so with acceptable selectivity, exposure, and safety?

As discussed in Chapters 2, 4, 5, and 6, this is a multi-objective problem. Small-molecule discovery is not simply a search for the compound with the strongest binding affinity or lowest assay IC₅₀.

8.2 From a hit to a credible lead

A *hit* is an initial compound with reproducible evidence of relevant activity. Depending on the programme, a hit may show binding to a target, inhibition of an enzyme, a desired cellular effect, or another useful signal.

Hits can arise from several sources:

- *High-throughput screening*, in which large compound collections are tested in an assay.
- *Fragment screening*, in which small and usually weakly binding compounds are identified and developed further.
- *Virtual screening*, in which computational methods prioritise compounds for experimental testing.
- *Natural products*, literature knowledge, or known compounds with relevant biological activity.
- *Structure-based design*, using a target structure or binding hypothesis to guide compound selection or design.

A screening hit is not automatically a useful starting point. It must first be confirmed and understood. Apparent activity can arise from assay artefacts, chemical instability, aggregation, non-specific reactivity, poor compound identity or purity, or activity against an unintended target.

A useful early question is:

Is this a reproducible and target-relevant starting point
that can plausibly be improved into a medicine?

A *lead* is a more mature compound, or a related *chemical series*, with enough evidence of target-relevant activity and developability to justify systematic optimisation. A credible lead does not need to be a finished drug candidate, but it should provide a realistic path toward improved potency, selectivity, exposure, and safety.

The transition from hit to lead often involves confirming activity in more than one assay context:

```
Initial hit
→ confirm chemical identity and reproducible activity
→ test target binding or mechanism
→ test relevant cellular activity
→ assess early selectivity and physicochemical properties
→ establish whether a credible optimisation path exists
```

8.3 Lead optimisation as multi-objective design

Lead optimisation is the iterative process of improving a chemical series until one compound has a sufficiently favourable overall profile to become a drug candidate.

The purpose is not simply to maximise potency. A compound must usually balance many competing requirements:

```
Target potency and functional activity
Selectivity and off-target profile
Solubility and permeability
Metabolic stability and pharmacokinetics
Exposure in the relevant tissue
Safety margin
Synthetic feasibility, formulation, and manufacturability
```

Improving one property can worsen another. For example, increasing lipophilicity may improve membrane permeability and hydrophobic interactions in a binding pocket, but it can also reduce solubility, increase non-specific binding, increase metabolic liability, or create safety risks.

Similarly, increasing potency against purified protein may not improve cellular activity if the compound cannot enter cells or is rapidly exported. Improving cellular potency may not translate into clinical usefulness if the compound cannot reach the target tissue at a safe exposure.

The practical process is iterative:

```
Define a chemical and biological hypothesis
→ design related compounds
→ synthesise and test them
→ measure target activity, cellular effects, PK, and safety-related properties
→ interpret the trade-offs
→ update the hypothesis
→ repeat
```

This is why small-molecule discovery is often described as a *design–make–test–analyse* cycle. The goal is a coherent and decision-relevant body of evidence, rather than one exceptionally strong result in a single assay.

8.4 Scaffolds, analogues, and structure–activity relationships

A *scaffold* is the central chemical framework of a molecule. A *chemical series* is a group of related compounds that share a scaffold or a common design logic.

An *analogue* is a compound closely related to another compound but modified in one or more structural features. For example, a medicinal chemist might replace one functional group, alter the position of a substituent on an aromatic ring, change a heterocycle, reduce molecular flexibility, or introduce a stereochemical change.

Testing related analogues produces a *structure–activity relationship*, or *SAR*: an evolving interpretation of how chemical changes affect measured properties.

The basic logic is:

```
Change one structural feature
→ measure the effect on potency, selectivity, solubility, PK, or safety
Compare related analogues
→ infer which features improve or worsen the overall profile
Use this evidence
→ design the next set of compounds
```

For example, adding a group that forms a new interaction with the target may improve biochemical potency. However, the same modification could increase molecular weight, reduce solubility, alter metabolism, or create off-target activity. SAR is therefore not limited to target potency; it includes every property relevant to the intended medicine.

SAR is also local and context-dependent. A modification that improves one chemical series may have the opposite effect in another. Measurements depend on assay conditions, protein state, cell type, and the broader molecular structure. Consequently, SAR should be treated as experimentally grounded evidence rather than a universal rule.

8.5 Medicinal chemistry and developability

Medicinal chemistry is the discipline of designing, synthesising, and analysing compounds in the context of biology, pharmacology, PK, safety, and practical development constraints.

A medicinal chemist does not optimise a molecule in isolation. Useful questions include:

```
Does this modification improve the desired biological effect?
Can the compound dissolve and be formulated appropriately?
Can it enter the relevant cells and reach the target tissue?
Is it metabolically stable enough to achieve useful exposure?
Does it affect related proteins or other problematic off-targets?
Can it be synthesised, purified, scaled, and manufactured reliably?
```

The term *ADMET* is often used to summarise several properties that must be considered during optimisation:

Letter	Meaning
A	Absorption: whether sufficient drug enters systemic circulation.
D	Distribution: whether drug reaches the relevant tissues and cells.
M	Metabolism: how the drug is chemically modified in the body.
E	Excretion: how the drug and its metabolites are removed.
T	Toxicity: harmful effects caused by intended or unintended biological activity.

A compound can fail because of any one of these dimensions. It may have poor oral absorption, insufficient tumour penetration, rapid clearance, problematic metabolites, interactions with cardiac ion channels, or clinically important interactions with other medicines.

The most important point is:

A potent compound
≠ a viable drug candidate
A viable drug candidate
≠ a medicine
A medicine must also show useful benefit and acceptable risk in patients.

8.6 Computational and ML support in the design–make–test–analyse cycle

Computational chemistry and molecular machine learning can help prioritise compounds, generate hypotheses, and guide experiments. They can be used to estimate properties such as likely target activity, solubility, permeability, metabolic stability, selectivity, or synthetic accessibility.

Methods may use molecular descriptors, fingerprints, molecular graphs, three-dimensional conformations, protein structures, experimental assay data, or combinations of these inputs.

Their practical role is:

Existing chemical and biological data
→ prioritise compounds or propose modifications
→ select experiments with higher expected value
→ generate new data
→ update the chemical hypothesis

These methods are most useful when they improve a real decision: for example, choosing which analogues to synthesise, identifying a likely liability, selecting compounds for a screen, or helping interpret a structure–activity relationship.

They do not replace experimental measurement. Assay values are context-dependent, and a model may perform well on familiar compounds while failing in a new chemical series or outside its training distribution. Predictions should therefore be treated as evidence with uncertainty, to be tested prospectively in the relevant experimental system.

8.7 KRAS as a case study: from “undruggable” to mutation-selective inhibition

KRAS is an intracellular signalling protein that illustrates how advances in structural biology, chemistry, and disease understanding can change the tractability of a target.

KRAS is a small *GTPase*: a protein that acts as a molecular switch by cycling between two nucleotide-bound states.

GTP, short for *guanosine triphosphate*, is a small molecule containing three phosphate groups. *GDP*, short for *guanosine diphosphate*, is the related molecule containing two phosphate groups.

In simplified terms:

KRAS bound to GTP
→ generally active
→ can activate downstream growth and survival signalling
KRAS bound to GDP
→ generally inactive
→ downstream signalling is reduced

Normally, KRAS switches between these states in response to upstream signals. Proteins called *guanine nucleotide exchange factors*, or *GEFs*, can promote replacement of GDP with GTP, turning KRAS on. Other proteins, called *GTPase-activating proteins*, or *GAPs*, help KRAS convert GTP to GDP, turning it off.

A simplified cycle is:

GDP-bound KRAS

- GEF promotes GDP release and GTP binding
- GTP-bound active KRAS
- GAP promotes GTP hydrolysis
- GDP-bound inactive KRAS

When active, KRAS can stimulate signalling pathways including the MAPK and PI3K pathways, which can promote cell growth, survival, and other cancer-associated behaviours.

Mutations in KRAS are common in several cancers, including pancreatic cancer, colorectal cancer, and non-small-cell lung cancer. These mutations can impair normal switching behaviour, causing KRAS to spend more time in its active GTP-bound state or to signal inappropriately.

For many years, KRAS was considered difficult to target with conventional small molecules because:

- Its surface appeared relatively smooth, with few obvious deep pockets for small molecules.
- KRAS binds GDP and GTP very tightly.
- GDP and GTP are present at substantial concentrations inside cells.
- KRAS function depends on conformational cycling and interactions with several other proteins.
- Selectively inhibiting mutant KRAS while preserving normal KRAS function was challenging.

The situation changed with identification of an exploitable pocket beneath the *switch II* region of KRAS. This *switch II pocket* is accessible in particular KRAS conformations and is an example of how protein dynamics can reveal therapeutic opportunities.

The KRAS G12C mutation replaces the normal amino acid at position 12, glycine, with *cysteine*. Cysteine contains a chemically reactive sulfur-containing group that can be targeted by appropriately designed covalent inhibitors.

KRAS G12C inhibitors, including sotorasib and adagrasib, are designed to bind the mutant cysteine when KRAS G12C visits its GDP-bound inactive state. They then form a covalent bond and stabilise a state that cannot signal normally.

The conceptual mechanism is:

KRAS G12C cycles into GDP-bound inactive state

- inhibitor binds the switch II pocket
- inhibitor forms a covalent bond with mutant cysteine
- KRAS is trapped in an inactive state
- downstream signalling is reduced
- tumour growth may be inhibited

This mechanism also explains an important limitation: KRAS G12C inhibitors depend on the mutant protein visiting the GDP-bound state. If signalling mechanisms keep more KRAS in the active GTP-bound state, or if tumour cells activate bypass pathways, inhibition may be incomplete.

Resistance can emerge through additional KRAS alterations, activation of upstream or parallel signalling pathways, downstream pathway changes, phenotypic adaptation, or tumour heterogeneity. KRAS-directed therapy may therefore be used in carefully selected patients and can motivate rational combination strategies.

KRAS illustrates several general lessons:

Biological importance does not guarantee immediate tractability
Protein dynamics can reveal previously hidden opportunities
A mutation can create a selectivity opportunity
A successful inhibitor can still face resistance,
incomplete pathway suppression, and safety constraints

8.8 Key takeaways

Small-molecule discovery is the process of turning an initial chemical starting point into a compound with a sufficiently balanced biological, pharmacological, and practical profile.

Several distinctions are especially useful to retain:

- Small molecules can often enter cells and are therefore especially valuable for many intracellular targets, but they must still achieve sufficient tissue exposure and selectivity.
- A hit is an initial reproducible starting point; a lead is a more credible chemical series suitable for systematic optimisation.
- Lead optimisation is a multi-objective process involving potency, selectivity, solubility, permeability, metabolic stability, PK, tissue exposure, safety, synthesis, and manufacturability.
- Scaffolds and analogues allow medicinal chemists to build structure–activity relationships and learn how chemical modifications affect the complete compound profile.
- ADMET properties can prevent a potent molecule from becoming a viable drug candidate.
- Computational and ML methods can support prioritisation and hypothesis generation, but their value depends on improving real experimental decisions and prospective validation.
- KRAS G12C demonstrates how a target once considered difficult to drug can become tractable through a mutation-selective covalent strategy that exploits protein conformation and chemistry.

When reading a small-molecule discovery paper, ask:

What evidence makes this a credible hit or lead?
Which properties were improved during optimisation?
What trade-offs remain unresolved?
Does the compound work in relevant cells and tissues,
not only against purified protein?
Is there evidence of target engagement, selectivity,
adequate exposure, and a plausible therapeutic window?

8.9 Super-short summary

Small molecules are particularly useful for many intracellular targets, but their development requires much more than finding a strong binder. A credible drug candidate must balance target activity, selectivity, cell entry, tissue exposure, PK, safety, and practical manufacturability.

Small-molecule discovery proceeds through iterative design, synthesis, testing, and analysis. KRAS G12C shows how a mutation-specific pocket and covalent chemistry can make a previously difficult target therapeutically tractable, while also illustrating the continuing challenges of incomplete inhibition and resistance.

8.10 Practice questions

How to use these questions. Try to answer each question without looking at the answer. Focus on the path from an initial active compound to a medicine, including the trade-offs that arise during optimisation.

1. Why are small molecules particularly useful for many intracellular targets?

Answer: Many small molecules can cross cell membranes and therefore reach intracellular proteins such as enzymes, kinases, signalling proteins, and some protein–protein interactions. However, they must still achieve sufficient intracellular and tissue exposure while avoiding unacceptable off-target effects.

2. What is the difference between a hit and a lead?

Answer: A hit is an initial compound with reproducible evidence of desired activity. A lead is a more mature compound or chemical series with enough evidence of target relevance, activity, and developability to justify systematic optimisation.

3. Why must an initial hit be confirmed before major optimisation begins?

Answer: Apparent activity can arise from assay artefacts, chemical instability, aggregation, non-specific reactivity, impurity, or an unintended mechanism. Confirming activity, chemical identity, and target relevance helps ensure that optimisation begins from a credible starting point.

4. Why is lead optimisation a multi-objective problem?

Answer: A useful candidate must balance potency, selectivity, solubility, permeability, metabolic stability, tissue distribution, safety, synthesis, formulation, and manufacturability. Improving one property can worsen another, so the aim is an appropriate overall profile rather than the best value for one measurement.

5. What are a scaffold, an analogue, and a structure–activity relationship?

Answer: A scaffold is a molecule’s central chemical framework. An analogue is a related compound with one or more structural changes. A structure–activity relationship, or SAR, is the evidence-based interpretation of how such changes affect properties such as potency, selectivity, solubility, PK, or safety.

6. Why can a potent molecule fail to become a drug candidate?

Answer: It may have poor solubility, insufficient cell entry, rapid metabolism, inadequate tissue exposure, problematic off-target activity, toxicity, poor formulation properties, or impractical synthesis and manufacturing requirements.

7. How can computational chemistry or ML add value during small-molecule discovery?

Answer: These methods can prioritise compounds, propose chemical modifications, estimate properties, identify possible liabilities, and guide experiments. Their value depends on whether they improve real design or testing decisions and whether their predictions are prospectively confirmed in relevant assays.

8. What are GDP and GTP, and how do they relate to KRAS activity?

Answer: GDP is guanosine diphosphate and GTP is guanosine triphosphate. KRAS generally acts as a molecular switch: GDP-bound KRAS is relatively inactive, whereas GTP-bound KRAS is active and can stimulate downstream growth and survival signalling.

9. Why was KRAS historically difficult to target with conventional small molecules?

Answer: KRAS appeared to have few suitable pockets, binds GDP and GTP tightly, functions through a dynamic nucleotide-switching cycle and multiple protein interactions, and was difficult to inhibit selectively without affecting normal KRAS biology.

10. How do KRAS G12C inhibitors work conceptually, and why can resistance still occur?

Answer: KRAS G12C inhibitors bind a pocket accessible when mutant KRAS visits its GDP-bound state, form a covalent bond with the mutant cysteine, and trap KRAS in an inactive state. Resistance can occur through additional KRAS changes, increased active-state signalling, bypass pathways, downstream alterations, phenotypic adaptation, or tumour heterogeneity.

9 Antibodies, trastuzumab, and antibody–drug conjugates

9.1 What antibodies are and how they bind targets

Antibodies are proteins used by the immune system to recognise particular molecular features. Therapeutic antibodies are engineered or selected to bind a chosen target with high specificity and to produce a desired pharmacological effect.

A *monoclonal antibody*, or *mAb*, is a population of identical antibody molecules designed to recognise the same region of a target. This recognised target region is called an *epitope*.

A conventional therapeutic antibody has a characteristic Y-like structure. It has two conceptually important functional regions:

Fab regions
→ bind the target antigen
Fc region
→ interacts with immune-system components
→ influences distribution and persistence in the body

The *Fab*, short for fragment antigen-binding, forms the two arms of the antibody. At the tip of each Fab region are variable amino-acid sequences that create a three-dimensional binding surface. This antibody-binding surface is sometimes called the *paratope*.

The paratope binds a complementary epitope on the target through many non-covalent interactions, including hydrogen bonds, electrostatic interactions, van der Waals interactions, and hydrophobic interactions. The interaction depends on complementarity between the shape and chemical properties of the antibody-binding surface and the target epitope.

Unlike a conventional small molecule, an antibody does not usually need to fit into a deep binding pocket. Its relatively large binding surface can recognise an exposed three-dimensional region on a protein. This makes antibodies particularly useful for soluble extracellular proteins and cell-surface proteins.

The *Fc*, short for fragment crystallisable, is the stem of the antibody. It can bind specialised proteins called *Fc receptors* on immune cells and can also interact with other components of the immune system. These interactions can recruit immune effector functions, influence tissue distribution, and help determine how long an antibody persists in circulation.

Conventional antibodies generally do not enter living cells efficiently. They are therefore usually best suited to:

Soluble extracellular target
→ antibody can neutralise or remove a signal
Cell-surface target
→ antibody can block, activate, recruit immunity,
or deliver a therapeutic payload
Intracellular target
→ conventional antibody targeting is usually difficult

This is a useful general rule rather than an absolute one. Specialised delivery systems and engineered approaches can sometimes extend antibody-based strategies beyond these usual limits.

9.2 What antibodies can do pharmacologically

Antibodies can act through several mechanisms. The relevant mechanism depends on the target, epitope, antibody format, Fc region, disease setting, and biology of the target cell.

An antibody may:

- *Neutralise a soluble ligand*, preventing it from reaching and activating its receptor.
- *Block a receptor*, preventing ligand binding, receptor activation, receptor dimerisation, or another required interaction.
- *Activate or cluster a receptor*, where receptor activation is therapeutically useful.
- *Alter receptor internalisation or degradation*, thereby changing the amount or location of target protein.
- *Recruit immune effector cells* through Fc-mediated mechanisms such as antibody-dependent cellular cytotoxicity.
- *Deliver a payload*, as in an antibody–drug conjugate.
- *Bring two targets together*, as in a bispecific antibody designed to connect an immune cell with a tumour cell or to affect two biological pathways at once.

The central point is:

Antibody binding
≠ one universal mechanism
The same target can be bound by different antibodies
at different epitopes, producing different biological effects.

For example, two antibodies against the same receptor may bind different exposed regions. One may block ligand binding, another may alter receptor interactions, and another may primarily recruit immune-mediated killing. This is why the target name alone does not fully describe an antibody's pharmacology.

9.3 HER2 biology

HER2, also called *ERBB2*, is a receptor tyrosine kinase (*receptor*: the protein participates in detecting information at the cell surface, *tyrosine kinase*: its intracellular enzyme region can transfer phosphate groups from ATP to specific amino acids called *tyrosines*) in the ERBB receptor family. Related family members include EGFR/HER1, HER3, and HER4.

HER2 is a cell-surface protein. Like other receptor tyrosine kinases, it has an extracellular region, a membrane-spanning region, and an intracellular kinase region. Its extracellular region can interact with other ERBB-family receptors, while its intracellular region can transmit growth and survival signals through phosphorylation and downstream signalling pathways.

Unlike several related ERBB receptors, HER2 has no clearly established direct soluble ligand that activates it in the same way that EGF activates EGFR. Instead, HER2 is an important preferred partner for receptor *dimerisation*: the association of two receptor molecules into a signalling complex.

A simplified picture is:

HER2 forms a complex with another ERBB-family receptor
→ intracellular kinase activity and phosphorylation increase
→ downstream signalling pathways are activated
→ cell growth and survival can be promoted

HER2-associated signalling can influence pathways including MAPK and PI3K–AKT signalling. These pathways are involved in cell growth, survival, metabolism, and other cell behaviours.

In a subset of breast, gastric, and other cancers, the *ERBB2* gene is amplified or HER2 protein is overexpressed. *Gene amplification* means that tumour cells have acquired extra copies of the gene, often resulting in production of unusually high amounts of HER2 protein.

High HER2 abundance can increase the probability of HER2-containing signalling complexes forming, leading to excessive growth and survival signalling. In these settings, tumour cells may become unusually dependent on HER2-associated biology.

HER2 is therefore both:

```
Therapeutic target
→ HER2-associated biology contributes to tumour growth
in relevant cancers
Predictive biomarker
→ HER2 testing helps identify patients more likely to benefit
from HER2-directed therapies
```

HER2 status is assessed using validated tests, commonly including immunohistochemistry to measure HER2 protein expression and in situ hybridisation methods to assess *ERBB2* gene amplification. The clinical interpretation of HER2 status depends on the disease setting, assay, scoring system, and specific treatment under consideration.

9.4 Trastuzumab

Trastuzumab is a humanised IgG1 monoclonal antibody directed against HER2. It was a landmark example of biomarker-guided cancer therapy because it showed that patients with HER2-positive tumours could derive substantial benefit from treatment directed at a specific tumour-associated target.

Trastuzumab binds an exposed extracellular region of HER2, commonly described as domain IV of the HER2 extracellular region. Its binding does not simply mean that HER2 is physically occupied. The binding event can alter HER2-associated biology and can mark HER2-expressing tumour cells for immune recognition.

Trastuzumab has several proposed and context-dependent mechanisms, including:

- Perturbation of HER2-associated receptor signalling and downstream pathway activity.
- Reduced formation of some biologically active HER2-associated receptor states or complexes.
- Inhibition of HER2 extracellular-domain shedding, a process that can generate a shorter membrane-associated HER2 fragment with altered signalling properties.
- Fc-mediated recruitment of immune effector cells, particularly through antibody-dependent cellular cytotoxicity.
- Potential effects on receptor internalisation, cell-cycle regulation, and tumour-cell survival that depend on the biological context.

The most accurate high-level description is:

```
Trastuzumab binds HER2 on tumour cells
→ perturbs HER2-associated biology
→ can recruit immune-mediated anti-tumour activity
→ may reduce tumour growth in susceptible HER2-positive settings
```

It is therefore incomplete to say that trastuzumab simply “switches off HER2.” HER2 exists within a complex signalling network, and trastuzumab’s clinical activity reflects both direct effects on HER2-associated biology and immune-mediated mechanisms.

9.5 Antibody-dependent cellular cytotoxicity

A major anti-tumour mechanism associated with trastuzumab is *antibody-dependent cellular cytotoxicity*, or *ADCC*.

After trastuzumab binds HER2 on a tumour cell, its Fc region remains exposed. Immune effector cells, especially *natural killer*, or *NK*, cells, express Fc receptors that can recognise and bind the Fc region of antibodies attached to a target cell.

When multiple Fc receptors are engaged, the immune cell can become activated and release molecules that damage or kill the antibody-coated tumour cell.

The simplified sequence is:

```
Trastuzumab binds HER2 on tumour cell
→ Fc region is exposed on cell surface
→ Fc receptors on NK cell recognise antibody Fc region
→ NK cell becomes activated
→ tumour cell is attacked and may be killed
```

This mechanism allows the antibody to connect a tumour-associated antigen with the immune system:

```
Fab region
→ recognises HER2 on tumour cell
Fc region
→ recruits immune effector function
```

The strength of ADCC can vary between patients and tumours. Relevant factors include HER2 density on tumour cells, accessibility of the tumour to immune cells, the number and state of NK cells and other immune cells, Fc-receptor biology, tumour microenvironment, concurrent treatments, and tumour heterogeneity.

Antibodies can also engage other Fc-dependent immune mechanisms, including interactions with macrophages and components of the complement system. However, the relative importance of each mechanism depends on the antibody and disease context. It is generally more useful to view Fc-mediated activity as one component of an integrated therapeutic mechanism rather than as an isolated guarantee of tumour-cell killing.

9.6 Why an ADC is more than an antibody plus chemotherapy

An *antibody–drug conjugate*, or *ADC*, is a therapeutic system that combines a targeting antibody with a highly potent payload.

Its basic architecture is:

```
Targeting antibody
+
Linker
+
Cytotoxic payload
```

The antibody binds a chosen antigen on the tumour-cell surface. The *linker* connects the antibody to the payload and is designed to balance two competing requirements: it should remain sufficiently stable in circulation to minimise premature payload release, while allowing efficient payload release after the ADC reaches an appropriate intracellular environment.

The *payload* is commonly a highly potent cytotoxic compound. Such molecules may be unsuitable for free systemic administration at the intracellular concentrations required to kill tumour cells. Conjugation to an antibody is intended to preferentially deliver the payload to antigen-expressing cells.

An ADC is therefore not simply “an antibody plus chemotherapy.” Its activity depends on the full sequence:

```
ADC reaches tumour tissue
→ antibody binds accessible target antigen
→ ADC is internalised or otherwise processed
→ linker is cleaved or payload is released
→ payload reaches its intracellular target
→ tumour cell is damaged or killed
```

Failure at any point can limit activity. For example, a tumour may express too little accessible antigen, internalise the ADC inefficiently, process the linker poorly, export the released payload, repair the payload-induced damage, or contain cells that do not depend on the affected biological process.

9.7 Trastuzumab deruxtecan

Trastuzumab deruxtecan, also called *T-DXd* or Enhertu, is a HER2-directed ADC. It combines a trastuzumab-based HER2-targeting antibody with a cleavable linker and a potent topoisomerase-I inhibitor payload.

Its principal components are:

Component	Function
Trastuzumab-based antibody	Recognises accessible HER2 on tumour-cell surfaces and retains antibody-associated HER2 targeting properties.
Cleavable linker	Connects antibody and payload; designed to remain sufficiently stable in circulation and permit release after cellular uptake and processing.
Deruxtecan-derived payload, DXd	Potent inhibitor of topoisomerase I, intended to damage DNA preferentially in cells receiving payload.

T-DXd also has a relatively high average *drug-to-antibody ratio*, or *DAR*, meaning that multiple payload molecules are attached to each antibody molecule. DAR is an engineered property rather than a simple measure of quality. It can influence potency, stability, aggregation, pharmacokinetics, tumour distribution, and safety.

The therapeutic logic is:

```
HER2 expression helps direct T-DXd to tumour cells
→ HER2 binding promotes intracellular delivery
→ DXd is released after ADC processing
→ topoisomerase-I inhibition can damage tumour-cell DNA
```

T-DXd and trastuzumab share HER2 recognition, but they should not be understood as the same therapy with different strength. T-DXd adds a distinct intracellular cytotoxic mechanism, has different requirements for activity, and has a different safety profile.

9.8 HER2 binding, internalisation, and intracellular trafficking

The trastuzumab component of T-DXd binds HER2 on the surface of an HER2-expressing cell.

```
T-DXd reaches an HER2-expressing cell
→ trastuzumab component binds accessible HER2
→ HER2-ADC complex can be internalised
```

This binding creates preferential delivery to cells with sufficient accessible HER2. However, it does not create absolute tumour specificity. HER2 can be present in normal tissues, and delivery to a tumour depends on many factors beyond antigen expression, including blood flow, tissue penetration, target accessibility, and intratumour heterogeneity.

After binding, the HER2–ADC complex can be taken into the cell by *receptor-mediated endocytosis*. During this process, a region of the cell membrane folds inward and forms a membrane-bound intracellular compartment containing the HER2–ADC complex.

The complex then moves through intracellular compartments, including *endosomes* and eventually, for many ADCs, *lysosomes*. Lysosomes are acidic intracellular compartments containing enzymes that help break down proteins and other biological material.

The simplified sequence is:

```
T-DXd binds HER2
→ HER2-ADC complex is internalised
→ complex traffics through endosomal compartments
→ reaches lysosomal processing environment
→ linker can be cleaved and payload released
```

Internalisation and intracellular trafficking are important because the ADC payload must be released in a form that can reach its intracellular target. Binding to a surface antigen alone is therefore insufficient for many ADCs.

9.9 Linker cleavage, topoisomerase-I inhibition, and cell death

The linker in T-DXd is designed to be cleaved after cellular uptake and processing, particularly in intracellular environments containing relevant proteolytic enzymes. Cleavage releases the DXd payload from the antibody-derived component.

Linker design is one of the central engineering challenges in ADC development:

```
Linker too unstable in circulation
→ premature payload release
→ reduced selectivity and increased systemic toxicity risk
Linker too stable or poorly processed in target cell
→ inadequate intracellular payload release
→ reduced anti-tumour activity
```

The released DXd payload inhibits *topoisomerase I*. Topoisomerase I is an enzyme that helps manage torsional stress in DNA. When DNA is unwound during replication or transcription, twisting strain can build up. Topoisomerase I temporarily cuts one DNA strand, allows the DNA to relax, and then reseals the break.

A simplified normal process is:

```
DNA twisting stress develops
→ topoisomerase I makes a temporary single-strand break
→ DNA can relax
→ topoisomerase I reseals the DNA strand
```

Topoisomerase-I inhibitors interfere with this process. They can stabilise an intermediate in which topoisomerase I remains associated with damaged DNA. When a replication machinery encounters this abnormal complex, DNA damage and replication stress can accumulate.

The therapeutic sequence is:

```
DXd released inside tumour cell
→ topoisomerase-I function is disrupted
→ replication stress and DNA damage increase
→ cell-cycle progression is disrupted
→ susceptible tumour cell may undergo cell death
```

The final effect depends on tumour-cell biology. Cells may differ in payload sensitivity, DNA-damage response capacity, cell-cycle state, drug-export mechanisms, and ability to repair or tolerate DNA damage. Thus, successful HER2 binding and payload delivery do not guarantee that every targeted tumour cell will die.

9.10 Bystander effect, tumour heterogeneity, and ADC safety

DXd is sufficiently membrane permeable that some released payload can diffuse from the initially targeted cell into nearby cells. This is called the *bystander effect*.

```
HER2-expressing cell internalises T-DXd
→ DXd payload is released
→ some released payload diffuses locally
→ neighbouring cells may also be exposed
```

The bystander effect can be useful in heterogeneous tumours, where HER2 expression differs among cells. A HER2-expressing tumour cell can internalise the ADC and release payload that may also affect nearby tumour cells with lower or absent HER2 expression.

However, this effect also means that payload exposure is not perfectly restricted to antigen-positive cells. The same property that may help overcome tumour heterogeneity can contribute to toxicity if released payload reaches susceptible normal cells.

ADC activity and safety therefore depend on the complete system:

```
Antigen expression, density, and accessibility
+
Antibody binding and tumour penetration
+
Internalisation and intracellular trafficking
+
Linker stability and cleavage
+
Payload potency, permeability, and mechanism
+
Tumour heterogeneity and resistance biology
+
Normal-tissue antigen expression and payload exposure
+
Overall therapeutic window
```

Important adverse effects can reflect target-related biology, payload-related toxicity, off-target payload release, or a combination of these factors. For T-DXd, clinically important toxicities include gastrointestinal and haematological effects associated with cytotoxic treatment, as well as a risk of interstitial lung disease or

pneumonitis. The specific benefit–risk balance, monitoring requirements, and approved use depend on the indication, line of therapy, patient characteristics, and current product information.

The central lesson is:

An ADC is a system-level medicine.
Its benefit and risk cannot be inferred from antigen expression
or payload potency alone.

9.11 Resistance to HER2-directed antibodies and ADCs

Resistance to HER2-directed therapy can arise through several mechanisms. These mechanisms differ between trastuzumab and T-DXd because the therapies share a target but have distinct pharmacological actions.

Possible resistance mechanisms include:

- Reduced HER2 expression, altered HER2 accessibility, or tumour-cell populations with low HER2 abundance.
- Changes in HER2-associated signalling or activation of alternative growth and survival pathways.
- Reduced immune-cell recruitment or an immune-suppressive tumour microenvironment, which may limit Fc-mediated anti-tumour activity.
- Reduced ADC internalisation, altered intracellular trafficking, or inadequate linker processing.
- Drug-export mechanisms that reduce intracellular payload concentration.
- Enhanced DNA-damage repair or altered sensitivity to topoisomerase-I inhibition.
- Clonal evolution and tumour heterogeneity, allowing resistant cell populations to survive and expand.

Resistance should not be interpreted as proof that HER2 was never relevant. A tumour can initially depend on HER2-associated biology and later adapt through changes that reduce the effectiveness of a particular intervention. Understanding whether resistance is target-related, payload-related, exposure-related, or driven by tumour heterogeneity is important for developing subsequent treatments and combinations.

9.12 Key takeaways

Antibodies and ADCs provide ways to address extracellular and cell-surface targets that may be inaccessible to conventional small molecules.

Several distinctions are especially useful to retain:

- The Fab region of an antibody recognises a target epitope, whereas the Fc region can interact with immune-system components and influence antibody behaviour in the body.
- Antibodies bind exposed three-dimensional protein surfaces and do not require a deep small-molecule-sized pocket.
- HER2 is a cell-surface receptor tyrosine kinase, a therapeutic target in some cancers, and a predictive biomarker for HER2-directed treatment.
- Trastuzumab binds HER2, perturbs HER2-associated biology, and can recruit immune effector mechanisms such as ADCC.
- An ADC combines target recognition, linker-controlled payload release, and intracellular cytotoxic activity; it is not simply an antibody with chemotherapy attached.

- T-DXd uses HER2 recognition to promote intracellular delivery of DXd, a topoisomerase-I inhibitor that can cause replication stress and DNA damage.
- The bystander effect can help address heterogeneous tumours but can also reduce the precision of payload delivery.
- ADC efficacy and safety depend on antigen expression, tumour penetration, internalisation, linker behaviour, payload biology, heterogeneity, normal-tissue exposure, and therapeutic window.

When reading about an antibody or ADC, ask:

Which epitope does the antibody recognise?
 What does antibody binding do biologically?
 Is immune-effector recruitment important?
 For an ADC, is target internalisation and payload release efficient?
 Which cells may be affected directly or through bystander exposure?
 What normal-tissue expression, payload toxicity, and resistance mechanisms could limit the therapeutic window?

9.13 Super-short summary

Antibodies use Fab regions to recognise exposed protein epitopes and Fc regions to interact with immune-system components. They are therefore especially useful for soluble extracellular targets and cell-surface proteins.

Trastuzumab binds HER2 and combines effects on HER2-associated biology with immune-mediated killing. T-DXd uses HER2 binding to deliver a topoisomerase-I inhibitor intracellularly. Its clinical effect depends on the complete sequence from antigen expression and internalisation to linker cleavage, payload action, tumour heterogeneity, and safety.

9.14 Practice questions

How to use these questions. Try to answer each question without looking at the answer. Focus on the causal chain from antibody binding to immune recruitment or payload delivery, and on why a shared target does not make two HER2-directed treatments pharmacologically equivalent.

1. What is a monoclonal antibody, and what is an epitope?

Answer: A monoclonal antibody is a population of identical antibody molecules designed to recognise the same target region. An epitope is the particular exposed molecular region on the target that the antibody recognises and binds.

2. What are the Fab and Fc regions of an antibody?

Answer: The Fab regions contain the antigen-binding surfaces that recognise a target epitope. The Fc region can interact with Fc receptors and other immune-system components, influencing immune effector functions, distribution, and persistence in the body.

3. Why do conventional antibodies not require a deep small-molecule binding pocket?

Answer: Antibodies are relatively large proteins with binding surfaces that can recognise broad exposed three-dimensional target regions. They bind epitopes through many non-covalent interactions and therefore do not need to fit into a small deep pocket. However, the epitope must be accessible, usually outside the cell or on the cell surface.

4. Why is HER2 both a therapeutic target and a predictive biomarker?

Answer: HER2 overexpression or ERBB2 amplification can contribute to tumour growth through excessive HER2-associated signalling, making HER2 a therapeutic target. HER2 testing also identifies patients whose tumours are more likely to benefit from HER2-directed therapies.

5. **Why is it incomplete to say that trastuzumab simply switches off HER2?**

Answer: Trastuzumab binds an extracellular HER2 epitope and can perturb several HER2-associated processes. It can also recruit immune effector cells through its Fc region. Its activity therefore reflects both effects on receptor-associated biology and immune-mediated mechanisms.

6. **What is ADCC?**

Answer: Antibody-dependent cellular cytotoxicity is a mechanism in which an antibody bound to a target cell is recognised through its Fc region by immune effector cells, especially NK cells. Fc-receptor engagement activates the immune cell, which can attack and kill the antibody-coated target cell.

7. **What are the three principal components of an ADC?**

Answer: An ADC consists of a targeting antibody, a linker that connects the components and controls payload release, and a potent cytotoxic payload.

8. **Why is internalisation important for many ADCs?**

Answer: Internalisation brings the ADC into intracellular compartments where it can be processed, the linker can be cleaved, and payload can be released. Surface binding alone may not provide sufficient intracellular payload delivery.

9. **How does the DXd payload in T-DXd cause tumour-cell damage?**

Answer: DXd inhibits topoisomerase I, an enzyme that helps relieve torsional stress in DNA during replication and transcription. Disrupting topoisomerase-I function can increase replication stress and DNA damage, interfere with cell-cycle progression, and lead susceptible cells to die.

10. **What is the bystander effect, and why can it be both beneficial and harmful?**

Answer: The bystander effect occurs when released payload diffuses from an initially targeted antigen-expressing cell into nearby cells. It may help treat heterogeneous tumours in which not all cancer cells express the target equally. However, it also reduces the precision of delivery and can expose nearby normal or antigen-low cells to cytotoxic payload.

11. **Why can T-DXd resistance occur even when HER2 is still present?**

Answer: Resistance can arise through reduced ADC internalisation, altered intracellular trafficking or linker processing, payload export, increased DNA-damage repair, reduced sensitivity to topoisomerase-I inhibition, bypass signalling, tumour heterogeneity, or insufficient tumour exposure. HER2 presence alone does not guarantee successful payload delivery and cell killing.

10 T cells, antigens, immune checkpoints, and pembrolizumab

10.1 The adaptive immune system

The immune system protects the body against infections, abnormal cells, and other threats. It includes many interacting cell types and signalling systems. The *adaptive immune system* is the part capable of highly specific recognition and immunological memory.

Two major adaptive immune-cell types are *B cells* and *T cells*. B cells can produce antibodies. T cells recognise abnormal or foreign molecular signals presented by other cells and can coordinate or directly carry out immune responses.

T cells develop in the *thymus*, an organ involved in T-cell maturation and selection. They then circulate through blood, lymph nodes, and tissues, where they can encounter cells displaying molecular evidence of infection, cellular damage, or abnormality.

A central feature of T-cell biology is that T cells do not generally recognise whole proteins directly. Instead, each T cell expresses a specialised *T-cell receptor*, or *TCR*, that can recognise a short protein fragment, called

a *peptide*, when that peptide is presented by a specialised cell-surface molecule called *major histocompatibility complex*, or *MHC*.

The basic recognition unit is therefore:

```
T-cell receptor
+
peptide
+
MHC molecule
```

This system allows T cells to inspect a sample of proteins being made, processed, or taken up by another cell. It is an indirect but powerful way for the immune system to detect virus-infected cells, abnormal cells, and some cancer cells.

10.2 Antigen presentation

An *antigen* is a molecular feature that can be recognised by the immune system. In the T-cell setting, the relevant antigen is often a peptide displayed by an MHC molecule.

Cells continually break down some of their proteins into short peptides. These peptides can be loaded onto MHC molecules and displayed at the cell surface for T cells to inspect.

A simplified process is:

```
Cell contains or takes up proteins
→ proteins are processed into short peptides
→ peptides are loaded onto MHC molecules
→ peptide-MHC complexes are displayed at cell surface
→ T cells inspect the displayed peptides
```

Two broad MHC classes are particularly important:

MHC class	Simplified role
MHC class I	Present on most nucleated cells; commonly displays peptides derived from proteins made within the cell. These peptide-MHC complexes are inspected mainly by CD8 T cells.
MHC class II	Present mainly on specialised antigen-presenting cells; commonly displays peptides derived from proteins taken up from outside the cell. These complexes are inspected mainly by CD4 T cells.

Antigen-presenting cells, or *APCs*, include dendritic cells, macrophages, and B cells. Dendritic cells are especially important for initiating T-cell responses because they can take up antigens, travel to lymphoid tissues, and provide the additional signals needed to activate suitable T cells.

Cancer cells may display peptides derived from proteins altered by tumour-associated mutations. These mutation-derived peptides are called *neoantigens*.

The simplified sequence is:

```
Tumour mutation changes protein sequence
→ altered protein is processed into peptides
→ mutant peptide is displayed by MHC
→ a T cell may recognise it as abnormal
```

Neoantigens are particularly interesting because they are absent from normal cells. However, not every mutation produces a neoantigen. A mutated peptide must be generated by protein processing, bind a patient's particular MHC molecules, be displayed at sufficient level, and be recognised by an appropriate T-cell receptor.

10.3 CD8 and CD4 T cells

Two broad T-cell categories are especially useful for understanding anti-tumour immunity: *CD8 T cells* and *CD4 T cells*.

CD8 T cells, often called *cytotoxic T cells*, primarily recognise peptides displayed by MHC class I. Because most nucleated cells express MHC class I, CD8 T cells can inspect many cells throughout the body for evidence of infection or abnormality.

When a CD8 T cell recognises an appropriate peptide–MHC complex and receives the necessary activation signals, it can kill the target cell. It can release molecules such as *perforin* and *granzymes*. Perforin helps create access to the target cell, while granzymes can trigger molecular processes that lead to target-cell death.

A simplified process is:

```
CD8 T cell recognises abnormal peptide-MHC class I
→ cytotoxic programme is activated
→ perforin and granzymes are delivered to target cell
→ target cell may undergo programmed cell death
```

CD4 T cells, often called *helper T cells*, primarily recognise peptides displayed by MHC class II. Their roles are diverse. They can produce cytokines, support the activation and persistence of CD8 T cells, help B cells produce antibodies, activate macrophages, and influence how antigen-presenting cells stimulate other immune cells.

The distinction is useful but simplified:

```
CD8 T cells
→ often directly kill abnormal target cells
CD4 T cells
→ coordinate, sustain, and shape immune responses
```

In cancer, effective immune responses can involve both cell types. CD8 T cells may directly kill tumour cells, while CD4 T cells can help sustain a productive immune response and improve the function of antigen-presenting cells and CD8 T cells.

10.4 T-cell activation requires more than recognition

A T cell does not become fully activated merely because its T-cell receptor recognises a peptide–MHC complex. This is an important safety feature: without additional regulation, T cells might respond inappropriately to harmless molecules or normal tissues.

A useful simplified model is that T-cell activation depends on several categories of signal:

```
Signal 1
→ T-cell receptor recognises a particular peptide-MHC complex
Signal 2
→ co-stimulatory signals indicate an appropriate immune context
Signal 3
→ cytokines and local environment shape T-cell behaviour
Inhibitory signals
→ restrain activation and limit harmful immune responses
```

A major co-stimulatory interaction involves *CD28* on the T cell and molecules such as *CD80* and *CD86* on an antigen-presenting cell. This interaction helps communicate that antigen recognition is occurring in an appropriate inflammatory or danger-associated context.

Cytokines are signalling proteins that influence immune-cell growth, survival, differentiation, movement, and function. They help determine whether a T cell becomes a short-lived effector cell, a memory cell, an exhausted or dysfunctional cell state, or another functional state.

Inhibitory pathways, often called *immune checkpoints*, help prevent excessive or prolonged immune activity. They are important for limiting damage to normal tissues and maintaining *immune tolerance*, meaning the ability of the immune system to avoid inappropriate attack on the body's own cells.

Cancer can exploit this regulation. A tumour may not need to eliminate every tumour-reactive T cell; it may only need to reduce T-cell activation or function enough to avoid effective immune destruction.

10.5 PD-1, PD-L1, and PD-L2

PD-1, short for programmed cell death protein 1, is an inhibitory receptor expressed on activated T cells and some other immune cells. PD-1 is part of a normal system for reducing immune activity after activation and limiting damage to healthy tissue.

Its main ligands are *PD-L1* and *PD-L2*. PD-L1 can be expressed by tumour cells, immune cells, and some normal tissues. PD-L2 has a more restricted pattern of expression but can also engage PD-1.

When PD-1 on a T cell binds PD-L1 or PD-L2, it sends inhibitory signals into the T cell. These signals can reduce aspects of T-cell activation, proliferation, cytokine production, metabolism, survival, and cytotoxic function.

A useful first approximation is:

```
PD-1
→ inhibitory receptor on activated T cell
PD-L1 or PD-L2
→ ligand that can engage PD-1
PD-1-ligand interaction
→ inhibitory or "brake" signal for T-cell activity
```

This pathway has a normal physiological role. For example, PD-L1 can increase in inflamed tissue as part of a feedback mechanism that helps prevent excessive immune-mediated damage.

Tumours can exploit the pathway in several ways. Some tumour cells express PD-L1 because of tumour-intrinsic signalling changes. In other cases, interferon-gamma and other inflammatory signals produced by nearby immune cells can induce PD-L1 expression. This latter pattern is sometimes called *adaptive immune resistance*: the tumour responds to immune attack by increasing an inhibitory signal that suppresses the attacking T cells.

The simplified tumour context is:

```
Tumour-reactive T cells enter tumour
→ inflammatory signals increase
→ PD-L1 may be expressed by tumour or immune cells
→ PD-1 signalling suppresses T-cell activity
→ anti-tumour immune response may weaken
```

PD-1 is only one of several inhibitory pathways. Other checkpoints and suppressive mechanisms can also limit anti-tumour immunity. PD-L1 expression is therefore biologically informative but does not alone determine whether a patient will benefit from PD-1 blockade.

10.6 Pembrolizumab

Pembrolizumab, marketed as Keytruda, is a humanised IgG4 monoclonal antibody that binds PD-1.

Its central pharmacological action is to block the interaction between PD-1 and both of its major ligands, PD-L1 and PD-L2. By occupying PD-1, pembrolizumab reduces PD-1-mediated inhibitory signalling in T cells.

```
Without pembrolizumab:
PD-L1 or PD-L2 binds PD-1
→ inhibitory signalling in T cell
→ reduced anti-tumour immune activity
With pembrolizumab:
Pembrolizumab binds PD-1
→ PD-L1 and PD-L2 cannot engage PD-1 normally
→ reduced PD-1-mediated inhibition
→ T cells may sustain or regain anti-tumour activity
```

Pembrolizumab does not directly bind or kill tumour cells in the way that a cytotoxic chemotherapy agent, a targeted small molecule, or an ADC payload can. Instead, it changes the immune context in which tumour cells are recognised and attacked.

The expected sequence is:

```
Tumour-associated antigens are presented
→ tumour-reactive T cells are present or can be generated
→ PD-1-mediated inhibition limits their activity
→ pembrolizumab blocks PD-1
→ anti-tumour T-cell activity may increase
→ tumour control may improve
```

The word *may* is important. Pembrolizumab does not create a complete anti-tumour immune response from nothing in every patient. It is most likely to help when a relevant immune response is present or can develop, and when PD-1-mediated inhibition is an important reason that response is ineffective.

Pembrolizumab's IgG4 antibody format is designed primarily to block PD-1 rather than strongly recruit Fc-mediated killing of PD-1-expressing immune cells. This differs conceptually from antibodies such as trastuzumab, where Fc-mediated immune effector functions can contribute to anti-tumour activity.

10.7 Why checkpoint inhibitors do not work for every patient

Checkpoint inhibitors do not automatically generate tumour-specific immunity. Their activity depends on multiple steps in the cancer-immunity cycle, from antigen generation and presentation through T-cell activation, tumour infiltration, recognition, and effective killing.

A useful simplified chain is:

```
Tumour contains potentially recognisable antigens
→ antigens are processed and presented on MHC
→ tumour-reactive T cells are activated
→ T cells reach and enter tumour tissue
→ T cells recognise tumour cells
→ inhibitory pathways are overcome sufficiently
→ tumour cells are killed
```

Failure at any step can limit response. Important factors include:

- *Antigenicity*: whether the tumour contains peptides that T cells can recognise, including possible neoantigens.
- *Antigen presentation*: whether tumour cells retain appropriate MHC expression and antigen-processing machinery.
- *T-cell priming*: whether antigen-presenting cells have effectively activated tumour-reactive T cells.
- *T-cell infiltration*: whether relevant immune cells can enter and persist within the tumour.
- *Immune suppression*: whether other checkpoints, suppressive immune cells, cytokines, metabolic conditions, or physical barriers inhibit T-cell function.
- *Tumour sensitivity to immune killing*: whether tumour cells remain vulnerable after being recognised by T cells.
- *Patient immune competence*: whether the patient can generate and sustain an effective immune response.

Tumours are sometimes described informally as *immune-inflamed*, *immune-excluded*, or *immune-desert*.

```
Immune-inflamed tumour
→ immune cells, including T cells, are present in tumour tissue
→ checkpoint blockade may be more plausible, but is not guaranteed to work
Immune-excluded tumour
→ immune cells are present near tumour but do not enter effectively
Immune-desert tumour
→ few relevant T cells are present
→ PD-1 blockade alone may have limited effect
```

These are useful conceptual categories, not complete biological diagnoses. A single tumour can contain different immune states in different regions or at different time points.

PD-L1 testing can support treatment decisions in some clinical settings, but it is an imperfect predictive biomarker. A PD-L1-positive tumour may fail to respond because of absent antigen presentation, poor T-cell infiltration, other immune-suppressive mechanisms, or resistance within the tumour. Conversely, some patients with low or negative PD-L1 test results can still benefit in particular indications and treatment settings.

10.8 Immune-related adverse events

PD-1 signalling normally helps restrain immune responses and maintain tolerance to normal tissues. Blocking PD-1 can therefore enhance beneficial anti-tumour immunity while also increasing the risk of harmful immune activity against healthy organs.

These toxicities are called *immune-related adverse events*, or *irAEs*. They can affect many organ systems, including the skin, gastrointestinal tract, liver, lungs, endocrine organs, kidneys, nervous system, heart, and other tissues.

The central trade-off is:

Reduced PD-1-mediated immune inhibition
→ stronger anti-tumour immune activity may become possible
Reduced PD-1-mediated immune inhibition
→ harmful immune activity against normal tissue may also increase

The timing, severity, and organ affected can vary substantially between patients. Some irAEs are mild and manageable, whereas others can be serious or life-threatening. Their clinical management requires prompt assessment and treatment by appropriately qualified healthcare professionals, often including temporary treatment interruption and immunosuppressive therapy when indicated.

This illustrates an important general principle in pharmacology and immunology: increasing a beneficial immune response is not equivalent to selectively increasing immunity only against cancer. The therapeutic challenge is to obtain sufficient anti-tumour activity while maintaining an acceptable safety profile.

10.9 Resistance and durability of response

Some patients experience durable responses to checkpoint inhibition, but others have primary resistance or develop acquired resistance after an initial response.

Primary resistance means that the tumour does not show meaningful benefit from the outset. *Acquired resistance* means that the tumour initially responds but later progresses.

Potential resistance mechanisms include:

- Loss or reduction of tumour antigens or neoantigens recognised by T cells.
- Loss of antigen-presentation capacity, for example through altered MHC expression or antigen-processing machinery.
- Reduced T-cell infiltration or exclusion of immune cells from tumour tissue.
- Activation of alternative inhibitory pathways or accumulation of immune-suppressive cells in the tumour microenvironment.
- Tumour-cell alterations that reduce susceptibility to immune-mediated killing.
- Clonal evolution and tumour heterogeneity, allowing immune-resistant subclones to survive and expand.

Resistance does not necessarily mean that the original immune response was absent or irrelevant. A tumour can initially be vulnerable to immune attack and later evolve mechanisms that evade or resist that attack.

This is why checkpoint blockade is often studied in combinations with other treatments, such as chemotherapy, radiation, targeted therapy, cancer vaccines, or other immunotherapies. The biological aim of a combination may be to increase antigen release, improve T-cell priming, promote immune-cell infiltration, reduce an additional suppressive pathway, or prevent the emergence of resistant tumour-cell populations. Whether a combination is clinically useful depends on evidence of added benefit relative to added toxicity.

10.10 Key takeaways

Checkpoint inhibitors alter the interaction between cancer and the immune system rather than directly poisoning tumour cells.

Several distinctions are especially useful to retain:

- T cells generally recognise short peptides displayed by MHC molecules, rather than recognising whole proteins directly.

- MHC class I commonly presents peptides to CD8 T cells, while MHC class II commonly presents peptides to CD4 T cells.
- CD8 T cells can directly kill recognised abnormal cells, whereas CD4 T cells coordinate and support immune responses.
- Full T-cell activation requires antigen recognition, co-stimulation, cytokine and tissue context, and regulation by inhibitory pathways.
- PD-1 is an inhibitory receptor on activated T cells; PD-L1 and PD-L2 can engage PD-1 and reduce T-cell activity.
- Pembrolizumab blocks PD-1-mediated inhibition, allowing existing or developing tumour-reactive T cells to sustain or regain activity.
- PD-L1 is an imperfect biomarker because response also depends on antigenicity, antigen presentation, T-cell priming, immune-cell infiltration, other suppressive mechanisms, and tumour sensitivity to immune killing.
- Blocking immune checkpoints can produce anti-tumour benefit but can also cause immune-related adverse events through increased immune activity against normal tissues.

When reading about checkpoint inhibition, ask:

Can tumour antigens be presented effectively?
 Are tumour-reactive T cells present or plausibly inducible?
 Can these T cells enter the tumour and recognise cancer cells?
 Is PD-1-mediated inhibition an important barrier?
 What other immune-suppressive or tumour-resistance mechanisms remain?
 What benefit-risk trade-off could result from stronger immune activity?

10.11 Super-short summary

T cells inspect peptide fragments displayed by MHC molecules. Cancer mutations can create neoantigens that may allow tumour cells to be recognised as abnormal. CD8 T cells can directly kill recognised cells, while CD4 T cells help coordinate and sustain immune responses.

Pembrolizumab blocks PD-1, an inhibitory receptor that can suppress tumour-reactive T cells after engagement by PD-L1 or PD-L2. It can produce durable benefit when tumour-directed immunity exists or can develop, but response depends on antigen presentation, T-cell activation and infiltration, immune suppression, tumour heterogeneity, and susceptibility to immune killing. The same immune activation that can control cancer can also cause immune-related adverse events.

10.12 Practice questions

How to use these questions. Try to answer each question without looking at the answer. Focus on the causal chain from tumour antigen presentation to T-cell activation, checkpoint-mediated inhibition, anti-tumour immunity, and immune-related toxicity.

1. What is the adaptive immune system?

Answer: The adaptive immune system is the component of immunity capable of highly specific recognition and immunological memory. T cells and B cells are central components. T cells recognise peptide fragments presented by MHC molecules, while B cells can produce antibodies.

2. What do T cells recognise?

Answer: T cells generally recognise a complex formed by a short peptide presented in an MHC molecule. The T-cell receptor contacts both the peptide and the MHC molecule rather than recognising a whole protein directly.

3. What is antigen presentation?

Answer: Antigen presentation is the process through which proteins are processed into peptides, loaded onto MHC molecules, and displayed at the cell surface for inspection by T cells.

4. What is the broad difference between MHC class I and MHC class II?

Answer: MHC class I is present on most nucleated cells and commonly presents peptides derived from proteins inside the cell to CD8 T cells. MHC class II is mainly expressed by specialised antigen-presenting cells and commonly presents peptides derived from proteins taken up from outside the cell to CD4 T cells.

5. What is a neoantigen?

Answer: A neoantigen is a mutation-derived tumour peptide that can be presented by MHC and recognised by T cells. It is absent from normal cells, but not every tumour mutation creates a peptide that is effectively processed, presented, and recognised.

6. What is the broad difference between CD8 and CD4 T cells?

Answer: CD8 T cells can directly kill cells displaying an appropriate peptide–MHC class I complex. CD4 T cells usually recognise peptide–MHC class II complexes and coordinate or support immune responses through cytokines and interactions with other immune cells.

7. Why does T-cell activation require more than peptide–MHC recognition?

Answer: Full activation also depends on co-stimulatory signals, cytokines, tissue context, and inhibitory pathways. This multi-signal system helps prevent harmful or inappropriate immune responses against normal tissues or harmless antigens.

8. What does PD-1 signalling do?

Answer: PD-1 is an inhibitory receptor expressed on activated T cells. When PD-L1 or PD-L2 binds PD-1, inhibitory signals reduce aspects of T-cell activation, proliferation, cytokine production, and cytotoxic activity. This can limit immune damage but can also suppress anti-tumour immunity.

9. How does pembrolizumab work?

Answer: Pembrolizumab binds PD-1 and prevents its normal interaction with PD-L1 and PD-L2. This reduces PD-1-mediated inhibition and may allow tumour-reactive T cells to sustain or regain anti-tumour activity. It does not directly kill tumour cells; it changes the immune conditions under which they are recognised and attacked.

10. Why might a PD-L1-positive tumour fail to respond to pembrolizumab?

Answer: PD-L1 expression alone does not guarantee that a productive anti-tumour immune response exists. The tumour may lack recognisable antigens, lose MHC-mediated antigen presentation, exclude T cells, use other suppressive mechanisms, resist immune killing, or contain resistant tumour-cell subclones.

11. Why can pembrolizumab cause immune-related adverse events?

Answer: PD-1 normally helps limit immune activity and maintain tolerance to normal tissues. Blocking PD-1 can strengthen anti-tumour immunity but can also permit harmful immune attack against healthy organs, producing immune-related adverse events.

12. Why can checkpoint-inhibitor resistance emerge after an initial response?

Answer: Tumours can evolve through antigen loss, impaired antigen presentation, reduced T-cell infiltration, alternative immune-suppressive pathways, reduced sensitivity to immune killing, or expansion of immune-resistant subclones. An initial response therefore does not guarantee durable tumour control.

11 Cancer vaccines, CAR-T, and cell therapies

11.1 Two ways to direct immunity against cancer

Cancer immunotherapy can work in different ways. Chapter 10 discussed immune-checkpoint inhibitors such as pembrolizumab, which reduce inhibitory signals that can suppress existing anti-tumour T-cell responses.

Cancer vaccines and engineered cell therapies address different parts of the same broader problem.

```
Cancer vaccine
→ aims to teach, expand, or strengthen immune recognition
of tumour-associated antigens
Checkpoint inhibitor
→ reduces an inhibitory signal that can restrain
tumour-reactive T cells
CAR-T cell therapy
→ gives T cells an engineered receptor that recognises
a selected cell-surface antigen directly
```

These approaches are related because each seeks to increase effective immune attack against cancer. However, they use different mechanisms, require different types of target antigen, and have distinct manufacturing, safety, and resistance challenges.

A central question is:

```
Can the immune system identify cancer cells accurately enough,
reach them effectively enough,
and destroy them without unacceptable injury to normal tissue?
```

11.2 Cancer vaccines

A *cancer vaccine* aims to stimulate or strengthen an immune response against cancer-associated antigens.

Unlike preventive vaccines against infectious disease, therapeutic cancer vaccines are generally given to patients who already have cancer or who are at high risk of cancer recurrence after treatment. The objective is not necessarily to prevent the initial development of cancer. Instead, it is to help the immune system recognise and control tumour cells that are already present, including microscopic residual disease that may remain after surgery.

A cancer vaccine may contain tumour antigens, peptides, proteins, nucleic acids such as mRNA, tumour cells, viral vectors, or specialised antigen-presenting cells. Different formats aim to deliver tumour-associated information to the immune system in different ways.

The intended biological sequence is:

```
Tumour-associated antigen is delivered or made in the body
→ antigen is processed and presented by antigen-presenting cells
→ suitable T cells are activated and expanded
→ T cells enter tumour tissue
→ T cells recognise tumour cells displaying relevant antigen
→ tumour cells may be attacked and killed
```

This is difficult because each step can fail. The selected antigen may not be sufficiently tumour-specific, antigen-presenting cells may not activate appropriate T cells effectively, tumour cells may not present the antigen through MHC, T cells may not enter the tumour, or suppressive features of the tumour microenvironment may limit immune activity.

A vaccine therefore does not guarantee that a patient will develop a useful anti-tumour response. It is an attempt to generate or improve one.

11.3 Tumour antigens and neoantigens

A *tumour antigen* is a molecular feature associated with cancer cells that may be recognised by the immune system. Tumour antigens can arise in several ways.

They may be:

- *Mutation-derived antigens*, created by tumour-specific DNA mutations.
- *Overexpressed normal proteins*, produced at unusually high levels by tumour cells.
- *Differentiation antigens*, associated with a particular tissue lineage from which the tumour arose.
- *Cancer-testis antigens*, proteins normally expressed mainly in reproductive tissues but abnormally expressed in some cancers.
- *Virus-derived antigens*, present in cancers associated with oncogenic viruses.

A *neoantigen* is especially attractive because it is created by a tumour-specific mutation and is absent from normal cells. A mutation can alter the amino-acid sequence of a tumour protein, creating a peptide that differs from the corresponding peptide in healthy cells.

The basic logic is:

```
Tumour mutation
→ altered protein sequence
→ altered peptide generated during protein processing
→ peptide displayed on MHC
→ potential recognition by T cells as abnormal
```

Neoantigens can be valuable therapeutic targets because T cells recognising them may have less risk of attacking normal tissues that lack the mutation. However, not every tumour mutation produces a useful neoantigen.

For a mutation to produce a potentially useful neoantigen, several conditions generally need to be satisfied:

```
Mutation is present in tumour cells
→ mutated gene is expressed
→ altered protein is produced and processed
→ relevant peptide binds patient's MHC molecule
→ peptide-MHC complex is displayed sufficiently
→ suitable T-cell receptor can recognise it
```

The patient's particular MHC molecules matter because MHC genes are highly variable between individuals. A peptide predicted to bind one patient's MHC molecules may not bind another patient's. This is one reason personalised neoantigen approaches require patient-specific analysis.

Tumour heterogeneity is another challenge. A neoantigen may be *clonal*, meaning it is present in most or all tumour cells descended from an early common ancestor, or *subclonal*, meaning it is present only in a subset of tumour cells. Clonal neoantigens may be more attractive targets because tumour cells cannot easily escape immune recognition simply by expansion of antigen-negative subclones.

11.4 Personalised neoantigen vaccines

A *personalised neoantigen vaccine* is designed using information from an individual patient's tumour. It aims to select mutation-derived antigens that are more likely to distinguish tumour cells from normal cells and stimulate tumour-reactive T-cell responses.

A typical workflow is:

```
Obtain tumour sample and matched normal sample
→ sequence tumour and normal DNA, often with tumour RNA
→ identify tumour-specific mutations
→ infer patient-specific MHC type
→ predict and prioritise candidate neoantigens
→ manufacture patient-specific vaccine
→ administer vaccine to stimulate T-cell responses
```

The computational and laboratory steps are substantial. They can include identifying *somatic mutations*, which are mutations found in tumour cells but not matched normal cells; determining whether mutated genes are expressed; inferring the patient's MHC alleles; predicting peptide processing and MHC presentation; and prioritising candidates using evidence such as mutation clonality, RNA expression, and predicted peptide–MHC binding.

However, neoantigen prediction is uncertain. A computationally predicted peptide may fail to be processed naturally, may not be displayed effectively by MHC, or may not stimulate a sufficiently strong T-cell response. Conversely, an immune-relevant neoantigen can be missed by imperfect prediction methods.

The practical challenge is therefore broader than prediction alone:

```
Accurate tumour and normal sequencing
+
Neoantigen prioritisation
+
Rapid personalised manufacturing
+
Effective vaccine delivery and immune priming
+
Durable tumour antigen presentation
+
Control of immune suppression and tumour evolution
```

Personalisation may improve tumour specificity, but it also creates logistical complexity. The therapy must be designed and manufactured quickly enough to remain clinically useful for the patient.

11.5 mRNA neoantigen vaccines: how they are intended to work

An *mRNA vaccine* contains messenger RNA encoding selected antigens. The mRNA is generally packaged in *lipid nanoparticles*, which are small lipid-based particles designed to protect the mRNA and help deliver it into cells after administration.

The mRNA does not alter the patient's DNA. It provides temporary instructions that cells can use to make selected protein fragments.

The intended sequence is:

mRNA enters suitable cells

- cellular ribosomes translate mRNA into encoded protein material
- protein material is processed into peptides
- peptides are displayed by MHC molecules
- T cells recognising the peptides can be activated or expanded

Antigen-presenting cells, particularly dendritic cells, are important because they can take up vaccine-derived material and provide the co-stimulatory signals needed to activate T cells effectively. Vaccine-derived antigen can also be transferred between cells and presented through processes that help prime CD8 T-cell responses.

A personalised mRNA neoantigen vaccine can encode several selected neoantigen sequences in one construct. This may broaden the immune response and reduce the chance that tumour control depends on recognition of one antigen alone.

The intended goal is:

Selected tumour neoantigen sequences

- vaccine-driven antigen presentation
- expansion of neoantigen-reactive CD4 and CD8 T cells
- improved recognition and killing of tumour cells

This still depends on tumour cells retaining the relevant mutation and presenting the associated peptide through MHC. If the tumour loses antigen expression, downregulates antigen presentation, or evolves antigen-negative subclones, the vaccine-induced response may become less effective.

11.6 Intismeran autogene

Intismeran autogene, also known as *mRNA-4157* or *V940*, is an investigational mRNA-based individualised neoantigen therapy. It is designed to encode up to 34 selected patient-specific neoantigens in a synthetic mRNA construct formulated in lipid nanoparticles.

Its intended development process is:

Tumour and matched normal samples are analysed

- tumour-specific mutations are identified
- candidate neoantigens are predicted and prioritised
- selected neoantigen sequences are encoded in mRNA
- personalised mRNA therapy is manufactured
- vaccine administration aims to expand tumour-reactive T cells

As of 26 August 2026, the most recent public company update was the 19 August 2026 topline announcement from Merck and Moderna on the Phase 3 *INTerpath-001* trial. According to that company announcement, intismeran autogene combined with pembrolizumab met its primary endpoint of recurrence-free survival and a key secondary endpoint of distant-metastasis-free survival compared with pembrolizumab alone in patients with completely resected high-risk cutaneous melanoma.

The announced trial enrolled patients with resected stage IIB–IV melanoma who had not received prior systemic therapy. The companies reported that the combination showed a statistically significant and clinically meaningful improvement at a pre-specified interim analysis, with no new safety signals reported in the topline announcement.

These findings should be interpreted with appropriate caution. At the time of writing, this is public topline information rather than a full peer-reviewed report. Detailed effect sizes, follow-up duration, subgroup analyses, statistical methods, complete safety data, quality-of-life data, and overall-survival results require assessment when presented in full. The therapy remains appropriately described as *investigational* unless and until approved by relevant regulatory authorities for a particular use.

The broader scientific importance of this trial is that it tests whether individualised tumour sequencing, neoantigen selection, mRNA delivery, and checkpoint inhibition can be combined into a clinically meaningful therapeutic strategy.

11.7 Why combine a neoantigen vaccine with pembrolizumab?

The rationale for combining a neoantigen vaccine with pembrolizumab follows directly from the mechanisms discussed in Chapters 10 and 11.

A vaccine aims to improve the *recognition* part of anti-tumour immunity by generating or expanding T cells that recognise selected tumour neoantigens. Pembrolizumab aims to reduce *PD-1-mediated inhibition* that may restrain activated T cells in lymphoid tissues or the tumour microenvironment.

The intended complementarity is:

```
Neoantigen vaccine
→ aims to generate or expand tumour-specific T cells
Pembrolizumab
→ reduces PD-1-mediated inhibition of activated T cells
Combination
→ aims to improve both tumour recognition
and sustained anti-tumour T-cell activity
```

The combination can still fail for many reasons. Neoantigens may be selected imperfectly; the vaccine may induce insufficient T-cell responses; tumour cells may lose antigen presentation; T cells may fail to enter the tumour; other inhibitory pathways may dominate; or resistant tumour subclones may survive and expand.

The combination principle is therefore not that one treatment automatically makes the other work. It is a biological hypothesis that two different barriers to immune control may be addressed simultaneously.

11.8 Cell therapies

Cell therapy uses living cells as the therapeutic product. In oncology, many cell therapies use immune cells, especially T cells, because T cells can recognise, migrate toward, and kill abnormal cells.

Cell therapies differ from conventional drugs in an important way: the administered cells can expand, persist, move between tissues, change functional state, and interact dynamically with the patient's immune system and tumour.

The overall outcome can therefore depend on:

```
Cell collection and starting-cell quality
+
Genetic modification or ex vivo expansion
+
Manufacturing consistency
+
Cell survival after infusion
+
Tissue trafficking and tumour infiltration
+
Persistence and functional state
+
Tumour antigen expression and immune environment
+
Safety and control of excessive immune activation
```

Some therapies use a patient's own cells. This is called *autologous* cell therapy.

Autologous therapy

- cells are collected from the patient
- modified or expanded outside the body
- returned to the same patient

Other approaches use cells from a donor. These are called *allogeneic* cell therapies.

Allogeneic therapy

- cells are obtained from a donor
- manufactured for use in another patient

Autologous therapy can reduce some risks of immune incompatibility because the cells originate from the patient. However, it requires patient-specific manufacturing and may be difficult when the patient's cells are limited in number or functionally impaired.

Allogeneic approaches may support more standardised, potentially more rapidly available products, but donor cells can create additional immunological challenges. These include rejection of the infused cells by the patient and, for some T-cell products, the risk that donor T cells attack the patient's normal tissues. Such risks require specialised engineering and clinical management.

11.9 CAR-T cells

A *CAR-T cell therapy* modifies T cells to express a *chimeric antigen receptor*, or *CAR*. A CAR is a synthetic receptor designed to redirect T-cell recognition toward a chosen cell-surface antigen.

The word *chimeric* reflects that the CAR combines functional elements originating from different biological systems.

A simplified CAR contains:

Extracellular antigen-recognition domain

- often derived from an antibody-binding region
- recognises a selected cell-surface antigen

Hinge and spacer region

- provides flexibility and helps position the recognition domain relative to the target antigen

Transmembrane region

- anchors the CAR in the T-cell membrane

Intracellular signalling domains

- activate the T cell after antigen binding

The antigen-recognition region is often based on a *single-chain variable fragment*, or *scFv*, derived from the variable regions of an antibody. This gives the engineered T cell an antibody-like ability to recognise an exposed cell-surface target.

The intracellular part commonly includes signalling from *CD3-zeta*, a component normally involved in T-cell receptor signalling. Many modern CARs also include a *co-stimulatory domain*, such as CD28 or 4-1BB, to support T-cell activation, expansion, metabolism, and persistence.

Unlike a conventional T-cell receptor, a CAR does not require the target antigen to be processed into peptides and displayed by MHC. Instead, it can bind an accessible molecule directly on the cell surface.

The key contrast is:

Conventional T-cell receptor

→ recognises peptide presented by MHC

CAR

→ recognises selected cell-surface antigen directly

→ does not require peptide-MHC presentation

This can be advantageous when tumour cells have impaired MHC antigen presentation. However, it also limits CAR-T therapy to targets that are accessible at the cell surface.

11.10 How CAR-T cells kill target cells

After CAR-T cells are infused, they must survive, circulate, enter the relevant tissues, find antigen-expressing cells, bind the target antigen, and activate sufficiently to kill.

The simplified sequence is:

CAR-T cell encounters target cell

→ CAR binds cell-surface antigen

→ CAR signalling activates T cell

→ CAR-T cell releases cytotoxic molecules and cytokines

→ target cell may undergo cell death

→ activated CAR-T cell can potentially engage another target cell

CAR-mediated activation can trigger T-cell proliferation and release of cytokines, as well as cytotoxic mechanisms involving perforin and granzymes. These are related to the mechanisms used by conventional CD8 T cells, but the antigen-recognition step is redirected by the synthetic CAR.

The strength and consequences of CAR signalling depend on many factors, including antigen density, epitope location, CAR design, co-stimulatory domain, T-cell state, tumour burden, local immune environment, and the physical accessibility of cancer cells.

A CAR-T cell may fail to control a tumour if it does not persist, becomes dysfunctional, cannot reach the tumour, encounters insufficient antigen, or faces an immune-suppressive microenvironment.

11.11 Choosing a CAR-T target antigen

Target selection is one of the most important decisions in CAR-T development. The ideal target antigen would be strongly and uniformly expressed on tumour cells, exposed at the cell surface, absent from essential normal tissues, and difficult for cancer cells to lose.

In practice, few targets satisfy all of these criteria.

A suitable CAR-T target antigen ideally:

- Is present on the surface of relevant tumour cells.
- Is expressed at sufficiently high density for reliable CAR recognition.
- Is absent, or expressed only at tolerable levels, on essential normal tissues.
- Is present across most clinically important tumour cells and lesions.
- Is retained during tumour evolution, so that antigen-negative cells cannot easily escape.
- Is physically accessible to infused CAR-T cells in the relevant tissue.

In some blood cancers, lineage-associated antigens can be suitable targets even when they are also expressed on a non-essential normal cell population. For example, loss of a normal B-cell population may be clinically manageable in some settings with monitoring and supportive care.

Solid tumours create a harder problem. Many cell-surface antigens are *tumour-associated* rather than truly tumour-specific: they are more abundant on cancer cells but are also present to some degree on normal tissues. In addition, antigen expression may vary between tumour regions, metastatic sites, and individual cancer cells.

This creates a central tension:

```
Target expressed broadly enough on tumour
→ improves chance of tumour recognition
Target expressed on essential normal tissue
→ increases risk of unacceptable toxicity
```

11.12 On-target, off-tumour toxicity

On-target, off-tumour toxicity occurs when a therapy correctly recognises its intended antigen but that antigen is also present on healthy tissue.

```
CAR-T cell recognises target antigen on cancer cell
→ intended killing
Same antigen expressed on normal cell
→ mechanistically on-target but clinically harmful killing
```

This is not an off-target effect in the molecular sense. The CAR is acting exactly as designed. The problem is that the selected antigen is not sufficiently restricted to tumour cells.

On-target, off-tumour toxicity is a major challenge in CAR-T therapy, particularly for solid tumours. It can also occur with antibodies, ADCs, bispecific antibodies, and other targeted therapies when the chosen target is expressed in normal tissue.

Potential strategies to improve selectivity include identifying more tumour-restricted antigens, requiring recognition of more than one antigen, tuning CAR affinity so that cells with high antigen density are preferentially recognised, using logic-gated designs, or incorporating controllable safety mechanisms. These approaches are scientifically promising but do not eliminate the need for careful experimental and clinical validation.

11.13 Other important CAR-T toxicities

CAR-T toxicity can also arise from strong immune activation even when antigen recognition is directed toward tumour cells.

One important complication is *cytokine release syndrome*, or *CRS*. CRS is a systemic inflammatory response that can occur when activated immune cells release large amounts of cytokines and activate other immune cells.

The simplified sequence is:

```
CAR-T cells encounter substantial antigen burden
→ strong CAR-T activation
→ cytokine release and activation of other immune cells
→ systemic inflammation
→ cytokine release syndrome may occur
```

Another important complication is *immune effector cell-associated neurotoxicity syndrome*, often abbreviated *ICANS*. This refers to neurological toxicity that can occur after immune effector cell therapies. Its mechanisms are incompletely understood and may involve systemic inflammation, effects on blood vessels and the blood–brain barrier, and complex interactions between immune cells and the nervous system.

The severity, timing, and management of these toxicities vary. CAR-T therapies require specialised clinical infrastructure, close monitoring, and experienced multidisciplinary care. The key conceptual point is:

CAR-T efficacy depends on strong immune activation against tumour
Strong immune activation can also create systemic toxicity

11.14 Why solid tumours remain particularly challenging

CAR-T therapy has shown major clinical benefit in several blood cancers, but solid tumours often present additional barriers.

These include:

- *Target specificity*: few tumour antigens are completely absent from essential normal tissues.
- *Antigen heterogeneity*: different tumour cells may express different amounts of the target antigen.
- *Physical access*: CAR-T cells may have difficulty entering dense tumour tissue or reaching all metastatic sites.
- *Tumour microenvironment*: low oxygen, limited nutrients, suppressive immune cells, inhibitory cytokines, and abnormal blood vessels can reduce T-cell function.
- *T-cell persistence and dysfunction*: engineered cells may not persist long enough or may become less functional over time.
- *Antigen escape*: tumour cells that lose or reduce the target antigen can survive and expand.

These challenges illustrate why a compelling cell-surface target alone is not enough. Successful cell therapy requires an integrated strategy involving target selection, cell design, manufacturing, tumour access, immune fitness, safety, and resistance management.

11.15 Key takeaways

Cancer vaccines and CAR-T therapies use immune recognition in different ways.

Personalised neoantigen vaccine
→ uses tumour mutations to teach or strengthen
peptide-MHC-specific T-cell recognition
CAR-T therapy
→ engineers T cells with an artificial receptor
that directly recognises a selected cell-surface antigen

Several distinctions are especially useful to retain:

- Neoantigens are mutation-derived peptides that may be displayed by MHC and recognised by T cells; not every tumour mutation produces a useful neoantigen.
- Personalised neoantigen vaccines require tumour analysis, patient-specific antigen selection, rapid manufacturing, effective immune priming, and continued tumour antigen presentation.
- mRNA vaccines provide temporary instructions for cells to produce selected antigenic material; they do not modify the patient's DNA.
- Combining a neoantigen vaccine with pembrolizumab aims to improve both tumour-specific T-cell recognition and resistance to PD-1-mediated immune inhibition.

- CARs use antibody-derived recognition to bind cell-surface antigens directly, without requiring peptide processing or MHC presentation.
- CAR-T efficacy depends on target expression, T-cell activation, trafficking, persistence, tumour access, and resistance biology.
- On-target, off-tumour toxicity results from correct recognition of a target antigen that is also expressed on normal tissue.
- CAR-T therapy can also cause systemic immune toxicities such as cytokine release syndrome and neurological toxicity.

When evaluating a cancer vaccine or cell-therapy approach, ask:

What antigen is being targeted, and how tumour-specific is it?
 Can the tumour retain and present that antigen?
 Can vaccine-induced or engineered T cells reach the tumour?
 What suppressive mechanisms or antigen-escape routes may remain?
 What normal tissues could be affected?
 How feasible are manufacturing, delivery, monitoring, and safety management?

11.16 Super-short summary

Cancer vaccines aim to help the immune system recognise tumour antigens, especially mutation-derived neoantigens. Personalised mRNA neoantigen vaccines use tumour sequencing to select patient-specific antigens and encode them in mRNA, with the goal of expanding tumour-reactive T-cell responses.

CAR-T therapy instead engineers T cells with a synthetic receptor that directly recognises a chosen cell-surface antigen without requiring MHC presentation. Both strategies depend on antigen selection, tumour heterogeneity, immune context, manufacturing, tumour access, persistence, and safety. Their central challenge is to create sufficiently strong and durable anti-tumour immunity without unacceptable damage to normal tissue.

11.17 Practice questions

How to use these questions. Try to answer each question without looking at the answer. Focus on the chain from tumour-antigen selection to immune recognition and tumour killing, and on why antigen specificity, tumour heterogeneity, manufacturing, and safety are central constraints.

1. What is the purpose of a therapeutic cancer vaccine?

Answer: A therapeutic cancer vaccine aims to stimulate or strengthen an immune response against cancer-associated antigens in patients who already have cancer or who are at risk of recurrence after treatment. It seeks to improve immune recognition and control of tumour cells rather than prevent the initial development of cancer.

2. What is the difference between a tumour antigen and a neoantigen?

Answer: A tumour antigen is any molecular feature associated with cancer cells that may be recognised by the immune system. A neoantigen is a tumour-specific antigen created by a mutation and absent from normal cells. Neoantigens can be especially attractive because they may allow more tumour-specific T-cell recognition.

3. Why does not every tumour mutation create a useful neoantigen?

Answer: The mutated gene must be expressed, the altered protein must be processed into a suitable peptide, the peptide must bind the patient's MHC molecule and be displayed at the cell surface, and a suitable T-cell receptor must recognise it. Failure at any step can prevent useful immune recognition.

4. What is the basic workflow for a personalised neoantigen vaccine?

Answer: Tumour and matched normal samples are analysed to identify tumour-specific mutations. Candidate neoantigens are predicted using information including tumour expression and the patient's MHC type. Selected neoantigen sequences are then incorporated into a personalised vaccine, which is administered to stimulate tumour-reactive T-cell responses.

5. How is an mRNA neoantigen vaccine intended to work?

Answer: The vaccine mRNA encodes selected neoantigen sequences. Cells take up the mRNA and translate it into protein material, which is processed into peptides and displayed by MHC molecules. Antigen-presenting cells can then help activate and expand T cells that recognise the selected neoantigens. The mRNA provides temporary instructions and does not modify the patient's DNA.

6. Why might a neoantigen vaccine be combined with pembrolizumab?

Answer: A neoantigen vaccine aims to generate or expand tumour-specific T cells. Pembrolizumab blocks PD-1-mediated inhibitory signalling that may suppress those activated T cells. The combination aims to improve both tumour recognition and sustained anti-tumour immune activity.

7. What is the difference between autologous and allogeneic cell therapy?

Answer: Autologous cell therapy uses cells collected from the same patient who will receive them. Allogeneic cell therapy uses donor-derived cells manufactured for use in another patient. Autologous therapy requires patient-specific production, whereas allogeneic therapy can introduce additional immune-compatibility challenges.

8. What are the main functional components of a CAR?

Answer: A CAR contains an extracellular antigen-recognition domain, often derived from an antibody-binding region; a hinge or spacer; a transmembrane region anchoring it in the T-cell membrane; and intracellular signalling domains, commonly including CD3-zeta and often a co-stimulatory domain such as CD28 or 4-1BB.

9. How do CAR-T cells differ from conventional T cells in antigen recognition?

Answer: Conventional T-cell receptors recognise peptides presented by MHC molecules. CARs bind selected accessible cell-surface antigens directly and therefore do not require peptide processing or MHC presentation. This can overcome some antigen-presentation defects but restricts CAR targets to cell-surface molecules.

10. What is on-target, off-tumour toxicity?

Answer: On-target, off-tumour toxicity occurs when a therapy correctly recognises its intended antigen on cancer cells but the same antigen is also expressed on healthy tissue. The therapy is acting through its intended mechanism, but target distribution is insufficiently tumour-specific.

11. What are cytokine release syndrome and ICANS in the context of CAR-T therapy?

Answer: Cytokine release syndrome is a systemic inflammatory response caused by strong immune-cell activation and cytokine release after therapy. ICANS is immune effector cell-associated neurotoxicity syndrome, a potentially serious neurological complication associated with immune effector cell therapies. Both illustrate that powerful anti-tumour immune activation can also create significant systemic toxicity.

12. Why are solid tumours particularly challenging for CAR-T therapy?

Answer: Solid tumours often lack completely tumour-specific surface antigens, contain heterogeneous target expression, restrict T-cell entry, create suppressive microenvironments, and enable antigen escape or CAR-T dysfunction. Effective treatment therefore requires more than identifying a tumour-associated cell-surface target.

Part III From target to medicine

12 The drug-discovery workflow

12.1 The end-to-end map

Drug discovery and development is the process of turning a hypothesis about disease biology into evidence that a treatment provides meaningful patient benefit with acceptable risk.

A simplified end-to-end pathway is:

```
Unmet medical need
→ disease hypothesis
→ target identification and validation
→ assay development
→ hit discovery
→ hit-to-lead
→ lead optimisation
→ candidate selection
→ preclinical development
→ Phase I
→ Phase II
→ Phase III
→ regulatory review
→ real-world evidence and lifecycle management
```

The workflow is not strictly linear. New evidence can send a programme back to an earlier stage, alter the intended patient population, change the therapeutic modality or dose, require a different biomarker strategy, or lead to programme termination.

Stopping a weak project early is often good scientific and portfolio decision-making rather than failure. The purpose of each stage is to reduce a particular uncertainty before larger amounts of time, money, patient participation, and organisational effort are committed.

The central principle is:

```
Every stage should answer a decision-relevant question.
The aim is not simply to generate more data,
but to reduce the uncertainty that matters
for the next decision.
```

12.2 Unmet medical need and disease hypothesis

The first decision is:

```
What patient problem are we trying to solve,
and what biological hypothesis explains how an intervention could help?
```

An *unmet medical need* is a patient problem that is not adequately addressed by existing prevention, diagnosis, monitoring, or treatment options. The need may arise because no effective therapy exists, because available therapies are too toxic, because benefit is not durable, because only a subset of patients responds, or because treatment is difficult to deliver.

A *disease hypothesis* is a proposed explanation of disease biology and how changing a particular biological process might improve patient outcomes.

Relevant evidence can include disease epidemiology, natural history, clinical outcomes, current standard of care, patient experience, genetics, pathology, molecular data, scientific literature, and knowledge of disease progression.

A useful disease hypothesis should specify:

```
Which patients have the greatest need?
What biological state is thought to cause or maintain disease?
Which intervention could change that state?
What patient-relevant outcome should improve if the hypothesis is correct?
```

Programmes can fail at this early conceptual stage because:

- The disease need was misunderstood.
- The therapeutic hypothesis was weak, incomplete, or biologically incorrect.
- The intended patient population was too broad or too heterogeneous.
- The expected treatment benefit was not meaningful relative to existing care.
- The treatment landscape changed during development, altering the relevant standard of care.

A compelling molecular mechanism is not enough. The intervention must address a meaningful patient problem in a setting where the expected benefit is large enough to justify its risks, burden, and cost.

12.3 Target identification and validation

The central decision is:

```
Which target should be pursued,
in which disease population,
and with what intended mechanism?
```

As discussed in Chapter 6, *target identification* proposes a biological entity that may matter in disease, whereas *target validation* builds evidence that changing that entity is likely to provide useful patient benefit.

Evidence may include human genetics, functional experiments, disease-tissue data, pathway knowledge, preclinical models, human biology, and early pharmacology.

A strong target hypothesis links several elements:

```
Disease-relevant biology
+
Target modulation
+
Feasible therapeutic modality
+
Defined patient population
+
Plausible biomarker strategy
```

For example, a programme might hypothesise that a molecularly defined subgroup of tumours depends on a particular signalling pathway, that inhibiting a key kinase can reduce that dependency, and that a biomarker can identify the patients most likely to benefit.

Common reasons for failure include weak causal relevance of the target, poor translation from models to humans, inability to achieve sufficient target modulation safely, lack of a viable modality, and absence of a practical strategy for identifying likely responders.

Target validation is not completed by one positive experiment. It requires convergent evidence that the target matters, that the intended mechanism is plausible, and that the target can be modulated at a tolerable exposure in the relevant patient population.

12.4 Assay development

The central decision is:

How will we measure whether compounds or biologics are acting on the relevant biology?

An *assay* is an experiment designed to measure a chemical, biological, cellular, pharmacological, or physico-chemical property. Assays can measure target binding, enzyme activity, cell viability, receptor signalling, protein phosphorylation, gene expression, immune-cell activation, drug concentration, or toxicity-related effects.

An assay should be reproducible, interpretable, sufficiently scalable for its purpose, and meaningfully connected to the programme's mechanism.

Different assays answer different questions:

Biochemical assay
→ does the intervention bind or alter purified target activity?
Cell-based assay
→ can it produce the expected effect in living cells?
Disease-relevant model
→ does it affect biology in a more complex context?
Clinical biomarker assay
→ can target engagement or pathway modulation be assessed in humans?

Important assay considerations include controls, dynamic range, measurement precision, reproducibility, biological relevance, sensitivity to artefacts, and the relationship between the assay readout and downstream decisions.

Failure can arise because an assay:

- Measures an irrelevant proxy rather than the biology of interest.
- Is too noisy, biased, or poorly reproducible.
- Does not distinguish true activity from experimental artefacts.
- Does not translate from purified protein to cells, tissues, or patients.
- Produces results that are difficult to interpret mechanistically.

An assay is not valuable merely because it is high-throughput or technically sophisticated. It is valuable when it provides reliable evidence relevant to the next programme decision.

12.5 Hit discovery

The decision is:

Which initial molecules or biologics show enough evidence of desirable activity to pursue?

For small molecules, discovery approaches may include high-throughput screening, fragment screening, structure-based design, virtual screening, natural products, literature-based discovery, and rational chemical design.

For biologics, approaches can include antibody discovery platforms, display technologies, protein engineering, immunisation strategies, screening of naturally occurring antibodies, or selection and engineering of immune-cell products.

A *hit* is an initial molecule or biological candidate with evidence of activity in a relevant assay. The precise definition depends on the programme. A small-molecule hit might inhibit a purified enzyme or produce a desired cellular effect. An antibody hit might bind a target epitope and block a relevant interaction. A cell-therapy hit might show antigen-specific activation and killing in an appropriate experimental system.

A credible hit package generally includes repeatable activity, confirmation in an independent or *orthogonal* assay, preliminary information about selectivity or mechanism, confirmation of compound or biological-product quality, and early evidence that the starting point is tractable.

Common causes of apparent hit activity that later fail include assay artefacts, compound aggregation, fluorescence interference, impurities, poor reproducibility, chemical instability, nonspecific toxicity, lack of cellular activity, and activity through an unintended target or mechanism.

The early objective is not to claim that a hit is a medicine. It is to establish whether the hit is a credible, reproducible starting point for further work.

12.6 Hit-to-lead and lead optimisation

The decision is:

Which chemical series or therapeutic design can be improved into a viable development candidate?

A *lead* is a more mature compound or therapeutic design with sufficient promise to justify systematic optimisation. For small molecules, this stage is often called *lead optimisation*, as discussed in Chapter 8.

Evidence considered at this stage includes potency, selectivity, cellular activity, PK, ADMET properties, structural understanding, synthetic feasibility, formulation considerations, tissue distribution, and preliminary in vivo evidence where appropriate.

For biologics or cell therapies, analogous optimisation may involve affinity, epitope selection, Fc properties, antigen specificity, internalisation, payload design, cell persistence, manufacturing process, and safety-related characteristics.

The central challenge is multi-objective optimisation:

Target activity
+
Selectivity
+
Exposure in relevant tissue
+
Safety margin
+
Practical manufacture and delivery

Common failures include:

- Inability to achieve required exposure.
- Poor selectivity or unacceptable off-target activity.
- Toxicity or inadequate safety margin.
- Insufficient tissue penetration.
- Chemical or biological instability.
- Difficult synthesis, manufacturing, storage, or formulation.
- Loss of activity in disease-relevant models.
- Inability to identify a dose and schedule that is both effective and tolerable.

A promising molecule or therapeutic design can fail at this stage even if it performs well in one assay. The purpose of optimisation is to build a balanced profile that can plausibly survive the transition from experimental system to patient.

12.7 Candidate selection

The decision is:

`Is there a sufficiently mature candidate whose expected benefit
justifies formal preclinical development?`

A *development candidate* requires a coherent overall package. This includes a plausible mechanism, adequate activity, selectivity, PK, safety margin, manufacturability, formulation, biomarker strategy, and an intended clinical-development plan.

Candidate selection is not simply choosing the molecule with the strongest potency value. A candidate may be highly potent but unsuitable because of poor exposure, safety risk, unstable chemistry, inadequate manufacturability, limited differentiation from existing treatments, or uncertainty about which patients should receive it.

The candidate-selection decision should integrate:

`Mechanism and target validation
+
Pharmacology and selectivity
+
PK, tissue exposure, and dose feasibility
+
Safety and therapeutic window
+
Manufacturing and formulation
+
Biomarker and clinical-development strategy`

Failure can occur because no candidate has an acceptable overall profile, even if individual candidates have excellent properties in one dimension.

Candidate selection is therefore a multidisciplinary decision. It requires input from biology, chemistry, pharmacology, safety, PK, clinical pharmacology, manufacturing, clinical development, regulatory strategy, and other relevant disciplines.

12.8 Preclinical development

The decision is:

Is the candidate sufficiently understood, safe, manufacturable,
and pharmacologically credible to test in humans?

Preclinical development builds the evidence package needed before clinical testing. The exact requirements depend on the modality, route of administration, indication, intended patient population, and applicable regulatory expectations.

Depending on the programme, preclinical work can include PK/PD studies, toxicology, safety pharmacology, tissue-distribution studies, formulation development, manufacturing controls, genotoxicity where relevant, immunogenicity assessment for biologics, and disease-model evidence.

Safety pharmacology studies investigate potentially harmful effects on major physiological systems, including cardiovascular, respiratory, and central nervous-system function.

Toxicology studies investigate adverse biological effects. They examine the relationship between dose, exposure, treatment duration, reversibility, affected organs or tissues, and potential safety margins.

Chemistry, manufacturing, and controls, often abbreviated *CMC*, concerns whether the therapeutic product can be made reproducibly to an appropriate quality standard. CMC includes the identity, purity, stability, formulation, manufacturing process, specifications, and quality controls of the product.

The preclinical objective is not to prove that a medicine will work in humans. It is to establish a sufficiently credible benefit–risk and product-quality package to justify carefully controlled first-in-human testing.

Common failures include toxicity, inability to reach target exposure, unexpected species-specific effects, weak translational evidence, formulation problems, product instability, and manufacturing difficulties.

12.9 Phase I clinical trials

The central decision is:

What dose and schedule can be administered safely,
and is there evidence of exposure, target engagement,
and early biological activity in humans?

Phase I trials are usually the first studies of a new therapy in humans. Historically, these studies often focused primarily on safety and dose escalation. Modern Phase I development commonly also examines PK, PD, target engagement, dose–exposure relationships, biomarker effects, and preliminary anti-tumour activity when relevant.

Phase I design depends on the treatment and disease setting. Some therapies are first tested in healthy volunteers, whereas oncology therapies with substantial expected toxicity are commonly first tested in patients with cancer.

Important evidence includes safety, tolerability, PK, PD, target engagement, dose escalation, dose-limiting toxicities, dose schedule, and early evidence of activity.

A *dose-limiting toxicity*, or *DLT*, is a treatment-related adverse effect severe enough to prevent further dose escalation or continued dosing under the trial rules. The *maximum tolerated dose*, or *MTD*, is sometimes estimated as the highest dose associated with an acceptable level of DLTs.

However, the MTD is not automatically the best dose. For targeted therapies, biologics, and immunotherapies, an appropriate *recommended Phase II dose* may be selected by considering the totality of evidence:

Safety and tolerability
+
PK and tissue exposure
+
Target engagement and PD effects
+
Evidence of biological or clinical activity
+
Feasibility of chronic or combination treatment

Failure can occur because of dose-limiting toxicity, inadequate exposure, lack of proof of mechanism, poor tolerability, or no credible activity signal in the intended setting.

12.10 Phase II clinical trials

The central decision is:

Is there enough evidence of meaningful activity
in an appropriately selected population
to justify confirmatory trials?

Phase II trials generally investigate whether a treatment has sufficient evidence of activity and acceptable safety in a defined patient population. They can refine dose, schedule, biomarker strategy, combination approach, and endpoint selection.

Phase II studies may be single-arm or comparative, depending on the disease, available standard of care, strength of the early evidence, and the intended regulatory and clinical-development strategy.

Relevant evidence can include tumour response, durability of response, progression-related outcomes, biomarker associations, dose selection, safety, comparator context, patient-reported outcomes, and other evidence of clinical benefit.

A response signal may be encouraging without being sufficient. For example, tumour shrinkage may not necessarily translate into durable disease control, improved quality of life, or longer survival. Conversely, a therapy may produce limited measurable tumour shrinkage yet provide meaningful long-term benefit in an appropriate context.

Failure can occur because early activity does not replicate, the wrong biomarker population was selected, resistance emerges quickly, treatment effect is not durable, safety limits treatment, or the benefit is not meaningful relative to current standard of care.

12.11 Phase III clinical trials

The decision is:

Does the treatment provide clinically meaningful benefit
compared with an appropriate standard of care?

Phase III trials are commonly confirmatory studies designed to test whether a treatment provides sufficient benefit and acceptable safety relative to an appropriate comparator in a defined patient population.

The comparator may be an established standard treatment, an active control, placebo in addition to standard care where ethically appropriate, or another clinically justified approach. The choice matters because clinical benefit is interpreted relative to what patients would otherwise receive.

Phase III studies generally use predefined endpoints and analysis plans. Depending on the disease setting, endpoints may include survival, progression-free survival, recurrence-free survival, symptoms, function, quality of life, disease response, hospitalisation, or another patient-relevant outcome.

Evidence considered includes efficacy, durability of benefit, safety, quality of life, subgroup consistency, statistical robustness, manufacturing consistency, and overall benefit–risk.

Common causes of failure include:

- Inadequate efficacy relative to standard of care.
- Insufficient durability of benefit.
- Unexpected safety signals.
- Poor adherence or treatment discontinuation.
- Inadequate exposure in the intended population.
- A changing treatment landscape.
- An inappropriate endpoint, comparator, dose, or patient population.

A statistically significant result is not automatically clinically meaningful. The magnitude and durability of benefit, safety burden, quality-of-life effects, feasibility of treatment, and available alternatives all matter for interpreting benefit–risk.

12.12 Regulatory review and lifecycle management

The decision is:

Does the total evidence support approval for a defined indication, and how should evidence continue to be generated after launch?

Regulators evaluate the quality, safety, efficacy, manufacturing, labelling, risk-management plan, and overall benefit–risk of a therapy for a defined use. Regulatory review asks not only whether the treatment can work, but whether its benefits justify its risks for the intended patients under defined conditions of use.

The approved indication specifies important boundaries, such as disease, patient population, biomarker status where relevant, line of therapy, combination partners, dose, and route of administration.

Approval is not the end of evidence generation. After approval, ongoing pharmacovigilance and *real-world evidence* can help assess effectiveness, safety, utilisation, adherence, rare adverse events, and outcomes in populations broader or more diverse than those represented in clinical trials.

Lifecycle management may include:

- Development in new indications or patient populations.
- New treatment combinations.
- Alternative formulations, routes of administration, or dosing schedules.
- Biomarker refinement.
- Strategies to address resistance or treatment sequencing.
- Additional safety, effectiveness, and quality-of-life evidence generation.

The continuing process is:

Clinical development
→ regulatory decision
→ real-world use
→ monitoring, additional evidence, and improvement

12.13 Key takeaways

The drug-discovery workflow is a sequence of increasingly expensive and consequential decisions. It connects a disease hypothesis to patient benefit through multiple stages of biological, chemical, pharmacological, preclinical, clinical, manufacturing, and regulatory evidence.

At every stage, ask:

What uncertainty are we trying to reduce?
What evidence is needed for the next decision?
What result would make us stop, change direction,
or invest further?

Several principles apply throughout the workflow:

- A promising target, molecule, or early clinical signal is not enough on its own; the complete causal chain from intervention to patient benefit must be supported.
- The relevant patient population, biomarker strategy, dose, schedule, comparator, and endpoint should be considered early rather than treated as late-stage details.
- Evidence from one system does not automatically translate to another. Binding, cellular activity, animal-model activity, target engagement, and clinical benefit answer different questions.
- Manufacturing, formulation, product quality, and deliverability are essential parts of developing a medicine rather than administrative steps after scientific discovery.
- Stopping or redirecting a programme when evidence is insufficient can protect patients and focus resources on stronger therapeutic opportunities.

12.14 Super-short summary

Drug discovery progresses from an unmet patient need and disease hypothesis through target work, assay development, hit discovery, optimisation, candidate selection, preclinical development, clinical trials, regulatory review, and post-approval evidence generation.

The workflow is iterative rather than linear. Each stage exists to reduce a specific uncertainty before greater investment and patient exposure. A successful medicine requires not only biological activity, but also appropriate exposure, safety, manufacturing quality, meaningful clinical benefit, and an acceptable overall benefit–risk profile.

12.15 Practice questions

1. Why is drug development not a strictly linear process?

Answer: New evidence can require changes to the target hypothesis, therapeutic design, dose, biomarker strategy, patient population, manufacturing approach, or trial design. A programme may return to an earlier stage or be terminated if the evidence is not sufficiently supportive.

2. What is an unmet medical need?

Answer: It is a patient problem not adequately addressed by existing options, for example because treatments are ineffective, too toxic, insufficiently durable, difficult to deliver, or unavailable for a particular patient subgroup.

3. Why is a disease hypothesis needed before target selection?

Answer: It defines the patient problem, relevant disease biology, and proposed mechanism through which an intervention could improve outcomes. Without a disease hypothesis, target selection may lack clinical relevance.

4. What makes an assay useful in a drug-discovery programme?

Answer: It should be reproducible, interpretable, sufficiently scalable, biologically relevant, and connected to the programme's mechanism and downstream decisions. Different assays are needed to answer different questions about binding, cellular activity, pharmacology, and clinical translation.

5. What is a hit?

Answer: A hit is an initial molecule or biological candidate with repeatable evidence of desired activity in a relevant assay. It must be confirmed and assessed for artefacts, quality, mechanism, and tractability before major optimisation begins.

6. Why is lead optimisation a multi-objective problem?

Answer: A candidate must balance target activity, selectivity, solubility, permeability, PK, tissue distribution, safety, synthesis, formulation, and manufacturability. Improving one property can worsen another, so the aim is a suitable overall profile rather than the best value in one assay.

7. What is the purpose of candidate selection?

Answer: Candidate selection determines whether a sufficiently mature therapeutic candidate has an acceptable overall package to justify formal preclinical development. The decision integrates mechanism, pharmacology, PK, safety, manufacturability, formulation, biomarkers, and the intended clinical plan.

8. What are the main goals of preclinical development?

Answer: Preclinical development assesses whether the candidate is pharmacologically credible, sufficiently safe, manufacturable to an appropriate quality standard, and suitable for carefully controlled testing in humans. It can include PK/PD, toxicology, safety pharmacology, tissue distribution, formulation, and CMC work.

9. What are the main goals of Phase I trials?

Answer: Phase I trials assess safety, tolerability, dose and schedule, PK, PD, target engagement, and early biological or clinical activity in humans. The goal is to identify a dose and schedule with a credible balance of exposure, pharmacological activity, and safety.

10. What is the central question in Phase II development?

Answer: The central question is whether there is enough evidence of meaningful and sufficiently durable activity, acceptable safety, and an appropriate patient-selection strategy to justify confirmatory trials.

11. Why is a statistically significant Phase III result not automatically clinically meaningful?

Answer: Clinical interpretation also depends on the magnitude and durability of benefit, safety burden, quality-of-life effects, feasibility of treatment, comparator, and available alternatives. Statistical significance alone does not establish a favourable benefit–risk profile.

12. Why does evidence generation continue after regulatory approval?

Answer: Real-world use can provide information about effectiveness, rare safety events, adherence, outcomes in broader populations, treatment sequencing, and opportunities for new indications, combinations, formulations, or improved dosing.

13 Clinical development and clinical evidence

13.1 Why preclinical success is not enough

Preclinical research can establish a plausible therapeutic hypothesis. It can show that an intervention binds a target, changes a pathway, affects cells, reaches tissues in animal models, or has an acceptable preliminary safety profile.

None of these findings alone establishes that a treatment will help patients.

Human disease is more heterogeneous than laboratory models. Patients differ in disease stage, tumour biology, prior treatments, organ function, age, comorbidities, immune status, concomitant medicines, and ability to tolerate or adhere to treatment. Tumours can also evolve during therapy, and the standard of care may change while a programme is being developed.

Clinical development therefore asks:

Does this intervention provide clinically meaningful benefit,
at an acceptable risk,
in a clearly defined patient population?

The causal chain remains important:

Therapeutic intervention
→ exposure in patients
→ target engagement and biological effect
→ change in disease course
→ patient-relevant benefit and toxicity

A programme can fail at any point in this chain. A drug may engage its target but not affect disease sufficiently. It may affect disease biology but require an intolerable dose. It may benefit a biomarker-defined subgroup but not the broader enrolled population. It may improve an imaging-based measure without improving symptoms, quality of life, or survival.

Clinical trials are designed to distinguish these possibilities as reliably as possible.

13.2 Indication and patient population

An *indication* is the specific disease setting for which a medicine is developed, studied, or approved. It is more precise than a disease name alone.

For oncology, an indication may specify:

Cancer type
+
Disease stage and clinical setting
+
Biomarker status
+
Resectable or unresectable disease
+
Treatment line and prior therapies
+
Combination partner, if applicable

For example, “HER2-positive metastatic breast cancer after prior therapy” is much more specific than simply “breast cancer.” Likewise, treatment after complete surgical removal of high-risk melanoma is biologically and clinically different from treatment of metastatic melanoma with measurable disease.

The *patient population* is the group of people to whom trial results are intended to apply. Defining this group carefully is essential because expected benefit and risk can differ substantially across disease stages, biomarker groups, prior treatments, disease burden, and patient characteristics.

A clinically useful question is:

Which patients are expected to have a favourable benefit-risk balance relative to the available alternatives?

The answer may depend on whether treatment is intended to cure, reduce recurrence risk, prolong survival, control symptoms, delay progression, or improve quality of life. The acceptable level of toxicity can differ substantially between these goals.

13.3 Inclusion and exclusion criteria

Inclusion criteria specify who can enter a clinical trial. *Exclusion criteria* specify who cannot.

Criteria may relate to age, diagnosis, disease stage, biomarker status, organ function, prior therapy, performance status, pregnancy, concurrent illness, infection risk, central nervous-system involvement, or use of interacting medicines.

These criteria serve two important purposes:

Protect patient safety
+
Make trial results interpretable

For example, excluding patients with severe organ impairment may reduce the risk that an investigational therapy causes avoidable toxicity. Requiring a particular biomarker may make it possible to test a treatment hypothesis in the population most likely to benefit.

However, restrictive eligibility criteria can limit *generalisability*, also called *external validity*: the degree to which trial results apply to broader real-world populations.

Trial participants may not fully represent older patients, patients with multiple chronic conditions, patients with impaired kidney or liver function, people receiving different concurrent medicines, or populations that have historically been underrepresented in research.

There is therefore a balance:

More restrictive eligibility criteria
→ may improve safety and interpretability
→ may reduce applicability to broader clinical practice
Broader eligibility criteria
→ may improve representativeness
→ may increase heterogeneity and operational complexity

13.4 Randomisation, control groups, and blinding

Clinical outcomes can change for reasons unrelated to the investigational treatment. Disease may improve or worsen naturally, patients may differ in prognosis, assessments may be influenced by expectation, and supportive care may vary between groups.

Well-designed comparative trials use methods intended to reduce these sources of bias.

Randomisation means assigning participants to treatment groups by a process based on chance. If performed appropriately, it helps make the groups comparable on average with respect to both measured and unmeasured baseline characteristics.

Randomisation

→ treatment groups should differ systematically
mainly in the assigned intervention

Observed outcome difference

→ can more credibly be attributed to treatment

A *control group* provides the comparison needed to interpret the investigational treatment's effect. Depending on the clinical setting, the control may receive standard of care, an active comparator, placebo plus standard care, or another ethically justified strategy.

Blinding, sometimes called *masking*, means that one or more parties do not know which treatment a participant received. Blinding can reduce bias in reporting symptoms, clinical decisions, imaging assessments, and outcome evaluation.

In some trials, particularly those involving surgery, complex cell therapies, or treatments with distinctive toxicities, full blinding may be impractical. Other methods, such as independent blinded review of imaging or predefined objective endpoints, can then help reduce bias.

Randomisation, control groups, and blinding do not guarantee that a trial is correct. However, they strengthen the ability to distinguish a real treatment effect from confounding, bias, chance variation, or differences in patient prognosis.

13.5 Biomarker stratification

Biomarker stratification means organising enrolment, randomisation, analysis, or interpretation according to biomarker-defined subgroups.

For example, a trial may enrol only patients whose tumours carry a specific mutation or express HER2 at a predefined level. Alternatively, it may enrol a broader population while prespecifying analyses in biomarker-defined subgroups.

A biomarker can play different roles:

Diagnostic biomarker

→ helps identify or classify disease

Prognostic biomarker

→ provides information about likely outcome
regardless of treatment

Predictive biomarker

→ identifies patients more or less likely to benefit
from a particular treatment

A biomarker strategy is strongest when it is prospective:

```
Biomarker hypothesis
+
Validated or appropriate test method
+
Predefined threshold
+
Patient-selection or stratification plan
+
Statistical analysis plan
All defined before analysing treatment results
```

This matters because retrospective subgroup analyses can produce apparently compelling findings by chance, especially when many biomarkers, cut-points, endpoints, or patient subgroups are examined.

Exploratory analyses remain valuable for hypothesis generation. However, an exploratory association should not automatically be interpreted as proof that the biomarker predicts treatment benefit. Prospective confirmation is generally needed before using a biomarker to guide clinical decisions or patient selection.

13.6 Standard of care and comparator

The *standard of care* is the treatment approach considered appropriate for a particular clinical situation based on available evidence, guidelines, clinical practice, and patient-specific factors.

A *comparator* is the treatment, placebo, or control strategy against which an investigational therapy is evaluated in a trial.

Clinical benefit is inherently relative:

```
Improvement versus placebo
≠ necessarily sufficient when an effective standard treatment exists
Improvement versus standard of care
→ must be interpreted alongside safety, durability,
quality of life, and patient need
```

The comparator determines the question that a trial can answer. For example, a new treatment may appear active in a single-arm study, but a randomised comparison may show that outcomes are similar to those achieved with existing care.

Comparator choice is particularly important when standard treatment changes rapidly. A trial designed against an older treatment may remain scientifically informative, but its relevance to current clinical practice can become more limited.

A useful question when reading a clinical trial is:

```
What would the participants have received
if they had not entered this trial,
and is the control group an appropriate representation of that care?
```

13.7 Endpoints and estimands

An *endpoint* is a predefined outcome used to assess treatment in a clinical trial. Endpoints can be clinical, biological, imaging-based, or patient-reported.

The most appropriate endpoint depends on the disease, treatment intent, patient population, comparator, and clinical question.

A *primary endpoint* is the main outcome on which a trial is designed and statistically powered. *Secondary endpoints* provide additional predefined evidence. *Exploratory endpoints* can generate hypotheses but generally require more cautious interpretation.

The key principle is:

Endpoint chosen before trial
→ determines what question the trial is designed to answer

A modern trial protocol should also define the *estimand*: the precise treatment effect that the study aims to estimate. This includes the target population, treatment comparison, outcome, timing of assessment, and approach to events that complicate interpretation.

Such events can include treatment discontinuation, use of a new anti-cancer therapy, crossover to another trial treatment, death, missing assessments, or interruption of treatment because of toxicity.

For example, a study may ask:

What is the effect of assigning patients to Treatment A rather than Treatment B, regardless of whether some patients later discontinue treatment or receive subsequent therapy?

Or it may ask a different question:

What would the effect have been if patients had remained on their assigned treatment without receiving an additional anti-cancer therapy?

These are not identical questions. Clear endpoint and estimand definitions help ensure that the reported result corresponds to a clinically meaningful and interpretable treatment effect.

13.8 Surrogate endpoints

A *surrogate endpoint* is a measurement expected to predict clinical benefit but is not itself necessarily the ultimate outcome that matters directly to patients.

For example, tumour shrinkage can provide evidence of anti-tumour activity, but it does not always imply improved survival, improved symptoms, reduced treatment burden, or improved quality of life.

A useful distinction is:

Surrogate endpoint
→ expected indicator of clinical benefit
Clinical outcome
→ direct patient-relevant outcome, such as survival, symptoms, function, or quality of life

Surrogate endpoints can support faster development, particularly in serious diseases with high unmet need. However, their interpretation depends on how reliably the surrogate predicts the outcome that matters to patients in the relevant disease, treatment setting, and therapeutic class.

A treatment can improve a surrogate endpoint without producing the expected clinical benefit. It can also create toxicity that offsets the value of a favourable surrogate result.

The appropriate question is therefore not simply:

Did the treatment improve the surrogate?

It is:

How strongly does improvement in this surrogate support a meaningful net benefit for patients in this setting?

13.9 Objective response rate and duration of response

In oncology, *objective response rate*, or *ORR*, is the proportion of patients whose tumours show a predefined amount of shrinkage according to standard response criteria.

A response may be:

- *Complete response*: Detectable disease disappears according to the relevant assessment criteria.
- *Partial response*: Tumour burden decreases substantially but incompletely according to the criteria.

The *duration of response*, or *DoR*, measures how long a response lasts among patients who achieve one. ORR and DoR should often be interpreted together.

High ORR with short DoR
→ substantial initial tumour shrinkage
→ but limited durability
Lower ORR with long DoR
→ fewer patients respond
→ but responses may be sustained in those who do

Tumour response can be clinically meaningful, especially when it relieves symptoms, reduces immediate disease burden, enables surgery, or provides a signal of substantial anti-tumour activity in a setting with few alternatives.

However, ORR alone does not establish how long the benefit lasts, whether survival improves, whether quality of life improves, or whether treatment toxicity is acceptable. Response assessment can also depend on imaging timing, assessment criteria, and whether scans are reviewed by investigators or independent reviewers.

13.10 Progression-free survival

Progression-free survival, or *PFS*, is the time from a defined starting point, often randomisation, until disease progression or death, according to the trial protocol.

PFS is often useful because it can be measured earlier than overall survival and may be less directly influenced by therapies received after disease progression.

However, interpretation requires care. PFS can be affected by imaging schedules, assessment method, response criteria, timing of scans, missed assessments, informative dropout, and the definition of progression.

An improvement in PFS may or may not translate into improvement in overall survival or quality of life. Its clinical importance depends on the magnitude and durability of the effect, toxicity, treatment burden, patient-reported outcomes, and the treatment setting.

For example:

Longer PFS
+
maintained quality of life
+
acceptable toxicity
→ may represent meaningful patient benefit
Longer PFS
+
substantial toxicity or reduced quality of life
→ benefit-risk may be less favourable

13.11 Overall survival

Overall survival, or *OS*, is the time from a defined starting point until death from any cause.

OS is often considered a highly clinically meaningful endpoint because it directly captures whether patients live longer. It is also relatively objective compared with endpoints that depend on imaging interpretation or subjective assessment.

However, OS may require long follow-up. Its interpretation can also be complicated by treatments received after progression, crossover between trial groups, changing standards of care, competing causes of death, and incomplete follow-up.

OS is important, but it is not the only outcome that matters to patients. In some settings, improved symptoms, preserved function, better quality of life, fewer hospitalisations, delayed recurrence, or avoidance of toxic treatment can also be highly meaningful.

The central point is:

Overall survival is highly important,
but clinical value should be assessed using the full benefit-risk picture.

13.12 Recurrence-free survival and distant-metastasis-free survival

Recurrence-free survival, or *RFS*, is commonly used after curative-intent treatment such as surgery. It measures time until disease recurrence or death according to the trial definition.

Distant-metastasis-free survival, or *DMFS*, measures time until distant metastatic recurrence or death according to a predefined protocol.

These endpoints are particularly relevant in *adjuvant* treatment settings. *Adjuvant therapy* is treatment administered after definitive local treatment, such as surgery, with the aim of reducing the chance that residual microscopic disease later causes recurrence.

The clinical logic is:

Curative-intent treatment removes visible disease
→ microscopic residual disease may remain
→ adjuvant therapy aims to reduce recurrence risk
→ RFS and DMFS assess recurrence-related outcomes

RFS and DMFS can be highly meaningful because recurrence, particularly distant metastatic recurrence, can substantially worsen prognosis and increase treatment burden. However, their interpretation should still consider toxicity, quality of life, duration of follow-up, subsequent treatment options, and whether the observed delay in recurrence is expected to translate into a durable patient-relevant advantage.

13.13 Patient-reported outcomes and quality of life

A treatment can affect patients in ways that are not captured adequately by imaging, laboratory measures, or survival alone. *Patient-reported outcomes*, often abbreviated *PROs*, are outcomes reported directly by patients without interpretation by a clinician or another observer.

PROs can assess symptoms, physical function, emotional wellbeing, treatment burden, ability to perform daily activities, and health-related quality of life.

Examples of clinically important questions include:

Does treatment reduce pain, fatigue, breathlessness, or nausea?
Does it preserve a patient's ability to work, move, or care for themselves?
Does longer disease control occur with maintained quality of life?
Does treatment toxicity outweigh the benefit experienced by patients?

Patient-reported outcomes are particularly important when treatments have similar survival outcomes but differ in toxicity, route of administration, monitoring burden, or impact on daily life.

PRO data require careful collection and interpretation. Missing questionnaires, unblinded treatment assignment, disease progression, and treatment discontinuation can complicate conclusions. Nevertheless, patient experience is an essential part of evaluating whether a statistically significant treatment effect is also clinically meaningful.

13.14 Adverse events and safety

An *adverse event*, or *AE*, is an unwanted medical occurrence in a trial participant after receiving treatment. An AE does not automatically mean that treatment caused the event.

A *treatment-related adverse event* is judged by investigators to be related, or possibly related, to treatment. Attribution can be difficult because participants may have underlying disease, prior treatment effects, other illnesses, and concurrent medications that could contribute to an event.

Safety evaluation considers frequency, severity, timing, reversibility, management, relationship to dose and exposure, impact on daily life, and effect on treatment continuation.

In oncology, adverse events are commonly graded using standard criteria. Higher grades generally represent more severe events, although the clinical importance of an event also depends on its type, duration, reversibility, and patient context.

Safety evaluation should consider more than the proportion of patients with any adverse event:

What adverse events occurred?
How severe, frequent, and durable were they?
Were they manageable or reversible?
Did they lead to dose reduction, treatment interruption, hospitalisation, or permanent discontinuation?
Did they affect quality of life or create delayed harm?

The acceptable safety profile depends strongly on context:

Advanced aggressive disease with few alternatives
→ higher toxicity may be acceptable
Earlier-stage disease with curative options
→ lower tolerance for severe, permanent, or delayed toxicity

13.15 Dose selection and treatment burden

Dose selection is discussed in Chapters 5 and 12. In clinical development, the main point is that the highest tolerable dose is not automatically the best dose.

Historically, especially in oncology, early development often focused on identifying the *maximum tolerated dose*, or *MTD*. The MTD is an estimate of the highest dose associated with an acceptable frequency of serious dose-limiting toxicities under the trial rules.

For many modern therapies, particularly targeted therapies, biologics, and immunotherapies, selecting a dose requires a broader evidence base:

```
Safety and tolerability
+
PK and exposure
+
PD and target engagement
+
Preliminary efficacy
+
Long-term treatment feasibility
+
Patient-reported outcomes
+
Overall benefit-risk
```

A lower dose may provide similar target engagement and clinical benefit with fewer adverse effects, less need for dose interruption, better adherence, and improved quality of life.

Treatment burden also matters. Route of administration, dosing frequency, infusion time, monitoring requirements, travel to treatment centres, laboratory testing, supportive medicines, and hospitalisations can all affect whether a therapy is practical and acceptable for patients.

13.16 Benefit-risk

Benefit-risk is the overall balance between a medicine's expected clinical benefits and harms.

It is not a fixed property of a molecule. It depends on disease severity, alternative treatments, patient population, dose, route of administration, treatment duration, monitoring requirements, treatment burden, and individual patient preferences.

For example:

```
High toxicity may be acceptable
→ aggressive, otherwise lethal disease with few alternatives
The same toxicity may be unacceptable
→ lower-risk disease with effective alternatives
```

Benefit-risk should incorporate the complete evidence package:

```
Magnitude and durability of efficacy
+
Survival and recurrence-related outcomes
+
Symptoms, function, and quality of life
+
Frequency and severity of toxicity
+
Reversibility and manageability of harm
+
Treatment burden and feasibility
+
Available alternatives
```

A statistically significant efficacy result can still have an unfavourable benefit–risk profile if the benefit is small, short-lived, uncertain, or offset by substantial toxicity. Conversely, a treatment with meaningful toxicity may still have a favourable profile in a serious disease with few alternatives if the clinical benefit is substantial and durable.

13.17 Why context matters

A mechanistically persuasive preclinical result can fail clinically for many reasons:

- The target is not sufficiently important in the enrolled patients.
- The treatment does not achieve adequate tissue exposure or target engagement.
- The required dose is not tolerable.
- The chosen biomarker is inadequate.
- Resistance emerges rapidly.
- The comparator is stronger than expected.
- The endpoint does not capture meaningful benefit.
- Patient heterogeneity dilutes a real effect confined to a subgroup.
- The disease model failed to reflect human biology.
- The treatment burden or safety profile prevents practical use.

This is why clinical development requires careful trial design rather than merely more patients or more data.

The central clinical question remains:

```
For which patients,
compared with what standard of care,
at which dose and schedule,
does the treatment create clinically meaningful benefit
that justifies its risks?
```

13.18 Key takeaways

Clinical evidence is the final test of whether a therapeutic hypothesis provides meaningful value to patients. Several distinctions are especially useful to retain:

-
- An indication defines a specific disease setting and patient population; it is not simply the broad disease name.
 - Inclusion and exclusion criteria protect participants and improve interpretability, but may limit generalisability to real-world patients.
 - Randomisation, control groups, and blinding help reduce bias and support causal interpretation of treatment differences.
 - Biomarker strategies are strongest when the test, threshold, hypothesis, and analysis plan are prospectively specified.
 - The comparator determines the clinical question a trial can answer and must be interpreted in the context of current standard of care.
 - Primary endpoints are predefined outcomes that a trial is designed to assess; secondary and exploratory endpoints provide additional evidence with different levels of certainty.
 - Surrogate endpoints, response rates, and progression-related endpoints can be valuable, but they are not automatically equivalent to longer survival, improved symptoms, or better quality of life.
 - Patient-reported outcomes help determine whether measured treatment effects correspond to benefits that patients can experience.
 - Dose selection should balance exposure, target engagement, efficacy, long-term tolerability, and treatment burden rather than relying only on the maximum tolerated dose.
 - Benefit–risk is context-specific and depends on the magnitude and durability of benefit, harms, alternatives, patient need, and feasibility of treatment.

13.19 Super-short summary

Preclinical evidence supports a therapeutic hypothesis, but clinical trials determine whether a treatment provides meaningful patient benefit at an acceptable risk.

Clinical evidence depends on the precise indication, patient population, biomarker strategy, control group, endpoint, dose, schedule, safety profile, patient-reported outcomes, and standard-of-care context. The central task is to establish not merely that a treatment has biological activity, but that it improves outcomes that matter to patients relative to available alternatives.

13.20 Practice questions

1. Why is preclinical success not enough to establish clinical benefit?

Answer: Human disease is more heterogeneous than laboratory models, and patient outcomes depend on exposure, tolerability, prior treatments, comorbidities, immune status, resistance, adherence, treatment burden, and standard-of-care context. A positive result in a laboratory system does not prove meaningful patient benefit.

2. What is an indication?

Answer: An indication is the specific disease setting for which a medicine is developed, studied, or approved. It can specify cancer type, disease stage, biomarker status, treatment line, prior therapies, and treatment combination or route.

3. Why are inclusion and exclusion criteria necessary?

Answer: They protect patient safety and help make trial results interpretable. However, highly restrictive criteria can reduce generalisability because enrolled participants may not fully represent patients treated in routine clinical practice.

4. Why are randomisation and a control group valuable in a clinical trial?

Answer: Randomisation helps ensure that treatment groups are comparable on average, while a control group shows what outcomes would be expected without the investigational treatment. Together, they make it more credible that an observed difference is caused by the assigned treatment rather than bias, prognosis differences, or natural disease variation.

5. What is biomarker stratification?

Answer: Biomarker stratification is the organisation of enrolment, randomisation, analysis, or interpretation according to biomarker-defined subgroups. It can help identify patients with meaningfully different expected benefit or risk.

6. Why is prospective biomarker planning stronger than retrospective subgroup analysis?

Answer: In prospective planning, the biomarker test, threshold, hypothesis, patient-selection strategy, and analysis plan are predefined. This reduces the risk that an apparent subgroup effect is a chance finding created by examining many possible subgroups after results are known.

7. What is the difference between standard of care and a comparator?

Answer: Standard of care is the treatment approach regarded as appropriate in clinical practice. A comparator is the specific treatment, placebo, or control strategy against which an investigational therapy is evaluated in a trial.

8. What is the difference between a primary, secondary, and exploratory endpoint?

Answer: The primary endpoint is the main predefined outcome used to design and power the trial. Secondary endpoints provide additional predefined evidence. Exploratory endpoints can generate hypotheses but generally require more cautious interpretation and often need confirmation.

9. What is a surrogate endpoint?

Answer: A surrogate endpoint is a measurement expected to predict patient benefit but is not itself necessarily a direct patient-relevant outcome. For example, tumour shrinkage may indicate anti-tumour activity but does not automatically prove longer survival, symptom improvement, or better quality of life.

10. Why are ORR and duration of response often interpreted together?

Answer: ORR indicates how many patients experience predefined tumour shrinkage, while duration of response indicates how long those responses last. A high response rate with very short responses may have different clinical value from a lower response rate with durable responses.

11. What is the difference between PFS and OS?

Answer: PFS measures time until disease progression or death, whereas OS measures time until death from any cause. PFS is often available earlier, while OS directly captures survival but may require longer follow-up and can be influenced by subsequent treatments.

12. Why are patient-reported outcomes important?

Answer: Patient-reported outcomes capture symptoms, function, treatment burden, and quality of life directly from patients. They help determine whether an observed treatment effect corresponds to an outcome that patients experience as meaningful.

13. Why is the maximum tolerated dose not necessarily the best clinical dose?

Answer: The best dose should balance safety, PK, PD, target engagement, efficacy, long-term tolerability, quality of life, and treatment burden. A lower dose may provide similar benefit with fewer adverse effects and fewer treatment interruptions.

14. **What is benefit–risk?**

Answer: Benefit–risk is the overall balance between a treatment’s expected clinical benefits and harms in a specific disease, patient population, dose, schedule, and treatment context. It depends on efficacy, durability, toxicity, quality of life, alternatives, and patient need.

15. **What is the central question of clinical development?**

Answer: For which patients, compared with what standard of care, at what dose and schedule, does the treatment provide clinically meaningful benefit that justifies its risks?

Glossary

Core biology

Amino acid A small chemical building block from which proteins are made. The sequence of amino acids influences a protein's structure and function.

Antigen A molecular feature that can be recognised by the immune system. For T cells, this is commonly a peptide displayed by an MHC molecule.

Antibody A protein used by the immune system, and adapted as a therapeutic modality, to bind a particular molecular target or *epitope*.

Antigen-presenting cell (APC) An immune cell that processes protein material and displays peptides on MHC molecules to help activate T cells. Dendritic cells are especially important APCs.

Cell The basic living unit of the body. Cells use energy, produce molecules, communicate, sense their environment, divide, repair damage, and die when appropriate.

Cytokine A signalling protein used by immune cells and other cells to influence inflammation, cell growth, movement, survival, and immune activity.

DNA Deoxyribonucleic acid. The molecule that stores genetic information.

Enzyme Usually a protein that accelerates a chemical reaction without being consumed by that reaction.

Gene A region of DNA that contains instructions for producing a functional product, commonly a protein or functional RNA molecule.

Genome The complete set of genetic material of an organism or cell.

Kinase An enzyme that commonly transfers a phosphate group from ATP to another molecule, often another protein.

Ligand A molecule that binds a biological target, such as a receptor. A ligand can be a natural signalling molecule, metabolite, or therapeutic drug.

mRNA Messenger RNA. A temporary RNA copy of genetic information that can be translated by ribosomes into protein.

Mutation A change in DNA sequence. Mutations can have little effect, alter protein structure or expression, or contribute to disease.

Pathway A connected set of biological events through which cells process a signal and change behaviour.

Phosphorylation Addition of a phosphate group to a molecule, commonly a protein. It can alter protein activity, location, stability, or interactions.

Protein A chain of amino acids folded into a three-dimensional structure. Proteins perform much of the molecular work in cells.

Receptor A protein that detects and responds to a particular signal, often by binding a ligand.

RNA Ribonucleic acid. A family of molecules involved in using, regulating, or processing genetic information.

Substrate The molecule on which an enzyme acts during a chemical reaction.

Target A biological entity that a therapy is intended to alter in order to create therapeutic benefit. Targets are often proteins, but can also be RNAs, cells, or broader pathways.

Cancer biology and biomarkers

Biomarker A measurable biological characteristic relevant to disease or treatment. Biomarkers can help diagnose disease, estimate prognosis, select therapy, monitor response, or detect toxicity.

Clonal mutation A mutation present in most or all cancer cells descended from a common tumour ancestor.

Driver mutation A genetic alteration that contributes causally to tumour development, maintenance, or progression.

Gene amplification An increase in the number of copies of a gene in a cell. It can result in unusually high production of the corresponding protein.

Heterogeneity Meaningful biological variation. In cancer, this can refer to differences between tumour cells within one lesion, between lesions, or between patients.

Metastasis Spread of cancer cells from the original tumour site to distant organs or tissues.

Oncogene A gene that can promote cancer when abnormally activated, overexpressed, amplified, or rearranged.

Oncogene addiction Strong tumour dependence on a particular oncogenic pathway or signal for continued growth or survival.

Passenger mutation A mutation present in a tumour that does not contribute substantially to the cancer phenotype.

Predictive biomarker A biomarker that identifies patients more or less likely to benefit from a particular treatment.

Prognostic biomarker A biomarker associated with likely patient outcome regardless of which treatment is received.

Resistance Reduced treatment sensitivity that is present before treatment or develops during treatment. Resistance can arise through target changes, bypass pathways, tumour evolution, or altered drug exposure.

Subclonal mutation A mutation present in only a subset of tumour cells.

Tumour microenvironment The cells, blood vessels, immune cells, signalling molecules, extracellular material, and physical conditions surrounding tumour cells.

Tumour suppressor gene A gene that normally restrains inappropriate cell growth, supports DNA repair, or promotes elimination of damaged cells.

Chemistry and small-molecule discovery

ADMET A common abbreviation for absorption, distribution, metabolism, excretion, and toxicity.

Affinity The strength with which a molecule binds its target under defined experimental conditions.

Allosteric site A binding site distinct from a protein's main functional site. Binding at an allosteric site can alter protein function indirectly.

Analogue A compound structurally related to another compound, usually made to investigate how a chemical change affects activity or other properties.

Chemical series A group of related compounds that share a scaffold or common design logic.

Covalent inhibitor A molecule designed to form a covalent bond with its target protein. Depending on its chemistry, this interaction may be irreversible or slowly reversible.

Cryptic pocket A binding pocket that is not obvious in one protein structure but becomes accessible in another protein conformation.

Drug candidate A therapeutic compound or product selected for formal preclinical development before clinical testing.

Druggability The practical possibility of modulating a target safely and effectively with a therapeutic modality.

Functional group A chemically meaningful arrangement of atoms that influences a molecule's properties, including binding, solubility, reactivity, and metabolism.

Hit An initial compound or biological candidate with reproducible evidence of desired activity.

Lead A more mature compound or chemical series with sufficient promise to justify systematic optimisation.

Lead optimisation The iterative improvement of a chemical series to balance potency, selectivity, PK, safety, formulation, and manufacturability.

Ligandability Whether a target contains a physical site that can bind a ligand with useful affinity and selectivity.

Lipophilicity The tendency of a molecule to partition into non-polar environments rather than water.

pKa A property related to the pH at which an ionisable chemical group is approximately half protonated and half deprotonated.

Potency The concentration or dose needed to produce a specified effect in a particular assay.

Scaffold The central chemical framework of a molecule.

Selectivity The degree to which a therapy acts on its intended target rather than other biological targets.

Solubility The extent to which a compound dissolves in a particular solvent under defined conditions.

Structure–activity relationship (SAR) An evidence-based understanding of how chemical modifications change activity and other measured properties.

Stereochemistry The three-dimensional arrangement of atoms in a molecule. Molecules with the same chemical formula and connectivity can have different stereochemistry and different biological effects.

Pharmacology, PK, and PD

Absorption Entry of a drug into systemic circulation after administration.

Agonist A molecule that binds and activates a receptor.

Antagonist A molecule that prevents or reduces receptor activation.

Area under the concentration–time curve (AUC) A measure of total systemic drug exposure over a defined time period.

Bioavailability The fraction of an administered dose that reaches systemic circulation in active form.

Clearance The efficiency with which the body removes a drug from systemic circulation.

$C_{\max}$ The highest observed drug concentration after dosing.

Distribution Movement of a drug from blood into tissues and body compartments.

Dose The amount of a drug administered.

EC50 The concentration producing 50% of a specified maximal effect in a particular assay.

Efficacy The maximum effect a treatment can achieve in a particular biological system.

Exposure Drug concentration in the body over time.

Half-life The approximate time required for drug concentration to decrease by half during an appropriate elimination phase.

IC50 The concentration producing 50% inhibition in a particular assay.

K_d Dissociation constant. A measure of binding affinity under defined conditions; lower values generally indicate tighter binding.

K_i An inhibition-related parameter estimated under a particular mechanistic model and assay conditions.

Metabolism Chemical modification of a drug by the body, often involving enzymes in the liver or other tissues.

Off-target effect A biological effect caused by activity at an unintended target.

Pharmacodynamics (PD) What a drug does to the body, including target engagement, pathway effects, efficacy, and toxicity.

Pharmacokinetics (PK) What the body does to a drug, including absorption, distribution, metabolism, and excretion.

Target engagement Evidence that a therapy reaches and interacts with its intended target in the relevant biological setting.

Therapeutic window The range of exposure expected to provide useful benefit without unacceptable toxicity.

Antibodies, ADCs, and immune therapies

Antibody–drug conjugate (ADC) A therapy consisting of a targeting antibody, linker, and potent payload intended for preferential delivery to antigen-expressing cells.

Antibody-dependent cellular cytotoxicity (ADCC) Immune-mediated killing in which immune effector cells recognise the Fc region of antibodies attached to a target cell and attack that cell.

Bystander effect Local exposure of nearby cells to payload released from an ADC-targeted cell.

CAR-T cell A T cell engineered to express a chimeric antigen receptor that recognises a chosen cell-surface antigen.

Chimeric antigen receptor (CAR) A synthetic receptor that combines an extracellular antigen-recognition region with intracellular T-cell signalling domains.

CD4 T cell A T cell that commonly recognises peptides presented by MHC class II and helps coordinate immune responses.

CD8 T cell A T cell that commonly recognises peptides presented by MHC class I and can directly kill recognised cells.

Checkpoint inhibitor A therapy that blocks an inhibitory immune pathway, with the aim of strengthening anti-tumour immune responses.

Cytokine release syndrome (CRS) A systemic inflammatory toxicity that can occur after strong immune-cell activation, including after some CAR-T therapies.

Drug-to-antibody ratio (DAR) The average number of payload molecules attached to each antibody molecule in an ADC.

Epitope The specific molecular region of a target recognised by an antibody or immune receptor.

Fab region The antigen-binding region of an antibody.

Fc region The antibody region that can interact with immune-system components and influence antibody persistence and immune effector functions.

Immune-related adverse event (irAE) An adverse event caused by excessive or misdirected immune activity after immunotherapy.

Major histocompatibility complex (MHC) Cell-surface molecules that display peptides for inspection by T cells.

MHC class I MHC molecules present on most nucleated cells that commonly present peptides to CD8 T cells.

MHC class II MHC molecules mainly present on specialised antigen-presenting cells that commonly present peptides to CD4 T cells.

Monoclonal antibody (mAb) A population of identical antibody molecules designed to recognise the same epitope.

Neoantigen A mutation-derived peptide found in tumour cells but absent from normal cells that may be presented by MHC and recognised by T cells.

On-target, off-tumour toxicity Harm caused when a therapy correctly recognises its intended target, but that target is also expressed on healthy tissue.

PD-1 An inhibitory receptor expressed on activated T cells and other immune cells.

PD-L1 and PD-L2 Ligands that can bind PD-1 and reduce aspects of T-cell activity.

Personalised neoantigen vaccine A vaccine designed using tumour-specific mutations identified in an individual patient's cancer.

T-cell receptor (TCR) A receptor on T cells that recognises a peptide presented by an MHC molecule.

Clinical development and evidence

Adjuvant therapy Treatment given after definitive local treatment, such as surgery, to reduce the risk of disease recurrence from residual microscopic disease.

Adverse event (AE) An unwanted medical occurrence in a clinical-trial participant after treatment. It does not automatically mean treatment caused the event.

Benefit–risk The overall balance between a treatment's expected benefits and harms in a defined clinical context.

Biomarker stratification Organisation of trial enrolment, randomisation, analysis, or interpretation according to biomarker-defined subgroups.

Blinding Keeping treatment assignment unknown to one or more parties in a trial to reduce bias.

Comparator The treatment, placebo, or control strategy against which an investigational therapy is evaluated.

Complete response (CR) Disappearance of detectable disease according to the relevant response criteria.

Dose-limiting toxicity (DLT) A treatment-related adverse effect severe enough to limit further dose escalation or influence dose selection.

Distant-metastasis-free survival (DMFS) Time until distant metastatic recurrence or death according to the trial definition.

Duration of response (DoR) The time for which a tumour response persists among patients who achieve a response.

Endpoint A predefined outcome used to assess a treatment in a clinical trial.

Estimand The precise treatment effect that a trial aims to estimate, including the population, treatment comparison, outcome, timing, and approach to events such as treatment discontinuation.

Generalisation or external validity The extent to which trial results apply to patients outside the study population.

Inclusion and exclusion criteria Predefined requirements that determine who can and cannot enter a clinical trial.

Indication The specific disease setting for which a medicine is studied, developed, or approved.

Maximum tolerated dose (MTD) An estimate of the highest dose associated with an acceptable frequency of serious toxicities under a trial's dose-escalation rules.

Objective response rate (ORR) The proportion of patients whose tumours show predefined shrinkage according to response criteria.

Overall survival (OS) Time from a defined starting point until death from any cause.

Partial response (PR) Substantial but incomplete tumour shrinkage according to the relevant response criteria.

Patient-reported outcome (PRO) Information on symptoms, function, wellbeing, or treatment burden reported directly by the patient.

Phase I trial An early clinical trial that commonly examines safety, tolerability, PK, PD, dose, schedule, target engagement, and early activity in humans.

Phase II trial A clinical trial that investigates activity, safety, dose, biomarkers, and patient-selection strategy in a defined population.

Phase III trial A generally confirmatory trial designed to establish benefit–risk relative to an appropriate comparator.

Progression-free survival (PFS) Time until disease progression or death according to the trial protocol.

Randomisation Assignment of trial participants to treatment groups by chance in order to improve comparability between groups.

Recurrence-free survival (RFS) Time until disease recurrence or death after curative-intent treatment.

Standard of care Treatment considered appropriate for a particular clinical situation based on available evidence and clinical practice.

Surrogate endpoint A measurement expected to predict clinical benefit but not necessarily itself a direct patient-relevant outcome.

Part IV Various things I wanted to learn more about

14 [26-08-28] How mRNA vaccines work: the example of COVID-19 vaccines

mRNA vaccines use messenger RNA to give cells temporary instructions for making a selected protein antigen. The immune system then learns to recognise that antigen, creating immune memory that can help it respond more quickly and effectively to a future infection.

The COVID-19 mRNA vaccines provide a concrete example. SARS-CoV-2, the virus that causes COVID-19, has a surface protein called the *spike protein*. Spike helps the virus attach to and enter human cells. The vaccines do not contain live SARS-CoV-2, and they do not instruct cells to make a complete virus. Instead, they contain mRNA encoding a modified form of the spike protein.

The central sequence is:

```
mRNA encoding SARS-CoV-2 spike protein
→ delivered into cells after vaccination
→ cells produce spike protein temporarily
→ immune system recognises spike as foreign
→ antibodies, T-cell responses, and immune memory develop
→ faster immune response after later viral exposure
```

The mRNA is packaged in *lipid nanoparticles*, often abbreviated *LNPs*. These are small particles made from lipids, or fat-like molecules. They protect the fragile mRNA from rapid degradation and help it enter cells near the injection site and in nearby lymphoid tissue.

After injection into muscle, some cells take up the lipid nanoparticles. The particles enter cells through a process called *endocytosis*, in which part of the cell membrane folds around external material and brings it inside. For vaccination to work, some mRNA must escape from these intracellular compartments into the *cytoplasm*, the part of the cell where ribosomes translate mRNA into protein.

```
Lipid nanoparticle enters cell
→ mRNA reaches cytoplasm
→ ribosomes read mRNA sequence
→ spike protein is produced temporarily
```

The vaccine mRNA does **not** need to enter the cell nucleus, where DNA is stored. It does not alter the person's DNA. Like normal cellular mRNA, it is temporary and is broken down by normal cellular processes after its instructions have been used.

The produced spike protein, or fragments of it, can be processed and displayed to the immune system. Antigen-presenting cells are particularly important because they can present spike-derived peptides to T cells and provide the signals needed to initiate a strong adaptive immune response.

B cells can bind exposed features of the spike protein. With appropriate support from CD4 T cells, selected B cells multiply and mature into cells that produce antibodies. Some of these antibodies can bind spike in a way that reduces the ability of the virus to attach to or enter cells. These are often called *neutralising antibodies*.

B cell recognises spike protein
→ receives support from helper T cells
→ B cells expand and mature
→ antibodies against spike are produced
→ some antibodies can reduce viral entry into cells

The vaccines can also stimulate T-cell responses. CD4 T cells help coordinate the immune response, including support for B cells and CD8 T cells. CD8 T cells can recognise and kill infected cells displaying spike-derived peptides on MHC class I molecules. Antibodies are especially important for reducing infection and early viral spread, whereas T-cell responses can contribute to control of infected cells and protection against severe disease.

Vaccination also produces *immune memory*. Memory B cells, long-lived antibody-producing cells, and memory T cells can persist after the initial response. If the vaccinated person later encounters SARS-CoV-2, the immune system can respond more rapidly than it could during a first encounter.

First spike exposure through vaccination
→ initial immune response and memory formation
Later exposure to SARS-CoV-2
→ faster recall response against spike
→ reduced risk of severe disease, depending on immunity and viral variant

mRNA vaccines also stimulate parts of the *innate immune system*, the body's rapid and less specific first line of defence. This early inflammatory signalling helps alert antigen-presenting cells and supports the development of adaptive immunity. However, the mRNA and lipid-nanoparticle formulation must be designed carefully: too little innate stimulation may produce a weak immune response, whereas excessive stimulation can reduce protein production or increase reactogenicity.

Common short-term effects after vaccination, such as pain at the injection site, fatigue, fever, headache, or muscle aches, often reflect this temporary immune activation. They do not directly measure how strong a person's protection will be, and their absence does not mean that vaccination failed.

Protection can change over time. Antibody concentrations commonly decline after vaccination, and viral evolution can alter the spike protein sufficiently to reduce recognition by antibodies generated against earlier variants. This is one reason updated COVID-19 vaccines have been developed. T-cell responses often recognise a broader set of spike-derived peptides and may remain more conserved across variants, although they are not completely unaffected by viral evolution.

The general platform can be adapted relatively quickly because changing the vaccine target mainly means changing the mRNA sequence. The same overall delivery and manufacturing logic can then be used for another antigen. This flexibility is one reason mRNA technology is being investigated not only for infectious diseases but also for personalised cancer vaccines, where the encoded sequences can be selected from tumour-specific neoantigens.

15 [26-08-28] How the immune system works: a short overview

The *immune system* protects the body against infections, abnormal cells, and other potential threats. It must solve a difficult problem: it needs to respond strongly enough to control pathogens and dangerous abnormal cells, while avoiding unnecessary damage to healthy tissue.

A useful high-level division is:

```
Innate immune system
→ rapid, broad, early defence
Adaptive immune system
→ slower initial response, highly specific recognition,
and immunological memory
```

These systems work together rather than independently. The innate immune system detects that something may be wrong and creates an early inflammatory response. The adaptive immune system then develops a more specific response against particular molecular targets.

15.1 The innate immune system

The *innate immune system* is the body's rapid first line of defence. It can respond within minutes to hours after detecting infection, tissue injury, or abnormal cellular signals.

Important components include physical barriers such as skin and mucosal surfaces; chemical defences such as stomach acid and antimicrobial molecules; blood proteins such as the *complement system*; and immune cells including neutrophils, macrophages, dendritic cells, natural killer cells, and mast cells.

Innate immune cells do not usually recognise one highly specific antigen in the way that T cells and B cells do. Instead, they use receptors that detect broad patterns associated with danger.

These patterns can include:

```
Pathogen-associated molecular patterns
→ molecular features common in microbes
Damage-associated molecular patterns
→ molecules released or exposed by stressed,
injured, or dying cells
```

For example, a macrophage may detect molecular features associated with bacteria or viruses. It can then engulf material, release signalling molecules, recruit other immune cells, and help initiate a broader immune response.

Inflammation is part of this coordinated response. Blood vessels may become more permeable, immune cells may enter affected tissue, and cytokines may alter local and whole-body physiology.

```
Danger signal detected
→ innate immune cells become activated
→ cytokines and chemokines are released
→ blood flow and immune-cell recruitment increase
→ infection or damaged tissue can be contained and cleared
```

Cytokines are signalling proteins that affect immune-cell activity, growth, survival, and movement. *Chemokines* are a related group of signals that help direct immune cells toward particular tissues or sites of inflammation.

Inflammation is useful when controlled and temporary. However, excessive, prolonged, or misdirected inflammation can damage healthy tissue and contribute to disease.

15.2 Phagocytes and antigen-presenting cells

Some innate immune cells are *phagocytes*: cells that can engulf and digest microbes, damaged cells, and cellular debris. Major phagocytes include neutrophils, macrophages, and dendritic cells.

Neutrophils are abundant, rapidly recruited immune cells that can help control infections through engulfment, antimicrobial molecules, and other mechanisms. They are important for early defence but can also contribute to tissue damage if their responses are excessive.

Macrophages are versatile immune cells that can remove microbes and dead cells, release cytokines, repair tissue, and influence inflammation. Their effects depend strongly on context. In cancer, some macrophages may support anti-tumour immunity, whereas others can contribute to immune suppression, blood-vessel formation, or tumour progression.

Dendritic cells are especially important because they connect innate and adaptive immunity. They can take up antigenic material in tissue, become activated by danger signals, travel to lymph nodes, and present antigen-derived peptides to T cells.

```
Dendritic cell encounters pathogen or abnormal-cell material
→ takes up and processes proteins
→ travels to lymph node
→ presents peptides to T cells
→ helps initiate adaptive immune response
```

15.3 The adaptive immune system

The *adaptive immune system* provides highly specific recognition and immunological memory. Its main cellular components are T cells and B cells.

Unlike innate immune cells, individual T cells and B cells are specialised to recognise particular molecular structures. Before exposure to a pathogen, the body contains many different lymphocytes, each with a distinct receptor. Most will not recognise a particular infection. The small number that do can be activated, multiply, and form a targeted immune response.

```
Rare lymphocyte recognises relevant antigen
→ receives activation signals
→ proliferates into many related cells
→ helps control threat
→ some cells persist as memory cells
```

This process is called *clonal expansion*. It is one reason the first adaptive immune response to a new pathogen usually takes time, whereas later responses to the same or a closely related pathogen can be faster and stronger.

15.4 B cells and antibodies

B cells are lymphocytes that can recognise molecular structures, including proteins or parts of proteins, in their native form. When appropriately activated, B cells can differentiate into *plasma cells*, which produce antibodies.

Antibodies can contribute to protection in several ways:

```
Neutralisation
→ antibody blocks a virus, toxin, or ligand
from interacting with its target
Opsonisation
→ antibody coats a target and helps immune cells
recognise and engulf it
Complement activation
→ antibody can help activate blood proteins
that support immune defence
Antibody-dependent cellular cytotoxicity
→ immune cells recognise antibodies attached to a target cell
and may attack that cell
```

B-cell activation is often strengthened by support from CD4 T cells. This support helps B cells produce antibodies with improved affinity and form long-lived memory B cells and plasma cells.

This is the basis of much vaccine protection: vaccination exposes the immune system to an antigen or instructions for producing an antigen, allowing B cells and T cells to establish memory before a dangerous infection occurs.

15.5 T cells and antigen presentation

T cells generally do not recognise whole proteins directly. They recognise short peptides displayed by *major histocompatibility complex*, or *MHC*, molecules at the cell surface.

Protein is processed into peptide
→ peptide is loaded onto MHC molecule
→ peptide-MHC complex appears at cell surface
→ T-cell receptor can inspect the complex

MHC class I is expressed on most nucleated cells and commonly presents peptides derived from proteins inside the cell. This allows CD8 T cells to inspect whether a cell may be infected by a virus or has become abnormal.

MHC class II is mainly expressed by specialised antigen-presenting cells, including dendritic cells, macrophages, and B cells. It commonly presents peptides derived from material taken up from outside the cell.

MHC class I
→ commonly presents peptides to CD8 T cells
MHC class II
→ commonly presents peptides to CD4 T cells

CD8 T cells can kill cells displaying an appropriate peptide-MHC class I complex. They commonly do this by releasing cytotoxic molecules, including perforin and granzymes, which can trigger programmed target-cell death.

CD4 T cells coordinate immune responses. They can produce cytokines, support B-cell antibody responses, activate macrophages, help dendritic cells stimulate CD8 T cells, and influence the strength and type of immune response.

15.6 Natural killer cells

Natural killer, or *NK*, cells are lymphocytes that belong to the innate immune system, although they share some functional similarities with cytotoxic T cells.

NK cells can kill stressed, infected, or abnormal cells without needing to recognise a specific peptide-MHC complex through a T-cell receptor. They integrate signals from activating and inhibitory receptors.

One important principle is that many healthy cells express MHC class I, which can provide an inhibitory signal to NK cells. Some virus-infected or cancer cells reduce MHC class I expression to evade CD8 T cells. This can remove an inhibitory signal and make them more susceptible to NK-cell attack.

NK cells can also recognise antibody-coated target cells through Fc receptors and contribute to antibody-dependent cellular cytotoxicity.

15.7 Immune memory and vaccination

After an adaptive immune response, some B cells and T cells persist as *memory cells*. These cells are prepared to respond more rapidly if they encounter the same antigen again.

```
First antigen exposure
→ initial adaptive immune response
→ memory B cells and memory T cells form
Later exposure to similar antigen
→ faster and often stronger immune response
```

Vaccines use this principle. They expose the immune system to an antigen, or instructions for making an antigen, in a controlled setting. The goal is to establish protective immune memory without requiring the person to experience the full disease first.

15.8 Immune tolerance and regulation

A strong immune response can be dangerous if directed against healthy tissue. The immune system therefore contains several mechanisms that promote *tolerance*: the ability to avoid inappropriate attack on the body's own cells and harmless environmental molecules.

During T-cell development, many strongly self-reactive T cells are removed or controlled. In peripheral tissues, additional mechanisms regulate immune responses after T cells leave the thymus.

These include:

```
Requirement for co-stimulation
→ helps prevent activation after inappropriate antigen recognition
Regulatory T cells
→ help suppress excessive immune responses
Inhibitory checkpoints
→ reduce T-cell activation and tissue damage
Resolution of inflammation
→ returns tissue toward normal state after threat is controlled
```

Immune checkpoints such as PD-1 are therefore not simply harmful. They are normal physiological brakes that limit immune-mediated damage. Cancer immunotherapies such as pembrolizumab work by reducing one such brake, but this can also cause immune-related adverse events when normal tissues become targets of increased immune activity.

15.9 The immune system and cancer

The immune system can sometimes recognise and destroy cancer cells. Tumour cells may contain mutations that create neoantigens, express abnormal levels of proteins, experience cellular stress, or display other features that make them distinguishable from healthy cells.

However, cancer can evolve mechanisms that limit immune control. Tumours may reduce antigen presentation, exclude T cells from tumour tissue, recruit suppressive immune cells, produce inhibitory signals such as PD-L1, create metabolically unfavourable conditions, or select tumour-cell populations that are less visible to the immune system.

```
Immune system recognises tumour-associated antigen
→ anti-tumour T cells may become activated
→ tumour attempts to suppress, exclude, or evade immunity
→ tumour control or immune escape depends on this balance
```

Cancer immunotherapies aim to change this balance. Checkpoint inhibitors reduce inhibitory signals. Cancer vaccines aim to improve antigen-specific recognition. CAR-T cells provide engineered recognition of selected cell-surface antigens. Antibodies can recruit immune effector functions or deliver cytotoxic payloads to tumour cells.

The central challenge is to create sufficiently strong and durable anti-tumour immunity while avoiding unacceptable damage to healthy tissue.

16 [26-08-28] T-cell engagers: directing T cells towards cancer cells

T-cell engagers, often abbreviated *TCEs*, are therapies designed to bring a patient's T cells into close contact with cancer cells and activate the T cells to kill those cells.

They are often *bispecific antibodies*: engineered proteins with two different binding functions.

```
One binding arm
→ binds a selected target on the cancer cell
Other binding arm
→ binds CD3 on a T cell
The T-cell engager
→ physically bridges the T cell and cancer cell
→ activates the T cell locally
→ promotes killing of the cancer cell
```

CD3 is part of the signalling machinery associated with the normal T-cell receptor. In ordinary adaptive immunity, a T-cell receptor recognises a particular peptide presented by an MHC molecule. A CD3-directed T-cell engager bypasses the need for the T cell's own receptor to recognise a specific tumour peptide-MHC complex.

Instead, the engager creates an artificial interaction:

```
Tumour cell expresses target antigen
+
T cell expresses CD3
+
T-cell engager binds both
→ close cell-cell contact
→ T-cell activation
→ cytotoxic attack on the tumour cell
```

The activated T cell can release cytotoxic molecules such as perforin and granzymes, which can induce target-cell death. It can also release cytokines, including interferon-gamma and tumour-necrosis factor alpha, which may strengthen local immune activity but can also contribute to toxicity.

T-cell engagers are therefore a way of *redirecting existing polyclonal T cells* towards cancer. Unlike CAR-T therapy, they do not require collecting a patient's T cells, genetically modifying them outside the body, expanding them, and reinfusing them. Instead, the therapeutic protein is administered directly and uses the patient's endogenous T cells.

```
CAR-T therapy
→ patient T cells are engineered before infusion
→ engineered receptor determines target recognition
T-cell engager
→ therapeutic protein is administered directly
→ it temporarily links endogenous T cells to target cells
```

This can make T-cell engagers more straightforward to manufacture and administer than personalised autologous cell therapy. However, their efficacy still depends on having functional T cells, sufficient target expression, tumour accessibility, appropriate dose and schedule, and a tumour microenvironment that does not completely suppress T-cell activity.

16.1 Why T-cell engagers can be effective

A conventional cytotoxic T cell normally needs to recognise a specific peptide–MHC complex through its own T-cell receptor. Tumours can escape this type of immune recognition by reducing MHC expression or disrupting antigen-processing pathways.

Many CD3-based T-cell engagers do not require peptide presentation by tumour-cell MHC. If the selected target is present on the tumour-cell surface and the engager can bind it, the engager can recruit a nearby T cell.

```
Conventional CD8 T-cell recognition
→ requires tumour peptide displayed by MHC class I
→ recognised by matching T-cell receptor
CD3-based T-cell engager
→ requires accessible tumour-cell surface target
→ links this target directly to CD3 on T cells
→ does not require tumour peptide-MHC recognition
```

This is potentially useful, but it creates demanding target-selection requirements. The tumour target should ideally be strongly expressed, accessible at the cell surface, present on most clinically important tumour cells, and absent or minimally expressed on essential normal tissues.

If the same target is expressed on normal tissue, the engager may recruit T cells against that tissue. This is called *on-target, off-tumour toxicity*.

A second major risk is *cytokine release syndrome*, or *CRS*. If many T cells are activated rapidly, inflammatory cytokines can be released systemically.

```
Strong, widespread T-cell activation
→ cytokine release
→ activation of additional immune cells
→ fever and systemic inflammation
→ cytokine release syndrome may occur
```

For this reason, many T-cell engagers use cautious dose escalation or *step-up dosing*, in which early doses are lower before a patient receives the intended treatment dose.

16.2 AZD5863: a CLDN18.2 × CD3 T-cell engager

The article you found in the *Journal for ImmunoTherapy of Cancer* describes *AZD5863*, an investigational bispecific T-cell engager that targets *claudin 18.2*, abbreviated *CLDN18.2*, on tumour cells and *CD3* on T cells.

Claudins are proteins involved in *tight junctions*, structures that help seal neighbouring epithelial cells together and regulate movement of molecules between cells. CLDN18.2 is a particular form, or isoform, of claudin 18 that is normally associated mainly with differentiated gastric tissue.

In healthy stomach tissue, CLDN18.2 is relatively sequestered within organised epithelial tight junctions. In some cancers, tissue architecture becomes disorganised and CLDN18.2 can become more accessible at the tumour-cell surface. It can also be expressed in several cancer types, particularly subsets of gastric, gastroesophageal-junction, oesophageal, and pancreatic cancers.

This makes CLDN18.2 an attractive but imperfect tumour-associated target:

```
CLDN18.2 expression in tumour cells
+
surface accessibility in disorganised tumour tissue
→ potential therapeutic target
Some normal gastric-tissue expression
→ possible on-target toxicity risk
```

AZD5863 is designed to bind CLDN18.2 with *high-affinity bivalent binding*: it has two CLDN18.2-binding regions, allowing strong binding when CLDN18.2 is present on a tumour-cell surface. It binds CD3 on T cells with *lower-affinity monovalent binding*: one CD3-binding interaction designed to be relatively weak in isolation.

The intended design logic is:

```
Strong binding to CLDN18.2 on tumour cell
+
weaker binding to CD3 on circulating T cell
→ reduce widespread T-cell activation away from tumour
Both targets engaged at tumour cell
→ stable tumour cell-T-cell bridge
→ local T-cell activation and tumour-cell killing
```

In other words, AZD5863 is intended to be more effective when both a CLDN18.2-positive target cell and a T cell are present together. The lower CD3 affinity is intended to reduce non-specific or peripheral T-cell activation and thereby improve the therapeutic window, particularly by reducing excessive cytokine release.

The basic proposed mechanism is:

```
AZD5863 binds CLDN18.2 on tumour cell
→ AZD5863 binds CD3 on nearby T cell
→ tumour cell and T cell are held together
→ CD3-associated signalling activates T cell
→ cytotoxic molecules and cytokines are released
→ CLDN18.2-positive tumour cell may be killed
```

The preclinical paper reported *T-cell-dependent cellular cytotoxicity* against CLDN18.2-expressing cancer cells and tumour-growth inhibition in preclinical models of gastric, gastroesophageal-junction, oesophageal, and pancreatic cancer.

It also reported a *bystander-killing* effect in experimental systems: CLDN18.2-negative tumour cells could be killed when they were near CLDN18.2-positive cells. This does not mean AZD5863 directly binds CLDN18.2-negative cells. Rather, inflammatory and death-inducing signals generated after local T-cell activation, including interferon-gamma, TNF-alpha, and Fas-FasL signalling, may affect nearby cells.

```
CLDN18.2-positive cell engages T cell through AZD5863
→ local T-cell activation and inflammatory signalling
→ neighbouring CLDN18.2-negative tumour cells
may also be affected
```

This could be useful in heterogeneous tumours, where not every cancer cell expresses CLDN18.2. However, it could also make toxicity and the biological effect less spatially precise. The clinical importance of this effect must be established in human studies rather than inferred from preclinical experiments.

As of **28 August 2026**, AZD5863 is being evaluated in the first-in-human Phase I/II study *NCT06005493*. The trial includes adults with advanced or metastatic CLDN18.2-positive gastric, gastroesophageal-junction,

oesophageal, and pancreatic cancers. It investigates intravenous and subcutaneous administration, including dose-escalation and dose-expansion components.

At this stage, the central clinical questions are:

What dose and schedule are tolerable?
Does AZD5863 achieve suitable exposure and pharmacological activity?
How frequent and severe are cytokine-mediated and gastric toxicities?
Does it produce meaningful and durable anti-tumour activity in appropriately selected patients?

The preclinical results are encouraging mechanistic evidence, but they do not yet establish clinical efficacy or an acceptable benefit–risk profile in patients.

16.3 Other important examples of T-cell engagers

The clearest clinical successes of CD3-based T-cell engagers have initially been in blood cancers. In these diseases, tumour cells and T cells are often more accessible to one another than in solid tumours, and several targets are well established.

Therapy	Target pair	Illustrative use and key point
Blinatumomab	CD19 × CD3	Used for B-cell precursor acute lymphoblastic leukaemia. It was the first widely recognised CD3-based bispecific T-cell engager and established the clinical feasibility of this approach.
Teclistamab	BCMA × CD3	Used in relapsed or refractory multiple myeloma. BCMA is expressed on malignant plasma cells and normal plasma-cell populations.
Elranatamab and linvoseltamab	BCMA × CD3	Other BCMA-directed T-cell engagers for relapsed or refractory multiple myeloma.
Talquetamab	GPRC5D × CD3	Used in relapsed or refractory multiple myeloma. It targets GPRC5D, providing an alternative to BCMA-directed therapy.
Mosunetuzumab	CD20 × CD3	Used for follicular lymphoma in appropriate treatment settings.
Epcoritamab and glofitamab	CD20 × CD3	Used in selected B-cell non-Hodgkin lymphomas, including diffuse large B-cell lymphoma. Glofitamab uses a distinctive 2:1 CD20:CD3 binding format.
Tarlatamab	DLL3 × CD3	Used for small-cell lung cancer in appropriate treatment settings. It is an important example of a CD3 engager directed at a solid-tumour-associated target.

Tebentafusp is another important related example, although its structure and tumour-recognition mechanism are different. It is a soluble bispecific fusion protein that combines a high-affinity T-cell receptor directed against a gp100 peptide presented by HLA-A*02:01 with an anti-CD3 component.

Tebentafusp
→ recognises gp100 peptide presented by a particular MHC molecule
→ binds CD3 on T cells
→ redirects T cells towards gp100-presenting tumour cells

Tebentafusp is used for unresectable or metastatic uveal melanoma in patients with the relevant HLA type. Unlike most conventional antibody-based T-cell engagers, it targets a *peptide–MHC complex*, allowing

it to address an intracellular protein indirectly. However, it requires the patient to express the relevant HLA-A*02:01 molecule and requires tumour cells to retain appropriate MHC antigen presentation.

These examples illustrate the main design principle shared by T-cell engagers:

```
Choose a tumour-associated target
+
Link that target to CD3 on T cells
+
Achieve sufficient tumour-selective activation
+
Avoid excessive cytokine release and normal-tissue injury
```

The difficult part is not merely constructing a molecule that binds two targets. The difficult part is finding the right target, achieving adequate tumour exposure and T-cell access, controlling systemic immune activation, overcoming tumour immune suppression, and producing a favourable clinical benefit–risk balance.

17 [26-08-28] Binding-to-release drug delivery: an alternative to conventional ADCs

The Nature paper *A binding-to-release system for internalization-independent drug delivery* introduces a new drug-delivery concept called *binding-to-release*, or *BTR*.

The central problem it addresses is that conventional antibody–drug conjugates, or ADCs, usually require their target to be efficiently *internalised*. That is, after the antibody binds an antigen on the cell surface, the entire ADC–target complex must be taken into the cell. The payload is then released, often after trafficking to lysosomes, where the linker can be cleaved or degraded.

The conventional ADC sequence is:

```
ADC binds cell-surface antigen
→ antigen-ADC complex is internalised
→ complex traffics through intracellular compartments
→ linker is processed and payload is released
→ intracellular payload damages or kills susceptible cell
```

This requirement is important because many potentially useful cell-surface targets do **not** internalise efficiently. An ADC may bind such a target very well but still deliver too little intracellular payload to be effective.

The new BTR approach separates *target binding* from *cellular internalisation*. Instead of relying on the cell to take up the conjugate, the authors designed a linker that is cleaved directly after the targeting ligand binds its protein target.

```
BTR conjugate binds target protein
→ target-proximal amino-acid residue reacts with linker
→ linker is cleaved at the target site
→ payload is released without requiring internalisation
```

The paper introduces a chemical strategy called *PhoPEX*, short for phosphorus(V)–phenol exchange. In their model system, binding near a suitable tyrosine residue in the target protein brings that residue into the correct position to react with and cleave the PhoPEX linker. The protein itself therefore helps trigger local payload release.

This is a form of *proximity-induced chemistry*:

Target-binding ligand positions conjugate in binding pocket
→ nearby target residue acts as a chemical reaction partner
→ linker cleavage occurs preferentially at target
→ active payload is released locally

The authors demonstrated the approach using targets including *fibroblast activation protein*, or *FAP*, and PD-L1. FAP is expressed by cancer-associated fibroblasts in the tumour microenvironment in several tumour types. In the FAP-targeting experiments, the released payload could act locally without requiring FAP itself to be internalised.

The conceptual difference from a conventional ADC is therefore:

Property	Conventional ADC	Binding-to-release approach
Targeting component	Usually an antibody binding a cell-surface antigen	Can use a target-binding ligand, including a small-molecule ligand
Required event after binding	Internalisation of the conjugate into the target cell	Target-proximal linker cleavage after binding
Typical payload release site	Intracellularly, often after lysosomal processing	Locally at or near the bound target, potentially extracellularly
Suitable targets	Targets with sufficient binding, internalisation, and trafficking	Potentially includes targets that bind well but internalise poorly

This does **not** mean that BTR is simply a “better ADC.” It is better understood as a broader alternative drug-delivery paradigm. It may be useful precisely where the usual ADC internalisation requirement is a limitation, but it introduces different questions. For example, the released payload must still reach susceptible cells, remain local enough to avoid unacceptable normal-tissue exposure, and produce a favourable therapeutic window.

The paper is a substantial scientific result because it proposes and demonstrates a potentially general solution to a long-standing target-selection constraint in targeted drug delivery:

Many cell-surface proteins can be bound
≠
many cell-surface proteins can support effective ADC internalisation
Binding-to-release
→ may allow productive payload delivery
even for poorly internalising targets

The work is particularly notable because it combines a new linker chemistry with a clear therapeutic rationale, mechanistic evidence, and results in preclinical tumour models, including patient-derived xenograft models. It also suggests that the platform could support both therapeutic payload delivery and imaging or diagnostic applications.

Publication in *Nature* indicates that independent peer reviewers and editors considered the work broadly important, novel, and of high scientific quality. It is therefore reasonable to view it as a potentially important *platform advance*.

However, it remains preclinical research. The crucial next questions are whether the approach can be reproduced across many targets and payloads, whether released drugs remain sufficiently local in humans, whether toxicity is manageable, and whether the platform produces clinical benefit beyond existing ADCs or other targeted therapies. A high-profile publication is strong evidence of scientific importance; it is not yet evidence that the approach will become a successful medicine.

18 [26-08-28] Disitamab vedotin in HER2-positive breast cancer with liver metastases

(*Disitamab vedotin for HER2-positive breast cancer with liver metastases: phase 3 results from the RC48-C006 phase 2/3 trial* , <https://www.nature.com/articles/s41467-026-75733-y>)

The paper reports results from the randomised RC48-C006 Phase 2/3 trial of *disitamab vedotin*, also called *RC48*, in patients with previously treated HER2-positive advanced breast cancer and liver metastases.

This is a clinically challenging population. Liver metastases are associated with a poor prognosis, and patients may have already received HER2-directed treatment and chemotherapy.

Disitamab vedotin is a HER2-targeted antibody–drug conjugate, or ADC. Like trastuzumab deruxtecan (T-DXd), it uses HER2 as an address for preferential delivery of a cytotoxic payload to HER2-expressing tumour cells.

Its basic mechanism is:

```
Disitamab vedotin binds HER2 on tumour cell
→ HER2-ADC complex is internalised
→ linker is processed inside the cell
→ MMAE payload is released
→ microtubules are disrupted
→ cell division is blocked and susceptible cell may die
```

MMAE, or *monomethyl auristatin E*, is a microtubule inhibitor. Microtubules are essential components of the cellular machinery required for chromosome separation during cell division. Disrupting them can cause cell-cycle arrest and cell death.

In the trial, disitamab vedotin was compared with *lapatinib plus capecitabine*. The reported median progression-free survival was **9.9 months** with disitamab vedotin versus **4.9 months** with lapatinib plus capecitabine. This corresponded to a hazard ratio of **0.561**, meaning that, under the trial’s analysis, the risk of progression or death was lower in the disitamab-vedotin group. Overall-survival data were still immature at the reported analysis.

18.1 How is disitamab vedotin different from T-DXd?

Although both treatments are HER2-targeted ADCs, they are not the same drug. They differ in their antibody component, payload, linker design, and approximate drug-to-antibody ratio.

Property	Disitamab vedotin	Trastuzumab deruxtecan (T-DXd)
Target	HER2	HER2
Antibody	Hertuzumab; binds a HER2 epitope distinct from trastuzumab	Trastuzumab-based antibody
Payload	MMAE, a microtubule inhibitor	Deruxtecan, a topoisomerase-I inhibitor
Main intracellular consequence	Disrupts microtubules and cell division	Causes replication-associated DNA damage through topoisomerase-I inhibition
Approximate drug-to-antibody ratio	About 4 payloads per antibody	About 8 payloads per antibody
Bystander effect	Possible, because released MMAE can cross cell membranes	Possible, because released deruxtecan payload can diffuse locally

The different HER2-binding epitope means that disitamab vedotin may interact with HER2 differently from trastuzumab-based ADCs. The different payload is also important: a tumour resistant to one payload class may not necessarily be resistant to another. However, this does **not** mean that disitamab vedotin will reliably overcome resistance to T-DXd, because resistance can arise through many mechanisms, including loss or reduction of HER2, impaired ADC internalisation, altered intracellular trafficking, drug efflux, and broader changes in cell state.

18.2 Is it better than T-DXd?

This study does **not** establish that disitamab vedotin is better than T-DXd. It did not compare the two treatments directly.

The RC48-C006 comparator was lapatinib plus capecitabine, not T-DXd. Therefore, the trial demonstrates that disitamab vedotin performed better than that specific comparator in the enrolled population; it cannot determine whether it is superior, equivalent, or inferior to T-DXd.

The result is nevertheless important because it provides randomised Phase 3 evidence that a HER2-targeted MMAE ADC can improve progression-free survival in a high-risk population with breast-cancer liver metastases.

Whether disitamab vedotin has a useful role relative to T-DXd depends on questions that require further evidence:

Does it retain activity after T-DXd treatment?
Does its distinct HER2 epitope or MMAE payload benefit particular resistance mechanisms or tumour subgroups?
How do efficacy, durability, safety, and quality of life compare with current HER2-directed standards of care?
Which treatment sequence provides the best overall benefit-risk?

The key lesson is that sharing the same surface target does not make two ADCs interchangeable. The antibody, epitope, linker, payload, drug loading, intracellular release, and resulting toxicity profile all contribute to the medicine's clinical behaviour.

19 [26-08-28] Allosteric disruption of RAS–RAF signalling

(<https://www.nature.com/articles/s41467-026-76337-2>)

The paper *Small-molecule RAS/RAF inhibitors target RAS-driven cancers via allosteric RAF disruption* describes a new way to inhibit a central cancer-signalling pathway: instead of binding directly to mutant RAS or inhibiting RAF's kinase active site, the compounds disrupt the physical interaction between *RAS* and *RAF*.

This is important because abnormal RAS signalling is one of the most common drivers of human cancer.

19.1 RAS-driven cancers

RAS proteins are small signalling proteins that act as molecular switches. They cycle between an inactive GDP-bound state and an active GTP-bound state.

RAS-GDP
→ mostly inactive state
RAS-GTP
→ active state
→ can bind downstream effector proteins such as RAF

Growth-factor signalling normally activates RAS temporarily. Active RAS then recruits and activates RAF, which helps transmit a growth-related signal through the MAPK pathway:

```
Growth-factor receptor
→ RAS-GTP
→ RAF
→ MEK
→ ERK
→ changes in gene expression, growth, survival, and division
```

Cancer-associated mutations in *KRAS*, *NRAS*, or *HRAS* can make RAS spend too much time in its active GTP-bound state. The result is excessive and persistent signalling through pathways that promote cell growth and survival.

A *RAS-driven cancer* is therefore a cancer in which abnormal RAS signalling is an important dependency. Common examples include many pancreatic cancers, colorectal cancers, and lung adenocarcinomas, as well as subsets of melanoma and other tumour types.

19.2 What is usually targeted?

There are several ways to inhibit this pathway. Some therapies target a specific mutant form of RAS directly, such as KRAS G12C inhibitors. Others target downstream kinases such as BRAF, MEK, or ERK.

However, each strategy has limitations. Direct RAS inhibitors can be mutation-specific and resistance can emerge. Conventional RAF kinase inhibitors target the enzyme region of RAF, but they can have complex effects depending on the RAS and RAF state of the tumour. In some RAS-mutant settings, certain RAF inhibitors can even produce *paradoxical activation* of MAPK signalling through RAF dimerisation.

The paper takes a different approach:

```
Rather than inhibit RAS directly
or block RAF catalytic activity,
disrupt the RAS-RAF protein-protein interaction
that is needed to transmit the signal.
```

19.3 What do the compounds do?

The researchers identified a small-molecule series beginning with *Kobe3708* and developed more active derivatives, including *RK440* and *RK447*.

These compounds bind RAF at a site that is distinct from the normal RAS-binding interface. They therefore do not simply occupy the place where RAS would normally bind RAF.

Instead, they act *allosterically*.

An *allosteric inhibitor* binds one part of a protein and changes the shape, dynamics, or function of another part of that protein. It is a form of molecular “remote control.”

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Compound binds allosteric site on RAF
→ RAF changes conformation
→ RAS-binding interface on RAF is disrupted
→ active RAS can no longer productively recruit RAF
→ RAF-MEK-ERK signalling is reduced
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The key idea is that RAF normally needs to recognise and bind active RAS at the cell membrane. By changing RAF's conformation from a separate binding site, the compounds reduce this critical protein–protein interaction.

Mutant RAS remains active
→ but cannot efficiently pass its signal to RAF
→ downstream MAPK signalling is weakened
→ RAS-dependent tumour cells may lose growth and survival support

This is why the paper calls the strategy *allosteric RAF disruption*: the compounds disrupt RAF's ability to function as a RAS effector without directly competing with RAS at the RAS–RAF interface and without relying primarily on inhibition of RAF's kinase active site.

19.4 Why could this be useful?

A potential advantage is breadth. The approach targets a common downstream step used by active KRAS, NRAS, and HRAS rather than one particular RAS mutation. In principle, it could therefore be relevant across several RAS-driven cancers.

The paper reports preclinical anti-tumour activity in models driven by different RAS mutations, including models of melanoma with resistance to existing BRAF-targeted treatment.

Mutation-specific direct RAS inhibitor
→ may address one defined mutant state
Allosteric RAS-RAF interaction inhibitor
→ may potentially address multiple active RAS proteins
that depend on RAF signalling

The approach could also avoid some limitations of conventional RAF kinase inhibitors, including mechanisms related to RAF dimerisation and paradoxical pathway activation. However, this possibility needs careful testing across relevant tumour and normal-tissue contexts.

This is a notable *preclinical* result because protein–protein interactions have often been difficult targets for small molecules: their binding surfaces can be broad and relatively flat compared with traditional enzyme active sites. The study shows that one can potentially disrupt the RAS–RAF interaction indirectly by exploiting an allosteric site on RAF.

It is not yet evidence of a clinical treatment. The important next questions are whether these compounds can achieve suitable human PK and tumour exposure, whether they inhibit MAPK signalling sufficiently and durably in patients, what toxicities arise from inhibiting normal RAS–RAF signalling, and whether resistance develops. Nevertheless, the paper provides an elegant example of how an apparently difficult target interaction may become tractable through protein conformational biology and allosteric drug design.

20 [26-08-28] Estrogen receptors, immunity, and new combination therapies in breast cancer

(<https://www.nature.com/articles/s41698-026-01670-0>)

The review *Novel immunological effects of estrogen receptors redefine breast cancer biology and therapy* argues that estrogen receptors should not be viewed only as regulators of tumour-cell growth. They can also influence the *tumour immune microenvironment*: the immune cells, signals, and physical conditions surrounding a tumour.

This matters particularly for *estrogen-receptor-positive*, or *ER-positive*, breast cancer. These cancers are often treated with endocrine therapy, which reduces oestrogen signalling or directly blocks or degrades the

estrogen receptor. Examples include aromatase inhibitors, tamoxifen, fulvestrant, and newer oral selective estrogen-receptor degraders, or SERDs.

The usual simplified model is:

```
Oestrogen activates ER in breast-cancer cell
→ gene-expression changes
→ tumour-cell growth and survival may be supported
Endocrine therapy reduces ER signalling
→ tumour-cell growth support is reduced
```

The review adds an important second layer: estrogen receptors are also present in, or can indirectly influence, immune cells such as T cells, B cells, macrophages, dendritic cells, and myeloid-derived suppressor cells. Therefore, oestrogen and ER signalling can affect not only tumour-cell behaviour, but also whether the tumour microenvironment supports or suppresses anti-tumour immunity.

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ER signalling in tumour and immune cells
→ changes cytokines, immune-cell recruitment,
activation, and differentiation
→ may alter whether the tumour is immunologically
"cold" or more responsive to immune attack
```

ER-positive breast cancers are often considered relatively *immunologically cold*. Compared with some other tumours, they may have fewer activated tumour-infiltrating T cells, weaker inflammatory signalling, and a more immunosuppressive microenvironment. This is one reason why immune checkpoint inhibitors, such as PD-1 or PD-L1 inhibitors, have generally shown less consistent benefit in unselected ER-positive breast cancer than in some other cancer types, including subsets of triple-negative breast cancer.

The therapeutic idea is therefore not simply:

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Endocrine therapy
+
checkpoint inhibitor
=
automatically better treatment
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Instead, the hypothesis is:

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Appropriate endocrine or ER-targeted treatment
→ may alter tumour and immune biology
More favourable immune microenvironment
→ may make checkpoint blockade,
vaccination, or another immunotherapy more effective
```

20.1 What new treatments could this lead to?

The most immediate possibilities are *rational combinations* of existing or emerging therapies, selected for biologically appropriate patient subgroups.

Endocrine therapy plus immune-checkpoint blockade is the most direct example. A patient might receive an aromatase inhibitor, fulvestrant, or an oral SERD together with a PD-1 or PD-L1 inhibitor. The endocrine component would suppress ER-driven tumour growth and might also reduce immune-suppressive effects of ER signalling. The checkpoint inhibitor would then aim to restore or sustain T-cell activity.

SERD or aromatase inhibitor
→ suppresses ER-dependent tumour biology
and may remodel immune microenvironment
PD-1 or PD-L1 inhibitor
→ reduces inhibitory signalling in T cells
Combination
→ aims to make anti-tumour immune responses
more likely or more durable

However, this will probably not be effective for every patient with ER-positive disease. It may require biomarkers indicating that the tumour has sufficient immune-cell infiltration, antigen presentation, interferon-related signalling, PD-L1 expression, a particular ER state, or another feature suggesting that immune activation is possible.

A second concrete strategy is **endocrine therapy plus CDK4/6 inhibition plus immunotherapy**. CDK4/6 inhibitors are already important treatments for many ER-positive breast cancers because they reduce tumour-cell proliferation. Preclinical work also suggests that they can affect antigen presentation, interferon signalling, and immune-cell behaviour. In principle, a three-drug combination could suppress ER signalling, block cell-cycle progression, and improve the ability of T cells to attack tumour cells.

Endocrine therapy
→ reduces ER-driven growth
CDK4/6 inhibitor
→ reduces cell-cycle progression
and may alter tumour immunogenicity
Checkpoint inhibitor
→ supports T-cell activity
Combination
→ may be useful in selected,
biomarker-defined ER-positive tumours

A third possibility is combining endocrine therapy with **cancer vaccines**. If endocrine therapy makes the tumour microenvironment less suppressive, a vaccine could provide or expand tumour-specific T-cell responses, while checkpoint blockade could prevent those T cells from being switched off. This creates a possible three-part strategy:

Cancer vaccine
→ generates or expands tumour-reactive T cells
Endocrine therapy
→ suppresses ER-driven biology
and may improve immune context
Checkpoint inhibitor
→ reduces inhibitory signals that restrain T cells

The review also points toward more experimental approaches involving **selective modulation of ER subtypes** or **GPER1**. ER α , ER β , and GPER1 are distinct estrogen-related receptors with partly different biological roles. Rather than broadly suppressing all estrogen-related signalling, future drugs might selectively alter a receptor or pathway that improves anti-tumour immunity while avoiding undesirable effects in tumour cells or normal tissues.

For example, a future treatment might combine an ER α degrader for direct tumour control with a GPER1-targeting agent intended to alter immune-cell recruitment or activation. This remains much more speculative than established endocrine therapies or checkpoint inhibitors and requires substantial preclinical and clinical validation.

The central clinical-development challenge is patient selection. A combination that is biologically plausible can still fail if it is tested in a population where most tumours lack the required immune context.

ER-positive breast cancer
≠ one immunological disease
Future strategy
→ identify ER-positive tumours with a plausible
immune-response mechanism
→ match endocrine and immunotherapy combination
to that subgroup

The review is therefore important not because it establishes a new standard treatment today, but because it reframes ER-positive breast cancer as an *immuno-endocrine* disease. This could lead to more precise trials of endocrine therapy, CDK4/6 inhibitors, SERDs, vaccines, and checkpoint inhibitors, guided by biomarkers that describe both tumour-cell ER biology and the immune microenvironment.

The crucial uncertainty remains clinical: these combinations must show that they improve patient-relevant outcomes, such as durable disease control, survival, symptoms, or quality of life, enough to justify additional toxicity, treatment burden, and cost.

21 [26-08-28] Why aggressive brain tumours are so difficult to treat

Brain tumours are not one disease. They include many different tumour types, with very different biology, treatment options, and prognosis. Some are benign or slow-growing, some can be cured or controlled for many years, and some arise from a cancer elsewhere in the body and spread to the brain.

However, the situation you describe is sadly consistent with an aggressive primary brain tumour such as *glioblastoma*, the most common malignant primary brain tumour in adults. Glioblastoma can occur in people who were previously healthy, highly active, and functioning at a very high level. Physical fitness does not protect someone from every form of cancer, particularly cancers driven by biological events that may develop silently before symptoms become apparent.

For aggressive brain tumours, symptoms can appear suddenly because even a relatively local tumour can disrupt essential brain functions or increase pressure within the skull. Symptoms may include headache, seizures, altered personality or cognition, weakness, speech difficulties, visual changes, or problems with balance and coordination. The apparent suddenness of diagnosis does not necessarily mean that the tumour developed only days or weeks earlier; it may have been growing without producing noticeable symptoms until it affected a critical brain region or caused swelling.

21.1 Why the brain is a uniquely difficult site for cancer

The brain is essential, highly specialised, and has very limited spare capacity in some functions. A tumour does not need to spread widely through the body to cause severe consequences. Growth in a small but critical area can affect speech, movement, memory, personality, vision, consciousness, or the ability to breathe and swallow.

The skull also creates a physical problem. Unlike many other tissues, the brain is enclosed within a rigid bony space. A growing tumour, surrounding inflammation, fluid accumulation, or bleeding can increase *intracranial pressure*.

Tumour growth or swelling within skull
→ increased pressure on normal brain tissue
→ impaired neurological function
→ potentially life-threatening complications

Treatment is difficult because the aim is not simply to kill tumour cells. It is to kill or control tumour cells while preserving normal brain tissue and neurological function.

21.2 Glioblastoma: the central clinical challenge

Glioblastoma is a highly aggressive tumour arising from glial-lineage cells or their precursors. Glial cells normally support, protect, and regulate neurons. Glioblastoma cells can grow rapidly, invade surrounding brain tissue, alter blood-vessel growth, suppress immune responses, and evolve under the selection pressure created by treatment.

A crucial feature is that glioblastoma is usually not a neatly bounded mass.

Visible tumour on MRI
≠ necessarily the full biological extent of disease
Tumour cells may already have infiltrated
surrounding brain tissue beyond the visible margin

This infiltrative growth makes complete surgical removal extraordinarily difficult. A neurosurgeon may remove all tumour visible on imaging, sometimes called a *gross total resection*, while microscopic tumour cells remain in surrounding tissue. Removing a wider margin may not be possible if it would damage areas responsible for speech, movement, vision, memory, or other essential functions.

The usual treatment framework for a fit adult with newly diagnosed glioblastoma commonly includes maximal safe surgery, radiotherapy, and the chemotherapy drug *temozolomide*. These treatments can improve outcomes for many patients, but recurrence remains common because residual tumour cells can survive and regrow.

Maximal safe surgery
→ reduces tumour burden and relieves pressure
Radiotherapy
→ damages DNA in remaining tumour cells
Temozolomide
→ adds DNA damage and can increase treatment effect
Residual resistant tumour cells
→ may survive, evolve, and cause recurrence

This does **not** mean that “there is nothing that can be done.” Treatment can prolong survival, preserve function, reduce symptoms, and provide valuable time. But for glioblastoma, current treatment still often does not provide durable control, which is why the disease remains one of the most urgent unmet needs in oncology.

21.3 Why aggressive brain tumours resist treatment

Several difficult biological and practical problems occur at the same time.

First, the tumour is often *heterogeneous*. Different regions of one tumour can contain cells with different mutations, gene-expression states, metabolic behaviour, growth rates, and treatment sensitivity.

Treatment kills sensitive tumour cells
→ resistant cells survive
→ surviving cells expand
→ recurrent tumour may differ from original tumour

Second, glioblastoma cells can infiltrate normal brain tissue. This makes local therapy difficult: surgery and radiation cannot simply remove or destroy a large margin of surrounding tissue in the way that may be possible for tumours in some other organs.

Third, many drugs have difficulty reaching the brain at sufficient concentrations. The *blood–brain barrier* is a specialised system of blood vessels and cellular barriers that protects the brain from toxins, pathogens, and fluctuations in blood chemistry. This protection is biologically valuable, but it can limit delivery of medicines.

Drug in bloodstream
→ must cross blood-brain barrier
→ must reach infiltrating tumour cells
→ must remain at adequate free concentration
→ must do so without unacceptable toxicity

The blood–brain barrier can be disrupted in some visible tumour regions, but it may remain relatively intact around infiltrating tumour cells. This means that a drug might reach the centre of a tumour more effectively than the microscopic cells responsible for later recurrence.

Fourth, the tumour microenvironment can suppress immune responses. Glioblastoma may contain relatively few effective tumour-reactive T cells, can recruit immune-suppressive myeloid cells, and can create metabolic and signalling conditions that reduce immune-cell function. The brain also has specialised immune biology, making it inappropriate to think of it as simply another tissue.

Finally, many therapies are limited by the need to protect normal brain function. Even a treatment that kills tumour cells may be unacceptable if it also causes substantial irreversible neurological damage.

21.4 Why the course can sometimes seem so rapid

The course of aggressive brain tumours can be frighteningly fast. This can reflect a combination of rapid tumour growth, tumour infiltration, brain swelling, seizures, treatment-resistant residual disease, and the fact that neurological decline can occur when a lesion affects an essential region.

It is important to avoid overgeneralising from individual cases. Survival varies substantially among people and depends on factors including age, overall health, tumour location, extent of surgery, molecular subtype, response to treatment, and access to specialised multidisciplinary care.

For glioblastoma, median survival statistics are often measured in months rather than decades, which is deeply distressing. But a median is not a deadline for an individual person: some patients do substantially better or worse than the median, and research is aimed specifically at shifting the whole survival distribution towards longer and better lives.

21.5 Is there hope for the future?

Yes, there is real scientific hope, but it should be expressed honestly. There is not yet a broadly transformative therapy that reliably cures glioblastoma. Progress has been slower than patients, families, clinicians, and researchers want. However, several promising strategies aim to address different parts of the problem.

The most plausible future is probably not one universal “magic bullet.” It may instead involve better biological classification, more effective local and systemic therapies, improved drug delivery, rational combinations, and treatment matched to a patient’s tumour.

21.6 Better molecular classification and precision treatment

Glioblastoma is increasingly understood as a collection of biologically distinct tumours rather than one uniform disease. Molecular features can affect prognosis, treatment sensitivity, and possible therapeutic vulnerabilities.

For example, *MGMT promoter methylation* is clinically important because it is associated with greater sensitivity to temozolomide in many patients. The MGMT protein can repair certain types of DNA damage caused by temozolomide; reduced MGMT expression can therefore make tumour cells less able to reverse that damage.

Temozolomide causes DNA damage
→ MGMT can repair part of this damage
MGMT promoter methylation
→ often reduces MGMT expression
→ may increase sensitivity to temozolomide

Other molecular alterations may identify smaller patient subgroups with more specific treatment opportunities. The long-term goal is to move from treating all glioblastomas similarly toward identifying biologically meaningful subtypes and matching therapies to the mechanisms that sustain each tumour.

21.7 Improved drug delivery to the brain

A major area of research is finding ways to increase drug concentration in tumour tissue while limiting exposure in the rest of the body.

One approach is *convection-enhanced delivery*, in which a treatment is infused directly into or around the tumour region through catheters. Rather than relying on the drug to cross the blood–brain barrier from the bloodstream, this approach attempts to distribute it locally through brain tissue.

Another approach is temporary and controlled opening of the blood–brain barrier, for example with focused ultrasound combined with circulating microbubbles. Ultrasound can potentially make parts of the barrier more permeable for a limited period, allowing therapeutic agents to enter tumour regions more effectively.

Focused ultrasound plus microbubbles
→ temporary local increase in barrier permeability
→ improved delivery of selected treatment
→ barrier subsequently recovers

Other strategies include implantable local drug-delivery systems, nanoparticles, engineered viruses, and molecules specifically designed for favourable brain penetration. These approaches must still demonstrate that they safely reach infiltrating tumour cells and improve patient outcomes, not merely drug concentrations in tissue.

21.8 Immunotherapy and therapeutic vaccines

Checkpoint inhibitors such as pembrolizumab have transformed treatment for some cancers, but glioblastoma has been much harder. A major reason is that many glioblastomas are immunologically suppressive and may not contain enough active tumour-reactive T cells for checkpoint blockade alone to be effective.

However, researchers are investigating whether immunotherapy can work better in carefully chosen combinations. These include therapeutic cancer vaccines, dendritic-cell vaccines, personalised neoantigen vaccines, oncolytic viruses, T-cell therapies, and combinations with radiation or checkpoint inhibitors.

A *personalised neoantigen vaccine* would aim to identify mutations in an individual's tumour, select candidate neoantigens, and stimulate T cells against those tumour-specific targets.

Tumour sequencing
→ identify candidate tumour mutations
→ select neoantigen targets
→ vaccinate to expand tumour-reactive T cells
→ combine with treatment that reduces immune suppression

An *oncolytic virus* is a virus designed or selected to preferentially infect tumour cells. It may kill infected tumour cells directly and also stimulate local inflammation that helps the immune system recognise the tumour.

Oncolytic virus infects tumour cells
→ tumour-cell damage and death
→ tumour antigens and danger signals are released
→ local immune activation may increase
→ combination immunotherapy may become more effective

These approaches are scientifically compelling, but glioblastoma presents difficult challenges: tumour heterogeneity, limited T-cell infiltration, immune suppression, steroid use to control brain swelling, and the need to avoid damaging normal brain tissue. Their value therefore remains an active area of clinical investigation rather than established standard care.

21.9 Cell therapies and T-cell engagers

CAR-T cells, T-cell receptor therapies, and other engineered immune-cell approaches are also being explored. A potential strategy is to engineer T cells to recognise a tumour-associated cell-surface target, such as EGFRvIII, HER2, IL13R α 2, or another antigen found in subsets of brain tumours.

The central challenge is target selection:

Useful tumour target
→ expressed strongly and broadly on tumour cells
Safe target
→ absent or minimally expressed on essential normal brain tissue
Durable target
→ difficult for tumour cells to lose or bypass

Glioblastoma heterogeneity makes this difficult. If only some tumour cells express the target, treatment can eliminate those cells while leaving antigen-negative cells to expand. Potential future solutions include targeting multiple antigens at once, combining cell therapy with vaccines or checkpoint blockade, or designing treatments that also reshape the tumour microenvironment.

T-cell engagers are another possibility. These therapies could link T cells to a tumour-cell surface target, but the same issues of target specificity, local immune suppression, cytokine toxicity, and brain safety apply.

21.10 Targeted therapies and pathway combinations

Many glioblastomas have abnormalities in growth and survival pathways, including alterations involving EGFR, PI3K–AKT–mTOR signalling, CDK4/6, DNA-damage repair, and other processes. Researchers are testing inhibitors directed at these pathways.

The main difficulty is that inhibiting one pathway may not be sufficient. Tumour cells can activate parallel pathways, adapt their cell state, or evolve resistant subclones. This makes rational combinations attractive, although combinations can also increase toxicity.

A future strategy may look like:

Molecular profiling identifies tumour vulnerability
→ targeted inhibitor suppresses key pathway
Improved brain delivery
→ adequate concentration at infiltrating tumour cells
Immunotherapy or second targeted agent
→ limits adaptation and resistance
Biomarker monitoring
→ detects response and emerging resistance

This is conceptually similar to treatment strategies in other cancers, but it is especially difficult in glioblastoma because the tumour is heterogeneous, infiltrative, and protected by the blood–brain barrier.

21.11 What meaningful progress would look like

Success does not need to mean an immediate universal cure. Important progress could include longer survival with preserved cognition and independence; fewer early recurrences; effective treatments for molecularly defined patient subgroups; improved control after recurrence; reduced need for high-dose steroids; and treatments that maintain quality of life rather than extending survival at an unacceptable neurological cost.

The central challenge is:

**Kill infiltrating, heterogeneous, treatment-resistant tumour cells
while preserving essential brain function**

That is an exceptionally difficult problem. It explains why aggressive brain tumours can feel so cruel and why progress has been slow. But it is also a problem receiving serious effort across neurosurgery, neuro-oncology, cancer biology, immunology, drug delivery, radiotherapy, genomics, imaging, and machine learning.

Better imaging and computational pathology can help characterise tumour heterogeneity. Molecular profiling can help identify treatment-relevant subgroups. Machine learning may support tumour segmentation, surgical planning, prognosis, biomarker discovery, response assessment, clinical-trial design, and integration of imaging, pathology, molecular, and clinical data. These tools will not replace biological and clinical breakthroughs, but they may help make those breakthroughs more targeted, testable, and usable.

The honest conclusion is therefore neither hopelessness nor false reassurance. Aggressive brain tumours such as glioblastoma remain among the most difficult cancers to treat, and current therapies are often inadequate. Yet the disease is not being ignored, and there are credible routes toward better outcomes: more precise tumour classification, improved delivery across the blood–brain barrier, local therapies, immunotherapy combinations, engineered cell therapies, targeted treatments, and biomarker-guided clinical trials. The goal is to turn a disease that is often rapidly fatal into one that can first be controlled for longer, then increasingly treated with durable and less harmful interventions.

22 [26-09-01] Other cancers with poor survival, and why progress is difficult

The term “aggressive cancer” can mean different things. It may describe a cancer that grows rapidly, spreads early, recurs frequently after treatment, is difficult to remove surgically, or has poor survival once it has spread. These features often overlap, but not always.

It is also important not to rank cancers using one number without context. Five-year relative survival is a useful population-level measure, but it depends strongly on stage at diagnosis, subtype, age, geography, access to specialist care, and the period in which patients were treated. It cannot predict an individual patient’s outcome.

Recent United States SEER estimates, based on cohorts diagnosed in approximately 2015–2022 depending on cancer type, illustrate several cancers with persistently poor all-stage five-year relative survival:

22.1 Why these cancers are particularly difficult

Several recurring biological and clinical problems explain why these cancers remain hard to treat. The first is *late diagnosis*. Pancreatic, ovarian, oesophageal, liver, and lung cancers can produce few specific symptoms while they are still localised. By the time symptoms become severe enough to trigger investigation, the cancer may already have invaded nearby structures or spread to distant organs.

The second is *anatomical and surgical difficulty*. Pancreatic tumours can grow around major blood vessels, making complete surgical removal impossible in many patients. Oesophageal surgery is complex and physiologically demanding. Liver cancers may arise in a damaged cirrhotic liver, so treatment must preserve

Cancer type	Approximate five-year relative survival	Important qualification
Glioblastoma	Around 7–8%	This is much lower than the average for the broad “brain and other nervous system” category, which includes less aggressive tumours.
Small-cell lung cancer	Around 8%	A particularly aggressive lung-cancer subtype that is commonly metastatic at diagnosis.
Mesothelioma	Around 10–15%	A rare cancer, usually arising from the pleura around the lungs, often associated with previous asbestos exposure.
Pancreatic cancer	Around 13–14%	Survival is substantially better for the minority diagnosed with localised, surgically removable disease.
Oesophageal cancer	Around 20–22%	Outcomes differ by histology, location, stage, and whether curative treatment is feasible.
Liver and intrahepatic bile-duct cancers	Around 20–22%	This grouped category includes biologically distinct cancers; cirrhosis and impaired liver function can complicate treatment.
Metastatic cancers of many primary sites	Often substantially below the all-stage average	Once distant spread has occurred, cure is much more difficult for most solid tumours.

Table 1: Illustrative recent population-level survival estimates. Exact values depend on registry definitions and diagnosis period; they should not be used as individual prognoses.

enough remaining liver function. Brain tumours may infiltrate regions responsible for speech, movement, vision, memory, or consciousness, limiting how much tissue can safely be removed.

The third is *biological heterogeneity and rapid evolution*. A cancer is not necessarily one uniform population of cells. It can contain multiple subclones with different mutations, cell states, metastatic ability, and drug sensitivities. Treatment may remove sensitive cells while allowing resistant populations to survive and expand.

The fourth is a difficult *tumour microenvironment*. Pancreatic cancer is a particularly clear example. Many pancreatic tumours contain dense fibrotic tissue, abnormal blood vessels, suppressive immune cells, and poor drug penetration. This can make it harder for drugs and anti-tumour immune cells to reach every cancer cell. Liver cancers also develop in a highly abnormal inflammatory and fibrotic tissue environment.

The fifth is a limited number of safe therapeutic vulnerabilities. Some cancers have a dominant, targetable oncogenic driver, such as EGFR-mutant lung cancer or BCR::ABL1-positive chronic myeloid leukaemia. Others are driven by complex combinations of tumour-suppressor loss, chromosomal instability, altered cell states, and interactions with surrounding tissue. These abnormalities may be biologically important but difficult to inhibit selectively with a drug.

Glioblastoma illustrates several of these problems simultaneously:

Diffuse infiltration of normal brain
→ complete surgery is usually impossible
Blood-brain barrier and tissue distribution constraints
→ many systemic drugs do not reach sufficient free concentration
Intratumour heterogeneity and plastic cell states
→ resistant populations can survive treatment
Immunosuppressive local environment
→ immune therapies may have limited activity
Critical normal brain function nearby
→ narrow margin for treatment toxicity

Small-cell lung cancer is difficult for somewhat different reasons. It often grows quickly, metastasises early, and is commonly diagnosed after spread has already occurred. It can respond impressively to initial chemotherapy, sometimes combined with immunotherapy, but relapse is frequent because residual resistant cells remain.

Pancreatic ductal adenocarcinoma is also difficult because it is frequently diagnosed late, is often not surgically removable, and commonly has a dense stromal microenvironment. Its common genomic alterations, including KRAS activation and loss of tumour-suppressor pathways, have historically offered fewer therapeutic opportunities than some other cancers. Progress in KRAS-targeted therapy is important, but it does not yet solve the broader challenge of pancreatic-cancer heterogeneity and drug resistance.

22.2 Is there hope for the future?

There is real reason for scientific optimism, but it should be expressed carefully. For the most difficult cancers, progress is more likely to come from multiple linked improvements than from one universally curative treatment.

Earlier detection could have a large effect because localised disease is often much more treatable than metastatic disease. This may involve improved access to proven screening where it already exists, surveillance of people at high risk, better imaging, blood-based biomarkers, and combinations of molecular and clinical risk signals. A screening test must, however, detect clinically important cancers early enough to improve outcomes while avoiding excessive false positives, unnecessary procedures, overdiagnosis, and inequitable access.

Precision oncology may identify smaller groups of patients whose tumours have an actionable dependency. Examples include mutation-selective kinase inhibitors, therapies directed at DNA-repair deficiencies, and biomarker-guided antibodies or antibody–drug conjugates. For pancreatic cancer, one important future direction is to identify biologically meaningful subgroups rather than treating all patients as though they have one disease.

Drug delivery is especially important for brain tumours. Potential approaches include designing molecules with improved brain penetration, temporarily altering or bypassing the blood–brain barrier, local delivery after surgery, convection-enhanced delivery, and radiopharmaceutical or nanoparticle strategies. Each approach must demonstrate not only that drug reaches the tumour, but that it reaches the relevant tumour cells at a useful and safe concentration.

Immunotherapy may also improve through better patient selection and combinations. A checkpoint inhibitor works best when a tumour contains recognisable antigens, can present them through MHC, contains or can recruit functional T cells, and relies materially on the particular inhibitory pathway being blocked. In “cold” tumours with few functional anti-tumour T cells, checkpoint blockade alone may be insufficient. Vaccines, T-cell engagers, engineered cell therapies, local radiation, targeted therapy, or microenvironment-modifying treatments might help create a more effective immune response, but these strategies remain technically and clinically challenging.

The most credible expectation is therefore:

Better biological classification
+ earlier detection where feasible
+ improved surgery and radiotherapy
+ more effective systemic treatment
+ rational combinations
+ measurement of resistance over time
→ stepwise improvement in outcomes

For some patients, especially those with a targetable subtype or disease detected early, advances may be substantial. For the population as a whole, however, major improvements require rigorous clinical trials that show a meaningful benefit in survival, symptoms, function, or quality of life.

23 [26-09-01] Cancer types in which survival has improved substantially

Cancer survival has improved substantially over the past several decades for many diseases. The reasons are not uniform. In some cancers, progress has come mainly from prevention or earlier diagnosis. In others, it has come from better surgery, radiotherapy, chemotherapy, endocrine therapy, molecularly targeted treatment, immunotherapy, supportive care, or more precise selection of treatment intensity.

A useful caution is that population survival can improve even when a new treatment has only a modest effect on advanced disease, because more cancers are diagnosed at an earlier stage. Conversely, a major therapeutic breakthrough in metastatic disease may transform outcomes for a subgroup but have only a limited immediate effect on the all-stage population survival statistic.

23.1 Childhood acute lymphoblastic leukaemia

Childhood acute lymphoblastic leukaemia, or ALL, is one of the clearest long-term success stories in cancer medicine. In the 1960s, long-term survival was uncommon. In high-income countries today, approximately 90% of children with ALL survive for at least five years, although outcomes remain worse for some high-risk molecular subtypes and in settings with limited access to specialist care.

This progress did not arise from one drug. It came from successive clinical trials that refined multi-agent chemotherapy, identified the importance of treatment phases such as induction, consolidation, and maintenance, and improved treatment of disease in the central nervous system. Risk stratification became increasingly important: chromosome-level alterations, molecular features, and measurement of minimal residual disease after early treatment can help identify patients who need more intensive therapy and those who may avoid unnecessary toxicity.

Supportive care was also essential. Better prevention and treatment of infection, safer blood-product support, improved management of tumour lysis syndrome, and specialist paediatric oncology care allowed more children to receive potentially curative therapy safely. More recently, targeted treatments and immune therapies, including bispecific T-cell engagers and CAR-T therapy, have created new options for some patients with relapsed or refractory disease.

23.2 Chronic myeloid leukaemia

Chronic myeloid leukaemia, or CML, is a particularly clear example of molecularly targeted therapy changing the natural history of a cancer. Most CML is driven by the BCR::ABL1 fusion gene, which produces an abnormally active tyrosine kinase.

The central logic is:

BCR::ABL1 fusion

- constitutively active kinase signalling
- excessive growth and survival of myeloid cells

ABL tyrosine-kinase inhibitor

- reduced BCR::ABL1 signalling
- durable disease control in many patients

The introduction of imatinib, followed by later-generation ABL inhibitors, transformed CML from a disease with a substantial risk of progression to one in which many patients can expect long-term control and near-normal life expectancy when appropriately monitored. This success depended on a strong causal target, a druggable kinase site, sensitive molecular monitoring, and the ability to develop alternative inhibitors for some resistance mutations.

23.3 Metastatic melanoma

The improvement in advanced melanoma since the early 2010s is one of the most striking recent examples in adult solid tumours. Historically, metastatic melanoma had few effective systemic treatments and a poor median survival. Checkpoint inhibitors, particularly antibodies targeting CTLA-4 and PD-1, demonstrated that a subset of patients can achieve deep and durable responses.

For tumours with activating BRAF mutations, combined BRAF and MEK inhibition created an additional targeted-treatment strategy. These approaches work differently:

BRAF-mutant melanoma

- BRAF/MEK inhibitors directly suppress oncogenic MAPK signalling
- Immunogenic melanoma with an effective anti-tumour immune response
- checkpoint inhibition can reduce inhibitory immune signalling
 - tumour-reactive T cells may produce durable control

Not every patient benefits, immune-related adverse events can be serious, and resistance remains important. Nevertheless, the change in long-term outcomes for a proportion of patients with previously lethal metastatic melanoma is genuine evidence that durable control of advanced solid tumours is possible.

23.4 Lung cancer

Lung cancer remains a leading cause of cancer mortality, but survival has improved meaningfully compared with the 1990s. The improvement has several causes. Reduced smoking prevalence has lowered incidence and mortality over time. Low-dose CT screening for people at sufficiently high risk can detect some lung cancers earlier, when surgery or curative-intent radiotherapy is possible.

Treatment has also changed substantially. Non-small-cell lung cancer is now divided into clinically relevant molecular subgroups. Some tumours contain actionable alterations in genes such as EGFR, ALK, ROS1, BRAF, MET, RET, NTRK, KRAS, or HER2. Patients with an appropriate alteration may receive a matched targeted therapy, while immunotherapy has improved outcomes in selected patients without such a driver or in combination approaches.

The key lesson is that “lung cancer” is not one disease:

Histology

- + genomic driver alterations
- + PD-L1 and other immune features
- + stage and resectability
- + patient fitness and coexisting disease
- treatment strategy and expected outcome

Despite this progress, small-cell lung cancer and metastatic disease without a durable treatment response remain major unmet needs.

23.5 HER2-positive breast cancer

HER2-positive breast cancer was historically associated with an aggressive clinical course because ERBB2 amplification or HER2 overexpression can drive strong growth and survival signalling. The development of trastuzumab showed that this biology could be exploited therapeutically.

Over time, the treatment landscape expanded to include dual HER2 blockade, antibody–drug conjugates such as trastuzumab emtansine and trastuzumab deruxtecan, and selected small-molecule HER2 inhibitors. These therapies have improved outcomes in early-stage disease and extended survival in metastatic disease.

This is an important example of a predictive biomarker becoming clinically useful:

ERBB2 amplification or HER2 overexpression

- identifies a tumour subtype more likely to benefit from HER2-directed treatment

HER2-directed therapy

- pathway perturbation, immune effector functions, and/or targeted cytotoxic payload delivery
- improved clinical outcomes in appropriate patients

The progress required reliable HER2 testing, well-designed adjuvant and metastatic clinical trials, sequential development of several therapeutic modalities, and improved management of toxicities. It also illustrates that a biomarker identifies a higher-probability group; it does not guarantee that every biomarker-positive tumour will respond.

23.6 Colorectal cancer

Colorectal cancer provides an example in which prevention, screening, surgery, and systemic treatment have all contributed. Colonoscopy can identify and remove some precancerous polyps, thereby preventing some cancers rather than merely detecting them earlier. Faecal blood testing and other screening approaches can also identify cancers at a more treatable stage.

For patients with established colorectal cancer, improved surgery, radiotherapy for selected rectal cancers, combination chemotherapy, biological drugs directed at VEGF or EGFR pathways in appropriate molecular contexts, and treatment guided by RAS, BRAF, mismatch-repair, and other biomarkers have improved outcomes. Immune checkpoint inhibitors have been particularly important for the subgroup with mismatch-repair-deficient or microsatellite-instability-high tumours.

Population-level improvement is real, but it should not obscure concerns about rising incidence of early-onset colorectal cancer in some countries, unequal screening access, and the continuing difficulty of treating metastatic disease.

23.7 Testicular germ-cell tumours

Testicular cancer is now among the most curable solid cancers, including many cases with metastatic spread. The central historical advance was cisplatin-based combination chemotherapy, introduced several decades ago and refined through clinical trials, together with surgery, tumour-marker monitoring, risk classification, and careful use of radiotherapy in selected settings.

This example is useful because it shows that cure of disseminated cancer is sometimes possible when several conditions align: the cancer is highly sensitive to systemic therapy, treatment can be delivered at sufficiently effective intensity, resistance is relatively limited, and specialised multidisciplinary management is available.

23.8 What the successful examples have in common

The successful examples do not imply that every cancer can be solved by the same strategy. Rather, they show several recurrent routes to progress:

```
Prevention and risk reduction
→ fewer cancers occur
Screening and early diagnosis
→ more cancers are treated while localised
Better pathology and molecular classification
→ patients are matched to more appropriate treatment
Effective systemic therapies
→ improved cure rates, longer survival, or durable control
Refined surgery and radiotherapy
→ stronger local control with less harm
Supportive care and specialist multidisciplinary treatment
→ more patients can complete effective therapy safely
Clinical trials and long-term follow-up
→ treatment strategies are progressively improved
```

The most important general lesson is that cancer progress is usually cumulative. A breakthrough drug matters, but its full benefit often depends on accurate diagnosis, biomarker testing, appropriate sequencing or combination with other treatments, management of toxicity, access to specialist care, and continued research into resistance.

24 [26-09-01] CD19-directed CAR-T therapy, and why severe immune toxicity can occur

(<https://www.fiercebiotech.com/biotech/novartis-bristol-myers-squibb-halt-car-t-cell-trials-due-immune->

The reported pauses of selected Novartis and Bristol Myers Squibb CAR-T trials followed serious immune-related events, including three deaths. At the time of writing, this is an evolving safety investigation. A trial pause is an appropriate precaution, but it does *not* by itself establish that the CAR-T product caused every death, nor does it mean that all CD19-directed CAR-T therapies have the same risk. Investigators, companies, regulators, and independent safety-monitoring groups need to determine the clinical sequence, alternative explanations such as infection or disease progression, product-specific factors, dose and treatment context, and whether there is a reproducible causal pattern.

24.1 What “CD19-directed” means

CD19-directed CAR-T therapy is the same basic technology described in Section 11, but with a specific target antigen: *CD19*. CD19 is a cell-surface protein expressed by most B cells, including many malignant B cells in B-cell leukaemias and lymphomas.

```
Patient T cells are collected
→ engineered to express a CAR recognising CD19
→ expanded and infused back into the patient
→ CAR-T cells bind CD19-positive cells
→ T-cell activation and killing of those cells
```

Thus, CD19 CAR-T cells can kill malignant B cells without requiring peptide presentation by MHC. They also usually kill normal CD19-positive B cells. This *on-target, off-tumour* effect is expected: it causes B-cell depletion, or *B-cell aplasia*. Loss of B cells can reduce antibody production and increase susceptibility to infection, sometimes requiring immunoglobulin replacement and long-term monitoring.

The same strategy is now also being explored in autoimmune diseases. The aim there is not to kill a tumour, but to remove pathological B-cell populations and allow the B-cell compartment to re-form in a less disease-promoting state.

24.2 How can CAR-T treatment go seriously wrong?

CAR-T cells are living, self-amplifying immune cells. Once they encounter abundant target antigen, they can expand rapidly and release large quantities of inflammatory signalling molecules, called cytokines. The intended immune response can therefore become excessive and systemic.

CAR-T cells encounter many CD19-positive cells
→ rapid T-cell activation and expansion
→ cytokine release and broad immune activation
→ fever, low blood pressure, low oxygen, organ stress,
neurological toxicity, or severe inflammatory syndromes

The best-known acute toxicities are *cytokine-release syndrome*, or CRS, and *immune effector cell-associated neurotoxicity syndrome*, or ICANS. Severe CRS can produce profound hypotension, capillary leak, respiratory failure, coagulopathy, kidney injury, cardiac stress, and multi-organ dysfunction. Less commonly, an extreme inflammatory state resembling haemophagocytic lymphohistiocytosis can occur. These syndromes can be fatal despite intensive monitoring and treatments such as corticosteroids and cytokine-directed therapy.

Risk is influenced by many factors: tumour burden or total amount of CD19-positive tissue, the CAR design and dose, the pace of CAR-T expansion, conditioning chemotherapy, infection, patient fitness and organ function, and the disease setting. A complication may also arise from the interaction of several factors rather than from one simple product defect.

The central lesson is that CAR-T therapy deliberately creates a powerful immune attack. Its effectiveness and its danger arise from the same property:

Strong CAR-T activation
→ potentially effective elimination of target-positive cells
Excessive, poorly controlled, or system-wide activation
→ potentially life-threatening inflammation and tissue injury

These events are therefore a reminder that target specificity does not guarantee complete safety. CD19 may direct CAR-T cells mainly towards B-lineage cells, but the immune response triggered after target recognition can affect the whole body. Careful patient selection, dose design, inpatient monitoring, rapid recognition of inflammatory toxicity, and continued investigation of severe events are essential parts of CAR-T development.

24.3 CD19 CAR-T cells in rheumatoid arthritis

(*CD19 CAR T cell therapy for treatment-refractory seropositive rheumatoid arthritis: a phase 1 trial*, <https://www.nature.com/articles/s41591-026-04603-3>)

This study uses the same basic CAR-T principle described in Section 11, but applies it to an autoimmune disease rather than cancer.

CD19 is a protein found on most B cells. In seropositive rheumatoid arthritis, B cells can contribute to disease by producing autoantibodies, presenting antigens to T cells, and supporting inflammatory immune responses. The investigators therefore used CD19-directed CAR-T cells to deplete B cells, including B cells in tissues such as synovium, bone marrow, and lymph nodes.

Patient's T cells are collected

- engineered to recognise CD19
- expanded and returned after lymphodepleting chemotherapy
- CAR-T cells kill CD19-positive B cells
- autoantibody-producing immune response may be reduced
- rheumatoid inflammation may improve

The study reports the Phase 1 part of the COMPARE trial. Six adults with severe, active, ACPA-positive rheumatoid arthritis that had not responded adequately to several previous treatments received one infusion of the autologous CD19 CAR-T product mivocabtagene autoleucel, or miv-cel.

The treatment produced profound depletion of CD19-positive B cells in blood and sampled tissues. Autoantibody levels fell substantially, and all six patients showed some improvement in disease activity. At the latest follow-up of 36–52 weeks, three patients had achieved both DAS28-CRP remission and an ACR70 response; several remained free of conventional disease-modifying treatment.

However, this was a very small, non-randomised Phase 1 study. Its primary aim was safety, not definitive proof of clinical benefit. All six patients developed low-grade cytokine-release syndrome, but there were no cases of ICANS or treatment-related serious adverse events. Larger, controlled studies with longer follow-up are needed to establish durability, infection risk, effects on normal immunity, and the overall benefit–risk balance.

24.4 Autogene cevumeran: an individualised mRNA cancer vaccine

(<https://www.biontech.com/int/en/home/mediaroom/news/statements/2026/08/BioNTech-Provides-Update-on-Phase-2-Clinical-Trial-of-Autogene-Cevumeran-in-Resected-Colorectal-Cancer.html>)

Autogene cevumeran, also called BNT122 or RO7198457, is an investigational individualised mRNA neoantigen vaccine. Like intismeran autogene, it is made separately for each patient using information from that patient's tumour.

Tumour and matched normal samples

- identify tumour-specific mutations
- select candidate neoantigens
- encode selected neoantigens in mRNA
- manufacture a patient-specific vaccine
- stimulate neoantigen-reactive T-cell responses

The intended mechanism is therefore similar to the personalised mRNA vaccine described in Section 11. The vaccine does not directly kill cancer cells. Instead, its mRNA provides temporary instructions for cells to produce selected tumour-derived antigenic material. This should help prime or expand T cells capable of recognising tumour cells that display the corresponding mutation-derived peptides through MHC.

However, the recent colorectal-cancer result was negative. BioNTech and Genentech/Roche terminated a Phase 2 trial of autogene cevumeran used alone after surgery for high-risk stage II or stage III colorectal cancer in patients with detectable circulating tumour DNA, or ctDNA. Detectable ctDNA after surgery suggests that microscopic residual cancer may remain and that the risk of recurrence is relatively high.

The independent data monitoring committee concluded that continuing was unlikely to demonstrate benefit. It had also observed a numerical imbalance in overall survival between the trial groups. The announcement stated that no new product-related safety signal had been identified, so the decision was based on lack of expected efficacy rather than an identified toxicity problem.

This result does *not* show that personalised mRNA cancer vaccines cannot work. It shows that this vaccine, as *monotherapy* in this particular colorectal-cancer setting, did not provide sufficient evidence of benefit. Colorectal cancer can have an immunosuppressive tumour environment, and many tumours are relatively

unresponsive to immunotherapy unless they have particular features, such as mismatch-repair deficiency or high microsatellite instability.

Other trials of autogene cevumeran continue, including a study combining it with checkpoint inhibition and chemotherapy after surgery for pancreatic ductal adenocarcinoma. The broader question remains whether personalised vaccines work best in selected tumour types, in minimal-residual-disease settings, or in combinations that both generate tumour-reactive T cells and help those T cells function effectively.

25 [26-09-01] Why did IL-6 inhibition fail in cardiovascular disease?

(<https://thecodon.substack.com/p/why-did-il-6-inhibition-fail>)

This is an instructive example of a drug programme with strong biological rationale, genetic support, and clear target engagement that nevertheless failed to improve the clinical outcome that mattered.

The drug was *ziltivekimab*, an investigational antibody that binds and neutralises the inflammatory cytokine interleukin-6, or IL-6. The Phase 3 *ZEUS* trial tested monthly ziltivekimab in more than 6,300 patients with established atherosclerotic cardiovascular disease, chronic kidney disease, and elevated high-sensitivity C-reactive protein, or hsCRP.

Atherosclerosis involves both cholesterol accumulation and inflammation. A simplified inflammatory pathway is:

```
Inflammatory stimuli
→ IL-1 $\beta$ 
→ IL-6 signalling
→ liver production of CRP
→ systemic inflammation and cardiovascular risk
```

IL-6 was an attractive target because higher IL-6 and CRP levels are associated with cardiovascular risk; genetic studies of the IL-6 receptor pathway suggested a possible causal role; and earlier evidence, including the CANTOS trial of IL-1 β inhibition, supported the broader inflammatory hypothesis.

Ziltivekimab did what it was designed to do. It reduced free IL-6 and hsCRP, confirming *target engagement* and pathway modulation. However, it did not reduce major adverse cardiovascular events: cardiovascular death, non-fatal myocardial infarction, or non-fatal stroke.

```
Ziltivekimab versus placebo in ZEUS:
Hazard ratio for major adverse cardiovascular events = 0.99
95% confidence interval = 0.88-1.11
Interpretation:
no evidence of reduced cardiovascular-event risk
in the studied population
```

The exact reason remains uncertain, especially because complete peer-reviewed trial data were not yet available as of 1 September 2026. However, several explanations are plausible.

First, a genetic effect acting throughout life is not equivalent to pharmacologically blocking a pathway in older patients with established, advanced vascular disease. IL-6 signalling may contribute to the slow development of plaques over decades, while late intervention may be unable to reverse sufficiently established plaque biology.

Second, the most prominent genetic evidence concerns *IL6R*, the gene encoding the IL-6 receptor, whereas ziltivekimab neutralises the *IL-6 ligand*. These interventions affect the same broad pathway but are not biologically identical. The IL-6 receptor exists in membrane-bound and soluble forms, and IL-6 can signal through both so-called classical signalling and trans-signalling pathways.

Third, hsCRP is an informative pharmacodynamic biomarker, but lowering it does not automatically mean that the particular inflammatory processes causing plaque rupture and thrombosis have been sufficiently changed.

```
Drug lowers IL-6
→ hsCRP falls
→ confirms pathway effect
But:
Biomarker improvement
≠ plaque stabilisation
≠ fewer heart attacks or strokes
```

Finally, inflammation is not one single therapeutic target. Different inflammatory pathways may have different roles in plaque formation, acute plaque instability, thrombosis, and tissue repair. The successful CANTOS trial with IL-1 β inhibition and mixed results with other anti-inflammatory interventions suggest that *which* pathway is modified, in *which* patients, at *which* stage of disease, matters greatly.

There was also a safety trade-off. Serious infections occurred more often with ziltivekimab, consistent with the normal role of IL-6 in host defence. Overall adverse-event rates and all-cause mortality were reportedly similar between trial groups.

The core lesson is:

```
Human genetics
+ plausible disease biology
+ reduced inflammatory biomarkers
+ successful target engagement
≠ established patient benefit
```

ZEUS does not show that inflammation is unimportant in atherosclerosis, or that IL-6 biology is irrelevant. It shows that neutralising IL-6 with ziltivekimab did not reduce cardiovascular events in this particular high-risk population. Two further ziltivekimab outcome trials, in heart failure and after acute myocardial infarction, were still ongoing and expected to report in 2027.

26 [26-09-01] Seeing tumour neoantigens and T cells in the same spatial neighbourhood

(*Single-nucleus multimodal spatial transcriptomics reveals spatial colocalization of neoantigen-expressing tumor cells and cognate T cells*, <https://www.nature.com/articles/s41587-026-03194-1>)

This paper describes a new experimental method, called *Slide-GoTags*, for studying an important question in cancer immunology:

```
Are T cells that can recognise a tumour neoantigen
actually located near tumour cells expressing that neoantigen?
```

This matters because a tumour sample may contain both mutation-derived neoantigens and many T cells, but those observations alone do not show that the relevant T cells are reaching the relevant cancer cells, recognising them, or functioning effectively.

26.1 The biological idea

A tumour mutation can alter a protein sequence and create a *neoantigen*. If the altered protein is expressed, processed into peptides, loaded onto MHC molecules, and displayed at the tumour-cell surface, a T cell with a suitable T-cell receptor, or TCR, may recognise it.

```
Tumour mutation
→ altered protein
→ mutation-derived peptide, or neoantigen
→ peptide displayed by MHC
→ recognition by a T cell with a compatible TCR
→ potential anti-tumour immune attack
```

A *cognate* T cell is a T cell whose TCR recognises a particular peptide–MHC complex. In this paper, the authors ask whether neoantigen-expressing tumour cells and their cognate T cells are spatially close together inside the actual tumour tissue.

26.2 What Slide-GoTags measures

Ordinary single-cell RNA sequencing can describe which cells are present and which genes they express, but cells must usually be removed from the tissue first. This loses their original spatial location.

Slide-GoTags instead combines several measurements from the same frozen tissue section:

```
Single-nucleus RNA sequencing
→ identifies cell types and cellular states
Targeted transcript genotyping
→ detects expression of selected tumour mutations
TCR sequencing
→ identifies individual T-cell clones
Spatial barcodes
→ estimate where each nucleus was located in the tissue
```

This allows the researchers to connect four kinds of information: *which tumour cells express a mutation, which T-cell clones are present, what functional state those cells are in, and where they are located relative to one another.*

26.3 What the paper found

In mouse and human tumours, expanded neoantigen-specific T-cell clones were not distributed randomly. They were enriched near tumour cells expressing their corresponding neoantigens.

In simple terms:

```
Neoantigen-expressing tumour-cell region
→ nearby enrichment of cognate T cells
→ local T-cell activation and clonal expansion
→ evidence of a spatially organised anti-tumour immune response
```

The paper also describes *immunogenic niches*: local tumour regions with interferon-associated signalling, neoantigen expression, activated or progenitor-exhausted T cells, and evidence of local immune engagement. These niches were more prominent in immunologically “hot” tumours than in “cold” tumours.

A progenitor-exhausted T cell is a T cell that shows some features of chronic stimulation and exhaustion, but retains proliferative capacity and may be capable of responding to checkpoint inhibition. This state is thought to be particularly important for sustained responses to PD-1-directed therapy.

26.4 Why this is interesting

The paper strengthens the idea that effective anti-tumour immunity is not only about whether a tumour contains T cells or whether it has mutations. It may depend on a local, spatially organised interaction:

```
Neoantigen is expressed
+ peptide is presented by MHC
+ cognate T cell enters the relevant tumour region
+ T cell remains functional enough to respond
+ local microenvironment permits killing
→ plausible effective anti-tumour immunity
```

This could eventually help explain why some tumours respond to checkpoint inhibitors while others do not, identify useful neoantigens for personalised cancer vaccines or TCR therapies, and improve methods for finding which tumour antigen a particular TCR recognises. The latter problem is often called *TCR deorphanisation*.

The work is primarily a powerful research and measurement advance, rather than a new treatment. Spatial proximity supports the hypothesis of a biologically meaningful interaction, but it does not by itself prove that every nearby T cell is killing tumour cells or that the interaction will produce clinical benefit.

27 [26-09-01] SHR-A1811 in early HER2-low breast cancer

(*Neoadjuvant anti-HER2 antibody-drug conjugate SHR-A1811 in Hormone Receptor-positive, HER2-low Breast Cancer: A Phase 2 Clinical Trial (HELEN-015)*, <https://www.nature.com/articles/s41467-026-77041-x>)

SHR-A1811, also called *trastuzumab rezetecan*, is an investigational HER2-directed antibody–drug conjugate, or ADC. It is conceptually similar to trastuzumab deruxtecan, or T-DXd (Enhertu): both use a trastuzumab-based antibody to bind HER2 and deliver a potent topoisomerase-I-inhibitor payload into HER2-expressing cells.

```
HER2-directed ADC binds HER2 on tumour cell
→ ADC is internalised
→ linker is processed
→ cytotoxic payload is released
→ topoisomerase-I inhibition causes DNA damage
→ susceptible tumour cell may die
```

The two ADCs therefore share the same broad therapeutic logic, but they are not identical products. T-DXd carries the topoisomerase-I inhibitor DXd, with an average drug-to-antibody ratio, or DAR, of approximately eight payload molecules per antibody. SHR-A1811 carries a related but chemically modified topoisomerase-I-inhibitor payload, called SHR169265, and has an average DAR of approximately six.

These design choices aim to retain potent payload delivery and a bystander effect while improving ADC stability and reducing free systemic payload exposure. This could potentially affect both efficacy and toxicity, including the risk of interstitial lung disease or pneumonitis. However, SHR-A1811 and T-DXd have not been proven equivalent, superior, or safer than one another in a direct randomised comparison.

The HELEN-015 study was a small, single-arm Phase 2 trial of SHR-A1811 given *before surgery*, described as neoadjuvant treatment, in 35 patients with hormone-receptor-positive, HER2-low early breast cancer. HER2-low generally means HER2 immunohistochemistry 1+, or 2+ without HER2 gene amplification. These tumours are not HER2-positive in the traditional sense, but may still express enough HER2 for an ADC to bind and deliver payload.

The study asked whether SHR-A1811 alone could shrink these tumours before surgery. The reported radiological objective response rate was 74.3%: 26 of 35 patients had a partial response. The remaining nine had stable disease. No patient had progressive disease during treatment.

However, none of the 35 patients achieved a pathological complete response, or pCR. A pCR means that no invasive cancer is found in the breast and sampled lymph nodes at surgery after neoadjuvant treatment.

Tumour shrinkage on imaging is encouraging evidence of biological activity, but it is not the same as eliminating all invasive tumour cells.

SHR-A1811 monotherapy before surgery
→ substantial tumour shrinkage in many patients
But:
0% pathological complete response
→ residual invasive tumour remained at surgery

This is an early, exploratory result rather than evidence that SHR-A1811 should replace standard treatment. In hormone-receptor-positive, HER2-low early breast cancer, endocrine therapy, chemotherapy when indicated, surgery, radiotherapy, and adjuvant systemic treatment are selected according to tumour stage, biology, and recurrence risk. A larger randomised trial would be needed to determine whether adding or substituting SHR-A1811 improves patient-relevant outcomes, such as recurrence-free or overall survival, with acceptable toxicity.

28 [26-09-01] AI-designed antibodies: a real scientific advance, but not automatically better medicines

(*The antibody revolution is real. The business, less so*, <https://couteiral.substack.com/p/the-antibody-revolution-is-real-the>)

This discussion concerns a recent apparent advance in computational antibody design. The central claim is nuanced:

Modern AI and structure-based methods may now be able to design
new antibodies against new target sites with useful experimental hit rates.
But:
Designing an antibody that binds a protein
≠ designing a useful medicine
≠ establishing a durable drug-discovery business

Antibodies are proteins that can bind a selected molecular target with high affinity and specificity. As discussed in Chapter 9, conventional therapeutic antibodies are especially useful for extracellular proteins and cell-surface targets, because they generally do not enter living cells efficiently.

They can work in several ways. An antibody may neutralise a soluble ligand such as TNF, block or activate a receptor, alter receptor internalisation, recruit immune effector cells through its Fc region, or deliver a cytotoxic payload as part of an antibody–drug conjugate, or ADC.

Antibody binds exposed target
→ target interaction, signalling, or cell behaviour changes
→ disease biology may be altered
→ patient may benefit

The important word is *may*. Binding is only the beginning of the therapeutic chain.

28.1 What is new about AI antibody design?

Historically, antibody discovery has often involved immunising animals, screening very large libraries of antibody variants, and experimentally refining promising hits. These approaches remain powerful, but finding an antibody with the required combination of binding, selectivity, functional activity, stability, manufacturability, pharmacokinetics, and safety can require substantial time and experimental work.

The newer computational approaches aim to start from a target protein structure, or a selected target region called an *epitope*, and generate antibody sequences predicted to bind it.

Target protein structure or desired epitope
→ computationally propose antibody-binding regions and sequences
→ manufacture a limited number of candidates
→ experimentally test binding and function
→ identify and refine credible leads

The potentially important advance is not simply that a computer can propose antibody sequences. Computational protein design and machine learning have supported protein engineering for many years. The more ambitious claim is that such methods can now produce experimentally active antibodies against *new* targets or epitopes, sometimes with relatively little subsequent optimisation.

Some early highly publicised claims were less convincing than their marketing suggested. For example, modifying a known antibody against an established target is scientifically different from designing a new antibody against a previously unaddressed target surface.

Optimising a known antibody
against a well-characterised target
≠
Creating a new antibody against a new target or epitope
with little prior experimental optimisation

Recent reports from several groups suggest that the latter capability may be becoming more credible. This would be a genuine technical achievement, especially if it proves reproducible across many targets and laboratories.

28.2 Why a strong binder is not automatically a drug

An antibody can bind its target tightly and still fail as a medicine. The target must be biologically important in the relevant patients, the antibody must produce the *right* functional effect, it must reach the relevant tissue, and its benefit must justify toxicity and treatment burden.

The epitope matters. Two antibodies can bind the same protein but have very different pharmacology. One may block a disease-relevant protein–protein interaction; another may bind without changing function; a third may stabilise an undesired protein state or activate an unwanted signal.

Same target protein
+ different antibody epitope
→ potentially different mechanism
→ potentially different efficacy, safety, and resistance

An antibody must also have suitable *developability*. It needs to be expressed and purified at practical scale, remain stable during manufacturing and storage, avoid excessive aggregation, support an appropriate formulation, and have acceptable pharmacokinetic behaviour.

Tissue distribution is another challenge. Antibodies often have a relatively long half-life in blood, but may penetrate solid tumours incompletely. A strong binder in a purified-protein assay may therefore fail to achieve enough free concentration throughout a heterogeneous tumour.

Safety remains decisive. An antibody may bind its intended target selectively but still cause harm if the target has an important normal physiological role or is expressed in essential healthy tissues. For an ADC, safety additionally depends on linker stability, payload release, payload distribution, and sensitivity of normal cells.

```
Target binding
≠ desired pharmacology
Desired pharmacology
≠ adequate tumour or tissue exposure
Adequate exposure
≠ acceptable safety
Acceptable safety
≠ meaningful clinical benefit
```

28.3 The target-selection problem remains

The most difficult question in drug discovery is often not “Can we make an antibody against this protein?” It is:

```
Does this target matter enough in the relevant disease and patients,
and can changing it produce a useful benefit at an acceptable risk?
```

Many extracellular proteins and cell-surface targets are technically accessible to antibodies. Far fewer have the right combination of causal disease relevance, suitable mechanism, biomarker strategy, therapeutic window, and clinical-development opportunity.

A computational design system may accelerate the step from a selected target to candidate antibody molecules. However, it does not itself validate the disease hypothesis or show which patient population will benefit.

```
Disease mechanism
→ target hypothesis
→ modality and desired mechanism
→ antibody design
→ exposure and pharmacology
→ clinical benefit
```

AI antibody design can contribute strongly to the *antibody design* step. It does not remove uncertainty at any of the other steps.

28.4 Why the business case is less obvious

A scientific capability does not automatically create a durable business advantage. If many companies can rapidly design acceptable antibodies against the same target, antibody generation itself may become less scarce and less differentiated.

The durable sources of value may instead remain:

```
Novel and clinically validated targets
+ proprietary biological and clinical data
+ differentiated therapeutic mechanisms
+ successful translation from model systems to patients
+ clinical-development expertise
+ manufacturing, regulatory, and commercial capability
```

An AI-antibody platform may therefore be most valuable when it generates a clearly differentiated therapeutic asset: for example, an antibody to an otherwise inaccessible but clinically important epitope, a molecule with a new functional mechanism, or a candidate with meaningfully improved developability or safety.

The strongest demonstration of this technology would not merely be a designed antibody that binds in vitro. It would be a repeatable path to medicines with meaningful clinical differentiation:

```
Novel target or epitope
+ reproducible binding and functional activity
+ differentiated pharmacology
+ suitable PK, tissue distribution, and developability
+ acceptable safety margin
+ meaningful patient benefit
→ clinically valuable medicine
```

The overall conclusion is therefore cautiously optimistic. The ability to generate new antibodies computationally may become an important enabling technology that makes discovery faster and expands the set of targetable protein surfaces. But it remains one component of drug discovery rather than a replacement for disease biology, target validation, pharmacology, clinical trials, and careful benefit–risk assessment.

29 [26-09-02] Personalised neoantigen vaccines in glioblastoma

(https://ascopubs.org/doi/pdf/10.1200/JCO.2026.44.16_suppl.2006, <https://www.cancernetwork.com/view/personalized-cancer-vaccine-pembrolizumab-feasible-effective-newly-diagnosed-gbm>, <https://www.nature.com/articles/s43018-026-01163-w>)

These reports describe a promising but still early approach to glioblastoma, or GBM: using a *personalised neoantigen vaccine* to generate T-cell responses against mutations unique to an individual patient’s tumour.

This connects directly to Section 11. The central idea is:

```
Sequence a patient’s tumour and matched normal sample
→ identify tumour-specific mutations
→ select candidate neoantigens
→ manufacture a patient-specific vaccine
→ activate or expand neoantigen-reactive T cells
→ aim for T cells to recognise and attack tumour cells
```

A neoantigen vaccine does not directly kill cancer cells in the way that chemotherapy or an ADC can. Instead, it aims to teach, expand, or sustain the patient’s own adaptive immune response. For this to work, the selected mutation-derived peptides must be produced by tumour cells, processed and presented on MHC molecules, recognised by suitable T-cell receptors, and reached by functional T cells inside the tumour.

29.1 Why this is particularly interesting in glioblastoma

Glioblastoma is an especially difficult cancer to treat, as described in the previous section on aggressive brain tumours. Complete surgical removal is usually impossible because tumour cells infiltrate normal brain; the blood–brain barrier can limit drug delivery; tumour cells are heterogeneous and evolve resistance; and the local tumour microenvironment is often strongly immunosuppressive.

GBM is also often described as an immunologically “cold” tumour. This means that it may contain relatively few active anti-tumour T cells, limited inflammatory signalling, impaired antigen presentation, and local mechanisms that suppress immune attack.

A vaccine strategy attempts to address one part of this problem:

Personalised neoantigen vaccine

→ generate or expand tumour-specific T cells

Checkpoint inhibitor, such as pembrolizumab

→ reduce PD-1-mediated inhibition of T cells

Combination

→ aim to create an immune response

and help it remain active in the tumour

This is conceptually similar to the rationale for combining intismeran autogene with pembrolizumab in melanoma. However, the tumour biology and clinical challenges in GBM are very different, and success in melanoma does not imply success in glioblastoma.

29.2 NeoVax plus pembrolizumab: a personalised *peptide* vaccine

The JCO conference abstract and associated reporting describe an early Phase 1 study of *NeoVax*, a personalised long-peptide neoantigen vaccine, combined with the PD-1-blocking antibody pembrolizumab in newly diagnosed GBM.

This is *not* an mRNA vaccine. After sequencing each patient's tumour, investigators selected up to approximately 20 candidate neoantigens and chemically synthesised corresponding long peptides. These peptides were given with poly-ICLC, an immune-stimulating adjuvant intended to improve T-cell priming.

Patient-specific tumour mutations

→ selected neoantigen sequences

→ synthetic long peptides plus immune adjuvant

→ antigen uptake and presentation by antigen-presenting cells

→ activation of neoantigen-reactive CD4 and CD8 T cells

The reported study found that this treatment was feasible to manufacture and administer, with evidence that it induced durable neoantigen-specific T-cell responses. The investigators also reported changes consistent with a more immunologically active tumour microenvironment.

Exploratory survival comparisons were encouraging but require great caution. Reported median overall survival was 36.9 months in MGMT-methylated patients treated with vaccine plus pembrolizumab, compared with 25.3 months in propensity-matched standard-of-care controls. In MGMT-unmethylated patients, the corresponding values were 19.0 versus 16.7 months.

These numbers are *hypothesis-generating*, not definitive evidence that the treatment improves survival. This was a small, non-randomised Phase 1 programme, and the comparison group was not a concurrently randomised control arm. Differences in patient selection, surgery, supportive care, tumour characteristics, steroid exposure, or other unmeasured factors can create apparent survival differences.

29.3 GNOS-PV01: a personalised *DNA* vaccine

The *Nature Cancer* paper describes a different personalised vaccine platform, called GNOS-PV01, in nine patients with newly diagnosed *MGMT-unmethylated* GBM.

This is also not an mRNA vaccine. Instead, the selected neoantigen sequences are encoded in a synthetic DNA plasmid. The DNA is injected and delivered by *electroporation*: brief electrical pulses temporarily make local cell membranes more permeable, increasing DNA uptake.

Patient-specific mutations from multiple tumour regions

- selection of up to 40 neoantigens
- neoantigen sequences encoded in DNA plasmid
- intramuscular injection plus electroporation
- cells produce vaccine-encoded antigenic material
- antigen presentation and neoantigen-specific T-cell priming

The vaccine is described as *multivalent* because it includes many neoantigens rather than one target. This may be particularly useful in GBM, where one region of a tumour can differ genetically from another. Targeting several mutations could reduce the chance that the tumour escapes simply by losing or downregulating one antigen.

The study reported that vaccine manufacture and administration were feasible. No dose-limiting toxicities or treatment-related serious adverse events were reported; most adverse events were mild injection-site reactions. In evaluable patients, the vaccine generated circulating neoantigen-reactive T-cell responses, and the authors found evidence of CD4 and CD8 T-cell infiltration in tumour tissue. One patient receiving dexamethasone, an immunosuppressive corticosteroid often needed to control brain swelling, had a weaker immune response.

This observation is biologically important:

Dexamethasone reduces harmful brain swelling

But:

Dexamethasone suppresses T-cell activation and function

- may weaken the intended effect of cancer immunotherapy

The reported median progression-free survival was 8.5 months and median overall survival was 16.3 months. Three of nine patients survived for more than 24 months, including one long-term survivor without recurrence for almost five years. These outcomes are encouraging for a very difficult GBM subgroup, but they cannot establish efficacy in such a small, single-arm Phase 1 study.

29.4 Why MGMT status matters

MGMT is a DNA-repair protein. Temozolomide, a standard chemotherapy used in GBM, damages DNA. If the *MGMT* gene promoter is methylated, tumour cells often produce less MGMT protein and may be less able to repair temozolomide-induced DNA damage.

MGMT promoter methylation

- often less MGMT DNA-repair activity
- tumour may be more sensitive to temozolomide

MGMT promoter unmethylation

- often more MGMT DNA-repair activity
- tumour may be less sensitive to temozolomide
- generally poorer prognosis

The GNOS-PV01 study therefore focused on a clinically important high-unmet-need population: patients with MGMT-unmethylated GBM, for whom standard chemoradiotherapy tends to be less effective.

29.5 Peptide, mRNA, and DNA personalised vaccines

These approaches share the same therapeutic goal but use different ways to provide the neoantigen information.

Peptide vaccine

Synthetic neoantigen peptides are administered directly, usually with an immune adjuvant.

mRNA vaccine

mRNA encoding selected neoantigens is delivered, often in lipid nanoparticles. Cells temporarily produce the encoded antigenic material.

DNA vaccine

DNA encoding selected neoantigens is delivered, here using electroporation. Cells transcribe the DNA into RNA and then produce antigenic material.

None of these approaches changes the patient's genome. Vaccine DNA is intended to provide temporary instructions for antigen production; it is not designed to integrate into chromosomal DNA.

Each platform has possible advantages and challenges. Peptide vaccines directly provide defined antigen material but depend on effective uptake and presentation. mRNA can be manufactured rapidly and avoids the need to deliver DNA into the nucleus. DNA vaccines can encode many antigens in one construct and can be stable, but efficient cellular delivery is more challenging, hence the use of electroporation in GNOS-PV01.

29.6 What should be concluded?

These studies provide evidence that personalised neoantigen vaccination in GBM is technically feasible and can generate measurable, tumour-relevant T-cell responses. This is important because GBM has historically been very difficult for immunotherapies.

However, the appropriate conclusion is not yet that these vaccines improve survival. The studies are small, early phase, and either single-arm or non-randomised. A convincing demonstration of clinical benefit will require larger controlled trials that test whether vaccination, with or without checkpoint inhibition, improves progression-free survival, overall survival, symptoms, neurological function, and quality of life relative to an appropriate standard of care.

The key unresolved challenge is the full causal chain:

Personalised vaccine generates neoantigen-reactive T cells

- T cells reach infiltrative tumour cells in the brain
- tumour cells present the relevant neoantigens through MHC
- local immunosuppression is overcome
- enough tumour cells are eliminated to delay recurrence
- meaningful patient benefit without harmful inflammation

The reported work is therefore a credible and encouraging step, especially in a cancer with few effective options, but it remains early clinical evidence rather than an established treatment.

30 [26-09-02] BHB810: targeting CDH17 with an antibody–drug conjugate

(<https://www.bighatbio.com/news/bighat-biosciences-announces-first-patient-dosed-in-phase-1-trial-evaluating-bhb810>)

BHB810 is an investigational antibody–drug conjugate, or ADC, that targets *cadherin-17* (CDH17). BigHat Biosciences announced that the first patient had received the drug in a first-in-human Phase 1 trial in advanced gastric and gastro-oesophageal-junction cancers.

CDH17 is a cell-surface adhesion protein expressed in several gastrointestinal cancers, including gastric, colorectal, and pancreatic cancers. It is an attractive ADC target not simply because it is present in tumours, but because its *location* differs between normal intestinal tissue and many cancers.

```
Normal intestinal epithelial cells:
CDH17 is mainly located on the basolateral surface,
at cell-cell contacts and relatively inaccessible from the bloodstream.
Many gastrointestinal cancer cells:
loss of normal epithelial organisation and polarity
→ CDH17 can become more broadly exposed on the cell surface
→ potentially more accessible to a circulating ADC
```

This creates a possible therapeutic window: CDH17-positive tumour cells may be more accessible to the ADC than normal intestinal cells. However, this is a biological hypothesis rather than a guarantee of safety. If BHB810 reaches normal CDH17-expressing tissue, it could still cause *on-target*, *off-tumour* toxicity.

BHB810 uses a smaller-than-usual antibody format, called *VHH-Fc*, linked site-specifically to the cytotoxic payload MMAE. MMAE disrupts microtubules, which are needed for cell division.

```
BHB810 binds CDH17 on tumour cell
→ ADC is internalised
→ payload is released inside the cell
→ MMAE disrupts microtubules
→ cell division fails and susceptible tumour cell may die
```

The smaller VHH-Fc format is intended to improve tumour penetration relative to a conventional full-size IgG antibody. The molecule was also designed to internalise efficiently, since internalisation is essential for intracellular release of an ADC payload.

At this stage, BHB810 has only entered Phase 1 testing. The immediate questions are safety, tolerability, pharmacokinetics, dose selection, evidence of CDH17-target engagement, and preliminary anti-tumour activity. Strong preclinical activity, including in tumour models, is encouraging but cannot establish clinical benefit or safety in humans.

31 [26-09-02] Restoring GTP hydrolysis in mutant RAS with a molecular glue

("Pharmacological restoration of GTP hydrolysis by mutant RAS", <https://www.nature.com/articles/s41586-024-08283-2>)

This paper describes an unusual strategy for inhibiting oncogenic RAS. Rather than only blocking RAS signalling, the compounds can help force mutant RAS to convert from its active, GTP-bound state back to its inactive, GDP-bound state.

This connects directly to Section 19. RAS normally behaves as a molecular switch:

```
RAS-GDP
→ predominantly inactive
RAS-GTP
→ active
→ binds effectors such as RAF
→ activates downstream growth and survival signalling
```

Normally, RAS has intrinsic GTP-hydrolysis activity: it can convert GTP to GDP and turn itself off. This reaction is accelerated by GTPase-activating proteins, or GAPs. Many cancer-driving RAS mutations, particularly at codon 12, impair this switching-off process. The mutant protein therefore spends too much time in the active RAS-GTP state.

Mutant RAS has reduced GTP hydrolysis

- persistent RAS-GTP
- persistent RAF-MEK-ERK and other signalling
- tumour-cell growth and survival

The compounds studied are called *tri-complex inhibitors*, or TCIs. They are a type of *molecular glue*: small molecules that promote a useful interaction between two proteins that would otherwise not bind strongly.

Here, the drug recruits the abundant cellular protein cyclophilin A, or CYPA, to active RAS-GTP:

Tri-complex inhibitor

- + active mutant RAS-GTP
- + cyclophilin A
- drug-RAS-CYPA tri-complex

This has two effects. First, the large recruited CYPA protein physically obstructs RAS from binding downstream effectors such as RAF and PI3K. This is the allosteric-disruption idea described in Section 19: RAS is still present, but its productive interactions with signalling partners are prevented.

Second, and more surprisingly, the tri-complex changes the local RAS configuration in a way that promotes GTP hydrolysis. It helps position the switch-II region, including the important Gln61 residue, in a catalytically favourable arrangement. The complex can therefore partly compensate for the defective switch-off mechanism caused by oncogenic RAS mutations.

Drug-RAS-CYPA tri-complex

- blocks RAS binding to RAF and other effectors
- promotes hydrolysis of RAS-bound GTP
- RAS-GDP accumulates
- RAS signalling is more durably reduced

This is particularly notable for *G12X* RAS mutations, including variants such as KRAS G12D and G12V, which are common in pancreatic, colorectal, and lung cancers. Unlike KRAS G12C inhibitors, this strategy is not based on forming a covalent bond with a mutant cysteine. It instead exploits a drug-induced protein-protein interaction and the underlying RAS switching mechanism.

A *molecular glue* is therefore different from a conventional inhibitor:

Conventional inhibitor

- usually binds one target protein
- blocks an active site or regulatory site

Molecular glue

- binds one or two proteins
- stabilises a new or weak protein-protein interaction
- changes protein function, location, interaction partners, or stability

Some molecular glues cause degradation by recruiting an E3 ubiquitin ligase to a target protein. The RAS TCIs are different: they recruit CYPA not to degrade RAS, but to sterically block its effectors and restore its GTPase switch-off activity.

The key conceptual advance is:

Oncogenic RAS was often viewed as locked in an "ON" state.
These compounds show that some mutant RAS proteins can be
pharmacologically encouraged to turn themselves "OFF" again.

This is compelling preclinical mechanistic work, but it does not remove the usual development challenges: adequate tumour exposure, mutation coverage, resistance, toxicity from inhibiting normal RAS signalling, and demonstration of meaningful clinical benefit.

32 [26-09-02] Single versus dual checkpoint inhibition in metastatic triple-negative breast cancer

("A meta-analysis of single agent and dual immune checkpoint inhibitors in metastatic triple negative breast cancer", <https://www.nature.com/articles/s41523-026-01042-8>)

This meta-analysis examines immune checkpoint inhibitors, or ICIs, in metastatic triple-negative breast cancer, or mTNBC. Triple-negative breast cancers lack expression of oestrogen receptor and progesterone receptor, and do not have HER2 amplification or overexpression by conventional criteria. This removes several highly effective targeted-treatment options used in other breast-cancer subtypes.

The paper compares *single-agent* checkpoint blockade, usually targeting PD-1 or PD-L1, with *dual* checkpoint blockade, typically combining PD-1 or PD-L1 inhibition with CTLA-4 inhibition.

This is distinct from the more established strategy of combining an ICI with chemotherapy. Here, "dual" means two immune-checkpoint drugs, not immunotherapy plus chemotherapy.

32.1 Why block PD-1/PD-L1?

As described in Section 10, PD-1 is an inhibitory receptor on activated T cells. When PD-1 binds PD-L1 or PD-L2, it reduces T-cell activity. Tumours can exploit this pathway to suppress anti-tumour immunity.

Tumour-reactive T cell encounters tumour antigen
→ PD-1-PD-L1 signalling inhibits the T cell
→ reduced anti-tumour response
PD-1 or PD-L1 inhibitor
→ reduces this inhibitory signal
→ existing tumour-reactive T cells may regain activity

The limitation is that PD-1 blockade works best when tumour-reactive T cells already exist, can enter the tumour, and recognise tumour antigens. Many metastatic TNBC tumours do not have enough of this pre-existing immune activity to respond well to PD-1 or PD-L1 inhibition alone.

32.2 Why might dual checkpoint inhibition be better?

CTLA-4 is another inhibitory receptor, but it acts mainly at an earlier stage of the immune response. Whereas PD-1 commonly restrains T cells in peripheral tissues and within the tumour microenvironment, CTLA-4 limits T-cell priming and expansion in lymph nodes.

The intended logic of the combination is therefore complementary:

CTLA-4 blockade

→ may increase priming and expansion of tumour-reactive T cells

PD-1 or PD-L1 blockade

→ may help these T cells remain active in the tumour

Combination

→ aims to generate more anti-tumour T cells

and reduce inhibition of their activity

In principle, this could help convert an immunologically “cold” tumour, with limited effective T-cell activity, into a more inflamed and immune-responsive tumour. This rationale has produced major clinical successes in some cancers, especially melanoma, but does not mean the same combination will work equally well in TNBC.

32.3 What does the meta-analysis suggest?

The analysis pools results from relatively small and heterogeneous studies. Overall, it suggests that dual checkpoint blockade may produce higher response rates than single-agent ICI treatment in metastatic TNBC, but the evidence remains limited and uncertain.

The key limitation is that most available studies are early phase, involve small patient numbers, include patients treated in different lines of therapy, and often do not provide direct randomised comparisons between single and dual ICI treatment. A pooled response rate can therefore be encouraging without proving that dual treatment improves survival.

Higher objective response rate

≠ proven overall-survival benefit

Promising pooled early-phase data

≠ sufficient evidence for routine clinical use

The paper should therefore be understood mainly as support for continued investigation, rather than proof that dual checkpoint blockade is a new standard treatment for mTNBC.

32.4 Why is dual checkpoint blockade more dangerous?

The same immune activation that may improve tumour control can also damage normal tissue. Blocking both CTLA-4 and PD-1/PD-L1 removes two partially distinct immune brakes, increasing the risk of immune-related adverse events.

These can include inflammatory toxicity affecting the skin, bowel, liver, lungs, endocrine organs, nervous system, heart, or other tissues. CTLA-4-containing combinations have generally produced more frequent and more severe immune toxicity than PD-1 or PD-L1 blockade alone.

More immune inhibition removed

→ potentially stronger anti-tumour immunity

But also:

More immune inhibition removed

→ greater risk of harmful immune attack on normal tissue

The clinical question is therefore not simply whether dual therapy creates more immune activity. It is whether any additional response or survival benefit is large and durable enough to justify the additional toxicity and treatment burden.

For metastatic TNBC, current evidence most strongly supports using checkpoint inhibition in carefully selected settings, often with chemotherapy and guided by biomarkers such as PD-L1 expression. Dual

checkpoint inhibition remains biologically plausible and potentially useful for selected patients, but requires stronger randomised evidence and better biomarkers to identify who is likely to benefit.

33 [26-09-02] Atezolizumab in early triple-negative breast cancer

("Atezolizumab in early triple-negative breast cancer: the randomized phase 3 NSABP B-59/GeparDouze trial", <https://www.nature.com/articles/s41591-026-04565-6>)

This Phase 3 trial tested whether adding *atezolizumab* to standard chemotherapy improves outcomes for patients with stage II–III triple-negative breast cancer, or TNBC, treated with curative intent.

Patients received chemotherapy before surgery, called *neoadjuvant* treatment, with either atezolizumab or placebo. After surgery, they continued the assigned study treatment for a further period. The main question was whether this strategy reduced the risk of recurrence, progression preventing surgery, a second primary cancer, or death: the trial's *event-free survival*, or EFS, endpoint.

Atezolizumab increased the pathological complete response, or pCR, rate:

Pathological complete response:
Atezolizumab + chemotherapy
→ 63%
Placebo + chemotherapy
→ 57%

A pCR means that no invasive cancer is detected in the breast and sampled lymph nodes at surgery. It is encouraging because patients who achieve pCR often have better long-term outcomes. However, pCR is not automatically a validated surrogate for recurrence-free or overall survival for every treatment and disease setting.

The trial's primary endpoint was not met. At approximately four years of follow-up, event-free survival was numerically higher with atezolizumab, but the difference was not statistically significant:

Four-year event-free survival:
Atezolizumab
→ 85.2%
Placebo
→ 81.9%
Hazard ratio = 0.80
95% confidence interval = 0.62–1.03
P = 0.08

Thus, the result is compatible with anything from a clinically worthwhile benefit to little or no benefit. It does *not* establish that adding atezolizumab improves event-free survival in the overall study population. Overall survival was also similar at this relatively early analysis.

33.1 What does atezolizumab do?

Atezolizumab is a humanised antibody that binds *PD-L1*, one of the ligands for the inhibitory T-cell receptor PD-1.

```
Without PD-L1 blockade:
PD-L1 on tumour or immune cell binds PD-1 on T cell
→ inhibitory signal
→ reduced T-cell activity
With atezolizumab:
Atezolizumab binds PD-L1
→ PD-L1 cannot bind PD-1
→ less inhibitory signalling
→ tumour-reactive T cells may remain more active
```

It also blocks interaction between PD-L1 and B7.1 (CD80). Its Fc region has been engineered to minimise immune-effector activity, reducing the risk that the antibody itself depletes PD-L1-expressing immune cells.

Atezolizumab is similar in overall purpose to anti-PD-1 antibodies such as pembrolizumab and nivolumab, but the exact target differs:

```
Anti-PD-L1 antibody, such as atezolizumab:
→ blocks PD-L1 interactions
→ does not directly block PD-L2 binding to PD-1
Anti-PD-1 antibody, such as pembrolizumab:
→ binds PD-1 on T cells
→ blocks both PD-L1-PD-1 and PD-L2-PD-1 interactions
```

This distinction may affect pharmacology, but it does not by itself prove that one class is inherently better. Outcomes also depend on the chemotherapy backbone, treatment schedule, patient population, biomarker strategy, duration of treatment, and trial design.

A relevant comparison is that pembrolizumab plus chemotherapy has shown event-free-survival benefit in the Phase 3 KEYNOTE-522 trial and is an established approach in high-risk early TNBC in many settings. This does *not* prove that the different outcome in this trial was caused solely by PD-L1 versus PD-1 targeting. Cross-trial comparisons cannot reliably isolate one explanation.

The main lesson is:

```
More pathological complete responses
≠ automatically improved long-term survival
Checkpoint-pathway blockade
≠ identical clinical benefit for every antibody,
chemotherapy backbone, or patient population
```

34 [26-09-02] Tumour-adjacent B cells and recurrence risk in localised prostate cancer

("Tumor-adjacent B cell infiltration stratifies recurrence risk in localized prostate cancer", <https://www.medrxiv.org/content/10.64898/2026.08.29.26361718v1.full.pdf>)

This preprint asks whether the *location* of B cells in prostatectomy tissue can help predict which patients with apparently localised prostate cancer are more likely to recur after surgery.

The important idea is that immune cells should not simply be counted across the whole tissue sample. Their spatial position may matter:

```
B cells inside the tumour
≠
B cells in non-malignant tissue immediately adjacent to the tumour
```

The authors report that B-cell infiltration in tissue adjacent to the tumour was associated with recurrence risk, adding biological information beyond conventional pathological measures such as tumour grade and stage. In other words, the immune environment around a prostate tumour may contain information about how clinically aggressive that tumour is.

This is interesting because prostate cancer is often considered immunologically “cold”: compared with cancers such as melanoma, it commonly has relatively limited productive anti-tumour T-cell activity and has been difficult to treat with checkpoint inhibition. Nevertheless, the local immune microenvironment is not irrelevant.

B cells can have several different roles. They can produce antibodies, present antigens to T cells, secrete cytokines, and participate in organised immune-cell aggregates called *tertiary lymphoid structures*. Depending on their state and local context, B cells may support anti-tumour immunity or contribute to inflammation and immune suppression.

Tumour-adjacent B-cell infiltration
→ marker of a particular local immune state
→ association with future recurrence risk
But:
Association
≠ proof that B cells directly cause recurrence

The main potential value is therefore *prognostic*, not immediately therapeutic. If independently validated, tumour-adjacent B-cell measurement might eventually help identify patients who need closer follow-up or more intensive adjuvant treatment, versus those who may safely avoid overtreatment.

However, this is a preprint and not yet peer reviewed. It should be regarded as an interesting biomarker hypothesis rather than a clinically established recurrence test. Independent validation in larger cohorts, standardised measurement methods, and evidence that it improves decisions beyond current clinical and genomic risk tools would be needed.

35 [26-09-02] T-effector biology in clear-cell renal-cell carcinoma

("A Visually Interpretable Histopathology-Based Immune Model Predicts T-effector Biology and Response to Immune checkpoint inhibition in Clear Cell Renal Cell Carcinoma Clinical Trial and Contemporary Real-World Datasets", <https://www.biorxiv.org/content/10.64898/2026.06.21.733614v2.full.pdf>)

This preprint concerns clear-cell renal-cell carcinoma, or ccRCC, the most common type of kidney cancer. Its main biological question is:

Does a tumour contain a local immune environment with active, cytotoxic T cells that might respond to checkpoint inhibition?

T-effector biology refers to the presence and activity of T cells, especially CD8 T cells, that are capable of recognising and attacking tumour cells. It is not simply a count of all lymphocytes in a tumour. It includes evidence that cytotoxic T cells are present, activated, and using their tumour-killing programme.

A simplified sequence is:

Tumour antigen is presented on MHC class I
→ recognised by a suitable CD8 T cell
→ T cell becomes activated
→ cytotoxic molecules and inflammatory cytokines are produced
→ tumour cell may be killed

The researchers use an RNA-based *T-effector*, or Teff, gene-expression signature as their biological reference. Such signatures measure expression of genes associated with cytotoxic T-cell presence and function. Typical components include:

```
CD8A
→ marker associated with CD8 T cells
EOMES
→ transcription factor associated with cytotoxic T-cell state
PRF1
→ perforin, used by cytotoxic lymphocytes to help kill target cells
IFNG
→ interferon- $\gamma$ , a cytokine produced by activated T cells
CD274
→ PD-L1; often induced by interferon signalling
and evidence of an adaptive immune-response environment
```

A high Teff score therefore suggests that a tumour contains an interferon-rich, cytotoxic T-cell-infiltrated immune microenvironment. It does *not* prove that T cells are successfully eliminating the tumour. In fact, high PD-L1 expression can reflect a tumour's attempt to suppress an existing anti-tumour T-cell response.

This explains the link to checkpoint inhibition:

```
Tumour-reactive T cells are present
+ PD-1/PD-L1 signalling inhibits them
→ anti-PD-1 or anti-PD-L1 therapy may release an existing immune brake
No meaningful tumour-reactive T-cell response is present
→ PD-1/PD-L1 blockade alone may have little to reactivate
```

The study used several experimental measurements to establish this biology. RNA sequencing measured the Teff gene-expression signature. Immunohistochemistry, or IHC, used stains for CD8 to locate cytotoxic T-cell-rich regions, PAX8 to identify tumour cells, and ERG to identify blood-vessel-associated endothelial cells. These measurements show not only whether a tumour contains immune cells, but whether CD8 T cells occur in biologically relevant locations relative to tumour and vascular tissue.

The paper then trained a model to recognise corresponding patterns in ordinary haematoxylin-and-eosin, or H&E, pathology slides. The machine-learning aspect is secondary to the biological point: microscopic tissue architecture contains visible correlates of an immune-active, CD8 T-cell-rich tumour microenvironment.

In clinical-trial and real-world ccRCC datasets, a high slide-derived immune score correlated with the RNA Teff signature and with better outcomes on immune-checkpoint inhibitors such as atezolizumab, nivolumab, or nivolumab plus ipilimumab. This is consistent with the biological expectation that checkpoint blockade is more likely to help when an anti-tumour T-cell response already exists.

However, this is a preprint rather than peer-reviewed clinical-validation evidence. The score should be regarded as a promising biomarker hypothesis, not an established test for selecting immunotherapy.

36 Other immune-checkpoint targets

PD-1/PD-L1 and CTLA-4 are the best-established checkpoint pathways, but they are not the only inhibitory signals used by tumours and immune cells.

The most clinically established additional target is *LAG-3*. LAG-3 is an inhibitory receptor expressed on activated and chronically stimulated T cells. It is often co-expressed with PD-1 in exhausted tumour-infiltrating T cells.

LAG-3 and PD-1 expressed on a tumour-reactive T cell
→ multiple inhibitory signals restrain T-cell activity
LAG-3 blockade + PD-1 blockade
→ aims to remove two distinct immune brakes

Relatlimab is an anti-LAG-3 antibody. In combination with the PD-1 antibody nivolumab, it is approved for unresectable or metastatic melanoma. This is therefore an example of a clinically established *dual checkpoint* combination that does not use CTLA-4.

Several other checkpoint targets are in clinical development, usually in combination with PD-1 or PD-L1 blockade:

TIGIT
→ inhibitory receptor on T cells and NK cells;
often co-expressed with PD-1 in dysfunctional or exhausted cells
TIM-3
→ inhibitory receptor associated with chronic stimulation
and more terminal T-cell exhaustion
VISTA
→ inhibitory pathway expressed especially by myeloid cells,
such as tumour-associated macrophages
NKG2A
→ inhibitory receptor on NK cells and some CD8 T cells;
interaction with HLA-E can protect tumour cells from killing
BTLA and CD96
→ additional inhibitory receptors under investigation

These targets are scientifically attractive because resistance to PD-1 blockade may involve several immune brakes at once. However, none should be assumed to work merely because it is another checkpoint. Clinical results for newer targets, particularly TIGIT, have been mixed, and most remain investigational.

There are also related approaches that are sometimes called immune-checkpoint therapy, but act on cells other than T cells. For example, the CD47–SIRP α pathway is a *myeloid checkpoint*: CD47 on tumour cells can send a “do not eat me” signal to macrophages.

CD47 on tumour cell binds SIRP α on macrophage
→ reduced phagocytosis
CD47–SIRP α blockade
→ aims to permit macrophages to engulf tumour cells

Thus, checkpoint immunotherapy is best understood as a broad strategy of releasing inhibitory immune signals. PD-1/PD-L1, CTLA-4, and LAG-3 are the most clinically established T-cell examples; many others are being tested, but their benefit–risk profiles and useful patient populations are still being defined.

37 [26-09-02] B cells: antibodies, immune memory, and their role in cancer

B cells are a major type of adaptive immune cell. Their best-known role is to make *antibodies*, but they can also present antigens to helper T cells and help organise local immune responses.

A useful contrast with T cells is:

B cell receptor, or BCR
→ can bind an intact molecular target directly
T-cell receptor, or TCR
→ recognises a short peptide only when displayed by MHC

Each B cell has a B-cell receptor with one particular binding specificity. If that B cell encounters its matching antigen and receives appropriate help from CD4 T cells, it can divide and differentiate.

```
B cell recognises its antigen
+ receives appropriate helper-T-cell signals
→ B-cell activation and expansion
Activated B cell
→ plasma cell, which secretes antibodies
Some activated B cells
→ memory B cells, which enable a faster response
if the same antigen is encountered again
```

Antibodies are soluble proteins released by plasma cells. They can bind targets outside cells, including microbes, toxins, and exposed proteins on abnormal cells. Antibodies can neutralise a target directly, mark a cell for uptake by phagocytes, activate complement, or recruit natural killer cells through antibody-dependent cellular cytotoxicity.

B cells also act as *antigen-presenting cells*. A B cell can bind an antigen through its BCR, take it inside, break it into peptides, and display those peptides on MHC class II to CD4 helper T cells. This creates a two-way interaction: helper T cells promote B-cell activation, while B cells can help sustain and shape the wider adaptive immune response.

37.1 B cells in cancer: potentially helpful or harmful

B cells are not automatically good or bad in cancer. Their effect depends on their subtype, activation state, antibodies produced, location, and interactions with other immune cells.

They can support anti-tumour immunity:

```
B cell recognises tumour-associated antigen
→ antibody production and antigen presentation
→ support for CD4 and CD8 T-cell responses
→ possible tumour-cell killing or improved immune control
```

B cells are often found in *tertiary lymphoid structures*, or TLS. These are organised immune-cell clusters that form in chronically inflamed tissues, including some tumours. They resemble small lymph nodes and may contain B cells, T cells, dendritic cells, and specialised blood vessels. In several cancer types, mature TLS are associated with stronger anti-tumour immune responses and better outcomes from checkpoint inhibition.

However, some B cells may instead promote tumour growth or suppress immune attack. So-called *regulatory B cells*, or Bregs, can produce immunosuppressive signals such as IL-10 and TGF- β . These signals may reduce effective T-cell activity and help create a tumour-supportive microenvironment.

```
B cells in an immune-active TLS
→ may support productive anti-tumour immunity
Immunosuppressive B-cell state
→ may reduce T-cell activity and support immune escape
```

This is why the prostate-cancer preprint is interesting but not yet definitive. B cells adjacent to a tumour may be a marker of a local immune state associated with recurrence risk. But a simple B-cell count cannot tell us whether those cells are antigen-presenting B cells, antibody-secreting plasma cells, TLS-associated B cells, or immunosuppressive B cells. Their exact location and biological state matter.

38 [26-09-05] A Frizzled7/PD-L1 bispecific antibody in NSCLC resistant to atezolizumab

(A novel bispecific antibody targeting *Frizzled7*/PD-L1 overcomes NSCLC resistance to atezolizumab by reprogramming tumor microenvironment, <https://www.nature.com/articles/s41698-026-01687-5>)

This paper describes a *bispecific antibody*: one antibody engineered to recognise two different targets. Here, the targets are *Frizzled7*, or FZD7, and *PD-L1*.

The central idea is to address two connected problems in some non-small-cell lung cancers, or NSCLC: tumour-cell signalling that supports an immune-suppressive tumour microenvironment, and PD-L1-mediated inhibition of anti-tumour T cells.

38.1 Why atezolizumab resistance can occur

Atezolizumab is a PD-L1-blocking antibody. It aims to prevent PD-L1 on tumour cells or other cells from engaging PD-1 on T cells, thereby reducing an inhibitory signal.

However, PD-L1 blockade alone may not work if the tumour microenvironment remains hostile to effective immunity. For example, tumours may exclude T cells, recruit suppressive immune cells, or activate signalling programmes that support tumour growth and immune escape.

The simplified problem is:

```
PD-L1 blockade
→ can reduce one inhibitory immune signal
But:
Tumour-intrinsic signalling
+ suppressive tumour microenvironment
→ may still prevent an effective T-cell response
```

38.2 What is Frizzled7?

FZD7 is a cell-surface receptor in the *Wnt signalling* system. Wnt signalling has important normal roles in development and tissue maintenance, but abnormal Wnt-pathway activity can support cancer-cell proliferation, survival, stem-like cell states, and immune evasion.

In this paper, FZD7 appears to be relevant not only as a tumour-cell target, but also because FZD7-associated signalling helps maintain a tumour microenvironment that is less favourable for anti-tumour immunity.

38.3 What does the bispecific antibody do?

The FZD7/PD-L1 bispecific antibody is intended to perform two actions at the same time:

```
FZD7-binding arm
→ interferes with FZD7-associated tumour-promoting
and immune-suppressive signalling
PD-L1-binding arm
→ blocks PD-L1-mediated inhibitory signalling to T cells
Combined effect
→ reprogrammes the tumour microenvironment
toward a state more permissive for anti-tumour immunity
```

Thus, it is not simply a more complicated version of atezolizumab. Atezolizumab primarily blocks the PD-L1 checkpoint interaction. The bispecific aims both to remove this immune brake *and* to alter FZD7-associated biology that may be contributing to resistance.

A further potential advantage of a bispecific format is that it may preferentially concentrate PD-L1 blockade at FZD7-positive tumour sites. This could improve the functional coupling of the two mechanisms, although the extent to which this produces a meaningful therapeutic window must be established experimentally.

38.4 How to interpret “reprogramming the tumour microenvironment”

In this context, reprogramming does not mean that the antibody directly converts every immune cell into an anti-tumour cell. It means that changing FZD7-associated signalling, together with PD-L1 blockade, shifts the local balance of signals and cell states.

The proposed causal chain is:

```
FZD7/PD-L1 bispecific antibody
→ reduced FZD7-associated pro-tumour signalling
+ reduced PD-L1-mediated T-cell inhibition
→ less immune-suppressive tumour microenvironment
→ improved anti-tumour immune activity
→ restored or increased sensitivity to checkpoint therapy
```

The important conceptual point is that resistance to checkpoint inhibition is often not caused by one checkpoint interaction alone. A tumour may evade immunity through several interacting mechanisms. This bispecific strategy attempts to address both a checkpoint and a tumour-associated signalling programme in one product.

38.5 What should be concluded?

The paper provides a biologically plausible strategy for overcoming atezolizumab resistance in a subset of NSCLC models. It supports the broader idea that effective immunotherapy combinations may need to alter both T-cell inhibition and the tumour microenvironment.

However, the evidence should be interpreted according to the experimental system. Improved tumour control in cell systems or animal models does not yet establish clinical benefit in patients. Important next questions include whether FZD7 is a useful predictive biomarker, whether the antibody has an acceptable safety profile given normal roles of Wnt/FZD7 signalling, whether benefit exceeds that of giving separate FZD7- and PD-L1-directed therapies, and which resistant NSCLC populations are most likely to benefit.

38.6 Super-short summary

Atezolizumab blocks PD-L1, but some NSCLCs remain resistant because their tumour microenvironment is still immune-suppressive. The FZD7/PD-L1 bispecific antibody aims to block PD-L1-mediated T-cell inhibition while also disrupting FZD7-associated tumour and microenvironmental signalling. The intended result is a more immune-permissive tumour environment and improved anti-tumour T-cell activity.

39 [26-09-05] Extracellular vesicles as biomarkers and decoys for CLDN18.2-targeted therapy

(Extracellular vesicle proteins monitor and interfere with CLDN18.2-targeted CAR-T cell and antibody therapies against gastrointestinal cancers, <https://www.nature.com/articles/s41467-026-77132-9>)

This paper concerns a potential resistance mechanism for therapies directed at *CLDN18.2*, including CLDN18.2-targeted antibodies and CAR-T cells in gastrointestinal cancers.

CLDN18.2 is a cell-surface protein that can be abnormally exposed or overexpressed in certain gastric and pancreatic cancers. This makes it an attractive target: an antibody or CAR-T cell can recognise CLDN18.2 on a tumour cell and then block, recruit immune killing, deliver a payload, or directly activate CAR-T-cell killing.

The complication described in the paper is that tumour cells do not only display proteins on their own surface. They can also release small membrane-bound particles called *extracellular vesicles*, or EVs.

Tumour cell releases extracellular vesicles
Extracellular vesicles carry tumour-associated proteins,
including CLDN18.2 and other proteins
These vesicles enter blood and the local tumour environment

39.1 How EV proteins can monitor the tumour

Because extracellular vesicles can carry proteins from the cells that released them, their protein content may provide a relatively accessible sample of tumour biology.

Instead of requiring repeated tissue biopsies, one might measure EV-associated proteins in blood or another body fluid. In principle, this could help monitor whether a tumour expresses CLDN18.2, how its biology changes during therapy, or whether features associated with resistance are emerging.

The important caution is that an EV measurement is a *biomarker candidate*, not automatically a clinically useful test. It must show reliable analytical measurement and, more importantly, demonstrate that it predicts treatment response or resistance in appropriately designed patient studies.

39.2 How EVs can interfere with CLDN18.2-directed therapies

The central mechanistic idea is that CLDN18.2-positive EVs can act as *decoys*. They display the same antigen that the therapy is designed to recognise, but they are not intact tumour cells.

CLDN18.2-positive extracellular vesicle
→ binds CLDN18.2-targeted antibody or CAR-T cell
Therapy is occupied away from tumour-cell surface
→ less effective tumour targeting may result

For an antibody, EV-associated CLDN18.2 may bind and sequester some of the administered antibody before it reaches tumour cells. This could reduce the amount of free antibody available for productive binding at the tumour.

For a CAR-T cell, CLDN18.2-positive vesicles may bind the CAR and provide repeated, non-productive stimulation. Such stimulation could alter CAR-T-cell behaviour, contribute to dysfunction or exhaustion, and reduce the number or functional capacity of CAR-T cells available to kill tumour cells.

The distinction is important:

Tumour-cell CLDN18.2 recognition
→ intended therapeutic interaction
EV-associated CLDN18.2 recognition
→ possible antigen decoy and non-productive activation

39.3 Why this matters for solid-tumour CAR-T therapy

This paper builds on a recurring theme in the document: a target can be biologically sensible and accessible at the tumour-cell surface, while the surrounding tumour system still limits treatment efficacy.

For CLDN18.2-targeted CAR-T therapy, relevant barriers include heterogeneous target expression, limited CAR-T-cell entry into solid tumours, immune suppression in the tumour microenvironment, antigen loss, and—as proposed here—interference by tumour-derived extracellular vesicles.

Thus, the relevant question is not only:

Do tumour cells express CLDN18.2?

but also:

How much accessible CLDN18.2 is present on tumour cells,
how much is released on extracellular vesicles,
and how does this affect the effective exposure
and function of the CLDN18.2-directed therapy?

39.4 What should be concluded?

The study supports the idea that extracellular vesicles can be both *informative* and *functionally important*. Their protein content may help monitor the changing tumour state, while vesicles carrying CLDN18.2 may reduce the effectiveness of CLDN18.2-targeted antibodies and CAR-T cells by acting as antigen-bearing decoys.

This does not mean that CLDN18.2 is an unsuitable target. Rather, it suggests that successful CLDN18.2 therapy may require attention to antigen distribution beyond intact tumour cells. Possible future strategies could include selecting patients using EV-based biomarkers, monitoring EV changes during treatment, engineering therapies less susceptible to decoy binding, or combining CLDN18.2-directed treatment with interventions that reduce EV production or EV-mediated immune effects.

The most important broader lesson is:

A cell-surface antigen is not found only on intact tumour cells.
It may also be released on extracellular vesicles,
which can change both biomarker measurements
and the real pharmacology of antigen-directed therapies.

40 [26-09-05] Tislelizumab plus chemotherapy in advanced oesophageal squamous-cell carcinoma

(*First-line tislelizumab plus chemotherapy versus placebo plus chemotherapy in unresectable advanced/metastatic esophageal squamous cell carcinoma: minimum 3-year survival follow-up of a randomized phase 3 trial (RATIONALE-306)*, <https://www.nature.com/articles/s41467-026-77327-0>)

This paper reports longer-term results from *RATIONALE-306*, a randomised Phase III trial in patients with previously untreated, unresectable locally advanced, recurrent, or metastatic *oesophageal squamous-cell carcinoma*, or ESCC.

Patients received either:

Tislelizumab + investigator-selected platinum-based chemotherapy
or
Placebo + the same type of chemotherapy

The key question was whether adding tislelizumab to standard first-line chemotherapy improves outcomes compared with chemotherapy alone.

40.1 Is tislelizumab different from the PD-1 inhibitors discussed earlier?

Not in its central therapeutic principle. Tislelizumab is an *anti-PD-1 immune-checkpoint inhibitor*, like pembrolizumab. It binds PD-1 on T cells and prevents PD-L1 or PD-L2 from activating this inhibitory receptor.

```
Tislelizumab binds PD-1 on T cells
→ PD-L1 and PD-L2 cannot engage PD-1 normally
→ reduced inhibitory T-cell signalling
→ anti-tumour T cells may remain active for longer
```

The important comparison is with *atezolizumab*, which binds *PD-L1*, rather than PD-1:

```
Tislelizumab or pembrolizumab
→ block PD-1
→ prevent inhibition mediated by both PD-L1 and PD-L2
Atezolizumab
→ blocks PD-L1
→ leaves possible PD-L2-PD-1 signalling intact
```

This distinction is biologically real, but it does *not* mean that PD-1 antibodies are automatically clinically superior to PD-L1 antibodies. Their benefit–risk profiles depend on the cancer type, treatment setting, combination partners, dosing, patient population, and clinical evidence.

Tislelizumab was also engineered to have reduced binding to certain Fc-gamma receptors on immune cells. The intended rationale is to reduce the possibility of unintended interactions with PD-1-expressing T cells through the antibody Fc region. This is a product-design feature, but the clinically important question remains whether the complete treatment regimen improves patient outcomes.

40.2 Why give chemotherapy with a checkpoint inhibitor?

Chemotherapy has direct anti-tumour effects, particularly by damaging or disrupting rapidly dividing cancer cells. It can reduce tumour burden quickly, which is important in aggressive metastatic disease.

It may also alter the immune context of the tumour. For example, tumour-cell death can release tumour material, and chemotherapy can reduce some immune-suppressive cell populations. These effects do not guarantee improved immunity, but they provide a rationale for combining chemotherapy with PD-1 blockade.

```
Chemotherapy
→ direct tumour-cell killing and tumour-burden reduction
Tislelizumab
→ reduces PD-1-mediated inhibition of T cells
Combination
→ aims to provide both immediate anti-tumour activity
and more sustained immune-mediated control
```

40.3 What did the longer follow-up show?

The minimum three-year follow-up continued to show an overall-survival benefit from adding tislelizumab to chemotherapy. In the intention-to-treat population, median overall survival was reported as approximately **17.2 months** with tislelizumab plus chemotherapy, compared with **10.6 months** with placebo plus chemotherapy.

The hazard ratio for death remained favourable for the tislelizumab combination. A hazard ratio below 1 means that, over the follow-up period, the risk of death was lower in the tislelizumab-plus-chemotherapy group than in the control group. It should not be read as a simple statement that every patient gains the same number of months.

The follow-up also supported sustained improvements in other efficacy measures, including progression-free survival, response, and duration of response. No new major safety signals were identified with longer

observation; however, the regimen still combines the toxicities of chemotherapy with immune-related adverse events possible from PD-1 blockade.

40.4 How should PD-L1 testing be understood here?

The study used a PD-L1 measurement called the *tumour area positivity*, or TAP, score. Higher PD-L1 expression can be associated with a greater likelihood of benefit from PD-1/PD-L1 blockade, but it is not a perfect predictor.

The trial found benefit in the overall randomised population, with results also supporting benefit in PD-L1-positive subgroups. This fits the earlier principle:

```
PD-L1 expression
→ potentially informative predictive biomarker
PD-L1 expression
≠ a complete explanation of response or resistance
```

Response still depends on such factors as tumour antigenicity, antigen presentation, T-cell presence and function, other immune-suppressive mechanisms, tumour heterogeneity, and the effects of chemotherapy.

40.5 What should be concluded?

The main message is not that tislelizumab has a fundamentally new checkpoint mechanism. It is a PD-1 inhibitor used in a clinically important first-line combination strategy for advanced ESCC.

The longer follow-up strengthens confidence that the original survival benefit was not only an early or short-lived result. In this setting, adding tislelizumab to chemotherapy produced a clinically meaningful improvement in overall survival compared with chemotherapy alone, with no unexpected long-term safety findings.

This is a useful example of the full causal chain discussed earlier:

```
PD-1 blockade has a plausible immune mechanism
→ combined with chemotherapy in a defined disease setting
→ tested against an appropriate control group
→ produces sustained evidence of improved survival
```

41 [26-09-05] Lymphodepletion to support CAR-T cells in recurrent brain tumours

(*Lymphodepletion mitigates anti-CAR immunity in pediatric and young adult patients with recurrent or refractory brain tumors: clinical trial results*, <https://www.medrxiv.org/content/10.64898/2026.08.27.26361261v1>)

This preprint reports clinical-trial results in paediatric and young-adult patients with recurrent or refractory brain tumours treated with CAR-T cells. Its central question is why CAR-T cells may disappear or lose activity after repeated administration, and whether *lymphodepletion* can reduce this problem.

The key issue is that CAR-T cells are living therapeutic products, but the patient's own immune system can sometimes recognise parts of the CAR as foreign. This is called *anti-CAR immunity*.

41.1 What is anti-CAR immunity?

A CAR contains an antigen-recognition component, often derived from an antibody. Some CAR components may be sufficiently different from normal human proteins that the patient's immune system recognises the engineered CAR-T cells as foreign.

CAR-T cells are administered

- patient's immune system recognises CAR-associated components as foreign in some patients
- anti-CAR T-cell and/or antibody responses develop
- subsequently infused CAR-T cells may be cleared more rapidly
- CAR-T-cell persistence and anti-tumour activity may decrease

This is especially relevant when CAR-T cells are given repeatedly. The first doses can effectively act as an immune exposure that primes the patient's immune system against the CAR product. Later doses may then be removed before they can persist and function effectively.

41.2 What does lymphodepletion mean?

Lymphodepletion, often abbreviated LD, is pre-treatment with chemotherapy intended to temporarily reduce the patient's lymphocytes: immune cells that include T cells, B cells, and natural killer cells.

It is not chemotherapy intended primarily to kill the brain tumour directly. In this setting, its main purpose is to prepare the immune environment for the infused CAR-T cells.

The study evaluates whether lymphodepletion can reduce anti-CAR immune responses and thereby improve CAR-T-cell persistence.

Lymphodepleting treatment

- temporary reduction of host lymphocytes
- Fewer host immune cells capable of recognising and eliminating CAR-T cells
- reduced anti-CAR immunity
- CAR-T cells are less rapidly cleared
- greater opportunity for persistence, expansion, and anti-tumour function

41.3 How does this help the CAR-T cells?

Lymphodepletion can help through at least two related mechanisms.

First, it reduces the number of *host T cells and B cells* that could mount an adaptive immune response against the CAR. B cells can contribute by producing antibodies, while host T cells can contribute to cellular immune rejection of the infused CAR-T cells.

Second, reducing the patient's existing lymphocyte population can make the immune environment more supportive of the transferred cells. Normal lymphocytes consume homeostatic cytokines, including signals such as IL-7 and IL-15 that support T-cell survival and proliferation.

Without lymphodepletion:

- many endogenous lymphocytes compete for survival signals
- CAR-T cells may have limited expansion and persistence

After lymphodepletion:

- fewer endogenous lymphocytes consume these signals
- CAR-T cells may have more favourable conditions for survival and expansion

This is sometimes described as reducing *cytokine sinks*: cells that consume cytokines needed by the therapeutic T cells.

41.4 What does the paper show?

The study supports the conclusion that lymphodepletion can mitigate anti-CAR immunity in this clinical setting. In practical terms, patients receiving lymphodepletion had less evidence that their immune system was rejecting the CAR-T product, helping permit more sustained CAR-T-cell exposure after treatment.

This does not mean lymphodepletion makes CAR-T therapy automatically effective against brain tumours. It addresses one important barrier: premature immune removal of the therapeutic cells. Other barriers remain, including whether CAR-T cells reach the tumour, persist in the central nervous system, recognise enough tumour cells, overcome the suppressive tumour microenvironment, and avoid unacceptable toxicity.

41.5 Why this is particularly relevant in brain tumours

Brain tumours are difficult CAR-T targets because they are heterogeneous, can have an immunosuppressive microenvironment, and may be treated by repeated locoregional CAR-T administration into cerebrospinal-fluid spaces. Repeated dosing is attractive because it may sustain local CAR-T-cell exposure, but it also increases the importance of preventing anti-CAR immune rejection.

The broader lesson is:

For cell therapy, the patient's immune system is not only a potential ally.
It can also recognise and remove the therapeutic cells.
Lymphodepletion temporarily reduces this rejection pressure,
giving CAR-T cells a better opportunity to persist and act.

Because this is a *preprint*, the findings should be interpreted cautiously until peer review and fuller clinical confirmation. Lymphodepletion also has important costs: it can cause low blood-cell counts and increase susceptibility to infection, so its intensity must be balanced against its potential benefit for CAR-T-cell persistence.

42 [26-09-05] Ivonescimab: combining PD-1 blockade with VEGF inhibition

(<https://www.fiercepharma.com/pharma/akeso-summit-ivonescimab-extends-lives-keytruda-head-head-trial-wh>
<https://smmmtx.com/ivonescimab-smt112/ivonescimab-overview/>)

Ivonescimab, also called AK112 or SMT112 in some territories, is a *bispecific antibody* that targets both *PD-1* and *VEGF*. It is therefore designed to combine an immune-checkpoint inhibitor with an anti-angiogenic treatment in one antibody.

The recent attention comes from *HARMONi-2*, a randomised Phase III trial in China that compared ivonescimab monotherapy directly with pembrolizumab monotherapy as first-line treatment for PD-L1-positive advanced non-small-cell lung cancer, or NSCLC.

42.1 What does ivonescimab do?

The PD-1-binding part works according to the mechanism already discussed for pembrolizumab and tislelizumab:

Ivonescimab binds PD-1 on T cells
→ reduces PD-1-mediated inhibitory signalling
→ tumour-reactive T cells may remain active
or regain anti-tumour activity

The additional target, VEGF, is important for more than blood-vessel formation. Tumours often release VEGF to stimulate *angiogenesis*: the formation of abnormal blood vessels that supply oxygen and nutrients to the growing tumour.

However, VEGF can also contribute to an immune-suppressive tumour microenvironment. It can impair effective immune-cell entry into tumours and support immune-suppressive cell states.

```
Tumour-derived VEGF
→ abnormal tumour blood vessels
+ tumour growth support
+ less favourable immune-cell infiltration and function
→ immune-suppressive tumour microenvironment
```

By binding and neutralising VEGF, ivonescimab aims to inhibit VEGF–VEGF-receptor signalling. The intended result is both anti-angiogenic activity and a tumour environment more permissive for T-cell-mediated immunity.

```
PD-1 blockade
→ releases an inhibitory brake on T cells
VEGF blockade
→ reduces pro-angiogenic and immune-suppressive VEGF effects
Combined action
→ aims to improve anti-tumour immune activity
while also disrupting tumour vascular support
```

42.2 Why use one bispecific rather than two separate antibodies?

PD-1 inhibitors and anti-VEGF drugs have already been combined clinically. The proposed advantage of ivonescimab is not simply that it hits two targets, but that it couples the two activities in one molecular product.

Its tetravalent design has two PD-1-binding sites and two VEGF-binding sites. Binding to one target can strengthen productive binding to the other, a feature described as *cooperative binding*. Since VEGF is often abundant in tumours, this may preferentially enhance PD-1 binding where both the immune and VEGF-associated biology are relevant.

```
VEGF-rich tumour environment
→ ivonescimab binds VEGF
VEGF-associated ivonescimab
→ enhanced opportunity to engage PD-1-positive T cells
Result
→ local coupling of VEGF inhibition
and PD-1 checkpoint blockade
```

This is a plausible design advantage, but it must not be assumed that a bispecific is automatically better than giving two separate drugs. That requires direct clinical comparison.

42.3 Why could it be better than pembrolizumab?

Pembrolizumab blocks PD-1, but it does not directly address VEGF-driven angiogenesis or VEGF-associated immune suppression. A tumour can therefore remain difficult for T cells to enter or function within even when PD-1 signalling is blocked.

Ivonescimab aims to address both barriers:

```
Pembrolizumab alone
→ removes PD-1-mediated T-cell inhibition
Ivonescimab
→ removes PD-1-mediated T-cell inhibition
+ neutralises VEGF-associated vascular and immune suppression
```

The Phase III HARMONi-2 study initially reported a substantial progression-free-survival benefit. Median progression-free survival was approximately **11.1 months** with ivonescimab and **5.8 months** with pembrolizumab, with a hazard ratio of approximately **0.51**.

On 2 September 2026, Akeso and Summit also announced that a pre-specified interim analysis had met the key secondary endpoint of *overall survival*. This is potentially important because overall survival is more directly patient-relevant than progression-free survival. However, the announcement was topline: detailed overall-survival data, follow-up, subgroup results, and a peer-reviewed publication are still needed for full interpretation.

42.4 What should be concluded?

Ivonescimab is not a fundamentally different type of immunotherapy from pembrolizumab on its PD-1 side. Its important difference is the addition of VEGF neutralisation within the same antibody.

The clinical results make the approach promising, particularly because ivonescimab was compared head-to-head with pembrolizumab rather than only with chemotherapy. However, it would be premature to conclude that it is universally better than pembrolizumab. The available head-to-head study was conducted in a defined first-line, PD-L1-positive advanced-NSCLC population in China, and the detailed overall-survival data have not yet been fully reported.

The central lesson is:

```
Checkpoint blockade may be limited not only by T-cell inhibition,
but also by the physical and immune-suppressive tumour environment.
Ivonescimab aims to address both:
PD-1 inhibition on T cells
and VEGF-driven vascular and microenvironmental resistance.
```

43 [26-09-05] G protein-coupled receptors: a major drug-target class

G protein-coupled receptors, or *GPCRs*, are a large family of cell-surface receptors. They detect signals outside a cell and convert them into changes in signalling inside the cell.

They are among the most important drug-target classes. The statement that they are targets of roughly one third of approved drugs is a useful approximation, although the exact proportion depends on how drugs and targets are counted.

A GPCR usually crosses the cell membrane *seven times*. It therefore has:

```
An extracellular region
→ can recognise signals outside the cell
Seven membrane-spanning segments
→ form the characteristic GPCR structure
An intracellular region
→ couples receptor activation to intracellular signalling
```

GPCRs respond to many normal biological signals, including hormones, neurotransmitters, lipids, peptides, metabolites, and sensory molecules. They help regulate heart rate, blood pressure, pain, mood, inflammation, metabolism, hormone release, smell, taste, and immune-cell movement.

43.1 How a GPCR sends a signal

The name comes from the receptor's interaction with a *heterotrimeric G protein*. This is a group of three intracellular proteins: a $G\alpha$ subunit, a $G\beta$ subunit, and a $G\gamma$ subunit.

In its inactive state, the $G\alpha$ subunit carries GDP. When an extracellular ligand activates the GPCR, the receptor changes shape and helps $G\alpha$ exchange GDP for GTP.

```
Ligand binds GPCR
→ receptor changes conformation
→ G protein is activated
Gα exchanges GDP for GTP
→ Gα and Gβγ regulate intracellular effectors
→ second messengers and signalling pathways change
→ cell behaviour changes
```

Different GPCRs can couple to different G proteins. A useful simplified picture is:

```
Gs
→ often increases cAMP signalling
Gi
→ often decreases cAMP signalling
Gq/G11
→ often activates phospholipase C,
calcium signalling, and protein kinase C
```

These are useful patterns, not complete rules. One receptor can couple differently depending on cell type, ligand, receptor state, and available signalling proteins.

GPCR signalling is also regulated by proteins called *beta-arrestins*. After prolonged activation, beta-arrestins can reduce further G-protein signalling and promote receptor internalisation. They can also initiate signalling pathways of their own.

```
Repeated GPCR activation
→ beta-arrestin recruitment
→ receptor desensitisation and/or internalisation
→ altered duration and type of signalling
```

This is important because a drug does not simply switch a receptor completely “on” or “off.” It can preferentially alter particular signalling outputs, a concept sometimes called *biased agonism*.

43.2 What GPCR-directed drugs do

A drug can bind a GPCR and activate, inhibit, or modify its signalling.

```
Agonist
→ activates the receptor
Partial agonist
→ activates the receptor, but less strongly
than a full agonist in that system
Antagonist
→ binds without activating sufficiently,
preventing the normal ligand from activating the receptor
Inverse agonist
→ reduces activity below the receptor's
baseline constitutive activity
Allosteric modulator
→ binds outside the main ligand-binding site
and changes the receptor's response to its normal ligand
```

For example, *salbutamol* is a β_2 -adrenergic receptor agonist used to relax airway smooth muscle in asthma. *Beta blockers* such as propranolol block β -adrenergic receptors and can reduce the effects of adrenaline-like signalling on the heart. *Semaglutide* activates the GLP-1 receptor, a GPCR involved in insulin secretion, appetite regulation, and glucose control.

Many GPCR drugs are small molecules because the natural ligand-binding pocket is often located within the membrane-spanning part of the receptor. A suitably designed small molecule can enter this pocket and stabilise an active or inactive receptor state.

43.3 Are targets already discussed in this document GPCRs?

Most of the receptors discussed earlier are *not* conventional GPCRs.

```
EGFR and HER2
→ receptor tyrosine kinases, not GPCRs
PD-1 and PD-L1
→ immune-checkpoint proteins, not GPCRs
VEGF
→ a soluble signalling ligand, not a receptor
VEGF receptors
→ receptor tyrosine kinases, not GPCRs
CLDN18.2
→ a tight-junction protein, not a GPCR
```

The closest example is *Frizzled7*, or FZD7, from the earlier FZD7/PD-L1 bispecific-antibody paper. Frizzled proteins belong structurally to the *class F GPCR* family because they have seven membrane-spanning segments.

However, FZD7 should not be understood as a typical GPCR such as the β_2 -adrenergic or GLP-1 receptor. FZD7 responds to Wnt ligands and is especially associated with Wnt signalling, including β -catenin-dependent and β -catenin-independent pathways. Its signalling involves proteins such as Dishevelled, and the contribution of conventional heterotrimeric G proteins is more context-dependent and less central than for many classical GPCRs.

```
Classical GPCR example:
adrenaline-like ligand binds beta-adrenergic receptor
→ heterotrimeric G-protein signalling
FZD7 example:
Wnt ligand binds Frizzled receptor
→ Wnt/Dishevelled-associated signalling
→ altered cell state, growth, and tissue behaviour
```

Thus, calling FZD7 a GPCR is structurally correct, but it does not mean its biology is identical to the more familiar GPCR examples.

43.4 Why are GPCRs difficult targets for antibodies?

GPCRs are very tractable for small molecules, but often difficult for conventional antibodies. This is mainly a problem of *physical access and molecular recognition*.

A typical antibody binds an exposed three-dimensional epitope on the outside of a protein. In many GPCRs, the most useful pharmacological pocket lies deep within the membrane-spanning structure, where a large antibody cannot reach.

In addition, the extracellular loops of many GPCRs are relatively small and flexible. This can provide only limited surface area for a selective, high-affinity antibody epitope. GPCRs are also dynamic proteins, existing in multiple active and inactive conformations, which complicates antibody discovery and characterisation.

```
Small molecule
→ can enter a pocket within the seven-transmembrane bundle
Conventional antibody
→ generally can bind only extracellular regions
Result
→ many GPCRs are easier to modulate with small molecules
than with antibodies
```

This does not make antibody targeting impossible. Some GPCRs have sufficiently accessible extracellular regions, particularly larger N-terminal regions or extracellular loops. Antibodies may block ligand binding, stabilise a particular receptor state, induce receptor internalisation, recruit immune effector functions, or deliver a payload.

The challenge is therefore not whether GPCRs matter biologically—they clearly do—but whether an individual GPCR presents an extracellular epitope that an antibody can bind productively, selectively, and safely.

43.5 The connection to drug discovery

GPCRs illustrate the distinction between a target's biological importance and the best modality for modulating it.

```
GPCR is important in disease
≠ antibody is the best modality
Accessible transmembrane binding pocket
→ small molecule may be especially suitable
Accessible extracellular epitope
→ antibody, ADC, or other biologic may be possible
```

Their long history as successful drug targets reflects several advantages: they sit at the cell surface, regulate powerful signalling pathways, can be activated or inhibited in multiple ways, and often contain pockets suitable for small molecules. Their biology can nevertheless be complex because one receptor may produce

different effects in different tissues, and therapeutic benefit must be separated from on-target effects in normal physiology.

44 [26-09-05] Benmelstobart plus antiangiogenic therapy in metastatic triple-negative breast cancer

(*Benmelstobart plus antiangiogenic therapy vs chemotherapy for metastatic triple-negative breast cancer: phase III trial*, <https://www.nature.com/articles/s41523-026-01045-5>)

This Phase III trial, *ETER901*, tested a chemotherapy-free combination in previously untreated recurrent or metastatic *triple-negative breast cancer*, or TNBC.

It compared:

Benmelstobart + anlotinib
versus
Nab-paclitaxel chemotherapy

Benmelstobart is an antibody against *PD-L1*. Anlotinib is an oral small-molecule *antiangiogenic* drug.

44.1 Is benmelstobart doing something new?

Its central checkpoint mechanism is familiar. Like atezolizumab, benmelstobart binds *PD-L1*, rather than PD-1.

PD-L1 on tumour or immune cells binds PD-1 on T cells
→ inhibitory T-cell signal
Benmelstobart binds PD-L1
→ PD-L1 cannot engage PD-1 normally
→ reduced inhibition of anti-tumour T cells

Thus, it is not a fundamentally new type of checkpoint inhibitor. The important difference in this study is its use with an antiangiogenic drug, without conventional chemotherapy.

As discussed earlier, a PD-L1 antibody differs from a PD-1 antibody such as pembrolizumab or tislelizumab:

PD-1 antibody
→ prevents PD-1 binding both PD-L1 and PD-L2
PD-L1 antibody
→ prevents PD-L1 binding PD-1
→ leaves PD-L2-PD-1 signalling possible

This biological distinction does not establish that one class is always clinically better. The relevant question is whether a particular regimen improves benefit–risk in a defined patient population.

44.2 What is antiangiogenic therapy?

Angiogenesis is the formation of new blood vessels. Growing tumours require blood vessels to obtain oxygen and nutrients, but tumour blood vessels are often abnormal: poorly organised, leaky, and inefficient.

Tumours commonly use signalling involving *vascular endothelial growth factor*, or VEGF, to promote this process. Anlotinib inhibits several receptor tyrosine kinases involved in angiogenesis, including VEGF receptors.

```
Tumour produces VEGF-associated signals
→ abnormal blood-vessel growth
→ tumour access to nutrients and oxygen
+ an immune-suppressive tumour microenvironment
Anlotinib inhibits angiogenesis-associated kinase signalling
→ aims to limit tumour vascular support
and improve the local tumour environment
```

Antiangiogenic therapy may sometimes partially “normalise” abnormal tumour vessels. This could improve immune-cell entry into the tumour and make local conditions less suppressive for T cells. However, this is a context-dependent hypothesis, not an automatic effect of every antiangiogenic drug in every tumour.

44.3 Why might the combination make sense?

The rationale is similar to the PD-1/VEGF strategy discussed for ivonescimab, but here the two mechanisms are supplied as separate medicines.

```
Benmelstobart
→ removes PD-L1-mediated inhibition of T cells
Anlotinib
→ reduces angiogenesis-associated tumour support
and may reduce VEGF-related immune suppression
Combination
→ aims to help T cells enter, persist, and function
within the tumour while reducing one checkpoint brake
```

In principle, this could control cancer without chemotherapy’s direct cytotoxic mechanism. That is the attraction of a chemotherapy-free regimen: if it gave comparable or superior durable benefit with an acceptable safety profile, it could reduce chemotherapy-related treatment burden for some patients.

44.4 Did this trial show that chemotherapy can be replaced?

No—not on the available Phase III evidence. This is the most important conclusion.

The combination showed numerically longer median progression-free survival, approximately **7.85 months** versus **5.55 months** with nab-paclitaxel. Median overall survival was also numerically longer, approximately **35.81 months** versus **21.03 months**.

However, neither the progression-free-survival nor overall-survival comparison reached statistical significance. The study enrolled only **147 patients**, after recruitment was disrupted and then terminated during the COVID-19 pandemic. This reduced its statistical power: even a real treatment difference may be difficult to establish reliably in a smaller-than-planned trial.

```
Numerically favourable result
≠ statistically confirmed treatment benefit
Statistically non-significant Phase III result
≠ proof that chemotherapy can be omitted
```

The experimental combination also had substantial toxicity. Severe treatment-related adverse events were more frequent than with nab-paclitaxel, notably including hypertension, which is consistent with VEGF-pathway inhibition.

44.5 What should be concluded?

Benmelstobart is a conventional *PD-L1 checkpoint inhibitor*; the novel clinical strategy is pairing it with anlotinib, an antiangiogenic multi-kinase inhibitor, as a chemotherapy-free first-line treatment.

The biological rationale is credible: antiangiogenic therapy may reduce vascular and immune barriers that limit checkpoint inhibition, while PD-L1 blockade may permit tumour-reactive T cells to act more effectively. But this particular trial did *not* provide statistically definitive evidence that the combination is better than nab-paclitaxel chemotherapy or that it can replace chemotherapy in metastatic TNBC.

The broader lesson is that a rational mechanism and encouraging numerical survival estimates are not enough. A chemotherapy-free regimen must still demonstrate robust benefit–risk in an adequately powered trial before it can be considered an established alternative.

45 [26-09-05] Daraxonrasib: a molecular-glue inhibitor of active RAS

(*Daraxonrasib in Previously Treated Advanced RAS-Mutated Pancreatic Cancer*, <https://www.nejm.org/doi/full/10.1056/NEJMoa2505783>, *Daraxonrasib for Previously Treated RAS-Mutant Non–Small-Cell Lung Cancer*, <https://www.nejm.org/doi/full/10.1056/NEJMoa2504059>)

Daraxonrasib, also called *RMC-6236*, is an oral small-molecule treatment designed for cancers driven by mutations in the *RAS* family of signalling proteins. The two studies report early clinical results in previously treated advanced pancreatic cancer and non-small-cell lung cancer, or NSCLC.

Its central importance is that it aims to inhibit multiple mutant forms of RAS, rather than one specific mutation such as KRAS G12C.

45.1 A reminder: what does RAS normally do?

RAS proteins, including KRAS, NRAS, and HRAS, are intracellular molecular switches. They move between:

```
RAS-GDP
→ predominantly inactive, or RAS(OFF)
RAS-GTP
→ active, or RAS(ON)
→ can bind downstream effector proteins
such as RAF and PI3K
→ growth and survival signalling
```

Normally, RAS is activated transiently in response to external signals, such as signalling from a receptor tyrosine kinase. It then returns to its GDP-bound state.

Many cancer-associated RAS mutations impair this switching-off process. The mutant protein can therefore spend too much time in its active GTP-bound state, driving sustained signalling through pathways such as RAF–MEK–ERK.

45.2 Is daraxonrasib a molecular glue?

****Yes.**** Daraxonrasib is often described as a *molecular glue*, although it works differently from the protein-degrading molecular glues discussed elsewhere.

A conventional molecular glue can bind two proteins and stabilise their interaction. Some molecular glues recruit an E3 ubiquitin ligase to a target protein, causing the target to be degraded. Daraxonrasib does *not* primarily work by degrading RAS.

Instead, it promotes formation of a non-productive three-component complex, or *tri-complex*, between active RAS and *cyclophilin A*, or CypA.

```
Daraxonrasib binds cyclophilin A
→ drug-CypA complex forms
Drug-CypA complex binds active RAS-GTP
→ RAS-daraxonrasib-CypA tri-complex forms
CypA occupies physical space near RAS effector-binding regions
→ RAS cannot effectively bind effectors such as RAF
Reduced RAS-effector interaction
→ reduced downstream growth and survival signalling
```

Thus, CypA acts as a deliberately recruited “blocker.” Daraxonrasib brings CypA close enough to active RAS that the larger complex sterically prevents RAS from making the protein–protein interactions needed to transmit its oncogenic signal.

45.3 How is this different from KRAS G12C inhibitors?

Earlier in the document, KRAS G12C inhibitors were described as covalent drugs that bind the mutant cysteine in a pocket accessible when KRAS is in its GDP-bound inactive state. They trap KRAS G12C in its OFF state.

Daraxonrasib uses a different strategy:

```
KRAS G12C inhibitor
→ covalently binds mutant KRAS G12C
in the GDP-bound RAS(OFF) state
Daraxonrasib
→ non-covalently recruits CypA to the GTP-bound
RAS(ON) state
→ inhibits multiple mutant RAS variants
by blocking downstream-effector binding
```

This matters because many clinically important RAS mutations do not contain the G12C cysteine required for the earlier covalent strategy. Daraxonrasib has activity across several KRAS, NRAS, and HRAS mutations, including variants at G12, G13, and Q61. It can also inhibit wild-type RAS signalling, which may contribute both to efficacy and to toxicity.

45.4 What did the pancreatic-cancer study show?

The pancreatic-cancer paper reports results from the Phase I/II *RMC-6236-001* study in previously treated advanced RAS-mutated pancreatic ductal adenocarcinoma.

In the subgroup with RAS G12-mutated pancreatic cancer who received the recommended Phase III dose of 300 mg daily in the second-line setting, the confirmed objective response rate was approximately **35%**. Median progression-free survival was **8.5 months**, and median overall survival was **13.1 months**.

These results are encouraging in a disease where treatment options after initial therapy are limited. However, the study was not a randomised comparison against a standard treatment. It therefore shows substantial anti-tumour activity and supports further testing, but does not by itself prove superiority over an appropriate control regimen.

45.5 What did the NSCLC study show?

The NSCLC paper studied previously treated RAS-mutant NSCLC, excluding KRAS G12C, in patients who had generally already received platinum-based chemotherapy and PD-(L)1 checkpoint inhibition.

Responses were observed across dose levels. In a clinically notable subgroup of *docetaxel-naïve* patients treated with 160–220 mg daily, the confirmed objective response rate was approximately **42%**, disease-control

rate was **89%**, median progression-free survival was **8.3 months**, and median overall survival was **16.0 months**.

Again, these are early-phase, non-randomised data. They are promising rather than definitive. The ongoing randomised Phase III *RASolve 301* trial is comparing daraxonrasib with docetaxel in previously treated metastatic RAS-mutant NSCLC.

45.6 Safety and the therapeutic challenge

The most common treatment-related adverse events included rash, diarrhoea, nausea, vomiting, and inflammation of the mouth lining, or mucositis. Some patients experienced severe adverse events or required dose modifications.

This safety profile is understandable in context. RAS signalling is important in normal tissues, particularly those that renew rapidly, such as skin and the gastrointestinal tract. A broadly acting RAS(ON) inhibitor may therefore affect both tumour RAS and normal RAS biology.

45.7 What should be concluded?

Daraxonrasib is a notable attempt to solve a long-standing drug-discovery problem: how to inhibit a broad range of active RAS proteins, which function through protein–protein interactions and were historically considered difficult to drug.

It is a molecular glue in the sense that it induces a new, inhibitory interaction between active RAS and cyclophilin A. It does not degrade RAS; rather, it uses CypA as a physical shield that blocks RAS from engaging its downstream effectors.

Mutant RAS remains in its active GTP-bound state
→ daraxonrasib recruits CypA to active RAS
→ RAS cannot productively bind RAF, PI3K,
and other effector proteins
→ oncogenic RAS signalling is reduced

The clinical findings in pancreatic cancer and NSCLC demonstrate meaningful early anti-tumour activity across RAS-mutant cancers. Whether this translates into a favourable benefit–risk profile relative to existing treatments must be determined by the ongoing randomised trials.

46 [26-09-06] Neoadjuvant PD-1 plus CTLA-4 blockade in recurrent glioblastoma

(*Neoadjuvant PD1 plus CTLA-4 immune checkpoint blockade in surgically resectable recurrent glioblastoma: a randomized surgical window-of-opportunity trial*, <https://www.nature.com/articles/s41467-026-77433-z>)

This paper reports a randomised *surgical window-of-opportunity* trial in patients with surgically resectable recurrent glioblastoma.

A window-of-opportunity trial gives treatment shortly before a planned operation. The tumour is then removed and studied directly. This can show whether the treatment reached the tumour and changed its immune environment, even when the trial is too small or too early to establish a survival benefit reliably.

The trial compared one pre-operative dose of:

Nivolumab + ipilimumab
versus
Nivolumab + placebo

Both groups then received the planned post-operative immunotherapy regimen. Thus, the key randomised question was whether adding *CTLA-4 blockade before surgery* changed the early immune response relative to PD-1 blockade alone.

46.1 A reminder: what PD-1 does

PD-1 is an inhibitory receptor expressed by activated T cells. When PD-1 binds PD-L1 or PD-L2, it reduces T-cell activity. This can help prevent excessive immune damage in normal physiology, but tumours can exploit the pathway to suppress anti-tumour T cells.

PD-1 engagement on an activated T cell
→ weaker T-cell activation and killing function
Nivolumab binds PD-1
→ reduces PD-1-mediated inhibition
→ can help tumour-reactive T cells sustain activity

PD-1 blockade acts mainly at the stage where an existing T cell is encountering antigen and is being restrained, often within the tumour microenvironment.

46.2 What does CTLA-4 do?

CTLA-4 is also an inhibitory receptor on T cells, but it acts somewhat earlier and in a different immune context from PD-1.

For a T cell to become fully activated, it generally needs both antigen recognition and a co-stimulatory signal. One important co-stimulatory receptor is *CD28* on the T cell. CD28 binds *CD80* or *CD86* on antigen-presenting cells.

CTLA-4 competes with CD28 for these same CD80/CD86 molecules, but it binds them more strongly. CTLA-4 therefore limits co-stimulation and restrains the activation or expansion of T cells.

T-cell receptor recognises peptide-MHC
→ signal 1
CD28 binds CD80/CD86 on an antigen-presenting cell
→ co-stimulatory signal 2
→ stronger T-cell activation
CTLA-4 binds CD80/CD86
→ reduces CD28 co-stimulation
→ restrains T-cell activation and expansion

Ipilimumab is an antibody that blocks CTLA-4. Its intended effect is to allow stronger co-stimulation and expansion of tumour-reactive T cells.

46.3 PD-1 versus CTLA-4

The distinction is simplified, but useful:

CTLA-4 blockade
→ can enhance early T-cell priming, activation, and expansion, particularly in lymphoid tissues
PD-1 blockade
→ can restore or sustain activity of T cells encountering antigen in tumours and peripheral tissues

In other words, CTLA-4 blockade may help generate or expand a broader anti-tumour T-cell response, while PD-1 blockade may help those activated T cells function when they reach the tumour.

Both checkpoints contribute to normal immune tolerance. Accordingly, blocking both can also cause more immune-related toxicity than blocking PD-1 alone.

46.4 Why treat before surgery?

Giving immunotherapy while the tumour is still present may provide a source of tumour antigens for T-cell priming. The immune system can potentially encounter tumour material, activate and expand relevant T-cell clones, and then retain some anti-tumour immune memory after the tumour is removed.

```
Intact recurrent tumour before surgery
→ contains tumour antigens
Neoadjuvant checkpoint blockade
→ aims to activate and expand tumour-reactive T cells
while antigen is available
Surgery removes tumour
→ resected tissue enables direct study of
T-cell infiltration, activation, and tumour immune biology
```

This is particularly useful in glioblastoma, where the tumour microenvironment is commonly immunosuppressive and systemic checkpoint inhibitors have historically shown limited clinical activity.

46.5 Why might dual blockade make sense?

The combination attempts to address two related but distinct barriers to effective anti-tumour immunity:

```
CTLA-4 blockade with ipilimumab
→ promotes T-cell priming and expansion
PD-1 blockade with nivolumab
→ reduces inhibition of activated T cells
Combination
→ aims to generate more tumour-reactive T cells
and help them retain function in the tumour
```

This does not mean that dual blockade must be better. It might increase the number and activation of relevant T cells, but glioblastoma can still evade immunity through poor antigen presentation, limited T-cell entry, corticosteroid exposure, myeloid-cell-mediated suppression, tumour heterogeneity, and local metabolic constraints. In addition, stronger immune activation creates a higher risk of immune-related adverse events.

46.6 What did the study show?

The main value of this study is mechanistic. The pre-operative treatment and subsequent tumour resection allowed the investigators to compare the local immune consequences of adding neoadjuvant ipilimumab to nivolumab.

The paper supports the idea that dual checkpoint blockade can alter the immune state of recurrent glioblastoma and promotes more active anti-tumour T-cell biology than PD-1 blockade alone in this setting. However, it should not be interpreted as establishing dual checkpoint blockade as an effective standard treatment for recurrent glioblastoma.

This was a relatively small surgical window study, designed principally to investigate biological effects in tumour tissue. Demonstrating a change in immune cells, gene expression, or T-cell receptor clones is important *proof-of-mechanism* evidence, but it is not the same as demonstrating durable tumour control or improved overall survival.

46.7 What should be concluded?

Nivolumab and ipilimumab are not two versions of the same drug. They release different inhibitory checkpoints at different stages of the T-cell response:

```
Ipilimumab blocks CTLA-4
→ aims to improve T-cell priming and expansion
Nivolumab blocks PD-1
→ aims to restore or sustain T-cell activity
Together
→ aim to create a stronger and more durable
anti-tumour immune response
```

For recurrent glioblastoma, the neoadjuvant approach is especially informative because it allows direct examination of the treated tumour. The study provides a rationale for further development, but the critical next question remains clinical: can this immune activation be translated into a sufficiently large and durable patient benefit to justify the additional toxicity of dual checkpoint blockade?

47 [26-09-06] PRMT7 as a brake on CD8 T-cell expansion

(*PRMT7 restricts CD8+ T cells expansion via the NF- κ B pathway*, <https://www.nature.com/articles/s41467-026-77155-2>)

This paper identifies *PRMT7* as an internal brake on CD8 T cells. When *PRMT7* was removed or inhibited, CD8 T cells expanded more strongly and showed increased anti-tumour activity in the experimental systems studied.

PRMT7 is a *protein arginine methyltransferase*: an enzyme that adds methyl groups to particular arginine residues on proteins. Such modifications can change protein interactions, location, stability, or function. In this case, *PRMT7* affects a central immune-signalling pathway called *NF- κ B*.

47.1 What is NF- κ B?

NF- κ B is a family of transcription factors that helps cells respond to activation signals, infection, stress, and inflammation. In T cells, appropriately regulated NF- κ B signalling helps support activation, proliferation, survival, cytokine production, and effector function.

A simplified chain is:

```
T-cell receptor and co-stimulatory signals
→ NF- $\kappa$ B activation
NF- $\kappa$ B enters the nucleus
→ expression of genes supporting
T-cell activation, expansion, and function
```

One important NF- κ B component is called *RelA*, also known as p65. For *RelA* to influence gene expression, it must move into the cell nucleus.

47.2 What does *PRMT7* do here?

The paper suggests that *PRMT7* associates with *RelA* and restricts its movement into the nucleus in activated CD8 T cells. In effect, *PRMT7* limits NF- κ B-dependent gene expression and therefore limits the size and functional activity of the CD8 T-cell response.

```
PRMT7 present in CD8 T cell
→ restricts RelA nuclear translocation
→ weaker NF-κB transcriptional activity
→ reduced T-cell expansion and effector function
PRMT7 deleted or inhibited
→ more RelA reaches the nucleus
→ stronger NF-κB-associated programme
→ increased CD8 T-cell expansion and activity
```

The reported consequences included greater proliferation and increased expression of T-cell activation and effector features, including inflammatory cytokines such as interferon- γ .

47.3 Why is this interesting for cancer immunotherapy?

The earlier checkpoint-inhibitor examples act mainly through *external inhibitory receptors* on T cells:

```
Anti-PD-1 or anti-PD-L1 treatment
→ blocks an inhibitory signal delivered
through the PD-1 checkpoint pathway
Anti-CTLA-4 treatment
→ increases the availability of CD28 co-stimulation
during T-cell activation
```

PRMT7 is different. It is an *intracellular regulator within the T cell itself*. Inhibiting PRMT7 could potentially make transferred or endogenous CD8 T cells more able to expand and function after they receive an activation signal.

```
Checkpoint blockade
→ removes inhibitory signals at the T-cell surface
PRMT7 inhibition
→ reduces an intracellular brake on
NF-κB-dependent T-cell activity
```

This suggests possible relevance for adoptive T-cell therapies, including CAR-T cells. A CAR provides an artificial activation signal when it recognises tumour antigen; reducing a downstream intracellular brake could potentially help CAR-T cells expand or persist.

47.4 What should be concluded?

The paper provides *mechanistic and preclinical* evidence that PRMT7 restrains CD8 T-cell expansion through the NF- κ B pathway. It identifies PRMT7 as a potentially useful immunotherapy target, but it does not establish that systemic PRMT7 inhibition will be safe or effective in patients.

This caution is particularly important because NF- κ B is a powerful and widely used signalling pathway. Stronger T-cell activity could improve tumour control, but uncontrolled immune activation could also cause toxicity, inflammation, or autoimmunity. Furthermore, PRMT7 can have different functions in different cell types, so the consequences of inhibiting it in a whole patient may differ from deleting it selectively in CD8 T cells.

The core idea is:

PRMT7 acts as an intracellular brake on CD8 T cells. Removing that brake increases NF- κ B-associated activation, allowing stronger T-cell expansion and anti-tumour function in the models studied.

48 [26-09-10] Omomyc: directly targeting the MYC transcription factor

(*MYC targeting by OMO-103 in solid tumors: a phase 1 trial*, <https://www.nature.com/articles/s41591-024-02805-1>, *MYC inhibition by Omomyc causes DNA damage and overcomes PARPi resistance in breast cancer*, <https://www.sciencedirect.com/science/article/pii/S2211124725013762?via%3Dihub>)

48.1 What does MYC do?

MYC is a transcription factor: a protein that changes which genes a cell transcribes into RNA. It is therefore not an enzyme with a conventional active site, but a regulator of broad gene-expression programmes.

MYC normally helps cells respond to growth signals. When active, it forms a complex with another protein, MAX, and the MYC-MAX complex binds particular DNA sequences, often called *E-boxes*. This can increase expression of many genes involved in cell growth, metabolism, protein production, cell-cycle progression, and DNA replication.

Growth signals
→ MYC expression or activity increases
→ MYC binds MAX
→ MYC-MAX alters gene expression
→ cell growth and division are promoted

In cancer, MYC can be overexpressed, amplified, or activated downstream of other oncogenic pathways. Persistent MYC activity can make tumour cells grow rapidly and tolerate substantial metabolic and replication stress. However, MYC itself does not usually contain the deep, well-defined pocket that makes many enzymes suitable for conventional small-molecule inhibitors. It was therefore long considered difficult to drug directly.

48.2 What are Omomyc and OMO-103?

Omomyc is an engineered, dominant-negative version of part of the MYC protein. OMO-103 is a 91-amino-acid mini-protein based on Omomyc that can enter cells and reach the nucleus.

Rather than inhibiting an enzyme active site, OMO-103 interferes with the protein-protein and protein-DNA interactions required for MYC function. It can form non-productive complexes with MYC and MAX, reducing formation and DNA binding of functional MYC-MAX complexes.

OMO-103 enters tumour cell and nucleus
→ interferes with MYC-MAX function
→ reduces MYC-dependent gene transcription
→ reduces programmes supporting growth, metabolism, replication, and stress tolerance

This is a useful example of a therapeutic strategy directed at a transcription factor and protein-protein interaction, rather than a conventional enzyme pocket.

48.3 What did the Phase I trial show?

The 2024 first-in-human Phase I trial tested OMO-103 in 22 heavily pre-treated patients with advanced solid tumours. Its principal purpose was to establish safety, pharmacokinetics, and a suitable dose range—not to demonstrate definitive clinical efficacy.

The study reported a generally manageable safety profile, with infusion-related reactions among the most common adverse events. It also provided evidence consistent with target engagement and reported preliminary anti-tumour activity, including disease stabilisation in several patients and one partial response.

The appropriate conclusion is:

The trial supports that direct MYC inhibition with a cell-penetrating mini-protein is feasible in humans.
It does not yet establish clinical benefit in a defined cancer population.

Larger, disease-specific and controlled studies are needed to determine which patients benefit, the best combination partners, and whether benefit is sufficiently durable relative to standard care.

48.4 Why might MYC inhibition help overcome PARP-inhibitor resistance?

The 2025 breast-cancer study investigated Omomyc in preclinical models, particularly triple-negative breast cancer models resistant to *PARP inhibitors*, or PARPi. Recall that PARP inhibition can exploit homologous-recombination deficiency, such as BRCA-associated DNA-repair impairment.

PARPi-resistant models in this study showed increased MYC transcriptional activity. MYC can drive rapid DNA replication and genomic stress, but it can also support expression of programmes that help tumour cells tolerate and repair DNA damage.

The reported logic is:

PARPi-resistant tumour cell
→ increased MYC-associated DNA-damage-response programmes
→ improved tolerance of replication-associated DNA damage
Omomyc inhibits MYC
→ DNA-damage-response capacity is reduced
→ DNA damage accumulates
Omomyc + PARPi
→ tumour cell faces greater damage
with less ability to manage it
→ increased cell death in preclinical models

The study found that Omomyc induced DNA damage and improved the activity of PARP inhibitors in cell-line and patient-derived xenograft models. This suggests a possible combination strategy and a potential role for a MYC transcriptional signature as a predictive biomarker.

However, this remains *preclinical* evidence. It does not yet show that OMO-103 plus a PARP inhibitor is safe or effective in patients with PARPi-resistant breast cancer.

48.5 What should be concluded?

MYC is a central regulator of cell-growth and stress-response gene expression, so MYC-dependent cancers may have an important vulnerability. OMO-103 is notable because it offers a way to interfere directly with this historically difficult target using a cell-penetrating mini-protein.

The Phase I study is encouraging evidence of feasibility and early biological activity. The PARPi-resistance study gives a plausible mechanistic rationale for combination therapy in selected breast cancers. But the key remaining clinical question is:

Which tumour types and biomarker-defined patients
are sufficiently dependent on MYC that MYC inhibition produces
a clinically meaningful benefit at an acceptable level of toxicity?

49 [26-09-10] Tislelizumab plus lenvatinib in fumarate-hydratase-deficient renal cell carcinoma

(*Tislelizumab plus lenvatinib in advanced fumarate hydratase-deficient renal cell carcinoma: a single-arm, phase 2 trial with metabolic biomarker analysis*, <https://www.nature.com/articles/s41467-026-77615-9>)

49.1 What is fumarate-hydratase-deficient renal cell carcinoma?

Fumarate hydratase, or *FH*, is an enzyme in the tricarboxylic-acid cycle, a central metabolic pathway through which cells extract energy and generate molecular building blocks.

FH normally converts fumarate into malate. When both functional copies of *FH* are lost in a tumour cell, fumarate accumulates. FH-deficient renal cell carcinoma, or *FHdRCC*, is an aggressive kidney-cancer subtype that can arise through inherited or tumour-acquired *FH* alterations.

Loss of FH activity
→ fumarate accumulates
→ cell metabolism and signalling are altered
→ tumour-promoting stress and growth programmes can emerge

Excess fumarate can chemically modify proteins, a process called *succination*. Detection of a succinated-protein product called *2SC* can therefore support identification of FH-deficient tumours. This is an example of a *metabolic biomarker*: a measurable consequence of the altered metabolism, rather than simply the underlying DNA alteration.

FH loss can also promote a state sometimes called *pseudohypoxia*. The cell activates parts of a low-oxygen response despite oxygen being present. This can increase expression of pro-angiogenic signals, including pathways involving *VEGF*, which promote formation of tumour blood vessels.

49.2 How does tislelizumab work?

Tislelizumab is a PD-1-blocking antibody, in the same broad therapeutic class as pembrolizumab discussed earlier.

PD-1 is an inhibitory receptor on activated T cells. When PD-L1 or PD-L2 binds PD-1, T-cell activity is reduced. Tumours can exploit this checkpoint to suppress anti-tumour immunity.

PD-L1 or PD-L2 binds PD-1 on T cell
→ inhibitory signalling
→ reduced anti-tumour T-cell activity
Tislelizumab binds PD-1
→ reduces PD-1-mediated inhibition
→ tumour-reactive T cells may function more effectively

Tislelizumab does not directly kill cancer cells. Its benefit depends on the presence or development of tumour-reactive immunity, antigen presentation, T-cell access to the tumour, and whether PD-1 signalling is an important barrier.

49.3 How does lenvatinib work?

Lenvatinib is an orally administered multi-target tyrosine-kinase inhibitor. Its important targets include VEGF receptors, which help regulate angiogenesis, as well as fibroblast-growth-factor receptors and several other receptor kinases.

Angiogenesis is formation of new blood vessels. Tumours need blood vessels to obtain oxygen and nutrients, remove waste, and grow beyond a limited size.

```
Tumour pro-angiogenic signals, including VEGF
→ VEGF receptors on vascular cells are activated
→ abnormal tumour blood vessels form
→ tumour growth and dissemination are supported
Lenvatinib inhibits VEGF-receptor signalling
→ reduced angiogenic support for the tumour
```

Lenvatinib is not selective only for tumour cells. Its inhibition of normal vascular and other signalling pathways helps explain toxicities such as hypertension, diarrhoea, fatigue, proteinuria, and hand-foot skin reactions.

49.4 Why does the combination make sense?

The combination uses two distinct, potentially complementary interventions.

FH-deficient tumours can have abnormal metabolism and a strong angiogenic phenotype. Lenvatinib aims to reduce the tumour's vascular support. In addition, VEGF-driven tumour vasculature can be disorganised and immunosuppressive, making it harder for T cells to enter and function within the tumour.

```
Lenvatinib
→ reduces VEGF-associated angiogenesis
→ may make the tumour microenvironment less immunosuppressive
→ may improve conditions for anti-tumour immunity
Tislelizumab
→ relieves PD-1-mediated inhibition of T cells
Combination
→ aims to impair tumour vascular support
and strengthen anti-tumour immune activity
```

This is a rationale, not proof that the drugs are synergistic in every patient. The clinical question is whether their combined effects improve benefit-risk compared with an appropriate alternative.

49.5 What did the Phase II study show?

This single-arm Phase II trial evaluated first-line tislelizumab plus lenvatinib in patients with advanced FHRCC and included metabolic-biomarker analyses. The study reported a high objective-response rate, with responses observed in patients with both inherited and tumour-acquired *FH* alterations.

The metabolic analyses are important because they connect the disease-defining loss of FH activity, fumarate-associated metabolic changes, and clinical behaviour. They may eventually help identify patients most likely to benefit or clarify mechanisms of resistance.

However, the evidence should be interpreted in context:

Single-arm Phase II result
→ encouraging evidence of activity
No randomised control group
→ cannot by itself establish comparative benefit,
long-term survival benefit, or the separate contribution
of each medicine

The regimen also requires careful safety management. Both drugs can contribute clinically meaningful toxicity, and lenvatinib dose interruption or reduction may be needed.

49.6 What should be concluded?

FHdRCC illustrates how a metabolic enzyme defect can shape tumour biology, biomarkers, and treatment hypotheses. Loss of FH causes fumarate accumulation and can promote angiogenesis and other cancer-supporting programmes. Lenvatinib targets important angiogenic signalling, while tislelizumab aims to restore anti-tumour T-cell activity.

The Phase II results are promising for a rare and aggressive cancer subtype, but confirmation with longer follow-up and comparative evidence remains important. The central question is:

Can a metabolic definition of FH-deficient disease identify patients
for whom angiogenesis inhibition plus PD-1 blockade produces
durable, clinically meaningful benefit at acceptable toxicity?

50 [26-09-10] Programmable particles for CAR-T-cell expansion

(*Programmable CAR-engaging particles rewire human CAR T-cell signaling for sustainable ex vivo expansion*, <https://www.nature.com/articles/s41467-026-76570-9>)

CAR-T therapy requires collection, engineering, and expansion of a patient's T cells outside the body, or *ex vivo*, before they are infused back into the patient. A manufacturing challenge is that strong or continuous stimulation can expand cells initially but also drive excessive differentiation, metabolic stress, and *T-cell exhaustion*. Exhausted T cells may persist and function less well after infusion.

This study describes *programmable CAR-engaging particles*, or *CAREps*: biodegradable microparticles coated with DNA scaffolds that can display defined combinations of molecules to stimulate CAR-T cells.

CAREp presents CAR target antigen
→ activates the CAR
CAREp presents anti-CD28 signal
→ provides co-stimulation
Controlled repeated stimulation
→ CAR-T-cell expansion with reduced exhaustion

Recall that T-cell activation normally requires more than antigen recognition. The CAR provides an engineered recognition and activation signal, while CD28 is an important co-stimulatory pathway that supports T-cell activation, proliferation, survival, and metabolism.

The key idea is not simply to stimulate CAR-T cells more strongly. CAREps allow the *timing*, *amount*, and *combination* of activating signals to be controlled. They act as a configurable artificial cell-like surface, rather than using living tumour cells or continuous non-specific stimulation.

The study found that repeated CAREp stimulation supported CAR-T-cell expansion for more than 100 days in *ex vivo* experiments. The expanded cells showed less terminal differentiation and improved mitochondrial fitness compared with conventional stimulation approaches.

Uncontrolled or persistent activation
→ rapid expansion but metabolic stress and exhaustion
Programmable intermittent stimulation
→ sustained expansion
+ more favourable metabolic state
+ potentially improved persistence and function

This is primarily a *cell-manufacturing* advance, rather than a new CAR-T treatment or proof of better patient outcomes. The important remaining question is whether CAR-T cells produced using this approach are safer, persist longer, and provide more durable anti-tumour benefit after infusion into patients.

51 [26-09-10] Multi-omic profiling for ADC target selection in ovarian cancer

(*Multi-omic profiling reveals antibody-drug conjugate targetability in ovarian cancer*, <https://www.nature.com/articles/s43856-026-01879-x>)

This study asks a practical question: *which cell-surface targets are present, sufficiently tumour-selective, and sufficiently consistent to support antibody-drug conjugate therapy in high-grade serous ovarian cancer (HGSC)?*

Recall that an ADC requires more than an antibody-binding target. The target should ideally be present on many tumour cells, exposed at the cell surface, relatively absent from essential normal tissues, and retained across tumour sites and over time.

Suitable ADC target
→ expressed on tumour-cell surface
→ sufficiently homogeneous and durable
→ antibody can bind and internalise
→ payload is released in susceptible cells
→ useful tumour killing with acceptable toxicity

51.1 What does “multi-omic” mean here?

The investigators analysed 867 tumour samples from 304 patients in the DECIDER cohort using complementary data types. Bulk RNA sequencing measures average gene expression across a tumour sample. Single-cell RNA sequencing helps identify *which cells* express a target, distinguishing cancer cells from immune and stromal cells. Whole-genome sequencing provides information about tumour genomic alterations.

Using several measurement types matters because high expression in bulk tumour tissue does not necessarily mean that cancer cells themselves express the target. It could instead reflect surrounding non-cancer cells.

51.2 What did the study find?

The study evaluated 11 targets of clinical ADCs. The most promising targets in HGSC were *FOLR1*, *TACSTD2*, and *ERBB2*.

FOLR1 encodes folate receptor alpha, a cell-surface protein already targeted by the ADC mirvetuximab soravtansine in selected folate-receptor-alpha-positive ovarian cancers. *TACSTD2* encodes Trop-2, another cell-surface protein targeted by several ADC programmes. *ERBB2* encodes HER2, discussed earlier in connection with trastuzumab and T-DXd.

FOLR1 and *TACSTD2* were particularly notable because single-cell data suggested relatively frequent expression in cancer cells rather than predominantly in the tumour microenvironment. Expression of ADC targets was generally more variable *between patients* than between samples from different sites or time points within the same patient.

Approximately 80% of patients had relatively homogeneous expression of at least one potentially actionable ADC target. Co-expression of FOLR1 and TACSTD2 was also frequent, suggesting possible opportunities for sequential treatment or dual-target strategies.

51.3 Why is this useful?

Ovarian cancer is heterogeneous, so a single universal ADC target is unlikely to be sufficient for every patient. This work supports a biomarker-led approach:

```
Profile an individual tumour
→ identify a sufficiently expressed surface target
→ choose an ADC with an appropriate target and payload
→ monitor for target loss, heterogeneity, and resistance
```

The finding that target expression can be relatively stable from diagnosis to relapse is encouraging, because it suggests an archival or early tumour sample may sometimes remain informative. However, this should not be assumed for every patient or every target.

51.4 What should not be concluded?

This is a targetability study, not a clinical trial showing that all target-positive patients benefit from an ADC. RNA expression is also not identical to cell-surface protein abundance, target internalisation, payload sensitivity, or a favourable safety window.

Thus, the study identifies promising *hypotheses and patient-selection opportunities*. Clinical evidence is still needed to establish which target, ADC design, biomarker threshold, treatment sequence, and patient population provide meaningful benefit.

52 [26-09-10] Disrupting CD73 to reduce adenosine-mediated immune suppression

(*CD73 disruption with tumor-specific CRISPR/Cas9 nanoparticles reduces adenosine-mediated immunosuppression and amplifies response to chemotherapy*, <https://www.nature.com/articles/s41467-026-77547-4>)

CD73 is a cell-surface enzyme encoded by the *NT5E* gene. It is part of an extracellular nucleotide-processing pathway that can convert inflammatory danger signals into an immunosuppressive signal.

When cells are stressed, damaged, or killed, they can release ATP into the tumour microenvironment. Extracellular ATP can support inflammation and immune activation. However, tumour cells and other cells in the tumour microenvironment can process ATP through two main enzymes:

```
Extracellular ATP or ADP
  CD39 → AMP
  CD73 → adenosine
```

CD73 performs the final step: it converts AMP into *adenosine*. Adenosine can bind receptors such as A2A receptors on T cells and natural-killer cells, reducing their anti-tumour activity. It can also support a more immunosuppressive tumour microenvironment.

```
High CD73 activity
→ more extracellular adenosine
→ reduced T-cell and NK-cell function
→ increased immune suppression
→ tumour immune escape
```

52.1 What does this study do?

This study uses tumour-targeted nanoparticles to deliver CRISPR/Cas9 gene-editing machinery intended to disrupt *NT5E* in tumour cells. In principle, successful editing reduces CD73 production more durably than temporarily blocking the enzyme with a conventional drug.

```
Tumour-targeted CRISPR/Cas9 nanoparticle
→ enters tumour cell
→ disrupts NT5E
→ less CD73 at the cell surface
→ less AMP-to-adenosine conversion
→ less adenosine-mediated immune suppression
```

The tumour specificity of delivery is essential. CD73 and adenosine signalling also have normal biological roles, so widespread editing outside the tumour could create unintended effects.

52.2 Why combine this with chemotherapy?

Some chemotherapy can cause *immunogenic cell death*: dying tumour cells release ATP and tumour antigens that may help recruit and activate immune responses.

However, in a CD73-high tumour, the released ATP can be converted into adenosine, turning a potentially immune-stimulating signal into an immune-suppressing one.

```
Chemotherapy kills tumour cells
→ ATP and tumour material are released
With high CD73:
ATP-derived AMP → adenosine
→ immune response is suppressed
With CD73 disruption:
less AMP → adenosine conversion
→ ATP-associated inflammatory signalling is retained
→ chemotherapy-associated anti-tumour immunity may increase
```

The study reports that CD73 editing reduced adenosine-mediated suppression and improved chemotherapy responses in preclinical models. The central idea is therefore to alter the tumour microenvironment, not simply to kill cancer cells directly.

52.3 What should be concluded?

This is an early *preclinical* therapeutic strategy. It provides a biologically coherent way to combine gene editing, nanoparticle delivery, chemotherapy, and cancer immunology.

It does not yet establish that tumour-specific CRISPR editing is safe, efficiently delivered, durable, or clinically beneficial in patients. Important remaining questions include the accuracy of tumour targeting, editing efficiency across heterogeneous tumours, off-target editing, immune responses to the delivery system or Cas9, and whether the added benefit justifies these complexities.

53 [26-09-10] GPC-3-targeted radical therapy for hepatocellular carcinoma

(*Generation of GPC-3-targeted radicals for precise therapy of hepatocellular carcinoma*, <https://www.nature.com/articles/s41467-026-77544-7>)

Hepatocellular carcinoma, or *HCC*, is the most common primary liver cancer. The study investigates an experimental precision-treatment strategy based on *glypican-3*, or *GPC-3*.

GPC-3 is a protein displayed on the surface of many HCC cells, while expression is usually low or absent in normal adult liver tissue. It is not necessarily the process driving the cancer; rather, it can serve as a relatively tumour-selective *address label* for delivering a therapeutic effect.

```
GPC-3 expressed on HCC-cell surface
→ GPC-3-targeting agent binds tumour cell
→ radical-generating system is localised near the tumour
→ local oxidative damage may kill the cell
```

53.1 What are radicals, and why are they useful?

A *free radical* is a highly reactive molecule with an unpaired electron. Reactive oxygen species, or *ROS*, include radicals and related reactive molecules. At sufficiently high local concentrations, they can damage DNA, proteins, lipids, and cellular membranes.

Cancer cells often already experience elevated oxidative and metabolic stress. Further local radical production may push them beyond the level of damage they can tolerate.

```
Local radical generation
→ oxidative stress and molecular damage
→ disrupted cell function and DNA damage
→ tumour-cell death
```

The therapeutic challenge is that radicals are inherently non-selective: they can damage normal cells as well as cancer cells. The proposed solution is therefore not a radical that recognises cancer by itself, but a system that generates radicals *only after localisation to GPC-3-positive tumour cells*.

53.2 What is the central innovation?

The work combines two components: a GPC-3-recognition component, which directs the system towards HCC cells, and a radical-generating component, which creates the cytotoxic effect locally.

This is conceptually related to an ADC, but differs in the nature of the payload:

```
ADC
→ antibody delivers a defined cytotoxic chemical payload
GPC-3-targeted radical strategy
→ targeting component localises a system
that generates short-lived reactive species nearby
```

Because radicals act over a short distance and react quickly, local generation could in principle limit systemic exposure compared with giving a broadly toxic agent throughout the body.

53.3 What should be concluded?

This is an early *preclinical* approach. It supports the broader idea that a cell-surface tumour antigen can be used not only for antibodies, ADCs, CAR-T cells, or T-cell engagers, but also to focus local chemical damage on cancer cells.

However, GPC-3 expression alone does not guarantee a useful treatment. Important remaining questions include whether the agent reaches all tumour deposits, whether enough tumour cells express GPC-3, how effectively radicals are generated in the human tumour microenvironment, and whether normal liver and other tissues can be protected sufficiently.

GPC-3 provides tumour localisation;
radical generation provides killing.
The clinical challenge is to achieve enough local damage in HCC
without unacceptable damage outside the tumour.

54 [26-09-10] Super-enhancer control of LGR5 in colorectal cancer

(A core transcriptional regulatory circuitry controls super-enhancer-driven activation of *LGR5* in colorectal cancer, <https://www.nature.com/articles/s41467-026-76761-4>)

LGR5 is a cell-surface receptor and a well-known marker of normal intestinal stem cells. In the intestine, it binds *R-spondin* proteins and strengthens Wnt signalling by counteracting negative Wnt regulators such as RNF43 and ZNRF3.

Recall that abnormal Wnt signalling is a central feature of many colorectal cancers, often following loss of the tumour-suppressor gene *APC*. *LGR5*-positive tumour cells can have stem-like properties: they can self-renew, generate diverse tumour-cell states, and potentially contribute to tumour maintenance and recurrence.

Wnt signalling
→ activation of stem-cell-associated gene programmes
→ increased *LGR5* expression
LGR5 signalling
→ further supports Wnt responsiveness
and a stem-like cell state

54.1 What is a super-enhancer?

An *enhancer* is a DNA regulatory region that helps increase expression of a gene, often by binding transcription factors and transcriptional co-activators. A *super-enhancer* is a large cluster of particularly active enhancers that work together to drive unusually strong expression of genes important for cell identity.

The important distinction is that the DNA sequence of the *LGR5* gene need not be mutated for its expression to become abnormally high. Cancer cells can instead alter the *regulatory machinery* around the gene.

Transcription factors + co-activators bind *LGR5* super-enhancer
→ high *LGR5* transcription
→ sustained *LGR5*-positive stem-like tumour state

54.2 What does “core transcriptional regulatory circuitry” mean?

The study identifies a self-reinforcing network of transcription factors that occupies the *LGR5* super-enhancer and regulates *LGR5* expression in colorectal cancer. A *core regulatory circuitry* is not a simple linear pathway. Its components support one another’s expression and activity, helping maintain a stable cell state.

Core transcription factors activate *LGR5* super-enhancer
→ *LGR5* expression and stem-like state are maintained
Core transcription factors also reinforce each other
→ the regulatory programme becomes more stable

This helps explain how cancer cells can maintain high *LGR5* expression even as conditions change. It also illustrates that cancer dependence can arise from abnormal gene regulation and chromatin organisation, not only from altered protein sequence.

54.3 Why is this therapeutically interesting?

LGR5 is an attractive but challenging target. As a cell-surface marker, it could potentially be recognised by antibodies, ADCs, CAR-T cells, or T-cell engagers. However, it is also expressed by normal intestinal stem cells, creating a risk of *on-target, off-tumour toxicity*.

An alternative strategy is to disrupt the transcriptional machinery maintaining the LGR5-positive state. The study suggests potential intervention points at the level of transcription factors, super-enhancer-associated co-activators, or chromatin regulation.

However, these are early mechanistic findings. Transcriptional regulators may also be needed in normal tissues, and shutting down a visible LGR5-positive population may not be sufficient if other colorectal-cancer cells can change state and regenerate it.

LGR5 is not simply a marker.
Its expression can be maintained by a self-reinforcing,
super-enhancer-associated gene-regulatory programme
that may support colorectal-cancer stemness and plasticity.

55 [26-09-10] AI models for personalised neoantigen cancer vaccines

(Genome language models can design cancer vaccines, <https://www.radicalnumerics.ai/blog/omnii-cancer-vaccines>)

The Moderna/Merck *intismeran autogene* programme discussed earlier uses tumour sequencing to select patient-specific mutations for an mRNA vaccine. A central difficulty is that a tumour may contain many mutations, but only a small fraction are likely to produce a useful *neoantigen*.

Recall the full biological chain:

Tumour mutation
→ mutated protein is produced
→ protein is processed into peptide
→ peptide binds patient's MHC molecule
→ peptide-MHC is displayed on tumour cell
→ an appropriate T-cell receptor recognises it
→ T cell can attack tumour cell

A mutation can fail at every step. It may not be expressed, its peptide may not be processed or displayed by MHC, the patient may have no T cells able to recognise it, or the tumour may prevent those T cells from entering or functioning. Therefore, selecting vaccine neoantigens is a difficult prediction problem rather than simply finding mutations.

55.1 What is the possible role of a model such as Omnii?

The Radical Numerics post describes *Omnii* as a genomic foundation model: an AI model intended to learn patterns in biological sequence and related molecular data. In principle, a model of this kind could help rank candidate neoantigens by integrating information that is usually considered separately.

For a personalised vaccine, potentially relevant inputs include the tumour DNA mutation, RNA expression, predicted mutant peptide sequence, patient MHC type, normal-tissue expression, mutation clonality, and features of antigen processing and presentation.

```
Tumour and normal sequencing
→ identify tumour-specific mutations
RNA and genomic context
→ estimate whether mutation is expressed
and present across tumour cells
MHC and peptide information
→ estimate presentation likelihood
Model-based ranking
→ prioritise a limited set of candidate neoantigens
for vaccine manufacture
```

The potential value is not that AI can “design an immune response” with certainty. Rather, it may improve the ordering of a large list of imperfect candidate peptides, helping researchers choose which candidates to test or include in a personalised vaccine.

55.2 Why could this matter clinically?

Current neoantigen selection already uses computational prediction, particularly for peptide–MHC binding. But MHC binding alone is not enough. A more useful model would better predict the full chain from mutation to displayed antigen to productive T-cell response.

If such predictions were sufficiently accurate, they could potentially improve vaccine quality, reduce the number of low-value vaccine components, shorten analysis time, and increase the probability that a vaccine contains several truly immunogenic targets.

This could complement, rather than replace, the clinical logic behind the Moderna/Merck approach:

```
Personalised mRNA vaccine
→ generates or expands neoantigen-reactive T cells
PD-1 blockade
→ reduces inhibition of those T cells
Improved neoantigen ranking
→ may increase the chance that vaccine-encoded targets
are genuinely relevant to that patient’s tumour
```

55.3 How impressive is this currently?

The underlying ambition is scientifically important, but claims for a general genomic AI model should be interpreted cautiously. A model can be technically impressive—for example, by predicting masked sequences or performing well on retrospective molecular benchmarks—without yet improving patient outcomes.

For practical neoantigen-vaccine use, the important evidence would be:

```
Independent benchmark performance
→ does it rank known immunogenic neoantigens
better than established methods?
Prospective experimental validation
→ do highly ranked candidates produce
stronger T-cell responses in patient samples?
Clinical validation
→ do model-selected vaccines improve meaningful outcomes,
such as recurrence-free survival, relative to a credible comparator?
```

The third standard is the decisive one. A model may become useful in practice before it proves superior in a Phase III trial—for example, by making candidate selection faster or more reliable within a validated workflow—but this still requires careful prospective testing.

55.4 What should be concluded?

The blog describes a plausible future use of genomic AI: improving selection of personalised cancer-vaccine targets by modelling more of the biology between tumour mutation and T-cell recognition.

However, this is not yet equivalent to a demonstrated new cancer-vaccine therapy. The clinically meaningful advance remains the combination of accurate target selection, feasible rapid manufacture, effective induction of tumour-reactive T cells, and an immune environment in which those T cells can control the tumour. The positive Phase III intismeran autogene announcement is important because it begins to test that complete chain in patients; an AI model would be valuable only if it measurably improves one or more links in that chain.

56 [26-09-10] AlphaGenome Atlas: predicting the effect of DNA variants

(*AlphaGenome Atlas: A predictive map of every possible DNA letter change in the human genome*, https://deepmind.google/blog/alphagenome-atlas-a-predictive-map-of-every-possible-dna-letter-change-in-the-human-genome/?utm_source=x&utm_medium=social&utm_campaign=&utm_content=)

The *AlphaGenome Atlas* is a large precomputed resource from Google DeepMind that predicts the molecular consequences of approximately 9 billion possible single-letter DNA changes across the human genome.

A single-nucleotide variant, or *SNV*, is a change of one DNA letter. For example, one person or tumour may have a G at a genomic position where the reference genome has an A.

The important idea is that many disease-relevant variants do not alter a protein's amino-acid sequence. They occur in *non-coding* DNA: regions that help control when, where, and how strongly genes are expressed.

DNA variant in regulatory region
→ altered transcription-factor binding
→ altered chromatin accessibility or enhancer activity
→ altered gene expression or RNA splicing
→ changed cell behaviour and, sometimes, disease risk

AlphaGenome predicts parts of this chain from local DNA sequence. It estimates how changing one DNA letter may affect molecular readouts such as gene expression, RNA splicing, chromatin accessibility, transcription-factor binding, and three-dimensional chromatin-associated features.

56.1 Why is this useful?

Human genome sequencing identifies very large numbers of variants. The hard problem is interpreting them: which are probably harmless, which alter molecular biology, and which may contribute to disease?

This is especially difficult outside protein-coding genes. A protein-coding mutation can sometimes be interpreted directly: for example, it may truncate a tumour-suppressor protein or change an amino acid in an enzyme active site. A non-coding variant may instead alter a distant enhancer whose relevant gene and cell type are unclear.

The Atlas provides an *AlphaGenome Variant Impact*, or *AVI*, score to help rank variants that are predicted to have substantial molecular effects.

Thousands of variants from sequencing
→ rank variants with predicted functional effect
→ identify plausible affected gene, pathway, and cell type
→ prioritise variants for experimental validation

This is potentially useful for rare-disease research, where a patient’s genome may contain a plausible but previously uninterpretable non-coding variant. It may also help interpret GWAS signals for common diseases, which often identify a genomic region associated with disease risk but not the actual causal variant, affected gene, or relevant cell type.

56.2 How could this matter in drug discovery?

The most immediate value is likely to be better *hypothesis generation* and experimental prioritisation, rather than direct clinical decision-making.

For example, a disease-associated non-coding variant might be predicted to increase expression of a gene specifically in an immune cell. This could support a more testable causal hypothesis:

```
Disease-associated variant
→ predicted enhancer disruption in relevant cell type
→ altered expression of candidate gene
→ altered disease pathway
→ candidate target or biomarker hypothesis
```

In cancer, the resource could help researchers investigate whether non-coding somatic mutations alter oncogene expression, tumour-suppressor regulation, immune escape, or treatment resistance. It could also help prioritise variants for functional assays such as CRISPR perturbation experiments.

Importantly, it does *not* establish that a variant causes disease, nor that the affected gene is a safe or effective drug target. It supplies predictive evidence at an early link in the larger causal chain.

56.3 Is it a big deal?

It is a meaningful advance because it makes a very large number of sequence-to-function predictions available in a consistent and accessible form. Previously, evaluating a non-coding variant often required specialised models, computational resources, and separate analysis for each question. Precomputing predictions for nearly every possible SNV makes variant prioritisation much more practical.

However, it is not a solved interpretation problem. AlphaGenome mainly models *cis*-regulatory effects: consequences of local DNA sequence near a variant. Biology also depends on cell state, environmental signals, distant genomic interactions, protein abundance, feedback loops, and *trans*-acting effects, such as changes in transcription-factor expression elsewhere in the genome.

Furthermore:

```
Predicted molecular effect
≠ proven effect in a patient
Proven molecular effect
≠ disease causality
Disease causality
≠ a useful therapeutic target
```

The predictions therefore require validation in relevant cells and experimental systems. The Atlas is a research resource, not a validated clinical diagnostic test.

56.4 Is it less important than AlphaFold?

Probably yes, at least for now. *AlphaFold* addressed a comparatively well-defined and widely reusable problem: predicting protein structure from amino-acid sequence. Protein structures have immediate relevance to basic biology, structural biology, and drug discovery, including understanding binding sites and designing molecules.

AlphaGenome addresses an arguably harder problem: predicting how DNA sequence changes affect gene regulation in different human cell types and disease contexts. Its outputs are highly useful, but further from a direct therapeutic decision.

A reasonable comparison is:

```
AlphaFold
→ predicts a structural property of a protein
AlphaGenome Atlas
→ predicts possible molecular consequences
of a DNA variant
Both generate hypotheses;
neither alone proves patient benefit.
```

AlphaGenome may become extremely useful in practice if its predictions reliably improve variant interpretation and experimental success rates. Its likely impact will be distributed across many genomics workflows—rare-disease research, functional genomics, target discovery, biomarker research, and cancer biology—rather than appearing as one immediate clinical transformation.

57 [26-09-10] Generative design of antigen-specific T-cell receptors

(*Generative design of antigen-specific T-cell receptor sequences with a conditional diffusion model*, <https://www.biorxiv.org/content/10.64898/2026.06.10.730756v2>)

A conventional T-cell receptor, or *TCR*, recognises a short peptide displayed by a particular MHC molecule. This differs from CAR-T therapy, where an engineered receptor binds a cell-surface antigen directly.

```
Tumour protein
→ processed into peptide
→ peptide displayed by patient's MHC
→ TCR recognises peptide-MHC complex
→ T cell may kill the tumour cell
```

TCR-based cell therapy can therefore potentially target *intracellular* tumour proteins, provided a suitable peptide is presented by MHC. This could include mutation-derived neoantigens, or cancer-associated antigens such as *MAGE-A3*. In contrast, CAR-T cells are restricted to cell-surface targets.

57.1 What problem does this study address?

Finding a useful tumour-reactive TCR is difficult. Researchers traditionally isolate and screen T cells from patients or donors, then identify receptors that recognise a chosen peptide–MHC target. This can be slow, and useful receptors may be rare.

The study presents *TCRDiff*, a generative model designed to propose new TCR sequences for a specified peptide–MHC target. It focuses particularly on the paired CDR3 regions of the TCR alpha and beta chains. These are highly variable receptor regions that make important contacts with peptide–MHC.

The model is conditioned on:

```
Chosen peptide-MHC target
+
TCR variable-gene context
+
predicted binding-interface geometry
→ candidate TCR alpha/beta sequences
```

The biological novelty is not that TCRs can recognise tumour antigens—that is the basis of normal anti-tumour immunity and TCR-T therapy. The interesting advance is the attempt to design candidate receptors *from the target backwards*, rather than relying entirely on finding an appropriate naturally occurring T cell.

57.2 What did the study show?

As a proof of concept, the authors generated candidate TCRs against a MAGE-A3 peptide–MHC target. Some designed receptors showed target-specific activation in laboratory assays, with limited cross-reactivity in the tested panels.

This is encouraging because TCR recognition is exceptionally difficult to predict. A TCR contacts both the peptide and the patient’s MHC molecule, and small sequence changes can alter affinity, specificity, signalling, expression, or unintended recognition.

However, it remains a *preprint* and early experimental work. A designed TCR that activates engineered cells in vitro is not yet a clinical therapeutic receptor.

57.3 Why is specificity the central safety issue?

TCRs are naturally somewhat cross-reactive: one receptor can recognise more than one peptide–MHC complex. This is biologically useful for immune coverage, but potentially dangerous in therapy.

A therapeutic TCR intended to recognise a tumour peptide might also recognise a similar peptide from an essential normal protein. If normal cells present that peptide on the same MHC type, engineered T cells could attack healthy tissue.

Desired recognition:
tumour peptide + patient’s MHC
→ tumour-cell killing
Dangerous cross-reactivity:
normal-tissue peptide + same MHC
→ on-target-like damage to healthy tissue

This is especially important for MAGE-A3. It is a cancer–testis antigen: expression is relatively restricted in normal adult tissues, but historic MAGE-A3-directed TCR programmes have demonstrated that unexpected peptide cross-reactivity can cause severe toxicity. Therefore, limited cross-reactivity in an initial laboratory panel is useful evidence, but cannot establish clinical safety.

57.4 Could this be useful in practice?

Potentially, yes. A mature approach could help identify TCR candidates faster for personalised neoantigens or shared tumour antigens, then feed them into a rigorous experimental selection process.

For example, it could complement personalised neoantigen vaccines:

Neoantigen vaccine
→ attempts to stimulate the patient’s own T cells
against selected tumour mutations
Designed TCR-T therapy
→ gives the patient T cells engineered with a receptor
chosen to recognise one selected peptide–MHC target

TCR-T may be valuable where the patient’s natural response is absent or too weak. Its practical limitations are substantial: the tumour must present the chosen peptide; the patient must have the relevant MHC type; antigen expression and presentation must persist; engineered T cells must reach and function in the tumour; and safety testing must rule out dangerous cross-reactivity.

57.5 What should be concluded?

This is a technically and biologically interesting step toward *programmable TCR discovery*. It could eventually reduce the time and experimental search required to find candidate receptors.

But AI generation will not remove the central medical problem: proving that an engineered receptor is sufficiently specific, functional, persistent, manufacturable, and safe in the intended patients. Its most realistic near-term role is likely to be improved candidate prioritisation—producing better starting points for extensive laboratory and safety testing—rather than directly designing clinical TCR therapies without experimental validation.

58 [26-09-10] Tozorakimab for prevention of COPD exacerbations

(*Tozorakimab demonstrated statistically significant and highly clinically meaningful reduction in COPD exacerbations in OBERON and TITANIA Phase III trials*, <https://www.astrazeneca.com/media-centre/press-releases/2026/tozorakimab-demonstrated-statistically-significant-highly-clinically-meaningful-reduction-copd-exacerbations> [html](https://www.astrazeneca.com/media-centre/press-releases/2026/tozorakimab-demonstrated-statistically-significant-highly-clinically-meaningful-reduction-copd-exacerbations.html), *Tozorakimab to Prevent COPD Exacerbations*, <https://www.nejm.org/doi/full/10.1056/NEJMoa2606998>)

Chronic obstructive pulmonary disease, or *COPD*, is a long-term lung disease in which airflow is persistently limited. It commonly includes a mixture of *chronic bronchitis*, involving airway inflammation and excess mucus, and *emphysema*, involving damage to the small air sacs, or alveoli, where oxygen enters the blood.

Smoking is the most important cause, although other inhaled exposures and genetic factors can contribute. COPD is not simply a matter of narrowed airways: it involves chronic inflammation, structural lung damage, mucus dysfunction, recurrent infection risk, and impaired repair.

Long-term inhaled injury, often smoking
→ airway and lung inflammation
+ mucus dysfunction
+ small-airway narrowing
+ alveolar damage
→ persistent breathlessness, cough, and reduced airflow

58.1 What is a COPD exacerbation?

An *exacerbation* is a sudden worsening of COPD symptoms, particularly breathlessness, cough, and sputum production. It can be triggered by viral or bacterial infection, air pollution, or other inflammatory stress.

Moderate exacerbations usually require treatment with systemic corticosteroids, antibiotics, or both. Severe exacerbations require emergency care or hospitalisation. Repeated exacerbations are important because they reduce quality of life, can accelerate loss of lung function, and increase the risk of hospitalisation and death.

58.2 What does tozorakimab do?

Tozorakimab is a monoclonal antibody that neutralises *interleukin-33*, or *IL-33*. IL-33 is an *alarmin*: a signalling molecule released or made available when epithelial and other structural cells are stressed or damaged.

In COPD, repeated exposure to smoke, infection, and other injury can activate IL-33-associated signalling. IL-33 can contribute to airway inflammation, mucus production, impaired epithelial repair, and activation of several immune-cell types.

```
Airway epithelial injury or stress
→ IL-33 signalling increases
→ inflammation, mucus dysfunction,
and impaired tissue repair
→ increased susceptibility to exacerbations
Tozorakimab binds IL-33
→ blocks IL-33 signalling
→ aims to reduce these disease-driving processes
```

A notable feature is that tozorakimab inhibits both *reduced* and *oxidised* forms of IL-33. This matters because IL-33 can exist in different chemical forms in inflamed tissue; blocking only one form might leave clinically relevant signalling active.

58.3 What did OBERON and TITANIA show?

OBERON and TITANIA were Phase III trials in patients with COPD who continued to have exacerbations despite inhaled standard-of-care treatment. Patients received tozorakimab 300 mg every four weeks or placebo in addition to that background therapy.

The primary analysis population was former smokers. Compared with placebo, tozorakimab reduced the annualised rate of moderate or severe exacerbations by ****29%**** in OBERON and ****34%**** in TITANIA. In the overall population, including current and former smokers, reductions were ****30%**** and ****29%****, respectively.

The results are particularly notable because benefit was reported across blood eosinophil levels. This suggests IL-33 may be relevant beyond the more eosinophil-associated inflammatory pattern targeted by some existing COPD biologics.

58.4 What should be concluded?

These are strong confirmatory results for a new biological mechanism in COPD: blocking the epithelial-injury alarmin IL-33 can reduce clinically important exacerbations on top of inhaled therapy. The benefit is not a cure for the underlying irreversible lung damage, and it does not mean every patient will respond, but preventing exacerbations can be highly meaningful for patients.

As of ****10 September 2026****, tozorakimab remains investigational. The reported Phase III benefit–risk profile appears encouraging, but regulatory review, detailed long-term safety assessment, and decisions about which patients should receive it remain necessary.

59 [26-09-10] A commensal fungus that can promote prostate-cancer progression

(*Commensal Nakaseomyces glabratus migrates into prostate tumors to accelerate cancer progression*, <https://www.nature.com/articles/s43018-026-01229-9>)

This study concerns the *mycobiome*: the community of fungi living in and on the human body. Most microbiome discussions focus on bacteria, but fungi can also influence inflammation, metabolism, immune responses, and disease biology.

Nakaseomyces glabratus, formerly called *Candida glabrata*, is a yeast that can live as a *commensal* organism—present without necessarily causing disease. Under particular conditions, however, it can behave as an opportunistic pathogen, especially when normal barriers or immune control are disrupted.

The central finding is that *N. glabratus* can migrate into prostate tumours and promote their progression in experimental models.

```
Commensal fungus outside the tumour
→ migration into prostate-tumour tissue
→ interaction with tumour and immune microenvironment
→ pro-tumour biological changes
→ faster tumour progression
```

59.1 Why is this biologically interesting?

A tumour is not only a mass of genetically altered cancer cells. As discussed earlier, it is an ecosystem containing immune cells, blood vessels, fibroblasts, extracellular matrix, and potentially microorganisms.

Microorganisms inside tumours could affect cancer through several non-exclusive mechanisms. They may cause persistent inflammation, alter local immune-cell activity, change tumour-cell signalling or metabolism, or affect response to treatment.

The important conceptual shift is:

```
Tumour biology
≠
tumour-cell genetics alone
Tumour biology
=
cancer cells
+ immune cells
+ surrounding tissue
+ metabolites
+ potentially bacteria and fungi
```

In this study, the fungal signal is not interpreted merely as an accidental contaminant or an association with advanced disease. The experimental work supports the stronger hypothesis that the fungus can contribute causally to progression in the studied models.

59.2 Does this mean that the fungus causes prostate cancer?

No. This distinction is essential.

```
Fungal presence may promote progression
≠
fungal infection caused the original prostate cancer
```

Prostate cancer is driven mainly by accumulated genetic, epigenetic, hormonal, and environmental changes in prostate cells. The study instead suggests that, after a tumour exists, fungal colonisation may create a more favourable environment for its growth or progression.

It is also possible that tumours create conditions that make fungal entry or persistence easier. Therefore, both directions may matter:

```
Tumour-associated changes
→ may permit fungal colonisation
Fungal colonisation
→ may further promote tumour progression
```


59.3 Why could this matter clinically?

The work raises several longer-term possibilities. Detection of tumour-associated fungal species might eventually contribute to prognosis or patient stratification. If a causal fungal dependency is confirmed in humans, antifungal treatment or prevention of tumour colonisation could become a possible combination strategy.

However, this is far from establishing that patients with prostate cancer should receive antifungal therapy. A useful clinical intervention would require evidence that the fungal signal is reproducible in independent patient cohorts, that it identifies a biologically distinct subgroup, and that removing or suppressing the fungus improves patient-relevant outcomes.

Interesting preclinical finding:
a commensal fungus may accelerate prostate-cancer progression.
Unproven clinical conclusion:
detecting or treating the fungus improves outcomes in patients.

The broader lesson is that cancer biology can extend beyond human tumour cells. The tumour microenvironment may include microbial components that modify immunity and disease progression, creating potential new biomarkers and therapeutic hypotheses.

60 [26-09-10] Inhaled siRNA therapy against RAGE in pulmonary inflammation

(*Inhaled siRNA therapy targeting RAGE for pulmonary inflammation: a first-in-human randomized trial*, <https://www.nature.com/articles/s41591-026-04607-z>)

Small interfering RNA, or *siRNA*, is a short RNA molecule designed to reduce production of one chosen protein. It does not alter the DNA sequence. Instead, it uses a natural cellular process called *RNA interference*, or RNAi.

Recall the basic biological flow:

DNA
→ messenger RNA
→ protein

An siRNA acts at the RNA step. After entering a cell, one strand of the siRNA is loaded into a protein complex called *RISC*. The complex uses the siRNA sequence as a guide to recognise complementary messenger RNA. The matching mRNA is then cut or degraded, so less target protein is produced.

Target-gene DNA
→ target mRNA
→ target protein
siRNA binds target mRNA
→ mRNA is degraded
→ less target protein is made

This is therefore different from a conventional small-molecule inhibitor or antibody. Those medicines usually bind an already-existing protein; siRNA reduces future production of the protein.

60.1 What is RAGE?

RAGE, short for *receptor for advanced glycation end-products*, is a cell-surface receptor encoded by the *AGER* gene. It is highly expressed in parts of the lung, particularly alveolar epithelial cells.

RAGE can bind several signals released by stressed, damaged, or inflamed tissue, including HMGB1 and certain S100 proteins. These are examples of *damage-associated molecular patterns*, or DAMPs: molecules that alert the immune system to injury.

Lung injury, infection, smoke exposure, or inflammation
→ danger signals are released
Danger signals bind RAGE
→ inflammatory pathways, including NF- κ B, are activated
→ cytokines, chemokines, and inflammatory-cell recruitment
→ further lung inflammation and tissue injury

RAGE signalling can be useful during an appropriate response to tissue injury. However, persistent or excessive RAGE activation may help maintain harmful inflammation in diseases such as asthma, COPD, and acute lung injury.

60.2 What does the inhaled siRNA treatment do?

The investigational treatment, ARO-RAGE, is inhaled so that the siRNA is delivered preferentially to lung epithelial cells rather than throughout the body. It is designed to reduce *AGER* mRNA and therefore reduce RAGE protein production in the lung.

Inhaled RAGE-targeting siRNA
→ uptake by lung epithelial cells
→ *AGER* mRNA degradation
→ reduced RAGE protein
→ reduced response to inflammatory danger signals

The Phase I randomised study primarily tested safety, tolerability, pharmacokinetics, and *target engagement*. The reported reduction in soluble RAGE in blood and bronchoalveolar-lavage samples provides evidence consistent with substantial and durable suppression of the RAGE pathway in the lung.

This is important because delivery is often the central challenge for RNA medicines. An siRNA can be designed relatively directly from a gene sequence, but it must still reach the right cells, enter them, avoid rapid degradation, and reduce the target sufficiently without causing problematic immune reactions or off-target effects.

60.3 Why could siRNA be useful in lung disease?

A key potential advantage is durability. One dose can reduce target mRNA for weeks or longer, because intracellular RISC complexes can continue degrading newly made target mRNA.

Conventional inhibitor
→ drug must remain present to block protein activity
siRNA
→ cell continues to degrade newly made target mRNA
→ target suppression may persist after drug exposure falls

For chronic inflammatory lung disease, this could permit relatively infrequent dosing. Inhalation also offers a plausible way to concentrate treatment in the affected organ while limiting systemic exposure.

60.4 What should be concluded?

This first-in-human study is an encouraging demonstration that inhaled siRNA can produce sustained RAGE knockdown in the lung. It establishes *proof of mechanism*, not yet proof of clinical benefit.

The remaining question is whether reducing RAGE will meaningfully improve patient outcomes, such as asthma control, small-airway function, COPD exacerbations, or lung injury, without impairing useful tissue-repair and host-defence functions.

Successful target knockdown
≠ proven clinical benefit.
The next step is to show that durable RAGE suppression improves
meaningful respiratory outcomes in the intended patient population.

61 [26-09-10] Tumour-associated proximity antigens: targeting protein neighbourhoods

(*Proximity-guided graph learning reveals tumour-associated proximity antigens*, <https://www.nature.com/articles/s41586-026-11003-7>)

Most target discovery asks whether a protein is more highly expressed on tumour cells than on normal cells. This study introduces a related idea: a useful cancer target may be defined not only by *which* proteins are present, but by *which proteins sit close together* on the cell surface.

The authors call these *tumour-associated proximity antigens*, or *TAPAs*. A TAPA is a cell-surface protein whose close spatial association with another antigen is characteristic of tumour cells.

Conventional target concept:
protein X is highly expressed on tumour cells
Proximity-antigen concept:
protein X is close to protein Y on tumour cells,
creating a tumour-associated surface neighbourhood

61.1 Why can proximity matter?

Cell-surface proteins are not distributed randomly. Receptors, adhesion molecules, and other proteins can cluster into local membrane domains, partly because they participate in the same signalling complexes, membrane structures, or cell state.

Two proteins may each be present on normal tissue and tumour tissue, making either one alone a poor target. But their *close co-localisation* may be much more common in tumour cells. This creates a possible molecular “AND gate” for therapeutic selectivity.

Antigen A alone
→ may be found on normal cells
Antigen B alone
→ may be found on normal cells
Antigen A near antigen B
→ may identify a more tumour-specific state

This is conceptually related to dual-antigen CAR-T cells or bispecific antibodies. The difference is that the antigen pair is selected not simply because both proteins are expressed in the same cell, but because they are physically close enough on the cell membrane to support productive joint targeting.

61.2 What did the study do?

The researchers mapped nanoscale protein neighbourhoods on tumour-cell surfaces and integrated these measurements into a broader map called *MetaMap*. Graph-learning methods were then used to identify recurring protein-proximity patterns across samples and prioritise candidate TAPAs.

The important biology is that this can reveal targets missed by ordinary RNA or protein-expression profiling:

RNA or bulk protein measurement
→ asks how much of a target is present
Single-cell measurement
→ asks which cells express the target
Proximity mapping
→ asks which surface proteins occupy
the same local membrane neighbourhood

All three questions can be relevant. High tumour-cell expression remains useful, but it does not tell us whether two proteins form a targetable surface configuration.

61.3 How could a therapy use a TAPA?

The study describes a bispecific antibody–drug conjugate strategy. One arm recognises *EGFR*, a well-known cell-surface receptor, and the other recognises a tumour-associated proximity partner, termed *TAPA-E1*. The programme is called *IDP-001*.

Bispecific ADC binds EGFR and TAPA-E1 in close proximity
→ preferential binding or clustering on tumour cell
→ internalisation of ADC
→ intracellular payload release
→ tumour-cell killing

The intended advantages are improved tumour selectivity, stronger internalisation, and reduced exposure of normal EGFR-expressing tissues to the cytotoxic payload. This is especially relevant because EGFR is expressed in several normal epithelial tissues, which can limit the therapeutic window of EGFR-directed treatments.

61.4 What should be concluded?

This is an interesting expansion of the target-discovery framework. It treats the tumour-cell surface as an organised spatial system rather than a list of independently expressed proteins.

However, a tumour-associated proximity pattern is not automatically a clinically useful target. It must be sufficiently frequent and stable across tumour cells and lesions, absent or uncommon in essential normal tissues, accessible to the therapeutic, and associated with adequate internalisation and payload sensitivity.

A tumour-associated protein neighbourhood
→ plausible precision-targeting hypothesis
A safe and effective medicine
→ still requires pharmacology, tissue-selectivity,
toxicology, and clinical benefit-risk evidence

The Phase I development of IDP-001 is therefore important as an early test of whether proximity-defined target pairs can translate from a compelling cell-surface biology concept into a meaningful therapeutic window in patients.

62 [26-09-10] TRI-611: degrading ALK in ALK-positive lung cancer

(*TRI-611*, a selective, brain-penetrant molecular glue degrader of *ALK*, <https://www.nature.com/articles/s41586-026-10998-3>)

ALK, short for *anaplastic lymphoma kinase*, is a receptor tyrosine kinase. Like other receptor tyrosine kinases, its normal function involves receiving or transmitting growth-related signals through phosphorylation.

In some cancers, especially a subset of non-small-cell lung cancers, a chromosome rearrangement creates an *ALK fusion*. A portion of the *ALK* gene becomes joined to another gene, often *EML4*. The fusion protein can be continuously active without its normal external signal.

ALK fusion protein
→ constitutive kinase activity
→ persistent downstream growth and survival signalling
→ tumour-cell dependence on ALK signalling

This is an example of *oncogene addiction*: despite many other alterations, some tumour cells remain strongly dependent on one abnormal driver signal. ALK fusions are therefore both a predictive biomarker and a therapeutic target.

62.1 What are ALK tyrosine-kinase inhibitors?

A *tyrosine-kinase inhibitor*, or *TKI*, is usually a small molecule that binds the kinase domain of a receptor or signalling protein and reduces its ability to transfer phosphate groups to substrate proteins.

For ALK-positive lung cancer, ALK TKIs such as crizotinib, alectinib, brigatinib, lorlatinib, and others can suppress the abnormal ALK-driven signalling and often produce substantial clinical responses.

ALK TKI binds ALK kinase domain
→ ALK phosphorylation activity decreases
→ downstream signalling decreases
→ ALK-dependent tumour cells may die or stop growing

However, resistance can emerge. Tumour cells may acquire mutations in the kinase domain that reduce TKI binding, activate bypass pathways, or evolve additional changes. Brain metastases are also an important clinical challenge, because a medicine must cross the blood–brain barrier to achieve useful exposure in the central nervous system.

62.2 How does a molecular glue differ from a TKI?

TRI-611 is a *molecular glue degrader*. Rather than merely inhibiting the ALK kinase domain while the drug is present, it brings ALK close to an E3 ubiquitin ligase called *cereblon*, or CRBN.

E3 ligases label selected proteins with *ubiquitin*. A chain of ubiquitin labels directs the tagged protein to the *proteasome*, the cell’s protein-recycling machinery, where it is broken down.

TRI-611 binds ALK at a site outside the kinase active site
+ recruits CRBN E3 ligase
→ ALK is ubiquitinated
→ proteasome degrades ALK
→ both ALK protein amount and kinase signalling decrease

The term “molecular glue” is appropriate because TRI-611 creates a productive interaction between two proteins that would not normally associate closely enough for CRBN to label ALK efficiently.

62.3 Why could this help after TKI resistance?

TKI-resistance mutations often occur in or near the ALK kinase domain, the region where conventional inhibitors bind. TRI-611 acts through a distinct binding site and removes the protein rather than only blocking its catalytic activity.

```
Conventional TKI
→ requires effective kinase-domain binding
TRI-611
→ binds elsewhere on ALK
→ recruits degradation machinery
→ removes the ALK fusion protein
```

The preclinical study reports degradation and anti-tumour activity against clinically relevant TKI-resistant ALK mutations, including compound mutations. It also reports tumour regression in cell-line and patient-derived models, including intracranial models.

This does not mean that every resistant ALK-positive tumour will respond. A cancer may become less dependent on ALK altogether, or use bypass signalling, in which case removing ALK may no longer be sufficient.

62.4 Why is brain penetration important?

ALK-positive lung cancer can spread to the brain. A brain-penetrant drug can cross the blood–brain barrier and reach tumour deposits in the central nervous system.

```
Adequate plasma exposure
≠ adequate brain exposure
Brain-penetrant TRI-611
→ potential to degrade ALK
in both systemic and brain tumour sites
```

This is a major practical property, but it must be demonstrated through human PK, pharmacodynamic, efficacy, and safety data.

62.5 What should be concluded?

TRI-611 is an important *preclinical* demonstration that an oncogenic fusion protein can be targeted by molecular-glue degradation rather than conventional kinase inhibition. Its key proposed advantages are activity against some TKI-resistant ALK variants and penetration into the brain.

As of September 2026, TRI-611 is in a first-in-human Phase I/II trial in ALK-positive NSCLC. The central unanswered question is whether the compelling mechanism and animal-model activity translate into durable benefit with an acceptable safety profile in patients, particularly after resistance to multiple ALK TKIs.

63 [26-09-10] PLA2G2D: an immune checkpoint in tumour-draining lymph nodes

(*PLA2G2D in tumour-draining lymph nodes regulates anti-tumour immunity*, <https://www.nature.com/articles/s41586-026-10954-1>)

A *tumour-draining lymph node*, or *TDLN*, is a lymph node that receives lymphatic fluid, tumour antigens, and immune cells from a nearby tumour. It is an important site where dendritic cells can present tumour-derived peptides to T cells and help initiate an anti-tumour immune response.

```
Tumour material reaches draining lymph node
→ dendritic cell presents tumour peptide on MHC
→ tumour-reactive T cells are activated and expand
→ T cells can travel to the tumour
→ T cells may recognise and kill tumour cells
```

This paper identifies *PLA2G2D* as an immunosuppressive regulator acting particularly in this early, lymph-node stage of the immune response.

63.1 What is PLA2G2D?

PLA2G2D, or phospholipase A2 group IID, is a secreted protein in the phospholipase A2 family. These enzymes can modify membrane lipids, releasing lipid components that can become inflammatory or immunoregulatory signalling molecules.

The unexpected finding here is that *PLA2G2D*'s relevant suppression of anti-tumour immunity appears to be largely *independent of its enzyme activity*. In other words, simply inhibiting its lipid-cleaving catalytic function may not be enough; the protein itself appears to provide an immunosuppressive signal through another, not yet fully resolved mechanism.

63.2 What did the spatial profiling show?

The researchers used spatial proteogenomic methods to examine not only which cells expressed particular molecules, but also *where those cells were located* and which cells they contacted within human melanoma TDLNs.

PLA2G2D was enriched in myeloid cells, particularly dendritic cells and macrophages, in the T-cell-rich *paracortex* of the lymph node. These *PLA2G2D*-positive myeloid cells were located near CD8 T cells that showed features of dysfunctional or exhaustion-prone states.

```
PLA2G2D-positive dendritic cells and macrophages
near tumour-reactive CD8 T cells in TDLN
→ reduced T-cell proliferation and effective activation
→ fewer functional anti-tumour T cells leave the lymph node
→ weaker immune attack in the tumour
```

Thus, *PLA2G2D* may suppress anti-tumour immunity *before* T cells reach the tumour. This differs from PD-1/PD-L1 blockade, which is often understood as restoring function of T cells already experiencing inhibitory signalling in or near the tumour.

63.3 Why could this be a useful checkpoint target?

A useful way to compare the two ideas is:

```
PLA2G2D blockade
→ may improve priming and expansion of tumour-reactive T cells
in tumour-draining lymph nodes
PD-1 blockade
→ reduces inhibitory signalling that limits activated T cells,
including in the tumour microenvironment
Combination
→ may increase the supply of functional T cells
and reduce inhibition of those T cells after they reach the tumour
```

In mouse models, PLA2G2D blockade enhanced anti-tumour immunity and improved activity of anti-PD-1 treatment, including in settings with resistance to PD-1 blockade. This provides a plausible rationale for combination therapy and for attempting to convert immunologically “cold” tumours into tumours more able to support productive T-cell responses.

63.4 What should be concluded?

The main conceptual advance is that immunotherapy targets need not operate only inside the tumour. The tumour-draining lymph node is a critical upstream site where the cancer can limit the generation of effective anti-tumour T cells.

PLA2G2D is a promising preclinical checkpoint hypothesis:
it may suppress anti-tumour CD8 T-cell responses
at their lymph-node starting point.
It is not yet a clinically validated cancer target.

Important remaining questions include the precise enzyme-independent mechanism, which cancers depend on this pathway, whether PLA2G2D expression predicts benefit, and whether sustained blockade can improve patient outcomes without causing unacceptable immune toxicity.

64 [26-09-10] De novo-designed CARs: binding a target is not enough

(Sequence and structural determinants of efficacious de novo chimaeric antigen receptors, <https://www.nature.com/articles/s41551-026-01790-9>)

CAR-T cells are usually engineered with an antibody-derived binding domain, often a *single-chain variable fragment*, or scFv. This external part of the CAR binds a chosen cell-surface antigen; intracellular CAR domains then activate the T cell.

This study asks whether *de novo* designed proteins—small binding proteins created computationally rather than derived from a natural antibody—can replace the usual antibody-derived binding domain.

CAR binding domain recognises cell-surface antigen
→ CAR molecules cluster
→ intracellular CAR signalling is activated
→ T cell kills the target cell

The authors screened 1,589 designed binders against the CAR-T targets BCMA, CD19, and CD22. The main lesson is that a binder which looks convincing computationally, or binds purified protein well, is not automatically a useful CAR.

64.1 What makes a designed CAR work?

The study identified three important failure modes.

First, some binders caused *tonic signalling*: unwanted CAR activation even when no target cell was present. This can happen if CARs cluster through self-association or aggregation of their external binding domains.

Binder self-association
→ CAR clustering without antigen
→ tonic signalling
→ chronic activation, exhaustion, and poorer CAR-T-cell function

Second, some binders recognised an epitope that was accessible on purified recombinant antigen but not on the natural protein at the cell surface. For CD19, for example, an epitope may be partly shielded by the neighbouring protein CD81, by glycosylation, or by the normal membrane context.

Third, some CARs responded to cells lacking the intended target. This *off-target activation* is especially concerning because it could lead to unintended killing of normal tissue.

64.2 Why is this biologically useful?

The paper shows that CAR design requires consideration of the entire biological setting, not only affinity for an isolated target protein:

Designed binder binds purified antigen
≠ antigen is accessible on a living cell
Binder has high affinity
≠ CAR gives useful activation
CAR activates target cells
≠ CAR is selective and safe

The researchers could improve several poor-performing binders by changing residues outside the main antigen-binding interface. These changes reduced aggregation and unwanted recognition while preserving productive target binding. This is a useful reminder that protein stability, geometry, epitope accessibility, and receptor clustering can determine CAR behaviour.

64.3 What is the medical significance?

Designed binders could eventually expand the range of usable CAR recognition domains. They may be smaller, more stable, easier to tune, and less constrained by the antibodies that happen to exist naturally. This could be particularly useful for difficult cell-surface targets or for multi-specific CAR designs.

However, this study is mainly a *platform and preclinical engineering* advance. It does not overcome the central challenges of CAR-T therapy: choosing an antigen that distinguishes tumour from essential normal tissue, achieving tumour access and persistence, overcoming immune suppression in solid tumours, and preventing systemic immune toxicity.

De novo CAR binders may improve how a CAR recognises a target.
They do not, by themselves, solve whether that target is safe,
tumour-selective, or clinically useful.

The important contribution is a clearer experimental roadmap: computational design can create many candidate CAR binders, but cell-based functional testing and broad specificity testing remain essential before a candidate can be considered for clinical development.

65 [26-09-14] A CD3×CD19 bispecific antibody for systemic lupus erythematosus

(A bispecific CD3×CD19 antibody for systemic lupus erythematosus: a phase 1 trial, <https://www.nature.com/articles/s41591-026-04572-7>)

65.1 What are CD3 and CD19?

CD3 is part of the *T-cell receptor complex* on T cells. When a T-cell receptor recognises a peptide–MHC complex, CD3 transmits the activating signal into the T cell. Binding CD3 with an appropriate therapeutic antibody can therefore activate a T cell, although this must be controlled carefully.

CD19 is a cell-surface protein expressed across much of the B-cell lineage. B cells are relevant in systemic lupus erythematosus, or *SLE*, because they can produce autoantibodies, present self-antigens to T cells, and contribute to inflammatory signalling. CD19 is therefore a practical marker through which to target many B cells.

65.2 What does a CD3×CD19 bispecific antibody do?

A CD3×CD19 bispecific antibody has one binding arm for CD3 on a T cell and another for CD19 on a B cell. It acts as a physical bridge:

```
CD3×CD19 antibody binds T cell and B cell
→ brings the cells into close proximity
→ activates the T cell locally
→ T cell kills the CD19-positive B cell
→ deep B-cell depletion may reduce autoimmune activity
```

This is a *bispecific T-cell engager*, or TCE. It uses the patient's existing T cells rather than engineering T cells outside the body, as in CAR-T therapy. Unlike normal T-cell recognition, this redirection does not require a particular peptide to be presented by MHC: the antibody creates the T-cell–B-cell interaction directly.

The intended idea in SLE is not simply temporary removal of circulating B cells. Deep depletion may reduce pathogenic B-cell populations in tissues and allow the immune system to re-establish a less inflammatory state. However, CD19-directed depletion may not eliminate every antibody-producing plasma cell, since some long-lived plasma cells have little or no CD19 expression.

65.3 What did the A-319 Phase 1 trial show?

The *Nature Medicine* paper reports an early, small Phase 1 study of *A-319*, a CD3×CD19 TCE, in 12 people with active, refractory SLE. Patients received priming and escalating doses three times weekly for three weeks.

The reported results were encouraging but preliminary. Among the 10 patients with 52-week follow-up, eight achieved *lupus low disease activity state*, or LLDAS, and six achieved DORIS remission. Disease activity and autoantibody measurements improved, while single-cell analyses suggested deep B-cell depletion in tissues and reduced type-I interferon signalling across immune-cell populations.

This resembles the proposed “immune reset” observed after CD19 CAR-T-cell therapy in severe autoimmune disease, but *A-319* is an *off-the-shelf* antibody product. It does not require collection and engineering of each patient's T cells, and the study did not use the lymphodepleting chemotherapy commonly given before CAR-T treatment.

65.4 Why is safety central?

T-cell engagement is powerful but can be hazardous. When many T cells activate at once, they can release inflammatory cytokines, causing *cytokine release syndrome*, or CRS. In this small study, CRS was common but reported as predominantly mild, with grade 1 CRS in 11 of 12 patients.

The central therapeutic challenge is therefore:

```
Enough T-cell redirection
→ sufficiently deep depletion of pathogenic B cells
But not so much systemic T-cell activation
→ unacceptable cytokine-mediated toxicity,
infection risk, or broader immune disruption
```

This was a Phase 1, single-arm study designed primarily to assess safety, dosing, and biological activity. It cannot establish comparative efficacy, long-term durability, or the full safety profile. Larger controlled

studies will be needed to determine whether CD3×CD19 TCEs can provide a practical and sufficiently safe alternative to CAR-T therapy or other B-cell-directed treatments in SLE.

66 [26-09-14] MPZL1: using amplified surface proteins to find CAR-T targets

(*Cancer-driver-agnostic targeting of amplified surface proteins identifies MPZL1 for selective CAR-T cell therapy*, <https://www.nature.com/articles/s41467-026-77465-5>)

This study proposes a useful target-discovery strategy for solid-tumour CAR-T therapy. Rather than starting with the question, “Which mutation is driving this cancer?”, it asks:

Which cell-surface proteins are highly expressed because their genes lie within recurrent genomic amplifications, while having limited expression in essential normal tissues?

The authors call this *cancer-driver-agnostic* targeting. The amplified gene does not need to be the mutation or oncogene that caused the cancer. Instead, its amplification can create a tumour-associated surface marker that CAR-T cells can use to recognise the tumour.

66.1 What is MPZL1?

MPZL1, or *myelin protein zero-like 1*, is a cell-surface protein. The study identified it as a candidate because its gene can be co-amplified within large chromosomal regions in several solid tumours, including hepatocellular carcinoma and triple-negative breast cancer.

The important proposed therapeutic feature is not necessarily that MPZL1 drives tumour growth. Rather, amplification may make tumour cells display substantially more MPZL1 than normal tissues:

Large genomic amplification in tumour
→ increased MPZL1 gene copy number
→ high MPZL1 surface expression
→ possible tumour-selective CAR-T recognition

This is analogous to using a visible “flag” on cancer cells. The flag may be a consequence of the cancer genome rather than its central biological dependency, but it can still be therapeutically useful if it distinguishes tumour cells from essential healthy cells.

66.2 What did the study show?

The researchers developed an antibody against the extracellular region of MPZL1 and used it to make MPZL1-directed CAR-T cells. In preclinical models, including tumour xenografts, patient-derived explants, and an autochthonous mouse liver-cancer model, these CAR-T cells infiltrated MPZL1-positive tumours and showed anti-tumour activity.

This supports the chain:

MPZL1-positive tumour cell
→ MPZL1 CAR binds surface antigen
→ CAR-T cell is activated
→ cytotoxic response is directed at tumour cell
→ tumour-cell killing and regression may occur

The results are therefore promising *preclinical* evidence for MPZL1 as a possible CAR-T target, rather than evidence that an MPZL1 CAR-T product is safe or effective in patients.

66.3 What is new here?

You have already encountered the central CAR-T mechanism: an engineered T cell recognises a chosen cell-surface antigen directly, without peptide–MHC presentation. You have also seen that target selection is the defining problem in solid tumours.

The particularly new idea is to use a likely *passenger* event as a targeting opportunity. Earlier, the distinction was:

```
Driver mutation
→ causally supports cancer growth or survival
Passenger mutation
→ present in cancer but not thought to drive its phenotype
```

This paper adds an important qualification: a passenger alteration can still be clinically useful if it creates a sufficiently tumour-selective and accessible antigen. The therapeutic objective here is not to inhibit MPZL1 because the tumour depends on it; it is to use MPZL1 expression to direct cell killing.

This potentially broadens precision oncology beyond direct targeting of oncogenic dependencies. A tumour may have an undruggable driver, but a co-amplified surface protein could provide an alternative way to identify and kill its cells.

66.4 The central remaining concern

The key safety question remains *on-target, off-tumour toxicity*. Even low-level MPZL1 expression in a vital normal tissue could be dangerous when recognised by highly potent CAR-T cells. Tumour heterogeneity also matters: cells with low or absent MPZL1 may survive and expand under treatment.

The essential conclusion is therefore:

```
Genomic amplification
≠ proof that MPZL1 drives cancer
But amplification
→ may create high tumour-selective surface expression
→ which may make MPZL1 a useful CAR-T address
```

The study provides a compelling framework for discovering solid-tumour cell-therapy targets, but clinical testing must establish the true expression pattern, therapeutic window, durability, and safety.

67 [26-09-14] Developing neoantigen cancer vaccines: where are we now?

(*Developing neoantigen cancer vaccines: where are we now?*, <https://www.nature.com/articles/s41467-026-77532-x>)

This review describes how personalised neoantigen vaccines have progressed from a compelling biological idea to an early but still unproven clinical strategy. The central aim is familiar from the earlier discussion of cancer vaccines: identify mutations unique to a patient's tumour, select those most likely to be visible to the immune system, and use a vaccine to generate or strengthen T-cell responses against them.

67.1 The central challenge: not every mutation is a useful neoantigen

A tumour can contain many mutations, but only a minority will produce a useful vaccine target. A candidate mutation must be expressed, processed into a peptide, presented by that patient's MHC molecules, and recognised by a T-cell receptor.

```
Tumour sequencing
→ identify tumour-specific mutations
Computational prioritisation
→ predict peptide processing, MHC presentation,
and likelihood of T-cell recognition
Personalised vaccine manufacture
→ stimulate neoantigen-specific T cells
```

The difficult step is *neoantigen selection*. Current computational models can help rank candidates, but they do not reliably predict every peptide that will generate a clinically useful T-cell response. This is an important setting in which improved data, experimental validation, and machine learning may help, but prediction is not yet a substitute for clinical evidence.

67.2 What has changed?

Earlier studies mainly showed that neoantigen vaccines could be manufactured and could induce measurable T-cell responses. The field is now asking the more demanding question: do these immune responses produce meaningful patient benefit?

The most important recent clinical signal is from the individualised mRNA vaccine *intismeran autogene* (mRNA-4157/V940) combined with pembrolizumab in resected high-risk melanoma. As discussed earlier, the reported Phase 3 topline results suggest improved recurrence-free and distant-metastasis-free survival compared with pembrolizumab alone. However, detailed peer-reviewed data, longer follow-up, and overall-survival evidence remain needed.

Thus, the field has moved from:

```
Can personalised vaccines generate T-cell responses?
→ increasingly, yes
Can they reproducibly improve patient outcomes?
→ promising early evidence, but still being established
```

67.3 Why combinations are likely to matter

A vaccine may generate tumour-reactive T cells, but those cells must still enter the tumour, recognise cancer cells, remain functional, and overcome inhibitory signals in the tumour microenvironment.

This explains the logic of combining a vaccine with a checkpoint inhibitor:

```
Neoantigen vaccine
→ generates or expands tumour-specific T cells
Checkpoint inhibitor
→ reduces inhibitory signalling, such as through PD-1
Combination
→ aims to provide both recognition and sustained function
```

Other combinations may try to increase antigen release, improve antigen presentation, alter the tumour microenvironment, or reduce immune suppression. The limitation is that each additional therapy can increase toxicity, complexity, and uncertainty about which component provides the benefit.

67.4 The practical and biological limitations

Personalised neoantigen vaccines are “one patient, one product” therapies. They require tumour and matched-normal sequencing, bioinformatic analysis, selection of multiple candidate neoantigens, manufacture, quality

testing, and delivery. This creates cost and time pressures, particularly for aggressive cancers that may progress before a vaccine is ready.

Tumour heterogeneity creates a second problem. A vaccine may target mutations present only in one subclone, allowing antigen-negative tumour cells to survive. Selecting multiple neoantigens, especially *clonal* neoantigens shared by most cancer cells, may reduce but not eliminate this risk.

Finally, some tumours are immunologically “cold”: they have limited T-cell infiltration, poor antigen presentation, or a strongly suppressive microenvironment. A well-designed vaccine may still fail if the resulting T cells cannot effectively reach or function within the tumour.

67.5 What should be concluded?

Neoantigen vaccines are not simply personalised versions of conventional cancer vaccines. They are an attempt to create patient-specific T-cell immunity against tumour mutations while avoiding the central tolerance problem associated with many normal self-antigens.

The field has strong mechanistic rationale and increasingly credible evidence of immunogenicity. Early clinical results—particularly in combination with checkpoint blockade and in settings of low residual disease—are encouraging. But these products remain investigational, and the key unanswered question is whether personalised immune responses can be made sufficiently reliable, scalable, durable, and clinically meaningful across different cancer types.

Personalised neoantigen vaccine
≠ guaranteed effective anti-tumour immunity
Induced T-cell response
≠ proven clinical benefit
Promising early trials
≠ established standard of care

68 [26-09-14] ROS-autocatalytic nanotherapy for glioblastoma: targeting tumour cells and supportive astrocytes

(*ROS-autocatalytic nanoplatform interfering astrocyte-cancer crosstalk and enhancing glioblastoma therapy*, <https://www.nature.com/articles/s41467-026-77395-2>)

This preclinical study describes a nanomedicine strategy for glioblastoma that aims to attack not only tumour cells, but also the harmful interaction between tumour cells and nearby *astrocytes*.

Astrocytes are abundant support cells in the brain. In healthy tissue, they help maintain neurons, regulate the local chemical environment, and contribute to the blood–brain barrier. Around glioblastoma, however, astrocytes can become *reactive*: their state and signalling change in response to injury, inflammation, and tumour-derived signals.

Glioblastoma cells alter nearby astrocytes
→ astrocytes become reactive
Reactive astrocytes support tumour survival,
invasion, and treatment resistance
→ glioblastoma further alters its local environment

This reciprocal communication is called *astrocyte–cancer crosstalk*. It is one example of the tumour microenvironment discussed earlier: cancer cells are influenced not only by their own mutations, but also by non-malignant cells and signals around them.

68.1 What are reactive oxygen species?

Reactive oxygen species, or ROS, are chemically reactive oxygen-containing molecules. Cells normally produce and control ROS as part of metabolism and signalling. At excessive levels, however, ROS can damage DNA, proteins, lipids, and cellular structures.

Cancer cells often live under increased oxidative stress because of rapid growth, abnormal metabolism, limited oxygen, and treatment pressure. They can therefore be vulnerable to a further increase in ROS:

Moderate ROS
→ can contribute to signalling and adaptation
Excessive ROS
→ oxidative damage overwhelms cellular defences
→ cell dysfunction and death may occur

The therapeutic challenge is selectivity. ROS damage is not inherently cancer-specific: excessive oxidative stress can also injure healthy brain cells.

68.2 What does “ROS-autocatalytic nanoplatform” mean?

A *nanoplatform* is a nanoscale engineered delivery system designed to carry, release, or generate therapeutic activity preferentially in a particular biological environment. In this case, the intended platform exploits tumour-associated oxidative stress to create a local positive-feedback loop.

Initial ROS in tumour environment
→ activates or changes the nanoplatform
Nanoplatform increases ROS-generating activity
→ further oxidative stress
Further oxidative stress
→ amplifies the intended anti-tumour effect

“Autocatalytic” therefore means that the system is designed so that an initial tumour-associated trigger helps generate conditions that reinforce subsequent therapeutic activity. This is not a claim that ROS production will continue indefinitely; in practice, the effect depends on delivery, local chemistry, dose, clearance, and cellular antioxidant defences.

The paper’s broader idea is to use this local oxidative damage to interfere with both glioblastoma cells and tumour-supportive reactive astrocytes. Reducing this crosstalk may make tumour cells less protected and more susceptible to therapy.

68.3 What is new relative to the earlier material?

You have already seen that glioblastoma is difficult to treat because of heterogeneity, the blood–brain barrier, treatment resistance, and an immunosuppressive microenvironment. This paper adds several related ideas.

First, *astrocytes* are a particularly important brain-specific part of the tumour microenvironment. They are not immune cells and are not cancer cells, but their altered state can nevertheless support tumour progression.

Second, the intervention is not a conventional small molecule, antibody, vaccine, or cell therapy. It is a *stimuli-responsive nanomedicine*: a delivery system intended to react differently in the abnormal chemical environment of the tumour.

Third, the goal is a form of *microenvironmental combination therapy* within one platform:

Direct oxidative injury to glioblastoma cells
+
Disruption of tumour-supportive astrocyte signalling
→ potentially greater anti-tumour effect

68.4 What should be concluded?

The study is an interesting example of targeting a disease-supporting ecosystem rather than only one tumour-cell protein. It also illustrates why nanomedicine is attractive in brain cancer: in principle, it may combine delivery, local activation, and more than one mechanism in a single product.

However, this remains a preclinical strategy. Strong activity in cell culture or animal models does not establish that a nanoplatform can reach human brain tumours safely, distribute through heterogeneous tumour tissue, avoid normal brain injury, and provide meaningful clinical benefit.

Promising local ROS generation
≠ established tumour selectivity
Disrupting astrocyte-cancer crosstalk in models
≠ proven clinical benefit in glioblastoma patients

69 [26-09-14] Intratumoral T-cell activation can kill tumours without tumour-specific T cells

(*Intratumoral T cell activation kills tumors regardless of T cell specificity*, <https://www.nature.com/articles/s41590-026-02642-z>)

This study challenges an important simplification in cancer immunology. Usually, T cells are thought to kill a tumour cell because their T-cell receptor recognises a particular tumour peptide presented by MHC. This is the mechanism used by conventional anti-tumour T-cell responses, neoantigen vaccines, and TCR-based cell therapies.

The paper shows in preclinical models that sufficiently strong *local* activation of T cells inside a tumour can also drive tumour killing when the activated T cells are not themselves tumour-specific.

69.1 What does “regardless of T-cell specificity” mean?

It does *not* mean that any activated T cell will safely kill any cell throughout the body. Rather, the proposed mechanism is local and indirect.

T cells are activated within tumour tissue
→ activated T cells release inflammatory signals
Signals act on nearby immune cells and tumour cells
→ local inflammatory anti-tumour response
Tumour cells die without needing direct recognition
by a tumour-specific T-cell receptor

Such T cells are called *bystander T cells*: T cells present in the tumour that may recognise unrelated antigens, such as viral antigens, rather than tumour antigens. They are often less exhausted than chronically stimulated tumour-specific T cells and may therefore retain substantial functional capacity.

69.2 How can this kill tumour cells?

Activated T cells release cytokines, including *interferon- γ* (IFN- γ) and *tumour necrosis factor* (TNF). These signals can recruit and activate innate immune cells, including myeloid cells, which can produce *nitric oxide*.

Together, these inflammatory signals can create a locally tumouricidal environment:

```
Intratumoral T-cell activation
→ IFN- $\gamma$  and TNF release
→ activation of nearby innate immune cells
→ nitric oxide and inflammatory signalling
→ caspase-dependent tumour-cell death
```

The paper relates this to *PANoptosis*: an inflammatory form of regulated cell death involving overlapping features of apoptosis, pyroptosis, and necroptosis. The key idea is that tumour killing can be mediated by a local cytokine-and-myeloid-cell network, rather than only by direct T-cell contact with an antigen-recognised tumour cell.

69.3 What is new here?

You have already seen that checkpoint inhibitors aim to restore or sustain tumour-reactive T cells, and that neoantigen vaccines aim to create T cells that recognise tumour peptide–MHC complexes. This work suggests a different possible strategy:

```
Conventional approach:
find or generate tumour-specific T cells
→ direct antigen-dependent killing
This study's idea:
activate available T cells within the tumour
→ local inflammatory bystander killing
```

In principle, this could help overcome some forms of immune escape, such as loss of one tumour antigen or reduced MHC presentation. It may also be relevant to tumours that contain T cells but lack enough effective tumour-specific T-cell responses.

69.4 What should be concluded?

This is an important mechanistic finding, but it does not remove the need for tumour selectivity. Systemic T-cell activation could cause serious inflammation and damage normal tissue. The therapeutic problem becomes how to activate T cells *inside the tumour*, strongly enough to generate local tumour killing while avoiding cytokine-mediated toxicity elsewhere.

The study therefore broadens the idea of what anti-tumour T cells can do. T-cell specificity remains highly important for safe immune targeting, but direct tumour-antigen recognition may not be the only route by which intratumoral T-cell activity contributes to tumour destruction.

70 [26-09-14] Hypoxia, ferroptosis, and chemotherapy resistance in pancreatic cancer

(*MMSDH facilitates ACSL4 propionylation to counteract ferroptosis upon hypoxia and impairs PDAC chemotherapy efficacy*, <https://www.nature.com/articles/s43018-026-01236-w>)

This study identifies a mechanism by which *pancreatic ductal adenocarcinoma*, or PDAC, cells may resist chemotherapy in poorly oxygenated tumour regions.

PDAC tumours often contain dense fibrotic tissue and abnormal blood vessels. This can restrict blood flow and create *hypoxia*: an environment with low oxygen availability. Hypoxia changes tumour-cell metabolism and can help cancer cells survive treatment.

70.1 What is ferroptosis?

Ferroptosis is a regulated form of cell death driven by iron-dependent damage to membrane lipids. It differs from apoptosis. In ferroptosis, reactive molecules attack certain membrane fats, creating lipid peroxides; once this damage exceeds the cell's antioxidant defences, the cell can die.

```
Iron and oxidising chemistry
→ lipid peroxidation
Insufficient cellular antioxidant defence
→ membrane damage accumulates
Excessive lipid damage
→ ferroptotic cell death
```

Some chemotherapy can contribute to oxidative stress and ferroptosis. Therefore, a tumour cell that suppresses ferroptosis may be more able to survive treatment.

70.2 What does ACSL4 do?

ACSL4 is an enzyme that helps place particular polyunsaturated fatty acids into cellular membranes. These lipids are especially susceptible to peroxidation. Consequently, ACSL4 can make cells more vulnerable to ferroptosis.

```
Higher ACSL4 activity
→ more oxidation-prone membrane lipids
→ greater potential sensitivity to ferroptosis
```

This does not mean that ACSL4 is simply “good” or “bad.” Its effect depends on cell type, metabolism, oxidative stress, and the balance of protective antioxidant systems. In this PDAC context, preserving ACSL4 would make chemotherapy-induced ferroptosis more likely.

70.3 What did the study find?

The paper proposes that hypoxia changes the activity of MMSDH, short for *methylmalonate-semialdehyde dehydrogenase*, a metabolic enzyme. Under hypoxia, MMSDH undergoes *lactylation*: the addition of a lactyl chemical group to the protein. This modification enables MMSDH to promote *propionylation* of ACSL4.

Propionylation is another post-translational modification: a propionyl group is added to a protein after it has been made. Such modifications can alter protein interactions, location, activity, or stability.

In this case, ACSL4 propionylation promoted its degradation:

```
Hypoxia
→ MMSDH lactylation
Lactylated MMSDH
→ ACSL4 propionylation
→ ACSL4 degradation
Less ACSL4
→ fewer oxidation-prone membrane lipids
→ reduced ferroptosis
→ greater chemotherapy resistance
```

Thus, the paper connects a low-oxygen tumour microenvironment to a specific metabolic and protein-modification pathway that protects PDAC cells from treatment-induced oxidative death.

70.4 What is new here?

Several concepts are new but build on earlier themes. You have seen that the tumour microenvironment can promote resistance, and that proteins can be modified after translation, for example by phosphorylation. This paper adds *metabolically linked protein modifications*: lactylation and propionylation. These modifications connect the metabolic state of a hypoxic tumour cell to the stability of a protein controlling ferroptosis sensitivity.

It also illustrates that therapy resistance need not result only from a target mutation or bypass signalling pathway. Cancer cells can become resistant by changing their metabolic state and their threshold for cell death.

70.5 What should be concluded?

The authors showed in preclinical models that a peptide designed to disrupt this MMSDH–ACSL4 mechanism restored ACSL4 and enhanced chemotherapy activity. This suggests a possible combination strategy:

```
Chemotherapy
+
blockade of hypoxia-induced ferroptosis protection
→ potentially greater tumour-cell killing
```

However, this remains preclinical. Ferroptosis is not tumour-specific, and interfering with oxidative-stress control could also damage normal tissues. The central unanswered questions are whether this mechanism is sufficiently important across patients with PDAC, whether an inhibitor can reach hypoxic tumour regions, and whether the therapeutic window is acceptable in humans.

71 [26-09-14] Teclistamab in high-risk smouldering multiple myeloma

(*Teclistamab versus lenalidomide-dexamethasone in high-risk smoldering multiple myeloma: a randomized phase 2 trial*, <https://www.nature.com/articles/s41591-026-04642-w>)

This randomised Phase 2 trial, called *ImmunoPRISM*, compared teclistamab with lenalidomide plus dexamethasone in people with *high-risk smouldering multiple myeloma*, or HR-SMM.

Smouldering multiple myeloma is an early precursor state in which abnormal plasma cells are present, but the person does not yet have the organ damage or other defining features of active multiple myeloma. Some patients have a high enough risk of progression that early treatment may be worthwhile, but this requires a particularly favourable benefit–risk profile because they may still feel well.

71.1 What does teclistamab do?

Teclistamab is a *bispecific T-cell engager*, similar in broad principle to the CD3×CD19 antibody discussed for lupus. It binds CD3 on T cells and *BCMA* on myeloma cells.

BCMA, or *B-cell maturation antigen*, is a cell-surface protein expressed mainly on plasma cells and many multiple-myeloma cells. Teclistamab bridges T cells to BCMA-positive myeloma cells:

```
Teclistamab binds CD3 on T cell
+
BCMA on myeloma cell
→ T cell and myeloma cell are brought together
→ T cell is activated locally
→ BCMA-positive myeloma cell may be killed
```

Thus, teclistamab does not directly poison the cancer cell. It redirects the patient's own T cells to recognise and kill BCMA-expressing plasma cells.

71.2 How does this differ from lenalidomide plus dexamethasone?

Lenalidomide and dexamethasone are not antigen-specific T-cell engagers.

Lenalidomide is an immunomodulatory drug. It can alter immune-cell activity, including T-cell and natural-killer-cell function, and also has direct anti-myeloma effects. Its mechanism includes binding the protein cereblon and changing degradation of particular regulatory proteins.

Dexamethasone is a corticosteroid. It reduces inflammation and can induce death of myeloma cells, but it also broadly suppresses aspects of immune activity.

The contrast is therefore:

```
Lenalidomide-dexamethasone
→ systemic anti-myeloma and immunomodulatory effects
Teclistamab
→ physically redirects T cells towards
BCMA-positive myeloma cells
```

71.3 What did the trial show?

Teclistamab produced deeper responses than lenalidomide–dexamethasone. Reported outcomes included higher rates of complete response and of *minimal residual disease*, or MRD, negativity, as well as encouraging progression-free survival.

MRD negativity means that highly sensitive testing cannot detect myeloma cells down to a predefined threshold. It is a strong sign of deep disease control, but it is still a *surrogate measure*: longer follow-up is needed to establish whether deep early responses translate into durable prevention of symptomatic myeloma, improved survival, and an acceptable long-term benefit–risk balance.

71.4 Why treat earlier?

T-cell-engaging therapy may work particularly well when disease burden is low and T cells are less impaired by advanced cancer or prior treatment.

```
Earlier high-risk disease
→ fewer myeloma cells
+ potentially fitter immune system
→ possibility of deeper immune-mediated clearance
```

However, the central trade-off remains important. Teclistamab can cause *cytokine release syndrome*, or CRS, because it activates T cells, and it can increase infection risk by depleting BCMA-positive normal plasma cells and impairing antibody-mediated immunity. In this study, CRS was common but mainly low grade; no high-grade CRS or ICANS was reported.

The key conclusion is that teclistamab is a promising early-intervention strategy, but this was a relatively small Phase 2 study. It supports the hypothesis that BCMA×CD3 T-cell redirection may achieve deeper disease control than lenalidomide–dexamethasone in HR-SMM; it does not yet establish a new standard of care.

72 [26-09-14] How PF4 becomes an antigen in heparin-induced thrombocytopenia

(A cryptic symmetry switch allosterically controls how the PF4 self-protein turns into a pathogenic antigen, <https://www.nature.com/articles/s41467-026-76387-6>)

This study explains a structural mechanism that can turn the normal human protein *platelet factor 4*, or PF4, into a target of harmful antibodies in *heparin-induced thrombocytopenia*, or HIT.

PF4 is released by activated platelets. Under normal conditions, it is a *self-protein*: the immune system is generally tolerant of it and does not make damaging antibodies against it. HIT can occur after exposure to heparin, an anticoagulant drug, when antibodies form against complexes involving PF4 and heparin.

Paradoxically, HIT causes both a low platelet count (*thrombocytopenia*) and a high risk of dangerous blood clots (*thrombosis*).

72.1 How does HIT occur?

Heparin is a strongly negatively charged molecule, or *polyanion*. PF4 has positively charged regions and can bind heparin. The resulting PF4–heparin complexes can be recognised by particular antibodies.

```
Heparin binds PF4
→ PF4 changes structural state
→ pathogenic antibody-binding region is exposed
Antibodies bind PF4-heparin complexes
→ immune complexes activate platelets
→ platelet consumption and thrombin generation
→ thrombocytopenia and thrombosis
```

Thrombin is a central enzyme in blood clotting. Thus, although platelets are depleted, the abnormal immune response can simultaneously promote excessive clot formation.

72.2 What is the “cryptic symmetry switch”?

PF4 normally forms a *tetramer*: a complex of four PF4 protein units. The study shows that this tetramer is dynamic rather than rigid. It can sample different structural arrangements.

A *cryptic* feature is one that is hidden in the usual state. The authors identified a symmetry-related structural switch that controls whether a pathogenic antibody-binding *epitope* is concealed or exposed.

```
PF4 predominantly samples closed state
→ pathogenic epitope is poorly exposed
→ PF4 remains immunologically inconspicuous
Heparin binds PF4 allosterically
→ open state is stabilised
→ pathogenic epitope is exposed for longer
→ anti-PF4 antibodies can bind
```

This is *allosteric control*: heparin binding at one region changes protein structure and antigenicity at another region. It builds directly on the earlier idea that proteins are dynamic and that a ligand can stabilise one conformation over another.

72.3 What is new here?

The new idea is not merely that heparin and PF4 form an antigenic complex. The study provides evidence that heparin shifts the dynamic structural ensemble of PF4, making a normally hidden pathogenic epitope more accessible.

Using solution NMR methods, the authors observed this conformational behaviour and showed that experimentally locking PF4 more strongly into its closed state reduced recognition by pathogenic HIT antibodies.

This is a useful example of:

```
Protein conformational change
→ altered epitope exposure
→ altered immune recognition
→ disease
```

72.4 What should be concluded?

This is a mechanistic structural-biology result, not yet a new treatment for HIT. However, it suggests possible future strategies: diagnostics that better identify pathogenic anti-PF4 antibodies, or interventions that prevent PF4 from adopting the antigenic open state.

The broader lesson is that a self-protein need not change its amino-acid sequence to become immunologically dangerous. A change in its binding partners and three-dimensional conformational state can expose a previously hidden epitope and create a pathogenic antigen.

73 [26-09-14] DNMT3A, DNA methylation, and tumour immune evasion

(*In vivo CRISPR screens identify dual functions of DNMT3A in mediating tumor immune evasion*, <https://www.nature.com/articles/s41467-026-77495-z>)

This paper uses *in vivo CRISPR screening* to identify DNMT3A as a regulator of tumour immune evasion. The central idea is that cancer cells can escape immune control not only through mutations in proteins or altered signalling pathways, but also through *epigenetic* changes: stable changes in gene activity that do not alter the DNA sequence itself.

73.1 What does DNMT3A do?

DNMT3A is a *DNA methyltransferase*. It adds a small chemical mark, a methyl group, to particular DNA sites. This process is called *DNA methylation*.

DNA methylation can alter whether genes are accessible to the cellular machinery that transcribes DNA into RNA. In simplified form, methylation near regulatory regions often contributes to reduced gene expression.

```
DNMT3A adds methyl groups to DNA
→ changes chromatin and gene accessibility
→ some genes become less likely to be expressed
→ cell behaviour changes
```

DNMT3A is therefore not a conventional receptor or signalling enzyme that acts mainly by changing one immediate pathway. It helps establish longer-lasting gene-expression states.

In cancer, this can have two relevant consequences. Within tumour cells, DNMT3A-mediated methylation may suppress genes that would make the tumour more visible or vulnerable to immune attack. Within T cells, DNMT3A can also contribute to the stable gene-expression programme associated with *T-cell exhaustion*, in which chronically stimulated T cells progressively lose proliferative capacity and anti-tumour function.

73.2 What is an *in vivo* CRISPR screen?

A CRISPR screen is a systematic experiment used to test the function of many genes at once. Researchers introduce a library of CRISPR guide RNAs into a large population of cells. Each guide directs Cas9 to disrupt one particular gene, so different cells in the population carry different gene knockouts.

An *in vivo* screen performs the selection step in a living animal, rather than only in a dish:

```
Large CRISPR guide-RNA library
→ tumour cells with different gene knockouts
Implant or generate tumours in animals
→ tumours encounter blood supply, immune cells,
stromal cells, and treatment pressure
Sequence guide RNAs in surviving tumour cells
→ identify gene knockouts enriched or depleted
```

For example, if cells lacking DNMT3A become less able to survive in an immunocompetent host, the corresponding guide RNA will become depleted from the recovered tumour population. This suggests that DNMT3A normally helps tumour cells resist immune attack.

The major advantage is biological context. An *in vitro* screen can identify genes needed for cancer-cell growth in culture, but it cannot fully capture interactions with T cells, myeloid cells, blood vessels, nutrients, and the tumour microenvironment.

73.3 What does “dual functions” mean here?

The study indicates that DNMT3A can support tumour immune evasion through more than one mechanism. One role is likely *tumour intrinsic*: DNA methylation helps cancer cells maintain a less immunogenic state. A second role concerns the tumour microenvironment, where DNMT3A-dependent programmes can influence immune-cell function and the ability of T cells to control the tumour.

The broad logic is:

```
DNMT3A-dependent methylation
→ reduced tumour visibility and/or immune susceptibility
DNMT3A-dependent immune suppression
→ weaker effective anti-tumour T-cell response
Together
→ greater tumour immune evasion
```

73.4 What should be concluded?

This paper is important because it illustrates that immune evasion can be controlled by reversible gene-regulatory states, not only fixed genetic mutations. It also shows why *in vivo* functional screens are useful: they can reveal genes that matter specifically when a tumour interacts with an intact immune system.

However, DNMT3A is biologically important in many normal cells. Broadly inhibiting DNA methylation could affect normal blood formation, immune cells, and other tissues. The therapeutic question is therefore not simply whether DNMT3A contributes to immune escape, but whether it can be modulated selectively enough, in the right tumour and treatment combination, to create an acceptable therapeutic window.

74 [26-09-14] The DNA damage response and cancer immunotherapy

(The DNA damage response and cancer immunotherapy, <https://www.nature.com/articles/s41568-026-00958-4>)

The *DNA damage response*, or DDR, is the collection of systems through which cells detect DNA damage, pause the cell cycle, repair damage when possible, or undergo cell death if the damage is too severe.

DNA is continually damaged by normal metabolism, replication errors, radiation, inflammation, and chemical exposures. Cancer cells often experience especially high DNA stress because they divide rapidly and have

unstable genomes. They therefore frequently depend on the remaining DDR pathways that they have not already lost.

74.1 What does the DNA damage response do?

Different forms of DNA damage need different responses. The main principle is:

```
DNA damage
→ damage sensing and signalling
→ cell-cycle pause
→ repair, tolerance, or cell death
```

For example, a *double-strand break* affects both strands of DNA and is particularly dangerous. Cells can repair it through *homologous recombination*, which is relatively accurate and involves proteins including BRCA1 and BRCA2, or through more error-prone processes such as *non-homologous end joining*.

Important DDR signalling proteins include ATM and ATR, which detect different forms of DNA stress and coordinate downstream responses. PARP proteins help detect and coordinate repair of other forms of DNA damage.

This builds on the earlier example of BRCA-associated homologous-recombination deficiency:

```
BRCA-associated repair defect
→ greater dependence on alternative DDR pathways
PARP inhibition
→ additional DNA-repair and replication stress
Damage becomes unmanageable
→ selective tumour-cell death may occur
```

This is the logic behind PARP inhibitors and *synthetic lethality*.

74.2 Why can DDR defects affect anti-tumour immunity?

DNA damage can make a tumour more visible to the immune system through at least two related routes.

First, genomic instability can generate more mutations. Some mutations may create *neoantigens*: altered peptides that can be displayed by MHC and recognised by tumour-reactive T cells.

Second, damaged DNA or chromosome fragments can appear in the cell cytoplasm, where DNA does not normally belong. The cell can interpret this as a danger signal.

```
Unrepaired DNA damage or chromosome mis-segregation
→ DNA fragments appear in cytoplasm
cGAS detects cytoplasmic DNA
→ activates STING signalling
Type-I interferon and inflammatory signals
→ immune-cell recruitment and anti-tumour immunity
```

The *cGAS-STING pathway* is therefore part of innate immune sensing. It is distinct from interferon- γ , which is mainly produced by activated T cells and natural-killer cells. Both types of interferon can help create a more inflamed tumour environment, but they arise through different mechanisms.

74.3 Why combine DDR-targeted therapy with immunotherapy?

The therapeutic hypothesis is that a DDR inhibitor can increase tumour-cell stress and immunogenic signalling, while a checkpoint inhibitor can help T cells respond to the resulting tumour antigens and inflammatory environment.

```
DDR-targeted therapy
→ DNA damage, genomic instability,
neoantigens, and/or cGAS-STING activation
Checkpoint blockade
→ reduces inhibitory signals on T cells
Combination
→ aims for stronger and more durable
anti-tumour immune activity
```

This is a plausible rationale, not a guarantee of clinical synergy. Some DNA-damaging treatments can also damage immune cells, increase toxicity, or select for tumour cells that avoid immune recognition. Tumours may have defective antigen presentation, impaired interferon signalling, or insufficient T-cell infiltration despite extensive DNA damage.

74.4 What should be concluded?

The important new connection is that DNA repair is not only a tumour-cell survival process. It can also influence whether a tumour is recognised as immunologically dangerous.

```
DDR deficiency
→ therapeutic vulnerability to selected DDR inhibitors
DDR deficiency or induced DNA damage
→ may also increase tumour immunogenicity
But increased immunogenicity
≠ reliable response to immunotherapy
```

Thus, DDR alterations can be both *predictive biomarkers* and possible treatment targets. The central clinical challenge is identifying which specific DDR defect, tumour type, drug schedule, and immunotherapy combination produce enough immune benefit to justify the added toxicity.

75 [26-09-14] ADC and checkpoint-inhibitor combinations: rationale and toxicity

(*Clinical toxicity of ADCs and ICI–ADC combinations: mechanisms, patterns and management*, <https://www.nature.com/articles/s41571-026-01188-1>)

This review considers the growing use of antibody–drug conjugates, or ADCs, together with immune-checkpoint inhibitors, or ICIs. The combination is biologically attractive because the two treatments can address different parts of the anti-tumour immune response, but it can also create overlapping and difficult-to-interpret toxicities.

75.1 Why should an ADC–ICI combination work?

An ADC delivers a potent cytotoxic payload preferentially to cells expressing its surface target. As discussed for trastuzumab deruxtecan, tumour-cell killing depends on target binding, internalisation, linker processing, payload release, and sensitivity to the payload.

Some payload-mediated killing may also be *immunogenic*: dying tumour cells can release tumour antigens and danger signals, called *damage-associated molecular patterns*, or DAMPs. These can help recruit and activate antigen-presenting cells, creating a more inflamed tumour microenvironment.

ADC kills susceptible tumour cells
→ tumour antigens and danger signals are released
→ antigen presentation and immune-cell recruitment may increase
→ more tumour-reactive T cells may be primed or activated

However, activated T cells can still be suppressed through checkpoints such as PD-1–PD-L1. An ICI may remove this inhibitory signal.

ADC
→ direct tumour-cell killing
+ possible immune priming
ICI
→ reduces inhibitory signalling on T cells
Combination
→ aims for both increased tumour immunogenicity
and more sustained T-cell activity

The combination may therefore be better than either therapy alone when the ADC helps convert an immunologically “cold” tumour into a more inflamed one, but the resulting T-cell response requires checkpoint blockade to function effectively.

75.2 Why does toxicity become more difficult?

An ADC is not perfectly tumour-specific. Toxicity can arise through three broad mechanisms:

Payload-mediated toxicity
→ cytotoxic payload reaches sensitive normal cells
On-target, off-tumour toxicity
→ target antigen is also present on normal tissue
Platform or host factors
→ antibody, linker, distribution, metabolism,
or patient-specific factors contribute to injury

An ICI can cause *immune-related adverse events*, or irAEs, when immune activity against normal tissues is no longer sufficiently restrained. These can affect skin, gut, liver, endocrine organs, lungs, and other tissues.

The difficulty is that ADCs and ICIs can injure some of the same organs through different mechanisms. For example, an ADC may cause direct payload-related tissue injury, whereas an ICI may cause immune-mediated inflammation. Together, the effects may be additive, interactive, or clinically indistinguishable.

75.3 Interstitial lung disease as an example

Interstitial lung disease, or ILD, is inflammation and damage affecting the lung interstitium, the tissue supporting the air sacs. Drug-related ILD or pneumonitis can cause cough, breathlessness, low oxygen levels, and potentially life-threatening respiratory failure.

Some ADCs, particularly those carrying topoisomerase-I-inhibitor payloads such as T-DXd, have an important ILD risk. PD-1 or PD-L1 blockade can also cause immune-mediated pneumonitis.

ADC-associated pulmonary injury
+
ICI-mediated lung inflammation
→ potentially increased or overlapping ILD risk

It can be difficult to determine whether lung toxicity is caused primarily by the payload, the immune checkpoint inhibitor, infection, tumour progression, or several factors together. This matters because the appropriate management can include treatment interruption, corticosteroids for suspected immune-mediated inflammation, supportive care, and careful decisions about whether treatment can safely restart.

75.4 What should be concluded?

ADC-ICI combinations have a coherent rationale, but synergy should not be assumed merely because one treatment kills tumour cells and the other activates immunity. Clinical benefit depends on the target, ADC design, payload, cancer type, biomarker-defined patient population, dose, schedule, and immune context.

The review's central practical message is that toxicity should be interpreted using all three layers: *immune activation*, *payload cytotoxicity*, and *target/platform-related effects*. Better clinical phenotyping and prospective safety monitoring are essential, particularly when combining treatments that can each cause lung, liver, gastrointestinal, skin, or blood-related toxicities.

Plausible immune synergy
≠ proven clinical benefit
Known toxicity of each individual drug
≠ predictable toxicity of the combination

76 [26-09-14] In vivo CAR-T therapy: engineering T cells inside the patient

(*In vivo* chimeric antigen receptor (CAR)-T cell therapy, <https://www.nature.com/articles/s41573-025-01291-5>)

Conventional CAR-T therapy requires a personalised manufacturing process. T cells are collected from one patient, genetically engineered and expanded in a specialised facility, tested, and then infused back into the same patient. This is called *ex vivo* CAR-T therapy because the engineering happens outside the body.

Patient blood collection
→ isolate T cells
Engineer T cells to express a CAR
→ expand and quality-test the product
Lymphodepleting treatment
→ infuse engineered CAR-T cells

In vivo CAR-T therapy aims to omit the collection and external manufacturing steps. Instead, a delivery system carries genetic instructions for a CAR directly into the patient and is designed to modify T cells *inside* the body.

76.1 What is the basic idea?

The central therapeutic product remains a CAR-expressing T cell. The difference is how that cell is made.

Targeted genetic-delivery vehicle enters patient
→ preferentially reaches T cells
T cells receive instructions encoding a CAR
→ CAR is produced within those T cells
In vivo-generated CAR-T cells
→ recognise and kill cells bearing the chosen antigen

For example, an in vivo approach could deliver instructions for a CD19 CAR. The patient's own T cells would then be converted into CD19-directed CAR-T cells and could deplete CD19-positive B cells, using the same broad recognition principle as conventional CD19 CAR-T therapy.

76.2 How can the CAR instructions be delivered?

Two broad delivery approaches are being developed.

Viral vectors, such as engineered lentiviral vectors, can deliver a DNA sequence encoding the CAR. If the sequence integrates into the T-cell genome, CAR expression may be stable and long-lasting. This could be useful when sustained anti-tumour activity is required, but integration creates important safety questions, including unintended genomic effects.

RNA-based delivery, often using lipid nanoparticles or related targeted particles, gives cells temporary instructions to make the CAR protein. Because RNA usually does not integrate into DNA, expression is transient. This may allow repeat dosing and reduce risks associated with permanent genetic modification, but it may also limit durability.

```
Integrating DNA delivery
→ potentially persistent CAR expression
+ potential insertional-genetic risk
Transient RNA delivery
→ temporary CAR expression
+ potentially repeatable and non-integrating treatment
```

76.3 Why is this attractive?

In vivo CAR-T could make CAR-T therapy more accessible, faster, and potentially less expensive. It could avoid delays caused by individualised manufacturing, product failure, and the need to obtain sufficient functional T cells from a heavily pre-treated patient.

It may be especially attractive in autoimmune disease. In severe SLE, for example, the desired effect may be a sufficiently deep but time-limited depletion of pathogenic B cells to enable an immune “reset,” rather than permanent persistence of a cancer-killing CAR-T population.

76.4 What is more difficult than it sounds?

The main challenge is *selective delivery*. A delivery vehicle must reach enough appropriate T cells while avoiding unintended transduction of other cells, particularly cells that could express the CAR in a harmful context.

Unlike ex vivo CAR-T, the final engineered T-cell product cannot be individually inspected, purified, or quality-tested before it acts in the patient. The dose, tissue distribution, type of T cell modified, extent of CAR expression, persistence, and immune response to the delivery vehicle must all be controlled in vivo.

```
Efficient T-cell targeting
≠ selective delivery only to desired T cells
Transient CAR expression
≠ automatically safe
In vivo CAR generation
≠ removal of CRS, infection, or on-target,
off-tumour toxicity risks
```

76.5 What should be concluded?

In vivo CAR-T is best understood as a new *manufacturing and delivery strategy*, rather than a fundamentally new form of CAR recognition. It aims to create CAR-T cells within the patient instead of in a facility.

The approach could substantially simplify cell therapy and extend it to cancer and autoimmune disease. However, it remains an early clinical and preclinical field. Its eventual value will depend on whether delivery can be made sufficiently T-cell-selective, controllable, reproducible, durable when needed, and safe.

77 [26-09-14] ATR inhibition can make Merkel cell carcinoma more immunogenic

(*ATR inhibition induces NF- κ B-dependent immunogenicity in Merkel cell carcinoma independent of cGAS-STING*, <https://www.nature.com/articles/s41698-026-01688-4>)

This study connects the DNA damage response, or DDR, to anti-tumour immunity in *Merkel cell carcinoma*, or MCC. MCC is a rare but aggressive skin cancer. Many MCC tumours are associated with Merkel cell polyomavirus, which provides viral antigens that can make the tumour visible to T cells. Nevertheless, some tumours remain poorly infiltrated by immune cells or become resistant to checkpoint inhibition.

The paper suggests that inhibiting ATR can make MCC cells more inflammatory and immunologically visible through NF- κ B signalling.

77.1 What does ATR do?

ATR, short for *ataxia telangiectasia and Rad3-related* protein, is a protein kinase in the DDR. Its central role is to respond to *replication stress*: problems that arise while a cell is copying its DNA before division.

For example, DNA replication can slow or stall because of DNA damage, limited nucleotides, abnormal oncogene signalling, or difficult-to-copy DNA regions. ATR helps stabilise these stalled replication forks, coordinate repair, and delay cell-cycle progression until the problem is managed.

```
DNA replication stress
→ ATR activation
ATR signalling
→ stabilises replication machinery
+ activates cell-cycle checkpoints
+ promotes repair and survival
```

Cancer cells often have unusually high replication stress, meaning that some become especially dependent on ATR. An ATR inhibitor removes part of this protective response:

```
ATR inhibition
→ unresolved replication stress
→ replication-fork collapse and DNA damage
→ tumour-cell death and/or altered immune signalling
```

77.2 What is NF- κ B?

NF- κ B is a family of transcription factors that controls many inflammatory and stress-response genes. When activated, it enters the nucleus and changes expression of cytokines, chemokines, survival signals, and immune-regulatory molecules.

In this study, ATR inhibition activated an NF- κ B-dependent inflammatory programme in MCC cells. This increased features associated with tumour *immunogenicity*: the ability of tumour cells to attract, activate, or be recognised by the immune system.

ATR inhibition in MCC cells
→ NF- κ B activation
→ inflammatory and immune-relevant gene expression
→ potentially more favourable conditions
for anti-tumour immune responses

77.3 Why is “independent of cGAS–STING” important?

Earlier, the DDR–immunity connection was described through cytoplasmic DNA activating the cGAS–STING pathway, which can induce type-I interferon and immune-cell recruitment.

This paper identifies a different route. The immunogenic effect of ATR inhibition in its MCC models depended on NF- κ B but did not require cGAS–STING.

Common proposed DDR–immunity route:
DNA damage
→ cytoplasmic DNA
→ cGAS–STING
→ inflammatory signalling
This study:
ATR inhibition
→ NF- κ B-dependent immunogenicity
→ without requiring cGAS–STING

This matters because tumours can have defective or suppressed cGAS–STING signalling. An alternative route to inflammatory tumour reprogramming could therefore be useful.

77.4 What should be concluded?

The paper gives a rationale for investigating ATR inhibitors with immunotherapy in MCC: ATR inhibition may both exploit a tumour-cell DDR dependency and make the tumour environment more supportive of immune attack.

However, this is not yet proof that ATR–checkpoint-inhibitor combinations improve patient outcomes. ATR inhibition can also harm normal proliferating cells and immune cells, causing toxicities such as myelosuppression. The practical question is whether a dose and schedule can create enough tumour-selective DNA stress and immune activation to justify those risks.

78 [26-09-14] PKM2 AP: a micropeptide degrader for vascular calcification

(PKM2 AP: a micropeptide inhibits vascular calcification via molecular glue-mediated ubiquitination of PKM2, <https://www.nature.com/articles/s41467-026-77693-9>)

This study applies the idea of a *molecular glue* outside cancer, to *vascular calcification*: the abnormal accumulation of calcium-containing mineral in blood-vessel walls.

Vascular calcification is not simply passive calcium deposition. In many settings, including chronic kidney disease, diabetes, and atherosclerosis, vascular smooth-muscle cells can change state and adopt features resembling bone-forming cells.

```
Vascular stress and inflammation
→ smooth-muscle-cell state change
Osteogenic-like programme
→ mineral deposition in vessel wall
Vessel stiffening
→ increased cardiovascular risk
```

78.1 What is PKM2?

PKM2, or *pyruvate kinase M2*, is a metabolic enzyme. Pyruvate kinase catalyses a late step in glycolysis, the process through which cells convert glucose into usable energy and metabolic building blocks.

PKM2 is notable because its role is not limited to energy production. Depending on its structure, location, and modifications, it can help cells adopt metabolic and gene-expression programmes associated with growth, stress responses, and cell-state change.

In vascular smooth-muscle cells, the study implicates PKM2 in the transition towards a calcification-promoting, osteogenic-like state:

```
Increased or dysregulated PKM2 activity
→ metabolic reprogramming
→ osteogenic-like smooth-muscle-cell state
→ vascular calcification
```

78.2 What is PKM2 AP?

PKM2 AP is a *micropeptide*: a very short protein-like molecule encoded by a small open reading frame. Micropeptides were historically overlooked because conventional gene annotation often focused on larger proteins. They can nevertheless have specific biological functions, including changing protein interactions, localisation, stability, or enzyme activity.

The paper reports that PKM2 AP inhibits vascular calcification by promoting degradation of PKM2 rather than merely blocking its enzymatic active site.

78.3 How does molecular-glue-mediated ubiquitination work?

Ubiquitination is a cellular protein-control system. An *E3 ubiquitin ligase* can attach ubiquitin molecules to a target protein. Certain ubiquitin patterns act as a label that directs the target to the proteasome, the cell's protein-degradation machinery.

A molecular glue promotes a useful interaction between two proteins that would otherwise interact weakly or not at all. In this case, PKM2 AP acts as a peptide-based glue that brings PKM2 into contact with a ubiquitination system.

```
PKM2 AP binds or stabilises an interaction involving PKM2
→ PKM2 is ubiquitinated
Ubiquitinated PKM2
→ proteasomal degradation
Less PKM2
→ reduced pro-calcific cell-state programme
→ less vascular calcification in models
```

This differs from a conventional enzyme inhibitor. An inhibitor must remain present at sufficient concentration to block activity. A degrader instead reduces the amount of the target protein itself, potentially affecting both its metabolic and non-metabolic functions.

78.4 What is new here?

You have encountered molecular glues in cancer, for example as a way to alter a protein interaction or promote degradation of a disease-relevant protein. This work extends the same principle to cardiovascular pathology and uses a *naturally derived micropeptide* rather than a conventional small molecule.

The conceptual advance is:

```
Disease-relevant protein
→ PKM2
Micropeptide
→ creates a degradation-promoting interaction
Protein degradation
→ altered pathological cell state
→ reduced tissue calcification
```

78.5 What should be concluded?

This is an interesting mechanistic and preclinical result. It suggests that micropeptides may provide templates for new targeted degraders, and that vascular calcification might be modified by intervening in smooth-muscle-cell metabolism and state change.

However, many questions remain before therapeutic use: whether a PKM2 AP-derived treatment can be delivered to diseased vessel walls, remain stable in the body, avoid disrupting beneficial PKM2 functions in normal tissues, and safely reduce clinically meaningful cardiovascular outcomes.

```
Reduced PKM2 and calcification in experimental models
≠ established treatment for vascular calcification
A naturally occurring molecular glue
≠ a readily deliverable medicine
```

79 [26-09-14] TET2 loss, single-stranded DNA gaps, and PARP-inhibitor sensitivity

(*TET2 loss triggers ssDNA gap accumulation and heightened PARP inhibitor sensitivity*, <https://www.nature.com/articles/s41467-026-77587-w>)

This study identifies *TET2 loss* as a possible source of vulnerability to PARP inhibitors. It extends the earlier idea of *synthetic lethality*: a tumour cell can tolerate one defect in DNA maintenance, but not that defect combined with inhibition of a compensating repair pathway.

TET2 is best known as an *epigenetic regulator*. It modifies methylated DNA and can influence gene expression. This paper adds a DNA-replication role: loss of TET2 can leave cells with more incompletely copied DNA after replication.

79.1 What is TET2?

TET2 is an enzyme that converts *5-methylcytosine* into oxidised forms, including *5-hydroxymethylcytosine*. These DNA chemical marks influence chromatin state and gene expression.

```
TET2 activity
→ DNA demethylation-related regulation
→ altered chromatin and gene-expression states
→ effects on cell identity and DNA-damage responses
```

TET2 is frequently mutated or lost in some blood cancers, especially myeloid malignancies, but altered TET2 function can be relevant beyond these diseases. Its consequences are context-dependent: loss can affect cellular differentiation, inflammation, metabolism, and genome stability.

79.2 What is an ssDNA gap?

DNA normally consists of two complementary strands. During DNA replication, the cell must copy both strands fully before division. A *single-stranded DNA gap*, or ssDNA gap, is a short region where one strand has not yet been copied or repaired, leaving the other strand temporarily unpaired.

This can occur when the replication machinery encounters a DNA lesion but continues copying downstream of the problem. This process is called *repriming*: rather than stopping permanently, replication restarts beyond the lesion, but leaves a gap behind.

```
Replication fork encounters DNA lesion
→ replication may restart downstream
Original lesion is bypassed temporarily
→ unfilled ssDNA gap remains behind the fork
Gap filling and repair
→ required before safe cell division
```

An ssDNA gap is not necessarily immediately lethal. Cells have systems to fill or repair such gaps. However, if many gaps persist, they create genome instability and can be converted into more serious damage, including double-strand breaks, during later replication or cell division.

79.3 How can TET2 loss create this vulnerability?

The paper reports that TET2 loss promotes accumulation of replication-associated ssDNA gaps. In broad terms, without TET2, cells copy or repair DNA less robustly under replication stress. They therefore enter a state with a greater burden of incompletely resolved DNA.

```
TET2 loss
→ altered replication-stress response
→ more post-replicative ssDNA gaps
→ increased dependence on gap-filling
and DNA-repair pathways
```

This is related to, but distinct from, the earlier BRCA example. BRCA loss primarily creates a defect in homologous-recombination repair. Here, the proposed vulnerability is linked more directly to persistent gaps left behind DNA replication.

79.4 Why does this increase sensitivity to PARP inhibitors?

PARP1 and PARP2 help coordinate repair of certain DNA lesions and replication-associated damage. PARP inhibitors both reduce PARP catalytic activity and, for some agents, trap PARP on DNA. This can make DNA replication more difficult and increase the burden of unfilled ssDNA gaps.

A TET2-proficient cell may have enough residual capacity to manage this burden. A TET2-deficient cell begins with more gaps and is therefore closer to a threshold of catastrophic DNA damage.

```
TET2 loss
→ elevated baseline ssDNA-gap burden
PARP inhibition
→ impaired repair and increased replication stress
Both together
→ gaps persist or accumulate
→ chromosome damage and cell death
```

This is the proposed synthetic-lethal relationship: neither perturbation necessarily kills every cell alone, but their combination can be selectively damaging to TET2-deficient cells.

79.5 What is new here?

The useful new distinction is that PARP-inhibitor sensitivity need not arise only from BRCA1, BRCA2, or a classic homologous-recombination defect. *Post-replicative ssDNA gaps* are increasingly understood as an important intermediate cause of PARP-inhibitor toxicity.

```
"PARP inhibitors kill because they create double-strand breaks"
→ useful but incomplete simplification
Replication-associated ssDNA gaps
→ can be an earlier and potentially decisive
source of PARP-inhibitor vulnerability
```

79.6 What should be concluded?

The work suggests that TET2 status could eventually help identify patients who may benefit from PARP inhibition, potentially beyond currently established biomarker groups. However, this is not yet a validated clinical biomarker. TET2 loss can have different effects in different tumour types and genetic backgrounds, and the therapeutic value depends on whether the proposed DNA-gap mechanism predicts a sufficiently large clinical benefit and safety margin.

80 [26-09-14] Tumour-targeted IL-10 plus IL-2: local cytokine cooperation

(*Tumor-targeted IL-10 synergizes with IL-2 to enhance antitumor immunity while minimizing immune-related adverse events*, <https://www.nature.com/articles/s41467-026-77461-9>)

This study examines a cytokine-based combination designed to increase anti-tumour immunity while limiting the systemic inflammation that can occur when immune-stimulatory cytokines circulate throughout the body.

Cytokines are signalling proteins used by immune cells. Their effects are strongly context-dependent: the same cytokine can be helpful or harmful depending on its dose, location, timing, receptor-expressing cells, and the other signals present.

80.1 What do IL-2 and IL-10 do?

IL-2 is a cytokine that promotes T-cell growth, survival, and activation. It can also activate natural-killer, or NK, cells. These are potentially useful effects for cancer therapy because CD8 T cells and NK cells can kill tumour cells.

However, IL-2 is not selectively anti-tumour. At systemic exposure, it can activate immune cells outside the tumour, expand immunosuppressive regulatory T cells in some contexts, and cause substantial toxicity, including capillary leak and inflammatory effects.

IL-10 is often introduced as an anti-inflammatory cytokine because it can suppress some antigen-presenting-cell and myeloid-cell inflammatory functions. This is true, but incomplete. IL-10 can also support cytotoxic CD8 T cells and NK cells, particularly in an already activated immune environment.

```
IL-2
→ promotes immune-cell proliferation and activation
IL-10
→ can restrain harmful inflammation
but can also support cytotoxic lymphocyte function
IL-2 + IL-10 in an appropriate tumour context
→ potentially stronger anti-tumour immune activity
with better control of systemic inflammation
```

80.2 What does “enhanced anti-tumour immune response” actually mean?

This phrase should be translated into measurable biological steps. In the study’s proposed mechanism, the combination increases the number, activation state, persistence, and/or killing capacity of immune cells within the tumour, especially cytotoxic CD8 T cells and NK cells.

```
Tumour-targeted IL-10
→ conditions the local tumour immune environment
IL-2
→ expands and activates cytotoxic lymphocytes
More functional CD8 T cells and NK cells in tumour
→ greater cytokine production,
tumour-cell recognition, and cytotoxic killing
More tumour-cell killing
→ reduced tumour growth in experimental models
```

Thus, “enhanced immunity” does not merely mean that generic inflammation increased. It means that immune cells expected to contribute to tumour control showed stronger local anti-tumour function and were associated with improved tumour control in the models tested.

80.3 Why target IL-10 to the tumour?

A tumour-targeted cytokine is designed to concentrate cytokine activity in tumour tissue, often by linking it to an antibody or targeting component that recognises a tumour-associated feature. The aim is to improve the therapeutic window:

```
Systemic cytokine exposure
→ broad immune activation
→ toxicity in normal tissues
Preferential tumour localisation
→ greater local immune stimulation
+ lower exposure elsewhere
→ potential improvement in benefit-risk
```

The word *preferential* is important. Tumour targeting is not absolute. Some treatment can still circulate, enter normal tissues, or activate immune cells outside the tumour.

80.4 Why might this reduce immune-related adverse events?

Immune-related adverse events occur when activated immunity damages normal tissues. The study's rationale is that tumour-localised IL-10 may support cytotoxic anti-tumour responses while also limiting excessive inflammatory activation in non-tumour tissues.

This is not the same as claiming that IL-10 makes immunotherapy intrinsically safe. Its anti-inflammatory effects could also, in some settings, weaken desirable anti-tumour immunity. The observed balance between immune stimulation and immune restraint depends on where IL-10 acts and which immune-cell populations are present.

80.5 What should be concluded?

The new idea is the apparently paradoxical use of IL-10 as part of an anti-cancer treatment. IL-10 is not simply an “immune suppressor”; it can have both regulatory and immune-supporting roles. This study aims to exploit that context dependence by concentrating IL-10 activity in the tumour and combining it with the T-cell- and NK-cell-stimulating effects of IL-2.

The findings provide a preclinical rationale for local cytokine combinations, but they do not yet establish clinical benefit or safety in patients.

More activated cytotoxic immune cells in a tumour model
≠ proven patient benefit
Tumour-targeted cytokine delivery
≠ elimination of systemic cytokine toxicity
Preclinical synergy
≠ a clinically viable dose, schedule,
or therapeutic window

81 [26-09-14] VAV1 molecular glues: degrading an immune-signalling hub

(Leveraging high-throughput proteomics and AI-based protein folding to accelerate VAV1 molecular glue discovery, <https://www.nature.com/articles/s41467-026-77657-z>)

This study concerns VAV1, an immune-cell-restricted signalling protein, and how molecular glues might be discovered more systematically. The medical goal is to reduce harmful immune activation in autoimmune and inflammatory disease.

VAV1 is expressed mainly in *haematopoietic cells*: cells of the blood and immune system. It helps transmit and organise signals downstream of receptors such as the T-cell receptor and B-cell receptor.

T-cell receptor or B-cell receptor detects antigen
→ VAV1 helps transmit and organise the signal
→ immune-cell activation, movement,
proliferation, and cytokine production

VAV1 has two related functions. It is a *guanine-nucleotide exchange factor*, helping activate small signalling proteins that regulate the cytoskeleton, and it acts as a *signalling scaffold*, bringing multiple proteins together. These activities help immune cells form an *immunological synapse*: the organised contact zone through which a T cell communicates with an antigen-presenting cell or target cell.

81.1 Why might degrading VAV1 help in disease?

In autoimmune disease, immune cells can respond inappropriately to self-antigens and produce inflammatory cytokines that injure normal tissue. Reducing VAV1 could dampen several parts of this response at once:

Less VAV1

- weaker receptor-driven immune signalling
- reduced T-cell and B-cell activation
- reduced inflammatory cytokine production
- potentially less tissue-damaging inflammation

This is not equivalent to eliminating immune cells. The intended effect is *immunomodulation*: reducing excessive immune activation while retaining enough normal immune function to control infection.

81.2 How does a VAV1 molecular glue work?

The study identifies compounds that bind the E3 ubiquitin ligase *cereblon*, or CRBN, and enable it to recognise VAV1. The compound is therefore a molecular glue: it creates a productive new protein–protein interaction.

Molecular glue binds CRBN

- CRBN-glue complex recruits VAV1
- VAV1 is ubiquitinated
- proteasome degrades VAV1
- VAV1-dependent immune signalling is reduced

This is potentially useful because conventional inhibitors would need to block both VAV1’s catalytic and scaffold functions. Degradation removes the protein itself and can therefore suppress both functions.

81.3 What did the study add?

The main advance is not yet a treatment result. It is an improved discovery approach for molecular glues. The researchers combined high-throughput proteomics, which can measure protein changes across many samples, with AI-based protein-structure prediction and experimental validation.

They identified a previously unrecognised structural feature in VAV1 that can act as a *degron*: a protein feature that enables recruitment to CRBN in the presence of the glue. This feature lies in a loop within a VAV1 SH3 domain, rather than in the more familiar degron patterns associated with cereblon-targeting compounds.

Large-scale protein measurements

+ structural prediction of CRBN-glue-VAV1 complexes

- identify plausible VAV1-recruiting compounds

Experimental testing

- confirm VAV1 degradation and mechanism

The biological lesson is that a target does not need a conventional enzyme pocket to be tractable. If it has an accessible structural surface that can be recruited to an E3 ligase, it may be degradable.

81.4 What should be concluded?

VAV1 degradation is a promising way to target a central immune-signalling node that has been difficult to inhibit conventionally. This may be relevant to diseases driven by pathological T-cell and B-cell activation.

However, VAV1 also contributes to normal immune defence. The central clinical question is whether a VAV1 degrader can suppress autoimmune pathology sufficiently while avoiding excessive infection risk or broader immune dysfunction. The paper primarily advances the *discovery logic* for such drugs; clinical benefit must still be demonstrated independently.

82 [26-09-18] A CA-IV-directed small-molecule shuttle for brain delivery of biologics

(CA-IV-directed small-molecule shuttle enables targeted brain delivery of biologics, <https://www.nature.com/articles/s41589-026-02294-y>)

The *blood–brain barrier*, or *BBB*, protects the brain by restricting movement of many substances from blood into brain tissue. This is helpful physiologically, but it creates a major drug-discovery problem: conventional antibodies, proteins, and nucleic-acid medicines generally have very limited access to the brain after systemic administration.

This paper describes *BrainCAB* (*brain access through carbonic-anhydrase-binder bioconjugation*), a delivery strategy intended to attach a brain-targeting small molecule to a larger biological therapeutic.

```
Biologic cargo, such as antibody or oligonucleotide
+
CA-IV-binding small-molecule shuttle
→ conjugate binds brain vascular endothelial cells
→ increased transport across blood-brain barrier
→ increased delivery into brain tissue
```

The target is *carbonic anhydrase IV*, or *CA-IV*, an enzyme displayed by endothelial cells lining brain blood vessels. CA-IV is particularly interesting because it appears relatively enriched at the BBB and is conserved across several mammalian species, including primates. This may offer a more brain-focused delivery route than targets with broad expression in peripheral tissues.

The authors attached a CA-IV-binding chemical group, derived from a carbonic-anhydrase inhibitor scaffold, to different biologic cargos. In cell systems, the resulting conjugates bound CA-IV and were internalised. In mouse and neonatal non-human-primate experiments, CA-IV-targeted antibody conjugates showed increased brain accumulation and measurable persistence in brain parenchyma after systemic dosing.

The central advance is therefore not a new therapeutic target for brain disease itself. CA-IV acts as a potential *delivery handle*: an address on BBB endothelial cells that may help transport another therapeutic product into the central nervous system.

```
Disease-relevant target is inside brain
+
Biologic would normally remain largely in blood
→ attach CA-IV-directed shuttle
→ improve probability of biologic reaching brain target
```

This is especially relevant to diseases such as glioblastoma, neurodegeneration, lysosomal-storage disorders, and other neurological conditions in which insufficient brain exposure has limited otherwise plausible therapeutic approaches.

However, this remains preclinical delivery research. Increased brain concentration does not by itself establish productive delivery into the relevant cell type, adequate target engagement, safety, or clinical benefit. Important remaining questions include the extent of transport in adult humans, peripheral CA-IV-related exposure, repeated-dose tolerability, immunogenicity of conjugated biologics, and whether the cargo remains functional after transport.

Improved BBB delivery
≠ effective treatment of brain disease
A useful medicine still requires:
appropriate brain distribution
+ functional target engagement
+ acceptable safety
+ meaningful patient benefit

83 [26-09-18] Mitochondrial ATP can help T cells become effective killers by opening chromatin

(Mitochondrial ATP promotes T cell differentiation through chromatin accessibility, [https://www.cell.com/cell/fulltext/S0092-8674\(26\)00996-7](https://www.cell.com/cell/fulltext/S0092-8674(26)00996-7))

This paper links three ideas already discussed in the document: T-cell activation, cellular metabolism, and gene regulation. Its central finding is that ATP made by mitochondria is not only general cellular “fuel.” It helps activated CD8 T cells acquire the gene-expression programme needed to become effective cytotoxic T cells.

A T cell must change substantially after recognising antigen. It needs to proliferate, produce cytokines, make cytotoxic molecules such as perforin and granzymes, and sustain this activity in tissues such as tumours. This requires not only external activation signals through the T-cell receptor and co-stimulatory receptors, but also an internal metabolic state capable of supporting the transition.

T-cell activation
→ metabolic reprogramming
→ mitochondrial ATP production
→ chromatin becomes accessible at effector-gene regions
→ cytotoxic T-cell programme is expressed
→ improved capacity to kill abnormal cells

83.1 What does “chromatin accessibility” mean?

DNA is not freely exposed inside the nucleus. It is wrapped around proteins called histones, forming *chromatin*. When chromatin near a gene is relatively closed, transcription factors and RNA-producing machinery have difficulty reaching that gene. When chromatin is more open or accessible, the gene can more readily be transcribed into RNA.

Closed chromatin
→ gene is difficult for transcription machinery to access
Open chromatin
→ transcription factors can bind more easily
→ gene expression becomes more possible

The important point is that opening chromatin is an active process. ATP-dependent chromatin-remodelling complexes use energy from ATP hydrolysis to move, remove, or rearrange nucleosomes. This can expose regulatory DNA regions needed to activate an effector T-cell programme.

83.2 Why mitochondrial ATP matters

Activated T cells are often associated with high glycolysis: rapid conversion of glucose into metabolic intermediates and ATP outside mitochondria. This supports rapid growth and proliferation. However, the paper indicates that glycolysis alone is insufficient for full cytotoxic differentiation.

Mitochondrial oxidative phosphorylation, or *OXPHOS*, produces ATP through the mitochondrial electron-transport system. The study supports the idea that mitochondrial ATP, together with the high-energy reserve molecule *phosphocreatine*, provides energy needed for chromatin remodelling during the transition towards an effector state.

Glycolysis
→ supports rapid growth and proliferation
Mitochondrial OXPHOS and ATP reserve
→ supports ATP-dependent chromatin remodelling
Chromatin remodelling
→ enables expression of cytotoxic effector genes

This is a more specific claim than simply saying that “healthy mitochondria are good for T cells.” The proposed mechanism is that mitochondrial ATP helps power the epigenetic changes required for T-cell differentiation.

83.3 What did the study suggest therapeutically?

The authors used D- α -hydroxybutyrate, or *DAHB*, to enhance mitochondrial ATP production and phosphocreatine-associated energy buffering. In their experimental systems, this promoted effector differentiation and improved anti-tumour activity of CD8 T cells.

This connects to the earlier discussion of T-cell exhaustion and cell therapy. A tumour can limit T-cell activity through checkpoint signalling, poor nutrient availability, hypoxia, and metabolic stress. Restoring or supporting mitochondrial function might therefore complement strategies such as PD-1 blockade, cancer vaccines, or CAR-T-cell engineering.

Checkpoint inhibitor
→ reduces an external inhibitory signal
Mitochondrial metabolic support
→ may improve the T cell’s internal capacity
to adopt and sustain an effector programme
Potential combination
→ stronger anti-tumour immunity,
if safety and clinical benefit are demonstrated

However, this remains an early mechanistic and preclinical direction. Increasing T-cell metabolic activity could also increase inflammation or affect other immune and normal cells. The central unanswered question is whether metabolic manipulation can improve durable anti-tumour immunity in patients without causing unacceptable toxicity.

More mitochondrial ATP
≠ automatically better immunotherapy
The therapeutic challenge is to enhance useful T-cell differentiation
in the right cells, tissues, and disease setting,
with an acceptable benefit-risk balance.

84 [26-09-18] Aligning protein language models with experimentally measured function

(Functional alignment of protein language models via reinforcement learning, <https://www.nature.com/articles/s41467-026-77557-2>)

This paper concerns *protein language models*, or *pLMs*: AI models trained on very large collections of natural protein sequences. Like a language model learns statistical patterns in text, a pLM learns which amino-acid sequences and sequence patterns are common in evolution.

This is useful because protein sequence strongly influences protein folding, stability, interactions, localisation, and function. However, a natural-looking protein sequence is not automatically a protein with the desired therapeutic or industrial property.

Protein language model learns:
Which amino-acid sequences resemble natural proteins?
But drug and protein engineering require:
Which sequence gives the desired measured function?

The paper's key idea is to use *reinforcement learning from experimental feedback*, called *RLXF*, to connect a generative protein model more directly to a chosen experimental objective.

84.1 What does “functional alignment” mean?

The word *alignment* here has the same broad meaning as in reinforcement learning from human feedback, or RLHF, for language models. A pretrained model has learned general patterns, but it is then adjusted towards a specific desired output.

For proteins, the desired output is not a helpful answer to a user. It is a measurable experimental property, such as protein fluorescence, binding, stability, catalytic activity, expression, or selectivity.

Natural protein sequences
→ pretrained protein language model
Experimental sequence-function data
→ reward model estimates desirable function
Reinforcement learning
→ protein language model generates sequences
with higher predicted reward

The model is therefore encouraged to propose amino-acid sequences that remain sufficiently protein-like while moving towards a defined functional goal.

84.2 What did the study show?

The authors applied this approach to protein-engineering tasks, including optimisation of the fluorescent protein *CreiLOV*. Fluorescence is not itself a therapeutic effect, but it is a useful test case because it can be measured quantitatively and depends on correct protein folding and chromophore-associated chemistry.

The RLXF-designed variants showed improved fluorescence compared with baseline sequence-generation approaches. This supports the central claim that experimental functional feedback can guide a pLM beyond merely reproducing patterns found in natural proteins.

Pretrained pLM alone
→ generates plausible protein sequences
pLM aligned with functional reward
→ preferentially generates plausible sequences
predicted to have a desired experimental property

84.3 What is new biologically?

The paper does not identify a new disease pathway, drug target, or therapeutic mechanism. Its main contribution is a *protein-engineering method*.

The biologically important idea is that protein function is not determined by one amino acid independently. It emerges from the whole three-dimensional protein structure, its dynamics, and interactions with other molecules. A model can therefore be useful if it learns combinations of mutations that preserve a viable protein while improving a measured property.

This builds on the document's earlier distinction:

```
Protein sequence
→ protein structure, dynamics, and interactions
→ biochemical function
But:
Predicted sequence quality
≠ experimentally confirmed function
```

84.4 Why could this matter for medicine?

In principle, similar methods could help engineer antibodies, enzymes, cytokines, receptors, viral vectors, or other biologics. For example, one might optimise an antibody for affinity, selectivity, stability, reduced aggregation, or a desired epitope; or optimise an enzyme for activity in a particular tissue environment.

However, the paper does not remove the main challenges of therapeutic development. A protein with improved activity in one assay may still have poor tissue exposure, immunogenicity, manufacturability, safety, or clinical relevance.

```
Improved experimental protein function
≠ useful medicine
A therapeutic protein still needs:
appropriate mechanism
+ tissue exposure
+ developability and manufacture
+ acceptable safety
+ meaningful patient benefit
```

The practical value of RLXF is therefore likely to be improved *candidate generation and prioritisation*. It may help researchers choose better protein variants to make and test, but experimental validation remains essential at every stage.

85 [26-09-18] OmniTCR: predicting and designing T-cell receptors for peptide–MHC targets

(*OmniTCR: a foundation model unifying T cell receptor recognition prediction and conditional sequence generation*, <https://www.biorxiv.org/content/10.64898/2026.09.10.750588v1.full.pdf>)

This preprint concerns *T-cell receptors*, or *TCRs*, and the difficult problem of predicting what they recognise.

Recall that a conventional T cell does not usually recognise a whole protein directly. Its TCR recognises a short peptide displayed by a particular *major histocompatibility complex*, or *MHC*, molecule:

```
Protein in tumour or infected cell
→ processed into peptide
→ peptide displayed by patient's MHC molecule
→ compatible TCR recognises peptide-MHC
→ T cell may become activated and kill the target cell
```

The recognition problem is therefore not simply “does this TCR bind this peptide?” A TCR contacts the combined three-dimensional surface formed by both the peptide and the MHC molecule. The same peptide can be presented differently by different MHC types, and a TCR may recognise one peptide–MHC combination but not another.

85.1 What does OmniTCR try to do?

OmniTCR is a *foundation model*: a large model trained on broad TCR-related data that can then be adapted to several related tasks. Its two central aims are:

```
Recognition prediction
→ does this TCR likely recognise this peptide-MHC target?
Conditional sequence generation
→ given a chosen peptide-MHC target,
generate candidate TCR sequences that may recognise it
```

The first task helps interpret naturally occurring T cells. For example, a tumour sample may contain many expanded T-cell clones, but researchers may not know which tumour antigen each clone recognises. Connecting a TCR sequence to a peptide–MHC target is often called *TCR deorphanisation*.

The second task is more ambitious. Rather than searching only for a naturally occurring TCR that recognises a chosen tumour target, the model attempts to propose candidate TCR sequences conditioned on that target.

85.2 Why could this matter in cancer?

TCR-based cell therapy may be able to target *intracellular* cancer proteins indirectly. A tumour mutation can create a neoantigen peptide, which is displayed by MHC. A patient's T cells could then be engineered with a TCR that recognises that neoantigen–MHC complex.

```
Tumour mutation
→ neoantigen peptide displayed by MHC
TCR recognising that peptide-MHC
→ potential engineered TCR-T-cell therapy
Unlike CAR-T:
target can originate from an intracellular protein
But:
target must be presented by the relevant MHC molecule
```

This could complement personalised neoantigen vaccines. A vaccine aims to activate the patient's own suitable T cells. TCR-T therapy instead supplies T cells engineered with a selected receptor, potentially when a natural anti-tumour T-cell response is absent or too weak.

85.3 Why is this difficult?

TCR recognition is highly specific but not perfectly specific. One TCR can sometimes recognise several related peptide–MHC complexes. This *cross-reactivity* is a normal feature of immunity, but it creates a major safety concern for therapeutic TCR design.

Desired recognition:
tumour peptide + patient's MHC
→ tumour-cell killing
Unintended recognition:
normal-tissue peptide + same MHC
→ potentially severe toxicity

A model prediction or a generated TCR sequence is therefore only an early hypothesis. It must be tested for expression, peptide–MHC binding, functional T-cell activation, recognition of tumour cells, and broad cross-reactivity against normal human proteins and cells.

85.4 What should be concluded?

The main advance is computational and experimental-tool oriented, rather than a new treatment. A model that can both predict TCR recognition and generate target-conditioned candidate receptors could make TCR discovery faster and more systematic.

However, the underlying biological constraints remain unchanged:

Generated TCR sequence
≠ functional receptor in a T cell
Functional recognition in vitro
≠ tumour selectivity in a patient
Tumour selectivity
≠ safe and clinically useful TCR–T therapy

Because this is a *preprint*, its performance and practical value need independent validation. The most useful near-term role is likely to be prioritising better TCR candidates for rigorous laboratory testing, rather than replacing experimental discovery or clinical safety assessment.

86 [26-09-18] Complete regression of hepatoblastoma after GPC3 CAR-T therapy with IL-15 and IL-21

(*Complete Regression of Hepatoblastoma after Interleukin-15- and Interleukin-21-Coexpressing CAR T-Cell Therapy*, <https://www.nejm.org/doi/full/10.1056/NEJMc2605958>)

This report describes an important but very early clinical result: complete regression of chemotherapy-resistant metastatic *hepatoblastoma* in one child after treatment with engineered CAR-T cells.

Hepatoblastoma is the most common primary liver cancer in young children. Many cases can be treated successfully with chemotherapy and surgery, but relapsed or treatment-resistant metastatic disease remains a major unmet need.

The CAR-T cells targeted *glypican-3*, or *GPC3*, a cell-surface protein expressed by many hepatoblastomas and hepatocellular carcinomas. GPC3 therefore acts as the tumour-cell address recognised by the CAR.

GPC3-positive hepatoblastoma cell
→ GPC3 CAR binds tumour-cell surface
CAR signalling activates T cell
→ cytotoxic molecules and cytokines are released
Tumour cell may be killed

The product was unusual because the CAR-T cells were also engineered to produce *interleukin-15*, or *IL-15*, and *interleukin-21*, or *IL-21*. These cytokines are intended to improve the internal fitness of the transferred T cells.

IL-15 can support T-cell survival, proliferation, and persistence. IL-21 can help sustain cytotoxic T-cell function and may support a less terminally exhausted T-cell state. Together, they aim to address a major limitation of CAR-T therapy in solid tumours: CAR-T cells may reach the tumour but fail to expand, persist, or remain functional in its hostile microenvironment.

Conventional CAR-T cell
→ recognises tumour antigen
but may become dysfunctional or disappear
CAR-T cell producing IL-15 and IL-21
→ antigen recognition
+ local T-cell survival and functional support
→ potentially more durable anti-tumour activity

The reported patient was a 3-year-old child with GPC3-positive hepatoblastoma that had progressed after several treatment lines and had metastatic disease in the lung. After two CAR-T-cell infusions, eight weeks apart, the patient had a partial response followed by complete regression of detectable metastatic disease. The report states that remission was maintained at one year after treatment.

The product also contained an *inducible caspase-9 safety switch*. If severe toxicity occurred, a separately administered activating drug could trigger caspase-9-mediated death of the engineered CAR-T cells.

Severe CAR-T toxicity occurs
→ safety-switch drug is administered
→ inducible caspase-9 is activated
→ therapeutic CAR-T cells are eliminated
→ immune activity can be reduced

No cytokine-release syndrome, dose-limiting toxicity, or major treatment-related safety signal was reported in this individual case. This is encouraging, especially because CAR-T therapy against solid tumours often causes concerns about cytokine toxicity, limited tumour access, antigen heterogeneity, and *on-target, off-tumour* injury.

However, the correct interpretation is cautious:

Complete response in one heavily pre-treated patient
→ strong proof that this therapeutic strategy can work
But:
One case
≠ established efficacy, safety, or durable benefit
across patients with hepatoblastoma

The result supports further study of GPC3-directed, cytokine-supported CAR-T cells in GPC3-positive paediatric solid tumours. The central remaining questions are whether responses can be reproduced, how long they last, whether all tumour sites are reached, whether GPC3-negative tumour cells escape, and whether the favourable safety profile persists in larger groups of patients.

87 [26-09-18] Assessing immunogenicity risk for peptide drugs with non-natural amino acids

(*Beyond natural amino acids: Extending immunogenicity risk assessment to non-canonical peptide drugs through chemical feature encoding*, <https://www.biorxiv.org/content/10.64898/2026.05.22.727138v1.full.pdf>)

This preprint addresses a practical limitation in computational immunogenicity assessment for peptide medicines.

Peptide drugs are often modified with *non-canonical amino acids*: building blocks beyond the approximately 20 standard amino acids used in normal human proteins. These modifications can improve stability, half-life, target binding, permeability, or resistance to enzymatic degradation. However, they can also change how the immune system sees the peptide.

A key immunogenicity question is whether a peptide can bind *MHC class II* molecules and be presented to CD4 T cells:

```
Peptide drug or peptide-derived fragment
→ uptake and processing by antigen-presenting cell
→ peptide binds MHC class II
→ peptide-MHC complex is displayed
→ CD4 T cell may recognise it
→ unwanted immune response may develop
```

Many existing prediction models are trained on natural peptide sequences. They represent each residue as one of the standard amino acids. This does not work well for a peptide containing a chemically modified or entirely non-natural residue, because the model has no meaningful way to represent its structure.

The paper proposes encoding each amino-acid position using *chemical features*, or molecular fingerprints, rather than treating each residue only as a letter in a biological sequence.

```
Conventional sequence encoding
→ amino acid represented mainly as a sequence symbol
Chemical feature encoding
→ residue represented by chemical properties
such as atoms, bonds, functional groups, and local structure
```

The most useful approach was *residue-level* chemical encoding. This preserves both the chemical identity of a modified residue and its position in the peptide. Position matters because MHC class II binding depends strongly on which residues occupy particular binding-pocket positions.

The authors showed that this approach performed comparably to conventional representations for natural peptides, while extending prediction to modified peptides. A citrullination case study illustrated why this matters: changing one residue can substantially alter MHC binding when it occurs at a key peptide-MHC contact position.

```
Chemical modification at non-critical peptide position
→ may have limited effect on MHC binding
Chemical modification at MHC-binding anchor position
→ may substantially alter presentation
and potential immunogenicity risk
```

The broader lesson is that peptide immunogenicity is not determined only by amino-acid sequence. For modified peptide medicines, the relevant input is the *chemical structure* of each building block, its position in the peptide, processing behaviour, MHC binding, and eventual T-cell recognition.

This is a useful early-discovery tool rather than a complete immunogenicity solution. Predicted MHC binding does not prove that a peptide will be processed, presented in patients, recognised by T cells, or cause clinically meaningful anti-drug immunity. The work is also a preprint, so it should be viewed as a promising methodological advance requiring independent validation.

88 [26-09-18] Using circulating tumour DNA to adapt adjuvant therapy in colon cancer

(Circulating tumor DNA-guided de-escalation or escalation of adjuvant therapy in high-risk stage II and stage III colon cancer: the phase 2 PEGASUS trial, <https://www.nature.com/articles/s43018-026-01237-9>)

This Phase II trial asks whether *circulating tumour DNA*, or *ctDNA*, can help decide how much chemotherapy a patient needs after surgery for high-risk stage II or stage III colon cancer.

After apparently curative surgery, no tumour may be visible on scans, but microscopic residual disease can remain. *Adjuvant therapy* aims to eliminate this residual disease and reduce recurrence risk. The difficulty is that standard clinicopathological features do not identify this risk perfectly: some patients receive substantial chemotherapy without needing it, while others recur despite treatment.

ctDNA is tumour-derived DNA fragments released into the bloodstream. A personalised ctDNA test looks for mutations identified in the patient's resected tumour.

```
Tumour is removed surgically
→ blood sample is tested for tumour-specific DNA
ctDNA detected after surgery
→ molecular evidence consistent with residual disease
and higher recurrence risk
ctDNA not detected
→ lower observed recurrence risk,
but not proof that no residual tumour cells remain
```

The PEGASUS strategy adapted treatment dynamically. Patients with detectable ctDNA initially received oxaliplatin-based chemotherapy, *CAPOX*. Those who remained ctDNA-positive could escalate to *FOLFIRI*, a different chemotherapy regimen. Patients without detectable ctDNA received a less intensive approach based on capecitabine alone, avoiding oxaliplatin exposure where possible.

```
ctDNA-positive
→ higher-risk molecular-residual-disease group
→ CAPOX, with possible escalation if ctDNA persists
ctDNA-negative
→ de-escalated treatment
→ reduced exposure to oxaliplatin-associated toxicity
```

The study confirmed that postoperative ctDNA status is strongly *prognostic*. Three-year disease-free survival was reported as approximately ****58.4%**** in ctDNA-positive patients and ****82.8%**** in ctDNA-negative patients.

The de-escalation strategy substantially reduced use of oxaliplatin-based therapy and reduced severe treatment-related adverse events. This matters because oxaliplatin can cause cumulative peripheral neuropathy: numbness, tingling, pain, or impaired sensation that can persist after treatment.

The escalation findings were also encouraging but preliminary. Some patients whose ctDNA remained detectable after CAPOX became ctDNA-negative after FOLFIRI and had not recurred at the reported analysis. This suggests that serial ctDNA may identify patients for whom standard adjuvant therapy has not eliminated molecular residual disease.

However, the trial did not meet its primary endpoint and was not definitive evidence that ctDNA-guided adaptation improves outcomes compared with standard management. In particular, ctDNA-negative status does not guarantee cure, and disappearance of ctDNA is a promising molecular response rather than proven survival benefit on its own.

ctDNA status
→ powerful prognostic biomarker
ctDNA-guided treatment change
≠ automatically proven to improve survival
The key clinical question:
can this strategy reduce unnecessary toxicity
while preserving or improving long-term cure rates?

The broader importance is that ctDNA may allow adjuvant therapy to become more personalised over time. Rather than using stage alone, clinicians may increasingly combine pathological risk, tumour biology, and repeated molecular-residual-disease measurements to decide who can safely receive less treatment and who may need more effective additional therapy.

89 [26-09-18] VES001: inhibiting sortilin to raise progranulin in frontotemporal dementia

(Preclinical and clinical investigation of VES001, an oral small molecule sortilin receptor inhibitor for the treatment of frontotemporal dementia: a first-in-human randomized placebo-controlled phase 1 trial, <https://www.nature.com/articles/s41467-026-77638-2>)

This study describes VES001, an oral, brain-penetrant small molecule designed for a genetically defined form of frontotemporal dementia, or *FTD-GRN*.

Some people with FTD carry one non-functional copy of the *GRN* gene. This causes *progranulin* levels to be too low. Progranulin is important for normal neuronal and lysosomal function; reduced progranulin can contribute to neurodegeneration, inflammation, and accumulation of abnormal proteins.

One defective *GRN* gene copy
→ reduced progranulin production
Reduced progranulin
→ impaired neuronal and lysosomal homeostasis
→ increased risk of FTD-GRN

Sortilin is a cell-surface receptor expressed by cells including neurons and microglia. One of its normal functions is to bind extracellular progranulin, bring it into the cell by endocytosis, and direct it towards intracellular degradation pathways.

Extracellular progranulin binds sortilin
→ sortilin-progranulin complex is internalised
→ progranulin is delivered towards lysosomal degradation
→ extracellular progranulin concentration decreases

In people who already produce too little progranulin because of a *GRN* mutation, this sortilin-mediated clearance can further reduce the amount available in the brain. VES001 binds sortilin and blocks its interaction with progranulin.

VES001 binds sortilin
→ less sortilin-mediated progranulin uptake
→ less intracellular progranulin degradation
→ higher extracellular progranulin in blood
and central nervous system

Thus, VES001 does **not** replace the missing *GRN* gene or directly make cells produce more progranulin. It aims to preserve the progranulin that the patient can still produce by reducing its receptor-mediated clearance.

The Phase 1 study was a randomised, placebo-controlled trial in healthy volunteers. It primarily assessed safety, pharmacokinetics, and *proof of mechanism*. The reported increase in progranulin in plasma and cerebrospinal fluid is evidence that oral VES001 reached the relevant compartment and changed the intended biology.

Oral dose
→ drug exposure in blood and brain
→ sortilin inhibition
→ increased progranulin in cerebrospinal fluid
→ proof of mechanism

This is encouraging, particularly because reaching the central nervous system is a major challenge for many neurological medicines. However, increased progranulin is a pharmacodynamic biomarker, not yet evidence that VES001 slows cognitive, behavioural, or functional decline in people with symptomatic FTD-GRN.

The crucial next question is whether sustained restoration of progranulin towards a more normal range can produce meaningful clinical benefit with acceptable long-term safety in the intended patient population.

90 [26-09-18] TGF- β 1-driven mesothelial-to-mesenchymal transition in mesothelioma

(*Single-cell transcriptomics reveals TGF- β 1-associated mesothelial-to-mesenchymal transition promoting tumor progression and immune suppression in mesothelioma*, <https://www.nature.com/articles/s41698-026-01704-7>)

This study uses *single-cell transcriptomics* to examine how malignant mesothelioma cells and their surrounding tissue environment differ across individual cells. Mesothelioma is an aggressive cancer arising from *mesothelial cells*, which normally form a thin protective lining around organs, including the lungs and abdominal organs.

The central finding is that a subset of tumour-associated mesothelial cells shows a *mesothelial-to-mesenchymal transition*, sometimes abbreviated *MMT*. This is related to the epithelial-to-mesenchymal transition, or EMT, discussed more broadly in cancer biology.

Mesothelial-like cell state
→ organised tissue-lining and cell-cell-contact features
TGF- β 1-associated transition
→ more mesenchymal, invasive, and matrix-remodelling features
Mesenchymal-like tumour state
→ tumour progression and immune suppression may be supported

TGF- β 1, or transforming growth factor beta 1, is a cytokine with strongly context-dependent effects. In normal tissue, it can contribute to wound healing, control of cell growth, extracellular-matrix production, and immune regulation. In established cancer, however, persistent TGF- β signalling can support fibrosis, invasion, tumour-cell plasticity, exclusion of T cells, and suppression of effective anti-tumour immunity.

TGF- β 1 signalling
→ mesothelial-to-mesenchymal cell-state change
→ extracellular-matrix remodelling and fibrotic tumour environment
→ more difficult T-cell entry and increased immune suppression
→ tumour progression may be supported

The *mesenchymal* state does not mean that a cell has become a completely different cell type. It means that the cell has changed its gene-expression programme and behaviour. Such cells may become more mobile, invasive, resistant to stress, and able to interact with fibroblasts, immune cells, and extracellular matrix in tumour-supportive ways.

Single-cell data are useful here because bulk tumour measurements can hide this variation. A high TGF- β signal in a whole tumour sample cannot show which cells produce the signal, which cells respond to it, or whether a particular mesenchymal-like cell population is present. Single-cell analysis can identify these distinct cellular states and their association with immune suppression.

The therapeutic implication is a hypothesis rather than an established treatment:

TGF- β 1-associated mesenchymal programme
→ possible biomarker of aggressive,
immune-suppressive mesothelioma
TGF- β pathway intervention
+ immunotherapy or another treatment
→ possible strategy to make the tumour
microenvironment more permissive to immune attack

However, TGF- β has important normal roles in tissue repair and immune regulation. Broad inhibition could therefore cause toxicity, and the single-cell associations do not by themselves prove that TGF- β 1 causes progression in every patient. The key next question is whether a defined TGF- β -associated tumour subgroup can benefit from a tolerable, biomarker-guided treatment strategy.

91 [26-09-18] Histone methylation can determine whether tumours are visible to immunity

(*Antagonistic roles of histone H3 lysine methyltransferases dictate tumor immune surveillance*, <https://www.nature.com/articles/s41467-026-76657-3>)

This paper links *epigenetic regulation* to tumour immune surveillance. Its central idea is that cancer cells can control whether they appear immunologically dangerous by changing which regions of their DNA are kept silent or accessible.

Recall that DNA is packaged into *chromatin*. Chemical marks on histones can influence whether genes and other DNA regions are accessible for transcription. Histone methyltransferases add methyl groups to selected amino acids on histones; these marks can help establish more active or more repressed chromatin states.

The study focuses on two enzymes that modify lysine 36 on histone H3:

NSD2
→ mainly adds H3K36 dimethylation, H3K36me2
SETD2
→ adds H3K36 trimethylation, H3K36me3

Although both act on the same histone position, they can have *antagonistic* effects on tumour immune surveillance. This is an important reminder that histone methylation is not one simple “on” or “off” signal: the exact amino acid, number of methyl groups, chromatin location, and interacting proteins all matter.

91.1 How can chromatin affect anti-tumour immunity?

Cancer-cell DNA contains many repetitive sequences, including endogenous retrovirus-like elements. These are normally kept silent. If they become transcribed, they can produce double-stranded RNA, or *dsRNA*, which resembles RNA generated during some viral infections.

Normally silent repetitive DNA becomes transcribed
→ double-stranded RNA accumulates
Cell detects dsRNA as a viral-like danger signal
→ interferon and inflammatory pathways activate
→ tumour becomes more visible to immune surveillance

This is called *viral mimicry*. The tumour cell is not actually infected with a virus. Instead, loss of epigenetic repression makes the cell produce RNA signals that trigger an antiviral-like innate immune response.

The paper suggests that reducing NSD2-associated H3K36me2, including through an H3K36M histone mutation in experimental models, can derepress these repetitive elements. The resulting dsRNA activates immune sensing and can promote immune-mediated tumour control.

91.2 Why is SETD2 important?

SETD2 is often described as a tumour-suppressor gene because it contributes to genome stability, transcriptional regulation, and DNA-repair-associated biology. However, the study shows that SETD2 is also required for the immune-activating effect caused by NSD2 attenuation.

This is the apparently paradoxical result:

Reduced NSD2-associated H3K36me2
+
remaining SETD2-associated H3K36me3 activity
→ altered chromatin balance
Altered chromatin balance
→ reduced EZH2-associated repression
at repetitive elements
Repetitive-element transcription
→ dsRNA sensing and viral mimicry
→ enhanced tumour immune surveillance

EZH2 is another histone methyltransferase. It adds the generally repressive mark H3K27me3. In this context, the balance between NSD2, SETD2, and EZH2 helps determine whether repetitive DNA remains silent or generates immune-stimulating RNA.

91.3 Why could this matter therapeutically?

This work provides a possible route to make some tumours more immunogenic. Rather than targeting a tumour antigen directly, an epigenetic intervention might make cancer cells generate an internal viral-like danger signal, increase interferon-associated biology, and become more susceptible to immune attack.

Tumour with weak immune surveillance
→ inhibit or reduce an appropriate epigenetic brake
→ viral mimicry and inflammatory signalling
→ improved conditions for T-cell-mediated control
→ possible combination opportunity with checkpoint blockade

However, this is primarily mechanistic and preclinical work. Histone methyltransferases regulate many genes in tumour and normal cells, so broad inhibition could cause substantial unintended effects. Viral mimicry may also not help if a tumour lacks antigen presentation, excludes T cells, or has other strong immune-evasion mechanisms.

Epigenetic immune activation
≠ proven patient benefit
The key future question is whether modulation of the
NSD2-SETD2-EZH2 chromatin balance can make selected tumours
more responsive to immunotherapy at an acceptable safety margin.

92 [26-09-21] GFH375: directly inhibiting KRAS G12D in advanced solid tumours

(Oral KRAS G12D inhibitor GFH375 for previously treated advanced solid tumors with KRAS^{G12D} mutations: a phase 1 trial, <https://www.nature.com/articles/s41591-026-04559-4>)

This Phase 1 trial evaluates GFH375, an oral small-molecule inhibitor of the cancer-driving mutation KRAS G12D. KRAS G12D occurs particularly often in pancreatic ductal adenocarcinoma, but also in colorectal cancer, non-small-cell lung cancer, and other solid tumours.

KRAS is a small signalling protein that normally acts as a molecular switch. It cycles between an inactive GDP-bound state and an active GTP-bound state. In response to normal growth-factor signals, active KRAS can stimulate pathways including RAF–MEK–ERK and PI3K–AKT, promoting controlled cell growth and survival.

Normal growth-factor signal
→ KRAS transiently switches "on"
KRAS signalling
→ controlled proliferation and survival
Signal ends
→ KRAS returns to "off" state

92.1 What does the G12D mutation do?

The G12D mutation replaces glycine with aspartate at amino-acid position 12 of KRAS. This interferes with normal KRAS inactivation, so a larger fraction of KRAS remains GTP-bound and active.

KRAS G12D mutation
→ impaired switching of KRAS back to inactive state
Persistently active KRAS signalling
→ RAF–MEK–ERK and other growth pathways remain active
→ tumour-cell proliferation and survival are supported

KRAS was historically considered difficult to drug because it binds GTP and GDP very tightly and has a relatively smooth protein surface with few obvious conventional drug-binding pockets. The clinical success of KRAS G12C inhibitors showed that direct KRAS inhibition is possible, but G12D lacks the cysteine residue exploited by covalent G12C drugs. It therefore requires a different drug-design strategy.

92.2 How does GFH375 work?

GFH375 is designed to bind selectively and *non-covalently* to mutant KRAS G12D. Rather than permanently attaching to a cysteine as many G12C inhibitors do, it binds through reversible molecular interactions with a pocket in KRAS.

An important feature is that GFH375 can inhibit both the GDP-bound “off” state and the GTP-bound “on” state of KRAS G12D. This matters because tumour cells continuously cycle some KRAS molecules between these states, and the GTP-bound form is the directly signalling-active population.

```
GFH375 binds KRAS G12D
→ interferes with productive KRAS interactions
with downstream signalling proteins
Reduced KRAS G12D signalling
→ less RAF-MEK-ERK pathway output
+ reduced tumour-cell growth and survival signalling
In susceptible tumour cells
→ tumour shrinkage may occur
```

Thus, GFH375 does not kill cancer cells through non-specific cytotoxicity, like conventional chemotherapy. It aims to remove a specific oncogenic dependency: continued signalling from mutant KRAS G12D.

92.3 What did the Phase 1 trial show?

Phase 1 trials primarily establish dose, pharmacokinetics, safety, and whether the drug reaches concentrations sufficient to affect its intended target. This study also reported preliminary anti-tumour activity in patients with previously treated advanced KRAS G12D-mutant cancers, including responses in pancreatic and lung cancer.

This is significant because KRAS G12D-mutant pancreatic cancer has historically had few effective targeted-treatment options. Objective responses provide early clinical evidence that direct inhibition of KRAS G12D can produce meaningful tumour regression in at least some cancers dependent on this mutation.

However, response rates in an early, non-randomised trial should be interpreted cautiously. They can vary by tumour type, prior therapy, co-occurring mutations, dose, and the definition of the efficacy-evaluable population.

92.4 Why will resistance still be expected?

A KRAS G12D mutation is important, but it is not the only biological feature of a tumour. Cancer cells can acquire resistance through additional KRAS alterations, activation of parallel signalling routes, reactivation of downstream pathways, or changes that reduce dependence on KRAS itself.

```
Initial KRAS G12D dependence
→ response to GFH375 may occur
Selective pressure from treatment
→ resistant tumour-cell populations may expand
Clinical implication
→ rational combination strategies will probably be needed
```

Possible combinations could target upstream KRAS regulators, downstream RAF-MEK-ERK signalling, PI3K-associated signalling, or tumour-type-specific vulnerabilities. Such combinations must be developed carefully because more complete pathway suppression can also increase toxicity in normal tissues.

92.5 What should be concluded?

GFH375 is an example of increasingly precise cancer pharmacology: a small molecule designed to inhibit a particular mutant form of a previously difficult drug target. Its ability to address both active and inactive KRAS G12D is intended to achieve more complete suppression of oncogenic KRAS signalling than approaches restricted to one nucleotide state.

```
KRAS G12D mutation
→ plausible predictive biomarker for GFH375 activity
Early tumour responses
→ clinical proof that KRAS G12D is druggable
But:
Phase 1 activity
≠ established long-term benefit, optimal dose,
comparative efficacy, or durable control across tumour types
```

The key next questions are which KRAS G12D-mutant tumours respond most reliably, how durable those responses are, which resistance mechanisms emerge, and whether combinations can extend benefit without narrowing the therapeutic window.

93 [26-09-21] CDK6 and STAT5B cooperate during initiation of ALK-positive lymphoma

(*Ablation of CDK6 and STAT5B rewires T cell developmental dynamics and prevents ALK+ lymphoma formation*, <https://www.nature.com/articles/s41467-026-77466-4>)

This study examines how *ALK-positive anaplastic large-cell lymphoma*, or ALK+ ALCL, can arise from developing T cells. Its central finding is that two proteins, *CDK6* and *STAT5B*, cooperate with oncogenic ALK signalling during early lymphoma formation. Removing both genes prevented lymphoma formation in the authors' mouse model.

ALK+ ALCL is a T-cell lymphoma often driven by a fusion protein such as *NPM-ALK*. A fusion joins parts of two genes that are normally separate. Here, the fusion causes the ALK kinase to remain abnormally active, sending persistent growth and survival signals without the usual external control.

```
NPM-ALK fusion
→ constitutively active ALK signalling
Persistent signalling
→ abnormal proliferation, survival,
and altered T-cell developmental programmes
Additional cooperating factors
→ lymphoma may emerge
```

93.1 What do CDK6 and STAT5B do?

CDK6 is a *cyclin-dependent kinase* that helps control progression through the cell cycle. In particular, it contributes to the transition from the G1 phase, when a cell grows and assesses conditions, towards DNA replication in S phase. CDK6 can also affect gene expression independently of its classical cell-cycle role.

STAT5B is a transcription factor. Cytokine and growth-factor signals can activate STAT5B, causing it to enter the nucleus and alter expression of genes involved in proliferation, survival, differentiation, and immune-cell function.

```
CDK6
→ cell-cycle progression
+ gene-regulatory effects
STAT5B
→ transmits cytokine-associated signals
to the nucleus
→ survival and differentiation gene expression
```

In this model, NPM–ALK does not act alone. It appears to exploit CDK6- and STAT5B-dependent programmes in immature T cells, allowing abnormal cells to survive and proliferate at a vulnerable developmental stage.

93.2 Why does T-cell development matter?

T cells develop in the thymus through defined intermediate stages. The study identifies immature *DN2/3 thymocytes* as a key cell population from which ALK-driven lymphoma can emerge in the model.

“DN” means *double negative*: these early thymocytes do not yet express the mature T-cell surface markers CD4 or CD8. At this stage, cells are actively proliferating and undergoing major developmental and gene-regulatory changes.

```
Immature DN2/3 thymocyte
+ oncogenic NPM-ALK signalling
+ CDK6- and STAT5B-dependent programmes
→ abnormal developmental trajectory
→ lymphoma initiation may occur
```

Single-cell analysis showed that ALK signalling creates a tumour-associated developmental trajectory rather than simply making a mature normal T cell divide faster. In other words, the cancer is linked to distortion of a particular normal developmental transition.

93.3 What happened when CDK6 and STAT5B were removed?

Deleting either *Cdk6* or *Stat5b* alone had limited or incomplete effects in the model. However, simultaneous deletion of both prevented NPM–ALK-driven lymphoma formation.

The combined loss altered the developmental dynamics of the susceptible immature T-cell population. The cells no longer followed the abnormal trajectory required for efficient tumour initiation.

```
NPM-ALK
+ CDK6
+ STAT5B
→ tumour-permissive immature T-cell state
→ ALK+ lymphoma formation
NPM-ALK
+ loss of CDK6 and STAT5B
→ developmental programme is rewired
→ tumour initiation is prevented in the model
```

This is an example of functional cooperation between oncogenic signalling, cell-cycle control, and transcriptional regulation. CDK6 and STAT5B have partially overlapping but non-identical roles; together, their loss crosses a threshold that the lymphoma-initiating cells cannot tolerate.

93.4 What does “synthetic lethality” mean here?

The study also found evidence that combined CDK6 and STAT5B disruption can be particularly harmful to human ALCL cells. This resembles *synthetic lethality*, discussed earlier for BRCA deficiency and PARP inhibition.

```
Tumour cell tolerates:
CDK6 loss alone
or STAT5B loss alone
But:
CDK6 loss
+
STAT5B loss
→ critical survival and proliferation programmes fail
→ selective tumour-cell vulnerability may emerge
```

The term should be used cautiously here. Genetic deletion in an experimental system is not equivalent to a clinically achievable drug combination. CDK6 can be inhibited pharmacologically by existing CDK4/6 inhibitors, but directly and selectively inhibiting STAT5B remains much more difficult.

93.5 What should be concluded?

The main contribution is mechanistic. The paper suggests that ALK+ lymphoma initiation depends not only on oncogenic ALK activity, but also on CDK6–STAT5B cooperation within a specific immature T-cell developmental context.

It provides a rationale for exploring combinations that suppress ALK signalling together with CDK6- and/or STAT5B-associated pathways. It also identifies a CDK6/STAT5B-dependent gene-expression signature that may have prognostic value in ALCL.

```
Genetic prevention of lymphoma in a mouse model
≠ established treatment in patients
CDK6/STAT5B dependency
≠ proof that available inhibitors can safely
reproduce the effect
The clinical challenge:
identify druggable ways to disrupt this cooperation
while preserving sufficient normal T-cell development
and immune function
```

94 [26-09-21] NCOA6 loss creates EGFR dependency in aggressive prostate cancer

(Loss of *NCOA6* protein causes aggressive prostate cancer with *EGFR* overexpression and dependency, <https://www.nature.com/articles/s41467-026-77701-y>)

This study identifies *NCOA6* as a tumour-suppressive regulator in prostate cancer. Loss of *NCOA6* can increase expression of *EGFR*, or epidermal growth-factor receptor, creating a tumour state that is more aggressive but also potentially dependent on EGFR signalling.

EGFR is a receptor tyrosine kinase at the cell surface. When activated by growth factors, it triggers pathways including RAS–RAF–MEK–ERK and PI3K–AKT, which can promote cell proliferation, survival, and migration.

```
Growth factor binds EGFR
→ EGFR signalling is activated
RAS-MAPK and PI3K-AKT pathways
→ cell growth and survival signals
Excessive EGFR activity
→ can support tumour progression
```

94.1 What does NCOA6 normally do?

NCOA6 is a transcriptional co-regulator. It does not simply switch one gene on or off; instead, it helps organise protein complexes at particular DNA regulatory regions and thereby influences chromatin state and gene expression.

In normal prostate cells, NCOA6 forms a complex with the transcription factor NFY near an *EGFR* regulatory region. This limits recruitment of *p300*, a histone acetyltransferase that adds the chromatin mark H3K27ac. H3K27ac is generally associated with active enhancers and promoters. Thus, NCOA6 helps keep the *EGFR* gene under appropriate transcriptional control.

```
NCOA6-NFY complex at EGFR regulatory DNA
→ limits p300 recruitment
→ limits H3K27 acetylation
→ restrains EGFR transcription
→ restrained EGFR signalling
```

94.2 What happens when NCOA6 is lost?

Without NCOA6, the regulatory brake is removed. p300 is recruited more effectively, H3K27 acetylation increases, and the *EGFR* gene becomes more transcriptionally active.

```
NCOA6 loss
→ increased p300 recruitment at EGFR regulatory DNA
→ increased H3K27ac
→ increased EGFR expression
→ stronger EGFR-driven growth and survival signalling
```

This is an epigenetic mechanism of oncogene activation. The tumour does not necessarily need an activating mutation in *EGFR*; it can increase EGFR pathway output by losing a protein that normally restrains *EGFR* transcription.

94.3 Why does PTEN matter?

The study found that NCOA6 loss cooperates with partial loss of *PTEN* in mouse models. PTEN is a tumour suppressor that normally restrains PI3K–AKT signalling.

```
NCOA6 loss
→ increased EGFR signalling
PTEN loss
→ reduced restraint of PI3K-AKT signalling
Both alterations
→ stronger growth and survival advantage
→ invasive prostate cancer may develop
```

This illustrates a general principle in cancer: aggressive disease often emerges through cooperating alterations rather than one mutation acting in isolation.

94.4 What does “EGFR dependency” mean?

An *EGFR dependency*, sometimes called EGFR addiction, means that NCOA6-deficient tumour cells become unusually reliant on EGFR signalling to maintain their malignant behaviour.

This creates a potential therapeutic vulnerability. If a tumour has low NCOA6 and high EGFR, EGFR-targeted treatment may have greater activity than in an unselected prostate-cancer population.

NCOA6-deficient prostate tumour
→ high EGFR expression
→ dependence on EGFR pathway activity
EGFR inhibition
→ preferential disruption of a key
tumour-cell growth and survival programme

This could help explain why EGFR-targeted drugs have not generally been successful in prostate cancer: EGFR expression alone may be insufficient, whereas the specific *NCOA6-low* / *EGFR-high* molecular state may define a more responsive subgroup.

94.5 What should be concluded?

The paper identifies an epigenetic route through which prostate cancer can activate EGFR signalling. It also suggests a biomarker-guided therapeutic hypothesis:

Low NCOA6
+
high EGFR
→ aggressive prostate-cancer subgroup
+ possible vulnerability to EGFR-targeted treatment

However, this is not yet evidence that EGFR inhibitors improve outcomes in NCOA6-deficient patients. The association with aggressive disease, the dependency observed in models, and the proposed treatment sensitivity must be validated prospectively in clinically relevant patient groups.

EGFR dependency in laboratory models
≠ proven clinical benefit from EGFR inhibition
The next step:
test whether NCOA6 loss, ideally combined with EGFR expression
and other genomic features such as PTEN status,
can select patients for an effective treatment strategy

95 [26-09-21] Broadly tumour-reactive $\gamma\delta$ TCRs in multiple myeloma

(Identification of broadly tumour-reactive $\gamma\delta$ TCRs from multiple myeloma, <https://www.nature.com/articles/s41586-026-11055-9>)

This study identifies a population of $\gamma\delta$ T cells associated with better responses in multiple myeloma, and characterises the T-cell receptors, or *TCRs*, that allow these cells to recognise tumour cells.

This builds on the earlier discussion of conventional $\alpha\beta$ TCRs. Conventional T cells generally recognise a specific short peptide presented by a particular MHC molecule. $\gamma\delta$ T cells are different: they use a TCR made of gamma and delta chains, rather than alpha and beta chains, and can recognise cellular stress- or transformation-associated features in less conventional ways.

```
Conventional  $\alpha\beta$  TCR:  
specific peptide  
+  
specific MHC molecule  
→ T-cell recognition  
 $\gamma\delta$  TCR:  
can recognise altered cellular features  
with less dependence on conventional peptide presentation  
→ potentially broader tumour recognition
```

95.1 What did the study find?

The authors analysed bone-marrow immune cells from patients with multiple myeloma. They found that responders to treatment had increased numbers of cytotoxic, granzyme-B-positive $\gamma\delta$ T cells.

Granzyme B is a protease released by cytotoxic lymphocytes. Once delivered into a target cell, it helps trigger cell death. Thus, granzyme-B-positive $\gamma\delta$ T cells are cells with an activated tumour-killing programme.

```
Cytotoxic  $\gamma\delta$  T-cell population  
→ expresses granzyme B and other killing-associated genes  
Greater presence in myeloma bone marrow  
→ associated with better treatment response
```

The researchers then cloned TCRs from these cells and transferred them into experimental T cells. Several receptors enabled recognition and killing of multiple myeloma-cell samples, including cells from different patients. This is what “broadly tumour-reactive” means: the receptors were not restricted to recognising one private tumour mutation in one individual patient.

95.2 What do the receptors recognise?

The identified $\gamma\delta$ TCRs reacted with *HLA-C*, a class-I HLA molecule. HLA molecules are normally best known for displaying intracellular peptides to conventional CD8 T cells, but here *HLA-C* itself appears to form an important part of the recognition surface for the $\gamma\delta$ TCR.

Importantly, the receptors could react with multiple *HLA-C* variants. This provides a possible explanation for their broad activity across myeloma samples.

```
Multiple myeloma cell expressing compatible HLA-C  
→ tumour-reactive  $\gamma\delta$  TCR engages target  
T-cell activation  
→ granzyme and cytotoxic activity  
→ myeloma-cell killing
```

At first, this creates an apparent safety problem. *HLA-C* is not unique to myeloma cells; it is expressed by many normal cells. Why then do these $\gamma\delta$ T cells not attack normal tissues broadly?

95.3 How can normal cells be spared?

The paper identifies an inhibitory “brake” involving *KIR3DL1*. *KIR3DL1* is an inhibitory receptor expressed by some $\gamma\delta$ T cells and natural-killer cells. It can recognise the *Bw4* structural motif carried by certain *HLA-A* and *HLA-B* molecules.

```
Tumour-reactive  $\gamma\delta$  TCR
→ activating signal through HLA-C recognition
KIR3DL1 binds Bw4 on normal cell
→ inhibitory signal
Inhibitory signal outweighs activation
→ normal cell is spared
```

Thus, tumour selectivity does not appear to result simply from the TCR recognising a molecule found only on cancer cells. Instead, it depends on the balance of an activating signal from HLA-C and inhibitory signals associated with normal-cell HLA expression.

This resembles the broader principle of NK-cell biology: immune killing is often determined by the integration of activating and inhibitory receptors, rather than by one receptor acting alone.

95.4 Why could this matter therapeutically?

Multiple myeloma is a cancer of antibody-producing plasma cells that grows mainly in the bone marrow. Existing immunotherapies include BCMA-directed CAR-T cells, bispecific antibodies, and the BCMA-directed ADC belantamab mafodotin. However, patients can relapse through antigen loss, T-cell dysfunction, or other immune-evasion mechanisms.

A broadly reactive $\gamma\delta$ TCR could offer a different therapeutic approach:

```
Introduce tumour-reactive  $\gamma\delta$  TCR
into therapeutic T cells
→ recognise a shared myeloma-associated
HLA-C-related target feature
Potential advantages:
less dependence on one tumour-specific mutation
+ possible activity across multiple patients
+ potential complementarity with existing myeloma therapies
```

The TCRs may also act as biomarkers. Their presence in a patient's bone marrow could indicate a pre-existing immune environment more capable of contributing to treatment response.

95.5 What should be concluded?

The important advance is the identification of human $\gamma\delta$ TCRs that can recognise myeloma across multiple samples, together with a proposed mechanism for balancing tumour reactivity against normal-tissue protection.

However, this is not yet an established cell therapy. HLA expression differs between individuals, tumours can alter HLA expression, and the KIR3DL1–Bw4 inhibitory system will not necessarily provide the same protection in every patient.

```
Broad myeloma-cell recognition in experimental systems
≠ safe and effective cell therapy in patients
Tumour reactivity
+ inhibitory self-tolerance mechanism
→ promising therapeutic design principle
But:
broad normal-tissue cross-reactivity, HLA genetics,
tumour escape, persistence, and clinical safety
must be tested rigorously
```

96 [26-09-21] Canakinumab and emavusertib did not improve anaemia in lower-risk MDS

(*Canakinumab or emavusertib for anemia in lower-risk MDS: results of the CANFIRE and LUCAS phase 2, open-label, multicenter trials*, <https://www.nature.com/articles/s43856-026-01885-z>)

This paper reports two small Phase II trials testing whether suppressing inflammation can improve anaemia in *lower-risk myelodysplastic neoplasms*, or MDS.

MDS is a group of bone-marrow disorders in which blood cells are produced inefficiently and often abnormally. In lower-risk MDS, the most important clinical problem is frequently persistent anaemia, causing fatigue, breathlessness, and need for red-blood-cell transfusions.

Abnormal bone-marrow stem and progenitor cells
→ ineffective blood-cell production
Reduced mature red blood cells
→ anaemia and possible transfusion dependence

Inflammatory signalling is thought to contribute to ineffective blood formation in some MDS cases. The trials therefore tested two drugs acting at different points in an inflammatory pathway.

96.1 What were the two drugs?

Canakinumab is an antibody that neutralises *interleukin-1 β* , or IL-1 β . IL-1 β is an inflammatory cytokine that can influence bone-marrow cells and their surrounding microenvironment.

Emavusertib inhibits *IRAK4*, a signalling protein downstream of innate-immune receptors, including IL-1 receptors and Toll-like receptors. IRAK4 helps activate inflammatory transcriptional programmes, including NF- κ B-associated signalling.

Inflammatory receptor activation
→ IRAK4 signalling
→ inflammatory cytokines and altered marrow environment
→ ineffective erythropoiesis may be reinforced
Canakinumab
→ blocks IL-1 β
Emavusertib
→ inhibits IRAK4-associated signalling

The hypothesis was that reducing this inflammation might allow red-blood-cell production, or *erythropoiesis*, to recover.

96.2 What did the trials show?

The CANFIRE trial tested canakinumab and enrolled 11 patients; 10 were evaluable for the primary response assessment. The LUCAS trial tested emavusertib in 36 patients.

Neither trial produced a *haematological improvement-erythroid*, or HI-E, response. HI-E means a clinically meaningful improvement in red-blood-cell production, such as increased haemoglobin or reduced transfusion need.

```
Canakinumab alone
→ no HI-E responses among evaluable participants
Emavusertib alone
→ no HI-E responses
Therefore:
single-pathway anti-inflammatory treatment
≠ sufficient correction of MDS-associated anaemia
```

LUCAS was stopped early after a planned futility analysis, meaning that continuing was unlikely to achieve the predefined efficacy goal. CANFIRE stopped early because recruitment was inadequate.

96.3 Why is the negative result useful?

The result does not show that inflammation is irrelevant in MDS. Emavusertib produced evidence of biological modulation in the marrow. Rather, it shows that changing IL-1 β or IRAK4 signalling alone did not translate into enough restored erythropoiesis over the study period.

This fits the broader biology of MDS:

```
Inflammation contributes to marrow dysfunction
But:
MDS-associated anaemia also reflects
clonal genetic abnormalities
+ abnormal stem-cell differentiation
+ cell death and ineffective maturation
+ altered marrow microenvironment
Therefore:
blocking one inflammatory pathway
may not overcome the full disease state
```

96.4 What should be concluded?

CANFIRE and LUCAS do not support canakinumab or emavusertib monotherapy as effective treatments for anaemia in unselected lower-risk MDS.

The work remains informative because it tests a plausible mechanism directly in patients. Future approaches may need more selective biomarker-defined populations, longer or differently scheduled treatment, or combinations that address both inflammatory signalling and the underlying malignant haematopoietic clone.

```
Evidence of target-pathway modulation
≠ clinical improvement in anaemia
Biologically plausible rationale
≠ effective monotherapy
```

97 [26-09-21] MAP2-driven cell plasticity and sorafenib resistance in hepatocellular carcinoma

(*MAP2-mediated integrin dysregulation reprograms cancer cell plasticity in hepatocellular carcinoma*, <https://www.nature.com/articles/s41467-026-77831-3>)

This study identifies *MAP2*, or microtubule-associated protein 2, as a driver of aggressive cell-state change in hepatocellular carcinoma, or HCC. MAP2 was enriched in a CD133-positive subset of tumour cells with stem-like properties.

Cancer-cell *plasticity* means that cells can change their gene-expression programme and behaviour without acquiring a new DNA mutation. Here, MAP2 promoted a transition towards a more *mesenchymal* and stem-like state: cells became less dependent on stable cell-cell attachment and more capable of migration, tumour initiation, survival under stress, and drug resistance.

High MAP2 expression
→ actin cytoskeleton remodelling
→ YAP activation
→ dysregulated integrin expression
→ mesenchymal and stem-like cell state

Integrins are cell-surface receptors that connect a cell to the surrounding extracellular matrix and transmit mechanical and biochemical signals in both directions. The study found broad loss of integrin- α expression after MAP2-associated cytoskeletal and YAP changes. This altered how HCC cells attached to, sensed, and responded to their environment.

YAP is a transcriptional co-regulator that responds to mechanical and cytoskeletal signals. When active, it can enter the nucleus and promote gene programmes associated with growth, survival, regeneration, and cancer-cell plasticity.

Altered integrin-mediated adhesion
→ altered mechanical signalling
YAP-driven gene expression
→ increased migration, tumour initiation,
and resistance to cell death

The MAP2-high state also reduced susceptibility to *sorafenib*, a multikinase inhibitor used in HCC. This does not necessarily mean that MAP2 directly blocks sorafenib binding to its targets. Rather, the mesenchymal, stem-like programme may give cells alternative survival mechanisms, allowing them to tolerate the signalling inhibition imposed by treatment.

The authors explored *estramustine phosphate*, or EMP, as a way to inhibit MAP2-associated biology. In experimental models, EMP increased sensitivity to sorafenib.

MAP2-high HCC cell state
→ plasticity and reduced sorafenib sensitivity
MAP2 inhibition
+ sorafenib
→ possible suppression of both
cell-state plasticity and growth signalling

The main implication is that treatment resistance in HCC may arise not only from mutations in a drug target, but also from reversible changes in cell identity. However, this is mechanistic and preclinical work. MAP2 expression or integrin dysregulation would need validation as biomarkers, and the safety and clinical value of MAP2-directed combination treatment remain unknown.

98 [26-09-21] BoltzMol-1: AI-guided hit discovery for a difficult cancer target

(Can BoltzMol-1 find a starting point on a target that resisted 350,000 compounds?, <https://boltz.com/boltzmol-wrn>)

This case study evaluates *BoltzMol-1*, an AI-based small-molecule hit-discovery system, against *WRN* helicase. *WRN* is an attractive but difficult cancer target: tumours with *microsatellite instability*, or MSI, can become particularly dependent on *WRN* for survival.

```
MSI-high tumour
→ replication-associated DNA stress
WRN helicase
→ helps resolve problematic DNA structures
WRN inhibition
→ selective tumour-cell death may occur
```

The difficulty is that a previous conventional high-throughput screen of roughly 350,000 compounds did not yield a useful WRN starting point. This does not mean that WRN is impossible to inhibit; rather, the relevant binding site, protein conformational state, and chemical requirements are difficult to find through relatively limited physical compound libraries.

BoltzMol-1 uses predicted protein–ligand structures and binding-related scores to prioritise compounds from much larger purchasable or make-on-demand chemical spaces. Instead of experimentally testing hundreds of thousands of molecules, researchers test a small AI-selected set.

```
Large chemical space
→ AI predicts compatible protein-ligand interactions
→ ranks a small number of candidates
→ compounds are made or purchased
→ experimental binding and activity tests
```

For WRN, the important result is that experimentally confirmed *micromolar hits* were obtained after testing only a small number of AI-prioritised compounds. A micromolar hit is not a drug: it is an initial chemical starting point that binds or inhibits sufficiently to support medicinal-chemistry optimisation.

```
Initial micromolar hit
≠ drug candidate
Hit optimisation still requires:
higher potency
+ selectivity
+ cell activity
+ pharmacokinetics
+ safety and in vivo efficacy
```

How impressive is this? ****It is genuinely interesting if the experimental data are robust****, because finding a real hit on a target that defeated a large conventional screen is a meaningful prospective validation. The value is not that AI “solved” WRN, but that it may have reduced the number of compounds needing to be synthesised and tested by orders of magnitude.

However, this should still be interpreted as an early discovery result and as a company case study rather than independent peer-reviewed validation. A small hit rate across a limited set of targets does not yet establish that the platform works reliably across drug targets, especially those involving highly flexible proteins, cryptic pockets, difficult cell permeability, or complex biology.

The practical near-term value is likely to be improved *hit generation and prioritisation*, not autonomous drug discovery.

```
AI-selected, experimentally confirmed hit
→ potentially faster start to medicinal chemistry
But:
A starting point
≠ a selective cellular probe,
preclinical candidate, or medicine
```


If repeated prospectively across diverse targets, with strong hit quality and successful progression into lead optimisation, this type of system could make a substantial practical difference by lowering the cost and time needed to begin discovery programmes.

99 [26-09-21] CellTrap: using intracellular protein abundance for cell-selective drug activity

(*Cell type-selective targeting by heterobifunctional protein binders via in-cell enrichment*, <https://www.nature.com/articles/s41467-026-77460-w>)

This paper introduces *CellTrap*, a strategy for making a small-molecule drug more active in one cell type than another by exploiting differences in intracellular protein abundance.

The authors use a *heterobifunctional binder*: one end binds a highly abundant intracellular “presenter” protein, FKBP12, and the other binds the cancer-relevant target BRD4.

```
CellTrap molecule binds FKBP12
+
CellTrap molecule binds BRD4
→ drug becomes concentrated inside cells
with abundant FKBP12
→ stronger BRD4 inhibition in those cells
```

99.1 What is different from a molecular glue or PROTAC?

A molecular glue or PROTAC usually recruits an E3 ubiquitin ligase to a target protein, causing target ubiquitination and degradation. CellTrap does not primarily aim to degrade BRD4.

Instead, FKBP12 acts as an intracellular *reservoir* or anchor. Binding to abundant FKBP12 increases local retention of the compound inside a cell. The BRD4-binding end can then inhibit BRD4 more effectively.

```
PROTAC or molecular glue
→ target recruitment to E3 ligase
→ target degradation
CellTrap
→ recruitment to abundant intracellular protein
→ intracellular compound enrichment
→ stronger target inhibition
```

99.2 Why does this create cell-type selectivity?

FKBP12 is broadly expressed, but its abundance differs between cell types. Cells with higher FKBP12 can accumulate more of the CellTrap compound and consequently experience more BRD4 inhibition.

In co-culture experiments, the approach preferentially impaired PC3 prostate-cancer cells relative to non-transformed PNT1-A prostate cells, consistent with differential intracellular enrichment.

```
High FKBP12 abundance
→ greater CellTrap retention
→ greater effective BRD4 inhibition
→ stronger growth inhibition or cell killing
Lower FKBP12 abundance
→ less enrichment
→ reduced drug effect
```

BRD4 is a transcriptional regulator that helps maintain expression of genes supporting cell growth and survival. It is a plausible cancer target, but conventional BRD4 inhibition can affect normal cells as well. CellTrap aims to widen this therapeutic window without requiring BRD4 itself to be uniquely expressed by tumour cells.

99.3 What should be concluded?

The key idea is that cell selectivity can arise from the abundance of a separate intracellular protein, rather than only from selective binding to the disease target itself.

Target protein present in tumour and normal cells
≠ necessarily unavoidable normal-tissue toxicity
If a tumour-enriched intracellular presenter is available:
→ local drug concentration may differ
→ target inhibition may become more selective

This is an early chemical-biology result. For therapeutic use, the required protein-expression difference must be robust in real patient tumours versus essential normal tissues, and the strategy must retain selectivity at clinically achievable drug exposures.

100 [26-09-21] KRASTAC: using KRAS-driven uptake to degrade surface proteins in cancer cells

(A lipoprotein-biomimetic nanoplatform for oncogenic KRAS mutation-specific lysosome-targeting protein degradation, <https://www.nature.com/articles/s41467-026-77697-5>)

This study describes *KRASTAC*, a nanoparticle platform designed to exploit a metabolic feature of cancer cells with oncogenic *KRAS* mutations. Rather than directly inhibiting mutant *KRAS*, it uses *KRAS*-driven cellular uptake to preferentially deliver selected surface proteins to the lysosome for degradation.

KRAS-mutant cells often have increased *macropinocytosis*: a form of bulk uptake in which cells engulf extracellular fluid, proteins, and particles. This can help highly proliferative cancer cells obtain nutrients.

Oncogenic *KRAS* signalling
→ increased macropinocytosis
Lipoprotein-like KRASTAC nanoparticle
→ preferential uptake by *KRAS*-mutant cell
Target-bound surface protein
→ lysosomal delivery and degradation

KRASTAC consists of a lipid nanoparticle coated with apolipoprotein E, or APOE, making it resemble a natural lipoprotein particle. It is coupled to an antibody that binds a chosen surface protein, such as *EGFR* or *PD-L1*.

The antibody binds its target at the cancer-cell surface. The *KRAS*-mutant cell then takes up the whole antibody–nanoparticle–target complex through macropinocytosis. Once internalised, the complex is routed towards lysosomes, acidic compartments that break down proteins.

```
Antibody on KRASTAC binds EGFR or PD-L1
→ complex is internalised by macropinocytosis
Lysosomal trafficking
→ target surface protein is degraded
Less EGFR
→ less growth signalling
Less PD-L1
→ potentially less local suppression of T cells
```

This is related to targeted protein degradation, but differs from the molecular glues and PROTAC-like approaches discussed earlier. Molecular glues usually recruit an intracellular protein to an E3 ubiquitin ligase and the proteasome. KRASTAC instead acts on *cell-surface proteins* and uses antibody binding, nanoparticle uptake, and lysosomal degradation.

The particularly interesting idea is the two-layer selectivity:

```
Antibody
→ selects the protein to be degraded
KRAS-driven macropinocytosis
→ preferential uptake by KRAS-mutant cells
Together
→ potential tumour-selective degradation
```

In experimental KRAS-mutant models, KRASTAC-mediated EGFR or PD-L1 degradation reduced tumour growth and was associated with increased anti-tumour immune activity. However, this remains preclinical work. KRAS mutations do not guarantee equally high macropinocytosis in every tumour, and nanoparticle distribution, uptake in normal tissues, immune clearance, repeated-dose tolerability, and manufacturability remain major translational questions.

```
Selective degradation in experimental models
≠ a clinically selective medicine
The platform is promising because it links
a cancer-cell metabolic phenotype
to lysosomal degradation of a chosen target,
but its therapeutic window must be demonstrated in patients.
```

101 [26-09-21] How ATR–ATRIP is activated and how ATR inhibitors bind

(*Mechanistic insights into phosphorylation-driven activation and therapeutic inhibition of human ATR–ATRIP*, <https://www.nature.com/articles/s41594-026-01887-4>)

This structural-biology study explains in greater detail how *ATR* becomes activated during replication stress, and how clinical ATR inhibitors block it.

Recall that ATR is a central kinase in the DNA-damage response, particularly when DNA replication forks stall. ATR forms a stable complex with *ATRIP*, which helps localise and regulate it.

```
Replication stress
→ ATR–ATRIP is recruited and activated
ATR signalling
→ stabilises stalled replication forks
+ delays cell-cycle progression
+ promotes repair and cell survival
```

101.1 How is ATR activated?

ATR does not act as a simple permanently active enzyme. Its kinase activity is controlled by structural changes and phosphorylation. The study used cryo-electron microscopy, or *cryo-EM*, to visualise different structural states of the human ATR–ATRIP complex.

A key activator is *TopBP1*. TopBP1 interacts with ATR–ATRIP and promotes phosphorylation-driven changes that organise the kinase domain into a more active state. Activated ATR can then phosphorylate downstream proteins, including CHK1.

```
Replication-stress signalling
→ TopBP1 engages ATR-ATRIP
ATR phosphorylation and structural rearrangement
→ kinase activity increases
ATR phosphorylates CHK1 and other substrates
→ replication checkpoint and repair response
```

Thus, phosphorylation here is not only a downstream marker that ATR has acted. It also helps regulate the structural transition through which ATR itself becomes fully active.

101.2 How do ATR inhibitors work?

The authors also resolved structures of ATR bound to several clinical inhibitors, including *gartisertib*, *berzosertib*, *ceralasertib*, and *elimusertib*.

These compounds bind the *ATP-binding pocket* of the ATR kinase domain. Protein kinases need ATP to transfer phosphate groups to their substrates. An ATP-competitive inhibitor occupies this pocket and prevents productive ATP use.

```
Active ATR requires ATP
→ ATR phosphorylates CHK1 and other proteins
ATR inhibitor occupies ATP-binding pocket
→ substrate phosphorylation is blocked
→ replication-stress response is weakened
Tumour cell with high replication stress
→ DNA damage may become unmanageable
→ cell death can occur
```

The structural data show how these inhibitors make particular contacts within ATR that support potency and selectivity. This matters because many kinases have similar ATP-binding pockets; a useful inhibitor must block ATR sufficiently while avoiding excessive inhibition of related kinases and normal proliferating tissues.

101.3 What should be concluded?

This paper does not establish a new ATR inhibitor or new clinical treatment strategy. Its contribution is a more detailed structural explanation of both ATR activation and drug binding.

It builds directly on the earlier ATR discussion:

```
Cancer-associated replication stress
→ increased ATR dependence
ATR inhibition
→ therapeutic vulnerability in selected tumours
This study:
→ explains the molecular structures underlying
ATR activation and ATP-competitive inhibition
```

Such structures can guide future inhibitor design, including attempts to improve potency, selectivity, pharmacological properties, or activity against resistance-associated ATR changes. However, a better structural understanding does not remove the central clinical challenge: ATR also protects normal dividing cells, so useful treatment requires a sufficient tumour-selective therapeutic window.

102 [26-09-21] A genomic framework for predicting response to BRAF-directed therapy

(A hierarchical genomic framework associated with response to BRAF-directed therapy across solid tumors, <https://www.nature.com/articles/s41698-026-01691-9>)

This study asks why cancers with a *BRAF* alteration can respond very differently to BRAF-directed treatment. The central message is that a BRAF mutation should not be treated as a single, uniform biomarker: its functional class, the tumour type, and other genomic alterations all influence whether the cancer is truly dependent on BRAF signalling.

BRAF is a kinase in the RAS–RAF–MEK–ERK, or *MAPK*, pathway:

```
Growth-factor and RAS signalling
→ BRAF
→ MEK
→ ERK
→ cell proliferation and survival
```

The best-known alterations are *BRAF V600* mutations, especially V600E. These can activate BRAF as a monomer, often creating strong MAPK-pathway dependence. This explains the major benefit of BRAF inhibitors, usually combined with MEK inhibitors, in diseases such as BRAF V600-mutant melanoma.

However, non-V600 BRAF mutations are biologically diverse. Some signal as active BRAF dimers, while others have impaired intrinsic kinase activity but activate MAPK signalling indirectly by promoting signalling through CRAF. These distinct mechanisms can respond differently, or even poorly, to conventional BRAF inhibitors.

```
BRAF V600 mutation
→ usually direct BRAF kinase activation
→ BRAF-MEK inhibition may be effective
Non-V600 BRAF alteration
→ may signal through dimers,
upstream RAS activity, or CRAF
Therefore:
"BRAF-mutant"
≠ one therapeutically uniform group
```

The authors developed a *hierarchical genomic framework* to classify tumours more precisely than by BRAF mutation alone. The framework considers the functional type of BRAF alteration and its genomic context, including co-alterations that may activate parallel signalling or reduce reliance on BRAF.

For example, a tumour with an activating BRAF V600 mutation but no strong competing MAPK-pathway driver may be highly dependent on BRAF. In contrast, co-occurring alterations in genes such as *RAS*, *NF1*, or receptor tyrosine kinases can change pathway wiring and potentially reduce sensitivity to a BRAF-directed drug.

```
BRAF alteration
+ tumour lineage
+ functional mutation class
+ co-occurring genomic alterations
→ more informative prediction of treatment response
```

The clinically important point is that genomic classification can help distinguish a *driver* alteration from a mutation that is present but not sufficient to define the tumour's key dependency. This is especially relevant in tumour-agnostic precision oncology, where the same mutation is treated across several cancer types.

However, such a framework is primarily a biomarker and treatment-selection approach. It does not mean that genomic profiling can perfectly predict response: drug exposure, tumour heterogeneity, adaptive resistance, and the immune microenvironment also matter.

```
More detailed genomic classification
→ potentially better patient selection
for BRAF-directed therapy
But:
A predicted BRAF dependency
≠ guaranteed and durable clinical response
```

103 [26-09-21] Progesterone can promote macrophage-driven immune suppression in hormone-receptor-positive breast cancer

(*Progesterone reshapes the tumor immune microenvironment toward macrophage-driven immune suppression in hormone receptor-positive breast cancer*, <https://www.nature.com/articles/s41523-026-01047-3>)

This study shows that progesterone can influence not only tumour-cell growth, but also the *tumour immune microenvironment* in hormone-receptor-positive breast cancer.

Hormone-receptor-positive tumours express receptors for oestrogen and/or progesterone. These receptors are transcription factors: when activated by their hormones, they alter gene expression. Endocrine therapies mainly aim to block oestrogen-receptor signalling, but this paper highlights a distinct immunological role for progesterone.

```
Progesterone signalling in hormone-receptor-positive tumour context
→ altered cytokine and immune-regulatory signals
→ macrophage accumulation or reprogramming
→ more immune-suppressive tumour microenvironment
```

103.1 Why are macrophages important?

Macrophages are innate immune cells that can remove pathogens and damaged cells, present antigens, and regulate inflammation. In tumours, however, they can be reprogrammed into *tumour-associated macrophages*, or TAMs, that support cancer rather than eliminate it.

Immune-suppressive macrophages can reduce effective T-cell activity, promote blood-vessel formation and tissue remodelling, and release growth- and survival-supporting signals.

```
Anti-tumour macrophage activity
→ inflammatory signalling
+ antigen presentation
+ support for T-cell responses
Immune-suppressive TAM programme
→ reduced effective T-cell activity
+ tumour-supportive tissue remodelling
→ tumour immune evasion may increase
```

The paper suggests that progesterone shifts the balance towards this second, macrophage-driven state.

103.2 How does this connect to endocrine therapy?

Hormone-receptor-positive breast cancers are often considered relatively immunologically “cold”: compared with some other breast-cancer subtypes, they may show lower cytotoxic T-cell infiltration and less consistent benefit from checkpoint inhibition.

This work provides one possible explanation. Progesterone signalling may help create an environment in which macrophage-mediated suppression restrains anti-tumour T cells.

```
Progesterone-associated signalling
→ immune-suppressive macrophage state
Immune-suppressive macrophages
→ weaker cytotoxic T-cell function
Weaker immune surveillance
→ possible reduced response to immunotherapy
```

It therefore extends the earlier examples of TGF- β and epigenetic regulation: tumour immunity depends not only on tumour antigens and checkpoints, but also on local cell-state programmes that determine whether immune cells are recruited, activated, or suppressed.

103.3 What should be concluded?

The main implication is that progesterone signalling may be relevant to breast-cancer immunity, not solely to direct tumour-cell proliferation. This raises the possibility that selected progesterone-receptor-directed approaches could alter the immune microenvironment, potentially alongside endocrine therapy or immunotherapy.

However, macrophages are diverse and plastic, and progesterone has important normal physiological roles. The paper therefore provides a mechanistic rationale, rather than proof that blocking progesterone signalling will improve patient outcomes.

```
Progesterone-associated immune suppression
→ possible therapeutic vulnerability
But:
Macrophage reprogramming in models
≠ proven clinical benefit from combining
progesterone-pathway targeting with immunotherapy
```

104 [26-09-21] Genome-wide CRISPR screens identify ways to improve human T cells in solid tumours

(*In vivo* genome-wide CRISPR screens of human T cells in solid tumours, <https://www.nature.com/articles/s41586-026-10906-9>)

This study develops a way to perform *genome-wide CRISPR knockout screens* directly in human T cells while they are infiltrating solid tumours. The goal is to identify genes whose removal makes T cells better able to enter tumours or retain anti-tumour function within the suppressive tumour microenvironment.

This extends the earlier *in vivo* CRISPR-screening example. Rather than editing tumour cells to identify immune-evasion mechanisms, the researchers edited the *T cells themselves*.

```
Human T cells receive genome-wide CRISPR library
→ each cell loses a different gene
Edited T cells enter solid tumour model
→ tumour environment selects more or less fit cells
Sequence recovered T cells
→ identify gene knockouts that improve
infiltration or effector function
```

A major technical challenge is scale. A genome-wide screen requires many cells carrying many different guide RNAs. The researchers used melanoma cells engineered to display an anti-CD3 binding fragment, allowing them to recruit large numbers of human T cells into tumours and recover enough cells for screening.

104.1 What targets were identified?

One screen identified the *P2RY8-Gα13-ARHGEF1* pathway as a brake on T-cell accumulation in tumours. Deleting *P2RY8* improved T-cell infiltration.

A second screen identified *GNAS*, which encodes the signalling protein Gαs, as a regulator of T-cell dysfunction. Gαs integrates suppressive signals in the tumour environment; deleting *GNAS* helped T cells retain effector functions such as interferon-γ production.

```
Delete P2RY8
→ improved T-cell trafficking and tumour infiltration
Delete GNAS
→ reduced response to suppressive signals
→ more sustained T-cell effector function
```

The important conceptual distinction is that solid-tumour cell therapy needs both properties. A T cell that is highly cytotoxic but cannot enter the tumour is ineffective; a T cell that enters but becomes exhausted or suppressed is also ineffective.

```
Effective solid-tumour T-cell therapy requires:
tumour access
+
persistence and functional fitness
+
recognition of an appropriate tumour target
```

This is particularly valuable because some relevant human T-cell genes, including *P2RY8*, are absent or function differently in standard mouse models. Human *in vivo* screening can therefore reveal dependencies that conventional mouse immunology may miss.

The study is a discovery platform rather than a clinical treatment result. Nonetheless, *P2RY8* or *GNAS* editing could eventually be explored in engineered T-cell products, such as CAR-T or TCR-T cells, to improve activity against solid tumours.

```
A favourable CRISPR-screen hit
≠ a clinically safe gene edit
Any therapeutic T-cell edit must still be tested for:
durable anti-tumour activity
+ normal-tissue safety
+ excessive inflammation
+ effects on infection control
```

105 [26-09-21] Lipid nanoparticles: delivering mRNA into cells

(*Lipid nanoparticles for mRNA delivery*, <https://www.nature.com/articles/s41578-021-00358-0>)

This review explains the delivery system underlying most mRNA vaccines, including the COVID-19 vaccines. The key point is that mRNA is not the medicine's final active product: it is a temporary set of instructions that cells use to make a protein.

For an mRNA vaccine, the encoded protein is a viral antigen. For other applications, the encoded protein could be a therapeutic enzyme, cytokine, gene-editing component, or a CAR.

```
mRNA enters a cell
→ ribosomes read its sequence
→ cell makes the encoded protein
mRNA is naturally degraded over time
→ protein production is temporary
```

Naked mRNA cannot readily cross cell membranes and is easily degraded by RNases, enzymes that break down RNA. It therefore needs a delivery vehicle. *Lipid nanoparticles*, or LNPs, are small lipid-based particles that protect mRNA in the body and help it enter cells.

105.1 What is inside an LNP?

A typical LNP contains four main lipid components. The most important is an *ionisable lipid*. It is largely uncharged in blood, which helps tolerability, but becomes positively charged in the acidic environment of an endosome after cell uptake.

```
Ionisable lipid
→ packages negatively charged mRNA
+ helps endosomal escape
Helper phospholipid and cholesterol
→ support particle structure and membrane fusion
PEG-lipid
→ helps control particle size and circulation behaviour
```

After injection, LNPs are taken up by cells through *endocytosis*: the cell encloses the particle in a membrane-bound compartment called an endosome.

The crucial and inefficient step is *endosomal escape*. As the endosome acidifies, the ionisable lipid becomes positively charged and interacts with negatively charged endosomal lipids. This destabilises the endosomal membrane, allowing some mRNA to reach the cytoplasm, where ribosomes can translate it into protein.

```
LNP is taken into endosome
→ endosome becomes acidic
Ionisable lipid becomes positively charged
→ endosomal membrane is disrupted
Small fraction of mRNA reaches cytoplasm
→ encoded protein is produced
```

Most internalised mRNA does *not* escape successfully and is eventually degraded. Improving endosomal escape is therefore a major LNP-design problem.

105.2 How did the COVID-19 mRNA vaccines work?

The vaccines delivered mRNA encoding the SARS-CoV-2 spike protein. Muscle and local immune cells took up the LNP-mRNA particles and temporarily produced spike protein or spike-derived antigen. Immune cells then processed this antigen and activated B cells, CD4 T cells, and CD8 T cells.

```
LNP delivers spike mRNA
→ host cells transiently make spike protein
Spike antigen is recognised by immune system
→ antibodies and memory immune cells develop
Later viral exposure
→ immune system responds faster and more effectively
```

The mRNA does not need to enter the nucleus, does not normally integrate into DNA, and is transient. The immune memory remains after the mRNA and most vaccine-produced protein have disappeared.

105.3 Is this used for *in vivo* CAR-T therapy?

Yes: this is broadly the same *RNA-delivery principle* discussed in the *in vivo* CAR-T section. An LNP can be loaded with mRNA encoding a CAR instead of viral spike protein.

The major difference is the intended destination. Standard vaccine LNPs are designed to be taken up mainly by cells near the injection site and draining lymph nodes. An *in vivo* CAR-T LNP must deliver its mRNA preferentially to *T cells*, while avoiding uptake by liver cells, other immune cells, or tumour cells.

```
mRNA vaccine:
LNP + antigen mRNA
→ transient antigen production
→ vaccination
In vivo CAR-T:
T-cell-targeted LNP + CAR mRNA
→ transient CAR expression in T cells
→ T cells can recognise and kill antigen-positive cells
```

Thus, the basic technology is similar, but *in vivo* CAR-T has a much harder targeting requirement. A vaccine can work when several local cell types take up the mRNA. By contrast, CAR mRNA expressed in the wrong cell type could be ineffective or potentially dangerous.

105.4 What should be concluded?

LNPs make mRNA therapeutically useful by solving three problems: they protect it from degradation, enable cellular uptake, and allow a fraction to escape into the cytoplasm for protein production.

The success of COVID-19 vaccines clinically validated this platform. However, making a protein temporarily is easier than delivering mRNA selectively to a particular cell type in the body.

```
Effective LNP delivery to cells near an injection site
≠ selective delivery to circulating or tissue-resident T cells
Transient CAR expression
≠ automatically safe or clinically effective in vivo CAR-T therapy
```

106 [26-09-21] “Out-of-distribution medicines”: generating biological data for AI drug discovery

(*Building Out-of-Distribution Medicines – Why AI-native drug discovery requires new kinds of biological data*, <https://www.waypointbio.com/writings/out-of-distribution-medicines>)

This article argues that AI drug discovery will not be transformed mainly by applying better models to existing biological datasets. Instead, it requires deliberately generating new experimental data that teach models how biological systems respond to interventions.

In machine learning, *out-of-distribution*, or OOD, means an input that differs substantially from the data on which a model was trained. A model trained mainly on known drugs, familiar targets, standard cell lines, or common disease states may work reasonably within that historical data distribution, but may fail when asked to predict a genuinely new therapeutic mechanism.

```
Historical biological and drug data
→ model learns correlations within known examples
Novel target, modality, cell state, or disease context
→ out-of-distribution prediction problem
Therefore:
better model alone
≠ reliable prediction beyond available data
```

The proposed solution is to generate more *causal*, intervention-rich data. Standard omics experiments may show that a disease-associated cell has a particular RNA or protein signature. However, they often cannot show which change would *cause* that cell to move towards a healthier state.

```
Descriptive data:
diseased cell has molecular state X
Perturbational data:
alter gene, protein, or drug exposure
→ measure whether cell state moves from X towards Y
This can teach a model:
which interventions may cause useful biological change?
```

Waypoint’s emphasis appears to be on high-content, spatial, and perturbational measurements. These aim to preserve information that conventional bulk assays can lose: which cells are affected, their local tissue environment, their morphology, and how they respond to an intervention over time.

The novelty is therefore not the basic idea that experimental data improve AI. Perturb-seq, high-content imaging, spatial biology, active learning, and closed-loop experimentation are already established directions. The more distinctive claim is that a drug-discovery company should build its data-generation system around the explicit goal of exploring biological states and interventions that are poorly represented in public datasets.

```
AI proposes informative perturbations
→ experiments measure cellular consequences
→ results retrain the model
→ next experiments become more informative
This is a closed experimental-computational loop.
```

How impressive is this? ****The underlying idea is important and credible, but not yet proof of a fundamentally new discovery paradigm.**** Many AI-drug-discovery claims focus on prediction from existing data; in practice, the strongest advantage may come from a proprietary experimental platform that generates consistent, high-quality causal data faster than competitors.

Its real value will depend on practical evidence: whether the platform identifies targets or drug mechanisms that conventional approaches miss, whether its predictions replicate across patient-relevant models, and whether resulting programmes advance into robust preclinical and eventually clinical candidates.

```
Novel biological data
+ useful AI model
+ rapid experimental feedback
→ potentially better target and drug hypotheses
But:
Sophisticated data platform
≠ validated medicine-discovery engine
The decisive evidence:
repeatable prospective discoveries
that progress beyond interesting cell-based results
```

107 [26-09-21] In vivo CAR-T: delivery platform determines the therapeutic product

(*In vivo* generation of CAR T cells: biology, delivery platforms, clinical promise, and translational challenges, <https://www.sciencedirect.com/science/article/pii/S3050597625000272>)

This review expands on the earlier *in vivo* CAR-T section. The central point is that *in vivo* CAR-T is not one technology: the final CAR-T-cell product depends strongly on the delivery platform, genetic payload, targeted T-cell subset, and duration of CAR expression.

```
Delivery vehicle
+ CAR genetic payload
+ targeted cell type
→ determines where, how long, and in which T cells
the CAR is expressed
→ determines efficacy and safety profile
```

The goal is to replace patient-specific *ex vivo* manufacturing with direct delivery of CAR instructions into the patient. This could shorten time to treatment, reduce cost and logistical complexity, and potentially make cell therapy available to patients unable to provide a suitable starting T-cell collection.

107.1 The main platform trade-off

As discussed for LNP-mediated mRNA delivery, non-viral delivery can produce transient CAR expression. This may be particularly suitable when temporary immune-cell depletion is sufficient, such as some autoimmune diseases.

Viral vectors can deliver DNA and potentially create persistent CAR expression. This could be useful in cancer, where durable tumour surveillance may be needed, but it introduces greater challenges around vector immunity, uncontrolled persistence, and—for integrating vectors—insertional genetic risk.

T-cell-targeted LNP + CAR mRNA
→ non-integrating, transient CAR expression
→ potentially controllable and repeat-dose compatible
T-cell-targeted viral DNA delivery
→ potentially durable CAR expression
→ greater requirements for long-term safety control

Thus, transient expression is not necessarily an inferior version of persistent CAR-T. It is a different therapeutic design, potentially preferable when the desired biological effect is a temporary immune reset rather than lifelong engineered-cell persistence.

107.2 Why is selective delivery the central challenge?

Conventional CAR-T cells are manufactured and characterised before infusion. In vivo CAR-T removes this control point. The delivery system must reach enough appropriate T cells while avoiding genetic delivery to other cells.

This includes avoiding CAR expression in tumour cells. A tumour cell accidentally expressing the CAR target antigen-recognition domain could potentially bind therapeutic CAR-T cells and interfere with their activity; more generally, off-target transduction creates unpredictable safety and biodistribution concerns.

Insufficient T-cell delivery
→ too few CAR-T cells for therapeutic activity
Delivery to unintended immune or non-immune cells
→ reduced efficiency or unexpected toxicity
Delivery to the right T-cell population
→ potential effective in vivo CAR-T product

The relevant T-cell state also matters. Naive and memory-like T cells may expand and persist differently from terminally differentiated or exhausted T cells. In vivo delivery therefore needs to control not only *which tissue* receives the payload, but ideally *which T-cell subsets* are engineered.

107.3 What should be concluded?

In vivo CAR-T could transform cell therapy from a personalised manufacturing process into a more medicine-like administered product. However, it does not eliminate familiar CAR-T risks, including cytokine-release syndrome, infection following B-cell depletion, and on-target, off-tumour toxicity.

No ex vivo manufacturing
≠ no manufacturing or quality challenge
In vivo generation of CAR-T cells
→ shifts quality control from the factory
into delivery-vector design and in-patient biology
The key requirement:
sufficiently selective, efficient, controllable,
and reproducible T-cell engineering in vivo

108 [26-09-21] Dual TIGIT and PD-1 blockade in advanced oesophageal squamous-cell carcinoma

(*Ociperlimab and tislelizumab in advanced esophageal squamous cell carcinoma (AdvanTIG-203): a phase II study*, <https://www.nature.com/articles/s41467-026-77705-8>)

This randomised Phase II study tested whether adding *ociperlimab* to *tislelizumab* improves outcomes for PD-L1-positive advanced oesophageal squamous-cell carcinoma, or ESCC.

Both drugs are immune-checkpoint antibodies. They do not directly kill tumour cells; they aim to restore anti-tumour T-cell activity by blocking distinct inhibitory signals.

108.1 How does tislelizumab work?

Tislelizumab binds *PD-1*, an inhibitory receptor expressed by activated T cells. When PD-1 binds PD-L1 or PD-L2, it reduces T-cell activation, cytokine production, and killing capacity.

Tumours can exploit this pathway by expressing PD-L1, particularly in an inflamed tumour environment already containing tumour-reactive T cells.

```
PD-1 on T cell binds PD-L1 on tumour or immune cell
→ inhibitory T-cell signal
→ weaker anti-tumour activity
Tislelizumab blocks PD-1
→ removes this inhibitory signal
→ T cells may recover anti-tumour function
```

108.2 How does ociperlimab work?

Ociperlimab binds *TIGIT*, another inhibitory receptor found on activated and chronically stimulated T cells, as well as natural-killer, or NK, cells.

TIGIT competes with the activating receptor *CD226* for ligands such as CD155. When TIGIT signalling dominates, T-cell and NK-cell activity is reduced.

```
TIGIT binds CD155
→ inhibitory immune-cell signalling
+ reduced CD226-associated activation
Ociperlimab blocks TIGIT
→ may restore T-cell and NK-cell activity
→ may improve immune control of tumour cells
```

108.3 Why combine TIGIT and PD-1 blockade?

PD-1 and TIGIT are often co-expressed on dysfunctional or exhausted tumour-infiltrating T cells. They represent separate but overlapping inhibitory mechanisms.

```
PD-1 blockade
→ removes one T-cell inhibitory signal
TIGIT blockade
→ removes another inhibitory signal
+ may enhance CD226-mediated activation
Combination
→ aims for deeper restoration of
anti-tumour T-cell and NK-cell function
```

Thus, the biological rationale resembles other checkpoint combinations: a tumour may use multiple brakes simultaneously, so releasing only PD-1 may leave TIGIT-mediated suppression intact.

108.4 What did AdvanTIG-203 show?

The combination produced a numerically higher objective response rate than tislelizumab alone: ****30.6%**** versus ****20.6%****. However, it did not improve the study's primary progression-free-survival endpoint; median progression-free survival was approximately ****3.4 months**** with the combination and ****3.5 months**** with tislelizumab plus placebo.

This distinction is important. Some additional patients had measurable tumour shrinkage, but this did not translate into clear evidence of longer disease control across the trial population.

Higher response rate
≠ improved progression-free survival
Plausible dual-checkpoint biology
≠ proven clinical benefit of the combination

The study therefore supports continued investigation of TIGIT biology and biomarker-defined subgroups, but does not establish ociperlimab plus tislelizumab as superior to PD-1 blockade alone in this setting. The key question is whether a more specific immune context—for example, tumours with high TIGIT, CD155, CD226, or a particular T-cell-inflamed state—can identify patients who gain a meaningful benefit from dual checkpoint blockade.

109 [26-09-21] GPC3 CAR-T cells armoured against TGF- β suppression in liver cancer

(GPC3-specific *dnTGF β RII*-armoured CAR T cells for hepatocellular carcinoma, <https://www.nature.com/articles/s41586-026-10786-z>)

This paper reports early clinical results for *C-CAR031* (also called *AZD7003*), an autologous CAR-T therapy for advanced hepatocellular carcinoma, or HCC.

The CAR recognises *glypican-3*, or GPC3, a cell-surface protein expressed by many HCC cells. This is the targeting component, similar to the GPC3 CAR-T approach described earlier for hepatoblastoma.

The extra feature is a *dominant-negative TGF- β receptor II*, abbreviated *dnTGF β RII*. This is what “armoured” means: the engineered CAR-T cells have an additional modification intended to protect them from the suppressive tumour microenvironment.

109.1 Why does TGF- β matter?

TGF- β is abundant in many solid tumours, including HCC. As discussed for mesothelioma, it can promote fibrosis, immune suppression, and T-cell exclusion. For a T cell, TGF- β signalling can reduce proliferation, cytokine production, cytotoxic function, and persistence.

TGF- β in tumour microenvironment
→ signalling through TGF- β receptor II in CAR-T cell
→ suppressive gene-expression programme
→ weaker CAR-T-cell expansion and killing

A conventional GPC3 CAR-T cell may therefore recognise HCC successfully but become less functional once it enters the TGF- β -rich tumour environment.

109.2 What is a dominant-negative receptor?

A *dominant-negative* protein is a deliberately altered version of a normal protein that interferes with the normal pathway.

dnTGF β RII retains the external part of the receptor that binds TGF- β , but lacks the intracellular signalling region needed to transmit the suppressive signal. It therefore acts as a non-signalling decoy receptor.

```
TGF- $\beta$  binds normal TGF- $\beta$  receptor II
→ inhibitory signal enters CAR-T cell
→ CAR-T-cell function is suppressed
TGF- $\beta$  binds dnTGF $\beta$ RII
→ no productive inhibitory signal
→ CAR-T cell is relatively protected
```

The receptor may also reduce locally available TGF- β by acting as a partial *sink*. However, its main intended effect is cell-intrinsic: making the engineered T cell less responsive to TGF- β .

109.3 Why combine this with a GPC3 CAR?

The two modifications solve different problems:

```
GPC3 CAR
→ tells the T cell what to attack
dnTGF $\beta$ RII armour
→ helps the T cell remain functional
after reaching a suppressive solid tumour
Together
→ tumour recognition
+ resistance to one important local inhibitory signal
```

This is a general strategy for solid-tumour cell therapy. Rather than only improving antigen recognition, CAR-T cells can be engineered to resist features of the tumour microenvironment, such as TGF- β , hypoxia, inhibitory checkpoints, or metabolic stress.

109.4 What did the trial show?

This first-in-human, single-arm Phase I study included heavily pre-treated advanced HCC. It reported an objective response rate of ****44.4%**** and median overall survival of ****14.2 months****, alongside a reported manageable safety profile.

These findings are encouraging because solid tumours have been much harder for CAR-T therapy than B-cell cancers. However, this was an early, non-randomised study without a control group.

```
GPC3-directed tumour killing
+
TGF- $\beta$ -resistant CAR-T-cell design
→ promising early activity in HCC
But:
Phase I responses
≠ proof of comparative efficacy, durable benefit,
or safety across the broader HCC population
```

The central idea is that a CAR-T cell can be engineered not only with a receptor for the tumour antigen, but also with molecular “armour” against the signals that would otherwise disable it inside the tumour.

110 [26-09-21] Neo-BCV: bacterial stimulation to reprogramme tumour-associated macrophages

(*Neo-BCV induces macrophage-associated immunometabolic remodeling and anti-tumor immunity in solid tumors*, <https://www.nature.com/articles/s41698-026-01709-2>)

This study examines *Neo-BCV*, a bacterial-component-based immunotherapy designed to make the tumour microenvironment less immunosuppressive. Rather than targeting a tumour mutation directly, it activates innate immunity, particularly *tumour-associated macrophages*.

Macrophages are highly plastic immune cells. In many solid tumours, they adopt an immunosuppressive, tissue-repair-like state, often simplified as “M2-like.” These macrophages can support blood-vessel formation, tumour growth, metastasis, and exclusion or suppression of cytotoxic T cells.

```
Immunosuppressive tumour macrophages
→ support tumour growth
+ restrain effective T-cell immunity
Neo-BCV stimulation
→ inflammatory macrophage reprogramming
→ improved anti-tumour immune environment
```

Neo-BCV contains inactivated bacterial components and a glucan-associated immune stimulant. These provide microbial “danger” signals that are recognised by innate immune receptors on macrophages and other myeloid cells. The intended effect is not vaccination against a specific tumour antigen; it is broad activation and reprogramming of innate immunity.

The study reports *immunometabolic remodelling*: Neo-BCV changed macrophage metabolism alongside their immune state. This matters because macrophage function is closely tied to metabolism. A suppressive macrophage and an inflammatory, tumour-controlling macrophage require and use nutrients differently.

```
Neo-BCV-associated innate immune activation
→ altered macrophage metabolic programme
+ reduced JAK2-STAT3-associated signalling
→ more inflammatory, anti-tumour macrophage state
Reprogrammed macrophages
→ improved antigen presentation
+ greater recruitment or activation of T cells and NK cells
→ increased tumour control in models
```

The study also supports combination with cisplatin chemotherapy. Cytotoxic therapy can kill tumour cells and release antigens, whereas Neo-BCV may help convert the resulting signals into a productive immune response rather than an immunosuppressive wound-healing response.

```
Chemotherapy
→ tumour-cell injury and antigen release
Neo-BCV
→ more inflammatory myeloid-cell response
Combination
→ potential increase in anti-tumour immunity
```

The main concept is related to the earlier discussion of tumour-targeted cytokines and epigenetic immune activation: successful immunotherapy may require changing the *tumour microenvironment*, not only releasing a checkpoint brake on T cells.

However, the M1/M2 distinction is a useful simplification, not a complete description of macrophage biology. Macrophage states exist along a spectrum, vary between tumour types, and can change over time. This

is also preclinical work: broad innate immune stimulation could cause systemic inflammation, and tumour control in mouse models does not establish a safe or effective treatment in patients.

Inflammatory macrophage remodelling in models
≠ proven clinical immunotherapy
The key translational question:
can Neo-BCV create sufficient local anti-tumour innate immunity
without unacceptable systemic inflammatory toxicity?

111 [26-09-21] **PLAU promotes pancreatic-cancer invasion through PI3K–AKT–mTORC1 signalling**

(*PLAU orchestrates pancreatic cancer progression via PI3K/AKT/mTORC1-mediated upregulation of MMP2/9 and FSCN1*, <https://www.nature.com/articles/s41698-026-01713-6>)

This paper identifies *PLAU* as a promoter of pancreatic-cancer progression. The central idea is that *PLAU* activates a growth and survival signalling pathway, PI3K–AKT–mTORC1, which then increases proteins that help cancer cells move through surrounding tissue.

PLAU encodes *urokinase-type plasminogen activator*, often called uPA. uPA is a secreted enzyme that converts *plasminogen* into *plasmin*, another enzyme that helps break down extracellular material around cells. *PLAU* can also signal through its cell-surface receptor, *uPAR*.

PLAU/uPA activity
→ extracellular-matrix remodelling
+ cell-surface signalling through uPAR
→ tumour cells can more readily invade
surrounding tissue

111.1 What is PI3K–AKT–mTORC1?

PI3K, AKT, and mTORC1 are parts of a major cell-growth signalling pathway. It normally responds to nutrients, growth factors, and other external signals. Cancer cells often activate it excessively.

External growth or survival signal
→ PI3K activation
→ AKT activation
→ mTORC1 activation
mTORC1
→ increased protein synthesis, growth,
metabolism, and survival-associated cell programmes

In this paper, *PLAU* is proposed to activate this pathway. The consequence is not only faster tumour-cell growth, but increased expression of proteins involved in invasion and migration.

111.2 What are MMP2, MMP9, and FSCN1?

MMP2 and *MMP9* are *matrix metalloproteinases*. These are enzymes that degrade components of the *extracellular matrix*: the protein-rich material surrounding cells.

A tumour cell needs to cross or remodel this matrix to invade nearby tissue, enter blood vessels, and eventually metastasise. *MMP2* and *MMP9* can therefore help create a physical path for tumour-cell movement.

FSCN1, or fascin-1, is different. It is an *actin-bundling protein*. Actin is part of the cell's internal skeleton. *FSCN1* helps form thin, protrusive structures that allow cells to attach, move, and push into surrounding tissue.

```
MMP2 and MMP9
→ break down external tissue barriers
FSCN1
→ reorganises internal cell skeleton
→ supports cell movement
Both together
→ increased invasive potential
```

111.3 What is the complete proposed mechanism?

```
Increased PLAU
→ PI3K-AKT-mTORC1 signalling
PI3K-AKT-mTORC1 signalling
→ increased MMP2, MMP9, and FSCN1
MMP2/9 remodel extracellular matrix
+
FSCN1 supports cellular movement
→ pancreatic-cancer invasion, progression,
and potentially metastasis
```

The useful distinction is that cancer invasion requires both an *external* change and an *internal* change. MMP2 and MMP9 help remove the barriers around a cell, whereas *FSCN1* helps the cell physically move through the newly permissive environment.

111.4 What should be concluded?

PLAU may be both a biomarker of aggressive pancreatic cancer and a possible treatment target. Blocking PLAU, its receptor-associated signalling, or the downstream PI3K–AKT–mTORC1 pathway could potentially reduce tumour invasion.

However, this pathway is biologically widespread. PI3K–AKT–mTORC1 is also important in normal cell metabolism and tissue maintenance, while matrix-remodelling enzymes participate in normal healing. Therefore, a mechanistic pathway linking PLAU to invasion does not automatically identify a safe or effective drug strategy.

```
PLAU-associated invasive programme
→ plausible therapeutic vulnerability
But:
Reduced invasion in experimental models
≠ proven reduction in metastasis
or clinical benefit in patients
```

112 [26-09-21] Anti-thymocyte globulin can induce TNF-dependent death in T-ALL

(*Anti-thymocyte globulin triggers TNF-dependent cell death in T-cell acute lymphoblastic leukemia*, <https://www.nature.com/articles/s41467-026-77027-9>)

This study repurposes *anti-thymocyte globulin*, or ATG, as a possible treatment strategy for T-cell acute lymphoblastic leukaemia, or T-ALL. ATG is already used clinically to deplete T cells, for example before stem-cell transplantation or to treat some forms of transplant rejection.

Unlike a single monoclonal antibody, ATG is a *polyclonal antibody mixture*. It contains many antibodies that recognise different proteins on human T cells, including CD3 and other surface molecules.

ATG binds multiple surface proteins on T-ALL cell
→ broad receptor engagement and signalling
→ TNF-associated cell-death programme
→ T-ALL cell death in susceptible cells

The key mechanism involves *tumour necrosis factor*, or TNF. TNF is an inflammatory cytokine with context-dependent effects. It can promote cell survival and inflammation, but it can also trigger cell death when the balance of intracellular signalling is altered.

The study found that ATG treatment induced TNF-dependent killing of T-ALL cells. This creates a rationale for combination with a *SMAC mimetic*, such as birinapant. SMAC mimetics inhibit proteins called IAPs, which normally help protect cells from programmed cell death.

ATG
→ TNF-associated death signal
SMAC mimetic
→ removes IAP-mediated protection from cell death
Combination
→ stronger TNF-dependent killing of T-ALL cells

This is conceptually different from using an antibody to recognise one tumour-specific antigen and recruit immune killing. ATG broadly binds T-cell-associated proteins and appears to push leukaemic T cells towards a cell-death state. Its lack of tumour specificity is both an advantage and a challenge: it can act across diverse T-ALL cells, but it may also deplete normal T cells and cause substantial immunosuppression.

The work is preclinical and does not establish ATG, alone or with a SMAC mimetic, as a new T-ALL treatment. However, it is an interesting example of using an existing immune-depleting medicine to expose a context-dependent death-pathway vulnerability in cancer.

ATG-induced T-ALL killing in experimental systems
≠ safe and effective treatment in patients
The key translational question:
can leukaemic-cell killing be achieved without
unacceptable depletion of normal T cells,
infection risk, or systemic TNF-related toxicity?

113 [26-09-21] Entinostat plus dual checkpoint blockade in HER2-negative breast cancer

(Sequencing analysis reveals evidence of immune activation in advanced HER2-negative breast cancer responders treated with the combination of entinostat, nivolumab, and ipilimumab, <https://www.nature.com/articles/s43856-026-01908-9>)

This study examines tumour biopsies from a Phase Ib trial of three immunologically distinct drugs in advanced HER2-negative breast cancer: *entinostat*, *nivolumab*, and *ipilimumab*.

The rationale is that checkpoint inhibitors can only work well if tumour-reactive T cells exist and can enter the tumour. Many breast tumours have an immune-suppressive, relatively “cold” microenvironment, so the three drugs aim to address different barriers.

113.1 What does each drug do?

Entinostat is an inhibitor of histone deacetylases, or HDACs. HDACs remove chemical marks from histones and can help maintain gene repression. In this context, entinostat is intended to make the tumour microenvironment more immune-permissive, including by altering suppressive myeloid-cell programmes and tumour-associated gene expression.

Nivolumab blocks PD-1, an inhibitory receptor on activated T cells. It helps restore function in T cells that have encountered persistent antigen and become suppressed or exhausted.

Ipilimumab blocks CTLA-4, another inhibitory receptor. CTLA-4 acts earlier during T-cell activation, particularly when T cells are initially primed by antigen-presenting cells. It can also reduce regulatory-T-cell-associated suppression.

```
Entinostat
→ aims to make tumour microenvironment
less immune suppressive
Ipilimumab: anti-CTLA-4
→ supports T-cell priming and expansion
Nivolumab: anti-PD-1
→ restores activity of tumour-exposed T cells
```

113.2 Why use all three?

The triplet is designed as a sequence of complementary steps, rather than three unrelated anti-cancer drugs.

```
Entinostat
→ increase conditions for productive anti-tumour immunity
Ipilimumab
→ help generate or expand tumour-reactive T cells
Nivolumab
→ prevent PD-1-mediated inhibition
of activated T cells in the tumour
Together
→ aim to convert a weak immune response
into sustained tumour-cell killing
```

Using both checkpoint inhibitors makes biological sense because CTLA-4 and PD-1 act at different stages and locations of the anti-tumour response. However, dual checkpoint blockade also increases the risk of immune-related adverse events, such as colitis, hepatitis, endocrine toxicity, or pneumonitis. Entinostat may add both immune-modulating benefit and its own toxicity, including fatigue and blood-count effects.

Three-drug regimens are therefore not unusual in oncology, especially where drugs have complementary mechanisms. But each additional drug must justify its added toxicity, cost, and practical complexity.

113.3 What did the sequencing analysis show?

The researchers compared tumour biopsies collected before treatment, after a two-week entinostat-only period, and after addition of nivolumab plus ipilimumab.

Responders tended to have evidence of a more immune-active tumour environment even before treatment. After triplet therapy, tumours showed increased expression of immune-associated pathways, including interferon- γ responses, inflammatory signalling, and IL-2/STAT5-associated T-cell activity.

```
Pre-existing immune activity
+ triplet treatment
→ stronger inflammatory and T-cell-associated
gene-expression programme
This was associated with response,
but does not prove that every observed pathway
caused the response.
```

The previously reported clinical trial showed an objective response rate of approximately **25%** overall, with higher activity in triple-negative breast cancer. However, this sequencing paper is mainly a *mechanistic correlative study*: it helps explain which immune changes accompanied response, rather than proving that the three-drug regimen is superior to entinostat plus one checkpoint inhibitor, or dual checkpoint blockade alone.

113.4 What should be concluded?

The useful model is:

```
Epigenetic and myeloid-cell modulation
+ improved T-cell priming
+ relief of T-cell exhaustion
→ potential multi-step route to stronger immunity
```

The study supports the idea that responders may need a partially pre-existing anti-tumour immune response for this regimen to amplify. The central clinical question is whether biomarkers can identify patients for whom the added activity of triplet therapy outweighs the added toxicity of dual checkpoint blockade.

114 [26-09-25] PUM3 protects homologous-recombination-deficient breast cancer from replication stress

(*PUM3 is an essential suppressor of replication stress in homologous recombination deficient breast cancer*, <https://www.nature.com/articles/s41467-026-77417-z>)

This study identifies *PUM3* as a potential vulnerability in homologous-recombination-deficient, or HR-deficient, breast cancer. This includes some cancers with *BRCA1* or *BRCA2* defects, which already have limited capacity to repair dangerous DNA double-strand breaks accurately.

As discussed for PARP inhibitors, HR-deficient cells can survive only because remaining DNA-protection pathways compensate for their underlying repair defect. This paper suggests that *PUM3* is one such protective factor.

```
BRCA1/2 or other HR defect
→ impaired accurate DNA repair
→ elevated replication-associated stress
PUM3 activity
→ helps tumour cells tolerate this stress
→ continued survival despite HR deficiency
```

PUM3 is an RNA-binding protein, but the relevant function here is linked to controlling *transcription-replication conflicts*. These occur when the machinery that copies DNA and the machinery that transcribes DNA into RNA obstruct one another.

Such conflicts can create *R-loops*: structures in which newly made RNA remains bound to one DNA strand, leaving the other DNA strand unpaired. Persistent R-loops can stall replication forks and cause DNA damage.

PUM3 supports TOP1 localisation to chromatin
→ reduced transcription-replication conflicts
+ fewer harmful R-loops
Loss of PUM3
→ increased replication stress and DNA damage

TOP1, or topoisomerase I, helps relieve twisting strain in DNA during transcription and replication. The study suggests that PUM3 is needed for appropriate TOP1 activity on chromatin. Without PUM3, HR-deficient cells accumulate more replication-associated damage than they can repair.

HR deficiency
→ reduced capacity to repair severe DNA damage
PUM3 loss
→ increased replication-associated DNA damage
Both together
→ catastrophic genome instability
→ selective tumour-cell death

This is another proposed example of *synthetic lethality*. It resembles the TET2–PARP-inhibitor example: the therapeutic opportunity arises because a cancer cell begins with one DNA-maintenance weakness and becomes unusually dependent on a compensatory pathway.

The work is especially interesting because PUM3 may represent a vulnerability distinct from PARP. In principle, PUM3 inhibition could provide an alternative or complementary strategy for HR-deficient tumours, including tumours that do not respond to, or become resistant to, PARP inhibition.

However, this is still mechanistic and preclinical work. PUM3 is not yet an established drug target, and an effective selective inhibitor would need to be developed. It would also be necessary to show that normal proliferating tissues can tolerate PUM3 inhibition sufficiently better than HR-deficient tumour cells.

PUM3 dependency in HR-deficient cancer models
≠ an available treatment
The central opportunity:
exploit a new replication-stress dependency
in BRCA/HR-deficient tumours
The central challenge:
achieve enough tumour selectivity
to create a useful therapeutic window

115 [26-09-25] Epigenetically silencing HBV DNA with an LNP-delivered CRISPR system

(Development of an investigational epigenetic silencer therapy to transcriptionally inactivate viral DNA in chronic hepatitis B, <https://www.nature.com/articles/s41551-026-01802-8>)

This paper describes *TUNE-401*, an investigational therapy designed to silence hepatitis-B-virus, or HBV, DNA in liver cells. It uses the LNP–mRNA delivery principle discussed for COVID-19 vaccines and in vivo CAR-T, but the protein made from the mRNA is an epigenetic gene-silencing system.

Chronic HBV is difficult to cure because viral DNA can persist in hepatocytes in two forms: *cccDNA*, a stable circular viral DNA reservoir, and HBV DNA integrated into the host-cell genome. Both can continue producing viral RNA and hepatitis B surface antigen, or HBsAg.

Persistent cccDNA and integrated HBV DNA
→ viral RNA and viral proteins, including HBsAg
Persistent viral antigen production
→ chronic infection and exhausted antiviral immunity

115.1 What does TUNE-401 do?

The treatment uses lipid nanoparticles to deliver mRNA to liver cells. The mRNA encodes a modified CRISPR-associated protein: *dead Cas9*, or dCas9, fused to gene-repressing epigenetic effector domains.

Unlike ordinary Cas9, dCas9 can bind a chosen DNA sequence but cannot cut it. Guide RNAs direct it to a conserved regulatory region in HBV DNA, sometimes described as a viral “master controller.”

LNP delivers mRNA and guide RNA to hepatocyte
→ cell temporarily makes dCas9 epigenetic silencer
Guide RNA directs dCas9 to HBV DNA
→ repressive epigenetic marks are deposited
HBV DNA remains present
but viral gene transcription is switched "off"

The effector domains promote DNA methylation and repressive chromatin formation. This is analogous to the earlier discussion of epigenetics in cancer: the DNA sequence is retained, but made less accessible to the transcription machinery.

115.2 Why silence rather than cut the viral DNA?

A DNA-cutting CRISPR approach might damage cccDNA, but cutting integrated HBV DNA could also create breaks in the patient’s own chromosome. This could cause unintended genomic rearrangements or mutations.

TUNE-401 instead aims to create a durable inactive state without cutting either viral or host DNA.

Conventional genome editing
→ DNA cleavage
→ possible permanent sequence change
+ risk from unintended DNA breaks
Epigenetic silencing
→ no intended DNA cleavage
→ reduced viral transcription
+ aims for durable repression without DNA damage

115.3 What should be concluded?

The therapeutic goal is a *functional cure*: sustained control of HBV, including loss of clinically relevant viral antigen, after a finite treatment course. It is not necessarily elimination of every viral-DNA molecule.

This is an important use of LNP–mRNA delivery beyond vaccination: the LNP provides temporary instructions, but the protein it produces may establish longer-lasting epigenetic repression of the viral reservoir.

However, the paper is primarily preclinical. Key questions include the durability and completeness of silencing in human liver cells, delivery to enough infected hepatocytes, possible off-target epigenetic effects, immune reactions to the CRISPR components, and whether viral suppression can be sustained after treatment stops.

Reduced HBV transcription in experimental models
≠ established functional cure in people
Transient mRNA delivery
→ can potentially produce a durable effect
if the induced epigenetic state persists

116 [26-09-25] The evolving landscape of drug targets

(*The evolving landscape of drug targets*, <https://www.nature.com/articles/s41573-026-01530-3>)

This review maps the proteins and other biological entities targeted by approved medicines, using drug approvals through June 2025. Its central message is that drug discovery has expanded beyond the historically dominant target classes, but approved medicines still act on a relatively small and non-random part of human biology.

Many successful small-molecule drugs act on targets with clear binding pockets, particularly enzymes, G-protein-coupled receptors, ion channels, and nuclear receptors.

Protein has accessible binding pocket
→ conventional small molecule can bind and alter function
No obvious pocket or intracellular scaffold function
→ conventional inhibition becomes more difficult

This connects to the molecular-glue and targeted-degradation examples in this document. New modalities can make previously difficult proteins more tractable:

Small molecule
→ often inhibits enzyme or receptor activity
Antibody or biologic
→ can bind extracellular proteins or cell-surface targets
Molecular glue or PROTAC-like strategy
→ can remove selected intracellular proteins

The review also shows that target choice is shaped by therapeutic area. Oncology and immunology have driven much recent target innovation, including kinases, immune checkpoints, cytokines, cell-surface antigens, and genetically defined cancer dependencies. However, a newly drugged target is not necessarily a highly novel biological discovery: it may be a better way to intervene in a well-known pathway.

The practical lesson is that a *target* is not simply a protein associated with disease. It must also be linked convincingly to disease biology, modifiable by a feasible drug modality, exposed in the relevant tissue, and alterable with an acceptable therapeutic window.

Disease association
≠ viable drug target
Viable target requires:
causal biological rationale
+ tractable intervention strategy
+ appropriate tissue exposure
+ sufficient safety margin

Thus, the expanding target landscape is encouraging, but it does not mean that all disease-relevant proteins are now readily druggable. The major opportunity is to match emerging modalities with biological mechanisms that conventional drugs could not previously address.

117 [26-09-25] Animal and human disease models: what they can, and cannot, predict

(*Can Animal Models of Disease Reliably Inform Human Studies?*, <https://journals.plos.org/plosmedicine/article?id=10.1371/journal.pmed.1000245>; *Role of animal models in biomedical research: a review*, <https://link.springer.com/article/10.1186/s42826-022-00128-1>; *Human disease models in drug development*, <https://www.nature.com/articles/s44222-023-00063-3>)

Together, these papers address a central problem in drug development: a treatment can produce a convincing result in a disease model but still fail in human trials. Animal models remain essential sources of biological understanding and safety information, but they are *models* of human disease rather than miniature patients.

```
Result in cell or animal model
→ evidence for a biological hypothesis
Result in patients
→ evidence of clinical benefit and safety
Therefore:
Preclinical efficacy
≠ reliable proof of human therapeutic benefit
```

117.1 Why are animal models used?

Animal models allow researchers to study a whole living organism: drug absorption and metabolism, blood flow, tissue interactions, immune responses, toxicity, tumour growth, and disease progression. They also allow controlled experiments that would be impossible or unethical in humans.

For example, a mouse cancer model can test whether a treatment reaches a tumour, engages its target, changes tumour growth, and produces obvious toxicity. A genetically engineered mouse can test whether a mutation such as *KRAS* or *BRCA1* contributes causally to cancer formation.

```
Alter gene or administer treatment in animal
→ observe molecular, cellular,
whole-body, and disease-level consequences
This can establish:
biological mechanism
+ preliminary efficacy
+ preliminary safety
```

The important point is that animal studies are often highly valuable for deciding *what to test next*. They are much less reliable for proving exactly how effective or safe a medicine will be in patients.

117.2 Why can animal results fail to translate?

Humans and laboratory animals share much biology, but differ in anatomy, lifespan, metabolism, immune systems, gene regulation, microbiomes, and disease context. A disease induced rapidly in a young, genetically similar laboratory mouse may not reproduce a disease that develops over decades in genetically diverse people with ageing, co-morbidities, prior treatments, and environmental exposures.

Human disease

→ heterogeneous genetics, age, environment,
co-morbidities, and disease history

Laboratory animal model

→ controlled genetics and environment,
simplified or induced disease process

Useful experimental control

≠ complete representation of patient biology

This issue is especially important in cancer and immunology. Human tumours evolve over time, contain diverse cancer-cell clones, and interact with a complex immune and stromal microenvironment. Mouse tumours may have different mutation patterns, growth rates, antigens, immune-cell composition, and drug sensitivities.

Likewise, a human immune-checkpoint antibody may not bind the corresponding mouse protein, requiring use of a mouse-adapted antibody or a specialised humanised model. Each modification can make the experiment more useful for one question while introducing new limitations.

117.3 What did the PLOS Medicine paper show?

The PLOS Medicine paper reviewed comparisons between animal experiments and subsequent human studies. Its key conclusion was that agreement was often incomplete or uncertain, with some interventions showing similar effects in animals and humans, but others showing weak correspondence or opposing results.

The paper does *not* show that animal research is useless. Rather, it challenges the assumption that a positive animal experiment automatically predicts a successful human trial.

A major concern was study quality. Many preclinical studies historically used small sample sizes, limited randomisation or blinding, selective reporting, and incomplete publication of negative results. These problems can exaggerate apparent treatment effects.

Small, non-randomised, unblinded animal study

→ increased risk of biased efficacy estimate

Publication mainly of positive results

→ apparent preclinical success may be inflated

Inflated preclinical effect

→ greater risk of failed human trial

Thus, poor translation can arise for two different reasons: the model may not sufficiently represent the relevant human biology, or the animal evidence itself may be methodologically unreliable. Often, both contribute.

117.4 What are human disease models?

The third paper discusses a growing set of *human-relevant models* intended to complement animal studies. These include patient-derived cells, induced pluripotent stem cells, organoids, tissue explants, organs-on-chips, and computational models.

An *organoid* is a three-dimensional cell culture system that reproduces some features of an organ or tumour. For example, a patient-derived tumour organoid may retain aspects of that patient's tumour genetics and drug sensitivity.

An *organ-on-chip* is a microfluidic device containing living human cells arranged to reproduce selected physical and biological features of an organ, such as fluid flow, tissue barriers, mechanical forces, or interactions between two tissue types.

```
Patient-derived cells or organoid
→ retains some human genetics and disease biology
Organ-on-chip
→ models selected human tissue interactions
under controlled physical conditions
Human model
→ may improve relevance for a specific question
```

These systems can be particularly useful for studying human-specific biology, including drug metabolism, liver toxicity, blood–brain-barrier transport, immune-cell interactions, or patient-to-patient variation. They also connect to the earlier discussion of AI-native discovery: informative human perturbation data may help identify causal mechanisms and prioritise better drug hypotheses.

117.5 Why will human models not simply replace animals?

Human models also have limitations. An organoid may preserve tumour cells but lack blood vessels, nerves, systemic metabolism, a complete immune system, and the complex tissue architecture of a patient. An organ-on-chip can model a carefully selected interaction, but not an entire human body over years of disease progression.

```
Animal model
→ whole-organism physiology
but imperfect human relevance
Human cell-based model
→ closer human molecular biology
but incomplete whole-body physiology
Best approach:
use complementary models for clearly defined questions
```

The practical goal is therefore not a single “best” disease model. It is a *model portfolio*: use the simplest system that can answer a mechanistic question, then test the finding across increasingly human-relevant systems before entering clinical trials.

For example, a cancer programme might progress through engineered cell lines, patient-derived organoids, immune co-cultures, mouse pharmacology and safety studies, and finally biomarker-rich early clinical trials.

117.6 What should be concluded?

Animal models have made major contributions to biomedical science, but they should be interpreted as evidence with defined scope rather than definitive forecasts of human outcomes. A strong translational package requires consistency across several systems, robust experimental design, a clear mechanism, clinically relevant drug exposure, and biomarkers that can be measured in patients.

```
Positive result across:
human cells
+ patient-derived models
+ appropriate animal studies
+ mechanistic biomarkers
→ stronger rationale for clinical testing
But:
Even extensive preclinical evidence
≠ proof of patient benefit
The definitive test of a medicine remains
a well-designed clinical study in people
```

118 [26-09-25] Trastuzumab plus nivolumab and chemotherapy in HER2-positive biliary-tract cancer

(*Trastuzumab plus nivolumab with gemcitabine and cisplatin as first-line therapy for HER2-positive unresectable biliary tract cancer: a phase 1b/2 trial*, <https://www.nature.com/articles/s41591-026-04679-x>)

This Phase Ib/II study tested four drugs as first-line treatment for *HER2-positive unresectable biliary-tract cancer*, or BTC. BTC includes cancers arising in the bile ducts and gallbladder. “Unresectable” means that surgery is not considered able to remove the cancer completely.

The combination addresses four different but connected aspects of tumour biology: HER2-driven growth, tumour-cell killing by chemotherapy, and anti-tumour immune activation.

```
Trastuzumab
→ targets HER2-driven tumour signalling
Gemcitabine + cisplatin
→ directly damage proliferating tumour cells
Nivolumab
→ releases PD-1-mediated inhibition of T cells
```

118.1 What does trastuzumab do?

HER2 is a cell-surface growth-factor receptor. In HER2-positive tumours, excess HER2 protein can cause persistent growth and survival signalling.

Trastuzumab is an antibody that binds HER2. It can reduce HER2-associated signalling and mark HER2-expressing tumour cells for immune-mediated killing, particularly through antibody-dependent cellular cytotoxicity.

```
HER2 overexpression
→ excessive tumour-cell growth signalling
Trastuzumab binds HER2
→ reduces HER2-associated signalling
+ recruits immune effector cells to HER2-positive tumour cells
```

This is analogous to HER2 targeting in breast and gastric cancer, but HER2 positivity occurs in only a subset of biliary-tract cancers. It is therefore a *biomarker-selected* strategy.

118.2 What do gemcitabine and cisplatin do?

Gemcitabine is a nucleoside analogue: it resembles a normal DNA building block. During DNA replication, tumour cells can incorporate it into DNA, which disrupts DNA synthesis and causes replication stress.

Cisplatin is a platinum chemotherapy that forms chemical crosslinks within DNA. These DNA lesions prevent normal copying and transcription of DNA.

```
Gemcitabine
→ impaired DNA synthesis and replication stress
Cisplatin
→ DNA crosslinks and DNA damage
Together
→ tumour cells cannot replicate or repair DNA effectively
→ cell death may occur
```

Gemcitabine plus cisplatin is a long-established chemotherapy backbone for advanced BTC. It is not selective for tumour cells, however, so it can also affect normal rapidly dividing tissues, causing effects such as low blood counts, infection risk, fatigue, nausea, and kidney toxicity.

118.3 What does nivolumab do?

Nivolumab is an antibody against *PD-1*, an inhibitory receptor on activated T cells. PD-1 signalling normally helps limit immune responses. Tumours can exploit it to restrain tumour-reactive T cells.

```
Tumour-associated PD-L1 engages PD-1 on T cell
→ reduced T-cell activity
Nivolumab blocks PD-1
→ removes one inhibitory signal
→ T cells may better recognise and kill tumour cells
```

Nivolumab does not directly kill tumour cells. It helps pre-existing or treatment-induced anti-tumour T-cell responses function more effectively.

118.4 Why combine all four drugs?

The logic is that chemotherapy reduces tumour burden and may release tumour antigens; trastuzumab blocks a defined HER2 growth driver and can recruit innate immune killing; and nivolumab helps sustain T-cell activity against the tumour.

```
Chemotherapy
→ direct killing and tumour-antigen release
Trastuzumab
→ HER2 pathway blockade
+ antibody-mediated immune recruitment
Nivolumab
→ reduced PD-1 immune inhibition
Combination
→ direct cytotoxicity
+ HER2-targeted pressure
+ enhanced anti-tumour immunity
```

This is a four-drug regimen because each drug has a different intended role. However, more drugs do not automatically mean a better treatment: the combination must improve efficacy enough to justify added toxicity, cost, and treatment complexity.

118.5 What did the trial show?

The HERBOT trial enrolled 40 patients. It reported an objective response rate of **55%**, including one complete response, with a median progression-free survival of **10.6 months** and median response duration of **12.6 months**. At the reported follow-up, median overall survival had not yet been reached.

Higher HER2 expression appeared associated with greater benefit, which is biologically plausible: trastuzumab requires sufficient HER2 on tumour cells to act effectively.

HER2-positive BTC
→ plausible target for trastuzumab-containing therapy
Early clinical activity of the four-drug regimen
→ encouraging
But:
Single-arm Phase Ib/II trial
≠ proof that the regimen is superior
to chemotherapy plus immunotherapy alone

The key next step is a comparative trial that determines whether adding trastuzumab provides enough additional and durable benefit in HER2-positive BTC to outweigh its added treatment burden and cardiac-monitoring requirements.

119 [26-09-25] p38 α links BRAF-mutant melanoma fitness to immune suppression

(p38 α orchestrates BRAF^{V600E}-melanoma progression and immunotherapy response via tumor fitness and immune microenvironment modulation, <https://www.nature.com/articles/s41467-026-77868-4>)

This study examines p38 α , also called MAPK14, in BRAF^{V600E}-mutant melanoma. It shows that p38 α can support both tumour-cell fitness and an immune-suppressive tumour microenvironment, making it a possible combination target with checkpoint blockade.

p38 α is a stress-activated protein kinase. It can respond to inflammation, DNA damage, oxidative stress, and oncogenic signalling. Its effects are strongly context-dependent: early in cancer formation, p38 α can help impose *oncogene-induced senescence*, a protective growth arrest. In established melanoma, however, the same pathway can be co-opted to support tumour progression.

BRAF^{V600E} signalling
→ abnormal MAPK-pathway activation
→ melanoma growth and survival
p38 α activity in established tumour
→ supports tumour-cell fitness
+ shapes an immune-suppressive microenvironment

The tumour-cell effect concerns *fitness*: the capacity to survive, proliferate, and tolerate stress in the tumour environment. Loss or inhibition of p38 α made BRAF^{V600E} melanoma cells less fit, reducing their ability to sustain tumour growth.

p38 α also influenced immune context. Tumour-cell p38 α activity was associated with reduced effective cytotoxic immune activity and with recruitment or maintenance of suppressive myeloid cells. Thus, it can help the tumour escape immunity indirectly, not only by changing cancer-cell-intrinsic signalling.

Tumour-cell p38 α activity
→ tumour-supportive inflammatory signals
+ less favourable environment for cytotoxic T cells
p38 α loss or inhibition
→ weaker tumour-cell fitness
+ more immune-permissive tumour microenvironment

This provides a rationale for combining p38 α inhibition with PD-1 checkpoint blockade. PD-1 inhibition releases an inhibitory brake on tumour-reactive T cells, whereas p38 α inhibition may make the tumour both intrinsically weaker and less able to create an immune-suppressive local environment.

```
Anti-PD-1 treatment
→ restores some T-cell activity
p38α inhibition
→ reduces tumour fitness
+ reduces local immune suppression
Combination
→ potential for improved immune-mediated
control of BRAF-mutant melanoma
```

The important lesson is that pathways active inside tumour cells can also shape the behaviour of nearby immune cells. However, this remains mainly mechanistic and preclinical evidence. Because p38α also contributes to normal stress and inflammatory responses, systemic inhibition could affect immune function and normal tissues.

```
Improved checkpoint response in experimental models
≠ established clinical benefit
The translational question:
can p38α be inhibited at a dose and schedule
that weakens melanoma and improves immunotherapy
without unacceptable toxicity?
```

120 [26-09-25] Public and private TCRs can recognise the same viral target differently

(Unveiling structural determinants of peptide recognition by public and private SARS-CoV-2-specific T-cell receptors, <https://www.nature.com/articles/s41467-026-77827-z>)

This structural-immunology study asks how different T-cell receptors, or TCRs, can recognise the same SARS-CoV-2 peptide-MHC target.

Recall that a conventional CD8 T cell recognises a short viral peptide displayed by a particular MHC class-I molecule. Here, the target was an immunodominant peptide from the SARS-CoV-2 protein ORF3a, presented by HLA-A*01:01.

```
Viral protein
→ processed into peptide
→ peptide displayed by HLA molecule
→ compatible TCR activates CD8 T cell
```

120.1 What are “public” and “private” TCRs?

A *public* TCR is a similar or identical receptor sequence found in multiple unrelated people after exposure to the same antigen. A *private* TCR is found only in one individual.

Public TCRs are more likely to arise when their gene segments can recombine relatively easily during normal T-cell development, and when they recognise a common viral target effectively.

```
Public TCR
→ appears in several people
→ reproducible immune solution to a shared antigen
Private TCR
→ unique to one individual
→ personalised immune solution
```

120.2 What did the study find?

The authors compared one public and one private SARS-CoV-2-specific TCR using cryo-electron microscopy. Both recognised the same peptide–MHC complex and used a very similar TCR beta chain, but they used different alpha chains.

Despite similar overall binding and T-cell function, they contacted the peptide–MHC surface differently.

```
Public TCR
→ more peptide-focused recognition
Private TCR
→ contacts distributed more broadly
across the peptide-MHC complex
Different recognition geometry
→ similar functional viral recognition
```

The key structural determinant was the TCR alpha chain. Changing the alpha chain altered how the full receptor was positioned over peptide–MHC, and therefore which parts of the peptide and MHC were contacted.

120.3 Why does this matter?

This builds on the earlier OmniTCR discussion. A TCR sequence cannot be understood simply as a direct measure of affinity for one peptide. Recognition depends on the combined structure of both TCR chains and the peptide–MHC surface.

```
Same viral peptide-MHC target
≠ one required TCR binding solution
Different TCR alpha-beta pairings
→ different recognition geometries
→ potentially different cross-reactivity profiles
```

This matters for TCR-T-cell therapy because two TCRs that both kill a desired target in vitro may have different unintended recognition of related peptides on normal tissues.

120.4 What should be concluded?

The paper shows that public and private T-cell responses are not simply stronger or weaker versions of one another. They can be distinct structural solutions to the same antigen-recognition problem.

This is primarily a mechanistic study, but it reinforces why therapeutic TCR design requires structural and functional testing for each full alpha–beta receptor pair.

```
TCR binding to intended peptide-MHC
≠ complete understanding of specificity
Therapeutic TCR development still requires:
broad cross-reactivity testing
+ functional testing in relevant human cells
+ careful clinical safety assessment
```

121 [26-09-25] Microglial CK2 may protect against obesity through NLRP3 inflammasome signalling

(Microglial protein kinase CK2 prevents obesity in mice potentially through NLRP3 inflammasome activation, <https://www.nature.com/articles/s41467-026-77895-1>)

This mouse study identifies *CK2*, a protein kinase in brain-resident immune cells called *microglia*, as a regulator of body-weight control. The central and slightly counterintuitive finding is that microglial CK2 activity may protect against diet-induced obesity partly by activating the *NLRP3 inflammasome*.

Microglia are the immune cells of the central nervous system. In regions such as the hypothalamus, they can respond to nutritional state, inflammation, and signals from neurons involved in appetite and energy expenditure.

High-fat diet and metabolic stress
→ altered hypothalamic immune and neuronal signalling
→ disrupted control of food intake
and energy expenditure
→ obesity risk

CK2 is a broadly acting kinase: it adds phosphate groups to proteins and can alter signalling, protein stability, and gene expression. In this study, loss of CK2 specifically in microglia promoted greater weight gain and metabolic dysfunction in mice.

Microglial CK2 activity
→ supports protective hypothalamic immune signalling
→ helps maintain energy-balance regulation
Loss of microglial CK2
→ impaired protective response
→ increased susceptibility to obesity in mice

The proposed mechanism involves the NLRP3 inflammasome. An *inflammasome* is a protein complex that detects particular forms of cellular stress and can activate inflammatory cytokines such as IL-1 β .

NLRP3 is usually discussed as a driver of harmful chronic inflammation in obesity, diabetes, and cardiovascular disease. This paper instead suggests that its effect depends on *cell type, anatomical location, timing, and degree of activation*. In microglia, a controlled CK2–NLRP3 response may support protective communication with hypothalamic circuits.

CK2 in microglia
→ NLRP3 inflammasome activation
→ local inflammatory signalling
In this specific mouse context
→ improved regulation of energy balance
→ protection from obesity

The main lesson is that inflammation is not intrinsically harmful or beneficial. An inflammatory pathway that contributes to chronic metabolic disease in peripheral tissues can have a distinct, potentially protective role in a particular brain-cell population.

However, this is mechanistic work in mice, not a treatment strategy. CK2 and NLRP3 have many functions throughout the body, and broad activation of either could produce harmful inflammation or other toxicity.

Microglial CK2–NLRP3 activity protects mice from obesity
≠ activating NLRP3 is a safe anti-obesity treatment
The translational challenge:
identify whether an equivalent human microglial pathway exists
and whether it can be modulated selectively
without systemic inflammatory harm

122 [26-09-25] A bacterial toxin domain repurposed to target BCAM-positive tumours

(Structural basis of laminin mimicry by *E. coli* CNF1 and its therapeutic potential for cancer immunotherapy, <https://www.nature.com/articles/s41467-026-77975-2>)

This study shows how a domain from the *E. coli* toxin CNF1 can mimic *laminin*, a normal component of the extracellular matrix, to bind the human receptor *BCAM*.

Laminin normally helps cells attach to their surrounding tissue environment. BCAM, also called Lutheran glycoprotein, is a cell-surface laminin receptor. The authors found that the D5 binding domain of CNF1 engages BCAM in a structurally similar way to laminin, but with sufficiently high affinity to compete with the normal ligand.

```
Normal situation:
Laminin binds BCAM
→ cell adhesion and extracellular-matrix interaction
Bacterial infection:
CNF1 D5 mimics laminin
→ binds BCAM
→ bacterial toxin gains access to host-cell pathway
```

The therapeutic idea is to retain the BCAM-binding D5 domain, while removing the toxin's harmful enzymatic functions. The resulting engineered protein becomes a tumour-targeting module rather than a toxin.

122.1 How was this used for immunotherapy?

BCAM is expressed by some tumour cells, including experimental ovarian-cancer models. The authors fused CNF1 D5 to a CD3-binding arm, creating a *T-cell engager*, or TCE.

As in other T-cell engagers, one binding arm attaches to a tumour-associated target and the other activates nearby T cells through CD3.

```
CNF1 D5 arm binds BCAM on tumour cell
+
CD3-binding arm binds T cell
→ tumour cell and T cell are brought together
→ immune synapse forms
→ T-cell cytotoxicity and granzyme-mediated killing
```

Thus, the unusual feature is not the basic TCE mechanism. It is the source of the tumour-binding domain: a bacterial protein fragment that has evolved to mimic a human extracellular-matrix ligand.

122.2 What should be concluded?

This is an elegant example of using pathogen biology as a source of therapeutic targeting tools. Structural information allowed the authors to isolate a BCAM-binding part of CNF1 and repurpose it for anti-tumour T-cell recruitment.

However, it remains preclinical. BCAM is not tumour-exclusive, and the therapeutic window will depend on its expression across patient tumours versus normal tissues, as well as the risk of cytokine-mediated toxicity and immunogenicity against the bacterial-derived binding domain.

Tumour control in experimental models
≠ safe clinical T-cell engager
Pathogen-derived target binder
→ promising targeting module,
but requires rigorous normal-tissue and immune-safety testing

123 [26-09-25] Persistent BCMA CAR-T cells and delayed enterocolitis

(*Persistence of mucosal CAR-T cells and inflammatory remodeling in enterocolitis associated with BCMA CAR-T cell therapy*, <https://www.nature.com/articles/s41591-026-04632-y>)

This study examines *enterocolitis*, or inflammation affecting the small intestine and colon, occurring after BCMA-directed CAR-T-cell therapy. It provides evidence that this complication can involve persistent CAR-T cells in the gut mucosa together with substantial remodelling of the local immune environment.

BCMA CAR-T cells are designed to recognise *B-cell maturation antigen*, or BCMA, which is highly expressed by plasma cells and many multiple-myeloma cells. Their intended activity is:

BCMA CAR-T cell recognises BCMA-positive myeloma cell
→ CAR-T activation
→ myeloma-cell killing

However, CAR-T cells can persist long after infusion. Persistent activity is often desirable for durable myeloma control, but it can also create delayed toxicity if CAR-T cells enter non-tumour tissues or induce prolonged immune disturbance.

CAR-T cells persist after treatment
→ continued anti-myeloma surveillance
But in some patients:
CAR-T cells accumulate in intestinal mucosa
→ local immune activation and inflammation
→ enterocolitis

The intestinal mucosa normally contains a highly regulated immune system. It must tolerate food, commensal microbes, and harmless environmental antigens while remaining able to control infections. The study suggests that BCMA CAR-T-associated enterocolitis involves disruption of this balance, with CAR-T-cell persistence and broader inflammatory changes involving other immune-cell populations.

This is not necessarily simple *on-target*, *off-tumour* toxicity. Although normal plasma cells can express BCMA, the proposed pathology appears to involve a more complex inflammatory tissue response rather than only direct killing of one BCMA-positive normal cell type.

Persistent CAR-T activity in gut
+ altered local immune-cell composition
+ inflammatory cytokine signalling
→ breakdown of normal mucosal immune regulation
→ chronic or delayed intestinal inflammation

The paper is important because it shows that CAR-T toxicities are not limited to early cytokine-release syndrome, neurotoxicity, or infection risk from B-cell/plasma-cell depletion. Some toxicities may arise later through the long-term tissue distribution and activity of the engineered cells.

Durable CAR-T persistence
→ potentially durable anti-cancer benefit
But:
Durable persistence
≠ uniformly beneficial in every tissue
Long-term follow-up
→ needed to identify delayed,
tissue-specific immune toxicities

The practical implication is that severe or persistent gastrointestinal symptoms after BCMA CAR-T therapy should be investigated for inflammatory enterocolitis rather than assumed to be infection, chemotherapy-related injury, or unrelated gastrointestinal disease. The key unanswered question is how to control pathological gut inflammation without unnecessarily eliminating CAR-T cells and compromising anti-myeloma benefit.

124 [26-09-25] Targeting ZMYND8 may restore IL-2 signalling in exhausted T cells

(Targeting ZMYND8 unleashes IL-2 signalling to override T cell exhaustion, <https://www.nature.com/articles/s41586-026-11059-5>)

This study identifies ZMYND8 as an internal molecular brake that helps drive CD8 T-cell exhaustion. It connects directly to the earlier discussion of IL-2: IL-2 supports T-cell proliferation, survival, and cytotoxic function, but chronically stimulated T cells in tumours can progressively lose their ability to respond to it effectively.

Persistent tumour-antigen stimulation
→ T-cell exhaustion programme
Exhausted T cell
→ weaker proliferation, cytokine production,
and tumour-cell killing

124.1 What does ZMYND8 do?

ZMYND8 is a chromatin-associated regulatory protein. Rather than acting as a cell-surface checkpoint such as PD-1, it acts inside the T cell to alter gene expression.

The study shows that ZMYND8 restrains expression of the IL-2 receptor by interfering with *p300*, a transcriptional co-activator. p300 helps make chromatin more permissive for gene expression, partly through histone acetylation.

ZMYND8 inhibits p300-associated gene activation
→ reduced IL-2-receptor expression
Reduced IL-2 signalling
→ weaker T-cell expansion and effector function
→ exhaustion is reinforced

Thus, ZMYND8 does not remove IL-2 itself. It reduces the T cell's capacity to receive and respond to IL-2.

124.2 Why could inhibiting ZMYND8 help?

Removing ZMYND8 in experimental CD8 T cells increased IL-2-receptor signalling and helped cells retain a more functional anti-tumour state despite chronic stimulation.

ZMYND8 inhibition or deletion

- p300 activity is less restrained
- IL-2-receptor expression and signalling increase
- exhausted T cells may regain function

This differs from PD-1 blockade. Nivolumab blocks an *external inhibitory receptor signal*; ZMYND8 targeting would alter an *internal chromatin and transcriptional programme* that contributes to exhaustion. The approaches could therefore be complementary.

124.3 What should be concluded?

The paper suggests that T-cell exhaustion is not fixed. It can potentially be reprogrammed by changing the gene-regulatory mechanisms that determine how T cells respond to cytokines such as IL-2.

However, this is mainly a genetic and preclinical result. Deleting *ZMYND8* experimentally is not the same as having a selective, safe drug that inhibits it in patient T cells. Increasing IL-2 responsiveness could also increase unwanted immune activation.

ZMYND8 loss improves anti-tumour T-cell function in models

≠ established treatment for cancer

Potential future strategy:

checkpoint blockade

+

carefully controlled IL-2 support

+

ZMYND8-targeted T-cell reprogramming

125 [26-09-25] Bitopic kinase inhibitors: one molecule engaging two sites on BCR::ABL1

(A design approach for bitopic kinase inhibitors, <https://www.nature.com/articles/s41586-026-11056-8>)

This study presents a systematic strategy for making *bitopic kinase inhibitors*: single molecules designed to bind two distinct sites on the same kinase at the same time.

The authors apply this to *BCR::ABL1*, the fusion kinase that drives chronic myeloid leukaemia, or CML. BCR::ABL1 continuously activates growth and survival signalling, and can acquire mutations that make it resistant to conventional kinase inhibitors.

BCR::ABL1 fusion kinase

- persistent growth signalling
- CML-cell survival and proliferation

Most kinase inhibitors bind the *ATP-binding pocket*, the site where kinases normally bind ATP to transfer phosphate groups to other proteins. This can be effective, but the ATP pocket is similar across many kinases, creating selectivity challenges. Mutations within this pocket can also reduce drug binding.

The authors instead link two established binding components:

```
Ponatinib component
→ binds BCR::ABL1 ATP pocket
Asciminib component
→ binds the distant allosteric myristoyl pocket
Link both components
→ one bitopic molecule binds both sites
```

The resulting compounds, termed *PonatiLinks*, can bridge the ATP pocket and the allosteric myristoyl pocket simultaneously. This is not simply equivalent to giving ponatinib and asciminib as two separate drugs: once one end of the linked molecule binds, the other end is held at high local concentration near its second binding site.

```
First binding interaction
→ anchors molecule to BCR::ABL1
Second binding interaction
→ becomes more likely because it is tethered nearby
Two-site binding
→ stronger and potentially more selective inhibition
```

The allosteric myristoyl pocket normally helps regulate ABL1 activity. Binding there can stabilise an inactive kinase conformation. Therefore, the bitopic inhibitor both obstructs ATP-dependent catalysis and reinforces an inactive structural state.

This dual interaction may also help address resistance. For example, the *T315I* gatekeeper mutation makes many ATP-site inhibitors ineffective, but ponatinib retains activity against it. A linked inhibitor can add the additional constraint of allosteric-pocket binding, potentially making it harder for the kinase to escape through a single resistance mutation.

```
Single-site inhibitor
→ one binding-site mutation may cause resistance
Bitopic inhibitor
→ resistance may require disruption of
two coordinated binding interactions
```

The broader contribution is a *design principle*, rather than an approved new medicine. Structure, linker length, flexibility, and the relative orientation of the two binding elements are all crucial: simply joining two inhibitors will not necessarily produce a cell-permeable, potent, selective, and drug-like compound.

```
Two ligands that bind individually
≠ successful bitopic drug when linked
A useful bitopic inhibitor still requires:
compatible geometry
+ appropriate linker
+ cell permeability
+ selectivity
+ pharmacokinetics and safety
```

This is an impressive chemical-biology advance because it offers a general way to exploit both conserved catalytic pockets and more selective regulatory pockets. Its clinical value remains to be established, but the approach could be particularly useful for kinases where resistance to single-site inhibition is a major limitation.

126 [26-09-25] Cholesterol-coated platinum nanodrug targets stromal fibroblasts to improve tumour penetration

(Surface-displayed cholesterol enhances penetration and anti-tumor activity of a platinum nanodrug by targeting NPC1L1-expressing cancer associated fibroblasts, <https://www.nature.com/articles/s41467-026-78085-9>)

This paper addresses a major obstacle in solid-tumour treatment: drugs often cannot penetrate far beyond tumour blood vessels. A dense tumour stroma, particularly in cancers such as pancreatic cancer, can act as a physical and biological barrier.

The authors modify a platinum-containing nanodrug with *cholesterol* displayed on its surface. This enables the particle to interact with *NPC1L1*, a cholesterol-uptake receptor expressed by a subset of *cancer-associated fibroblasts*, or CAFs.

```
Cholesterol-coated platinum nanoparticle
→ binds NPC1L1-positive CAFs
CAF targeting
→ altered tumour-stroma barrier
→ deeper nanoparticle penetration into tumour
More platinum reaches tumour cells
→ increased DNA damage
→ enhanced anti-tumour activity in models
```

126.1 Why target CAFs rather than only cancer cells?

CAFs are non-malignant cells within the tumour microenvironment. They can produce extracellular matrix, including collagen, and contribute to a dense, poorly perfused tissue structure. This can limit entry of chemotherapy, immune cells, antibodies, and nanoparticles.

```
CAF activity
→ dense extracellular matrix and stromal barrier
Dense stroma
→ reduced drug movement through tumour tissue
Targeting an appropriate CAF population
→ may improve access to cancer cells
```

The strategy is therefore indirect. NPC1L1 is not mainly used as a tumour-cell target; it is used as a way to direct the nanodrug towards stromal cells that restrict drug distribution.

126.2 What does the platinum component do?

Platinum drugs damage DNA by forming crosslinks between DNA strands. This impairs DNA replication and transcription, eventually causing cell death if the damage cannot be repaired.

```
Platinum reaches tumour cell
→ DNA crosslinks form
→ DNA replication and repair are disrupted
→ tumour-cell death may occur
```

The proposed advantage is not that cholesterol changes platinum's fundamental DNA-damaging mechanism. It changes *where the drug goes within the tumour* and may therefore increase the fraction of cancer cells exposed to an effective dose.

126.3 What should be concluded?

This is an example of using a tumour-microenvironment feature to improve delivery of a conventional cytotoxic mechanism.

More drug administered
≠ necessarily better tumour treatment
Better intratumoural distribution
→ may increase efficacy without proportionally increasing systemic exposure

However, this remains preclinical nanomedicine research. NPC1L1 expression, CAF populations, stromal architecture, and nanoparticle distribution may differ substantially between animal models and patient tumours. A useful clinical product would still need reproducible manufacturing, adequate tumour selectivity, acceptable platinum-related toxicity, and evidence of meaningful benefit in patients.

127 [26-09-25] Replication stress: a cancer vulnerability and therapeutic opportunity

(*Replication stress in cancer: origins, consequences and therapeutic opportunities*, <https://www.nature.com/articles/s41568-026-00979-z>)

This review explains *replication stress*: problems that arise when a cell is copying its DNA before division. Cancer cells commonly experience high replication stress because oncogenic signalling drives rapid, poorly coordinated proliferation.

Oncogene activation, such as MYC or RAS
→ excessive drive to enter DNA replication
Limited nucleotides, DNA lesions, or difficult DNA regions
→ replication forks slow or stall
→ replication stress

A *replication fork* is the moving molecular machinery that separates and copies DNA. If a fork stalls and cannot be stabilised or restarted, it may collapse, producing DNA breaks, chromosome instability, and potentially cell death.

Cancer cells can tolerate this stress because they activate protective DNA-damage-response pathways, particularly ATR–CHK1 and WEE1 signalling. These pathways slow the cell cycle, stabilise stalled forks, coordinate repair, and give the cell time to complete DNA replication.

Replication stress
→ ATR–CHK1 and WEE1 checkpoint activation
→ fork stabilisation, cell-cycle delay, and repair
→ cancer-cell survival despite DNA damage

This creates a therapeutic opportunity. A cancer cell with unusually high baseline replication stress may depend more strongly than normal cells on these protective pathways.

Tumour cell with high replication stress
+ ATR, CHK1, WEE1, or related inhibitor
→ unresolved DNA replication problems
→ fork collapse and chromosome damage
→ selective tumour-cell death may occur

This is related to the earlier examples of BRCA deficiency, PARP inhibition, TET2-associated ssDNA gaps, and PUM3 dependency. In each case, a tumour begins with impaired DNA maintenance and therefore becomes reliant on a remaining compensatory pathway.

Replication stress can also affect anti-tumour immunity. Unresolved DNA damage and chromosome mis-segregation can generate cytoplasmic DNA, potentially activating cGAS-STING and inflammatory signalling. However, as discussed for ATR inhibition in Merkel cell carcinoma, DDR-targeted therapies can alter immunity through several mechanisms, and increased DNA damage does not guarantee useful immune activation.

Replication-stress-targeting therapy
→ direct tumour-cell damage
+ possible inflammatory or immunogenic signalling
But:
DNA-damage induction
≠ reliable response to immunotherapy

The main clinical challenge is the therapeutic window. Normal tissues that divide rapidly, especially bone marrow and gastrointestinal tissue, also rely on replication-stress responses. ATR, CHK1, and WEE1 inhibitors can therefore cause toxicities such as myelosuppression.

The review's central message is that replication stress is both a hallmark of cancer and a potential selective weakness. The goal is to identify tumours with sufficient dependence on a particular replication-stress pathway, then inhibit that pathway at a dose and schedule that damages tumour cells more than normal proliferating tissues.

128 [26-09-25] TRIM21 suppresses breast cancer by degrading OGT

(*TRIM21-mediated OGT degradation inhibits breast cancer tumorigenesis and metastasis*, <https://www.nature.com/articles/s41523-026-01057-1>)

This study identifies *TRIM21* as a tumour-suppressive regulator in breast cancer. Its central mechanism is degradation of *OGT*, an enzyme that promotes a protein modification called *O-GlcNAcylation*.

OGT adds a small sugar-like chemical group, O-GlcNAc, to proteins. Like phosphorylation, this can alter a protein's stability, location, interactions, or activity. Cancer cells can use increased O-GlcNAcylation to support growth, metabolic adaptation, and metastasis.

High OGT activity
→ increased O-GlcNAcylation of cellular proteins
→ tumour-supportive signalling and cell-state programmes
→ growth and metastatic potential

TRIM21 is an *E3 ubiquitin ligase*: it helps attach ubiquitin to selected proteins, which can label them for destruction by the proteasome.

TRIM21 binds OGT
→ OGT ubiquitination
Ubiquitinated OGT
→ proteasomal degradation
Less OGT
→ reduced O-GlcNAcylation
→ reduced breast-cancer growth and invasion in models

The key idea is that TRIM21 controls OGT at the protein level. It does not need to prevent OGT gene transcription; it reduces the amount of OGT protein available in the cell.

This connects to the earlier molecular-glue and protein-degradation examples. Here, the endogenous ubiquitin-proteasome system naturally uses TRIM21 to regulate OGT. A future therapy might aim to restore TRIM21 function, inhibit OGT directly, or develop a degrader that recruits OGT to another suitable E3 ligase.

TRIM21 loss or reduced activity
→ OGT accumulates
→ pro-tumour O-GlcNAcylation increases
Restored OGT degradation
→ possible anti-tumour vulnerability

This is primarily a mechanistic result. OGT modifies many proteins in normal cells as well as cancer cells, so broadly removing it could be toxic. The therapeutic question is whether the TRIM21-OGT axis identifies a breast-cancer subgroup in which OGT can be targeted with a sufficient therapeutic window.

129 [26-09-25] Evorpacept plus trastuzumab, ramucirumab, and paclitaxel in HER2-positive gastric cancer

(*Evorpacept plus trastuzumab, ramucirumab and paclitaxel in HER2-positive gastric cancer: a randomized phase 2 trial*, <https://www.nature.com/articles/s41591-026-04700-3>)

This randomised Phase II study, called *ASPEN-06*, tested whether adding *evorpacept* to trastuzumab, ramucirumab, and paclitaxel improves treatment of previously treated HER2-positive advanced gastric or gastro-oesophageal-junction cancer.

The four drugs address four related aspects of tumour biology: HER2-driven growth, blood-vessel support, direct tumour-cell killing, and macrophage-mediated immune evasion.

129.1 What does each drug do?

Trastuzumab is an antibody against *HER2*, a receptor that can be overexpressed by tumour cells and drive growth signalling. It blocks aspects of HER2 function and labels HER2-positive cells for immune attack.

Ramucirumab is an antibody against *VEGFR2*, a receptor involved in responding to VEGF signals that promote tumour blood-vessel formation. Blocking VEGFR2 aims to limit the abnormal vasculature that supplies tumour growth.

Paclitaxel is conventional cytotoxic chemotherapy. It disrupts microtubules, structures needed for cell division, causing dividing cells to arrest and die.

Evorpacept blocks the CD47-SIRP α innate-immune checkpoint. CD47 on a cancer cell acts as a “don’t eat me” signal by binding SIRP α on macrophages.

Trastuzumab
→ targets HER2-driven tumour cells
+ labels them for immune recognition
Ramucirumab
→ inhibits VEGFR2-mediated angiogenesis
Paclitaxel
→ disrupts tumour-cell division
Evorpacept
→ blocks CD47 “don’t eat me” signalling
→ enables macrophage phagocytosis

129.2 Why does the four-drug combination make sense?

The most direct immune synergy is between trastuzumab and evorpacept. Trastuzumab binds HER2-positive cancer cells through its antigen-binding region, while its Fc region can engage Fc receptors on macrophages. This provides a positive “eat this cell” signal, called antibody-dependent cellular phagocytosis, or ADCP.

However, CD47 can still inhibit that macrophage response. Evorpacept removes this opposing brake.

```
Trastuzumab coats HER2-positive tumour cell
→ macrophage receives pro-phagocytic signal
Tumour-cell CD47 binds macrophage SIRPα
→ inhibitory "don't eat me" signal
Add evorpacept
→ block CD47-SIRPα
→ trastuzumab-tagged cell is more readily engulfed
```

Paclitaxel contributes direct cytotoxic pressure against dividing cancer cells. It may also make tumour cells more vulnerable to immune clearance through stress and cell-death signals. Ramucirumab addresses a different tumour dependency: VEGF-VEGFR2-driven blood-vessel formation. Anti-angiogenic treatment can restrict tumour support and may alter the abnormal tumour vasculature and immune environment.

Thus, the intended logic is not simply “more drugs must be better”:

```
HER2 signalling and tumour-cell marking
→ trastuzumab
Macrophage immune evasion
→ evorpacept
Cell division
→ paclitaxel
Tumour vascular support
→ ramucirumab
Together
→ multiple non-identical constraints
on tumour survival and immune escape
```

129.3 Why not give only two or three drugs?

A two-drug regimen of trastuzumab plus evorpacept would test the cleanest mechanistic immune pair, but may leave substantial HER2-independent growth and vascular support untreated in an advanced, previously treated cancer.

Paclitaxel plus ramucirumab is an established treatment backbone in later-line gastric cancer. The trial therefore asked a practical question: can evorpacept improve this active standard-like backbone in the biomarker-defined HER2-positive subgroup, while trastuzumab preserves HER2 targeting?

The drawback is increased complexity and potentially overlapping toxicity. Paclitaxel can cause myelosuppression and neuropathy; ramucirumab can cause hypertension, bleeding, thrombosis, impaired wound healing, and proteinuria. Adding drugs should only be justified if the added benefit is sufficiently large.

129.4 What did the trial show?

Adding evorpacept increased objective responses compared with trastuzumab, ramucirumab, and paclitaxel alone. The benefit appeared particularly substantial in tumours that remained HER2-positive and had high CD47 expression.

This biomarker pattern fits the proposed mechanism:

```
HER2 retained
→ trastuzumab can still mark tumour cells
High CD47
→ stronger macrophage "don't eat me" brake
Evppacept added
→ greater opportunity to restore
trastuzumab-supported phagocytosis
```

Evppacept has an engineered inactive Fc region. This is important because normal red blood cells also express CD47. An Fc-active CD47 antibody could potentially label red blood cells for immune clearance; evppacept is designed to block the inhibitory signal without independently creating a strong Fc-mediated “eat me” signal.

```
Strong Phase II activity, especially in CD47-high disease
→ supports the CD47 biomarker hypothesis
But:
Four-drug Phase II result
≠ proof that every component is necessary
or that the regimen is established standard care
```

The central next steps are confirming benefit in a larger trial, validating retained HER2 expression and CD47 level as patient-selection biomarkers, and determining whether the clinical benefit justifies the added treatment burden beyond the existing trastuzumab–ramucirumab–paclitaxel backbone.

130 [26-09-25] UpTCR: using AI to predict TCR–antigen recognition

(*UpTCR: a unified progressive knowledge transfer foundation model for robust T-cell receptor-antigen binding recognition*, <https://www.nature.com/articles/s41467-026-78075-x>)

This paper describes an AI method for predicting whether a T-cell receptor, or *TCR*, can recognise a particular antigen target. It builds directly on the earlier OmniTCR discussion, but its main contribution is a computational framework rather than a new T-cell mechanism or therapy.

For conventional T cells, the biologically relevant target is usually not an antigen alone. It is a peptide displayed by a particular MHC molecule:

```
Protein from virus or tumour
→ processed into peptide
→ peptide displayed by MHC
TCR recognises peptide-MHC surface
→ T-cell activation may occur
```

The challenge is that recognition depends jointly on the TCR’s alpha and beta chains, the peptide sequence, and the patient’s MHC type. Small changes in any of these can abolish recognition or create unintended cross-reactivity.

130.1 What is UpTCR trying to predict?

UpTCR aims to learn general patterns connecting TCR sequences with their antigen-recognition behaviour. Given a TCR and a candidate peptide–MHC target, it predicts whether binding or functional recognition is likely.

```
TCR sequence
+
peptide sequence
+
MHC context
→ model estimates probability of recognition
```

This could help identify which T-cell clones in a tumour or blood sample are likely to recognise a viral antigen, tumour-associated antigen, or neoantigen. It could also prioritise candidate TCRs for TCR-T-cell therapy.

130.2 What does “progressive knowledge transfer” mean?

Experimental TCR–antigen datasets are limited, uneven, and often biased towards well-studied viruses, peptides, and HLA types. A model trained only on one small dataset may work poorly for a new cancer antigen or different patient population.

UpTCR therefore uses a *foundation-model* approach: it first learns broader sequence and recognition-related patterns from large datasets, then transfers this knowledge progressively to more specific prediction tasks.

```
Broad TCR and antigen data
→ learn general sequence patterns
Progressive transfer to labelled TCR-antigen examples
→ learn recognition-associated features
Specific prediction task
→ prioritise likely TCR-target pairs
```

The intended advantage is greater robustness when data are scarce or when the model encounters TCRs and antigens not closely represented in its training set.

130.3 Why could this matter medically?

For cancer, a model like this could speed up *TCR deorphanisation*: identifying what an expanded T-cell clone from a tumour is likely to recognise. It could also help prioritise receptors for therapeutic engineering against a patient-specific neoantigen.

```
Tumour sequencing identifies candidate neoantigen
TCR sequencing identifies tumour-infiltrating T-cell clones
→ UpTCR prioritises plausible TCR-neoantigen pairs
→ selected pairs are tested experimentally
→ possible starting point for vaccine or TCR-T development
```

The same logic could support infectious-disease research, such as identifying TCRs that recognise viral epitopes shared across patients.

130.4 What should be concluded?

UpTCR addresses an important bottleneck in personalised immunotherapy: there are many possible TCRs and many possible antigens, but experimentally testing every pairing is impractical.

However, it is a prioritisation tool, not a replacement for experimental immunology.

```
Predicted TCR-antigen recognition
≠ confirmed T-cell activation
Confirmed recognition in vitro
≠ safe therapeutic TCR
A therapeutic TCR still requires:
functional testing
+ peptide-MHC specificity assessment
+ broad normal-tissue cross-reactivity testing
+ clinical safety evaluation
```

The underlying biological lesson remains that TCR recognition is highly context-dependent. AI may make the search for useful receptors more efficient, but it cannot remove the need to establish whether a receptor is functional, tumour-selective, and safe in real human cells.

131 [26-09-25] GDF15 and weight loss: increasing energy expenditure rather than mainly reducing food intake

(*GDF15 promotes weight loss by enhancing energy expenditure in muscle*, <https://www.nature.com/articles/s41586-023-06249-4>)

This study examines *growth differentiation factor 15*, or GDF15, a stress-responsive hormone that can reduce body weight. Its important new contribution is that GDF15 did not only reduce food intake in mouse models: it also increased energy expenditure, particularly through skeletal muscle.

This differs from the main mechanism of *semaglutide* (Ozempic/Wegovy), a *GLP-1 receptor agonist*. Semaglutide does not target GLP-1 itself; it mimics the natural gut hormone GLP-1 and activates the GLP-1 receptor for longer than the native hormone would.

131.1 How does semaglutide cause weight loss?

GLP-1 receptor activation acts mainly through the gut–brain axis and the central nervous system. It reduces hunger, increases fullness after eating, and slows stomach emptying, particularly early in treatment.

```
Semaglutide activates GLP-1 receptor
→ reduced appetite and food reward
+ increased satiety
+ slower gastric emptying
Less energy consumed
→ weight loss
```

Thus, semaglutide mainly changes the *intake* side of energy balance. This is highly effective clinically, but weight loss normally triggers a compensatory fall in energy expenditure: the body requires fewer calories at lower body weight and may adapt further to resist continued weight loss.

131.2 How does GDF15 differ?

GDF15 acts through *GFRAL*, its receptor in a restricted region of the hindbrain. GDF15–GFRAL signalling was already known to reduce food intake, often in response to physiological stress.

This paper adds a second pathway. In mice, GDF15–GFRAL signalling increased sympathetic, noradrenaline-associated signalling in skeletal muscle. This promoted *futile calcium cycling*: muscle cells consumed ATP to move calcium ions without producing useful mechanical work.

GDF15 activates GFRAL in hindbrain
→ increased sympathetic signalling to muscle
→ calcium cycling consumes ATP
→ energy is released as heat
→ increased energy expenditure

GDF15 therefore has potential effects on both sides of energy balance:

Semaglutide
→ mainly decreases energy intake
GDF15
→ decreases food intake
+ may increase energy expenditure
Combination hypothesis
→ reduce calorie intake
+ counteract adaptive metabolic slowing

131.3 Would GDF15 be better than semaglutide?

Not necessarily. The muscle thermogenesis finding is interesting because increasing energy expenditure could, in principle, complement appetite suppression and help limit weight-loss plateaus or regain. However, this paper is primarily mechanistic work in mice. It does not establish that therapeutic GDF15 activation produces a large, sustained increase in human energy expenditure.

There is also an important tolerability concern. Natural GDF15 rises in settings such as infection, cancer, mitochondrial disease, and some drug toxicities. High GDF15 signalling can cause marked appetite suppression, nausea, food aversion, and weight loss that may be harmful. Therefore, stronger GDF15 activity is not automatically desirable.

Increased energy expenditure in mice
≠ superior or safe human obesity treatment
Potential advantage:
target both energy intake and expenditure
Potential risk:
excessive GDF15 signalling may cause
nausea, aversion, or unhealthy loss of lean mass

The practical implication is that GDF15-based therapy could be *complementary* to GLP-1-based treatment rather than a replacement. Its eventual value depends on whether a GDF15–GFRAL treatment can increase clinically meaningful, durable fat loss in people while maintaining acceptable tolerability, nutrition, and lean-mass preservation.

132 [26-09-25] Circulating tumour DNA as a large-scale map of actionable metastatic breast-cancer genetics

(Actionable genetic landscape and dynamics in circulating DNA from 12,827 patients with metastatic breast cancer, <https://www.nature.com/articles/s41467-026-75976-9>)

This study analyses *circulating tumour DNA*, or ctDNA, from 12,827 people with metastatic breast cancer. ctDNA consists of small fragments of DNA released into blood by tumour cells. Sequencing a blood sample can therefore provide a relatively non-invasive view of some cancer mutations.

```
Metastatic tumour cells release DNA fragments into blood
→ blood sample contains ctDNA
Sequence ctDNA
→ identify tumour-associated genetic alterations
without requiring a new tissue biopsy
```

The main value of a study at this scale is to map which potentially *actionable* alterations occur in real-world metastatic breast cancer, how often they occur, which alterations co-occur, and how the genomic profile changes after treatment.

An actionable alteration is one that may inform treatment selection, trial eligibility, or resistance monitoring. Examples in breast cancer can include alterations in *PIK3CA*, *ESR1*, *BRCA1/2*, *ERBB2* (HER2), and selected alterations affecting the AKT–PI3K pathway or DNA repair.

```
ctDNA identifies PIK3CA mutation
→ possible rationale for PI3K-pathway-targeted treatment
ctDNA identifies acquired ESR1 mutation
→ evidence of endocrine-treatment resistance
+ may influence next endocrine strategy
ctDNA identifies germline or somatic BRCA-associated biology
→ possible relevance of PARP inhibition
```

The particularly useful feature of blood-based testing is that it can be repeated over time. A conventional tumour biopsy samples one lesion at one time point. In metastatic disease, different lesions can contain different resistant subclones, and these populations can change under treatment pressure.

```
Initial therapy reduces sensitive tumour cells
→ resistant subclones may survive and expand
Repeat ctDNA testing
→ emerging resistance mutation may become detectable
before or during clinical progression
```

Thus, ctDNA can act as a form of *molecular monitoring*. It may help identify which resistance mechanisms repeatedly emerge after particular therapies, guide enrolment into biomarker-defined trials, and support more adaptive treatment decisions.

This is potentially a very useful resource because **12,827 patients is unusually large** for metastatic breast-cancer genomic analysis. It can reveal uncommon but clinically relevant alterations that smaller cohorts may miss, and it can provide more reliable estimates of how genomic patterns differ by breast-cancer subtype, previous treatment, and disease stage.

However, it is not a complete representation of every patient's cancer. ctDNA shedding varies substantially: some tumours release little detectable DNA into blood, and a negative ctDNA result does not prove that a mutation is absent. Blood testing can also detect age-related mutations arising in normal blood cells, called *clonal haematopoiesis*, which must not be mistaken for tumour mutations.

```
ctDNA alteration detected
→ potentially useful treatment or resistance information
No ctDNA alteration detected
≠ proof that the tumour lacks an actionable mutation
Large observational genomic dataset
≠ proof that changing treatment on the basis of each
detected mutation improves patient outcomes
```

The study is therefore a significant *resource and precision-oncology map*, rather than a new treatment by itself. Its greatest practical value is likely to be helping clinicians and drug developers understand which genomic testing should be prioritised, which resistance mechanisms require new drugs, and which smaller patient subgroups may be suitable for biomarker-guided trials.

133 [26-09-25] Minerva: using a genome language model to find hidden functional RNA elements

(*Coevolutionary mining of prokaryotic non-coding elements with a genome language model*, https://evodesign.org/Minerva_manuscript_joined.pdf)

This preprint introduces *Minerva*, a genome-language-model approach for finding functional *non-coding elements* in bacterial and archaeal genomes.

Non-coding does not mean non-functional. A DNA region may not encode a protein but can still produce a functional RNA, regulate a nearby gene, form a structured RNA molecule, or interact with proteins or other nucleic acids. Examples include ribosomal RNAs, CRISPR RNAs, riboswitches, and many poorly understood small RNAs.

```
Genome sequence
→ protein-coding regions
+ regulatory DNA
+ structured non-coding RNAs
+ repetitive and mobile-genetic elements
Much of the non-coding functional landscape
→ remains incompletely annotated
```

133.1 What problem are the authors trying to solve?

Many non-coding RNAs are defined more by their *structure* than by an obvious shared sequence. For example, two RNA positions that pair with each other in a stem must remain chemically compatible as evolution changes the sequence.

```
RNA base pair:
G-C
One position mutates:
G-C
→ A-C  unstable
Compensatory mutation:
A-C
→ A-U  paired again
```

This is *coevolution*: changes at one sequence position are statistically linked to changes at another because the two positions must continue to work together.

Conventionally, researchers detect such relationships by collecting many related sequences, aligning them, and looking for correlated mutations. This can work very well, but it requires enough suitable homologous sequences and a good multiple-sequence alignment. It can be difficult for rare, rapidly evolving, poorly conserved, or previously unknown elements.

133.2 What does Minerva do differently?

Minerva uses a *genome language model*, trained on large amounts of prokaryotic DNA and protein sequence. Like a protein language model learns statistical regularities in amino-acid sequences, a genome language model learns which DNA patterns and combinations of positions are plausible in genomes.

The authors then interrogate the model to ask whether two sequence positions appear interdependent. The resulting pairwise scores form a two-dimensional *coevolutionary map*.

```
Raw genomic sequence
→ genome language model
Model detects statistical dependency
between position A and position B
→ potential RNA base pair,
protein contact, repeat relationship,
or other functional interaction
```

Thus, the model does not simply identify a familiar sequence motif. It tries to infer hidden *relationships within the sequence*, potentially revealing structural or functional constraints even when the exact sequence varies.

133.3 What did the authors find?

Across 150 bacterial genomes, Minerva recovered known structured elements, including ribosomal RNA and CRISPR-associated repeat systems. This is an important positive control: a method intended to find functional non-coding structures should recover examples already known to biology.

It also identified many predicted interactions in intergenic regions that were not present in existing annotations. The authors estimate that ****84.3%**** of the detected intergenic base-pairing interactions fell outside known annotations.

They further used the method to extend the predicted structure of a known *TwoAYGGAY* non-coding-RNA family in *Pseudomonas*, and to identify structured, sequence-diverse non-coding RNA arrays associated with an *Unknown Group 27* prophage reverse-transcriptase system.

```
Known structured RNA recovered
→ supports that the model detects real biology
Previously unannotated structured regions predicted
→ candidate new non-coding elements
But:
Model prediction
≠ experimentally confirmed RNA structure or function
```

133.4 Why did this attract attention?

The result is genuinely interesting because it suggests that a language model can extract a type of evolutionary and structural information that traditionally requires explicit sequence alignments across many homologues.

If robust, this could make it much easier to search huge microbial-genome and metagenomic datasets for functional RNA elements. Microbial genomes contain enormous unexplored diversity, including systems relevant to phage biology, antimicrobial resistance, microbial metabolism, RNA biology, and potentially new biotechnology tools.

It is also conceptually important. The model was not explicitly told the locations of every RNA stem, protein-coding region, or protein contact. Some of these features appear to emerge from learning the statistical organisation of genomic sequences.

```
Traditional approach:
many homologous sequences
→ alignment
→ identify coevolution
Minerva approach:
one genomic sequence
+ patterns learned from many genomes
→ infer candidate coevolutionary interactions
```

133.5 How big a deal is it really?

This is best viewed as a promising *genome-annotation and hypothesis-generation advance*, not yet as proof that AI has solved non-coding-genome interpretation.

The potentially major practical value is scale. If Minerva can reliably find structured non-coding elements from individual sequences, it could prioritise a much larger set of candidates for experimental testing than alignment-based methods alone. This would be especially useful for poorly characterised microbes and metagenomic sequences, where related sequences may be sparse or difficult to align.

However, several reasons justify caution. The work is a *preprint*, the estimated number of new interactions depends on model thresholds and prediction accuracy, and many predicted elements will require direct validation. Researchers still need to show that a candidate is transcribed, adopts the predicted structure, interacts with the proposed partner, and has a biological function.

```
AI-predicted coevolutionary interaction
→ strong candidate for biological follow-up
But:
Predicted RNA structure
≠ demonstrated function
Demonstrated function in one bacterium
≠ immediate medical or drug-discovery application
```

The near-term impact is likely to be better discovery of microbial genetic systems and non-coding RNAs. Its longer-term importance will depend on independent validation, performance against established alignment-based methods, and whether the newly discovered elements reveal useful biological mechanisms or engineerable tools.

134 [26-09-25] SpinForecast: assigning NMR signals in intrinsically disordered proteins without a sequential “walk”

(*SpinForecast: chain-free probabilistic backbone assignment of intrinsically disordered proteins from NMR chemical shifts*, <https://www.biorxiv.org/content/10.64898/2026.07.31.740808v1.full.pdf>; <https://bindresearch.org/science/>)

This preprint describes *SpinForecast*, a computational tool for analysing *intrinsically disordered proteins*, or IDPs, using solution-state NMR spectroscopy.

IDPs lack one stable three-dimensional structure. Instead, they fluctuate between many conformations. They are important in cell signalling, transcription, phase separation, cancer, and neurodegeneration, but are difficult to study and drug because conventional structure-based approaches assume a relatively stable binding pocket.

```
Structured protein
→ often has one dominant folded shape
+ potentially stable drug-binding pocket
Intrinsically disordered protein
→ samples many changing conformations
→ no single structure fully describes it
```

134.1 What is the NMR-assignment problem?

NMR detects signals from particular atomic nuclei in a protein. A *chemical shift* is the precise frequency of one such signal; it depends on the atom's local chemical environment.

Before researchers can interpret an NMR experiment biologically, they need to know which signal belongs to which amino acid in the protein sequence. This is called *backbone assignment*.

```
NMR spectrum contains many signals
→ identify which signal belongs to residue 1,
residue 2, residue 3, and so on
Assigned chemical shifts
→ study local structure, dynamics,
binding, and conformational change
```

This resembles receiving many unnamed measurements and needing to match each one to its correct location in a long protein sequence.

134.2 Why are IDPs difficult?

For a folded protein, each residue often has a sufficiently distinct local environment that its NMR signals are relatively well separated. For an IDP, many residues occupy similar flexible, solvent-exposed environments. Their signals can therefore overlap heavily.

The conventional approach uses several NMR experiments to connect one residue to its adjacent residues, building a sequential assignment “walk” along the protein chain.

```
Conventional assignment:
identify residue A
→ connect it to neighbouring residue B
→ connect B to C
→ gradually build assignment along sequence
Overlapping or missing signals
→ sequential walk can break down
```

This is especially problematic for IDPs, which often have repeated amino-acid patterns, narrow chemical-shift ranges, and proline-rich regions that interrupt standard backbone experiments.

134.3 What does SpinForecast do differently?

SpinForecast does not require construction of one continuous sequential chain. Instead, it treats each observed NMR signal set as a probabilistic matching problem.

It uses chemical-shift distributions learned from experimentally observed *disordered* protein regions in the Biomolecular Magnetic Resonance Data Bank, or BMRB. The authors filtered these reference data using AlphaFold pLDDT predictions to enrich for genuinely disordered sequence regions.

```
Observed NMR chemical shifts
+
protein sequence
+
reference shifts for disordered residues
→ calculate probability that signal belongs
to each plausible residue position
High probability
→ confident assignment
Several plausible positions
→ report ranked candidates and uncertainty
```

This is a Bayesian approach: rather than claiming every peak has one certain identity, it explicitly estimates the relative plausibility of alternative assignments.

134.4 How is this different from the other AI methods discussed?

SpinForecast is *not* a structure-prediction model like AlphaFold, a protein-generation method like the pLM work, or a small-molecule discovery platform such as BoltzMol. It does not design a drug or predict a complete protein structure.

It is an AI- and data-assisted *experimental-analysis tool*. Its purpose is to help turn raw NMR measurements into assigned residue-level information, which researchers can then use to study IDP conformations and how a potential drug or binding partner changes them.

```
Protein language model
→ generate or score protein sequences
Structure-prediction model
→ predict molecular three-dimensional arrangement
SpinForecast
→ interpret experimental NMR signals
for disordered proteins
```

134.5 Why might this matter for drug discovery?

BIND Research's broader aim is to make IDPs more experimentally tractable and eventually more druggable. If an NMR assignment can be obtained more rapidly, researchers can more easily measure which regions of an IDP respond to a small molecule, peptide, protein partner, or post-translational modification.

```
Candidate molecule binds IDP
→ local NMR chemical shifts change
SpinForecast-assisted assignment
→ identify affected residues
Affected residues
→ infer possible binding region
or conformational effect
```

This could be useful because IDP binders may work by shifting a flexible conformational ensemble rather than occupying one deep, stable pocket.

134.6 How significant is it?

This is a technically useful and fairly specialised advance, rather than a broad breakthrough comparable to solving protein-structure prediction. Its main novelty is combining an IDP-specific reference dataset with

chain-free, uncertainty-aware assignment, potentially avoiding a major practical bottleneck in IDP NMR experiments.

The reported performance is encouraging: when the method could not make one confident assignment, the correct residue was reportedly included among its candidate assignments in over 97% of tested cases. However, this is a *preprint*; performance needs independent testing across more diverse IDPs and realistic experimental datasets.

Faster, probabilistic IDP NMR assignment
→ potentially more efficient structural
and binding studies of difficult proteins
But:
Better NMR analysis
≠ direct demonstration that IDPs are now druggable
or that a therapeutic molecule has been discovered

The near-term impact is likely to be enabling researchers to study disordered proteins more efficiently. Its longer-term importance depends on whether this improved measurement capability helps produce reproducible IDP-binding mechanisms and ultimately useful drug-discovery programmes.

135 [26-09-25] Claude and ART: an AI-guided discovery of a CRISPR-like phage enzyme system

(*Claude discovers a novel enzyme system with CRISPR-like repeats*, <https://www.anthropic.com/news/claude-discovers-novel-enzyme-system>)

Anthropic reports that Claude helped identify a previously uncharacterised phage-associated genetic system termed *array-associated reverse transcriptase*, or ART. The system was found by searching a very large set of microbial and phage genome sequences for unusual reverse transcriptases and their neighbouring DNA.

A *reverse transcriptase*, or RT, is an enzyme that copies RNA into DNA. RTs occur in many biological systems, including retroviruses, mobile genetic elements, and bacterial anti-phage defence systems such as retrons.

ART contains three notable components:

Reverse transcriptase gene
+
nearby partner gene of unknown function
+
long array of repeated DNA sequences
→ ART system

The repeat array resembles a CRISPR array. In CRISPR systems, repeated DNA sequences are interspersed with variable “spacers”; the array is transcribed and processed into short guide RNAs that direct a CRISPR-associated protein to a matching nucleic-acid target.

```
CRISPR array
→ repeated sequences + variable spacers
→ short guide RNAs
→ RNA-guided targeting by Cas protein
ART array
→ repeated sequences near reverse transcriptase
→ short RNAs detected experimentally
But:
ART function
≠ yet known to be CRISPR-like
```

The important experimental result is limited but real: Anthropic’s laboratory work found that the ART repeat array is transcribed into distinct short RNAs. This supports the idea that the array is a functional genetic element rather than random repetitive DNA. However, it is not yet known what those RNAs do, whether they guide the RT or partner protein to a target, or whether ART acts in phage defence, phage replication, gene regulation, or something else.

135.1 What did Claude actually do?

Anthropic researchers gave many Claude agents a high-level task: inspect a large database of reverse-transcriptase sequences and their genomic neighbourhoods for interesting, undercharacterised patterns. The agents searched approximately 200,000 candidate RTs, identified unusual examples, examined adjacent DNA, measured repeat spacing, compared sequences, and searched relevant literature.

One agent highlighted a striking repeat array near an unusual RT. Human researchers then reviewed the analysis, recognised it as a plausible new system, and performed follow-up laboratory experiments.

```
Large genomic database
→ Claude agents search for unusual patterns
Candidate ART locus identified
→ human expert review
Wet-lab experiments
→ confirm repeat-derived short RNAs
This is AI-assisted discovery,
not autonomous proof of biological function
```

135.2 Why are people excited?

The most impressive aspect is the *workflow*, not a demonstrated new biotechnology tool. Modern microbial and metagenomic databases contain enormous amounts of sequence data, much of it poorly annotated. Scientists cannot manually inspect every unusual gene and genomic neighbourhood.

Claude appears to have combined several ordinary but important scientific tasks: large-scale filtering, inspection of local genomic context, pattern recognition, literature review, calculation, hypothesis formation, and concise reporting. The final candidate was then validated experimentally.

This resembles the Minerva work discussed earlier:

```
Huge sequence datasets
→ many potentially functional systems
remain unrecognised
AI-assisted search and prioritisation
→ more plausible systems selected
for human review and experiments
```

If this workflow can be made reproducible, it could accelerate discovery of microbial enzymes, RNA systems, phage-defence mechanisms, and potentially useful molecular tools.

135.3 Why are biologists pushing back?

The scepticism is justified. Bacterial and phage genomes contain many unusual RTs, repeat arrays, non-coding RNAs, and defence systems. Discovering an additional candidate system is interesting, but it is not automatically a biological or technological breakthrough.

ART is also related in broad principle to known systems such as *retrons*, which combine an RT with a non-coding RNA and can contribute to bacterial anti-phage defence. The distinctive feature of ART is its CRISPR-like repeat array, but its biological activity has not yet been established.

```
Unusual genomic architecture
+ short RNA expression
→ credible new biological hypothesis
But:
Newly named genetic system
≠ demonstrated molecular mechanism
Demonstrated mechanism
≠ useful genome-editing tool or medicine
```

CRISPR became transformative only after researchers established its RNA-guided targeting mechanism and showed that it could be engineered to cut chosen DNA sequences. ART has not yet reached even the first of those steps.

135.4 How big a deal is this?

The appropriate interpretation is: **a credible and encouraging proof of concept for AI-assisted biological discovery, but not yet a CRISPR-scale discovery.**

For the biology, ART is a promising early finding. Its value depends on follow-up experiments showing what the RT, repeat-derived RNAs, and partner protein do during phage or bacterial biology. Many newly discovered microbial systems will prove specialised, redundant, or difficult to engineer; a small minority may become broadly useful tools.

For AI, the result is more interesting as evidence that an agentic system can help navigate a large and messy research space and generate a laboratory-testable hypothesis. However, it was guided by humans, used curated biological resources and standard analytical tools, and required expert review plus wet-lab validation. The reported discovery process was also not necessarily reproducible in every rerun.

```
One experimentally supported AI-generated hypothesis
→ evidence that this discovery workflow can work
But:
One discovery
≠ evidence that AI will reliably produce
high-value biological discoveries at scale
```

The strongest future test is not whether ART becomes useful, although that would be valuable. It is whether the same overall approach repeatedly identifies genuinely novel systems, assigns them plausible functions, and produces candidates that independent laboratories can validate. If it does, AI could become a powerful way to convert the vast amount of unexamined sequence data into experimentally testable biology.

136 [26-09-25] “Progressable binders”: designing biologics that can become medicines

(*Progressable Binders: The Binders that Matter*, <https://www.xaira.com/news/progressable-binders-the-binders-that-matter>)

This article argues that the important question in AI-based antibody or protein design is not simply, “can the model create a molecule that binds the target?” It is whether it can create a *progressable binder*: a binder that has enough of the required properties to advance into drug development.

A binder is any molecule that attaches to a target. This could be an antibody, antibody fragment, or designed protein. Binding is necessary for a targeted biologic, but it is only the beginning.

```
Designed binder binds target
≠ therapeutic candidate
Therapeutic candidate requires:
target binding
+ desired biological effect
+ developability
+ selectivity
+ low immunogenicity risk
+ appropriate pharmacology
```

For example, a designed antibody may bind its target tightly in one laboratory assay but fail because it is poorly expressed, aggregates during manufacturing, binds unrelated proteins, is unstable at body temperature, is rapidly cleared from blood, or triggers an unwanted immune response.

Xaira therefore proposes reporting a *zero-shot progressable rate*: the proportion of designs that meet all predefined criteria without extensive rounds of experimental optimisation.

```
Zero-shot hit rate:
What fraction of designed molecules bind?
Zero-shot progressable rate:
What fraction bind
+ express well
+ remain stable and non-aggregated
+ avoid non-specific binding
+ have acceptable sequence and immunogenicity properties?
```

Their suggested thresholds include confirmed binding of roughly 100 nM affinity or better, sufficient expression in mammalian cells, thermal stability, monodispersity, low polyreactivity, and avoidance of sequence liabilities such as motifs prone to chemical degradation.

Polyreactivity means unintended binding to many unrelated molecules. It is undesirable because it can cause poor specificity, rapid clearance, and toxicity. *Developability* is the collective term for properties needed to make, formulate, store, dose, and reliably manufacture a medicine.

```
Strong target affinity
+ poor stability or high polyreactivity
→ likely dead end
Moderate-to-strong affinity
+ robust developability and specificity
→ more realistic starting point
for lead optimisation
```

This is closely related to the earlier discussion of protein language models aligned to experimental function. The key extension is that the desired function cannot be one property alone. A medicine must satisfy many constraints simultaneously, and improvement in one property can worsen another.

136.1 How novel or important is this?

The underlying principle is **not new**. Antibody discovery teams have always assessed affinity, specificity, expression, aggregation, stability, pharmacokinetics, and immunogenicity risk before progressing a molecule. The novelty is mainly in applying this full *multi-parameter optimisation* standard to evaluate AI-generated binders from the first design round, rather than celebrating binding hit rate alone.

This is an important and welcome standard because early AI-design reports can make a field look more mature than it is by focusing on the most favourable metric: a molecule that binds in vitro. A high zero-shot progressable rate would be substantially more meaningful evidence that AI can reduce experimental iteration and accelerate biologic discovery.

However, this Xaira article is a company perspective, not independent proof that the stated framework reliably produces clinical candidates. Even a binder that passes all early developability filters can fail through insufficient cellular activity, poor tissue distribution, unexpected toxicity, or lack of patient benefit.

High zero-shot progressable-binder rate
→ potentially major improvement in
biologic discovery efficiency
But:
Progressable in discovery assays
≠ preclinical candidate
Preclinical candidate
≠ safe, effective medicine

The approach is therefore *practically important rather than conceptually revolutionary*. Its real significance will depend on prospective, transparent evidence that AI-designed binders meeting these criteria repeatedly progress through lead optimisation, animal studies, and eventually clinical development more efficiently than conventional discovery approaches.

137 [26-09-25] Lila's RNA Discovery Engine: closed-loop optimisation of therapeutic mRNA

(From *Prompt to CAR-T: Inside Lila's RNA Discovery Engine*, <https://www.lila.ai/news/inside-lilas-rna-discovery-engine>)

This article describes Lila's *RNA Discovery Engine*: an AI-driven experimental platform for designing mRNA sequences with improved stability and protein expression.

An mRNA medicine gives a cell temporary instructions to make a chosen protein. For example, an *in vivo* CAR-T approach can deliver mRNA encoding a CAR into T cells inside the patient. The T cell then transiently makes the CAR and can recognise the chosen target.

However, the protein-coding sequence is only part of an mRNA medicine. Features including the 5' untranslated region, 3' untranslated region, codon usage, RNA secondary structure, and chemical modifications affect how long the mRNA survives and how much protein it produces.

```
mRNA sequence and structure
→ cellular stability, translation,
and innate-immune sensing
More stable, efficiently translated mRNA
→ more protein made for longer
→ potentially stronger therapeutic effect
```

Lila's approach is a closed experimental–computational loop. The company reports physically synthesising and testing more than 950,000 mRNA sequences in living cells, then feeding the measured results back into its model. The model can then propose the next sequences expected to improve a specified property.

```
AI proposes mRNA designs
→ automated experiments measure expression
and stability in cells
Experimental results
→ retrain model
→ improved next-round designs
```

This is related to the earlier discussion of “out-of-distribution medicines.” The potential advantage is not an AI model trained only on public RNA data, but a large proprietary dataset linking deliberate mRNA sequence changes to experimentally measured cellular outcomes.

The company used this platform to optimise mRNA for several proteins, including EPO, factor IX, and IL-10. It also applied it to an in vivo CAR-T proof-of-concept. In non-human primates, their optimised mRNA formulation reportedly gave deeper and more durable B-cell depletion than benchmark compositions.

```
mRNA encoding CD19 CAR delivered in vivo
→ T cells transiently express CAR
More effective mRNA design
→ stronger or longer CAR expression
More CAR activity
→ deeper B-cell depletion in primates
```

This is genuinely interesting because it addresses a major practical constraint of RNA medicines: obtaining sufficient protein expression in the right cells without simply increasing dose. It also demonstrates a relatively demanding biological endpoint—in vivo immune-cell reprogramming and B-cell depletion in primates—rather than only reporter-protein expression in cultured cells.

However, the result should not yet be interpreted as a clinical CAR-T breakthrough. The reported programme was a platform demonstration rather than a candidate intended for human trials. The final effect depends on much more than mRNA sequence: delivery-particle design, tissue distribution, cell-type selectivity, dose, innate immune activation, persistence, and CAR safety all remain critical.

```
Improved mRNA performance in cells and primates
→ encouraging platform validation
But:
Transient B-cell depletion in primates
≠ safe, controllable, durable in vivo CAR-T therapy
in patients
```

The underlying concept—high-throughput experimental optimisation guided by AI—is not entirely new. What is potentially impressive is the reported scale, the use of living-cell data, and translation to a complex primate in vivo endpoint within a short development cycle. Its real importance will depend on reproducibility across many payloads and delivery systems, independent validation, and whether similar programmes can progress into safe and effective human medicines.

138 [26-09-25] Owkin's "biology-first AGI" bet: patient data as the missing layer for scientific AI

(*Biology is the most important step to AGI. We can get there faster*, <https://www.linkedin.com/pulse/doing-one-thing-well-yes-biology-iterating-speed-light-clozel-md-6zhte/>)

Owkin's central claim is that powerful general AI will not become genuinely useful for medicine merely by reading more scientific papers or scaling language models. It must learn from *multimodal human biological data*: the combination of pathology, genomics, RNA expression, clinical records, treatments, and patient outcomes.

Their practical focus is precision medicine and drug development, particularly in oncology. The aim is to answer questions such as:

```
Why did this patient's tumour respond or resist treatment?
Which molecular and tissue features define a disease subgroup?
Which biomarker, target, or trial population
is most likely to produce clinical benefit?
```

138.1 What is Owkin's main bet?

Owkin's main bet is that the decisive advantage in biological AI will come from generating and learning from patient-linked data, not only from a more capable general-purpose reasoning model.

Different data types describe complementary layers of disease biology:

```
Genomics
→ mutations and inherited risk
Transcriptomics
→ which genes are active
Digital pathology
→ tissue architecture, tumour cells,
immune cells, and spatial relationships
Clinical outcomes
→ what happened to real patients
after a particular treatment
```

The intended value is to connect these layers. For example, a model might identify a tissue pattern associated with a tumour genotype, an immune-cell state, resistance to a treatment, and a particular patient outcome.

138.2 Why federated learning?

Patient data are held separately by hospitals and cannot easily be transferred to one central database because of privacy, governance, and legal constraints.

With *federated learning*, Owkin sends a model to a hospital, where it is trained or evaluated locally. Only model updates or aggregated results are returned; the raw patient data remain under the hospital's control.

```
Model goes to hospital
→ learns locally from patient data
→ aggregated update is shared
Raw patient data
→ remain under local institutional control
```

This could create access to a larger and more diverse patient-data network. Diversity is important because a model trained at one institution may learn local biases in patient population, sample handling, imaging equipment, and clinical practice that do not generalise elsewhere.

138.3 How does this compare with Anthropic and OpenAI?

Anthropic, OpenAI, and similar companies are building increasingly capable *general-purpose foundation models*. Their potential advantage is broad reasoning, coding, tool use, scientific-literature synthesis, planning, and agentic workflows at enormous scale.

As illustrated by Claude’s ART discovery, a powerful general model can search biological databases, recognise interesting patterns, review literature, generate hypotheses, and help direct laboratory follow-up.

```
General AI company
→ broad reasoning and tool-use capability
+ rapidly improving models
+ large-scale compute and engineering
Potential biology application
→ search literature and public data
+ generate and prioritise hypotheses
```

Owkin’s proposed advantage is not necessarily that its base model reasons better than the leading general models. Rather, it is that it has a more specialised *data layer and biological context*.

```
Owkin’s proposed edge:
multimodal patient data
+
federated hospital network
+
models trained on disease and outcome data
+
wet-lab and clinical validation
General-model company’s proposed edge:
stronger general reasoning
+
software engineering and agentic tool use
+
ability to absorb broad scientific knowledge
```

A model can read every public paper about a cancer target, but it cannot infer all human treatment-response biology from literature alone. The most valuable data—how molecular features, tissue state, treatment exposure, toxicity, and survival relate within real patients—are often fragmented, private, and difficult to standardise.

Thus, Owkin’s bet is that general models may become broadly available or commoditised, whereas high-quality longitudinal patient data and the infrastructure to use them responsibly could remain a durable advantage.

138.4 Is Owkin’s approach fundamentally different?

It is different in emphasis, but not completely different in principle. Large pharmaceutical companies already use pathology, genomics, biomarkers, real-world evidence, AI, and hospital collaborations. Anthropic and OpenAI can also partner with hospitals, acquire specialised data, fine-tune biological models, and build laboratory automation.

Therefore, Owkin does not have an automatic permanent moat merely because it uses federated learning or multimodal data.

Its potential edge is more specific:

```
Hospital partnerships
+
privacy-preserving data infrastructure
+
patient-level multimodal datasets
+
specialised biological models
→ potential advantage in discovering
human-relevant biomarkers and disease subgroups
```

This could matter particularly where conventional preclinical models are weak predictors of clinical response. Human tumours contain complex pathology, immune environments, treatment histories, co-morbidities, and genetic diversity that are difficult to reproduce in cell cultures or animal models.

138.5 How impressive is the claim?

The strategy is credible and potentially important. Better biological AI will probably require both components: increasingly capable general reasoning systems *and* reliable, intervention-linked human biological data.

Owkin’s vision is therefore not that it can ignore advances from Anthropic or OpenAI. More plausibly, it aims to combine leading general AI capabilities with a specialised patient-data and validation engine.

```
General AI reasoning
+ human biological data
+ experimental validation
→ potentially more useful scientific AI
But:
Association in patient data
≠ causal disease mechanism
Strong biomarker
≠ effective drug target
AI-supported decision
≠ improved clinical success rate
```

Terms such as “biological AGI” should still be treated as an ambitious long-term vision, not an established capability. The decisive test is prospective evidence: whether Owkin repeatedly identifies biomarkers that improve trial design, avoids failed programmes earlier, discovers useful targets, or produces medicines with a higher probability of clinical success.

Owkin may have a meaningful specialised advantage in patient-derived multimodal data and federated-learning infrastructure. Anthropic and OpenAI may have a major advantage in general model capability, compute, engineering scale, and agentic systems. The eventual winners may be organisations that combine both rather than those relying on only one.

139 [26-09-25] Bioptimus: faster hypotheses need biological models grounded in real tissue data

(*Faster Hypotheses vs. Real Biology: My Take on Anthropic’s Claude Science*, <https://www.linkedin.com/pulse/faster-hypotheses-vs-real-biology-my-take-anthropics-claude-vert-yiwre/>)

The CEO of Bioptimus contrasts two complementary forms of AI for biology. Anthropic’s Claude can accelerate *hypothesis generation*: it can search literature and genomic databases, operate analytical tools, recognise unusual patterns, and produce plausible candidates for scientists to investigate. The ART example—a CRISPR-like repeat array associated with a phage reverse transcriptase—illustrates this capability.

Biopimus’s argument is that this is useful, but insufficient. Biology is not only a large collection of sequences, papers, and known facts. It is a complex, context-dependent physical system involving cells, tissues, molecular states, spatial organisation, and patient outcomes.

```
General AI agent
→ search, reason, synthesise evidence,
and generate hypotheses quickly
But:
Hypothesis
≠ biological mechanism
Biological mechanism
≠ useful diagnostic or medicine
```

The central claim is therefore not that Claude’s ART discovery is unimportant. It is that the limiting factor in medicine will increasingly be the ability to test, interpret, and learn from *real biological data*, particularly data derived from human tissues.

139.1 What is Biopimus’s main bet?

Biopimus is building *foundation models for biology*: models trained directly on large biological datasets, rather than general-purpose language models trained principally on text and code.

Its initial strength is digital pathology. A pathology slide contains information about tissue architecture: tumour-cell morphology, immune-cell infiltration, fibrosis, necrosis, blood vessels, and spatial relationships between cell populations. These features can reflect molecular biology and clinical behaviour, even when they are difficult to describe explicitly in text.

```
Pathology image
→ tissue morphology and spatial organisation
RNA and genomic data
→ molecular state and genetic alterations
Clinical outcome data
→ treatment response, progression, and survival
Combined model
→ attempt to connect tissue appearance,
molecular state, and clinical consequence
```

Its model *H-Optimus-1* is a pathology-image foundation model. Biopimus is also developing more multimodal models, intended to combine histopathology with bulk RNA sequencing and spatial transcriptomics. Spatial transcriptomics measures gene activity while preserving each cell’s approximate position in a tissue section.

This matters because the same mutation or gene-expression signature can have different implications depending on *where* it occurs and which cells surround it. For example, an immune cell at the tumour edge may have a different role from the same cell type deep within the tumour.

139.2 How is this different from Anthropic or OpenAI?

Anthropic and OpenAI primarily build powerful *general* models. Their strength is broad reasoning, coding, tool use, literature synthesis, planning, and agentic workflows. They can help a researcher explore an enormous space of existing knowledge and data.

Biopimus aims to build a more specialised representation of biological state, particularly human tissue biology.

```
Anthropic or OpenAI:
"What hypotheses can be inferred from
literature, sequence data, and available tools?"
Bioptimus:
"What does this tissue state mean biologically,
molecularly, and clinically?"
Both:
"Which hypothesis should be tested next?"
```

This distinction is useful but not absolute. General AI companies can acquire biological data, train specialised models, and partner with laboratories. Conversely, Bioptimus can use general AI systems for reasoning and workflow orchestration. The likely successful system will combine both capabilities.

139.3 What is Bioptimus's potential edge?

Bioptimus's potential advantage is not mainly a better chatbot or agent. It is a specialised data and model stack built around high-dimensional, multimodal human biology.

Their *STELA* initiative aims to generate standardised data linking pathology images, spatial transcriptomics, other molecular measurements, and clinical information across many patient samples. This is conceptually similar to Owkin's patient-data thesis, but Bioptimus places greater emphasis on foundation models that learn from the spatial and visual organisation of tissue.

```
Large, diverse, well-annotated tissue datasets
+
multimodal biological foundation model
+
experimental and clinical validation
→ potential ability to identify patterns
not obvious from one data type alone
```

A durable edge could arise if Bioptimus obtains data at a scale and quality that competitors cannot readily reproduce, and if its models generalise across hospitals, diseases, tissue-processing methods, and patient populations.

139.4 How impressive is this?

The strategy is credible and important. Human tissue is a major missing layer between molecular experiments and patient outcomes, and multimodal models may help reveal relationships that are invisible in isolated genomics, RNA, or image datasets.

However, this is not yet proof of a "world model" that understands biology causally. A model may learn a reliable association between tissue morphology and a molecular feature without knowing why that relationship exists or whether changing the feature will change disease outcome.

```
Multimodal pattern predicts treatment response
≠ proven causal mechanism
Causal mechanism
≠ druggable target
Accurate model in retrospective data
≠ improved prospective patient outcome
```

The near-term impact could be substantial in pathology assistance, biomarker discovery, patient stratification, and trial design. The longer-term drug-discovery value depends on prospective evidence that these models

identify biologically actionable patient subgroups, select more effective treatment strategies, or improve clinical-trial success rates.

The post's core message is therefore reasonable: ****general AI can make hypothesis generation dramatically faster, but progress in medicine ultimately depends on whether those hypotheses are anchored in, and repeatedly tested against, real biology.****

140 [26-09-30] Vepdegestrant: a PROTAC that degrades mutant oestrogen receptor in breast cancer

(FDA approves vepdegestrant for ER-positive, HER2-negative, ESR1-mutated advanced or metastatic breast cancer, <https://www.fda.gov/drugs/resources-information-approved-drugs/fda-approves-vepdegestrant-er-positive-her2-negative-esr1-mutated-advanced-or-metastatic-breast>; Vepdegestrant, a PROTAC Estrogen Receptor Degradar, in Advanced Breast Cancer, <https://www.nejm.org/doi/10.1056/NEJMoa2505725>)

Vepdegestrant is the first FDA-approved *PROTAC*, or *proteolysis-targeting chimera*, medicine. It is an oral treatment for ER-positive, HER2-negative, *ESR1*-mutated advanced or metastatic breast cancer that has progressed after endocrine therapy.

ER-positive breast cancer depends on signalling through the *oestrogen receptor*, or ER. ER is a transcription factor: when activated, it changes gene expression in ways that can promote cancer-cell survival and proliferation.

Oestrogen binds ER
→ ER activates growth-associated genes
→ ER-positive breast-cancer growth is supported

140.1 How do standard endocrine therapies work?

Aromatase inhibitors, such as letrozole, reduce the body's production of oestrogen. With less oestrogen available, there is less activation of ER.

Fulvestrant binds ER, inhibits its activity, and can promote some ER degradation. It is a *selective oestrogen-receptor degrader*, or SERD.

Both approaches aim to reduce ER-driven tumour growth. However, cancer cells can evolve resistance, particularly through acquired mutations in *ESR1*, the gene encoding ER.

140.2 Why are ESR1 mutations important?

ESR1 mutations often arise after treatment with an aromatase inhibitor. Common examples include Y537S and D538G. These mutations stabilise ER in an active shape, allowing it to signal even when little oestrogen is present.

Aromatase inhibitor
→ low oestrogen
→ ER signalling should decrease
Acquired ESR1 mutation
→ ER becomes active with little or no oestrogen
→ endocrine-treatment resistance

Therefore, *ESR1*-mutated tumours remain ER-dependent, but have become less vulnerable to simply lowering oestrogen. They need a treatment that more completely suppresses the abnormal receptor itself.

140.3 How do PROTACs work?

A PROTAC is a two-ended small molecule connected by a linker. One end binds the disease-relevant target protein; the other binds an *E3 ubiquitin ligase*, part of the cell's normal protein-disposal system.

```
PROTAC binds target protein
+
PROTAC binds E3 ubiquitin ligase
→ target and E3 ligase are brought together
E3 ligase adds ubiquitin to target
→ proteasome recognises ubiquitinated target
→ target protein is degraded
```

Unlike a conventional inhibitor, which must continuously occupy a target's active or binding site, a PROTAC can act *catalytically*: after helping destroy one target-protein molecule, it can potentially recruit another.

Vepdegestrant binds ER at one end and the E3 ligase *cereblon*, or CRBN, at the other. It therefore creates an ER–vepdegestrant–CRBN complex that leads to ER ubiquitination and proteasomal destruction.

```
Vepdegestrant binds ER
+ recruits CRBN E3 ligase
→ ER ubiquitination
→ proteasomal ER degradation
→ reduced ER-driven gene expression
```

This is directly related to the molecular-glue examples discussed earlier. Both use the ubiquitin–proteasome system to remove a protein. The difference is that a molecular glue is usually one small molecule that creates a new protein–protein interaction, whereas a PROTAC has two separately binding components joined by a linker.

140.4 Why was benefit larger in ESR1-mutated disease?

The Phase III *VERITAC-2* trial compared vepdegestrant with fulvestrant. In the full study population, progression-free survival was not significantly improved. However, the prespecified ESR1-mutated subgroup showed a clear benefit.

```
ESR1-mutated subgroup:
Vepdegestrant
→ median PFS: 5.0 months
Fulvestrant
→ median PFS: 2.1 months
```

The hazard ratio was ****0.57****, corresponding to a ****43% reduction**** in the relative risk of progression or death during the study. Objective response rates were ****19%**** with vepdegestrant and ****4%**** with fulvestrant.

This pattern fits the biology. ESR1-mutated tumours are more strongly selected for continued signalling from an abnormal, ligand-independent ER. Vepdegestrant can degrade these mutant ER proteins, rather than relying mainly on reducing oestrogen or functionally antagonising the receptor.

```
ESR1 mutation
→ tumour remains dependent on abnormal ER
Vepdegestrant
→ removes mutant ER protein itself
Therefore:
ESR1 mutation
→ predictive biomarker for greater
vepdegestrant benefit
```

The FDA approval therefore includes *ESR1*-mutated disease and uses a blood-based companion diagnostic, Guardant360 CDx, to identify eligible patients.

140.5 What should be concluded?

This is an important clinical validation of targeted protein degradation. It shows that a PROTAC can be orally delivered, achieve sufficient target degradation in patients, and improve a clinically relevant outcome in a biomarker-defined cancer subgroup.

However, it does not mean that PROTACs are inherently superior to conventional inhibitors or SERDs. Their benefit depends on the target, disease biology, ability to recruit a suitable E3 ligase, drug exposure, resistance mechanisms, and safety.

```
Clinical success of one PROTAC
→ validates targeted degradation
as a therapeutic modality
But:
PROTAC mechanism
≠ automatic superiority for every target
The key lesson:
remove the relevant disease-driving protein
in the patient subgroup most dependent on it
```

141 [26-09-30] StriMap: modelling the full TCR–peptide–HLA recognition problem

(A structure-informed deep learning framework for modeling TCR-peptide-HLA interactions, <https://www.nature.com/articles/s41467-026-78063-1>)

This paper presents *StriMap*, an AI framework designed to predict recognition between a T-cell receptor, or *TCR*, a peptide, and a particular HLA molecule.

This builds directly on the earlier OmniTCR and UpTCR papers. The central biological point is that a TCR does not recognise an antigenic peptide in isolation. It recognises the three-dimensional surface formed by the *peptide–HLA complex*.

```
Protein from tumour, virus, or bacterium
→ processed into peptide
Peptide binds patient's HLA molecule
→ peptide-HLA surface is displayed
Compatible TCR binds this combined surface
→ T-cell activation may occur
```

A different peptide, HLA allele, or TCR sequence can substantially change recognition.

141.1 What does structure-informed mean?

Many earlier models use sequence information: the amino-acid letters in the TCR, peptide, and HLA. StriMap also incorporates predicted structural and physicochemical features at the contact surfaces.

```
Sequence-only model
→ asks whether amino-acid patterns resemble
known TCR-peptide-HLA interactions
Structure-informed model
→ also considers shape, charge,
and possible contacts at binding interfaces
```

This makes biological sense because binding depends on whether the three molecules can fit together physically and form favourable interactions. However, predicted structure remains an approximation: TCRs, peptides, and HLA molecules are dynamic rather than rigid objects.

141.2 Why is this medically useful?

In cancer, StriMap could help prioritise *neoepitopes*: mutant tumour peptides that may be presented by a patient's HLA molecules and recognised by T cells. These predictions could support personalised cancer vaccines or discovery of TCRs for engineered TCR-T-cell therapy.

In autoimmunity, the same logic can help identify possible *molecular mimics*. A microbial peptide may resemble a self-peptide sufficiently that a TCR primed against the microbe also reacts against normal tissue.

```
Microbial peptide activates T cell
→ TCR may cross-react with similar self peptide-HLA
Cross-reactive T-cell response
→ possible contribution to autoimmune disease
```

The study applied this approach to ankylosing spondylitis, screening bacterial proteins for peptides predicted to activate disease-associated TCRs. Candidate mimics were then experimentally shown to activate relevant T cells.

141.3 What should be concluded?

StriMap is useful because it treats antigen recognition as the coupled interaction it really is: TCR + peptide + HLA, rather than only TCR + peptide.

```
Predicted TCR-peptide-HLA interaction
→ better candidate prioritisation
But:
Prediction
≠ confirmed antigen recognition
Confirmed T-cell activation
≠ proof of disease causation
or safe therapeutic TCR specificity
```

Its likely value is to narrow a very large experimental search space. Laboratory testing remains essential to establish peptide presentation, T-cell function, cross-reactivity against normal tissues, and clinical relevance.

142 [26-09-30] CRISPR–Cas9: programmable, RNA-guided genome editing

(*Genome engineering using the CRISPR-Cas9 system*, <https://www.nature.com/articles/nprot.2013.143>; *The new frontier of genome engineering with CRISPR-Cas9*, <https://www.science.org/doi/10.1126/>

[science.1258096](https://doi.org/10.1016/j.cell.2015.09.054); *Past, present, and future of CRISPR genome editing technologies*, [https://www.cell.com/cell/fulltext/S0092-8674\(24\)00111-9](https://www.cell.com/cell/fulltext/S0092-8674(24)00111-9))

CRISPR is a programmable biological system originally used by bacteria and archaea to defend themselves against viruses, especially bacteriophages. Scientists adapted it into a technology for making targeted changes to DNA.

The key idea is simple: an RNA molecule provides an address, and a CRISPR-associated protein provides an activity at that address.

```
Guide RNA contains sequence complementary to target DNA
→ guide RNA directs Cas protein to matching DNA sequence
Cas protein acts at target site
→ DNA can be cut, modified, or regulated
```

142.1 What does CRISPR mean in nature?

CRISPR stands for *clustered regularly interspaced short palindromic repeats*. A natural CRISPR locus in a bacterial genome contains repeated DNA sequences separated by unique *spacers*.

The spacers often originate from DNA of viruses that previously infected the bacterium. They act as a molecular memory of past infection.

```
Virus infects bacterium
→ fragment of viral DNA is captured
Viral DNA fragment inserted as new spacer
into bacterial CRISPR array
→ inherited record of prior infection
Same virus infects again
→ CRISPR system recognises matching sequence
→ viral DNA is destroyed
```

Thus, CRISPR is often described as an adaptive immune system for bacteria. The bacterium stores sequence information from an invading virus and later uses that information to identify the same or a closely related virus.

142.2 How does CRISPR–Cas9 work?

In the widely used CRISPR–Cas9 system, the CRISPR array is transcribed into a long RNA. This RNA is processed into shorter *CRISPR RNAs*, or crRNAs. Each crRNA contains one virus-derived spacer sequence.

A second RNA, the *tracrRNA*, helps Cas9 bind and use the crRNA. For laboratory genome editing, the crRNA and tracrRNA are usually combined into one engineered *single-guide RNA*, or sgRNA.

```
Engineered guide RNA
=
target-matching sequence
+
Cas9-binding RNA structure
Guide RNA binds Cas9
→ Cas9-guide complex searches DNA
Matching DNA sequence found
→ Cas9 cuts both DNA strands
```

The guide RNA determines *where* Cas9 goes. Cas9 provides the DNA-cutting activity.

Cas9 also requires a short nearby DNA sequence called a *PAM*, or protospacer-adjacent motif. For the commonly used *Streptococcus pyogenes* Cas9, this is usually NGG, where N can be any DNA base.

```
Target DNA matches guide RNA
+
nearby PAM sequence is present
→ Cas9 can bind and cut
Target DNA matches guide RNA
but no compatible PAM
→ Cas9 usually cannot cut
```

The PAM is important biologically because it helps bacteria distinguish foreign viral DNA from their own CRISPR DNA. It is also important technically because it limits which genomic positions a particular Cas protein can edit.

142.3 How does cutting DNA create an edit?

Cas9 makes a *double-strand break*. The cell then attempts to repair the broken DNA, and genome editing exploits this repair process.

The most common repair route is *non-homologous end joining*, or NHEJ. It reconnects broken DNA ends, but often introduces small insertions or deletions, called indels. These can disrupt a gene.

```
Cas9 cuts target gene
→ NHEJ repairs break imperfectly
→ small insertion or deletion
Coding sequence disrupted
→ gene may be knocked out
```

If researchers supply a DNA repair template, cells can sometimes use *homology-directed repair*, or HDR, to copy a chosen sequence into the cut site.

```
Cas9 cut
+
supplied DNA template with desired sequence
→ HDR-mediated repair
Desired sequence copied into genome
→ precise change may be introduced
```

HDR is useful but often less efficient than NHEJ, especially in non-dividing cells. This is one reason why many newer CRISPR technologies try to edit DNA without requiring a full double-strand break.

142.4 What are base editing and prime editing?

CRISPR has expanded beyond simply cutting DNA.

Base editors combine a Cas protein modified so that it does not fully cut both DNA strands with an enzyme that chemically changes one DNA base into another. For example, some base editors can convert a C–G base pair into a T–A base pair.

Prime editors combine a modified Cas protein with a reverse transcriptase and a specialised guide RNA, called a prime-editing guide RNA. They can install a broader range of small substitutions, insertions, or deletions.

```
Standard Cas9 editing
→ double-strand DNA break
→ cell repair creates or installs edit
Base editing
→ chemically changes one DNA letter
without full double-strand break
Prime editing
→ writes a specified small DNA change
using reverse-transcription-based repair
```

These approaches can potentially reduce unwanted large DNA changes associated with double-strand breaks, but each has its own constraints involving delivery, editing efficiency, sequence context, and unintended edits.

142.5 Why is CRISPR useful medically?

CRISPR can be used to study disease genes, engineer research models, modify immune cells, or potentially correct disease-causing genetic changes.

For example, CRISPR can edit a patient's blood stem cells outside the body. The edited cells can then be returned to the patient. This basic strategy is used in approved treatment for some people with sickle-cell disease and transfusion-dependent beta-thalassaemia.

CRISPR can also be used in discovery screens, as discussed for tumour cells and human T cells:

```
Large CRISPR guide-RNA library
→ each cell receives disruption of one gene
Cells experience treatment or tumour environment
→ some edits improve or impair survival
Sequence guide RNAs afterwards
→ identify genes controlling the phenotype
```

Thus, CRISPR is both a possible therapeutic modality and a powerful tool for identifying therapeutic targets.

142.6 What does CRISPR have to do with Anthropic's ART discovery?

Anthropic's proposed *array-associated reverse transcriptase*, or ART, system was described as *CRISPR-like* because it contains a long DNA repeat array with short variable regions between repeats.

That architecture resembles a CRISPR array:


```

Natural CRISPR:
repeats
+
variable spacer sequences
→ transcribed into guide-like RNAs
→ Cas protein can target matching nucleic acid
ART:
repeats
+
variable sequences
+
reverse transcriptase
+
partner protein of unknown function
→ short RNAs are produced
But:
ART mechanism
≠
established CRISPR mechanism

```

The key difference is that CRISPR–Cas systems have been shown to use their guide RNAs to recognise matching foreign nucleic acid, and particular Cas proteins have defined cutting or modifying activities.

For ART, researchers have only early evidence that the repeat array produces short RNAs. It is *not yet known* whether these RNAs guide the reverse transcriptase or partner protein to a DNA or RNA target, whether ART functions in phage defence, or whether it can be engineered as a genome-editing tool.

```

ART has CRISPR-like genomic architecture
→ interesting hypothesis of RNA-directed function
But:
CRISPR-like repeats
≠
demonstrated programmable genome editing

```

This explains both the excitement and the scepticism around the ART report. If ART eventually proves to be a programmable RNA-guided enzyme system, it might reveal new biology and perhaps a useful biotechnology tool. At present, however, it is much earlier than the point at which CRISPR–Cas9 became a practical genome-editing technology.

```

CRISPR discovery
→ mechanism established
→ programmable targeting demonstrated
→ genome-editing technology developed
ART discovery so far
→ unusual repeat-associated enzyme system
+ short RNA expression
→ biological mechanism still unknown

```

143 [26-09-30] CXCR6-positive T cells may be required for CD3×CD20 bispecific-antibody activity in B-cell lymphoma

(*CXCR6+ T cells are required for CD3xCD20 Bispecific Antibody Efficacy in B cell Lymphomas*, <https://www.medrxiv.org/content/10.1101/2024.07.02.24309792v2.full.pdf>)

This preprint examines why CD3×CD20 *bispecific antibodies* work well in some B-cell lymphomas but not others. Its central finding is that a particular T-cell population expressing *CXCR6* appears important for effective treatment response.

A CD3×CD20 bispecific antibody has two binding arms. One binds CD20 on malignant B cells; the other binds CD3, part of the T-cell receptor complex, on T cells. It physically brings a T cell into contact with a lymphoma cell and activates cytotoxic killing.

```
CD20×CD3 bispecific antibody binds lymphoma cell
+
binds nearby T cell
→ forced immune synapse
T-cell activation
→ granzyme- and perforin-mediated killing
of CD20-positive lymphoma cell
```

Thus, unlike a CAR-T cell, the drug does not supply engineered T cells. It depends on the patient's own T cells being present, sufficiently functional, and able to enter the lymphoma tissue.

143.1 What is CXCR6?

CXCR6 is a *chemokine receptor*: a cell-surface receptor that helps direct immune-cell movement and tissue localisation. Its ligand, *CXCL16*, can be displayed or released by cells in inflamed tissues and tumour environments.

CXCR6-positive T cells are often associated with tissue residence and sustained local immune activity. In this study, they appear to represent a functionally important T-cell state that can support lymphoma killing after CD3×CD20 engagement.

```
CXCL16 in lymphoma microenvironment
→ recruitment or retention of CXCR6-positive T cells
CXCR6-positive T cells near CD20-positive lymphoma cells
+
CD3×CD20 bispecific antibody
→ productive tumour-cell killing
```

The important idea is that a bispecific antibody cannot work optimally if there are too few suitable T cells in the relevant tumour site. The drug can create a bridge, but it still needs effective immune cells on one side of that bridge.

143.2 Why might CXCR6-positive T cells be especially useful?

The study suggests that CXCR6-positive T cells have a more favourable combination of tumour localisation, cytotoxic capacity, and persistence than other available T-cell populations.

This resembles the distinction between an immune-infiltrated and immune-excluded tumour. Two patients may both have CD20-positive lymphoma and receive the same bispecific antibody, but the patient with appropriate tumour-associated T cells may be much more likely to respond.

```
CD20-positive lymphoma
≠ guaranteed response to CD3×CD20 bispecific antibody
CD20-positive lymphoma
+
sufficient functional CXCR6-positive T cells
→ greater potential for treatment response
```

143.3 What should be concluded?

The paper provides a possible *response biomarker* and therapeutic principle for CD3×CD20 bispecific antibodies: efficacy may depend on the quality and location of the T-cell compartment, not only on CD20 expression by the lymphoma.

It also suggests possible combination strategies that increase or preserve useful CXCR6-positive T cells. However, this remains a preprint and an association with response does not yet establish a clinically validated biomarker.

```
CXCR6-positive T-cell abundance
→ promising candidate indicator of bispecific activity
But:
Candidate biomarker
≠ validated patient-selection test
Improved T-cell localisation or function
≠ proven improvement in clinical outcome
```

144 [26-09-30] Sequence-selective translation inhibitors: stopping synthesis of MYC at the ribosome

(Halting translation by the letter with sequence-selective small molecules, <https://www.nature.com/articles/s41589-026-02318-7>)

This study describes *sequence-selective translation inhibitors*, termed *Interdictors*. These small molecules aim to reduce production of a chosen protein by causing the ribosome to stall while it is making that protein.

The ribosome is the cell's protein-making machine. It reads an mRNA sequence and joins amino acids together, one at a time, to produce the corresponding protein.

```
mRNA sequence
→ ribosome reads codons
Amino acids added sequentially
→ growing protein chain emerges
→ completed protein is released
```

Most translation inhibitors block ribosomes broadly, preventing synthesis of many proteins and causing substantial toxicity. The unusual claim here is that a small molecule can inhibit translation *selectively*, depending on the amino-acid sequence of the growing protein chain.

144.1 How do Interdictors work?

The compounds bind in the ribosome's *peptidyl-transferase centre*, the region that catalyses amino-acid addition. However, their activity depends on interactions with the newly forming protein, or *nascent chain*, within the ribosome.

```
Interdictor binds ribosome
+
specific nascent amino-acid sequence is being produced
→ productive ribosome-drug-nascent-chain interaction
→ ribosome stalls
→ target protein is not completed
```

The study focuses on *MYC*, an oncogenic transcription factor that drives cell growth and proliferation in many cancers. *MYC* has been difficult to inhibit directly because it is largely intrinsically disordered and lacks a stable conventional small-molecule binding pocket.

Rather than binding finished *MYC* protein, these compounds recognise features of a particular acidic amino-acid stretch while *MYC* is being synthesised.

MYC mRNA reaches target sequence
→ corresponding nascent *MYC* peptide enters ribosome
Interdictor binds ribosome and engages nascent chain
→ *MYC* translation stalls
→ less *MYC* protein is produced

This is conceptually distinct from targeted degradation. A PROTAC such as vepdegestrant removes an already made protein by recruiting an E3 ubiquitin ligase. An Interdictor acts earlier: it prevents completion of the protein during translation.

PROTAC
→ degrade existing target protein
Sequence-selective translation inhibitor
→ reduce synthesis of new target protein

The reported compounds reduced *MYC* production, inhibited growth of *MYC*-dependent cancer cells, and showed oral anti-tumour activity in triple-negative breast-cancer mouse models.

The work is genuinely interesting because it suggests a possible way to drug proteins that lack a conventional pocket, using the combination of the ribosome and a short nascent peptide sequence as a temporary, druggable binding environment.

However, this remains early preclinical work. A short amino-acid motif can occur in proteins other than *MYC*, so apparent sequence selectivity must be tested carefully across the whole cellular proteome. The therapeutic window will depend on whether enough *MYC* suppression can be achieved without unintended inhibition of other essential proteins or broader ribosomal toxicity.

Selective *MYC* translation stalling in models
≠ clinically selective *MYC* inhibition
The key advance:
use a growing protein sequence to create
a transient drug-binding context in the ribosome
The key challenge:
obtain sufficient target selectivity and safety
for therapeutic use

145 [26-09-30] Reprogramming glioma stem cells into dendritic-cell-like cells

(*In situ* reprogramming of glioma stem cells into dendritic cell-like cells for glioblastoma immunotherapy, <https://www.nature.com/articles/s41467-026-78070-2>)

This study proposes an unusual immunotherapy strategy for *glioblastoma*: rather than only trying to kill tumour cells directly, it aims to convert some tumour-propagating *glioma stem cells* into dendritic-cell-like cells within the tumour.

Glioma stem cells are a treatment-resistant tumour-cell population that can help glioblastoma regrow after surgery, radiation, or chemotherapy. They also contribute to the highly immune-suppressive, or “cold”, glioblastoma microenvironment.

```
Glioma stem cell
→ tumour growth, recurrence, and immune suppression
In situ reprogramming
→ dendritic-cell-like state
Dendritic-cell-like tumour-derived cell
→ antigen presentation and local T-cell activation
```

145.1 Why make dendritic-cell-like cells?

Dendritic cells are specialised *antigen-presenting cells*. They take up proteins, process them into peptides, display these on MHC molecules, and provide co-stimulatory signals that activate T cells.

The authors use a set of transcription factors—PU.1, IRF8 and BATF3—to force glioma stem cells towards a dendritic-cell-like programme. These factors normally help establish aspects of dendritic-cell identity.

```
Forced expression of PU.1, IRF8 and BATF3
→ glioma stem cell changes gene-expression programme
Tumour-derived dendritic-cell-like cell
→ presents tumour antigens
+ produces inflammatory signals
+ recruits or activates cytotoxic T cells
```

The useful feature is that the converted cells already contain tumour antigens. In principle, they can become local antigen-presenting cells displaying the same tumour-derived material that T cells need to recognise.

145.2 Why might this help in glioblastoma?

Glioblastoma often has few functional tumour-reactive T cells, limited antigen presentation, and strong local immune suppression. Checkpoint blockade alone may therefore have little immune response to release.

This strategy tries to create the missing first step: productive local T-cell priming and recruitment.

```
Poor antigen presentation in glioblastoma
→ weak anti-tumour T-cell response
Tumour-cell reprogramming into APC-like cells
→ increased antigen presentation and inflammation
More tumour-reactive T cells
→ possible improved immune control
```

It therefore has a potentially complementary rationale with checkpoint inhibition: reprogramming could help generate or recruit tumour-reactive T cells, while PD-1 blockade could prevent those cells from becoming inhibited within the tumour.

145.3 What should be concluded?

The central conceptual advance is that a harmful tumour-cell state may be converted into a cell type that helps expose the tumour to immunity. This is more than killing cancer cells: it aims to change their identity and function within the tumour microenvironment.

However, this remains preclinical gene- and cell-reprogramming research. Efficiently delivering reprogramming factors throughout a human brain tumour, ensuring stable and functional conversion, avoiding effects in normal brain cells, and controlling inflammation are substantial translational challenges.

Dendritic-cell-like conversion and immune control in models
≠ established glioblastoma immunotherapy
The key challenge:
achieve sufficiently selective and extensive
intratumoural reprogramming without harmful
brain inflammation or off-target effects

146 [26-09-30] Cytokine-armoured CAR-T cells: turning engineered T cells into local immune “micropharmacies”

(Progress in the development of cytokine armoured CAR T cells, <https://www.nature.com/articles/s41577-026-01280-8>)

An *armoured CAR-T cell* is a CAR-T cell engineered with additional features intended to help it function in a hostile disease environment. A conventional CAR gives a T cell a new antigen-recognition system; an armoured CAR adds further biological capabilities, such as cytokine secretion, resistance to suppressive signals, improved trafficking, or a safety switch.

This is particularly relevant for solid tumours. Unlike many blood cancers, solid tumours can be difficult for CAR-T cells to enter and contain suppressive immune cells, poor nutrient availability, hypoxia, dense extracellular matrix, and inhibitory signals such as TGF- β and PD-L1.

Conventional CAR-T cell
→ recognises chosen surface antigen
→ may kill antigen-positive tumour cell
Solid-tumour microenvironment
→ poor CAR-T infiltration, exhaustion,
and immune suppression
Armoured CAR-T cell
→ CAR recognition
+ added function to resist or remodel
the tumour environment

146.1 What is a cytokine-armoured CAR-T cell?

A *cytokine-armoured* CAR-T cell is engineered to produce one or more immune-signalling proteins, usually cytokines. The goal is to support the CAR-T cell itself and/or make the local tumour microenvironment more favourable for immune attack.

CAR-T cell encounters tumour antigen
→ CAR activation
Engineered cytokine production
→ supports CAR-T survival or killing
+ recruits or activates other immune cells
+ counteracts local immune suppression

The desired effect is local and conditional: the CAR-T cell should release its cytokine mainly after reaching and recognising tumour cells, rather than producing it continuously throughout the body.

146.2 Which cytokines are used?

Different cytokines solve different problems.

IL-15 can improve T-cell and NK-cell survival, proliferation, persistence, and cytotoxic function. It is intended to help CAR-T cells remain active rather than disappearing or becoming terminally dysfunctional. The GPC3 CAR-T case report discussed earlier used co-expression of IL-15 and IL-21 for this purpose.

IL-21 can support cytotoxic T-cell function and a less exhausted T-cell state. It may therefore complement IL-15 by helping transferred T cells remain functional over time.

IL-12 is a potent inflammatory cytokine. It can activate T cells and NK cells, increase interferon- γ production, improve antigen presentation, and reprogramme suppressive myeloid cells. Its potency also creates substantial systemic-toxicity risk.

IL-18 can enhance inflammatory immune activity, including T-cell and NK-cell function, and may help recruit or activate other immune cells in tumours.

```
IL-15 or IL-21
→ mainly support CAR-T-cell fitness,
persistence, and cytotoxic function
IL-12 or IL-18
→ stronger local inflammatory remodelling
+ recruitment or activation of endogenous immunity
All cytokines:
→ potential benefit
+ potential systemic inflammatory toxicity
```

146.3 Why can cytokine armour help with antigen heterogeneity?

A conventional CAR-T cell kills only cells expressing the chosen CAR target. If some tumour cells lack that antigen, they can survive and expand; this is *antigen escape*.

A cytokine-armoured CAR-T cell may recruit and activate the patient's own T cells, NK cells, dendritic cells, or macrophages. These endogenous immune cells can potentially recognise other tumour antigens, broadening the anti-tumour response beyond the original CAR target.

```
CAR-T cell kills antigen-positive tumour cell
→ tumour antigens and inflammatory signals released
Cytokine armour
→ activates local endogenous immune cells
Endogenous immune response
→ may also attack antigen-negative tumour cells
```

This is an attractive hypothesis, especially for solid tumours, but it requires an immune system capable of responding and a tumour environment that can be sufficiently reprogrammed.

146.4 How is cytokine activity controlled?

The main challenge is that systemic cytokine exposure can cause severe inflammation, cytokine-release syndrome, capillary leak, organ injury, or damage to normal tissues. More cytokine is therefore not necessarily better.

Some designs place cytokine expression under the control of activation-responsive genetic elements, often based on *NFAT*. NFAT becomes active after CAR signalling, so cytokine release is intended to occur mainly when the CAR-T cell recognises its target.

```
Constitutive cytokine expression
→ continuous cytokine production
→ higher risk of systemic toxicity
and CAR-T exhaustion
Antigen-induced cytokine expression
→ cytokine produced after CAR activation
→ potentially greater local activity
and improved therapeutic window
```

Other strategies include membrane-tethered cytokines, cytokine receptors engineered into the CAR-T cell, tumour-responsive logic circuits, or inducible safety switches that can eliminate CAR-T cells if severe toxicity occurs.

146.5 Armoured CAR-T cells more broadly

“Armoured” does not only mean cytokine-producing. Other armour designs include a *dominant-negative TGF- β receptor*, which prevents suppressive TGF- β signalling; a *switch receptor*, such as a PD-1 extracellular domain linked to a stimulatory intracellular domain; added chemokine receptors to improve tumour trafficking; and gene edits that reduce exhaustion or suppressive signalling.

```
Cytokine armour
→ add local immune stimulation
TGF- $\beta$  resistance
→ protect CAR-T cells from suppressive cytokine
Chemokine receptor armour
→ improve tumour trafficking
Checkpoint switch receptor
→ convert inhibitory tumour signal
into activation signal
```

The wider principle is that CAR-T cells can be programmed as living therapies with multiple functions, rather than only as antigen-directed killers.

146.6 What should be concluded?

Cytokine armour is a logical response to the central challenge of solid-tumour CAR-T therapy: target recognition alone is usually insufficient. CAR-T cells must also reach the tumour, persist, withstand local suppression, and ideally recruit broader immunity.

However, most cytokine-armoured CAR-T approaches remain preclinical or in early clinical development. Each added engineering feature can improve function but also creates new uncertainty about manufacturing, dose control, persistence, toxicity, and product consistency.

```
Cytokine-armoured CAR-T cell
→ potentially stronger and broader
anti-tumour immune activity
But:
More engineered immune activity
≠ safer or more effective therapy
The central design problem:
maximise tumour-local immune support
while retaining precise control over
systemic inflammation and long-term persistence
```

147 [26-09-30] Obesity can reduce CAR-T-cell durability in triple-negative breast cancer

(*Obesity impairs the antitumor activity of CAR-T cells in triple-negative breast cancer*, <https://www.nature.com/articles/s41523-026-01055-3>)

This preclinical study shows that obesity can reduce the anti-tumour activity of CAR-T cells in triple-negative breast cancer, or TNBC. It is an important reminder that CAR-T efficacy depends not only on the CAR target and engineered T-cell product, but also on the patient's systemic metabolic state and tumour microenvironment.

The study used CAR-T cells directed against *B7-H3*, a cell-surface protein expressed by many TNBC cells. In obesity-associated models, CAR-T cells could still kill tumour cells in simplified *in vitro* experiments, but their tumour control was weaker *in vivo*.

```
CAR-T cells + tumour cells in culture
→ preserved killing capacity
CAR-T cells in obesity-associated tumour environment
→ reduced persistence and anti-tumour control
```

This difference matters because a tumour is not simply a collection of cancer cells. It is an environment containing nutrients, fats, cytokines, blood vessels, extracellular matrix, and suppressive immune cells.

147.1 How might obesity impair CAR-T cells?

Obesity creates chronic low-grade inflammation and can alter nutrient availability, lipid metabolism, and immune-cell function. The study found that CAR-T cells in obese models had impaired metabolic fitness, including reduced glycolytic capacity.

As discussed for mitochondrial ATP and T-cell differentiation, metabolism is not merely a source of fuel. T cells need metabolic flexibility to proliferate, migrate, make cytokines, and sustain cytotoxic activity in a hostile tumour environment.

```
Obesity-associated metabolic and inflammatory environment
→ altered CAR-T-cell gene-expression programme
+ reduced metabolic fitness
Reduced metabolic fitness
→ weaker expansion, persistence,
and sustained tumour killing
```

The CAR-T cells also showed changes associated with dysfunction or terminal differentiation, including increased expression of genes such as *TOX2*, *PRDM1*, and *BATF*. These do not mean that every CAR-T cell is completely inactive; rather, they suggest a shift towards a less durable functional state.

147.2 Why is tissue-resident memory important?

A notable finding was reduced formation of *tissue-resident memory T cells*, or T_{RM} cells, in obese models. T_{RM} cells remain locally within a tissue after an immune response and can respond rapidly if the same threat returns.

```
Effective CAR-T response
→ tumour killing
+ formation of durable local memory-like T cells
Obesity-associated impairment
→ fewer  $T_{RM}$ -like CAR-T cells
→ weaker protection against tumour recurrence
```

Thus, obesity may affect not only the initial response but also its durability after apparent tumour control.

147.3 What should be concluded?

The paper does *not* show that CAR-T therapy cannot work in people with obesity. It shows that obesity is a biologically important variable that can alter CAR-T-cell fitness and treatment durability in TNBC models.

It also fits the earlier discussion of armoured CAR-T cells. In solid tumours, improving antigen recognition alone may be insufficient; CAR-T cells may need protection against local TGF- β , metabolic stress, hypoxia, and systemic metabolic inflammation.

```
Appropriate tumour antigen
≠ guaranteed CAR-T-cell efficacy
CAR-T-cell product
+
tumour microenvironment
+
host metabolic state
→ determines the final treatment response
```

Potential solutions could include metabolic CAR-T engineering or combinations that alter suppressive tumour environments. However, these remain hypotheses until clinical studies establish whether obesity predicts CAR-T outcomes and whether such interventions safely improve them.

148 [26-09-30] Werner helicase inhibition: exploiting an MSI-specific DNA-maintenance dependency

(*Werner helicase inhibitor for advanced microsatellite instability solid tumors: a phase 1 trial*, <https://www.nature.com/articles/s41591-026-04657-3>)

This Phase I trial tests *VVD-214*, also called RO7589831, an oral inhibitor of *Werner helicase*, or WRN. It is designed for tumours with high microsatellite instability, or *MSI-H*, and/or deficient mismatch repair, or *dMMR*.

WRN is a DNA *helicase*: an enzyme that uses energy from ATP to unwind DNA and help resolve difficult DNA structures during replication and repair. It is named after Werner syndrome, a rare inherited premature-ageing disorder caused by loss-of-function mutations in the *WRN* gene.

148.1 Why are MSI-H cancers dependent on WRN?

Microsatellites are short repeated DNA sequences, for example:

```
Normal sequence:
CACACACACACA
Microsatellite:
short DNA motif repeated many times
```

These repeats are prone to copying errors during DNA replication. Normally, the *mismatch-repair*, or MMR, system detects and corrects many of these errors.

DNA replication error in microsatellite
→ mismatch-repair system corrects error
→ repeat length remains relatively stable

If MMR is defective, repeat lengths change over time. This is microsatellite instability. MSI-H tumours accumulate many such alterations and can develop abnormal DNA structures that are difficult to copy safely.

Mismatch-repair deficiency
→ microsatellite instability
→ unstable repetitive DNA
→ increased replication-associated DNA problems
WRN helicase
→ helps resolve these problematic structures
→ allows MSI-H tumour cells to survive

Normal cells also use WRN, but MSI-H cancer cells appear to be unusually dependent on it. This creates a possible *synthetic-lethal* relationship.

148.2 How does VVD-214 work?

VVD-214 is a covalent allosteric WRN inhibitor. “Allosteric” means it binds outside WRN’s main ATP-binding site and changes the enzyme’s function through a structural effect. “Covalent” means it forms a durable chemical bond with WRN.

MSI-H tumour cell
→ high burden of problematic repeat-associated DNA structures
VVD-214 inhibits WRN
→ these structures are not resolved adequately
→ replication stress and chromosome damage accumulate
→ tumour-cell death may occur

This resembles the earlier examples of PARP inhibition in BRCA-deficient cancer, ATR inhibition under replication stress, and PUM3 dependency in HR-deficient breast cancer. In each case, a tumour acquires one defect and becomes unusually reliant on a remaining DNA-maintenance pathway.

Mismatch-repair deficiency
+ WRN inhibition
→ selective collapse of DNA maintenance
This is not simply:
"WRN inhibitor damages all dividing cells equally."
It is intended as:
"MSI-H cells have a greater dependence on WRN than most normal cells."

148.3 What did the trial show?

The study enrolled heavily pre-treated patients with MSI-H/dMMR advanced solid tumours. It reported early anti-tumour activity, including durable disease control and partial responses in cancers such as colorectal, ovarian, and endometrial cancer, including some patients previously treated with checkpoint inhibitors.

Among an early efficacy-evaluable group, the objective response rate was approximately **14%**, with disease control in around **two-thirds** of patients. The safety profile was considered manageable; gastrointestinal adverse events such as nausea, diarrhoea, and vomiting were common but largely low grade.

This is an important early clinical result because it supports the underlying biological hypothesis: MSI-H status may identify a cancer-selective dependency on WRN that can be targeted pharmacologically.

148.4 What should be concluded?

WRN inhibition is an example of precision oncology based on a *DNA-repair phenotype*, rather than an activating kinase mutation or a cell-surface antigen.

```
MSI-H/dMMR status
→ predictive biomarker hypothesis
WRN inhibitor
→ exploits resulting replication-stress dependency
Early clinical responses
→ first evidence that this dependency
can be targeted in patients
```

However, this remains an early Phase I result. MSI-H status alone may not identify every responder, and resistance is likely to emerge through changes that reduce WRN dependence or restore tolerance of repeat-associated DNA damage. The central next questions are which MSI-H tumour subgroups benefit most, how durable responses are, and whether WRN inhibitors can be combined safely with immunotherapy or other DNA-damage-response treatments.

149 [26-09-30] RNA-binding proteins: controlling the life of RNA inside cells

(*The cellular landscape of druggable RNA-binding proteins*, <https://www.nature.com/articles/s41573-026-01531-2>)

RNA-binding proteins, or RBPs, are proteins that bind RNA molecules and control what happens to them. They are important because RNA is the intermediate between DNA and protein:

```
DNA gene
→ RNA transcript
→ protein
RNA-binding proteins
→ control how, when, where, and whether
that RNA produces protein
```

After a gene is transcribed into messenger RNA, or mRNA, the mRNA is not simply translated immediately. It must be processed, transported, stabilised, and eventually degraded. Different RBPs bind particular RNA sequences or RNA structures and influence these steps.

```
RBP binds RNA
→ can alter RNA splicing,
localisation, stability, translation, or degradation
Changed RNA behaviour
→ changed amount, timing,
or location of protein production
```

For example, an RBP may stabilise an mRNA encoding a growth-promoting protein, allowing more of that protein to be made. Another RBP may repress translation until an RNA reaches the correct part of a cell.

RBPs are therefore major regulators of cell identity, stress responses, development, immunity, and neuronal function.

149.1 Why are RBPs relevant to disease?

Abnormal RBP activity can change many RNAs simultaneously. In cancer, an RBP may stabilise RNAs that promote proliferation, survival, metastasis, or drug resistance. In neurodegenerative disease, RBPs such as TDP-43 or FUS can mislocalise or aggregate, disrupting normal RNA regulation.

Dysregulated RBP
→ altered RNA programme
Many proteins change together
→ cancer-cell fitness, inflammation,
or neuronal dysfunction may be promoted

This makes RBPs attractive therapeutic targets, but also potentially risky: inhibiting one RBP can affect many RNAs and many cell types.

149.2 Why have RBPs been difficult to drug?

Many RBPs lack the deep, well-defined binding pockets found in enzymes or receptors. Their RNA-binding surfaces are often broad, relatively flat, and dynamic. This resembles the challenge of targeting transcription factors or intrinsically disordered proteins.

Conventional drug target:
deep pocket
→ small molecule can bind selectively
Many RBP-RNA interfaces:
broad, flexible surface
→ conventional inhibition is more difficult

However, “difficult” does not mean impossible. Potential strategies include small molecules that alter RBP–RNA binding, antisense oligonucleotides that block a particular RNA interaction, molecular glues or degraders that remove the RBP, and drugs that bind the RNA itself.

149.3 What does the “cellular landscape” mean?

The review considers RBPs in their cellular context: which cells express them, which RNAs they bind, where they act inside cells, and how this changes across tissues or disease states.

This matters because the same RBP can have different functions in different cells. A broadly expressed RBP may be a poor target if normal tissues depend on it, whereas an RBP that is unusually important in a tumour cell state may offer a better therapeutic window.

RBP is associated with disease
≠ suitable drug target
Useful target requires:
disease-relevant RNA programme
+ tractable way to intervene
+ selective enough effect in diseased cells
+ acceptable safety margin

The broader connection to this document is that RBPs sit between *genetic information* and *protein output*. They offer a way to change a pathological programme upstream of many individual proteins, but this broad influence also makes selectivity and toxicity central challenges.

150 [26-10-02] Tolinapant plus pembrolizumab: making tumour cells more vulnerable to immune-mediated death

(*IAP antagonist tolinapant (ASTX660) in combination with pembrolizumab in advanced solid tumors: a phase I trial*, <https://www.nature.com/articles/s41467-026-77552-7>)

This Phase I study examines *tolinapant*, an oral small-molecule *IAP antagonist*, in combination with pembrolizumab, an anti-PD-1 checkpoint inhibitor.

The combination addresses two connected problems. Pembrolizumab aims to restore the ability of T cells to attack a tumour, whereas tolinapant aims to make tumour cells less able to resist the cell-death signals produced during that immune attack.

150.1 What are IAP proteins?

IAPs, or *inhibitor-of-apoptosis proteins*, are proteins that can protect cells from programmed cell death. Important IAPs include XIAP, cIAP1, and cIAP2.

Apoptosis is a controlled form of cell death involving enzymes called *caspases*. IAP proteins can restrain caspase activity and alter signalling downstream of inflammatory cytokines such as tumour necrosis factor, or TNF.

Death signal, such as TNF
→ cell can activate survival or death pathways
IAP proteins active
→ favour survival signalling
+ restrain caspase-mediated apoptosis
IAP proteins inhibited
→ cell becomes more susceptible
to apoptosis and related cell-death pathways

Cancer cells can exploit these protective pathways to survive inflammatory stress, chemotherapy, and immune-cell attack.

150.2 How does tolinapant work?

Tolinapant mimics the action of *SMAC*, a natural cellular protein that antagonises IAPs. It binds and inhibits XIAP, cIAP1, and cIAP2.

Inhibition of cIAP1/2 can also cause their degradation, changing TNF-associated signalling. The overall result is that, in a susceptible tumour cell, inflammatory death signals are more likely to result in cell death rather than survival.

Tolinapant inhibits XIAP and cIAP1/2
→ less protection from caspase activation
+ altered TNF-associated survival signalling
TNF or other death signal present
→ apoptosis and/or related programmed
cell-death pathways become more likely

This does *not* mean tolinapant directly kills every cancer cell by itself. Its activity depends on cellular context, including whether the tumour has an intact death-signalling pathway and whether inflammatory signals such as TNF are present.

150.3 Why combine it with pembrolizumab?

Pembrolizumab blocks PD-1, an inhibitory receptor on activated T cells. PD-1 blockade can restore T-cell cytokine production and cytotoxic activity against tumour cells.

Activated T cells can produce TNF and other inflammatory signals. Therefore, pembrolizumab may increase the very immune pressure to which IAP inhibition makes tumour cells more vulnerable.

```
Pembrolizumab blocks PD-1
→ more functional anti-tumour T cells
→ increased cytotoxicity and TNF-associated signals
Tolipapant inhibits IAP-mediated survival
→ tumour cells are less able to resist
those immune death signals
Combination
→ stronger immune attack
+ greater tumour-cell susceptibility to death
```

Tolipapant may also make tumour-cell death more *immunogenic*: dying cells can release antigens and danger signals that support antigen presentation and recruit further immune activity. This creates a possible positive feedback loop, although it is not guaranteed in every tumour.

150.4 What should be concluded?

The combination has a coherent immunobiological rationale. It is related to the ATG–SMAC-mimetic example discussed earlier: both strategies attempt to expose a TNF-dependent cell-death vulnerability by removing IAP-mediated protection.

However, this remains early clinical development. TNF and IAP pathways also have important roles in normal immune and tissue biology, so systemic IAP inhibition may cause toxicity. Moreover, a tumour can resist the combination if it lacks effective T-cell infiltration, has defective antigen presentation, or has lost downstream components needed for programmed cell death.

```
PD-1 blockade
+ IAP antagonism
→ plausible combination of immune activation
and tumour-cell death sensitisation
But:
Plausible mechanism
≠ proven clinical synergy
The key question:
which tumour types and biomarkers identify cancers
in which immune-derived death signals can be converted
into a meaningful and durable clinical response?
```

151 [26-10-02] Semaglutide slows kidney-disease progression in type 2 diabetes

(*Effects of semaglutide on kidney disease in type 2 diabetes: a randomized placebo-controlled trial*, <https://www.nature.com/articles/s41591-026-04674-2>)

This paper reports the *FLOW* trial, which tested weekly semaglutide in people with type 2 diabetes and chronic kidney disease, or CKD. It is important because semaglutide is usually associated with blood-glucose control and weight loss, but this trial tested whether it also improves major kidney outcomes.

Diabetes can damage small blood vessels and filtering structures in the kidney. Over time, this can cause albumin to leak into urine, progressive loss of filtration capacity, kidney failure, and increased cardiovascular risk.

Type 2 diabetes
→ high glucose, metabolic stress,
inflammation, and vascular injury
Chronic kidney damage
→ albuminuria and declining eGFR
→ kidney failure and cardiovascular risk

eGFR, or estimated glomerular filtration rate, is a blood-test-based estimate of kidney filtering capacity. A lower eGFR generally indicates worse kidney function. *Albuminuria* means albumin is detected in urine, indicating damage to the kidney's filtration barrier.

Semaglutide is a *GLP-1 receptor agonist*. It mimics the gut hormone GLP-1, increasing insulin secretion when glucose is elevated, reducing glucagon, suppressing appetite, and often producing weight loss.

Semaglutide activates GLP-1 receptor
→ improved glucose regulation
+ reduced food intake and body weight
Reduced metabolic and cardiovascular stress
→ may indirectly protect kidney function

The trial found that semaglutide reduced the risk of major kidney-disease events by approximately ****24%**** compared with placebo. The composite kidney endpoint included kidney failure, a major sustained decline in eGFR, or death from kidney-related or cardiovascular causes. Semaglutide also slowed the average decline in eGFR.

The kidney benefit is probably not explained by one direct effect alone. Better glucose control, weight loss, lower blood pressure, reduced inflammation, and improved cardiovascular health may all reduce stress on the kidney.

Semaglutide treatment
→ improved metabolic control
+ weight loss
+ cardiovascular benefit
+ slower loss of kidney function
Therefore:
GLP-1 receptor activation
→ can affect more than blood glucose alone

This does not mean semaglutide replaces established CKD treatments. Patients in FLOW received standard care, commonly including renin–angiotensin-system inhibition and, where appropriate, SGLT2 inhibitors. Semaglutide is best understood as an additional treatment option for selected people with type 2 diabetes and CKD.

Reduced risk of composite kidney outcome
≠ reversal of established kidney damage
The key conclusion:
semaglutide can slow progression of diabetic CKD
and reduce cardiovascular risk,
when used alongside appropriate standard kidney care